



mathematics of evolution & phylogeny

EDITED BY
OLIVIER GASCUEL



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Olivier Gascuel
Montpellier-Montréal, June 2004

INTRODUCTION

Olivier Gascuel

The subject of this volume is evolution, which is considered at different scales: sequences, genes, gene families, organelles, genomes, and species. The focus is on the mathematical and computational tools and concepts, which form an essential basis of evolutionary studies, indicate their limitations and, inevitably, give them orientation. Recent years have witnessed rapid progress in this area, with models and methods becoming more realistic, powerful, and complex. This expansion has been driven by the phenomenal increase in genomic data available. Databases now contain tens of billions of sequence base pairs. Hundreds of species' genomes, including most notably the human genome, have been completely sequenced. This flood of data demands the development and use of formal mathematical, statistical, and computational methods. Tools derived from an evolutionary perspective are not the only ones, but they play a central part. Indeed, Nature did not explore all physical and chemical possibilities open to her. All components of life (e.g. proteins) have specific histories, which are of a great help for understanding their functions and mechanisms. Simple comparisons are often enough to obtain deep insight into the structure, function, and role of sequences, while chemical and physical approaches (e.g. energy minimization) are more problematic and can only be applied at a late confirmatory or refinement stage. It is no accident that many of the most widely used bioinformatics tools, for example, BLAST [2] and Neighbour Joining [39], have an evolutionary basis.

Research in evolution and genetics has also been a driving force in mathematics, statistics, and computer science [41]. Recall that R.A. Fisher, the founder of so many central concepts in statistics, was primarily a geneticist. Branching processes were first seen in the field of particle physics, but were also investigated by Yule to model the speciation process [49], and recently have been the subject of much work in the field of evolution, with important results on random trees [1, 30]. The first studies of tree metrics were partly conducted from an evolutionary perspective [9, 10, 20, 50]. Later developments, generally motivated by problems in evolution, have led to fundamental results in combinatorics [3], geometry [4], and probability theory [43]. As a final example, the recent profusion of research into genome rearrangements has undoubtedly promoted a new vision and understanding of permutations of finite sets [23].

This volume follows a conference organized at Institut Henri Poincaré (Paris, June 2003). Following enthusiastic feedback from the participants, we asked the speakers to write survey chapters based on the research they had presented, with the aim of compiling a compact summary of the state-of-art mathematical

techniques and concepts currently used in the field of molecular phylogenetics and evolution. The key to the success of this conference lay in the scientific relevance and timeliness of the subjects presented (e.g. [45]), and their multidisciplinary nature.

Evolutionary patterns, processes, and history

Evolutionary studies most often have multiple aims: determining the rates and patterns of change occurring in DNA sequences, proteins, organelles or genomes, and reconstructing the evolutionary history of those entities and of organisms and species. A general goal is to infer process from pattern: the processes of organism evolution deduced from patterns of DNA or genomic variation, and processes of molecular or genomic evolution inferred from the patterns of variations in the DNA or genome itself. Given patterns observed today, the aim is then to reconstruct the history (typically a phylogenetic tree) and to understand the processes that govern evolution. Consequently, a large part of this volume is devoted to mathematical (mostly Markov) models of sequence and genome evolution. These models are used to reconstruct phylogenetic trees or networks, for example using maximum-likelihood or Bayesian approaches. The aim is not only to obtain accurate reconstructions but also to check the models' fidelity in reflecting the evolution of the sequences or genomes. Model design has therefore been thoroughly researched during recent years, both at the sequence (e.g. [21, 48]) and genome (e.g. [31, 46]) levels, with a subsequent dramatic improvement in accuracy of phylogenetic reconstruction.

Comparative and functional genomics

One of the central goals in bioinformatics is to infer the function of proteins from genomic sequences. To this end, alignment methods are nowadays the most refined and used. Sequence alignment attempts to reconstruct evolution by postulating substitution, insertion, and deletion events that occurred in the past [40]. The mutation process is described by Markov models such as the famous Dayoff [11] and JTT matrices [25]. Related or "homologous" proteins are assumed to share a common ancestor and usually have similar structure and function. We distinguish paralogous proteins (separated by one or more duplication event) from orthologous proteins (derived through speciation only) [18]. Since duplication is one of the major evolutionary processes triggering functional diversification [32], only orthologous proteins are likely to share the same function. Assessing orthology is a complicated task that requires phylogenetic analysis of an extensive set of homologous proteins [44].

When the first genomes were fully sequenced, one of the main surprises was that only about half of the proteins of an organism were considered homologous to proteins already in databases. Alignment therefore gives indications of the function of only 50% of proteins in a genome. This limit has encouraged the development of new methods that exploit the information contained within the full genomic sequence. Phylogenomic profiling [14] is one of the major

non-alignment-based methods. It is designed to infer a likely functional relationship between proteins, and is based on the assumption that proteins involved in a common metabolic pathway, or constituting a molecular complex, are likely to evolve in a correlated manner. Each protein is given a phylogenetic profile denoting the presence or absence of that protein in various genomes with a known phylogeny. Similar or complementary function can then be assigned to proteins if they have a similar phylogenetic profile. A number of other approaches have been proposed. For example, conservation of gene clusters between genomes allows the prediction of functional coupling between genes [26, 33]. Phylogenetic footprinting [5] is a method for the discovery of regulatory elements in a set of homologous regulatory regions, making use of the phylogenetic relationships among those sequences. The detection of lateral gene transfer from multi-gene or genome sequence analysis gives insight on genome adaptation [29]. These methods are examples of the pervasiveness of the feedback loops between genomic analysis and evolutionary studies, and are grouped into the new field of “phylogenomics” [13].

Tree of Life

The genomics database GenBank has information on about 100,000 species. More than 4 million species of organisms have been discovered and described, and it is estimated that tens of millions remain to be discovered. Placing these species in the Tree of Life is among the most complex and important problems facing biology [45]. Since the mid-1980s, there has been an exponential growth in the number of phylogenetic papers published each year. Recently, the Deep Green consortium achieved a first draft of the phylogeny of all green plants [7, 35]. The Tree of Life project therefore promises to be a substantial, international research program involving thousands of biologists, computer scientists, and mathematicians. The scientific aim is to understand the origins of life, the shape of its evolution, the extent of modern biodiversity, and its vulnerability to existing or possible threats. Indeed, phylogenetic analysis is playing a major role in discovering new life forms. For example, many microorganisms cannot be cultivated and studied in the laboratory, thus the principal road to discovery is to isolate their DNA from samples collected from water or soils. The DNA samples are then sequenced and identified using phylogenetic analyses based on sequences of previously described organisms. This has led to the discovery of major microbial lineages, especially in the Archaea group. Phylogenetic analysis is also of primary importance in epidemiology. Understanding how organisms, as well as their genes and gene products, are related to one another has become a powerful tool for identifying disease organisms, for tracing the history of infections, and for predicting outbreaks. Phylogenetic studies have been crucial in identifying emerging viruses such as SARS [28]. Many other examples (e.g. in agriculture) could be given to illustrate the relevance of the Tree of Life project. Most important is the fact that phylogenetic knowledge is increasingly invaluable to the effort to mine, organize, and exploit the enormous amount of biological data held in numerous databases worldwide.

Biodiversity, ecology, and comparative biology

In the near future the Tree of Life should become the most natural way to represent biodiversity. With initiatives to sequence all the biota on the horizon [47], the amount of sequence data in public domain is rapidly accumulating, and it could even be that an organism's place in the Tree of Life will often be one of the few things known about it. Moreover, phylogenies provide new ways to measure biodiversity, to survey invasive species and to assess conservation priorities [27]. Notably, dated interspecies phylogenies contain information about rates and distributions of species extinctions and about the nature of radiations after previous mass extinctions [6]. Phylogenetic comparative approaches have also modelled extinction risk as a function of species' biological characteristics [36], which could be used as a basis for evaluating the status of species with unknown extinction risk. Comparative studies in biology also make an extensive use of phylogenetics when investigating adaptative traits and circumstances of adaptation [16, 24]. Indeed, species descended from a common ancestor are expected to resemble each other simply because they are related, and not necessarily because their common traits have common adaptive functions. We thus need phylogenies to infer which species are related; we need to know ancestral traits so that we can figure out what has evolved and when; and we need to know evolutionary dynamics to predict how often we should expect "chance" (i.e. non-adaptive) associations.

The goal of this volume is not to describe the numerous applications of phylogenetics and of other approaches that aim at reconstructing specific aspects of evolution. A large number of textbooks discuss the subjects rapidly surveyed above (e.g. [17, 22, 34]). Here, we concentrate on the fundamental mathematical concepts and research into current reconstruction methods. We describe a number of (probabilistic or combinatorial) models that address evolution at different scales, from segments of DNA sequences to whole genomes. We detail methods and algorithms that exploit such models for reconstructing phylogenetic trees and networks, and other mathematical techniques for various evolutionary inferences, for example, molecular dating. We explain how these reconstructions can be tested in a statistical sense and what are the inherent limits of these reconstructions. Finally, we present a number of mathematical results which give an in-depth understanding of the phylogenetic tools.

This volume is organized in fourteen chapters:

1 The minimum evolution distance-based approach of phylogenetic inference

Distance-based methods such as UPGMA [42] and Neighbour Joining [39] were among the first techniques used to reconstruct phylogenies. These methods are still widely used as they combine reasonable accuracy and computational speed. This chapter presents the most recent developments of distance-based methods, with a focus on the minimum evolution principle, which forms the basis of Neighbour Joining and other improved inference algorithms [12].

2 Likelihood calculation in molecular phylogenetics

Likelihood estimation was first introduced in molecular phylogenetics by Felsenstein [15], and is now widely used due to its accuracy and to the fact that it makes explicit the assumptions about the evolutionary model. This chapter outlines the basic probabilistic model and likelihood computation algorithm, as well as extensions to more realistic models and strategies of likelihood optimization. It surveys several of the theoretical underpinnings of the likelihood framework: statistical consistency, identifiability, effect of model misspecification, as well as advantages and limitations of likelihood ratio tests.

3 Bayesian inference in molecular phylogenetics

The Bayesian approach to phylogenetic inference was first introduced by Rannala and Yang [37], and is now widely used, thanks, in part, to the MrBayes software [38]. The main advantage of this approach is its ability to accommodate uncertainty, for example, by inferring several alternative phylogenies (instead of a single one) and estimating their respective posterior probabilities. This chapter introduces Bayesian statistics through comparison with the likelihood method. It discusses Markov chain Monte Carlo algorithms, the major modern computational methods for Bayesian inference, as well as two applications of Bayesian inference in molecular phylogenetics: estimation of species phylogenies and estimation of species divergence times.

4 Statistical approaches to test involving phylogenies

Statistical testing is an important issue in phylogenetics, for example to measure the support of a clade or to decide which evolutionary model is best. This chapter presents both the classical framework with the use of sampling distributions involving the bootstrap and permutation tests, and the Bayesian approach using posterior distributions. It contains a review of literature on parametric tests in phylogenetics and some suggestions for non-parametric tests. A number of open problems are discussed, mainly related to the non-conventional nature of tree space.

5 Mixture models in phylogenetic inference

The standard models of sequence evolution presume that sites evolve according to a common model or allow rates of evolution to vary across sites. This chapter discusses how a general class of approaches known as “mixture models” can be used to accommodate heterogeneity across sites in the patterns of sequence evolution. Mixture models fit more than one model of evolution to the data but do not require *a priori* knowledge of the evolutionary patterns across sites or of any site partitioning. The approach is illustrated on a concatenated alignment of 22 genes used to infer the phylogeny of mammals.

6 Hadamard conjugation: an analytic tool for phylogenetics

Phylogenetic inference is the process of estimating an unknown phylogeny from the evolutionary patterns that are observed in a set of aligned homologous sequences, thus inverting the mechanism which generated these patterns. For most models this inversion cannot be analysed directly. This chapter considers simple models of nucleotide substitution where this inversion is possible, thanks to “Hadamard conjugation” (or “phylogenetic spectral analysis”). Hadamard conjugation provides an analytic tool that gives insight into the general phylogenetic inference process. This chapter describes the basics of Hadamard conjugation, together with illustrations of how it can be applied to analyse a number of related concepts, such as the inconsistency of Maximum Parsimony or the determination of Maximum Likelihood points.

7 Phylogenetic networks

Phylogenetic networks are a generalization of phylogenetic trees that permit the representation of conflicting signal or alternative phylogenetic histories. Networks are clearly useful when the underlying evolutionary history is non-treelike, for example, when there has been recombination, hybridization, or lateral gene transfer. Even in cases where the underlying history is treelike, phenomena such as parallel evolution, model heterogeneity, and sampling error can make it difficult to represent the evolutionary history by a single tree, and networks can then provide a useful tool. This chapter reviews some methods for network reconstruction that are based on the representation of bipartitions or splits of the data set in question. As we shall see, these methods are based on a theoretical foundation that naturally generalizes the theory of phylogenetic trees.

8 Reconstructing the duplication history of tandemly repeated sequences

Tandemly repeated sequences can be found in all of the genomes that have been sequenced so far. However, their evolution is only beginning to be understood. In contrast to previous chapters, which study the evolution of orthologous sequences within a number of distant species, the objective in this chapter is to reconstruct the evolutionary history of paralogous sequences that are tandemly repeated within a single genome. This chapter presents a model, first proposed by Fitch [19], which assumes that duplications are caused by unequal recombination during meiosis. Duplication histories are then constrained by this model and duplication trees constitute a proper subset of phylogenetic trees. This chapter demonstrates strong biological support for this model, provides extensive mathematical and combinatorial characterizations of duplication trees, and describes various algorithms to infer tandem duplication trees from sequences.

9 Conserved segment statistics and rearrangement inferences in comparative genomics

This chapter continues the study of genome evolution, but at a much larger scale. Full genomes are compared in order to study genome rearrangements. It is shown that this field has evolved along with the biological methods for producing pertinent data, with each new type of data suggesting new questions and leading to new analyses. The development of conserved segment statistics is traced, from the mouse linkage/human chromosome assignment data analysed by Nadeau and Taylor in 1984, through the comparative gene order information on organelles (late 1980s) and prokaryotes (mid-1990s), to higher eukaryote genome sequences, whose rearrangements have been recently studied without prior gene identification.

10 The reversal distance problem

Among the many genome rearrangement operations, signed inversions stand out for many biological and computational reasons. Inversions, also known as reversals, are widely identified as one of the common rearrangement operations on chromosomes, they are basic to the understanding of more complex operations such as translocations, and they offer many computational challenges. This chapter presents an elementary treatment of the problem of sorting by inversions. It describes the “anatomy” of signed permutations, gives a complete proof of the Hannenhalli–Pevzner duality theorem [23], and details efficient and simple algorithms to compute the inversion distance.

11 Genome rearrangement with gene families

The major focus of the first genome rearrangement approaches has been to infer the most economical scenario of elementary operations transforming one linear order of genes into another. Implicit in most of these studies is that each gene has exactly one copy in each genome. This hypothesis is clearly unsuitable for divergent species containing several copies of highly paralogous genes, such as multigene families. This chapter reviews the different algorithmic methods that have been developed to account for multigene families in the genome rearrangement context, in the phylogenetic context, and when reconstructing ancestral genomes.

12 Reconstructing phylogenies from gene-content and gene-order data

This chapter continues to deal with genome rearrangements, but the focus shifts to phylogenetic reconstruction from gene-content and gene-order data, whereas standard phylogeny methods exploit DNA or protein sequences. Indeed such data offer low error rates, the potential to reach further back in time, and immunity from the so-called gene-tree versus species-tree problem. This chapter surveys

the state-of-the-art techniques that use such data for phylogenetic reconstruction, focusing on recent work that has enabled the analysis of insertions, duplications, and deletions of genes, as well as inversions of gene subsequences. It concludes with a list of research questions that will need to be addressed in order to realize the full potential of this type of data.

13 Distance-based genome rearrangement phylogeny

Evolution operates on whole genomes through mutations, such as inversions, transpositions, and inverted transpositions. This chapter details a Markov model of genome evolution, assuming these three rearrangement operations. The mathematical derivation of various statistically based evolutionary distance estimators is described, and it is shown that the use of these new distance estimators with methods such as Neighbour Joining [39] and Weighbor [8] can result in improved reconstructions of evolutionary history.

14 How much can evolved characters tell us about the tree that generated them?

This chapter reviews some recent results that shed light on a fundamental question in molecular systematics: how much phylogenetic “signal” can we expect from extant data? Both sequence and gene-order data are examined, and evolution is modelled using Markov processes. Results presented here apply to most of the approaches discussed throughout this volume. They provide upper bounds on the probability of accurate tree reconstruction, depending on the number of species, data, and model parameters. The chapter also discusses transition phase phenomena, which make phylogenetic reconstruction impossible when substitution rates exceed a critical value.

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THE MINIMUM EVOLUTION DISTANCE-BASED APPROACH TO PHYLOGENETIC INFERENCE

Richard Desper and Olivier Gascuel

Distance algorithms remain among the most popular for reconstructing phylogenies, especially for researchers faced with data sets with large numbers of taxa. Distance algorithms are much faster in practice than character or likelihood algorithms, and least-squares algorithms produce trees that have several desirable statistical properties. The fast Neighbor Joining heuristic has proven to be quite popular with researchers, but suffers somewhat from a lack of a statistical foundation. We show here that the balanced minimum evolution approach provides a robust statistical justification and is amenable to fast heuristics that provide topologies superior among the class of distance algorithms. The aim of this chapter is to present a comprehensive survey of the minimum evolution principle, detailing its variants, algorithms, and statistical and combinatorial properties. The focus is on the balanced version of this principle, as it appears quite well suited for phylogenetic inference, from a theoretical perspective as well as through computer simulations.

1.1 Introduction

In this chapter, we present recent developments in distance-based phylogeny reconstruction. Whereas character-based (parsimony or probabilistic) methods become computationally infeasible as data sets grow larger, current distance methods are fast enough to build trees with thousands of taxa in a few minutes on an ordinary computer. Moreover, estimation of evolutionary distances relies on probabilistic models of sequence evolution, and commonly used estimators derive from the maximum likelihood (ML) principle (see Chapter 2, this volume). This holds for nucleotide and protein sequences, but also for gene order data (see Chapter 13, this volume). Distance methods are thus model based, just like full maximum likelihood methods, but computations are simpler as the starting information is the matrix of pairwise evolutionary distances between taxa instead of the complete sequence set.

Although phylogeny estimation has been practiced since the days of Darwin, in the 1960s the accumulation of molecular sequence data gave unbiased

sequence characters (in contrast with subjective morphological characters) to build phylogenies, and more sophisticated methods were proposed. Cavalli-Sforza and Edwards [9] and Fitch and Margoliash [19] both used standard least-squares projection theory in seeking an optimal topology. While statistically sound, the least-squares methods have typically suffered from great computational complexity, both because finding optimal edge lengths for a given topology was computationally demanding and because a new set of calculations was needed for each topology. This was simplified and accelerated by Felsenstein [18] in the FITCH algorithm [17], and by Makarenkov and Leclerc [35], but heuristic least-squares approaches are still relatively slow, with time complexity in $O(n^4)$ or more, where n is the number of taxa.

In the late 1980s, distance methods became quite popular with the appearance of the Neighbor Joining algorithm (NJ) of Saitou and Nei [40], which followed the same line as ADDTREE [42], but used a faster pair selection criterion. NJ proved to be considerably faster than least-squares approaches, requiring a computing time in $O(n^3)$. Although it was not clear what criterion NJ optimizes, as opposed to the least-squares method, NJ topologies have been considered reasonably accurate by biologists, and NJ is quite popular when used with resampling methods such as bootstrapping. The value of NJ and related algorithms was confirmed by Atteson [2], who demonstrated that this approach is statistically consistent; that is, the NJ tree converges towards the correct tree when the sequence length increases and when estimation of evolutionary distances is itself consistent. Neighbor Joining has spawned similar approaches that improve the average quality of output trees. BIONJ [21] uses a simple biological model to increase the reliability of the new distance estimates at each matrix reduction step, while WEIGHBOR [5] also improves the pair selection step using a similar model and a maximum-likelihood approach.

The 1990s saw the development of minimum evolution (ME) approaches to phylogeny reconstruction. A minimum evolution approach, as first suggested by Kidd and Sgaramella-Zonta [31], uses two steps. First, lengths are assigned to each edge of each topology in a set of possible topologies by some prescribed method. Second, the topology from the set whose sum of lengths is minimal is selected. It is most common to use a least-squares method for assigning edge length, and Rzhetsky and Nei [39] showed that the minimum evolution principle is statistically consistent when using ordinary least-squares (OLS). However, several computer simulations [11, 24, 33] have suggested that this combination is not superior to NJ at approximating the correct topology. Moreover, Gascuel, Bryant and Denis [25] demonstrated that combining ME with *a priori* more reliable weighted least-squares (WLS) tree length estimation can be inconsistent.

In 2000, Pauplin described a simple and elegant scheme for edge and tree length estimation. We have proposed [11] using this scheme in a new “balanced” minimum evolution principle (BME), and have designed fast tree building algorithms under this principle, which only require $O(n^2 \log(n))$ time and have been implemented in the FASTME software. Furthermore, computer

simulations have indicated that the topological accuracy of FASTME is even greater than that of best previously existing distance algorithms. Recently, we explained [12] this surprising fact by showing that BME is statistically consistent and corresponds to a special version of the ME principle where tree length is estimated by WLS with biologically meaningful weights.

The aim of this chapter is to present a comprehensive survey of the minimum evolution principle, detailing its variants, mathematical properties, and algorithms. The focus is on BME because it appears quite well suited for phylogenetic inference, but we shall also describe the OLS version of ME, since it was a starting point from which BME definitions, properties, and algorithms have been developed. We first provide the basis of tree metrics and of the ME framework (Section 1.2). We describe how edge and tree lengths are estimated from distance data (Section 1.3). We survey the agglomerative approach that is used by NJ and related algorithms and show that NJ greedily optimizes the BME criterion (Section 1.4). We detail the insertion and tree swapping algorithms we have designed for both versions of ME (Section 1.5). We present the main consistency results on ME (Section 1.6) and finish by discussing simulation results, open problems and directions for further research (Section 1.7).

1.2 Tree metrics

We first describe the main definitions, concepts, and results in the study of tree metrics (Sections 1.2.1 to 1.2.5); for more, refer to Barthélemy and Guénoche [4] or Semple and Steel [43]. Next, we provide an alternate basis for tree metrics that is closely related to the BME framework (Section 1.2.6). Finally, we present the rationale behind distance-based phylogenetic inference that involves recovering a tree metric from the evolutionary distance estimates between taxa (Section 1.2.7).

1.2.1 Notation and basics

A *graph* is a pair $G = (V, E)$, where V is a set of objects called *vertices* or *nodes*, and E is a set of *edges*, that is, pairs of vertices. A *path* is a sequence $(v_0, v_1, \dots, v_k)$ such that for all i , $(v_i, v_{i+1}) \in E$. A *cycle* is a path as above with $k > 2$, $v_0 = v_k$ and $v_i \neq v_j$ for $0 \leq i < k$. A graph is *connected* if each pair of vertices, $x, y \in V$ is connected by a path, denoted p_{xy} . A connected graph containing no cycles is a *tree*, which shall be denoted by T .

The degree of a vertex v , $\deg(v)$, is defined to be the number of edges containing v . In a tree, any vertex v with $\deg(v) = 1$ is called a *leaf*. We will use the letter L to denote the set of leaves of a tree. Other vertices are called internal. In phylogenetic trees, internal nodes have degree 3 or more. An internal vertex with degree 3 is said to be *resolved*, and when all the internal vertices of a tree are resolved, the tree is said to be fully resolved.

A *metric* is a function with certain properties on unordered pairs from a set. Suppose X is a set. The function $d: X \times X \rightarrow \mathfrak{R}$ (the set of real numbers) is

a metric if it satisfies:

1. $d(x, y) \geq 0$ for all x, y , with equality if and only if $x = y$.
2. $d(x, y) = d(y, x)$ for all x, y .
3. For all x, y , and z , $d(x, z) \leq d(x, y) + d(y, z)$.

For the remainder of the chapter, we shall use d_{xy} in place of $d(x, y)$. We will assume that $X = L = [n] = \{1, 2, \dots, n\}$ and use the notation $\text{Met}(n)$ to denote the set of metrics on $[n]$.

Phylogenies usually have lengths assigned to each edge. When the molecular clock holds [49], these lengths represent the time elapsed between the endpoints of the edge. When (as most often) the molecular clock does not hold, the evolutionary distances no longer represent times, but are scaled by substitution rates (or frequencies of mutational events, for example, inversions with gene order data) and the same holds with edge lengths that correspond to the evolutionary distance between the end points of the edges.

Let $T = (V, E)$ be such a tree, with leaf set L , and with $l: E \rightarrow \mathbb{R}^+$ a length function on E . This function induces a *tree metric* on L : for each pair $x, y \in L$, let p_{xy} be the unique path from x to y in T . We define

$$d_{xy}^T = \sum_{e \in p_{xy}} l(e).$$

Where there is no confusion about the identity of T , we shall use d instead of d^T .

In standard graph theory, trees are not required to have associated length functions on their edge sets, and the word *topology* is used to describe the shape of a tree without regard to edge lengths. For our purposes, we shall reserve the word topology to refer to any unweighted tree, and will denote such a tree with calligraphic script $\mathcal{T}$, while the word “tree” and the notation T shall be understood to refer to a tree topology with a length function associated to its edges.

In evolutionary studies, phylogenies are drawn as branching trees deriving from a single ancestral species. This species is known as the *root* of the tree. Mathematically, a rooted phylogeny is a phylogeny to which a special internal node is added with degree 2 or more. This node is the tree root, and is denoted as r ; when r has degree 2, it is said to be resolved.

Suppose there is a length function $l: E \rightarrow \mathbb{R}^+$ defining a tree metric d . Suppose further that all leaves of T are equally distant from r , that is, there exists a constant c such that $d_{xr} = c$ for all leaves x . Then d is a special kind of tree metric called spherical or *ultrametric*. When the molecular clock does not hold, this property is lost, and the tree root cannot be defined in this simple way.

1.2.2 Three-point and four-point conditions

Consider an ultrametric d derived from a tree T . Let x, y , and z be three leaves of T . Let xy, xz , and yz be defined to be the least common ancestors of x and y , x and z , and y and z , respectively. Note that $d_{xy} = 2d_{x(xy)}$ and analogous equalities hold for d_{xz} and d_{yz} . Without loss of generality, xy is not ancestral

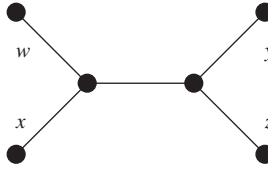


FIG. 1.1. Four-point condition.

to z , and thus $xz = yz$. In this case, $d_{xz} = 2d_{x(xz)} = 2d_{y(yz)} = d_{yz}$. In other words, the two largest of d_{xy} , d_{xz} , and d_{yz} are equal. This can also be written as: for any $x, y, z \in L$,

$$d_{xy} \leq \max\{d_{xz}, d_{yz}\}.$$

This condition is known as the ultrametric inequality or the three-point condition. It turns out [4] that the three-point condition completely characterizes ultrametrics: if d is any metric on any set L satisfying the three-point condition, then there exists a rooted spherical tree T such that $d = d^T$ with L the leaf set of T .

There is a similar characterization of tree metrics in general. Let T be a tree, with tree metric d , and let $w, x, y, z \in L$, the leaf set of T . Without loss of generality, we have the situation in Fig. 1.1, where the path from w to x does not intersect the path from y to z . This configuration implies the (in)equalities:

$$d_{wx} + d_{yz} \leq d_{wy} + d_{xz} = d_{wz} + d_{xy}.$$

In other words, the two largest sums are equal. This can be rewritten as: for all $w, x, y, z \in L$,

$$d_{wx} + d_{yz} \leq \max\{d_{wy} + d_{xz}, d_{wz} + d_{xy}\}.$$

As with the three-point condition, the four-point condition completely characterizes tree metrics [8, 52]. If d is any metric satisfying the four-point condition for all quartets w, x, y , and z , then there is a tree T such that $d = d^T$.

1.2.3 Linear decomposition into split metrics

In this section, we consider the algebraic approach to tree metrics. It is common to represent a metric as a symmetric matrix with a null diagonal. Any metric d on the set $[n]$ can be represented as the matrix $\mathbf{D}$ with entries $d_{ij} = d(i, j)$. Let $\text{Sym}(n)$ be the space of symmetric n by n matrices with null diagonals. Note that every metric can be represented by a symmetric matrix, but $\text{Sym}(n)$ also contains matrices with negative entries and matrices that violate the triangle inequality. It is typical to call $\text{Sym}(n)$ the space of *dissimilarity matrices* on $[n]$, and the corresponding functions on $[n]$ are called *dissimilarities*. Let $\mathcal{A}_n$ denote the vector space of dissimilarity functions.

For all $1 \leq i < j \leq n$, let $\mathbf{E}^{(ij)}$ be the matrix with $e_{ij}^{(ij)} = e_{ji}^{(ij)} = 1$, and all other entries equal zero. The set $E = \{\mathbf{E}^{(ij)}: 1 \leq i < j \leq n\}$ forms the standard basis for $\text{Sym}(n)$ as a vector space. We shall also express these matrices as vectors

indexed by pairs $1 \leq i < j \leq n$, with $d^{(ij)}$ being a vector with 1 in the (ij) entry, and zero elsewhere. In the following discussion, we will consider other bases for $\text{Sym}(n)$ that have natural relationships to tree metrics.

The consideration of the algebraic structure of tree metrics starts naturally by considering each edge length as an algebraic unit. However, as edges do not have a meaning in the settings of metrics or matrices, our first step is to move from edges to *splits*. A split, roughly speaking, is a bipartition induced by any edge of a tree. Suppose $X \cup Y$ is a non-trivial bipartition of $[n]$; that is, $X \neq \emptyset \neq Y$, and $X \cup Y = [n]$. Such a bipartition is a split, and we will denote it by the notation $X|Y$.

Given the split $X|Y$ of $[n]$, Bandelt and Dress [3] defined the split metric, $\sigma^{X|Y}$ on $[n]$ by

$$\sigma_{ab}^{X|Y} = \begin{cases} 1, & \text{if } |X \cap \{a, b\}| = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Any tree topology is completely determined by its splits. Let $e = (x, y)$ be an edge of the topology $\mathcal{T}$. Then define $U_e = \{u \in L : e \in p_{xu}\}$, the set of leaves closer to y than to x , and define $V_e = L \setminus U_e$. We define the set $\mathcal{S}(\mathcal{T})$ to be the set of splits that correspond to edges in $\mathcal{T}$: $\mathcal{S}(\mathcal{T}) = \{U_e | V_e : e \in E(\mathcal{T})\}$. For the sake of simplicity, we shall use σ^e to denote $\sigma^{U_e|V_e}$. This set shall prove to be useful as the natural basis for the vector space associated with tree metrics generated by the topology $\mathcal{T}$.

Suppose X is a set of objects contained in a vector space. The vector space generated by X , denoted $\langle X \rangle$, is the space of all linear combinations of elements of X . Given a tree topology $\mathcal{T}$, with leaf set $[n]$, let $\text{Met}(\mathcal{T})$ be the set of tree metrics from trees with topology $\mathcal{T}$, and let $\mathcal{A}(\mathcal{T}) = \langle \text{Met}(\mathcal{T}) \rangle$. Any tree metric can be decomposed as a linear sum of split metrics: if d is the metric corresponding to the tree T (of topology $\mathcal{T}$),

$$d = \sum_{e \in E(\mathcal{T})} t^T(e) \sigma^e.$$

Thus $\mathcal{A}(\mathcal{T})$ is a vector space with standard basis $\Sigma(\mathcal{T}) = \{\sigma^e : e \in E(\mathcal{T})\}$.

Note that $\dim \mathcal{A}(\mathcal{T}) = |\Sigma(\mathcal{T})| = |E(\mathcal{T})| \leq 2n - 3$ (with equality when $\mathcal{T}$ is fully resolved), and $\dim \mathcal{A}_n = n(n - 1)/2$, and thus for $n > 3$, $\mathcal{A}(\mathcal{T})$ is strictly contained in $\mathcal{A}_n$. Note also that many elements of $\mathcal{A}(\mathcal{T})$ do not define tree metrics, as edge lengths in tree metrics must be non-negative. In fact, the tree metrics with topology $\mathcal{T}$ correspond exactly to the positive cone of $\mathcal{A}(\mathcal{T})$, defined by linear combinations of split metrics with positive coefficients.

1.2.4 Topological matrices

Let $\mathcal{T}$ be a tree topology with n leaves and m edges, and let $e_1, e_2, \dots, e_m$ be any enumeration of $E(\mathcal{T})$. Consider the $n(n - 1)/2$ by m matrix, $\mathbf{A}^{\mathcal{T}}$, defined by

$$a_{(ij)k}^{\mathcal{T}} = \begin{cases} 1, & \text{if } e_k \in p_{ij}, \\ 0, & \text{otherwise.} \end{cases}$$

Suppose T is a tree of topology $\mathcal{T}$. Let l be the edge length function on E , let $\mathbf{B}$ be the vector with entries $l(e_i)$. Then

$$\mathbf{A}^T \times \mathbf{B} = \mathbf{D}^T,$$

where $\mathbf{D}^T$ is the vector form with entries $d_{(ij)}^T$. This matrix formulation shall prove to be useful as we consider various least-squares approaches to edge length estimation.

1.2.5 Unweighted and balanced averages

Given any pair, X, Y , of disjoint subsets of L , and any metric d on L , we use the notation $d_{X|Y}$ to denote the (unweighted) average distance from X to Y under d :

$$d_{X|Y} = \frac{1}{|X||Y|} \sum_{x \in X, y \in Y} d_{xy}, \quad (1.1)$$

where $|X|$ denotes the number of taxa in the subset X . The average distances shall prove to be useful in the context of solving for optimal edge lengths in a least-squares setting. Given a topology $\mathcal{T}$ with leaf set L , and a metric d on L , it is possible to recursively calculate all the average distances for all pairs A, B of disjoint subtrees of $\mathcal{T}$. If $A = \{a\}$, and $B = \{b\}$, we observe that $d_{A|B} = d_{ab}$. Suppose one of A, B has more than one element. Without loss of generality, B separates into two subtrees B_1 and B_2 , as shown in Fig. 1.2, and we calculate

$$d_{A|B} = \frac{|B_1|}{|B|} d_{A|B_1} + \frac{|B_2|}{|B|} d_{A|B_2}. \quad (1.2)$$

It is easy to see that equations (1.1) and (1.2) are equivalent. Moreover, the same equations and notation apply to define $\delta_{A|B}$, that is, the (unweighted) average of distance estimates between A and B .

Pauplin [38] replaced equation (1.2) by a “balanced” average, using $1/2$ in place of $|B_1|/|B|$ and $|B_2|/|B|$ for each calculation. Given a topology $\mathcal{T}$, we recursively define $d_{A|B}^{\mathcal{T}}$: if $A = a$, and $B = b$, we similarly define $d_{A|B}^{\mathcal{T}} = d_{ab}$, but

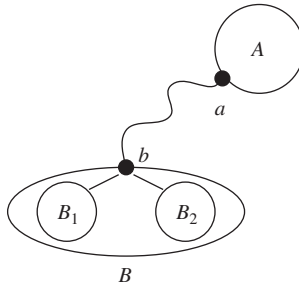


FIG. 1.2. Calculating average distances between subtrees.

if $B = B_1 \cup B_2$ as in Fig. 1.2,

$$d_{A|B}^{\mathcal{T}} = \frac{1}{2}d_{A|B_1}^{\mathcal{T}} + \frac{1}{2}d_{A|B_2}^{\mathcal{T}}. \quad (1.3)$$

For any fully resolved topology $\mathcal{T}$, consideration of these average distances leads us to a second basis for $\mathcal{A}(\mathcal{T})$, which we consider in the next section.

The balanced average uses weights related to the topology $\mathcal{T}$. Let τ_{ab} denote the topological distance (i.e. the number of edges) between taxa a and b , and τ_{AB} the topological distance between the roots of A and B . For any topology $\mathcal{T}$, equation (1.3) leads directly to the identity:

$$d_{A|B}^{\mathcal{T}} = \sum_{a \in A, b \in B} 2^{\tau_{AB} - \tau_{ab}} d_{ab}, \quad (1.4)$$

where

$$\sum_{a \in A, b \in B} 2^{\tau_{AB} - \tau_{ab}} = 1.$$

We thus see that the balanced average distance between a pair of subtrees places less weight on pairs of taxa that are separated by numerous edges; this observation is consistent with the fact that long evolutionary distances are poorly estimated (Section 1.2.7).

1.2.6 Alternate balanced basis for tree metrics

The split metrics are not the only useful basis for studying tree metrics. Desper and Vingron [13] have proposed a basis related to unweighted averages, which is well adapted to OLS tree fitting. In this section, we describe a basis related to balanced averages, well suited for balanced length estimation.

Let e be an arbitrary internal edge of any given topology $\mathcal{T}$, and let w, x, y , and z be the four edges leading to subtrees W, X, Y , and Z , as in Fig. 1.3(a). Let B^e be the tree with a length of 2 on e and length $-1/2$ on the four edges w, x, y , and z . Let β^e be the dissimilarity associated to B^e , which is equal to

$$\beta^e = 2\sigma^e - \frac{1}{2}\sigma^w - \frac{1}{2}\sigma^x - \frac{1}{2}\sigma^y - \frac{1}{2}\sigma^z. \quad (1.5)$$

Now consider e as in Fig. 1.3(b), and let B^e be defined to have a length of $\frac{3}{2}$ on e , and a length of $-\frac{1}{2}$ on y and z . Let β^e be the dissimilarity associated with B^e ,

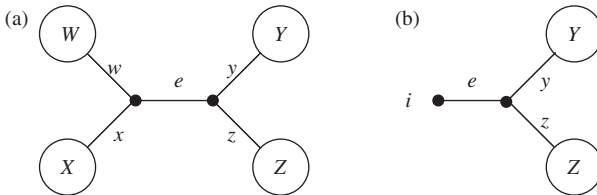


FIG. 1.3. Internal and external edge configurations.

that is,

$$\beta^e = \frac{3}{2}\sigma^e - \frac{1}{2}\sigma^y - \frac{1}{2}\sigma^z. \quad (1.6)$$

Let $\beta_{U_e|V_e}^{e'}$ be the balanced average distance between the sets of the bipartition $U_e | V_e$ when the dissimilarity is $\beta^{e'}$, where e' is any edge from $\mathcal{T}$. It is easily seen that

$$\beta_{U_e|V_e}^{e'} = 1 \quad \text{when } e = e', \quad \text{else } \beta_{U_e|V_e}^{e'} = 0. \quad (1.7)$$

Let $\mathcal{B}(\mathcal{T}) = \{\beta^e: e \in E(\mathcal{T})\}$. Then $\beta(\mathcal{T})$ is a set of vectors that are mutually independent, as implied by equation (1.7). To prove independence, we must prove that $v = \sum_e c_e \beta^e = 0$ implies $c_e = 0$ for all e . Let e' be any edge of $\mathcal{T}$ and consider the balanced average distance in the e' direction: $v_{U_{e'}|V_{e'}} = \sum_e c_e \beta_{U_{e'}|V_{e'}}^e = c_{e'} = 0$. Thus, $c_{e'} = 0$ for all e' , and independence is proven. Since $\mathcal{B}(\mathcal{T})$ is a linearly independent set of the correct cardinality, it forms a basis for $\mathcal{A}(\mathcal{T})$. In other words, any tree metric can be expressed uniquely in the form

$$d = \sum_e d_{U_e|V_e}^{\mathcal{T}} \beta^e, \quad (1.8)$$

which is another useful decomposition of tree metrics. From this decomposition, we see that the length of T is the weighted sum of lengths of the B^e s, that is,

$$l(T) = \sum_e d_{U_e|V_e}^{\mathcal{T}} l(B^e).$$

Note that $l(B^e) = 0$ for any internal edge e , while $l(B^e) = 1/2$ for any external edge e . Thus

$$l(T) = \frac{1}{2} \sum_{i \in L} d_{\{i\}|L \setminus \{i\}}^{\mathcal{T}}. \quad (1.9)$$

Returning to the expressions of equation (1.5) and equation (1.6), we can decompose d as

$$\begin{aligned} d = & \sum_{e \text{ external}} d_{U_e|V_e}^{\mathcal{T}} \left(\frac{3}{2}\sigma^e - \frac{1}{2}\sigma^y - \frac{1}{2}\sigma^z \right) \\ & + \sum_{e \text{ internal}} d_{U_e|V_e}^{\mathcal{T}} \left(2\sigma^e - \frac{1}{2}\sigma^w - \frac{1}{2}\sigma^x - \frac{1}{2}\sigma^y - \frac{1}{2}\sigma^z \right), \end{aligned}$$

that is,

$$\begin{aligned} d = & \sum_{e \text{ external}} \left(\frac{3}{2}d_{U_e|V_e}^{\mathcal{T}} - \frac{1}{2}d_{U_y|V_y}^{\mathcal{T}} - \frac{1}{2}d_{U_z|V_z}^{\mathcal{T}} \right) \sigma^e \\ & + \sum_{e \text{ internal}} \left(2d_{U_e|V_e}^{\mathcal{T}} - \frac{1}{2}d_{U_w|V_w}^{\mathcal{T}} - \frac{1}{2}d_{U_x|V_x}^{\mathcal{T}} - \frac{1}{2}d_{U_y|V_y}^{\mathcal{T}} - \frac{1}{2}d_{U_z|V_z}^{\mathcal{T}} \right) \sigma^e. \end{aligned} \quad (1.10)$$

Because the representation given by equation (1.8) is unique, equation (1.10) gives us formulae for edge lengths: for internal edges,

$$l(e) = 2d_{U_e|V_e}^{\mathcal{T}} - \frac{1}{2}d_{U_w|V_w}^{\mathcal{T}} - \frac{1}{2}d_{U_x|V_x}^{\mathcal{T}} - \frac{1}{2}d_{U_y|V_y}^{\mathcal{T}} - \frac{1}{2}d_{U_z|V_z}^{\mathcal{T}}, \quad (1.11)$$

and for external edges,

$$l(e) = \frac{3}{2}d_{U_e|V_e}^{\mathcal{T}} - \frac{1}{2}d_{U_y|V_y}^{\mathcal{T}} - \frac{1}{2}d_{U_z|V_z}^{\mathcal{T}}. \quad (1.12)$$

We shall see that these formulae (1.9, 1.11, 1.12) correspond to the estimates found by Pauplin via a different route. We shall also provide another combinatorial interpretation of formula (1.9) due to Semple and Steel [44].

1.2.7 Tree metric inference in phylogenetics

Previous sections (1.2.1 to 1.2.6) describe the mathematical properties of tree metrics. Inferring the tree corresponding to a given tree metric is simple. For example, we can use the four-point condition and closely related ADDTREE algorithm [42] to reconstruct the tree topology, and then formulae (1.11) and (1.12) to obtain the edge lengths. However, in phylogenetics we only have evolutionary distance estimates between taxa, which do not necessarily define a tree metric. The rationale of the distance-based approach can thus be summarized as follows [16].

The true Darwinian tree T is unknown but well defined, and the same holds for the evolutionary distance that corresponds to the number of evolutionary events (e.g. substitutions) separating the taxa. This distance defines a tree metric d corresponding to T with positive weights (numbers of events) on edges. Due to hidden (parallel or convergent) events, the true number of events is unknown and greater than or equal to the observed number of events. Thus, the distance-based approach involves estimating the evolutionary distance from the differences we observe today between taxa, assuming a stochastic model of evolution. Such models are described in this volume, in Chapter 2 concerning sequences and substitution events, and in Chapter 13 concerning various genome rearrangement events.

Even when the biological objects and the models vary, the basic principle remains identical: we first compute an estimate Δ of $\mathbf{D}$, the metric associated with T , and then reconstruct an estimate $\hat{T}$ of T using Δ . The estimated distance matrix Δ no longer exactly fits a tree, but is usually very close to a tree. For example, we extracted from TreeBASE (www.treebase.org) [41] 67 Fungi sequences (accession number M520), used DNADIST with default options to calculate a distance matrix, and used NJ to infer a phylogeny. The tree $\hat{T}$ obtained in this (simple) way explains more than 98% of the variance in the distance matrix (i.e. $\sum_{i,j}(\delta_{ij} - d_{ij}^{\hat{T}})^2 / \sum_{i,j}(\delta_{i,j} - \bar{\delta})^2$ is about 2%, where $\bar{\delta}$ is the average value of δ_{ij}). In other words, this tree and the distance matrix are extremely close, and the mere principle of the distance approach appears fully justified in

this case. Numerous similar observations have been made with aligned sequences and substitution models.

In the following, we shall not discuss evolutionary distance estimation, which is dealt with in other chapters and elsewhere (e.g. [49]), but this is clearly a crucial step. An important property that holds in all cases is that estimation of short distances is much more reliable than estimation of long distances. This is simply due to the fact that with long distances the number of hidden events is high and is thus very hard to estimate. As we shall see (Section 1.3.7 and Chapter 13, this volume), this feature has to be taken into account to design accurate inference algorithms. Even if the estimated distance matrix Δ is usually close to a tree, tree reconstruction from such an approximate matrix is much less obvious than in the ideal case where the matrix perfectly fits a tree. The next sections are devoted to this problem, using the minimum evolution principle.

1.3 Edge and tree length estimation

In this section, we consider edge and tree length estimation, given an input topology and a matrix of estimated evolutionary distances. We first consider the least-squares framework (Sections 1.3.1 to 1.3.3), then the balanced approach (Sections 1.3.5 and 1.3.6), and finally show that the latter is a special case of weighted least-squares that is well suited for phylogenetic inference.

For the rest of this section, Δ will be the input matrix, $\mathcal{T}$ the input topology, and $\mathbf{A}$ will refer to the topological matrix $\mathbf{A}^T$. We shall also denote as $\hat{l}$ the length estimator obtained from Δ , $\hat{T}$ the tree with topology $\mathcal{T}$ and edge lengths $\hat{l}(e)$, $\hat{\mathbf{B}}$ the vector of edge length estimates, and $\hat{\mathbf{D}} = (\hat{d}_{ij})$ the distance matrix corresponding to the tree metric $d^{\hat{T}}$. Depending on the context, Δ and $\hat{\mathbf{D}}$ will sometimes be in vector form, that is, $\Delta = (\delta_{ij})$ and $\hat{\mathbf{D}} = (\hat{d}_{ij})$.

1.3.1 The least-squares (LS) approach

Using this notation, we observe that $\hat{\mathbf{D}} = \mathbf{A}\hat{\mathbf{B}}$, and the edge lengths are estimated by minimizing the difference between the observation Δ and $\hat{\mathbf{D}}$. The OLS approach involves selecting edge lengths $\hat{\mathbf{B}}$ minimizing the squared Euclidean fit between Δ and $\hat{\mathbf{D}}$:

$$\text{OLS}(\hat{T}) = \sum_{i,j} (\hat{d}_{ij} - \delta_{ij})^2 = (\hat{\mathbf{D}} - \Delta)^t (\hat{\mathbf{D}} - \Delta).$$

This yields:

$$\hat{\mathbf{B}} = (\mathbf{A}^t \mathbf{A})^{-1} \mathbf{A}^t \Delta. \quad (1.13)$$

However, this approach implicitly assumes that each estimate δ_{ij} has the same variance, a false supposition since large distances are much more variable than short distances (Section 1.2.7). To address this problem, Fitch and Margoliash [19], Felsenstein [18], and others have proposed using a WLS

approach, that is, minimizing

$$\text{WLS}(\hat{T}) = \sum_{i,j} \frac{(\hat{d}_{ij} - \delta_{ij})^2}{v_{ij}} = (\hat{\mathbf{D}} - \Delta)^t \mathbf{V}^{-1} (\hat{\mathbf{D}} - \Delta),$$

where $\mathbf{V}$ is the diagonal $n(n-1)/2 \times n(n-1)/2$ matrix containing the variances v_{ij} of the δ_{ij} estimates. This approach yields

$$\hat{\mathbf{B}} = (\mathbf{A}^t \mathbf{V}^{-1} \mathbf{A})^{-1} \mathbf{A}^t \mathbf{V}^{-1} \Delta. \quad (1.14)$$

OLS is a special case of WLS, which in turn is a special case of generalized least-squares (GLS) that incorporates the covariances of the δ_{ij} estimates [7, 47]. When the full variance-covariance matrix is available, GLS estimation is the most reliable and WLS is better than OLS. However, GLS is rarely used in phylogenetics, due to its computational cost and to the difficulty of estimating the covariance terms. WLS is thus a good compromise. Assuming that the variances are known and the covariances are zero, equation (1.14) defines the minimum-variance estimator of edge lengths.

Direct solutions of equations (1.13) and (1.14) using matrix calculations requires $O(n^4)$ time. A method requiring only $O(n^3)$ time to solve the OLS version was described by Vach [50]. Gascuel [22] and Bryant and Waddell [6] provided algorithms to solve OLS in $O(n^2)$ time. Fast algorithms for OLS are based on the observation of Vach [50]: If $\hat{T}$ is the tree with edge lengths estimated using OLS equation (1.13), then for every edge e in $E(\hat{T})$ we have:

$$\hat{d}_{U_e|V_e} = \delta_{U_e|V_e}. \quad (1.15)$$

In other words, the average distance between the components of every split is identical in the observation Δ and the inferred tree metric.

1.3.2 Edge length formulae

Equation (1.15) provides a system of linear equations that completely determines edge length estimates in the ordinary least squares framework. Suppose we seek to assign a length to the internal edge e shown in Fig. 1.3(a), which separates subtrees W and X from subtrees Y and Z . The OLS length estimate is then [39]:

$$\hat{l}(e) = \frac{1}{2} [\lambda(\delta_{W|Y} + \delta_{X|Z}) + (1 - \lambda)(\delta_{W|Z} + \delta_{X|Y}) - (\delta_{W|X} + \delta_{Y|Z})], \quad (1.16)$$

where

$$\lambda = \frac{|W||Z| + |X||Y|}{|W \cup X||Y \cup Z|}. \quad (1.17)$$

If the same way, for external edges (Fig. 1.3(b)) the OLS length estimate is given by

$$\hat{l}(e) = \frac{1}{2} (\delta_{\{i\}|Y} + \delta_{\{i\}|Z} - \delta_{Y|Z}). \quad (1.18)$$

These edge length formulae allow one to express the total length of all edges, that is, the tree length estimate, as a linear sum of average distances between pairs of subtrees.

1.3.3 Tree length formulae

A general matrix expression for tree length estimation is obtained from the equations in Section 1.3.1. Letting $\mathbf{1}$ be a vector of 1s, we then have

$$\hat{l}(T) = \mathbf{1}^t (\mathbf{A}^t \mathbf{V}^{-1} \mathbf{A})^{-1} \mathbf{A}^t \mathbf{V}^{-1} \Delta. \quad (1.19)$$

However, using this formula would require heavy computations. Since the length of each edge in a tree can be expressed as a linear sum of averages between the four subtrees incident to the edge (presuming a fully resolved tree), a minor topological change will leave most edge lengths fixed, and will allow for an easy recalculation of the length of the tree. Suppose T is the tree in Fig. 1.3(a) and T' is obtained from T by swapping subtrees X and Y across the edge e , which corresponds to a nearest neighbour interchange (NNI, see Section 1.5 for more details). Desper and Gascuel [11] showed that the difference in total tree lengths (using OLS estimations) can be expressed as

$$\begin{aligned} \hat{l}(T) - \hat{l}(T') = & \frac{1}{2} [(\lambda - 1)(\delta_{W|Y} + \delta_{X|Z}) - (\lambda' - 1)(\delta_{W|X} + \delta_{Y|Z}) \\ & - (\lambda - \lambda')(\delta_{W|Z} + \delta_{X|Y})], \end{aligned} \quad (1.20)$$

where λ is as in equation (1.17), and

$$\lambda' = \frac{|W||Z| + |X||Y|}{|W \cup Y||X \cup Z|}.$$

We shall see in Section 1.5 that equation (1.20) allows for very fast algorithms, both to build an initial tree and to improve this tree by topological rearrangements.

1.3.4 The positivity constraint

The algebraic edge length assignments given in Sections 1.3.1 and 1.3.2 have the undesirable property that they may assign negative “lengths” to several of the edges in a tree. Negative edge lengths are frowned upon by evolutionary biologists, since evolution cannot proceed backwards [49]. Moreover, when using a pure least-squares approach, that is, when not only the edge lengths are selected using a least-squares criterion but also the tree topology, allowing for negative edge lengths gives too many degrees of freedom and might result in suboptimal trees using negative edge lengths to produce a low apparent error. Imposing positivity is thus desirable when reconstructing phylogenies, and Kuhner and Felsenstein [32] and others showed that FITCH (a pure LS method) has better topological accuracy when edge lengths are constrained to be non-negative.

Adding the positivity constraint, however, removes the possibility of using matrix algebra (equations 1.13 and 1.14) to find a solution. One might be tempted to simply use matrix algebra to find the optimal solution, and then set negative lengths to zero, but this jury-rigged approach does not provide an

optimal solution to the constrained problem. In fact, the problem at hand is non-negative linear regression (or non-negative least-squares, that is, NNLS), which involves projecting the observation Δ on the positive cone defined by $\mathcal{A}(\mathcal{T})$, instead of on the vector space $\mathcal{A}(\mathcal{T})$ itself as in equations (1.13) and (1.14). In general, such a task is computationally difficult, even when relatively efficient algorithms exist [34]. Several dedicated algorithms have been designed for tree inference, both to estimate the edge lengths for a given tree topology [4] and to incorporate the positivity constraint all along tree construction [18, 26, 29, 35]. But all of these procedures are computationally expensive, with time complexity in $O(n^4)$ or more, mostly due to the supplementary cost imposed by the positivity constraint.

In contrast, minimum evolution approaches do not require the positivity constraint. Some authors have suggested that having negative edges might result in trees with underestimated length, as tree length is obtained by summing edge lengths. In fact, having internal edges with negative lengths tends to give longer trees, as a least-squares fit forces these negative lengths to be compensated for by larger positive lengths on other edges. Trees with negative edges thus tend to be discarded when using the minimum evolution principle. Simulations [22] confirm this and, moreover, we shall see in Section 1.3.5 that the balanced framework naturally produces trees with positive edge lengths without any additional computational cost.

1.3.5 The balanced scheme of Pauplin

While studying a quick method for estimating the total tree length, Pauplin [38] proposed to simplify equations (1.16) and (1.18) by using weights $\frac{1}{2}$ and the balanced average we defined in Section 1.2.6. He obtained the estimates for internal edges:

$$\hat{l}(e) = \frac{1}{4}(\delta_{W|Y}^{\mathcal{T}} + \delta_{X|Z}^{\mathcal{T}} + \delta_{W|Z}^{\mathcal{T}} + \delta_{W|Y}^{\mathcal{T}}) - \frac{1}{2}(\delta_{W|X}^{\mathcal{T}} + \delta_{Y|Z}^{\mathcal{T}}), \quad (1.21)$$

and for external edges:

$$\hat{l}(e) = \frac{1}{2}(\delta_{\{i\}|Y}^{\mathcal{T}} + \delta_{\{i\}|Z}^{\mathcal{T}} - \delta_{X|Y}^{\mathcal{T}}). \quad (1.22)$$

Using these formulae, Pauplin showed that the tree length is estimated using the simple formula

$$\hat{l}(T) = \sum_{\{i,j\} \subset L} 2^{1-\tau_{ij}} \delta_{ij}. \quad (1.23)$$

In fact, equations (1.21), (1.22), and (1.23) are closely related to the algebraic framework introduced in Section 1.2.6. Assume that a property dual of Vach's [50] theorem (15) for OLS is satisfied in the balanced settings, that is, for every edge $e \in E(\mathcal{T})$:

$$\hat{d}_{U_e|V_e}^{\mathcal{T}} = \delta_{U_e|V_e}^{\mathcal{T}}.$$

We then obtain from equation (1.8) the following simple expression:

$$\hat{\mathbf{D}} = \sum_e \hat{d}_{U_e|V_e}^T \beta^e = \sum_e \delta_{U_e|V_e}^T \beta^e.$$

As a consequence, equations (1.9), (1.11), and (1.12) can be used as estimators of tree length, internal edge length, and external edge length, respectively, simply by turning the balanced averages of $\mathbf{D}$ into those of Δ , that is, $d_{X|Y}^T$ becomes $\delta_{X|Y}^T$. These estimators are consistent by construction (if $\Delta = \mathbf{D}$ then $\hat{\mathbf{D}} = \mathbf{D}$) and it is easily checked (using equations (1.3) and (1.4)) that these estimators are the same as Pauplin's defined by equations (1.21), (1.22), and (1.23). The statistical properties (in particular the variance) of these estimators are given in Section 1.3.7.

Moreover, we have shown [11] that the balanced equivalent of equation (1.20) is

$$\hat{l}(T) - \hat{l}(T') = \frac{1}{4}(\delta_{W|X}^T + \delta_{Y|Z}^T - \delta_{W|Y}^T - \delta_{X|Z}^T). \quad (1.24)$$

Equation (1.24) implies a nice property about balanced edge lengths.

Suppose we use balanced length estimation to assign edge lengths corresponding to the distance matrix Δ to a number of tree topologies, and consider a tree T such that $\hat{l}(T') > \hat{l}(T)$ for any tree T' that can be reached from T by one nearest neighbour interchange (NNI). Then $\hat{l}(e) > 0$ for every internal edge $e \in T$, and $\hat{l}(e) \geq 0$ for every external edge of T .

The proof of this theorem is obtained using equations (1.24) and (1.21). First, consider an internal edge $e \in T$. Suppose e separates subtrees W and X from Y and Z as in Fig. 1.3(a). Since T is a local minimum under NNI treeswapping, the value of equation (1.24) must be negative, that is, $\delta_{W|X}^T + \delta_{Y|Z}^T < \delta_{W|Y}^T + \delta_{X|Z}^T$. A similar argument applied to the other possible NNI across e leads to the analogous inequality $\delta_{W|X}^T + \delta_{Y|Z}^T < \delta_{W|Z}^T + \delta_{W|Y}^T$. These two inequalities force the value of $\hat{l}(e)$ to be positive according to equation (1.21). Now, suppose there were an external edge e with $\hat{l}(e) < 0$. Referring to equation (1.22), it is easy to see that a violation of the triangle inequality would result, contradicting the metric nature of Δ implied by the commonly used methods of evolutionary distance estimation.

1.3.6 *Simple and Steel combinatorial interpretation*

Any tree topology defines circular orderings of the taxa. A circular ordering can be thought of as a (circular) list of the taxa encountered in order by an observer looking at a planar embedding of the tree. For example (Fig. 1.4), the tree $((1, 2), 3, (4, 5))$ induces the four orderings $(1, 2, 3, 4, 5)$, $(1, 2, 3, 5, 4)$, $(2, 1, 3, 4, 5)$, and $(2, 1, 3, 5, 4)$.

As one traverses the tree according to the circular order, one passes along each edge exactly twice—once in each direction. Thus, adding up the leaf-to-leaf distances resulting from all pairs of leaves adjacent in the circular order will yield a sum equal to twice the total length of the tree. For example, using

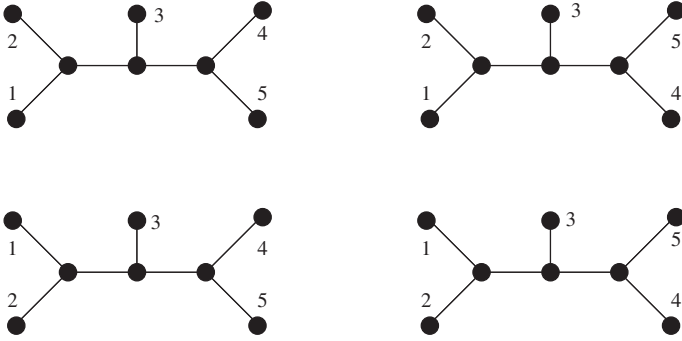


FIG. 1.4. Circular orders of a five-leaf tree.

(1, 2, 3, 4, 5) (which results from the tree in the upper left of Fig. 1.4), we get $l(T) = (d_{12} + d_{23} + d_{34} + d_{45} + d_{51})/2$.

In general, this equality holds for each circular order: given an order $o = (o(1), o(2), \dots, o(n))$,

$$l(T) = l(d, o) = \frac{1}{2} \left(d_{o(1)o(n)} + \sum_{i=1}^{n-1} d_{o(i)o(i+1)} \right).$$

As we average over $o \in C(T)$, the set of circular orders associated with the tree T , we observe

$$l(T) = \frac{1}{|C(T)|} \sum_{o \in C(T)} l(d, o). \quad (1.25)$$

Simple and Steel [44] have shown that this average is exactly equation (1.9), which becomes Pauplin's formula (1.23) when substituting the d_{ij} s with the δ_{ij} estimates. Moreover, they showed that this result can be generalized to unresolved trees. Let u be any internal node of $\mathcal{T}$, and $\deg(u)$ be the degree of u , that is, 3 or more. Then the following equality holds:

$$l(T) = \sum_{\{i,j\} \subset L} \lambda_{ij} d_{ij}, \quad (1.26)$$

where

$$\begin{aligned} \lambda_{ij} &= \prod_{u \in p_{ij}} (\deg(u) - 1)^{-1}, \quad \text{when } i \neq j, \\ &= 0 \quad \text{otherwise.} \end{aligned}$$

1.3.7 BME: a WLS interpretation

The WLS approach (equation (1.14)) takes advantage of the variances of the estimates. It is usually hard (or impossible) to have the exact value of these variances, but it is well known in statistics that approximate values are sufficient to obtain reliable estimators. The initial suggestion of Fitch and Margoliash [19], and the default setting in the programs FITCH [18] and PAUP* [48], is to

assume variances are proportional to the squares of the distances, that is, to use $v_{ij} \propto \delta_{ij}^2$. Another common approximation (e.g. [21]) is $v_{ij} \propto \delta_{ij}$. However, numerous studies [7, 36, 37, 47] suggest that variance grows exponentially as a function of evolutionary distance and, for example, Weighor [5] uses this more suitable approximation.

Desper and Gascuel [12] recently demonstrated that the balanced scheme corresponds to $v_{ij} \propto 2^{\tau_{ij}}$, that is, variance grows exponentially as a function of the topological distance between taxa i and j . Even when topological and evolutionary distances differ, they are strongly correlated, especially when the taxa are homogeneously sampled, and our topology-based approximation is likely capturing most of above-mentioned exponential approximations. Moreover, assuming that the matrix $\mathbf{V}$ is diagonal with $v_{ij} \propto 2^{\tau_{ij}}$, Pauplin's formula (1.23) becomes identical to matrix equation (1.19) and defines the minimum variance tree length estimator. Under this assumption, the edge and tree lengths given by BME are thus as reliable as possible. Since we select the shortest tree, reliability in tree length estimation is of great importance and tends to minimize the probability of selecting a wrong tree. This WLS interpretation then might explain the strong performance of the balanced minimum evolution method.

1.4 The agglomerative approach

In this section, we consider the agglomerative approach to tree building. Agglomerative algorithms (Fig. 1.5) work by iteratively finding pairs of neighbours in the tree, separating them from the rest of the tree, and reducing the size of the problem by treating the new pair as one unit, then recalculating a distance matrix with fewer entries, and continuing with the same approach on the smaller data set.

The basic algorithms in this field are UPGMA (unweighted pair group method using arithmetic averages) [45] and NJ (Neighbor Joining) [40]. The UPGMA algorithm assumes that the distance matrix is approximately ultrametric, while the NJ algorithm does not. The ultrametric assumption allows UPGMA to be quite simple.

1.4.1 UPGMA and WPGMA

Given an input distance matrix Δ with entries δ_{ij} ,

1. Find i, j such that $i \neq j$, δ_{ij} is minimal.
2. Create new node u , connect i and j to u with edges whose lengths are $\delta_{ij}/2$.

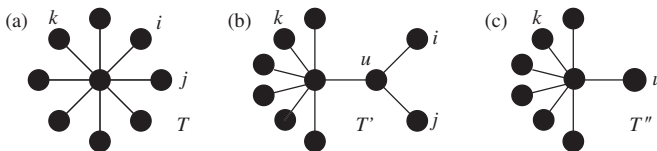


FIG. 1.5. Agglomerative algorithms: (a) find neighbours in star tree; (b) insert new node to join neighbours; (c) continue with smaller star tree.

3. If i and j are the only two entries of Δ , stop and return tree.
4. Else, build a new distance matrix by removing i and j , and adding u , with δ_{uk} defined as the average of δ_{ik} and δ_{jk} , for $k \neq i, j$.
5. Return to Step 1 with smaller distance matrix.

Step 4 calculates the new distances as the average of two distances that have been previously calculated or are original evolutionary distance estimates. In UPGMA, this average is unweighted and gives equal weight to each of the original estimates covered by the i and j clusters, that is, $\delta_{uk} = (|i|\delta_{ik} + |j|\delta_{jk}) / (|i| + |j|)$, where $|x|$ is the size of cluster x . In WPGMA the average is weighted (or balanced) regarding original estimates and gives the same weight to each cluster, that is, $\delta_{uk} = (\delta_{ik} + \delta_{jk})/2$. Due to ambiguity (weight of the clusters/weight of the original distance estimates), these two algorithms are often confused for one another and some commonly used implementations of “UPGMA” in fact correspond to WPGMA. In biological studies it makes sense to use a balanced approach such as WPGMA, since a single isolated taxon often gives as much information as a cluster containing several remote taxa [45].

However, the ultrametric (molecular clock) assumption is crucial to Step 1. If Δ is a tree metric but not an ultrametric, the minimal entry might not represent a pair of leaves that can be separated from the rest of the tree as a subtree. To find a pair of neighbours, given only a matrix of pairwise distances, the Neighbor Joining algorithm of Saitou and Nei [40] uses a minimum evolution approach, as we shall now explain.

1.4.2 *NJ as a balanced minimum evolution algorithm*

To select the pair of taxa to be agglomerated, NJ tests each topology created by connecting a taxon pair to form a subtree (Fig. 1.5(b)) and selects the topology with minimal length. As this process is repeated at each step, NJ can be seen as a greedy algorithm minimizing the total tree length, and thus complying with the minimum evolution principle. However, the way the tree length is estimated by NJ at each step is not well understood. Saitou and Nei [40] showed that NJ’s criterion corresponds to the OLS length estimation of the topology shown in Fig. 1.5(b), assuming that every leaf (cluster) contains a unique taxon. Since clusters may contain more than one taxon after the first step, this interpretation is not entirely satisfactory. But we shall see that throughout the process, NJ’s criterion in fact corresponds to the balanced length of topology as shown in Fig. 1.5(b), which thus implies that NJ is better seen as the natural greedy agglomerative approach to minimize the balanced minimum evolution criterion.

We use for this purpose the general formula (1.26) of Semple and Steel to estimate the difference in length between trees T and T' in Fig. 1.5. Each of the leaves in T and T' is associated to a subtree either resulting from a previous agglomeration, or containing a single, original, taxon that has yet to be agglomerated. In the following, every leaf is associated to a “subtree.” Each of these leaf-associated subtrees is binary and identical in T and T' , and we can thus define the balanced average distance between any subtree pair, which

has the same value in T and T' . Furthermore, the balanced average distances thus defined correspond to the entries in the current distance matrix, as NJ uses the balanced approach for matrix reduction, just as in WPGMA Step 4. In the following, A and B denote the two subtrees to be agglomerated, while X and Y are two subtrees different from A and B and connected to the central node (Fig. 1.5). Also, let r be the degree of the central node in T , and a, b, x , and y be any original taxa in A, B, X , and Y , respectively.

Using equation (1.26), we obtain:

$$\hat{l}(T) - \hat{l}(T') = \sum_{\{i,j\} \subset L} (\lambda_{ij} - \lambda'_{ij}) \delta_{ij},$$

where the coefficients λ and λ' are computed in T and T' , respectively. The respective coefficients differ only when the corresponding taxon pair is not within a single subtree A, B, X , or Y ; using this, the above equation becomes:

$$\begin{aligned} \hat{l}(T) - \hat{l}(T') = & \sum_{\{a,b\}} (\lambda_{ab} - \lambda'_{ab}) \delta_{ab} + \sum_{\{a,x\}} (\lambda_{ax} - \lambda'_{ax}) \delta_{ax} \\ & + \sum_{\{b,x\}} (\lambda_{bx} - \lambda'_{bx}) \delta_{bx} + \sum_{\{x,y\}} (\lambda_{xy} - \lambda'_{xy}) \delta_{xy}. \end{aligned}$$

Using now the definition of the λ 's and previous remarks, we have:

$$\begin{aligned} \hat{l}(T) - \hat{l}(T') = & ((r-1)^{-1} - 2^{-1}) \delta_{AB}^T + ((r-1)^{-1} - (2(r-2))^{-1}) \\ & \times \sum_X (\delta_{AX}^T + \delta_{BX}^T) + ((r-1)^{-1} - (r-2)^{-1}) \sum_{\{X,Y\}} \delta_{XY}^T. \end{aligned}$$

Letting I and J be any of the leaf-associated subtrees, we finally obtain:

$$\begin{aligned} \hat{l}(T) - \hat{l}(T') = & -2^{-1} \delta_{AB}^T + 2^{-1} (r-2)^{-1} \left(\sum_{I \neq A} \delta_{AI}^T + \sum_{I \neq B} \delta_{BI}^T \right) \\ & + ((r-1)^{-1} - (r-2)^{-1}) \sum_{\{I,J\}} \delta_{IJ}^T. \end{aligned}$$

The last term in this expression is independent of A and B , while the first two terms correspond to Studier and Keppler's [46] way of writing NJ's criterion [20]. We thus see that, all through the process, minimizing at each step the balanced length of T' is the same as selecting the pair A, B using NJ's criterion. This proves that NJ greedily optimizes a global (balanced minimum evolution) criterion, contrary to what has been written by several authors.

1.4.3 Other agglomerative algorithms

The agglomerative approach to tree metrics was first proposed by Sattath and Tversky [42] in ADDTREE. This algorithm uses the four-point condition (Section 1.2.2) to select at each step the pair of taxa to be agglomerated, and is therefore relatively slow, with time complexity in $O(n^4)$. NJ's $O(n^3)$ was thus

important progress and the speed of NJ, combined with its good topological accuracy, explains its popularity. To improve NJ, two lines of approach were pursued.

The first approach was to explicitly incorporate the variances and covariances of δ_{ij} estimates in the agglomeration scheme. This was first proposed in BIONJ [21], which is based on the approximation $v_{ij} \propto \delta_{ij}$ (Section 1.3.7) and on an analogous model for the covariances; BIONJ uses these (co)variances when computing new distances (Step 4 in algorithm of Section 1.4.1) to have more reliable estimates all along the reconstruction process. The same scheme was used to build the proper OLS version of NJ, which we called UNJ (Unweighted Neighbor Joining) [22], and was later generalized to any variance-covariance matrix of the δ_{ij} s [23]. Weighbor [5] followed the same line but using a better exponential model of the variances [36] and, most importantly, a new maximum-likelihood based pair selection criterion. BIONJ as well as Weighbor then improved NJ thanks to better statistical models of the data, but kept the same agglomerative algorithmic scheme.

The second approach that we describe in the next section involves using the same minimum evolution approach as NJ, but performing a more intensive search of the tree space via topological rearrangement.

1.5 Iterative topology searching and tree building

In this section, we consider rules for moving from one tree topology to another, either by adding a taxon to an existing tree, or by swapping subtrees. We shall consider topological transformations before considering taxon insertion, as selecting the best insertion point is achieved by iterative topological rearrangements. Moreover, we first describe the OLS versions of the algorithms, before their BME counterparts, as the OLS versions are simpler.

1.5.1 Topology transformations

The number of unrooted binary tree topologies with n labelled leaves is $(2n-5)!!$, where $k!! = k * (k-2) * \dots * 1$ for k odd. This number grows large far too quickly (close to n^n) to allow for exhaustive topology search except for small values of n . Thus, heuristics are typically relied upon to search the space of topologies when seeking a topology optimal according to any numerical criterion. The following three heuristics are available to users of PAUP* [48]. Tree bisection reconnection (TBR) splits a tree by removing an edge, and then seeks to reconnect the resulting subtrees by adding a new edge to connect some edge in the first tree with some edge in the second tree. Given a tree T , there are $O(n^3)$ possible new topologies that can be reached with one TBR. Subtree pruning regrafting (SPR) removes a subtree and seeks to attach it (by its root) to any other edge in the other subtree. (Note that an SPR is a TBR where one of the new insertion points is identical to the original insertion point.) There are $O(n^2)$ SPR transformations from a given topology. We can further shrink the search space by requiring the new insertion point to be along an edge adjacent to the original insertion

point. Such a transformation is known as an NNI, and there are $O(n)$ NNI transformations from a given topology. Although there are comparatively few NNIs, this type of transformation is sufficient to allow one to move from any binary topology to any other binary topology on the same leaf set simply by a sequence of NNIs.

1.5.2 A fast algorithm for NNIs with OLS

Since there are only $O(n)$ NNI transformations from a given topology, NNIs are a popular topology search method. Consider the problem of seeking the minimum evolution tree among trees within one NNI of a given tree. The naive approach would be to generate a set of topologies, and separately solve OLS for each topology. This approach would require $O(n^3)$ computations, because we would run the $O(n^2)$ OLS edge length algorithm $O(n)$ times.

Desper and Gascuel [11] have presented a faster algorithm for simultaneously testing, in $O(n^2)$ time, all of the topologies within one NNI of an initial topology. This algorithm, FASTNNI, is implemented in the program FASTME. Given a distance matrix Δ and a tree topology T :

1. Pre-compute average distances Δ_{avg} between non-intersecting subtrees of T . Initialize $h_{\min} = 0$. Initialize $e_{\min} \in E(T)$.
2. Starting with $e_{\min}$, loop over edges $e \in E(T)$. For each edge e , use equation (1.20) and the matrix Δ_{avg} to calculate $h_1(e)$ and $h_2(e)$, the relative differences in total tree length resulting from each of the two possible NNIs. Let $h(e)$ be the greater of the two. If $h_i(e) = h(e) > h_{\min}$, set $e_{\min} = e$, $h_{\min} = h(e)$, and the indicator variable $s = i$.
3. If $h_{\min} = 0$, stop and exit. Otherwise, perform NNI at $e_{\min}$ in direction pointed to by the variable s .
4. Recalculate entries of Δ_{avg} . Return to Step 2.

Step 1 of FASTNNI can be achieved in $O(n^2)$ time using equation (1.2). Each calculation of equation (1.20) in Step 2 can be done in constant time, and, because there is only one new split in the tree after each NNI, each recalculation of Δ_{avg} in Step 4 can be done in $O(n)$ time. Thus, algorithm requires $O(n^2)$ time to reach Step 2, and an additional $O(n)$ time for each NNI. If s swaps are performed, the total time required is $O(n^2 + sn)$.

1.5.3 A fast algorithm for NNIs with BME

The algorithm presented in Section 1.5.2 can be modified to also be used to search for a minimum evolution tree when edges have balanced lengths. The modified algorithm, FASTBNNI, is the same as FASTNNI, with the following exceptions:

1. Instead of calculating the vector of unweighted averages, we calculate the vector Δ_{avg}^T of balanced averages.

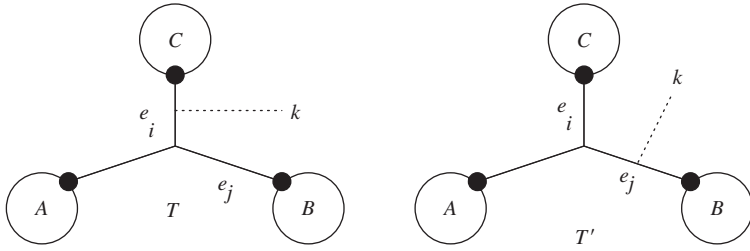


FIG. 1.7. Inserting a leaf into a tree; T' is obtained from T by NNI of k and A .

select a topology from the uniform distribution on the space of binary topologies, we would expect $\text{diam}(T) = O(\sqrt{n})$, while the more biologically motivated Yule-Harding distribution [28, 51] on the space of topologies would lead to an expected diameter in $O(\log n)$. Thus, s iterations of FASTBNNI would require $O(n^2 + sn \log n)$ computations, presuming a tree with a biologically realistic diameter.

1.5.4 Iterative tree building with OLS

In contrast to the agglomerative scheme, many programs (e.g. FITCH, PAUP*, FASTME) use an approach iteratively adding leaves to a partial tree. Consider Fig. 1.7. The general approach is:

1. Start by constructing T_3 , the (unique) tree with three leaves.
2. For $k = 4$ to n ,
 - (a) Test each edge of T_{k-1} as a possible insertion point for the taxon k .
 - (b) Based on optimization criterion (e.g. sum of squares, minimum evolution), select the optimal edge $e = (u, v)$.
 - (c) Form tree T_k by removing e , adding a new node w , and edges (u, w) , (v, w) , and (w, k) .
3. (Optional) Search space of topologies closely related to T_n using operations such as NNIs or global tree swapping.

Insertion approaches can vary in speed from very fast to very slow, depending on the amount of computational time required to test each possible insertion point, and on how much post-processing topology searching is done. The naive approach would use any $O(n^2)$ algorithm to recalculate the OLS edge lengths for each edge in each test topology. This approach would take $O(k^2)$ computations for each edge, and thus $O(k^3)$ computations for each pass through Step 2(a). Summing over k , we see that the naive approach would result in a slow $O(n^4)$ algorithm.

The FASTME program of Desper and Gascuel [11] requires only $O(k)$ computations on Step 2(a) to build a greedy minimum evolution tree using OLS edge lengths. Let Δ be the input matrix, and Δ_{avg}^k be the matrix of average distances

between subtrees in T_k .

1. Start by constructing T_3 , the (unique) tree with three leaves; initialize Δ_{avg}^3 , the matrix of average distances between all pairs of subtrees in T_3 .
2. For $k = 4$ to n ,
 - (a) We first calculate $\delta_{\{k\}|A}$, for each subtree A of T_{k-1} .
 - (b) Test each edge $e \in T_{k-1}$ as a possible insertion point for k .
 - i. For all $e \in E$, we will let $f(e)$ to be the cost of inserting k along the edge e .
 - ii. Root T_{k-1} at r , an arbitrary leaf, let e_r be the edge incident to r .
 - iii. Let $c_r = f(e_r)$, a constant we will leave uncalculated.
 - iv. We calculate $g(e) = f(e) - c_r$ for each edge e . Observe $g(e_r) = 0$. Use a top-down search procedure to loop over the edges of T_{k-1} . Consider $e = e_j$, whose parent edge is e_i (see Fig. 1.7). Use equation (1.20) to calculate $g(e_j) - g(e_i)$. (This is accomplished by substituting A, B, C , and $\{k\}$ for W, X, Y , and Z , respectively.) Since $g(e_i)$ has been recorded, this calculation gives us $g(e_j)$.
 - v. Select $e_{\min}$ such that $g(e_{\min})$ is minimal.
 - (c) Form T_k by breaking $e_{\min}$, adding a new node w_k and edges connecting w_k to the vertices of $e_{\min}$ and to k . Update the matrix Δ_{avg}^k to include average distances in T_k between all pairs of subtrees separated by at most three edges.
3. FASTNNI post-processing (Section 1.5.2).

Let us consider the time complexity of this algorithm. Step 1 requires constant time. Step 2 requires $O(k)$ time in 2(a), thanks to equation (1.2), constant time for each edge considered in 2(b)iv for a total of $O(k)$ time, and $O(k)$ time for 2(c). Indeed, updating Δ_{avg}^k from $\Delta_{\text{avg}}^{k-1}$ only requires $O(k)$ time because we do not update the entire matrix. Thus T_k can be created from T_{k-1} in $O(k)$ time, which leads to $O(n^2)$ computations for the entire construction process. Adding Step 3 leads to a total cost of $O(n^2 + sn)$, where s is the number of swaps performed by FASTNNI from the starting point T_n .

1.5.5 From OLS to BME

Just as FASTBNNI is a slight variant of the FASTNNI algorithm for testing NNIs, we can easily adapt the greedy OLS taxon-insertion algorithm of Section 1.5.4 to greedily build a tree, using balanced edge lengths instead of OLS edge lengths. The only differences involve calculating balanced averages instead of unweighted averages.

1. In Step 2(a), we calculate $\delta_{\{k\}|A}^{\mathcal{T}_{k-1}}$ instead of $\delta_{\{k\}|A}$, using equation (1.3).
2. In Step 2(b)iv, we use equation (1.24) instead of equation (1.20) to calculate $g(e_j)$.
3. In Step 2(c), we need to calculate $\delta_{X|Y}^{\mathcal{T}_k}$ for each subtree X containing k , and each subtree Y disjoint from X .

4. Instead of FASTNNI post-processing, we use FASTBNNI post-processing.

The greedy balanced insertion algorithm is a touch slower than its OLS counterpart. The changes to Step 2(a) and 2(b) do not increase the running time, but the change to Step 2(c) forces the calculation of $O(k \text{ diam}(T_k))$ new average distances. With the change to FASTBNNI, the total cost of this approach is $O(n^2 \text{ diam}(T) + sn \text{ diam}(T))$ computations, given s iterations of FASTBNNI. Simulations [11] suggest that $s \ll n$ for a typical data set; thus, one could expect a total of $O(n^2 \log n)$ computations on average.

1.6 Statistical consistency

Statistical consistency is an important and desired property for any method of phylogeny reconstruction. Statistical consistency in this context means that the phylogenetic tree output by the algorithm in question converges to the true tree with correct edge lengths, when the number of sites increases and when the model used to estimate the evolutionary distances is the correct one. Whereas the popular character-based parsimony method has been shown to be statistically inconsistent in some cases [15], many popular distance methods have been shown to be statistically consistent. We first discuss positive results with the OLS and balanced versions of the minimum evolution principle, then provide negative results, and finally present the results of Atteson [2] that provide a measure of the convergence rate of NJ and related algorithms.

1.6.1 Positive results

A seminal paper in the field of minimum evolution is the work of Rzhetsky and Nei [39], demonstrating the consistency of the minimum evolution approach to phylogeny estimation, when using OLS edge lengths. Their proof was based on this idea: if T is a weighted tree of topology $\mathcal{T}$, and if the observation Δ is equal to d^T (i.e. the tree metric induced by T), then for any wrong topology $\mathcal{W}$, $\hat{l}(\mathcal{W}) > \hat{l}(\mathcal{T}) = l(T)$. In other words, T is the shortest tree and is thus the tree inferred using the ME principle. Desper and Gascuel [12] have used the same approach to show that the balanced minimum evolution method is consistent.

The circular orders of Section 1.3.6 lead to an easy proof of the consistency of BME (first discussed with David Bryant and Mike Steel). Assume $\Delta = d^T$ and consider any wrong topology $\mathcal{W}$. Per Section 1.3.6, we use $C(\mathcal{W})$ to denote the set of circular orderings of $\mathcal{W}$, and let $\hat{l}(\Delta, o, \mathcal{W})$ be the length estimate of $\mathcal{W}$ from Δ under the ordering o for $o \in C(\mathcal{W})$. The modified version of equation (1.25) yields the balanced length estimate of $\mathcal{W}$:

$$\hat{l}(\mathcal{W}) = \frac{1}{|C(\mathcal{W})|} \sum_{o \in C(\mathcal{W})} \hat{l}(\Delta, o, \mathcal{W}).$$

If $o \in C(\mathcal{W}) \cap C(\mathcal{T})$, then $\hat{l}(\Delta, o, \mathcal{W}) = \hat{l}(\Delta, o, \mathcal{T}) = l(T)$. If $o \in C(\mathcal{W}) \setminus C(\mathcal{T})$, then some edges of T will be double counted in the sum producing $\hat{l}(\Delta, o, \mathcal{W})$.

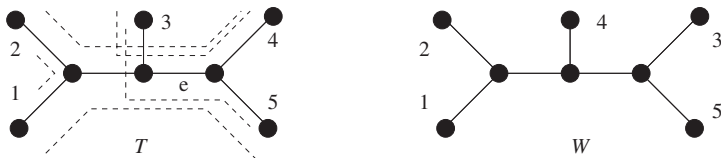


FIG. 1.8. Wrong topology choice leads to double counting edge lengths.

For example, if $\mathcal{T}$ and $\mathcal{W}$ are as shown in Fig. 1.8, and $o = (1, 2, 4, 3, 5) \in C(\mathcal{W}) \setminus C(\mathcal{T})$, then $\hat{l}(\Delta, o, \mathcal{W})$ (represented in Fig. 1.8 by dashed lines) counts the edge e twice. It follows that $\hat{l}(\mathcal{W}) > \hat{l}(\mathcal{T})$.

1.6.2 Negative results

Given the aforementioned proofs demonstrating the statistical consistency of the minimum evolution approach in selected settings, it is tempting to hope that minimum evolution would be a consistent approach for any least-squares estimation of tree length. Having more reliable tree length estimators, for example, incorporating the covariances of the evolutionary distance estimates, would then yield better tree inference methods based on the ME principle. Sadly, we have shown [25] that this is not always the case. Using a counter-example we showed that the ME principle can be inconsistent even when using WLS length estimation, and this result extends to various definitions of tree length, for example, only summing the positive edge length estimates while discarding the negative ones. However, our counter-example for WLS length estimation was artificial in an evolutionary biology context, and we concluded, “It is still conceivable that minimum evolution combined with WLS good practical results for realistic variance matrices.” Our more recent results with BME confirm this, as BME uses a special form of WLS estimation (Section 1.3.7) and performs remarkably well in simulations [12].

On the other hand, in reference [25] we also provided a very simple 4-taxon counter-example for GLS length estimation, incorporating the covariances of distance estimates (in contrast to WLS). Variances and covariances in this counter-example were obtained using a biological model [36], and were thus fully representative of real data. Using GLS length estimation, all variants of the ME principle were shown to be inconsistent with this counter-example, thus indicating that any combination of GLS and ME is likely a dead end.

1.6.3 Atteson’s safety radius analysis

In this section, we consider the question of algorithm consistency, and the circumstances under which we can guarantee that a given algorithm will return the correct topology $\mathcal{T}$, given noisy sampling of the metric d^T generated by some tree T with topology $\mathcal{T}$. As we shall see, NJ, a simple agglomerative heuristic approach based on the BME, is optimal in a certain sense, while more sophisticated algorithms do not possess this particular property.

Given two matrices $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ of identical dimensions, some standard measures of the distance between them include the L_p norms. For any real value of $p \geq 1$, the L_p distance between $\mathbf{A}$ and $\mathbf{B}$ is defined to be

$$\|\mathbf{A} - \mathbf{B}\|_p = \left(\sum_{i,j} (a_{ij} - b_{ij})^p \right)^{1/p}.$$

For $p = 2$, this is the standard Euclidean distance, and for $p = 1$, this is also known as the “taxi-cab” metric. Another related metric is the L_∞ norm, defined as

$$\|\mathbf{A} - \mathbf{B}\|_\infty = \max_{i,j} |a_{ij} - b_{ij}|.$$

A natural question to consider when approaching the phylogeny reconstruction problem is: given a distance matrix Δ , is it possible to find the tree T such that $\|d^T - \Delta\|_p$ is minimized? Day [10] showed that this problem is NP-hard for the L_1 and L_2 norms. Interestingly, Farach *et al.* [14] provided an algorithm for solving this problem in polynomial time for the L_∞ norm, but for the restricted problem of ultrametric approximation (i.e. $\|d^T - \Delta\|_\infty$ is minimized over the space of ultrametrics). Agarwala *et al.* [1] used the ultrametric approximation algorithm to achieve an approximation algorithm for the L_∞ norm: if $\epsilon = \min_T \|d^T - \Delta\|_\infty$, where d^T ranges over all tree metrics, then the single pivot algorithm of Agarwala *et al.* produces a tree T' whose metric $d^{T'}$ satisfies $\|d^{T'} - \Delta\|_\infty \leq 3\epsilon$.

The simplicity of the L_∞ norm also allows for relatively simple analysis of how much noise can be in a matrix Δ that is a sample of the metric d^T while still allowing accurate reconstruction of the tree T . We define the *safety radius* of an algorithm to be the maximum value ρ such that, if e is the shortest edge in a tree T , and $\|\Delta - d^T\|_\infty < \rho l(e)$, then the algorithm in question will return a tree with the same topology as T .

It is immediately clear that no algorithm can have a safety radius greater than $\frac{1}{2}$: consider the following example from [2]. Suppose $e \in T$ is an internal edge with minimum length $l(e)$. Let W , X , Y , and Z be four subtrees incident to e , such that W and X are separated from Y and Z , as in Fig. 1.9. Let d be a metric:

$$\begin{aligned} d_{ij} &= d_{ij}^T - \frac{l(e)}{2}, & \text{if } i \in W, j \in Y \quad \text{or} \quad i \in X, j \in Z, \\ d_{ij} &= d_{ij}^T + \frac{l(e)}{2}, & \text{if } i \in W, j \in X \quad \text{or} \quad i \in Y, j \in Z, \\ d_{ij} &= d_{ij}^T, & \text{otherwise.} \end{aligned}$$

d is graphically realized by the network N in Fig. 1.9, where the edge e has been replaced by two pairs of parallel edges, each with a length of $l(e)/2$.

Moreover, consider the tree T' which we reach from T by a NNI swapping X and Y , and keeping the edge e with length $l(e)$. Then it is easily seen that

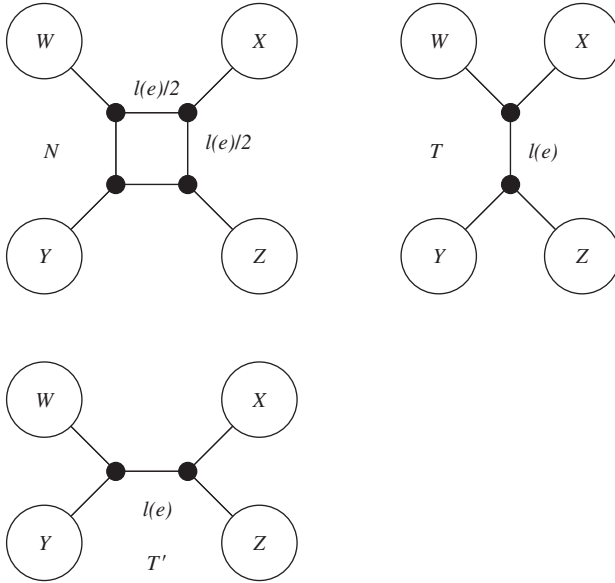


FIG. 1.9. Network metric equidistant from two tree metrics.

$\|d^T - d\|_\infty = l(e)/2 = \|d^{T'} - d\|_\infty$. Since d is equidistant to d^T and $d^{T'}$, no algorithm could guarantee finding the correct topology, if d is the input metric.

Atteson [2] proved that NJ achieves the best possible safety radius, $\rho = \frac{1}{2}$. If d^T is a tree metric induced by T , Δ is a noisy sampling of d^T , and $\epsilon = \max_{i,j} |d_{ij}^T - \delta_{ij}|$, then NJ will return a tree with the same topology as T , providing all edges of T are longer than 2ϵ . In fact, this result was proven for a variety of NJ related algorithms, including UNJ, BIONJ, and ADDTREE, and is a property of the agglomerative approach, when this approach is combined with NJ's (or ADDTREE's) pair selection criterion. An analogous optimality property was recently shown concerning UPGMA and related agglomerative algorithms for ultrametric tree fitting [27]. In contrast, the 3-approximation algorithm only has been proven to have a safety radius of $\frac{1}{8}$.

1.7 Discussion

We have provided an overview of the field of distance algorithms for phylogeny reconstruction, with an eye towards the balanced minimum evolution approach. The BME algorithms are very fast—faster than Neighbor Joining and sufficiently fast to quickly build trees on data sets with thousands of taxa. Simulations [12] have demonstrated superiority of the BME approach, not only in speed, but also in the quality of output trees. Topologies output by FASTME using the balanced minimum evolution scheme have been shown to be superior to those produced by BIONJ, WEIGHBOR, and standard WLS (e.g. FITCH or PAUP*), even though FASTME requires considerably less time to build them.

The balanced minimum evolution scheme assigns edge lengths according to a particular WLS scheme that appears to be biologically realistic. In this scheme, variances of distance estimates are proportional to the exponent of topological distances. Since variances have been shown to be proportional to the exponent of evolutionary distances in the Jukes and Cantor [30] and related models of evolution [7], this model seems reasonable as one expects topological distances to be linearly related to evolutionary distances in most data sets.

The study of cyclic permutations by Semple and Steel [44] provides a new proof of the validity of Pauplin’s tree length formula [38], and also leads to a connection between the balanced edge length scheme and Neighbor Joining. This connection, and the WLS interpretation of the balanced scheme, may explain why NJ’s performance has traditionally been viewed as quite good, in spite of the fact that NJ had been thought to not optimize any global criterion. The fact that FASTME itself more exhaustively optimizes the same WLS criterion may explain the superiority of the balanced approach over other distance algorithms.

There are several mathematical problems remaining to explore in studying balanced minimum evolution. The “safety radius” of an algorithm has been defined [2] to be the number ρ such that, if the ratio of the maximum measurement error over minimum edge length is less than ρ , then the algorithm will be guaranteed to return the proper tree. Although we have no reason to believe BME has a small safety radius, the exact value of its radius has yet to be determined. Also, though the BME approach has been proven to be consistent, the consistency and safety radius of the BME heuristic algorithms (e.g. FASTBNNI and the greedy construction of Section 1.5.5) have to be determined. Finally, there remains the question of generalizing the balanced approach—in what settings would this be meaningful and useful?

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LIKELIHOOD CALCULATION IN MOLECULAR PHYLOGENETICS

David Bryant, Nicolas Galtier, and Marie-Anne Poursat

Likelihood estimation is central to many areas of the natural and physical sciences and has had a major impact on molecular phylogenetics. In this chapter we provide a concise review of some of the theoretical and computational aspects of likelihood-based phylogenetic inference. We outline the basic probabilistic model and likelihood computation algorithm, as well as extensions to more realistic models and strategies of likelihood optimization. We survey several of the theoretical underpinnings of the likelihood framework, reviewing research on consistency, identifiability, and the effect of model mis-specification, as well as advantages, and limitations, of likelihood ratio tests.

2.1 Introduction

Maximum likelihood (ML) estimation is arguably the most widely used method for statistical inference. The framework was introduced in the early 1920s by the pioneering statistician and geneticist, R.A. Fisher [18]. Likelihood based estimation is now routinely applied in almost all fields of the biological sciences, including epidemiology, ecology, population genetics, quantitative genetics, and evolutionary biology.

This chapter provides a concise survey of computational, statistical, and mathematical aspects of likelihood inference in phylogenetics. Readers looking for a general introduction to the area are encouraged to consult Felsenstein [15] or Swofford *et al.* [49]. A detailed mathematical treatment is provided by Semple and Steel [42].

Likelihood starts with a model of how the data arose. This model gives a probability $\mathbb{P}[D|\theta]$ of observing the data, given particular values for the parameters of the model (here denoted by the symbol θ). In phylogenetics, the parameters θ include the tree, branch lengths, the sequence evolution model, and so on. The key idea behind likelihood is to choose the parameters that maximize the probability of observing the data we have observed. We therefore define a *likelihood function* $L(\theta) = \mathbb{P}[D|\theta]$ (sometimes written as $L(\theta|D) = \mathbb{P}[D|\theta]$) that captures how “likely” it is to observe the data for a given value of the parameters θ . A high likelihood indicates a good fit. The *maximum likelihood estimate* is the value of

θ that maximizes $L(\theta)$. In our context, we will be searching for the maximum likelihood estimate of a phylogeny.

For the remainder of the chapter we will assume that the reader is comfortable with the concepts and terminology of likelihood in general statistics. Background material on likelihood (and related topics in statistics) can be found in Edwards [11] and Ewens and Grant [12].

Molecular phylogenetics is the field aiming at reconstructing evolutionary trees from DNA sequence data. The maximum likelihood (ML) method was introduced to this field by Joe Felsenstein [14] in 1981, and since become increasingly popular, particularly following recent increases in computing power.

Maximum likelihood has an important advantage over the still popular *maximum parsimony* (MP) method: ML is *statistically consistent* (see Section 2.6). As the size of the data set increases, ML will converge to the true tree with increasing certainty (provided, of course, that the model is sufficiently accurate). Felsenstein showed that Maximum Parsimony is *not* consistent, particularly in the case of unequal evolutionary rates between different lineages [13].

While the basic intuition behind likelihood inference is straightforward, the application of the framework is often quite difficult. First there is the problem of model design. In molecular phylogenetics, the evolution of genetic sequences is usually modelled as a Markov process running along the branches of a tree. The parameters of the model include the tree topology, branch lengths, and characteristics of the Markov process. As in all applied statistics there is a pay-off between more complex, realistic models, and simpler, tractable models. More complex models result in a better fit, but are more vulnerable to random error.

The second major difficulty with likelihood based inference is the problem of computing likelihood values and optimizing the parameters. Likelihood in molecular phylogenetics is made possible by the dynamic programming algorithm of Felsenstein [14]. We outline this algorithm in Section 2.3. However, nobody has found an efficient and exact algorithm for optimizing the parameters. The techniques most widely used are surprisingly basic.

The third difficulty with likelihood is the interpretation and validation of the results of a likelihood analysis: assessing which results are significant and which analyses are reliable.

In this chapter, we will discuss all three aspects. First (Section 2.2) we describe the basic Markov models central to likelihood inference in molecular phylogenetics. Second we present the fundamental algorithm of Felsenstein (Section 2.3), as well as extensions to more complex models (Section 2.4), and a survey of optimization techniques used (Section 2.5). Third we review the theoretical underpinnings of the likelihood framework. In particular, we discuss the consistency of maximum likelihood estimation in phylogenetics, and the conditions under which maximum likelihood will return the correct tree (Section 2.6). Finally, we show how the likelihood framework can guide us in the development of improved evolutionary models, and outline the theoretical justification for the standard likelihood ratio tests already in wide use in phylogenetics (Section 2.7).

2.2 Markov models of sequence evolution

Before any likelihood analysis can take place we need to formulate a probabilistic model for evolution. In reality, the process of evolution is so complex and multifaceted that there is no way we can completely determine accurate probabilities. Our descriptions of the basic model will involve assumption built upon assumption. It is a wonder of phylogenetics that we can get so far with the basic models that we do have. Of course, this phenomenon is in no way unique to phylogenetics.

The reliance of likelihood methods on explicit models is sometimes seen as a weakness of the likelihood framework. On the contrary, the need to make explicit assumptions is a strength of the approach. Likelihood methods enable both inferences about evolutionary history and assessments of the accuracy of the assumptions made. “The purpose of models is not to fit the data, but to sharpen the questions.”¹ While the basic models we describe in this section do an excellent job explaining much of the random variation in molecular sequences, shortcomings of the models (e.g. with respect to rate variation) have led to better models, a better understanding of sequence evolution, and a host of “sharper and sharper” questions on the relationship between rate variation, structure, and function.

More detailed reviews of these models can be found in references. [15, 49].

2.2.1 Independence of sites

Our first simplifying assumption is the perhaps unrealistic assertion that sites evolve independently. Thus the probability that sequence A evolves to sequence B equals the product, over all sites i , that the state in site i of A evolves to the state in site i of B . This simplifies computation substantially. In fact it is almost essential for tractability (though can be stretched a little—see Section 2.4). With this assumption made, we spend the rest of the section focusing on the evolution of an individual site.

2.2.2 Setting up the basic model

Consider the cartoon representation of site evolution in Fig. 2.1. Over a time period t , the state A at the site is replaced by the state T . There are a number of random mutation events (in this case, three) that are randomly distributed through the time period. One of these is redundant, with A being replaced by A . We consider these redundant mutations more for mathematical convenience than anything else. The mutations from A to G and from G to T are said to be *silent*. We do not observe the change to G , only the beginning and end states.

Let $\mathcal{E}$ denote the set of states and let $c = |\mathcal{E}|$. For DNA sequences, $\mathcal{E} = \{A, C, G, T\}$, while for proteins, $\mathcal{E}$ equals the set of amino acids. For convenience, we assume that the states have indices 1 to $|\mathcal{E}|$. The mutation events occur according to a *continuous time Markov chain* with state set $\mathcal{E}$. The number

¹Samuel Karlin, 11th R.A. Fisher Memorial Lecture, Royal Society 20, April 1983.

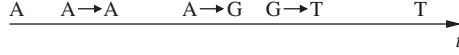


FIG. 2.1. Redundant and hidden mutations. Over time t , the site has a redundant mutation, followed by a mutation to G and then to T . The mutation to G is non-detectable, so is called *silent*. The timing (and number) of the mutation events is modelled by a Poisson process.

of these events has Poisson distribution: the probability of k mutation events is

$$\mathbb{P}[k \text{ events}] = \frac{(\mu t)^k e^{-\mu t}}{k!}.$$

Here μ is the rate of these events, so that the expected number of events in time t is μt . When there is a mutation event, we let $\mathbf{R}_{xy}$ denote the probability of changing to state y given that the site was in state x . Since redundant mutations are allowed, $\mathbf{R}_{xx} > 0$. Putting everything together, the probability of ending in state y after time t given that the site started in state x is given by the xy th element of $\mathbf{P}(t)$, where $\mathbf{P}(t)$ is the matrix valued function

$$\mathbf{P}(t) = \sum_{k=0}^{\infty} (\mathbf{R}^k) \frac{(\mu t)^k e^{-\mu t}}{k!}. \quad (2.1)$$

This formula just expresses the probabilities of change summed over the possible values of k , the number of mutation events.

Let $\mathbf{Q}$ be the matrix $\mathbf{R} - \mathbf{I}$, where $\mathbf{I}$ is the $c \times c$ identity matrix. After some matrix algebra, equation (2.1) becomes

$$\mathbf{P}(t) = \sum_{k=0}^{\infty} \frac{(\mathbf{R} - \mathbf{I})^k (\mu t)^k}{k!} = \sum_{k=0}^{\infty} \frac{(\mathbf{Q} \mu t)^k}{k!} = e^{\mathbf{Q} \mu t}. \quad (2.2)$$

The matrix $\mathbf{Q}$ is called the *instantaneous rate matrix* or *generator*. Here, $e^{\mathbf{Q} \mu t}$ denotes the *matrix exponential*. There is a standard trick to compute it.

First, diagonalize the matrix $\mathbf{Q}$ as $\mathbf{Q} = \mathbf{A} \mathbf{D} \mathbf{A}^{-1}$ with $\mathbf{D}$ diagonal (e.g. using Singular Value Decomposition, see [27]). For any integer k , we have that

$$\begin{aligned} (\mathbf{Q})^k &= (\mathbf{A} \mathbf{D} \mathbf{A}^{-1})(\mathbf{A} \mathbf{D} \mathbf{A}^{-1}) \cdots (\mathbf{A} \mathbf{D} \mathbf{A}^{-1}) \\ &= \mathbf{A} (\mathbf{D})^k \mathbf{A}^{-1}. \end{aligned}$$

Taking the powers of diagonal matrices is just a matter of taking the powers of its entries. It follows that

$$e^{\mathbf{Q} \mu t} = \mathbf{A} e^{\mathbf{D} \mu t} \mathbf{A}^{-1},$$

where $e^{\mathbf{D}}$ is a diagonal matrix and, for each x , $(e^{\mathbf{D}})_{xx} = e^{\mathbf{D}_{xx}}$.

As an example, consider the F81 model of Felsenstein [14]. We assume that the states in $\mathcal{E}$ are ordered A, C, G, T . The model is defined in reference [49] in terms of its rate matrix

$$\mathbf{Q} = \begin{bmatrix} -(\pi_Y + \pi_G) & \pi_C & \pi_G & \pi_T \\ \pi_A & -(\pi_R + \pi_T) & \pi_G & \pi_T \\ \pi_A & \pi_C & -(\pi_Y + \pi_A) & \pi_T \\ \pi_A & \pi_C & \pi_G & -(\pi_R + \pi_C) \end{bmatrix}. \quad (2.3)$$

Rows in $\mathbf{Q}$ indicate the initial state, and columns the final state, states being taken in the A, C, G, T alphabetic order. $\pi_A, \pi_C, \pi_G, \pi_T$ are probabilities that sum to one (see the next section), $\pi_R = \pi_A + \pi_G$ and $\pi_Y = \pi_C + \pi_T$. This model is equivalent to one with discrete generations occurring according to a Poisson process, and (single event) transition probability matrix

$$\mathbf{R} = \begin{bmatrix} 1 - (\pi_Y + \pi_G) & \pi_C & \pi_G & \pi_T \\ \pi_A & 1 - (\pi_R + \pi_T) & \pi_G & \pi_T \\ \pi_A & \pi_C & 1 - (\pi_Y + \pi_A) & \pi_T \\ \pi_A & \pi_C & \pi_G & 1 - (\pi_R + \pi_C) \end{bmatrix}.$$

The corresponding transition probability matrix, for a given time period t , is obtained by diagonalizing $\mathbf{Q}$ and taking the exponential. The resulting matrix can be expressed simply by

$$\mathbf{P}_{xy}(t) = \begin{cases} \pi_y + (1 - \pi_y)e^{-\mu t}, & \text{if } x = y, \\ \pi_y(1 - e^{-\mu t}), & \text{if } x \neq y. \end{cases} \quad (2.4)$$

2.2.3 Stationary distribution

We have described here a *continuous time Markov chain*, the continuous time analogue of a *Markov chain*. We will also assume that this Markov process is *ergodic*. This means that as t goes to infinity, the probability that the site is in some state y is non-zero and independent of the starting state. That is, there are positive values $\pi_1, \dots, \pi_c$ such that, for all x, y in EE

$$\lim_{t \rightarrow \infty} \mathbf{P}_{xy}(t) = \pi_y.$$

The values $\pi_1, \dots, \pi_c$ comprise a *stationary distribution* (also called the *equilibrium distribution* or *equilibrium frequencies*) for the states. For all $t \geq 0$ these values satisfy

$$\pi_y = \sum_{x \in \mathcal{E}} \pi_x \mathbf{P}_{xy}(t). \quad (2.5)$$

If we sample the initial state from the stationary distribution, then run the process for time t , then the distribution of the final state will equal the stationary distribution. A consequence of equation (2.5) is that

$$0 = \sum_{x \in \mathcal{E}} \pi_x \mathbf{Q}_{xy},$$

so that we can recover the stationary distribution directly from $\mathbf{Q}$. We use $\mathbf{\Pi}$ to denote the $c \times c$ diagonal matrix with π_x 's down the diagonal.

For the F81 model we see from equation (2.4) that

$$\lim_{t \rightarrow \infty} \mathbf{P}_{xy}(t) = \pi_y,$$

for $x = y$ or $x \neq y$. The values $\pi_A, \pi_C, \pi_G, \pi_T$ make up the stationary distribution for this model. Hence π_R is the stationary probability for purines (A or G) and π_Y is the stationary probability for pyrimidines (C or T). The matrix $\mathbf{\Pi}$ is given by

$$\mathbf{\Pi} = \begin{bmatrix} \pi_A & 0 & 0 & 0 \\ 0 & \pi_C & 0 & 0 \\ 0 & 0 & \pi_G & 0 \\ 0 & 0 & 0 & \pi_T \end{bmatrix}.$$

2.2.4 Time reversibility

The next common assumption is of *time reversibility*. This is not exactly what it sounds like. We do not assume that the probability of going from state x to state y is the same as the probability of going from state y to state x . Instead we assume that the probability of sampling x from the stationary distribution and going to state y is the same as the probability of sampling y from the stationary distribution and going to state x . That is, for all $x, y \in \mathcal{E}$ and $t \geq 0$ we have

$$\pi_x \mathbf{P}_{xy}(t) = \pi_y \mathbf{P}_{yx}(t).$$

One can show that this corresponds to the condition that

$$\pi_x \mathbf{Q}_{xy} = \pi_y \mathbf{Q}_{yx},$$

that is, the matrix $\mathbf{\Pi Q}$ is symmetric.

The F81 model is time reversible even though $\mathbf{P}(t)$ is not symmetric. To see this, consider arbitrary states x, y with $x \neq y$. Then

$$\pi_x \mathbf{P}_{xy}(t) = \pi_x \pi_y (1 - e^{-\mu t}),$$

$$\pi_y \mathbf{P}_{yx}(t) = \pi_y \pi_x (1 - e^{-\mu t}).$$

Time reversibility makes it much easier to diagonalize $\mathbf{Q}$. Since $\mathbf{\Pi Q}$ is symmetric, so is

$$\mathbf{\Pi}^{-1/2} \mathbf{\Pi Q \Pi}^{-1/2} = \mathbf{\Pi}^{1/2} \mathbf{Q \Pi}^{-1/2}.$$

Finding eigenvalues of a symmetric matrix is, in general, far easier than finding eigenvalues of a non-symmetric matrix [27]. Hence we first diagonalize

$$\mathbf{\Pi}^{1/2} \mathbf{Q \Pi}^{-1/2}$$

to give a diagonal matrix $\mathbf{D}$ and invertible matrix $\mathbf{B}$ such that

$$\mathbf{\Pi}^{1/2} \mathbf{Q \Pi}^{-1/2} = \mathbf{B D B}^{-1}.$$

Setting $\mathbf{A} = \mathbf{\Pi}^{-1/2} \mathbf{B}$ gives $\mathbf{Q} = \mathbf{A D A}^{-1}$. This approach is used by David Swofford when computing the exponential matrices of general rate matrices in

PAUP [48]. Time reversibility also makes it easier to compute likelihoods on a tree, since the likelihood becomes independent of the position of the root [14].

2.2.5 Rate of mutation

In molecular phylogenetics, time is measured in *expected mutations per site*, rather than in years. The reason is that the rate of evolution can change markedly between different species, different genes, or even different parts of the same sequence.

Recall that our model of site evolution has mutation events occurring according to a Poisson process, with an expected number of events equal to μt . However, some of these mutation events are nothing more than mathematical conveniences—the mutations from a state to itself. If we assume that the distribution of the initial state equals the stationary distribution, then the probability that a mutation event gives a redundant mutation is

$$\sum_{x \in \mathcal{E}} \pi_x \mathbf{R}_{xx} = \text{trace}(\mathbf{\Pi R}).$$

Hence the probability that the mutation event is not redundant is

$$1 - \text{trace}(\mathbf{\Pi R}) = -\text{trace}(\mathbf{\Pi Q}).$$

The expected number of these in unit time ($t = 1$) is then

$$-\mu \text{trace}(\mathbf{\Pi Q}). \quad (2.6)$$

This is the mutation rate for the process. Care must be taken when comparing two different models in case their underlying mutation rates differ. Given a rate matrix $\mathbf{Q}$ we choose μ such that the overall rate of mutation $-\mu \text{trace}(\mathbf{\Pi Q})$ is one. In this way the length of the branch corresponds to the expected number of mutations per site along that branch, irrespective of the model.

Applying equation (2.6) to the F81 model we obtain a rate of

$$-\mu \text{trace}(\mathbf{\Pi Q}) = \mu(\pi_A(1 - \pi_A) + \pi_C(1 - \pi_C) + \pi_G(1 - \pi_G) + \pi_T(1 - \pi_T)),$$

so, given $\pi_A, \dots, \pi_T$ we would set

$$\mu = [\pi_A(1 - \pi_A) + \pi_C(1 - \pi_C) + \pi_G(1 - \pi_G) + \pi_T(1 - \pi_T)]^{-1}$$

to normalize the rates.

2.2.6 Probability of sequence evolution on a tree

We now extend the model for sequence evolution to evolution on a phylogeny. We are still concerned, at this point, with the evolution of a single site. Because of independence between sites, the probability of a set of sequences evolving is just the product of the probabilities for the individual sites.

Each site i in a sequence determines a *character* on the leaves: a function χ_i from the leaf set to the set of states $\mathcal{E}$. An *extension* $\hat{\chi}_i$ of a character χ_i is an assignment of states to *all* of the nodes in the tree that agrees with χ_i on the leaves.

We define the *probability of an extension* as the probability of the state at the root (given by the stationary distribution) multiplied by the probabilities of all the changes (or conservations) down each branch in the tree. If we use b_{uv} to denote the length of the branch between node u and node v , and let $\hat{\chi}_i(v)$ denote the state assigned to v , then we have a probability

$$\mathbb{P}[\hat{\chi}_i | \theta] = \pi_{\hat{\chi}_i(v_0)} \prod_{\text{branches } \{u, v\}} \mathbf{P}_{\hat{\chi}_i(u)\hat{\chi}_i(v)}(b_{uv}). \quad (2.7)$$

Here, v_0 is the root of the tree. The probability of site i is then the *marginal probability* over all the extensions $\hat{\chi}_i$ of χ_i :

$$\mathbb{P}[\chi_i | \theta] = \sum_{\hat{\chi}_i \text{ extends } \chi_i} \mathbb{P}[\hat{\chi}_i | \theta]. \quad (2.8)$$

The probability of the complete alignment is simply the product of the probabilities of the sites. The next section gives the details of this fundamental calculation.

Equations (2.7) and (2.8) are perhaps better understood if we consider the problem of *simulating* sequences on a tree. To simulate a site in a sequence we first sample a state at the root from the stationary distribution. Then, working down the tree, we sample the state at the end of a branch (furthest from the root) using the value x already sampled at the beginning of the branch, the length of the branch b , and the probabilities in row x of the transition matrix $\mathbf{P}(b)$. The states chosen (eventually) at the leaves then give the character for one site of our simulated sequences. The probability $\mathbb{P}[\chi_i | \theta]$ equals the probability that the character χ_i could have been generated using this simulation method.

2.3 Likelihood calculation: the basic algorithm

Here we describe the basic algorithm for computing the likelihood $L(\theta) = \mathbb{P}[\chi_i | \theta]$ of a site, given a (rooted) tree, branch lengths, and the model of sequence evolution. The likelihood of an alignment is computed by multiplying the likelihoods for each of the n sites

$$L(\theta) = \prod_{i=1}^n \mathbb{P}[\chi_i | \theta]. \quad (2.9)$$

Remember that χ_i is the character (column) corresponding to the i th site in a sequence alignment. Let v be an internal node of the tree, and let $L_i^v(x)$, $x \in \mathcal{E}$ denote the *partial conditional likelihood* defined as:

$$L_i^v(x) = \mathbb{P}[\chi_i^v | \theta, \hat{\chi}_i(v) = x],$$

where χ_i^v is the restriction of the character χ_i to descendants of node v and $\hat{\chi}_i(v)$ is the ancestral state for site i at node v (Fig. 2.2). The value $L_i^v(x)$ is the likelihood at site i for the subtree underlying node v , conditional on state x at v .

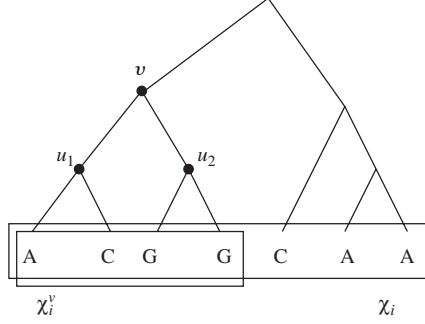


FIG. 2.2. Illustration of a node v , its children u_1, u_2 , the character χ_i and its restriction χ_i^v to the subtree rooted at v .

The likelihood of the complete character χ_i can be expressed as:

$$\mathbb{P}[\chi_i | \theta] = \sum_{x \in \mathcal{E}} \mathbb{P}[\hat{\chi}(v_0) = x] L_i^{v_0}(x), \quad (2.10)$$

where v_0 is the root node. The probability $\mathbb{P}[\hat{\chi}(v_0) = x]$ equals the probability for x under the stationary distribution, π_x .

The function $L_i^v(x)$ satisfies the recurrence

$$L_i^v(x) = \left(\sum_{y \in \mathcal{E}} \mathbf{P}_{xy}(t_1) L_i^{u_1}(y) \right) \left(\sum_{y \in \mathcal{E}} \mathbf{P}_{xy}(t_2) L_i^{u_2}(y) \right), \quad (2.11)$$

for all internal nodes v , where u_1 and u_2 are the children of v and t_1, t_2 are the lengths of the branches connecting them to v . Equation (2.11) results from the independence of the processes in the two subtrees below node v . For leaf l , we have

$$L_i^l(x) = \begin{cases} 1, & \text{if } \chi_i(l) = x, \\ 0, & \text{otherwise.} \end{cases}$$

Note that equation (2.11) can be easily extended to nodes v with more than two children.

The transition probabilities $P_{xy}(t_1)$ and $P_{xy}(t_2)$ are determined from equation (2.2). As observed above, this requires the diagonalization of the rate matrix $\mathbf{Q}$. However we need only perform this diagonalization once, after which point it only takes $O(c)$ operations, where c is the size of the state set, to evaluate each probability.

The above calculation was defined on a rooted tree. For a time reversible, stationary process, however, the location of the root does not matter: the likelihood value is independent of the position of the root [14]. As well, the logarithm of the likelihood is usually computed rather than the likelihood itself. The product in equation (2.9) becomes a summation if the log-likelihood is computed.

Calculating the log-likelihood of a tree therefore involves

- (i) diagonalization of $\mathbf{Q}$;
- (ii) for each branch of the tree, taking the exponential of $\mathbf{Q}t$, where t is the branch length;
- (iii) for every site and every possible state, applying equation (2.11) using a post-order traversal of the tree;
- (iv) taking the logarithm and summing over sites.

Recall that c is the number of states, m the number of leaves, and n the number of sites. Step (i) can be performed in $O(c^3)$ time using standard numerical techniques. Step (ii) takes $O(mc^3)$ time. Step (iii) takes $O(mnc^2)$ time, and step (iv) takes $O(n)$ time. The whole algorithm therefore takes $O(mc^3 + mnc^2)$ time. Step (iii) is the most computationally expensive step in virtually every application.

2.4 Likelihood calculation: improved models

The calculation presented above applies to standard Markov models of sequence evolution, assuming a single, common process to all sites and in all lineages, and independent sites. Actual molecular evolutionary processes often depart from these assumptions. We now introduce likelihood calculation under more realistic models of sequence evolution, with the aim of improving phylogenetic estimates and of learning more about the evolutionary forces that drive sequence variation.

2.4.1 Choosing the rate matrix

The choice of rate matrix (generator) $\mathbf{Q}$ is an important part of the modelling process. The rate matrix has $c(c-1)$ non-diagonal entries, where c is the number of states. Thus the number of off-diagonal entries equals 12 for DNA sequences, 180 for amino-acid sequences, and 3660 for codon sequences. This number is halved if we also have time reversibility. Allowing one free parameter per rate is not appropriate; one has to introduce constraints in order to reach a reasonable number of free parameters, preferably representing biologically meaningful features of evolutionary processes.

In practice, the features of $\mathbf{Q}$ are determined empirically. For example, in DNA sequences it has been observed that *transitions* (mutations between A and G or between C and T) are more frequent than *transversions* (other mutations). The HKY model [29] incorporates this observation into the rate matrix:

$$\mathbf{Q} = \begin{bmatrix} -(\pi_Y + \kappa\pi_G) & \pi_C & \kappa\pi_G & \pi_T \\ \pi_A & -(\pi_R + \kappa\pi_T) & \pi_G & \kappa\pi_T \\ \kappa\pi_A & \pi_C & -(\pi_Y + \kappa\pi_A) & \pi_T \\ \pi_A & \kappa\pi_C & \pi_G & -(\pi_R + \kappa\pi_C) \end{bmatrix}.$$

As before, $\pi_R = \pi_A + \pi_G$ and $\pi_Y = \pi_C + \pi_T$.

This matrix is the same as that for F81, except for an extra parameter κ affecting the relative rate of mutations within purines or within pyrimidines.

When $\kappa = 1.0$ we obtain the F81 model again. When $\kappa > 1.0$ the rate of transitions is greater than the rate of transversions. A large body of literature discusses the merits of various parameterizations of rate matrices for DNA, protein and codon models (e.g. [49]). We do not review this issue here. The above-described basic likelihood calculation procedure applies whatever the parameterization.

Non-homogeneous models of sequence evolution, in which distinct branches of the tree have distinct rate matrices, have been introduced for modelling variations of the selective regime of protein coding genes [60], or variations of base composition in DNA (RNA) sequences [22]. The calculation of transition probabilities along branches ($\mathbf{P}_{xy}(t)$ in equation (2.11)) should be modified accordingly, using the appropriate rate matrix for each branch. When the distinct rate matrices have unequal equilibrium frequencies [22, 59], the process becomes non-stationary: stationary frequencies are never reached because they vary in time. In this case, the likelihood function becomes dependent on the location of the root, and the ancestral frequency spectrum ($\mathbb{P}[\hat{\chi}(v_0) = x]$ in equation (2.10)) becomes an additional parameter: it can no longer be deduced from the evolutionary model.

2.4.2 Among site rate variation (ASRV)

A strong and unrealistic assumption of the standard model is that sites evolve at the same rate. In real data sets there are typically fast and slowly evolving sites, mostly as a consequence of variable selective pressure. Functionally important sites are conserved during evolution, while unimportant sites are free to vary.

Yang first introduced likelihood calculation incorporating variable rates across sites [55]. He proposed that the variation of evolutionary rates across sites be modelled by a continuous distribution: the rate of a specific site i is not a constant, but a random variable $r(i)$. The likelihood for site i is calculated by integrating over all possible rates:

$$\mathbb{P}[\chi_i | \theta] = \int_0^\infty \mathbb{P}[\chi_i | r(i) = r, \theta] f(r) dr, \quad (2.12)$$

where f is the probability density of the assumed rates distribution, and where $\mathbb{P}[\chi_i | r(i) = r, \theta]$ is the likelihood for character χ_i conditional on rate $r(i) = r$ for this site. The latter term is calculated by applying recurrence (2.11) after multiplying all of the branch lengths in the tree by r . Typically, a Gamma distribution is used for $f(r)$. Its variance and shape are controlled by an additional parameter that can be estimated from the data by the maximum-likelihood method.

The integration in equation (2.12) must be performed numerically, which is time consuming. In practice, this calculation can be completed only for small trees. For this reason, Yang proposed to assume a discrete, rather than continuous, distribution of rates across sites [56]:

$$\mathbb{P}[\chi_i | \theta] = \sum_{j=1}^g \mathbb{P}[\chi_i | r(i) = r_j, \theta] p_j, \quad (2.13)$$

where g is the assumed number of rate classes and p_j the probability of rate class j . Yang [56] uses a discretized Gamma distribution for the probabilities p_j . The complexity of the likelihood calculation under the discrete-Gamma model of rate variation is $O(mc^3g + mnc^2g)$, that is, essentially g times the complexity of the equal-rate calculation. Using ASRV models typically leads to a large increase of log-likelihood, compared to constant-rate models. The extension of this approach to heterogeneous models of site evolution is the subject of Chapter 5, this volume.

Note that sites are not assigned to rate classes in this calculation. Rather, all possible assignments are considered, and the conditional likelihoods averaged. Sites can be assigned to rate classes following likelihood calculation. The *posterior probability* of rate class j for site Y_i can be defined as:

$$\mathbb{P}[\text{site } i \text{ in class } j] = \frac{p_j P[\chi_i | r(i) = r_j, \theta]}{P[\chi_i | \theta]}, \quad (2.14)$$

where the calculation is achieved using the maximum likelihood estimates of parameters (tree, branch lengths, rate matrix, gamma shape). This equation does not account for the uncertainty in the unknown parameters, an approximate procedure called “empirical Bayesian” [61].

2.4.3 Site-specific rate variation

In models of between-site rate variation, the (relative) rate of a site is constant in time: a slow site is slow, and a fast site fast, in every lineage of the tree. In reality, evolutionary rate might, however, vary in time, if the level of constraint applying to a specific site changes. The notion that the evolutionary rate of a site can evolve was first introduced by Fitch [19], and subsequently modelled by Tuffley and Steel [52] and Galtier [21]. This process has been named *covariation* (for COncomitantly VARIable codON [19]), heterotachy, or site-specific rate variation.

Covariation models typically assume a compound process of evolution. The rate of a given site evolves along the tree according to a Markov process defined in the space of rates. Thus the site evolves in the state space according to a Markov process whose local rate is determined by the outcome of the rate process. A site can be fast in some parts of the tree, but slow in other parts. Such processes are called *Markov-modulated Markov processes* or *Cox processes*. The state process is modulated by the rate process.

Existing models use a discrete rate space: a finite number g of Gamma distributed rates are permitted, just like in discretized ASRV models (see above). Let $\mathbf{r} = (r_j)$ be the vector of allowed rates (size g), let $\text{diag}(\mathbf{r})$ be the diagonal matrix with diagonal entries r_j , and $\mathbf{G}$ be the rate matrix of the rate process, indexed by the rate classes. Let $\mathbf{Q}$ be the rate matrix of the state process. The compound process can be seen as a single process taking values in $\{r_j\} \times \mathcal{E}$, a compound space of size $g \cdot c$. The rate matrix, $\mathbf{Z}$, of this process can be expressed using the Kronecker operand $\otimes$. If $\mathbf{A}$ is an $m \times m$ matrix and $\mathbf{B}$ is an $n \times n$

matrix then $\mathbf{A} \otimes \mathbf{B}$ is the $mn \times mn$ matrix

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} \mathbf{A}_{11}\mathbf{B} & \dots & \mathbf{A}_{1m}\mathbf{B} \\ \vdots & \ddots & \vdots \\ \mathbf{A}_{m1}\mathbf{B} & \dots & \mathbf{A}_{mm}\mathbf{B} \end{bmatrix}.$$

The rate matrix $\mathbf{Z}$ can then be expressed as

$$\mathbf{Z} = \text{diag}(\mathbf{r}) \otimes \mathbf{Q} + \mathbf{G} \otimes \mathbf{I}_c, \quad (2.15)$$

where $\mathbf{I}_c$ is the $c \times c$ identity matrix [23]. Likelihood calculation under this model is therefore achieved similarly to the standard model, using a rate matrix of size $g \cdot c$. The complexity of the algorithm becomes $O(mc^3g^3 + mnc^2g^2)$.

As an example, consider the basic covarion model of Tuffley and Steel [52]. This model uses only two different rates: “on” ($r_1 = 1$) and “off” ($r_2 = 0$). The switching between rates is controlled by the rate matrix

$$\mathbf{G} = \begin{bmatrix} -s_1 & s_1 \\ s_2 & -s_2 \end{bmatrix}.$$

To apply the covarion approach with the F81 model we plug in the rate matrix $\mathbf{Q}$ from equation (2.3) to give the rate matrix for the compound process of

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Q} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} + \begin{bmatrix} -s_1\mathbf{I}_c & s_1\mathbf{I}_c \\ s_2\mathbf{I}_c & -s_2\mathbf{I}_c \end{bmatrix} = \begin{bmatrix} * & \pi_C & \pi_G & \pi_T & s_1 & 0 & 0 & 0 \\ \pi_A & * & \pi_G & \pi_T & 0 & s_1 & 0 & 0 \\ \pi_A & \pi_C & * & \pi_T & 0 & 0 & s_1 & 0 \\ \pi_A & \pi_C & \pi_G & * & 0 & 0 & 0 & s_1 \\ s_2 & 0 & 0 & 0 & * & 0 & 0 & 0 \\ 0 & s_2 & 0 & 0 & 0 & * & 0 & 0 \\ 0 & 0 & s_2 & 0 & 0 & 0 & * & 0 \\ 0 & 0 & 0 & s_2 & 0 & 0 & 0 & * \end{bmatrix}.$$

The values along the diagonal are chosen so that the row sums are all zero. The state set for this process is $\{(A, \text{on}), (C, \text{on}), (G, \text{on}), (T, \text{on}), (A, \text{off}), (C, \text{off}), (G, \text{off}), (T, \text{off})\}$.

2.4.4 Correlated evolution between sites

Independence between sites is a fundamental assumption of standard Markov models of sequence evolution, expressed in equation (2.9). The sites of a functional molecule, however, do not evolve independently in the real world: biochemical interactions between sites are required for stabilizing the structure, and achieving the function, of biomolecules.

Pollock *et al.* proposed a model for relaxing the independence assumption [35].

Consider the joint evolutionary process of any two sites of a protein. The space state for the joint process is $\mathcal{E} \times \mathcal{E}$. Under the assumption of independent

sites, the rate matrix for the joint process is constructed from that of the single-site process (assume reversibility):

$$\begin{aligned}\overline{\mathbf{Q}}_{xx',yy'} &= \mathbf{Q}_{xy} = \mathbf{S}_{xy}\pi_y, \\ \overline{\mathbf{Q}}_{xx',xy'} &= \mathbf{Q}_{x'y'} = \mathbf{S}_{x'y'}\pi'_{y'}, \\ \overline{\mathbf{Q}}_{xx',yy'} &= 0,\end{aligned}\tag{2.16}$$

for $x \neq y$ and $x' \neq y'$, where $\overline{\mathbf{Q}}_{xx',yy'}$ is the rate of change from x to y at site 1, and from x' to y' at site 2 (in $\mathcal{E} \times \mathcal{E}$), where $\mathbf{Q}_{xy}$ and π_x are the rate matrix and stationary distribution for the single-site process with state space $\mathcal{E}$ and where $\mathbf{S} = \mathbf{\Pi}^{-1}\mathbf{Q}$ is a symmetric matrix. The joint rate matrix $\overline{\mathbf{Q}}$ has dimension c^2 .

Modelling non-independence between the two sites involves departing from equation (2.16). This is naturally achieved by amending stationary frequencies. It is easy to show that the stationary frequency $\overline{\pi}_{xx'}$ of state $(x, x') \in \mathcal{E}$ is equal to the $\pi_x\pi'_{x'}$ product under the independence assumption. Non-independence can be introduced by rewriting the above equation as:

$$\begin{aligned}\overline{\mathbf{Q}}_{xx',yy'} &= \mathbf{S}_{xy}\overline{\pi}_{yx'}, \\ \overline{\mathbf{Q}}_{xx',xy'} &= \mathbf{S}_{x'y'}\overline{\pi}_{xy'}, \\ \overline{\mathbf{Q}}_{xx',yy'} &= 0,\end{aligned}\tag{2.17}$$

where $\overline{\pi}_{xx'}$'s are free parameters (possibly some function of π_x 's). This formalization accounts for the existence of frequent and infrequent combinations of states between the two sites, perhaps distinct from the product of marginal site-specific frequencies. Pollock *et al.* applied this idea in a simplified, two-state model of protein evolution [35], to be applied to a specific site pair of interest. The same idea was used by Tillier and Collins [51] when they introduced a model dedicated to paired sites in ribosomal RNA. From an algorithmic point of view, accounting for co-evolving site pairs corresponds to a squaring of the state space size c .

Other models aim at representing the fact that two sites have correlated evolutionary rates [17, 57]. Such models are extensions of the ASRV model in which the distribution of site-specific evolutionary rates are not independent among sites. More specifically, these two studies propose a model in which neighbouring sites have correlated rates, introducing an autocorrelation parameter. The idea was extended by Goldman and coworkers when they assumed distinct categories of rate matrices among amino acid sites, and correlated probabilities of the various categories between neighbouring sites [36, 50].

2.5 Optimizing parameters

So far we have not considered what is really the most difficult and limiting aspect of likelihood analysis in phylogenetics: parameter optimization. The problem of finding the maximum likelihood phylogeny combines continuous and discrete optimization. The optimization of branch lengths (and sometimes other

parameters) on a fixed tree is a continuous optimization problem, while the problem of finding the maximum likelihood tree is discrete. Both components are difficult computationally, and computational biologists have not got much past simple heuristics in either case. While these heuristics are proving highly effective, faster and more accurate algorithms are still needed.

2.5.1 *Optimizing continuous parameters*

Given a fixed tree, it is a non-trivial problem to determine the branch lengths giving the maximum likelihood score. On a hundred taxa tree, there are 197 branches, so we are faced with optimizing a 197 dimensional, non-linear, generally non-convex, function. Chor *et al.* [10] have shown that the function can become almost arbitrarily complex. There can be infinitely many local (or global) optima, even when there are only four taxa and two states. Rogers and Swofford [38] observe that multiple optima arise only infrequently in practice. This was not confirmed by our own, preliminary investigations, where we found it relatively easy to generate situations with multiple optima, especially when there was a slight violation of the evolutionary model.

Almost all of the widely used phylogeny programs improve branch lengths iteratively and one at a time. The general approach is to

1. Choose initial branch lengths (here represented as a vector **b**).
2. Repeat for each branch k :
 - (a) Find a real number λ_k so that replacing the length b_k of branch k with $b_k + \lambda_k$ gives the largest likelihood.
 - (b) Replace b_k with $b_k + \lambda_k$ and update the partial likelihood computations (see, for example, the updating algorithm of [1]).
3. If λ_k was small for all branches then return the current branch lengths, otherwise go to step 2.

Implementations differ with respect to the one-dimensional optimization technique used to determine λ_k . The technique used most often is Newton's method (also known as the Newton-Raphson method). The intuitive idea behind Newton's method is to use first and second derivatives to approximate the likelihood function (varying along that branch) by a quadratic function. The branch length is adjusted to equal the minimum of this quadratic function, a new quadratic function is fitted, and the procedure repeats until convergence. The search is constrained so as to maintain non-negative branch lengths. PUZZLE, PAUP*, and PHYLML use Brent's method for one-dimensional optimization [4], thereby avoiding the need for partial derivatives. This method is similar to Newton's method, but is more robust. PHYLIP uses a numerical approximation to Newton's method.

Two software packages, NHML and PAUP*, differ from the standard approach and implement a multi-dimensional search, so that more than one branch length is changed at a time. A (fiddly) modification of the pruning algorithm of Section 2.3 can be used to compute the gradient vector and Hessian matrix for a particular set of branch lengths in $O(mnc^3)$ and $O(m^2nc^3)$

time respectively. Hence, *multi-dimensional* Newton Rhapson and quasi-Newton methods can be implemented fairly efficiently (see [25] for an excellent survey of multi-dimensional optimization methods). A combination of full dimensional and single branch optimization is also possible. One complication is the constraint that branch lengths be non-negative. NHML handles this by defaulting to the *simplex method* (see [25]) when one branch length becomes zero.

Surprisingly, there appears to be no published experimental comparison between single branch and multi-dimensional optimization techniques for likelihood. Our preliminary simulation results indicate that the more sophisticated algorithms will occasionally find better optima, but the increased overhead makes the simple, one branch at a time, approach preferable for extensive tree searches.

2.5.2 Searching for the optimal tree

By far the most widely used method for finding maximum likelihood trees is local search. Using one of several possible methods, we construct an initial tree. We then search through the set of all minor modifications of that tree (see Swofford *et al.* [49] for a survey of these modifications). If we find a modified tree with an improved likelihood, we switch to that tree. The process then continues, each time looking for improved modifications, finally stopping when we reach a local optimum. In practice, users will typically constrain some groups of species, searching only through the smaller set of trees for which these groups are monophyletic (i.e. trees containing these groups as clusters).

There are five standard methods for obtaining an initial tree. Refer to Swofford *et al.* [49] for further details.

- **Randomly generated tree** Used to check for multiple local optima.
- **Distance based tree** Compute a distance matrix for the taxa and apply a distance based method such as Neighbor Joining [39] or BioNJ [24].
- **Sequential insertion** Randomly order the taxa. Construct a tree from the first three taxa. Thereafter, insert the taxa one at a time. At each insertion, place the taxon so that the likelihood is maximized. Some implementations perform local searches after each insertion. One advantage of random sequential insertion is that multiple starting trees can be obtained by varying the insertion order.
- **Star decomposition** Start with a tree with all of the taxa and no internal edges. At each step, choose a pair of nodes to combine, continuing until the tree is fully resolved.
- **Approximate likelihood** Perform a tree search using a criterion that is computationally less expensive than likelihood but chooses similar trees.

A typical maximum likelihood search will involve multiple runs of the starting tree and local search combination. As in all optimization problems there is a risk of getting stuck in a local optimum. To avoid this, it is sometimes desirable to occasionally, and randomly, move to trees with lower likelihood scores. This idea has been formalized in search strategies based on simulated annealing [40], as well

as approaches using genetic algorithms [3, 31]. Vinh and von Haeseler [54] have shown recently that deleting and re-inserting taxa can also help avoid getting trapped in local optima. When multiple searches are run in parallel, information can be communicated between the different searches in order to more rapidly locate areas of tree space with higher likelihoods [30].

2.5.3 *Alternative search strategies*

There has been only a small number of likelihood search methods proposed that differ significantly from the local search framework described above. NJML [34] combines a distance-based method (Neighbor Joining) with maximum likelihood. A partially resolved tree (i.e. a tree with some high degree nodes) is obtained by taking the consensus of a number of NJ bootstrap trees. The method then searches for the tree with maximum likelihood among all trees that contain all of the groups in this partially resolved tree, PhyML [28] gains considerable efficiency by not optimizing all branch lengths for every tree examined. Instead, the algorithm combines moves that improve branch lengths and moves that improve the tree. The advantage of this approach is a considerable gain in speed, as well as the potential to avoid being trapped in some local optima.

A quite different strategy is proposed by Friedman *et al.* [20]. They treat a phylogenetic tree as a graph with vertices and edges. One can estimate the expected mutations between any pair of vertices, then rearrange the tree by removing and adding edges between different pairs of vertices. While the approach has not yet gained widespread acceptance, it represents a completely new way to look at likelihood optimization on trees.

The optimization algorithms implemented in the most widely used phylogenetics packages are summarized in Table 2.1.

2.6 Consistency of the likelihood approach

In this section, we focus on the theoretical underpinnings of the likelihood approach. First we consider the question of consistency: if we have sufficiently long sequences, and the sequence evolution model is correct, will we recover the true tree? As we mentioned above, this does not hold for maximum parsimony. It turns out that maximum likelihood is consistent in most cases. As we shall see, to establish consistency we need to verify an identifiability condition, which ensures that we can distinguish two models from infinite length sequences. We also discuss the robustness of the likelihood approach in coping with model mis-specifications.

2.6.1 *Statistical consistency*

Recall that χ_i represents the character corresponding to the i th site observed in the m sequences and assume that the n sites are independent. The vector of parameters θ includes the tree topology, branch lengths and the parameters of the Markov evolution process. The maximum likelihood estimator $\hat{\theta}_n$ maximizes the

TABLE 2.1. Likelihood algorithms implemented in different software packages. The asterisk indicates that the package implements an algorithm, even if it is not the default algorithm used (as is the case, for example, in PAUP*)

Data	Nucleotides			Proteins and nucleotides					
	PAUP* [48]	fastDNAm1 [32]	NHML [22]	PHYLIP [16]	MOLPHY [1]	PAML [58]	Tree-Puzzle [47]	PHYML [28]	IQPNNI [54]
<i>Approach to branch length optimization</i>									
Single branch per iteration	*	*		*	*	*	*	(a)	*
Multiple branches per iteration									
Newton's method			*						
BFGS (see [25])	*								
Brent's multi-dimension algorithm	*								
Simplex method	*								
<i>Algorithm for one-dimension optimization</i>									
Newton's method (or approximation)	*	*		(b)	*	*			
Brent's one-dimension algorithm	*						*	*	
Subdivision algorithm		*							
<i>Algorithm for the initial tree</i>									
Distance method	*							*	
Random tree	*								
Sequential insertion	*	*	*	*		*			
Star decomposition	*	*		*	*				
Approximate likelihood	*			*	*		*		*
<i>Hill climbing</i>	*	*		*				*	*

Data: which kind of sequence data is analysed. Approach to branch length optimization: whether branches are optimized individually or all at once, and which method is used. Algorithm for one-dimension optimization: which algorithm is used for optimizing a single branch or, in the case of multidimensional optimization, which line search algorithm is used (see [25] for more on line search methods). Algorithm for initial tree: the method used to select the tree (or initial tree when searching). Hill climbing: implements local search that uses the likelihood optimization criterion.

Notes: (a) PHYML combines branch optimization with tree optimization. (b) PHYLIP uses a numerical approximation for first and second derivatives in Newton's method.

likelihood $L(\theta) = \prod_{i=1}^n Pr(\chi_i | \theta)$, or equivalently the normalized log-likelihood

$$l_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log Pr(\chi_i | \theta).$$

If the estimator $\hat{\theta}_n$ is used to estimate the true parameter θ_0 , then it is certainly desirable that the sequence $\hat{\theta}_n$ converges in probability to θ_0 as n tends to ∞ . If this is true, we say that $\hat{\theta}_n$ is *statistically consistent* for estimating θ_0 .

Clearly, the “asymptotic value” of $\hat{\theta}_n$ depends on the asymptotic behaviour of the random functions l_n . There typically exists a deterministic function $l(\theta)$ such that, by the law of large numbers,

$$l_n(\theta) \xrightarrow{P} l(\theta), \quad \text{for every } \theta.$$

What is expected is that the maximizer $\hat{\theta}_n$ of l_n converges to a unique point θ_0 which, moreover, is the maximum of the function l . This requires two conditions:

(1) *Model identifiability*

A model is said to be *identifiable* if the probability of the observations is not the same under two different values of the parameter:

$$l(\theta) \neq l(\theta_0) \quad \text{for } \theta \neq \theta_0.$$

Identifiability is a natural and a necessary condition: If the parameter is not identifiable then consistent estimators cannot exist.

(2) *Convergence of the likelihood function*

Consistency requires an appropriate form of the functional convergence of l_n to l to ensure the convergence of the maximum of l_n to a maximum of l . There are several situations under which this always holds. The “classical” approach of Wald relies on a continuity argument and a suitable compactification of the parameter set [53]. In the phylogenetic context, Wald’s conditions can be adapted for binary trees [7, 37]. In particular, the continuity of the likelihood reconstruction, with respect to the topology parameter, relies on an argument of Buneman [6].

In a variety of situations of parametric statistical inference, identifiability is trivially fulfilled or it implies restrictive but natural conditions on the parameter space. For most models in the phylogenetic setting, identifiability considerations are the principal difficulty in establishing the consistency of maximum likelihood. As long as the model is identifiable, maximum likelihood estimators are typically consistent.

Note, however, that consistency guarantees identification of the correct parameter values (e.g. the tree topology) with infinite length sequences. In real data situations, the sequence length is finite and no method can be sure to recover the correct parameter values.

2.6.2 Identifiability of the phylogenetic models

In earlier sections, we assumed that there was the same evolutionary model for each branch of the tree. We generalize this here by assigning a different rate matrix $\mathbf{Q}^{(b)}$ to each branch b . The evolutionary *scenario* then comprises the tree topology and the Markov transition matrices across the branches,

$$\mathbf{P}^{(b)}(t) = \exp(\mathbf{Q}^{(b)}t^{(b)}),$$

where $t^{(b)}$ is the length of branch b . Let π^v be the marginal distribution of a site at node v , $\pi^v(x) = \mathbb{P}[\hat{\chi}_i(v) = x]$.

Identifiability requires that two different scenarios (differing in topology or transition matrices or both) cannot induce the same joint distribution of the sites with infinite sequences; if two scenarios were indistinguishable from infinite sequences, there will be no hope that they could be distinguished from observed finite sequences and that maximum likelihood could consistently recover the correct scenario. Here we review what is and what is not known about the identifiability of Markov evolutionary scenarios.

Identical evolution of the sites. Suppose that each site evolves according to the same Markov process, that is, the characters χ_i are independent and identically distributed. Conditions under which identifiability of the full scenario (topology and transition matrices) holds were first established formally by Chang [7].

Identifiability of the topology

Assumption (H): There is a node v with $\pi^v(x) > 0$ for all x , and $\det(\mathbf{P}(b)) \notin \{-1, 0, 1\}$ for all branches b .

Under assumption (H), the topology is identifiable from the joint distribution of the pairs of sites. Assumption (H) is a mild condition which ensures that transition matrices are invertible and not equal to permutation matrices. It enables us to construct an additive tree distance from the character distribution. The so-called *LogDet transform* is a good distance candidate and the tree can be recovered using distance-based methods like those reviewed in Chapter 1, this volume. Identifiability just of the tree was proved by Chang and Hartigan [9] and Steel *et al.* [45] and is more thoroughly discussed in Semple and Steel [42].

Identifiability of the transition matrices Chang showed that we cannot just consider pairwise comparisons of sequences to reconstruct the transition matrices, and that the distribution of triples of sites is required to ensure the identifiability of the full scenario. More precisely, under assumption (H), if moreover the underlying evolutionary tree is binary and the transition matrices belong to a class of matrices that is *reconstructible from rows*, then all of the transition matrices are uniquely determined by the distribution of the triples of sites. Chang's additional condition is somewhat technical: a class of matrices is reconstructible from rows if no two matrices in the class differ only by a permutation of rows. An example of such a class is that in which the diagonal element is always the largest element in a row.

The situation is greatly simplified under the assumptions that the evolution process is stationary and reversible with equilibrium measure π . In this restricted class of Markov models, the distribution of the pairs of sites is enough to determine the full scenario:

Under assumption (H), if the rate matrix is identical on all branches $\mathbf{Q}^{(b)} \equiv \mathbf{Q}$, if it is reversible and the node distribution is the stationary distribution $\pi^v \equiv \pi$, then the (unrooted) topology and the transition matrix is identifiable from the pairwise comparisons of sequences.

In summary, the parameters are identifiable (and hence ML is consistent) not only for the basic models described above, but for far more general scenarios of sequence evolution.

Sites evolving according to different processes. Models that allow different sites to evolve at different rates can be seen as mixtures of Markov models (see Chapter 5, this volume). The difficulty with such heterogeneous models is that a mixture of Markov models is generally not a Markov model and the existence of an additive distance measure to reconstruct a topology, heavily relies upon the Markov property. Baake [2] established that if a rate factor varies from site to site, different topologies may produce identical pairwise distributions. Consequently, identifiability of the topology is lost on the basis of pairwise distributions, even if the distribution of rate factors is known. However, the maximum likelihood method makes use of the full joint distribution of the sites; it can still be expected that conditions of identifiability may be recovered from the complete information of infinite sequences in general heterogeneous models. Nothing has been proved in the general context yet.

Identifiability issues have been discussed under the stationary and reversible assumption. Results have been established by Chang [8], Steel *et al.* [46] and are summarized in Semple and Steel [42].

Suppose that the Markov process is stationary and time reversible, and that on every branch b , all sites evolve according to the same rate matrix $\mathbf{Q}$ multiplied by a rate factor r selected according to a probability distribution $f(r)$. The transition matrix for the sites evolving at rate factor r is

$$\mathbf{P}(b) = \exp\left(r\mathbf{Q}t^{(b)}\right), \text{ } r \text{ drawn with distribution } f.$$

Under assumption (H), the topology and the full scenario are identifiable if

- f is completely specified up to one or several free parameters, or
- f is unknown but a molecular clock applies, that is, all of the leaves of the tree are equidistant from the root.

The case with f completely specified is formally identical to the situation with constant rates, if the LogDet transform is replaced by an appropriate tree distance based on the moment-generating function of f . One tractable case is where f is a Gamma distribution and its density function is governed by one

parameter estimated from the data (see Section 2.4.2). Without a parameterized form of the distribution f or without strong assumptions such as a molecular clock, different choices of f and the transition matrices may be combined with any tree to produce the same joint distribution of the sites.

Tuffley and Steel [52] analysed a simple covarion model and compared it with the rates-across-sites model of Yang [55]. They showed that the two models cannot be distinguished from the pairwise distribution of the sites but argued that the two models could indeed be identified from the full joint distribution, provided the number of leaves is at least four. A proof of the identifiability of Site-specific Rate Variation models (see Section 2.4.3) remains to be done. However, these models are already implemented [21] and experience indicates that they should be identifiable.

2.6.3 *Coping with errors in the model*

Current implementations are restricted to stationary and reversible models: homogeneous or ASRV models, including mixed invariable sites and gamma-distributed rates. In these cases, the models are identifiable under mild conditions, and maximum likelihood will consistently estimate the tree topology, the branch lengths and the parameters of the Markov evolution process.

Several authors have published examples where maximum likelihood does not recover the true tree [8, 15]. However, none of these constitute a counterexample to the consistency of maximum likelihood methods since, in each case, the basic conditions for consistency are not fulfilled. They either lack identifiability, or the true model is not a member of the class of models considered.

We have stressed several times that the models used in likelihood analysis are simplifications of the actual processes. For this reason, it is essential that we consider the effect of *model misspecification*. Suppose we postulate a model $\{\mathcal{P}_\theta, \theta \in \Theta\}$; however, the model is misspecified in that the true distribution P that generated the data does not belong to the model. For instance, we can perform a maximum likelihood reconstruction with a single stationary Markov model whereas the observations were truly generated by a mixture of Markov models (Chapter 5, this volume). If we use the postulated model anyway, we obtain an estimate $\hat{\theta}_n$ from maximizing the likelihood. What is the asymptotic behaviour of $\hat{\theta}_n$?

Under conditions (1) and (2) (Section 2.6.1), we can prove that $\hat{\theta}_n$ converges to the value θ_0 that maximizes the function $\theta \rightarrow l(\theta)$. The model $\mathcal{P}_{\theta_0}$ can be viewed as the “projection” of the true underlying distribution P on the family $\{\mathcal{P}_\theta\}$ using the so-called *Kullback–Leibler divergence* as a distance measure. If the model $\mathcal{P}_{\theta_0}$ is not too far off from the truth, we can hope that the estimator $\mathcal{P}_{\hat{\theta}}$ is a reasonable approximation for the true model P . At least, this is what happens in standard classical models, which are nicely parametrized by Euclidean parameters [53].

In the phylogenetic setting, things are complicated by the presence of a discrete non-Euclidean tree parameter. The standard theory does not extend in a straightforward manner. It is not surprising that the above-cited

“counterexamples” all display tree topologies where long branches are separated by short branches; these situations typically favour a lack of robustness. To what extent can likelihood reconstructions recover the true topology when the evolution model is misspecified? A better understanding of the uncertainty in tree estimation is an important direction for future work, so that we can quantify the robustness of likelihood methods and improve testing procedures (see Chapter 4, this volume).

2.7 Likelihood ratio tests

Once a model is developed and the likelihood is optimized, that model may be used to carry out many different statistical tests. In traditional hypothesis testing one often chooses a *null hypothesis* H_0 defined as the absence of some effect; this can be viewed as testing whether some parameter values are equal to zero. For example, testing whether the proportion of invariant sites is zero, or whether there is no rate heterogeneity between sites. If the increase in log-likelihood from raising the proportion of invariant sites from its value under H_0 , that is, 0, to its maximum likelihood estimation is “significant” in some sense, then H_0 is rejected at level α (where α is the probability of rejecting H_0 when it is indeed true). Otherwise, we say that the data at hand do not allow us to reject H_0 ; the proportion of invariant sites may indeed be positive, but we cannot detect this.

Suppose that H_0 is derived from a full alternative H_1 by setting certain parameter values to 0. We can then define sets Θ_0 and Θ_1 such that H_0 corresponds to the situation that the true parameter θ is in $\Theta_0 \subseteq \Theta_1$, and H_1 corresponds to the case $\theta \in \Theta_1 - \Theta_0$. A natural testing idea is to compare the values of the log-likelihood computed under H_0 and H_1 , respectively. The corresponding normalized test statistic is called the (log)*likelihood ratio statistic*.

$$\text{LR} = -2 \left[\max_{\theta \in \Theta_0} \log(L(\theta)) - \max_{\theta \in \Theta_1} \log(L(\theta)) \right].$$

The statistic LR is asymptotically chi-squared distributed under the null hypothesis. The decision rule becomes: reject H_0 if the value of the likelihood ratio statistic exceeds the upper α -quantile of the chi-square distribution. Likelihood ratio tests turn out to be the most powerful tests in an asymptotic sense and in special cases. Thus they are widely used as byproducts of maximum likelihood estimation. However, it is important to realize that their validity heavily relies on two main conditions: H_0 is a simpler model *nested* within the full model H_1 and the correct model belongs to the full model H_1 . For example, in testing whether the proportion of invariant sites is zero, the latter condition implies that the estimated topology is correct and the true rate distribution belongs to gamma + invariant distributions.

Several papers have recently documented the incorrect use and interpretation of standard tests in phylogenetics, due to improper specifications of the test hypotheses [26], or to biases in the asymptotic test distributions [33] or to model misspecification [5]. Ewens and Grant [12] present examples where an

inappropriate use of the LR statistic can cause problems. We review here the assumptions that have to be fulfilled to ensure the validity of likelihood ratio tests and we make precise some restrictions on their applicability. In particular, tests comparing tree topologies cannot use directly the asymptotic framework of likelihood ratio testing.

2.7.1 When to use the asymptotic χ^2 distribution

Suppose a sequence of maximum likelihood estimators $\hat{\theta}_n$ is consistent for a parameter θ that ranges over an open subset of R^p . This is typically true under Wald's conditions and identifiability (see Section 2.6). The next question of interest concerns the order at which the discrepancy $\hat{\theta}_n - \theta$ converges to zero. A standard result says that the sampling distribution of the maximum likelihood estimator has a limiting normal distribution

$$\sqrt{n}(\hat{\theta}_n - \theta) \rightarrow \mathcal{N}(0, \mathbf{i}^{-1}(\theta)), \quad \text{as } n \rightarrow \infty,$$

where $\mathbf{i}(\theta)$ is the Fisher *information matrix*, that is, the $p \times p$ matrix whose elements are the negative of the expectation of all second partial derivatives of $\log L(\theta)$. The convergence in distribution means roughly that $(\hat{\theta}_n - \theta)$ is $\mathcal{N}(0, (n\mathbf{i}(\theta))^{-1})$ -distributed for large n . It implies that the maximum likelihood estimator is asymptotically of minimum variance and unbiased, and in this sense optimal [53].

Suppose we wish to test the null hypothesis H_0 that is nested in the full parameter set of the model of the analysis, say H_1 . Write $\hat{\theta}_{n,0}$ and $\hat{\theta}_n$ for the maximum likelihood estimators of θ under H_0 and H_1 , respectively. The likelihood ratio test statistic is

$$\text{LR} = -2 \left[\log L(\hat{\theta}_{n,0}) - \log L(\hat{\theta}_n) \right].$$

If both H_0 and H_1 are “regular” parametric models that contains θ as an inner point, then, both $\hat{\theta}_{n,0}$ and $\hat{\theta}_n$ can be expected to be asymptotically normal with mean θ and we obtain the approximation under H_0

$$\text{LR} \sim \sqrt{n}(\hat{\theta}_n - \hat{\theta}_{n,0})^t \mathbf{i}(\theta) \sqrt{n}(\hat{\theta}_n - \hat{\theta}_{n,0}).$$

Then the likelihood ratio statistic can be shown to be asymptotically distributed as a quadratic form in normal variables. The law of this quadratic form is a chi-square distribution with $p - q$ degrees of freedom, where p and q are the dimensions of the full and null hypotheses.

The main conditions for this theory to apply are that the null and full hypothesis H_0 and H_1 are equal to R^q and R^p (or are locally identical to those linear spaces), and that the maximum likelihood estimator finds a non-boundary point where the likelihood function is differentiable.

2.7.2 Testing a subset of real parameters

The requirement that the parameters of interest be real numbers is not met if the tree topology is estimated as part of the maximizing procedure. Thus for the moment we assume that the tree topology is given. θ represents here the

scalar parameters, that is, the branch lengths and/or parameters of the evolution process.

Suppose that we wish to test a general linear hypothesis $H_0: A\theta = 0$, where A is a contrast matrix of rank k (i.e. there are $p - k$ free parameters to estimate under H_0). For example, $A\theta = 0$ could correspond to the situation where a particular parameter is zero, in which case $k = 1$. For large n , it can be assumed in this case that LR has a chi-square distribution with k degrees of freedom under H_0 . LR is typically computed by examining successively more complex models, for example, to test whether increasing the number of parameters of the rate matrix $\mathbf{Q}$ yields a significant improvement in model fitting, with respect to the chosen topology.

The LR test is based on the assumption that the tree topology and the evolutionary model are correct. If it is not the case, the induced model bias can make tests reject H_0 too often, or too rarely [5]. In practice, phylogenetic models are always misspecified to a degree. This means that one has to be cautious in interpreting test results for any real data, even if the test is well-founded with respect to theory.

2.7.3 Testing parameters with boundary conditions

We have assumed that the topology is given; even under this restriction, the chi-square approximation fails in a number of simple examples. The “local linearity” of the hypotheses H_0 and H_1 mentioned above is essential for the chi-square approximation. If H_0 defines a region in the parameter space where some parameters are not specified, there is no guarantee in general that the distribution of the test statistic is the same for all points in this region. In tests of one-sided hypotheses, the null hypothesis is no longer locally linear at its boundary points. In this case, however, the testing procedure can be adapted: the asymptotic null distribution of the LR statistic is not chi-squared, but the distribution of a certain functional of a Gaussian vector [41].

A related example arises when some parameters of interest lie on the boundary of the parameter space Θ_1 . Usual boundary conditions are that the branch lengths, the proportion of invariant sites or the shape of a gamma distribution of site substitution rates have non-negative values and difficulties occur when testing whether those parameters are zero. Boundary related problems can also affect tests of the molecular clock. Ota *et al.* [33] derived the appropriate corrections to the asymptotic distributions of the likelihood ratio test statistics, which turn out to be a mixed combination of chi-square distributions and the Dirac function at 0.

2.7.4 Testing trees

When the tree topology is estimated as part of the testing procedure, the conditions derived at the end of Section 2.7.1 are not fulfilled. This is essentially because the tree topology is not a real parameter. Moreover, phylogenetic models displaying different tree topologies are in general not nested. For all these reasons,

tests involving estimated topologies are simply outside the scope of the likelihood ratio tests theory.

Tests involving topologies are thoroughly discussed in Chapter 4, this volume, and alternatives to the classical LR testing procedure are proposed. Another promising testing framework is provided by the likelihood-based tests of multiple tree selection developed in the papers by Shimodaira *et al.* [43, 44]. The model selection approach aims at testing which model is better than the other, while the object of the likelihood ratio test is to find out the correct model. This offers a more flexible approach to model testing, where different topologies combined with different evolution processes can be compared.

2.8 Concluding remarks

Molecular phylogeny is a stimulating topic that lies at the boundary of biology, algorithmics, and statistics, as illustrated in this chapter. The three domains have progressed considerably during the last twenty years: data sets are much bigger, models much better, and programs much faster. Some problems, however, still have to be solved. Not every model that we would want to use permits feasible likelihood calculation. Models for partially relaxing the molecular clock, for example, are highly desirable but currently not tractable in the ML framework. As far as algorithmics is concerned, we have already stressed the probable non-optimality of the optimization algorithms used in the field, a problem worsened by the fact that not all algorithms are published. The statistics of phylogeny also require some clarification, as illustrated in Sections 2.6 and 2.7. The problem of model choice, for example (which model to choose for a given data set), is not really addressed in a satisfactory way in current literature.

An important issue, finally, is the problem of combining data from different genes (the supertree problem). Most approaches to this question have come from combinatorics, while a statistical point of view should be the appropriate one. This would require research into the parametrization of the multi-gene model, and the ability of ML methods to cope with missing data. Recent progress in this area is surveyed in Chapter 5, this volume.

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BAYESIAN INFERENCE IN MOLECULAR PHYLOGENETICS

Ziheng Yang

The Bayesian method of statistical inference combines the prior for parameters with the data to generate the posterior distribution of parameters, upon which all inferences about the parameters are based. The method has become very popular due to recent advances in computational algorithms. In molecular evolution and phylogenetics, Bayesian inference has been applied to address fundamental biological problems under sophisticated models of sequence evolution. This chapter introduces Bayesian statistics through comparison with the likelihood method. I will discuss Markov chain Monte Carlo algorithms, the major modern computational methods for Bayesian inference, as well as two applications of Bayesian inference in molecular phylogenetics: estimation of species phylogenies and estimation of species divergence times.

3.1 The likelihood function and maximum likelihood estimates

The probability of observing the data D , when viewed as a function of the unknown parameters θ with the data given, is called the likelihood function: $L(\theta; D) = f(D | \theta)$. According to the likelihood principle, the likelihood function contains all information in the data about the parameters. The best point estimate of θ is given by the θ that maximizes the likelihood L or the log likelihood $\ell(\theta; D) = \log\{L(\theta; D)\}$. Furthermore, the likelihood curve provides information about the uncertainty in the point estimate. In this chapter, I use estimation of the distance between two sequences under the Jukes and Cantor model [23] as an example to contrast the likelihood and Bayesian methodologies (see Chapter 2, this volume for more about likelihood methods in phylogenetics).

Suppose x of the n sites are different between the two sequences, with the proportion of different sites to be x/n . The distance is the expected number of nucleotide substitutions per site, $\theta = \lambda t$, where λ is the substitution rate and t is the time that separates the two sequences—since rate and time are confounded, we estimate one single parameter θ using the data x . The probability that a site is different between two sequences separated by distance θ is

$$p = \frac{3}{4}(1 - e^{(-4/3)\theta}). \quad (3.1)$$

Thus the likelihood, or the probability of observing x differences out of n sites, is given by the binomial probability

$$L(\theta; x) = f(x | \theta) = Cp^x(1 - p)^{n-x}, \quad (3.2)$$

where $C = n!/[x!(n-x)!]$ is constant (independent of parameter θ) and can be ignored. By setting $dL/d\theta = 0$ or $d\ell/d\theta = 0$, one can determine that the likelihood is maximized at

$$\hat{\theta} = -\frac{3}{4} \log \left(1 - \frac{4}{3} \times \frac{x}{n} \right). \quad (3.3)$$

Thus $\hat{\theta}$ is the maximum likelihood estimate (MLE) of θ . This is the familiar Jukes–Cantor distance formula [23]. In most problems in molecular phylogenetics to which maximum likelihood is applied, the solution is not analytical and numerical algorithms are needed to find the MLEs.

The MLEs are invariant to transformations or re-parametrizations. The MLE of a function of parameters is the same function of the MLEs of the parameters: $\hat{h}(\theta) = h(\hat{\theta})$. For example, we can use the expected proportion of different sites p as the parameter; this is still a measure of distance although it is non-linear with time. Its MLE is $\hat{p} = x/n$ from the binomial likelihood (equation (3.2)). We can then view θ as a function of p through equation (3.1), and obtain its MLE $\hat{\theta}$, as given in equation (3.3). Whether we use p or θ as the parameter, the same inference is made, and the same log likelihood is achieved: $\ell(\hat{p}) = \ell(\hat{\theta}) = x \log(x/n) + (n-x) \log((n-x)/n)$.

As an example, suppose $x = 10$ differences are observed out of $n = 100$ sites. The log-likelihood curves are shown in Fig. 3.1(a) and (b) for parameters θ and p , respectively. The log likelihood is maximized at $\hat{\theta} = 0.107326$ and $\hat{p} = x/n = 0.1$, with $\ell(\hat{\theta}) = \ell(\hat{p}) = -32.508$.

Two approaches can be used to calculate a confidence interval for the MLE. The first relies on the theory that $\hat{\theta}$ is asymptotically normally distributed around the true θ when the sample size $n \rightarrow \infty$. This is equivalent to using a quadratic function to approximate the log likelihood around the MLE. The variance of the asymptotic normal distribution can be calculated using the curvature of the log-likelihood surface around the MLE:

$$\text{var}(\hat{\theta}) = - \left[\frac{d^2 \ell}{d\theta^2} \right]^{-1} = \frac{9\hat{p}(1-\hat{p})}{(3-4\hat{p})^2 n}. \quad (3.4)$$

Thus an approximate 95% confidence interval for θ can be constructed as $\hat{\theta} \pm 1.96 \sqrt{\text{var}(\hat{\theta})}$. For our example of $x = 10$ differences in $n = 100$ sites, we have $\text{var}(\hat{\theta}) = 0.001198$, and the 95% confidence interval is $0.10733 \pm 1.96 \times 0.06784$ or $(0.03948, 0.17517)$. Similarly, $\text{var}(\hat{p}) = \hat{p}(1-\hat{p})/n = 0.0009$, so that the 95% confidence interval for p is $(0.04120, 0.15880)$. Note that those two intervals do not match each other.

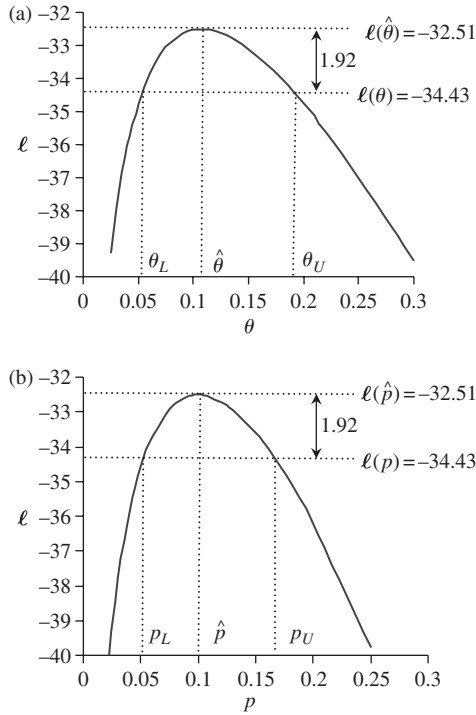


FIG. 3.1. Log-likelihood curves for estimation of sequence distance θ or p under the JC69 model [23]. Log-likelihood curves as a function of the sequence distance θ (a) or p (b). The data are two sequences, each of length $n = 100$ with $x = 10$ different sites. The likelihood interval is constructed by lowering the log likelihood ℓ from the optimum value by 1.92.

A second approach is based on the result that the likelihood ratio test statistic, $2[\ell(\hat{\theta}) - \ell(\theta)]$, where θ is the true parameter and $\hat{\theta}$ is the MLE, has a χ^2_1 distribution in large samples. Thus, we can lower the log likelihood by, say, $\frac{1}{2}\chi^2_{1,5\%} = 3.84/2 = 1.92$ from $\ell(\hat{\theta})$, to construct a 95% likelihood interval (θ_L, θ_U) (Fig. 3.1(a)). Thus at $\ell = \ell(\hat{\theta}) - 1.92 = -34.43$, the likelihood interval is found to be $(0.05327, 0.19119)$ for θ . Note that this interval is asymmetrical and is shifted to the right compared with the interval based on the normal approximation, due to the steeper drop of log likelihood and thus more information on the left side of $\hat{\theta}$ than on the right side. The corresponding likelihood interval for p is $(0.05142, 0.16876)$. This approach in general gives more reliable intervals than the normal approximation to MLEs. The normal approximation works well for some parameterizations but not for others; the use of the likelihood interval is equivalent to using the best parametrization.

The likelihood method may run into problems when the model involves too many parameters. If the number of parameters increases without bound with

the increase of the sample size, the MLEs may not even be consistent. Dealing with the so-called nuisance parameters is also a difficult area for likelihood. For example, if we are interested in the sequence distance under the substitution model of Kimura [24], we might consider distance θ as the parameter of interest, while the transition/transversion rate ratio κ is a nuisance parameter. Similarly, if our interest is in the phylogeny for a group of species, branch lengths as well as all parameters in the substitution model are nuisance parameters. Perhaps the biggest problem for the application of likelihood to molecular phylogeny reconstruction is the unconventional nature of the tree topology parameter, and the resulting difficulties in attaching a confidence interval for the maximum likelihood tree [51] (see Chapter 4, this volume).

3.2 The Bayesian paradigm

The central idea of Bayesian inference is that parameters θ have distributions. Before the data are observed, θ have a prior distribution $f(\theta)$. This is combined with the likelihood or the probability of the data given the parameters, $f(D | \theta)$, to give the posterior distribution, $f(\theta | D)$, through the Bayes theorem

$$f(\theta | D) = \frac{f(\theta)f(D | \theta)}{f(D)} = \frac{f(\theta)f(D | \theta)}{\int f(\theta)f(D | \theta) d\theta}. \quad (3.5)$$

The marginal probability of the data, $f(D)$, is a normalizing constant, to make $f(\theta | D)$ integrate to one. Equation (3.5) thus says that the posterior $f(\theta | D)$ is proportional to the prior $f(\theta)$ times the likelihood $f(D | \theta)$. Or equivalently, the posterior information is the sum of the prior information and the sample information.

The posterior distribution is the basis for all Bayesian inference concerning θ . For example, the mean, median, or mode of the distribution can be used as the point estimate. For interval estimation, one can use the interval encompassing the highest 95% of the density mass as the 95% highest posterior density (HPD) interval. This works even if there are multiple peaks in the distribution; the interval may include disconnected regions. For a single-moded posterior density, the 2.5% and 97.5% quantiles can be used to construct the 95% equal-tail credibility interval (CI). In general, the posterior expectation of any function of the parameters, $h(\theta)$, is constructed as $E[h(\theta) | D] = \int h(\theta)f(\theta | D) d\theta$.

Consider estimation of sequence distance θ under the JC69 model [23] using the data of $x = 10$ differences out of $n = 100$ sites. Suppose we use an exponential prior $f(\theta) = \mu^{-1}e^{(-\theta/\mu)}$, with mean $\mu = 0.1$. The posterior distribution of θ is

$$f(\theta | x) = \frac{f(\theta)f(x | \theta)}{f(x)} = \frac{f(\theta)f(x | \theta)}{\int f(\theta)f(x | \theta) d\theta}, \quad (3.6)$$

where the likelihood $f(x | \theta)$ is given in equation (3.2). It seems awkward, although possible, to calculate the integral for $f(x)$ in equation (3.6) analytically. Instead I use Mathematica to evaluate it numerically. Figure 3.2 shows the resulting posterior density, plotted together with the prior and scaled likelihood. In this

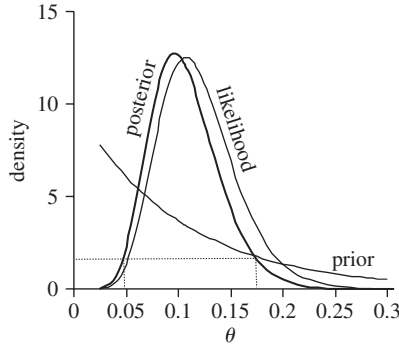


FIG. 3.2. Prior and posterior densities for sequence distance θ under the JC69 model. The likelihood is also shown, rescaled to match up with the posterior density. The data are two sequences, each of length $n = 100$ with $x = 10$ different sites. The 95% highest posterior density interval is (0.04758, 0.17260), indicated on the graph.

case the posterior is dominated by the likelihood. The posterior mean is found to be 0.10697, with standard deviation 0.03290. The 95% equal-tail credibility interval is (0.05284, 0.18077), while the 95% HPD interval is (0.04758, 0.17260).

The Bayesian paradigm also provides a natural way of dealing with nuisance parameters. Let $\theta = \{\lambda, \eta\}$, with λ to be the parameters of interest and η the nuisance parameters. The joint conditional distribution of λ and η given the data is

$$f(\lambda, \eta | D) = \frac{f(\lambda, \eta)f(D | \lambda, \eta)}{f(D)} = \frac{f(\lambda, \eta)f(D | \lambda, \eta)}{\iint f(\lambda, \eta)f(D | \lambda, \eta) d\lambda d\eta} \quad (3.7)$$

from which the (marginal) posterior density of λ can be obtained as

$$f(\lambda | D) = \int f(\lambda, \eta | D) d\eta. \quad (3.8)$$

3.3 Prior

Specification of the prior distribution for parameters, and indeed the need for such specification is where all controversies surrounding Bayesian inference lies. If the physical process can be used to model uncertainties in the quantities of interest, it is standard in the likelihood framework to treat such quantities as random variables, and derive their conditional probability distribution given the data. An example relevant to this chapter is the use of the Yule branching process [5] and the birth–death process [34] to specify the probability distributions of phylogenies. The parameters in the models are the birth and death rates, estimated from the marginal likelihood, which averages over the tree topologies and branch lengths, while the phylogeny is estimated from the conditional

probability distribution of phylogenies given the data. The controversy arises when no physical model is available to specify the distribution of parameters, and when subjective beliefs or diffuse distributions are used as “vague” priors. Modern terminology does not distinguish whether or not the prior is based on a model of the physical process; in either case the quantities of interest are considered parameters, the approach considered Bayesian, and the conditional probability is known as the posterior probability.

Approaches for specifying the prior include (1) use of a physical model, as mentioned above, (2) use of past observations of the parameters in similar situations, and (3) subjective beliefs of the researcher. To avoid undue influence of the prior on the posterior, uniform distributions are often used as vague priors. For a discrete parameter that can take m possible values, this means assigning probability $1/m$ to each element. For a continuous parameter, this means a uniform distribution over the range of the parameters. However, saying that distance θ is equally likely to be any value between 0 and 10 is not the same as saying that nothing is known about θ , so one should not consider any prior as entirely non-informative. Another criticism is that unlike the MLEs, the prior is not invariant to reparametrizations. For example, a uniform prior for parameter p is very different from a uniform prior for θ (see below).

Another class of priors is the conjugate priors. Here the prior and the posterior have the same distributional form, and the role of the data or likelihood is to update the parameters in that distribution. Well-known examples include (1) the binomial (n, p) distribution of data with a beta prior for the probability parameter p ; (2) poisson(λ) distribution of data with a gamma prior for the rate parameter λ ; and (3) normal distribution of data $N(\mu, \sigma^2)$ with a normal prior for the mean μ . In our example of estimating sequence distance under the JC69 model, if we use the probability of different sites p as the distance, we can assign a beta prior $\text{beta}(\alpha, \beta)$. When the data have x differences out of n sites, the posterior distribution of p is $\text{beta}(\alpha + x, \beta + n - x)$. This result also illustrates the information contained in the beta prior: $\text{beta}(\alpha, \beta)$ is equivalent to observing α differences out of $\alpha + \beta$ sites. Conjugate priors are possible only for special combinations of the prior and likelihood. They are theoretically convenient as the integrals are tractable analytically, but they may not be realistic models for the problem at hand. Conjugate priors have not found a use in molecular phylogenetics (except for the trivial one above), as the problem is typically too complex.

When the prior distribution involves unknown parameters, one can assign priors for them, called hyper-priors. Unknown parameters in the hyper-prior can have their own priors. This is known as the hierarchical or full Bayesian approach. Typically one does not go beyond two or three levels, as the effect will become unimportant. For example, the mean μ in the exponential prior in our example of distance calculation under JC69 in equation (3.6) can be assigned a hyper-prior. An alternative is to estimate the hyper-parameters from the marginal likelihood, and use them in posterior probability calculation for parameters of interest. This is known as the empirical Bayesian approach. For example, μ

can be estimated by maximizing $f(x \mid \mu) = \int f(\theta \mid \mu)f(x \mid \theta) d\theta$, and the estimate can be used to calculate $f(\theta \mid x)$ in equation (3.6). Empirical Bayesian approach has been used widely in molecular phylogenetics, for example, to estimate evolutionary rates at sites [55], to reconstruct ancestral DNA or protein sequences on a phylogeny [52], to identify amino acid residues under positive Darwinian selection [31], to infer secondary structure categories of a protein sequence [13], and to construct sequence alignments under models of insertions and deletions [46, 47].

An important question in real data analysis is whether the posterior is sensitive to the prior. It is always prudent to assess the influence of the prior. If the posterior is dominated by the data, the choice of the prior is inconsequential. When this is not the case, the effect of the prior has to be assessed carefully and reported. Due to advances in computational algorithms (see below), the Bayesian methodology is now very powerful and allows the researcher to fit sophisticated parameter-rich models. As a result, the researcher might be tempted to add parameters that are barely identifiable [33], and the posterior may be unduly influenced by some aspects of the prior even without the knowledge of the researcher. In our example of distance estimation under the JC69 model, identifiability problems will arise if we attempt to estimate both the substitution rate λ and time t instead of one parameter θ . It is thus important for the researcher to understand which aspects of the data provide information about the parameters, what parameters are knowable and what are not, to avoid overloading the model with parameters.

3.4 Markov chain Monte Carlo

Until recently, computational difficulties had prevented the use of the Bayesian method as a general inference methodology. For most problems, the prior and the likelihood are easy to calculate, but the marginal probability of the data $f(D)$, that is, the normalizing constant, is hard to calculate. Except for trivial problems such as cases involving conjugate priors, analytical results are unavailable. We have noted above the difficulty of calculating the marginal likelihood $f(D)$ (in equation (3.6)) in our extremely simple problem of distance estimation. More complex Bayesian models can involve hundreds or thousands of parameters and high-dimensional integrals have to be evaluated (see equations (3.7) and (3.8)). For example, to calculate posterior probabilities for phylogenetic trees, one has to evaluate the marginal probability of data $f(D)$, which is a sum over all possible tree topologies and integration over all branch lengths in those trees and over all parameters in the substitution model. The breakthrough is the development of Markov chain Monte Carlo (MCMC) algorithms, which provide a powerful method for achieving Bayesian computation.

3.4.1 Metropolis–Hastings algorithm

Here we describe the algorithm of Metropolis *et al.* [30]. The goal is to generate a Markov chain, whose states are the parameters θ , and whose steady-state

(stationary) distribution is $\pi(\theta) = f(\theta \mid D)$, the posterior distribution of θ . Suppose the current state of the Markov chain is θ . The algorithm proposes a new state θ^* through a proposal density or jumping kernel $q(\theta^* \mid \theta)$, which is symmetrical: $q(\theta^* \mid \theta) = q(\theta \mid \theta^*)$. For example, one can use a uniform distribution around θ , so that $\theta^* = U(\theta - w/2, \theta + w/2)$, with w controlling the size of steps taken. This is a sliding window with window size w . The candidate state θ^* is accepted with probability

$$\alpha = \min \left(1, \frac{\pi(\theta^*)}{\pi(\theta)} \right). \quad (3.9)$$

If the new state θ^* is accepted, the chain moves to θ^* . If it is rejected, the chain stays at the current state θ . Both acceptance and rejection are counted as an iteration, and the procedure is repeated for many iterations. The values of θ over iterations generated this way form a Markov chain, as they satisfy the Markovian property that “given the present, the future is independent of the past.” This Markov chain has $\pi(\theta)$ as the stationary distribution as long as the proposal density $q(\cdot \mid \cdot)$ specifies an irreducible and aperiodic chain. In other words, $q(\cdot \mid \cdot)$ should allow the chain to reach any state from any other state, and that the chain should not have a period.

Intuitively, one may think of the algorithm as describing a wanderer climbing a hill, the height at location θ being the target density $\pi(\theta)$. A random step in a random direction is chosen from the current location. If the step is uphill, that is, if $\pi(\theta^*) > \pi(\theta)$, it is always taken. However, if the step is downhill, it is not rejected straightaway but instead accepted with probability $\pi(\theta^*)/\pi(\theta) < 1$. If the wanderer is allowed to wander around for a very long time, he will explore the hill extensively and spend time in each location θ in proportion to the height of that location $\pi(\theta)$. Thus a sample of his visits can be used to estimate the target distribution $\pi(\theta)$.

Hastings [18] extended the Metropolis algorithm to allow the use of asymmetrical proposal densities, that is, if $q(\theta^* \mid \theta) \neq q(\theta \mid \theta^*)$. This involves a simple correction in calculation of the acceptance probability

$$\alpha = \min \left(1, \frac{\pi(\theta^*)q(\theta \mid \theta^*)}{\pi(\theta)q(\theta^* \mid \theta)} \right). \quad (3.10)$$

We might suppose that the wanderer has a tendency to move north, and takes a northward step three times as likely as a southward step. Then by accepting northward moves only $\frac{1}{3}$ times as often as southward moves, the Markov chain will still recover the correct target distribution $\pi(\theta)$ even if the proposal density is biased. The correction term, $q(\theta \mid \theta^*)/q(\theta^* \mid \theta)$, is called the proposal ratio or the Hastings ratio.

When the MCMC algorithm is used to approximate the posterior distribution of parameters θ , we have $\pi(\theta) = f(\theta \mid D) = f(\theta)f(D \mid \theta)/f(D)$, so that

$$\frac{\pi(\theta^*)}{\pi(\theta)} = \frac{f(\theta^*)f(D \mid \theta^*)}{f(\theta)f(D \mid \theta)}.$$

Importantly note that the normalizing constant $f(D)$ in equation (3.5) cancels. The acceptance probability is thus

$$\begin{aligned}\alpha &= \min \left(1, \frac{f(\theta^*)}{f(\theta)} \times \frac{f(D | \theta^*)}{f(D | \theta)} \times \frac{q(\theta | \theta^*)}{q(\theta^* | \theta)} \right) \\ &= \min(1, \text{prior ratio} \times \text{likelihood ratio} \times \text{proposal ratio}).\end{aligned}\quad (3.11)$$

In typical applications of MCMC algorithms to molecular phylogenetics, the prior ratio $f(\theta^*)/f(\theta)$ is easy to calculate. The likelihood ratio $f(D | \theta^*)/f(D | \theta)$ is often easy to calculate as well even though computationally expensive. The proposal ratio $q(\theta | \theta^*)/q(\theta^* | \theta)$ affects greatly the efficiency of the MCMC algorithm. So much of practical effort is spent on developing good proposal algorithms.

Here we use the example of distance estimation under the JC69 model to explain MCMC algorithms. Those who have not written any Bayesian MCMC program are invited to implement the algorithm below, using any programming language such as C/C++, Java, Basic, or Mathematica. The data are $x = 10$ differences out of $n = 100$ sites. We use an exponential prior

$$f(\theta | \mu) = \frac{1}{\mu} e^{-(1/\mu)\theta}$$

with $\mu = 0.1$. The proposal algorithm uses a sliding window of size w .

1. Initialize: $n = 100$, $x = 10$, $w = 0.01$.
2. Initial state $\theta = 0.5$.
3. Propose a new state as $\theta^* \sim U(\theta - w/2, \theta + w/2)$. That is, generate a $U(0, 1)$ random number r , and set $\theta^* = \theta - w/2 + wr$. If $\theta^* < 0$, set $\theta^* = -\theta^*$.
4. Calculate the acceptance probability, using equations (3.1) and (3.2) to calculate the likelihood $f(x | \theta)$.

$$\alpha = \min \left(1, \frac{f(\theta^* | \mu)}{f(\theta | \mu)} \times \frac{f(x | \theta^*)}{f(x | \theta)} \right).$$

5. Accept or reject the proposal θ^* . Draw $r \sim U(0, 1)$. If $r < \alpha$ set $\theta = \theta^*$. Otherwise set $\theta = \theta$.
6. Go to step 3.

Figures 3.3(a) and (b) show the first 500 iterations of five independent chains, starting from different initial values and using different window sizes. Figure 3.3(c) shows the posterior probability density estimated from a long chain with 10 million iterations. This is indistinguishable from the distribution calculated using numerical integration (Fig. 3.2).

A number of variations to the general Metropolis–Hastings algorithm exist. Below we mention three commonly used ones: the single-component Metropolis–Hastings algorithm, the Gibbs sampler, and Metropolis-coupled MCMC or MC³.

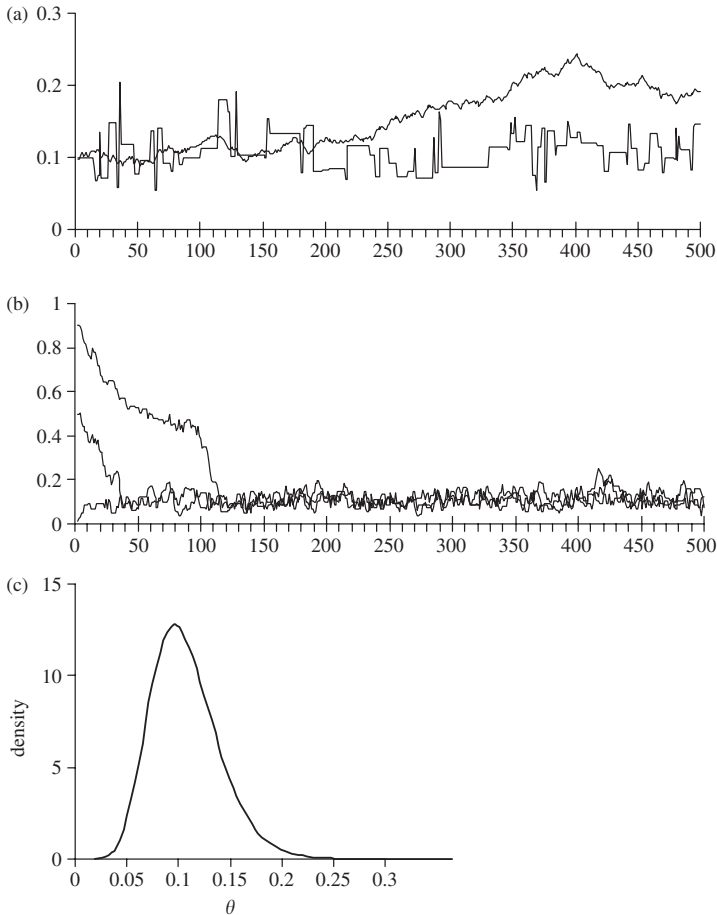


FIG. 3.3. MCMC runs for estimating sequence distance θ under the JC69 substitution model. The data consists of $x = 10$ differences between two sequences of $n = 100$ sites. (a) Two chains with the window size either too small ($w = 0.01$) or too large ($w = 1$). Both chains started at $\theta = 0.1$. The chain with $w = 0.01$ has an acceptance rate of 97%, so that almost every proposal is accepted. However, this chain takes tiny baby steps and mixes very poorly. The other chain, with $w = 1$, has an acceptance rate of 20%, so that 80% of proposals are rejected. The chain often stays at the same state for many iterations without a move. This window size is slightly too large. Further experiment shows that the window size $w = 0.2$ leads to an acceptance rate of 48%, and is near optimum (see text). (b) Three chains started from $\theta = 0.01, 0.5,$ and 1 . The window size is 0.1 , with an acceptance rate of 70%. It appears that after about 120 iterations, the three chains become indistinguishable and have reached stationarity, so that a burn-in of 200 iterations should be sufficient for those chains. (c) Posterior density estimated from a long chain (with 10,000,000 iterations) with window size $w = 0.1$, estimated by kernel density smoothing [40].

3.4.2 *Single-component Metropolis–Hastings algorithm*

Simple single-parameter problems are straightforward to deal with using the likelihood methodology. The advantage of Bayesian inference mostly lies in the ease with which it can deal with sophisticated multi-parameter models. In particular, Bayesian “marginalization” of nuisance parameters (equation (3.8)) provides an attractive way of accommodating variation in the data that we are not really interested in. In MCMC algorithms for such multi-parameter models, it is often unfeasible or computationally too complicated to update all parameters in θ simultaneously. Instead, it is more convenient to divide θ into components or blocks, of possibly different dimensions, and then update those components one by one. Different proposals are often used to update different components. This is known as “blocking.” Many models have a structure of conditional independence, and blocking often leads to computational efficiency.

A variety of strategies are possible concerning the order of updating the components. One can use a fixed order, or a random permutation of the components. There is no need to update every component in every iteration. One can also select components for updating with fixed probabilities. However, the probabilities should be fixed and not dependent on the current state of the Markov chain, as otherwise the stationary distribution may no longer be the target distribution $\pi(\cdot)$. It is advisable to update highly correlated components more frequently. It is also advantageous to group into one block components that are highly correlated in the posterior density, and update them simultaneously using a proposal density that accounts for the correlation (see below).

3.4.3 *Gibbs sampler*

The *Gibbs sampler* [11] is a special case of the single-component Metropolis–Hastings algorithm. The proposal distribution for updating the i th component is the conditional distribution of the i th component given all the other components. This proposal leads to an acceptance probability of 1; that is, all proposals are accepted. The Gibbs sampler has been widely used, especially in linear models involving normal prior and posterior densities. However, it has not been used in molecular phylogenetics as it is in general impossible to obtain the conditional distributions analytically.

3.4.4 *Metropolis-coupled MCMC*

If the target distribution has multiple peaks, separated by low valleys, the Markov chain may have difficulties in moving from one peak to another. As a result, the chain may get stuck on one peak and the resulting samples will not approximate the posterior density correctly. This is a serious practical concern for phylogeny reconstruction, as multiple local peaks are known to exist in the tree space during heuristic tree search under the maximum parsimony (MP), maximum likelihood (ML), and minimum evolution (ME) criteria, and the same can be expected for stochastic tree search using MCMC. Some strategies have been proposed to improve mixing of Markov chains in presence of multiple local

peaks in the posterior density. One such algorithm is the Metropolis-coupled MCMC or MCMCMC (MC³) algorithm suggested by Geyer [12].

In this algorithm, m chains are run in parallel, with different stationary distributions $\pi_j(\cdot)$, $j = 1, 2, \dots, m$, where $\pi_1(\cdot) = \pi(\cdot)$ is the target density, while $\pi_j(\cdot)$, $j = 2, 3, \dots, m$ are chosen to improve mixing. For example, one can use incremental heating of the form

$$\pi_j(\theta) = \pi(\theta)^{1/[1+\lambda(j-1)]}, \quad \lambda > 0, \quad (3.12)$$

so that the first chain is the cold chain with the correct target density, while chains $2, 3, \dots, m$ are heated chains. Note that raising the density $\pi(\cdot)$ to the power $1/T$ with $T > 1$ has the effect of flattening out the distribution, similar to heating a metal. In such a distribution, it is easier to traverse between peaks across the valleys than in the original distribution. After each iteration, a swap of states between two randomly chosen chains is proposed through a Metropolis–Hastings step. Let $\theta^{(j)}$ be the current state in chain j , $j = 1, 2, \dots, m$. A swap between the states of chains i and j is accepted with probability

$$\alpha = \min \left(1, \frac{\pi_i(\theta_j)\pi_j(\theta_i)}{\pi_i(\theta_i)\pi_j(\theta_j)} \right). \quad (3.13)$$

At the end of the run, output from only the cold chain is used, while those from the hot chains are discarded. Heuristically, the hot chains will visit the local peaks rather easily, and swapping states between chains will let the cold chain occasionally jump valleys, leading to better mixing. However, if $\pi_i(\theta)/\pi_j(\theta)$ is very unstable, proposed swaps will seldom be accepted; this is the reason for using several chains which differ only incrementally. An obvious disadvantage of the algorithm is that m chains are run but only one chain is used for inference. MC³ is ideally suited to implementation on parallel machines or network workstations, since each chain will in general require about the same amount of computation per iteration, and interactions between chains are minimal.

3.5 Simple moves and their proposal ratios

The proposal ratio is separate from the likelihood or the prior and is solely dependent on the proposal algorithm. Thus simple proposals can be used in a variety of Bayesian inference problems. As mentioned earlier, the proposal density has only to specify an aperiodic recurrent Markov chain to guarantee convergence of the MCMC algorithm. One can easily construct such chains and it is also typically easy to verify that the proposal density satisfies those conditions. For a discrete parameter that takes a set of values, calculation of the proposal ratio often amounts to counting the number of candidate elements in the source and target states, which is easy. Calculation for continuous parameters requires more care. In this section, I list a few commonly used proposals and their proposal ratios. I may use x instead of θ to represent the state of the chain.

Two results are particularly useful in deriving proposal ratios. So I mention them in the form of two theorems, before describing the proposals. The first result

concerns the distribution of functions of random variables (see, for example, [15]: pp. 107–112).

Theorem 3.1 (a) *If x is a random variable with density $f(x)$, and $y = y(x)$ and $x = x(y)$ is a one-to-one mapping between x and y , then the random variable y has the density*

$$f(y) = f(x(y)) \times \left| \frac{dx}{dy} \right|. \quad (3.14)$$

(b) *The multivariate version is very similar. Suppose random variables $\mathbf{x} = \{x_1, x_2, \dots, x_m\}$ and $\mathbf{y} = \{y_1, y_2, \dots, y_m\}$ constitute a one-to-one mapping through $y_i = y_i(\mathbf{x})$, and $x_i = x_i(\mathbf{y})$, $i = 1, 2, \dots, m$, and that $\mathbf{x}$ has probability density $f(\mathbf{x})$. Then $\mathbf{y}$ has density*

$$f(\mathbf{y}) = f(\mathbf{x}(\mathbf{y})) \times |J(\mathbf{y})|, \quad (3.15)$$

where $|J(\mathbf{y})|$ is the absolute value of the Jacobian determinant of the transform

$$J(\mathbf{y}) = \frac{\partial \mathbf{x}}{\partial \mathbf{y}} = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \cdots & \frac{\partial x_1}{\partial y_m} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \cdots & \frac{\partial x_2}{\partial y_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_m}{\partial y_1} & \frac{\partial x_m}{\partial y_2} & \cdots & \frac{\partial x_m}{\partial y_m} \end{vmatrix}. \quad (3.16)$$

As an example, suppose that the probability of different sites p has a uniform prior distribution $f(p) = 4/3$, $0 \leq p < 3/4$. What is the distribution of the sequence distance θ ? From equation (3.1), we have $dp/d\theta = e^{(-4/3)\theta}$. Thus the distribution of θ is $f(\theta) = 4/3 \times e^{(-4/3)\theta}$, $0 \leq \theta < \infty$. This is the exponential distribution with mean $3/4$.

The second useful result gives the proposal ratio when the proposal is made though transformed variables.

Theorem 3.2 *Suppose the Markov chain is run using the original variables $x_1, x_2, \dots, x_m$, but the proposal is through transformed variables $y_1, y_2, \dots, y_m$. Then*

$$\frac{q(\mathbf{x} \mid \mathbf{x}^*)}{q(\mathbf{x}^* \mid \mathbf{x})} = \frac{q(\mathbf{y} \mid \mathbf{y}^*)}{q(\mathbf{y}^* \mid \mathbf{y})} \times \frac{|J(\mathbf{y}^*)|}{|J(\mathbf{y})|}. \quad (3.17)$$

The proposal ratio in the original variables is the proposal ratio in the transformed variables times the ratio of the Jacobian.

The statement can be proved by noting that

$$q(\mathbf{y}^* \mid \mathbf{y}) = q(\mathbf{y}^* \mid \mathbf{x}) = q(\mathbf{x}^* \mid \mathbf{x}) \times J(\mathbf{y}^*). \quad (3.18)$$

The first equation is because conditioning on $\mathbf{y}$ is equivalent to conditioning on $\mathbf{x}$ due to the one-to-one mapping. The second equation applies Theorem 3.1(b) to derive the density of $\mathbf{y}^*$ as functions of $\mathbf{x}^*$.

3.5.1 *Sliding window using uniform proposal*

This proposal chooses the new state x^* as a random variable from a uniform distribution around the current state x :

$$x^* \sim U\left(x - \frac{w}{2}, x + \frac{w}{2}\right). \quad (3.19)$$

The window size w is a fixed constant, chosen to achieve a reasonable acceptance rate. The proposal ratio is 1 since $q(x^* | x) = q(x | x^*)$. If x is constrained in the interval (a, b) and x^* is outside the range, the excess is reflected back into the interval; that is, if $x^* < a$, x^* is reset to $a + (a - x^*) = 2a - x^*$, and if $x^* > b$, x^* is reset to $b - (b - x^*) = 2b - x^*$. The proposal ratio is 1 even with reflection, because if x can reach x^* through reflection, x^* can reach x through reflection as well. The window size w should be smaller than the range $b - a$. Note that it is incorrect to simply set the unfeasible proposed values to a or b .

3.5.2 *Sliding window using normally distributed proposal*

This algorithm uses a normal proposal density centred around the current state; that is, x^* has a normal distribution with mean x and variance σ^2 , with σ controlling the step size

$$x^* \sim N(x, \sigma^2). \quad (3.20)$$

As $q(x^* | x) = (1/(\sigma\sqrt{2\pi}))\exp\{-(x^* - x)^2/(2\sigma^2)\} = q(x | x^*)$, the proposal ratio is 1. This proposal works also if x is constrained in the interval (a, b) . If x^* is outside the range, the excess is reflected back into the interval, and the proposal ratio remains one. Both with and without reflection, the number of routes from x to x^* is the same as from x^* to x , and the densities are the same in the opposite directions, even if not between the routes. Note that sliding window algorithms using either uniform or normal jumping kernels are Metropolis algorithms with symmetrical proposals.

How do we choose σ ? Suppose the target density is the standard normal $N(0, 1)$, and the proposal is $x^* \sim N(x, \sigma^2)$. A large σ will cause most proposals to be in unreasonable regions of the parameter space and be rejected. The chain then stays at the same state for a long time, causing high correlation. A σ too small means that the proposed states are very close to the current state, and most proposals will be accepted. However, the chain baby-walks in the same region of the parameter space for a long time, leading again to high correlation. Proposals that minimize the auto correlations are thus optimal.

More formally, consider the sample mean $\hat{\theta} = (1/N) \sum x^{(t)}$, where $x^{(t)}$ is the state in iteration t , with $t = 1, 2, \dots, N$. With independent sampling, $\text{var}(\hat{\theta}) = 1/N$. The large-sample variance of a dependent sample is

$$\text{var}(\hat{\theta}) = \frac{1}{N}[1 + 2(\rho_1 + \rho_2 + \rho_3 + \dots)], \quad (3.21)$$

where ρ_k is the autocorrelation of the Markov chain at lag k . In effect, a dependent sample of size N is equivalent to an independent sample of size

$N/[1 + 2(\rho_1 + \rho_2 + \rho_3 + \dots)]$. By minimizing $\text{var}(\hat{\theta})$ in equation (3.21), Gelman *et al.* [9] found the optimum σ to be about 2.4. Thus if the target density is a general normal density $N(\mu, \tau^2)$, the optimum proposal density should be $N(x, \tau^2 \sigma^2)$ with $\sigma = 2.4$. As τ is unknown, one can monitor the acceptance rate or jumping probability, which is slightly below 0.5 at the optimum σ .

3.5.3 Sliding window using normal proposal in multidimensions

If the target density is a m -dimensional standard normal with density $N_m(0, I)$ where I is a $m \times m$ identity matrix, one can use the proposal density $q(\mathbf{x}^* | \mathbf{x}) = N_m(\mathbf{x}, \mathbf{I}\sigma^2)$. The proposal ratio is one. The Gelman *et al.* [9] analysis suggests that the optimum scale factor σ is 2.4, 1.7, 1.4, 1.2, 1, 0.9, 0.7 for $m = 1, 2, 3, 4, 6, 8, 10$, respectively, with an optimal acceptance rate of about 0.26 for $m > 6$. It is interesting to note that at low dimensions, the optimal proposal density is over-dispersed relative to the target density, suggesting that one should take big steps, while at high dimensions, one should use under-dispersed proposal densities and take small steps. In general one should try to achieve an acceptance rate of about 20–70% for 1-D proposals, and 15–40% for multi-dimensional proposals.

Those results are more useful than for just standard normal densities. When the target density is $\mathbf{x} \sim N_m(\mu, \mathbf{S})$, with variance–covariance matrix $\mathbf{S}$, several strategies can be used. One is to reparametrize the model using $\mathbf{y} = \mathbf{S}^{-1/2} \mathbf{x}$ as parameters, where $\mathbf{S}^{-1/2}$ is the square root of $\mathbf{S}^{-1}$. Note that $\mathbf{y}$ has unit variance, and the above proposal can be used. The second strategy is to propose new states using the transformed variables $\mathbf{y}$, that is, $q(\mathbf{y}^* | \mathbf{y}) = N_m(\mathbf{y}, \mathbf{I}\sigma^2)$, and then derive the proposal ratio in the original variables $\mathbf{x}$. The proposal ratio is one according to Theorem 3.2. A third approach is to simply use the proposal $x^* \sim N_m(\mathbf{x}, \sigma^2 \mathbf{S})$, where σ^2 is chosen according to the above discussion. The three approaches are equivalent and all of them take care of possible differences in the scales and possible correlations among the variables. In real data analysis, $\mathbf{S}$ is unknown. One can perform short runs of the Markov chain to obtain an estimate $\hat{\mathbf{S}}$ of the variance–covariance matrix in the posterior density, and then use it in the proposal. If $\mathbf{S}$ is estimated in the same run, samples taken to estimate $\mathbf{S}$ should be discarded. If the normal distribution is a good approximation to the posterior density, those guidelines should work well.

3.5.4 Proportional shrinking and expanding

For a variable that is always positive or always negative, this proposal multiplies the current value by a random number that is around 1. Let

$$\begin{aligned} c &= e^{\epsilon(r-1/2)}, \\ x^* &= cx, \end{aligned} \tag{3.22}$$

where $r \sim U(0, 1)$ and $\epsilon > 0$ is a small finetuning parameter. Note that x is shrunk or expanded depending on whether r is $<$ or $>$ $1/2$. To calculate the proposal ratio, derive the proposal density $q(x^* | x)$ through variable transform, noting that r and x^* are random variables while ϵ and x are constants. Since

$r = 1/2 + \log(x^*/x)/\epsilon$, and $dr/dx^* = 1/(\epsilon x^*)$, we have from Theorem 3.1(a)

$$q(x^* | x) = f(r(x^*)) \times \left| \frac{dr}{dx^*} \right| = \frac{1}{\epsilon |x^*|}. \quad (3.23)$$

Similarly $q(x | x^*) = 1/\epsilon |x|$, so the proposal ratio is $q(x | x^*)/q(x^* | x) = c$.

This proposal can be used to shrink or expand many variables by the same factor c : $x_i^* = cx_i$, $i = 1, 2, \dots, m$. This is useful for variables with a fixed order, such as the ages of nodes in a phylogenetic tree [48]. It is also effective in bringing all variables, such as branch lengths on a phylogeny, into the right scale if all of them are either too large or too small. Although all m variables are altered, the proposal is really in one dimension (along a line in the m -D space). We can derive the proposal ratio using the transform: $y_1 = x_1, y_i = x_i/x_1$, $i = 2, 3, \dots, m$. The proposal changes y_1 , but $y_2, \dots, y_m$ remain unchanged. The proposal ratio in the transformed variables is c . The Jacobian is $J(y_1, y_2, \dots, y_m) = |\partial \mathbf{x} / \partial \mathbf{y}| = y_1^{m-1}$. The proposal ratio in the original variables is thus $c \times (y_1^*/y_1)^{m-1} = c^m$, according to Theorem 3.2. Similarly, if the proposal multiplies m variables by c and divides n variables by c , the proposal ratio is $c^{m/n}$.

3.6 Monitoring Markov chains and processing output

3.6.1 Diagnosing and validating MCMC algorithms

An MCMC algorithm can suffer from two problems: slow convergence and poor mixing. The former means that it takes very long for the chain to reach stationarity. The latter means that the sampled states are highly correlated and the chain is very inefficient in exploring the parameter space. While it is often obvious that the proposal density $q(\cdot | \cdot)$ satisfies the required regularity conditions so that the MCMC is in theory guaranteed to converge to the target distribution, it is much harder to determine in real data problems whether the chain has reached stationarity. A number of heuristic methods have been suggested to diagnose the Markov chain. However, those diagnostics are able to reveal problems but unable to prove the correctness of the algorithm or implementation. Model misspecification, programming errors, and slow convergence all pose difficulties to program validation. A Bayesian MCMC program is notably harder to debug than a maximum likelihood program implementing a similar model. In a likelihood iteration, the convergence is to a point while in Bayesian MCMC, it is to a statistical distribution. In likelihood iteration, the log likelihood should always go up (at least if the optimizer is non-decreasing), and the gradient converges to zero. In a Bayesian MCMC algorithm, no statistics have a fixed direction of change. It is usually hard to independently calculate the posterior probability distribution. The temptation to use sophisticated models with excessive parameters in Bayesian modelling adds further difficulty. Often when the algorithm converges slowly or mixes poorly, it is difficult to decide whether this is due to faulty theory, buggy program, or inefficient but correct algorithm.

The following are some of the commonly used strategies for diagnosing and validating an MCMC program. (1) One can plot parameters of interest or their functions against the iterations. Such time-series plots can often reveal lack of convergence and/or poor mixing (see, for example, Figs. 3.3(a) and (b)). Often the chain appears to have converged with respect to some parameters but not to others. (2) The acceptance rate for each proposal should be neither too high nor too low. (3) It is advisable to run multiple chains from different starting points and make sure that the chains all converge to the same distribution. Gelman and Rubin's [10] statistic can be used to analyse multiple chains; see the next section. (4) Another technique is to run the chain without data, that is, to fix $f(D | \theta) = 1$ in equation (3.11). The posterior should then be the prior, which might be analytically available for comparison. (5) Simulation is also commonly used to validate MCMC algorithms. For example, Wilson *et al.* [49] simulated data under the prior to calculate the "hit probability" and "coverage probability" to validate their BATWING program. The former is the probability that the $100\alpha\%$ posterior credibility interval of a parameter includes the correct value. This should equal α . The latter is the average, across data replicates, of posterior coverage probability of a fixed interval. If this fixed interval has $100\alpha\%$ coverage probability in the prior, the average posterior coverage probability should also equal α [37, 49]. This is a more precise criterion for assessing interval coverage than the hit probability.

3.6.2 Gelman and Rubin's potential scale reduction statistic

Gelman and Rubin [10] suggested a diagnostic statistic called estimated "potential scale reduction," based on variance-components analysis of samples taken from several chains run using "over-dispersed" starting points. The idea is that after convergence, the within-chain variance should be indistinguishable from the between-chain variation while before convergence, the within-chain variance should be too small and the between-chain variance should be too large. The statistic can be used to monitor any or every parameter of interest. Let this be x , and its variance in the target distribution be τ^2 . Suppose there are m chains, each run for n iterations, after the burn-in is discarded. Let x_{ij} be the parameter sampled at the j th iteration from the i th chain. Gelman and Rubin [10] defined the between-chain variance

$$B = \frac{n}{m-1} \sum_{i=1}^m (\bar{x}_{i.} - \bar{x}_{..})^2, \quad (3.24)$$

and the within-chain variance

$$W = \frac{1}{m(n-1)} \sum_{i=1}^m \sum_{j=1}^n (x_{ij} - \bar{x}_{i.})^2, \quad (3.25)$$

where $\bar{x}_{i.} = (1/n) \sum_{j=1}^n x_{ij}$ is the mean within the i th chain, and $\bar{x}_{..} = (1/m) \sum_{i=1}^m \bar{x}_{i.}$ is the overall mean. If all the m chains have reached stationarity and x_{ij} are samples from the same target density, both B and W are

unbiased estimates of τ^2 , and so is their weighted mean

$$\hat{\tau}^2 = \frac{n-1}{n}W + \frac{1}{n}B. \quad (3.26)$$

If the m chains have not reached stationarity, W will be an *underestimate* of τ^2 since each chain has not traversed the whole parameter space and does not contain enough variation, while B will be an overestimate as the chains are from overdispersed starting points. Gelman and Rubin [10] showed that in this case $\hat{\tau}^2$ is also an overestimate of τ^2 . The estimated “potential scale reduction” is defined as

$$\hat{R} = \frac{\hat{\tau}^2}{W}. \quad (3.27)$$

This should get smaller and approach one when the parallel chains reach the same target distribution. In real data problems, values of $\hat{R} < 1.1$ or 1.2 indicate convergence.

3.6.3 Processing output

Before we process the output, the beginning part of the chain before it has converged to the stationary distribution is discarded as “burn-in.” Some programs do not sample every iteration but instead only takes a sample for every certain number of iterations. This is known as “thinning” the chain, as the thinned samples have reduced autocorrelations across iterations. While in theory sampling every iteration is more efficient (with smaller variances) than thinned samples, MCMC algorithms easily produce huge output files and it is often necessary to thin the chain to reduce the disk requirement.

After the burn-in, the samples taken from the MCMC can be summarized in a straightforward way. The sample mean, median, or mode can be used as a point estimate of the parameter, while the HPD or equal-probability credibility intervals can be constructed from the sample as well. For example, a 95% CI can be constructed by sorting the MCMC output for the variable and then using the 2.5% and 97.5% percentiles. The whole posterior distribution can be estimated by using a histogram, perhaps with further smoothing [40].

3.7 Applications to molecular phylogenetics

MCMC algorithms have been widely used in population genetics to analyse genetic data (DNA sequences, micro-satellites, etc.) under the coalescent models of variable complexity. Such applications include estimation of mutation rates (e.g. [4]), inference of population demographic processes or gene flow between subdivided populations (e.g. [3, 49]), and estimation of ancestral population sizes [35, 50], to name a few. See recent reviews by Griffiths and Tavaré [14] and Stephens and Donnelly [42]. Here I will discuss two major applications of Bayesian inference to molecular phylogenetics: estimation of phylogenetic trees and estimation of species divergence times under stochastic models of evolutionary rate change.

3.7.1 *Estimation of phylogenies*

Brief history. The Bayesian method was introduced to molecular phylogenetics by Rannala and Yang [34, 53], Mau and Newton [29], and Li *et al.* [28]. Those early studies assumed a constant rate of evolution (the molecular clock) as well as equal-probability prior for rooted trees either with or without ordered node ages (rooted trees or labelled histories). Since then, much more efficient MCMC algorithms have been implemented in the computer programs BAMBE [27] and MrBayes [21, 36]. The clock constraint is also relaxed, enabling phylogenetic inference under more realistic evolutionary models. A number of innovations have been introduced in those programs, adapting tree perturbation algorithms used in heuristic tree search (such as nearest-neighbour interchange, NNI, and subtree pruning and regrafting, SPR [44]), into flexible and efficient MCMC proposal algorithms for moving around in the tree space. In particular, MrBayes 3 has essentially incorporated all evolutionary models developed for likelihood inference, and can accommodate heterogeneous data sets from multiple gene loci in a combined analysis. A Metropolis-coupled MCMC algorithm (MC³) is implemented in MrBayes to overcome multiple local peaks in the tree space. The parallel algorithm is efficient on network workstations that are becoming accessible to empirical biologists [2, 36]. MrBayes is now widely used in phylogeny reconstruction and is the top-cited paper in August 2002 in the whole field of computer science!

General framework. To formulate the problem of phylogeny reconstruction in the general framework of Bayesian inference described requires no more than definition of symbols. Let D be the sequence data. Let θ include all parameters in the model, with a prior distribution $f(\theta)$. Let τ_i be the i th tree topology, $i = 1, 2, \dots, N(s)$, where $N(s)$ is the total number of tree topologies for s species. Usually a uniform prior $f(\tau_i) = 1/N(s)$ is assumed. Let b_i be branch lengths on tree τ_i , with prior probability $f(b_i)$. MrBayes 3 assumes that branch lengths have independent uniform or exponential priors with the parameter (upper bound for the uniform or mean for the exponential) set by the user. The posterior probability of tree τ_i is then

$$P(\tau_i | D) = \frac{\iint f(\theta)f(b_i | \theta)f(\tau_i | \theta)f(D | \tau_i, b_i, \theta) db_i d\theta}{\sum_{j=1}^{N(s)} \iint f(\theta)f(b_j | \theta)f(\tau_j | \theta)f(D | \tau_j, b_j, \theta) db_j d\theta}. \quad (3.28)$$

Note that calculating the denominator, the marginal probability of the data $f(D)$, would involve summing over all possible tree topologies and, for each tree topology τ_j , integrating over all branch lengths b_i and parameters θ , a virtually impossible task except for very small trees. The MCMC algorithm avoids direct calculation of $f(D)$, but integrates over branch lengths b_i and parameters θ through MCMC.

Summarizing output. It is straightforward to summarize the posterior probability distribution of trees, and several summaries are provided by MrBayes. One can take the tree with the maximum posterior probability (MAP) as a point

estimate, the so-called MAP tree [34]. This should be identical or very similar to the maximum likelihood tree under the same model. An approximate 95% credibility set of trees can be constructed by including trees with the highest posterior probabilities until the total probability exceeds 95%. Similarly to summarizing bootstrap support values for clades (subtrees) [8], posterior clade probabilities can also be collected and shown on a majority-rule consensus tree [27]. It may be noted that the branch lengths on the consensus tree produced by MrBayes 3 should be ignored as those are averages over different tree topologies; branch lengths are meaningful only on a fixed topology and their posterior probabilities should be calculated by running the MCMC on the fixed tree topology.

Comparison with likelihood. In terms of computational efficiency, stochastic tree search by MrBayes appears to be more efficient than heuristic tree search under likelihood using David Swofford's PAUP program [45]. Nevertheless, running time of the MCMC algorithm is proportional to the number of iterations the algorithm is run for. In general, longer chains are needed to achieve convergence in larger data sets due to the increased number of parameters to be averaged over. However, many users ran shorter chains for larger data sets because larger trees require more computation per iteration. As a result, it is not always certain that the MCMC algorithm has converged in Bayesian analyses of very large data sets. Furthermore, dramatic improvements to heuristic tree search under likelihood are still being made [16]. So it seems possible that for the purpose of obtaining a point estimate, likelihood heuristic search using numerical optimization can be faster than Bayesian stochastic search using MCMC. However, no one knows how to use the information in the likelihood tree search to attach a confidence interval or some other measure of sampling errors in the maximum likelihood tree—as one can use the local curvature or Hessian matrix calculated in a non-linear programming algorithm to construct a confidence interval for a conventional parameter. As a result, one currently resorts to bootstrapping. Bootstrapping under likelihood is an expensive procedure, and appears slower than Bayesian MCMC.

To many, Bayesian inference of molecular phylogenies enjoys a theoretical advantage over maximum likelihood with bootstrapping. Posterior probabilities have an easy interpretation: the posterior probability of a tree or clade is the probability that the tree or clade is correct given the data and the model [27, 34]. In contrast, the interpretation of bootstrap in phylogenetics has been controversial (e.g. [6, 19], Chapter 4, this volume). As a result, posterior probabilities of trees can be used in a straightforward manner in a variety of phylogeny-based evolutionary analyses to accommodate phylogenetic uncertainty; for example, they were used in comparative analysis to average the results over phylogenies [20, 22].

It has been noted that Bayesian posterior probabilities calculated from real data sets using MrBayes are often extremely high. One may observe that while bootstrap clade proportions are shown on published trees only if they are $>50\%$ (as otherwise the relationships may not be considered trustable), posterior clade

probabilities are reported only if they are $<100\%$ (as most of them are 100% !). Recently a number of simulation studies suggested that the posterior probabilities are often misleadingly high (e.g. [1, 7, 43]). Some of the high posterior probabilities from real data sets may be genuine and indicate high but correct confidence in the phylogenetic relationship. Some may be due to lack of convergence of the MCMC algorithm or inadequate evolutionary model, which could be resolved by running longer chains or implementing more realistic substitution models. However, the problem seems more serious. Extremely high probabilities were observed by Rannala and Yang [34], who studied only small trees and used numerical integration, in which case algorithm performance is not an issue. Yang and Rannala [54] note that the posterior probabilities of trees vary widely over simulated replicate data sets and that they can be unduly influenced by the prior on the internal branch lengths. It is easy to see that high posterior probabilities will decrease when the internal branch lengths assumed in the prior get smaller; in the extreme when internal branch lengths are assumed to be 0, all trees will have the same probability. It is not clear to what extent the high posterior probabilities observed in real data sets can be attributed to this sensitivity. The problem raises serious practical concern about the methodology and further investigation is urgently needed.

3.7.2 *Estimation of species divergence times*

Bayesian inference has also been successfully applied by Thorne and co-workers [26, 48] to estimate species divergence times under models of rate change, that is, when the evolutionary rate itself evolves. Traditionally the molecular clock has been assumed for divergence time estimation. However, in many data sets, especially when the species are not closely related, the clock assumption is seriously violated. Because the sequence data contain information only about the branch length, which is the product of time and rate, but not about time and rate individually, incorrectly assuming that the clock can lead to seriously biased time estimates.

The likelihood approach to this problem has been to classify the branches on the tree into a few rate classes and then to estimate the divergence times as well as those few branch rates by maximum likelihood [25, 32, 57]. The methods have the drawback of requiring the researcher to assign branches to rate groups, although ideas of heuristic rate smoothing [38, 39] can be used to automate that process. The likelihood method has also been extended to incorporate fossil calibration information at multiple nodes on the phylogeny and to account for the heterogeneity in evolutionary process of multiple gene loci in combined analysis [56]. Yang and Yoder [56] emphasized the importance of such combined analysis as a way of circumventing the serious confounding effect between time and rate; the rates vary over lineages in different ways among gene loci, but the divergence times are shared, so that the internal constraints in the model might lead to reliable estimation of divergence times even when the clock is violated in every gene.

The Bayesian method specifies a prior distribution $f(t)$ of divergence times (t) and a prior distribution $f(r)$ of evolutionary rates (r). Let θ be all parameters in the model, with prior $f(\theta)$. The joint posterior distribution of times and rates are then

$$f(t, r \mid D) = \frac{\int f(\theta) f(t \mid \theta) f(r \mid t, \theta) f(D \mid t, r, \theta) d\theta}{\iiint f(\theta) f(t \mid \theta) f(r \mid t, \theta) f(D \mid t, r, \theta) dr dt d\theta}. \quad (3.29)$$

This is approximated by the MCMC algorithm. The marginal posterior of divergence times

$$f(t \mid D) = \int f(t, r \mid D) dr \quad (3.30)$$

can be constructed from the samples taken from the MCMC.

Thorne *et al.* [48] and Kishino *et al.* [26] used a recursive procedure to specify the prior for the rates, proceeding from the root of the tree towards the tips. The rate at the root is assumed to have a gamma prior. Then the rate at each node is specified conditioning on the rate at the ancestral node. Specifically, given the log rate, $\log(r_A)$, of the ancestral node, the log rate of the current node, $\log(r)$, follows a normal distribution with mean $\log(r_A) - c$ and variance νt , where t is the time duration separating the two nodes. The correction term c in the mean is to remove any trend in the rate but is unimportant to the present description. Parameter ν controls how quickly the rate drifts and determines how clock-like the tree is *a priori*. This is a geometric Brownian motion model.

The prior for divergence times is specified using another recursive procedure [26], starting from the root and moving towards the tips. The age of the root has a gamma prior. Then each path from a tip to the root or an ancestral node is broken into random segments, corresponding to branches on the path, with the segment lengths having a Dirichlet density with equal probabilities (see [48]). Fossil calibration information is incorporated in the prior for times as constraints on node ages.

Thorne's program implements an efficient algorithm for divergence time estimation under the models of Thorne *et al.* [48] and Kishino *et al.* [26]. It incorporates fossil information at multiple nodes as lower and upper bounds. The likelihood is calculated using a normal approximation to the branch lengths estimated without the clock assumption, to achieve computational efficiency. Recent extensions made the method suitable for combined analysis of multiple data sets. The method and program has been used extensively to date divergences of major species groups, such as the radiation of mammals [17, 41].

While many factors including the substitution model can potentially affect divergence time estimation in the Bayesian method, the most difficult and important of those appear to be the priors for rates and times. An infinite amount of sequence data combined with a perfectly correct substitution model will reduce the errors in branch lengths to zero, but the errors in time estimates will persist as long as there is uncertainty in the fossil calibrations, or mismatch between the model and prior on one hand and reality on the other. Yoder and Yang [58]

described a case where species sampling had a major effect on Bayesian divergence time estimation. The authors estimated divergence times on a tree of mammals, when either two or nine mouse lemur species are included in the data. The estimated age of the mouse lemur clade in the bigger data set was 25% older than in the small data set. The reason appears to be the assumed prior model of times. As discussed above, the method assumes similar branch lengths on the tree. However, branches within the mouse lemurs are very short, and inclusion of more mouse lemur species in the large data set made the prior rather unrealistic and pushed back the age of the mouse lemur clades.

In sum, recent developments in Bayesian and likelihood frameworks make it possible to estimate divergence times without the molecular clock through integrated analysis of heterogeneous genetic data sets incorporating multiple fossil calibrations. However, one has to bear in mind that estimation of divergence times without a clock is an extremely difficult problem whatever method is used, and should critically assess the effects of assumptions about rates and times on time estimates. The quality of fossils is critically important.

3.8 Conclusions and perspectives

The Bayesian method, especially combined with MCMC algorithms, provides exciting opportunities to model-based analysis in molecular phylogenetics. Use of the likelihood function makes it straightforward to conduct integrated analysis of heterogeneous data sets from multiple loci while accommodating differences in their evolutionary characteristics, obliterating the need for ad hoc approaches such as supermatrix and supertree analyses. However, a number of computational and theoretical problems remain, which will no doubt prompt active research in the future. Computational problems include development of ingenious and efficient proposal mechanisms that will lead to improved mixing of the MCMC algorithms. While likelihood and Bayesian algorithms will probably never be fast enough to scale up with the ever-increasing sizes of real data sets analysed by molecular systematists, any gain in performance is highly beneficial. Theoretical problems include understanding the power and limitations of the Bayesian methods and its robustness to assumptions in the prior and in the substitution model. The complexity of likelihood estimation of phylogeny has been extensively discussed (Chapter 2, this volume). That complexity appears to apply also in the Bayesian framework, and it remains an open question whether Bayesian posterior probabilities will be the ultimate answer to molecular phylogeny reconstruction.

Program availability

The programs mentioned in this chapter are available at the following web sites: MrBayes: <http://morphbank.ebc.uu.se/mrbayes/>; Divergence time estimation by Bayesian methods (T³: Thornian Time Traveller): <ftp://abacus.gene.ucl.ac.uk/pub/T3/> and <http://statgen.ncsu.edu/thorne/multidivtime.html>;

Tree reconstruction by likelihood:

PAUP: <http://paup.csit.fsu.edu/>;

Time estimation by likelihood:

PAML: <http://abacus.gene.ucl.ac.uk/software/paml.html>.

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STATISTICAL APPROACH TO TESTS INVOLVING PHYLOGENIES

Susan Holmes

This chapter reviews statistical testing involving phylogenies. We present both the classical framework with the use of sampling distributions involving the bootstrap and permutation tests and the Bayesian approach using posterior distributions.

We give some examples of direct tests for deciding whether the data support a given tree or trees that share a particular property, comparative analyses using tests that condition on the phylogeny being known are also discussed.

We introduce a continuous parameter space that enables one to avoid the delicate problem of comparing exponentially many possible models with a finite amount of data. This chapter contains a review of the literature on parametric tests in phylogenetics and some suggestions of non-parametric tests. We also present some open questions that have to be solved by mathematical statisticians to provide the theoretical justification of both current testing strategies and as yet underdeveloped areas of statistical testing in non-standard frameworks.

4.1 The statistical approach to phylogenetic inference

From our point of view, as statisticians, we see the phylogenetic inference as both estimation and testing problems that are set in an unusual space. In most standard statistical theory, the parameter space is either the real line $\mathbb{R}$ or an Euclidean space of higher dimension, $\mathbb{R}^d$ for instance. One notable exception for which there are a number of available statistical models and tests are ranked data. These sit in the symmetric group $\mathfrak{S}_n$ of permutations of n elements. See [58] for a book long treatment on statistics in such spaces, see [15] for some examples of data and relevant statistical analyses based on decompositions of the space, and [27] on the use of distances and their applications in that context. Of course other relevant high dimensional parameters that statisticians use are probability distributions themselves (non-parametric statistics). The authors of [16] use them to show conditions on consistency for Bayes estimates. Thus, as opposed to some authors in systematics, statisticians actually do believe that both distributions and trees can be true parameters. Although some references [4, 76, 80] do not agree with this approach, we will confer the status of true parameters to both the

branching pattern or “tree topology,” that we will denote by τ and the rooted binary tree with edge lengths and n leaves denoted $\mathcal{T}_n$. The inner edge lengths are often denoted $\theta_1, \dots, \theta_{n-2}$ and considered **nuisance** parameters. One of the difficulties in manipulating such parameters is the lack of a natural ordering of trees.

The main focus here will be the subject of hypothesis testing using phylogenies, the method chosen to estimate these phylogenies is not the focus, so that much of what is discussed is relevant whether we use maximum likelihood (MC), parsimony- or distance-based estimates. We will review the different paradigms, frequentist and Bayesian and emphasize their different approaches to the question of testing a hypothesis H_0 (either composite or simple) versus either a simple alternative H_1 or a set of alternatives $\mathcal{H}_A$. We cannot cover many interesting aspects of the discussion between proponents of both perspectives and refer the reader to an extensive literature on the general subject of frequentist versus Bayesian approaches [6, 7, 48]. We will not go as far as a discussion of finding the best tests for each situation but will insist more on correct tests. The reader interested in the more sophisticated statistical theory of **uniformly most powerful tests** is referred to [50]. A serious attempt at applying the statistical theory of most powerful tests to model selection was made recently by [4]. We will comment on his findings, but insist that statistical tests should be able to adjust to cases where the evolutionary model is unknown or misspecified. Thus in Section 4.4 we concentrate on proposing non-parametric alternatives to existing tests.

Section 4.2 will give the statistical terminology and present some of the issues involved in statistical testing, the meaning of p -values and their comparison to Bayesian alternatives in the context of tests involving phylogenetic trees and the classical approaches to comparing tests. Section 4.3 concentrates on certain tests already in use by the community, with emphasis on their assumptions. Section 4.4 introduces a geometric interpretation of current problems in phylogeny, and proposes a non-parametric multivariate approach. Finally in the conclusion we note how many theoretical aspects of hypothesis testing remain unresolved in the phylogenetic setting. Most papers justify their results by analogy [22, 69] or by simulation [82]. To be blunt, apart from Chang [12, 13] and Newton [64] there are practically no statistical theorems justifying current tests used in systematics literature and this area is a wide open field for further researchers interested in the interface between multivariate statistics and geometry.

4.2 Hypotheses testing

For background on classical hypothesis tests [68] is a clear elementary introduction and [50] is an encyclopedic account.

4.2.1 Null and alternative hypotheses

We will consider tests of a null hypothesis H_0 , usually a statement involving an unknown parameter. For example, $\mu = \mu_0$, where μ_0 is a predefined value, such as 4 for a real valued parameter (a **simple hypothesis**), or of the type $H_0: \mu \in \mathcal{M}$,

with $\mathcal{M}$ a subset of the parameters, this is a **composite hypothesis**. The **alternative** is usually defined by the complementary set $\overline{\mathcal{M}}$: $H_A: \mu \in \mathcal{M}$. In the case of the Kishino–Hasegawa (KH) test [34] for instance the parameter of interest is the difference in log **likelihoods** of the two trees to be compared $\delta = \log L(D \mid \mathcal{T}_1) - \log L(D \mid \mathcal{T}_2)$ (for an extensive discussion of likelihood computations in the context of phylogenetic trees see Chapter 2, this volume). This difference δ in much of the literature, suggesting that this is the parameter of interest, however there is already slippage of the classical paradigm here since the parameter involves the data, so the definition of the exact parameter that is being tested in the KH test is unclear.

4.2.2 Test statistics

Suppose for the moment that H_0 is simple. Given some observed Data $D = \{x_1, x_2, \dots, x_n\}$, it is often impossible to test the hypothesis directly by asking whether the p -value $P(D \mid H_0)$ is small, so we will use some feature of the data, or test statistic S such that the distribution of this test statistic under the null hypothesis (the **null sampling distribution**) is known. Thus, if the observed value of S is denoted s , $P(s \mid H_0)$ can be computed. We call $P(D \mid H_0)$ as it varies with the data D the sampling distribution, the quantity $P(D \mid H)$ as a function of H is called the **likelihood** of H for the data D .

Some authors [4] identify trees with distributions, this is possible supposing a fixed Markovian evolutionary model and verification of certain identifiability constraints [12]. Thus, the parameters of interest become the distributions and a test for whether the k topologies forming $\mathcal{M}_k = \{\tau_1, \tau_2, \dots, \tau_k\}$ are equidistant from topology h is stated using the Kullback–Leibler distance between distributions [4].

In this survey, we also encourage the use of a distance between trees, but have tried to enlarge our outlook to encompass more general evolutionary models so that we no longer have the identification between trees and distributions. Not all test statistics are created equal, and in the case of the bootstrap it is always better to have a **pivotal test statistic** [23], that is a statistic whose distribution does not depend on unknown parameters. For this reason, it is preferable to centre and rescale the statistic so that the null distribution is centred at 0 and has a known variance, at least asymptotically.

4.2.3 Significance and power

Statisticians take into account two kinds of error:

Type I error or Significance This is the probability of rejecting a hypothesis when in fact it is true.

Type II error or (1-Power) This is the probability of not rejecting a hypothesis that is in fact false.

Usually the type I error is fixed at a given level, say 0.05 or 0.01 and then we might explore ways of making the type II error as small as possible, this is equivalent to maximizing what is known as the power function: the

probability of rejecting the null hypothesis H_0 given that the alternative is true $P(\text{reject } H_0 \mid H_A)$. We often use the rejection region R to denote the values of the test statistic s that lead to rejection, for a one-sided test $H_A: \mu > \mu_0$ at the 5% level the rejection region will be given by a half line of the form $[c_{0.95}, +\infty]$, where $c_{0.95}$ is the 95th percentile of the distribution of the test statistic under the null hypothesis.

The power of the test depends on the alternative H_A which can sometimes be defined as $\mu \in \overline{\mathcal{M}}$, then the power function written as a function of the rejection region is

$$P(S(D) \in R \mid \mu \in \overline{\mathcal{M}}).$$

Trying to find tests that are powerful against all alternatives (Uniformly Most Powerful, UMP) is not realistic unless we can use parametric distributions such as exponential families for which there is a well understood theory [50]. In the absence of analytical forms for the power functions, authors [4] are reduced to using simulation studies to compute the power function. In general the power will be a function of many things: the variability of the sampling distribution, the difference between the true parameter and the null hypothesis. In the case of trees, a power curve is only possible if we can quantify this difference with a distance between trees. Aris-Brosou [4] uses the Kullback–Leibler distance. As a substitute for the more general non-parametric setup, we suggest using a geometrically defined distance.

Parametric tests use a specific distributional form of the data, non-parametric tests are valid no matter what the distribution of the data are. Tests are said to be robust when their conclusions remain approximately valid even if the distributional assumptions are violated. Reference [4] shows in his careful power function simulations that the tests he describes are not robust.

Classical statistical theory (in particular the Neyman Pearson lemma) ensures that the most powerful test for testing one simple hypothesis H_0 versus another H_A is the likelihood ratio test based on the test statistic $S = P(D \mid H_0)/P(D \mid H_A)$.

Frequentists define the p -value of a test as the probability

$$P(S(D) \in \mathcal{S} \mid H_0),$$

where $\mathcal{S}$ is the random region constructed as the values of the statistic “as extreme as” the observed statistic $S(D)$, the definition of the region $\mathcal{S}$ depends also on the alternative hypothesis H_A , for instance for a real valued test statistic S and a two-sided alternative, $\mathcal{S}$ will be the union of two disjoint half lines bounded by what are called the **critical points**, for a one-sided alternative, $\mathcal{S}$ will only be a half line. If we prespecify a type I error to be α , we can define a **rejection region** R_α for the statistic $S(D)$ such that

$$P(S(D) \in R_\alpha \mid H_0) = \alpha.$$

We reject the null hypothesis H_0 if the observed statistic S is in the rejection region. This makes the link between confidence regions and hypothesis tests

which are often seen as dual of each other. The confidence region for a parameter μ is a region $\mathcal{M}_\alpha$ such that

$$P(\mathcal{M}_\alpha \ni \mu) = 1 - \alpha.$$

The usual image useful in understanding the reasoning behind the notion of confidence regions (and very nicely illustrated in the Cartoon Guide to Statistics [31]) is the archer and her target. If we know the precision with which the archer hits the target in the sense of the distribution of her arrows in the large circle. We can use it if we are standing behind the target to go back from a single arrow head seen at the back (where the target is invisible and all we see is a square bale of hay) to estimating where we think the centre was.

In particular, if we are lucky enough to have a sampling distribution with a lot of symmetry, we can look at the centre of the sampling distribution and find a good estimate of the parameter and hypothesis testing through the dual confidence region statement is easy.

For the classical hypothesis testing setup to work at all, there are many procedural rules that have to be followed. The main one concerns the order in which the steps are undertaken:

- State the null hypothesis.
- State the alternative.
- Decide on a test statistic and a significance level (Type I error).
- Compute the test statistic for the data at hand.
- Compute the probability of such a value of the test statistic under the null hypothesis (either analytically or through a bootstrap or permutation test simulation experiment).
- Compare this probability (or p -value, as it is called) to the type I error that was pre-specified, if the p -value is smaller than the preassigned type I error, reject the null hypothesis.

In looking at many published instances, it is surprising how often one or more of these steps are violated, in particular it is important to say whether the trees involved in the testing statements are specified prior to consulting the data or not. Data snooping completely invalidates the conclusions of tests that do not account for it (see [30] for a clear statement in this context).

There are ways of incorporating prior information in statistical analyses, these are known as Bayesian methods.

4.2.4 *Bayesian hypothesis testing*

I will not go into the details of Bayesian estimation as the reader can consult Yang, Chapter 3, this volume, who has an exhaustive treatment of Bayesian estimation for phylogenetics in a parametric context. Bayesian statisticians have a completely different approach to hypothesis testing. Parameters are no longer fixed, but are themselves given distributions. Before consulting the data, the parameter is said to have a prior distribution, from which we can actually write

statements such as $P(H_0)$ or $P(\tau \in \mathcal{M})$, which would be meaningless in the classical context. After consulting the data D , the distributions becomes restricted to the conditional $P(H_0 \mid D)$ or $P(\tau \in \mathcal{M} \mid D)$.

The most commonly used Bayesian procedure for hypothesis testing is to specify a prior for the null hypothesis, H_0 , say for instance with no bias either way, one conventionally chooses $P(H_0) = 0.5$ [48].

Bayesian testing is based on the ratio (or posterior odds)

$$\frac{P(H_0 \mid D)}{P(\overline{H}_0 \mid D)} = \frac{P(D \mid H_0)}{P(D \mid \overline{H}_0)} \times \frac{P(H_0)}{P(\overline{H}_0)}$$

to decide whether the hypothesis H_0 should be rejected, the first ratio on the right is called the Bayes factor; it shows how the prior odds $P(H_0)/P(\overline{H}_0)$ are changed to the posterior odds, if the Bayes factor is small, the null hypothesis is rejected. It is also possible to build sets B with given posterior probability levels: $P(\tau \in B \mid D) = 0.99$, these are called Bayesian **credibility sets**. A clear elementary discussion of Bayesian hypothesis testing is in Chapter 4 of [48].

An example of using the Bayesian paradigm for comparing varying testing procedures in the phylogenetic context can be found in [3]. The author proposes two tests. One compares models two by two using Bayes factors $P(D \mid \mathcal{T}_i)/P(D \mid \mathcal{T}_j)$ and suggests that if the Bayes factor is larger than 100, the evidence is in favour of $\mathcal{T}_i$. However, in real testing situations the evidence is often much less clear cut. In a beautiful example of Bayesian testing applied to the “out-of-Africa” hypothesis, Huelsenbeck and Imennov [44] show cases where the Bayes factor equal to 4.

Another test also proposed by Aris-Brosou [3] uses an average

$$\int_{\mathcal{T}, \Omega} p(D \mid \mathcal{T}, \theta) \frac{dP(\mathcal{T}, \theta)}{p(D \mid \mathcal{T}_i)}$$

for which there is not an exact statement of existence as yet, as integration over treespace is undefined. However by restricting himself to a finite number of trees to compare with, this average can be defined using counting measure. Of course the main advantage in the Bayesian approach is the possibility of integrating out all the nuisance parameters, either analytically or by MCMC simulation (see Chapter 3, this volume, for details). The software [47] provides a way of generating sets of trees under two differing models and thus some tests can use distances between the distributions of trees under competing hypotheses and the posterior distribution given the data.

4.2.5 Questions posed as functions of the tree parameter

In all statistical problems, questions are posed in terms of unknown parameters for which one wants to make valid inferences. In the current presentation, our parameter of choice is a semi-labelled binary tree. Sometimes the parameter itself appears in the definition of the null hypothesis,

H_0 : The true phylogenetic tree topology τ belongs to a set of trees $\mathcal{M}$.

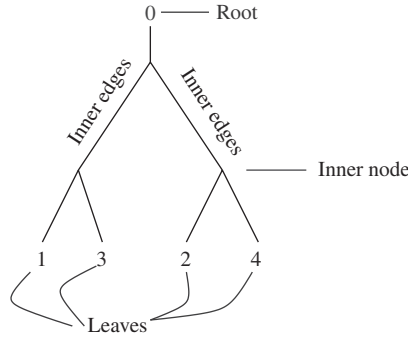


FIG. 4.1. The tree parameter is rooted with labelled leaves and inner branches.

For instance the set of trees containing a given clade, or a specific set of trees $\mathcal{M} = \{\tau_1, \tau_2, \dots, \tau_k\}$ as in reference [4].

The parameter space is not a classical Euclidean space, thus introducing the need for many non-standard techniques. The discrete parameter defined as the branching order of the binary rooted tree with n leaves, τ , can take on one of $(2n-3)!!$ values [70] (where $(2n-3)!! = (2n-3) \times (2n-5) \times (2n-7) \times \dots \times 3 \times 1$). $\mathcal{T}_n$ is the branching pattern with the $n-2$ inner branch lengths often considered as nuisance parameters $\theta_1, \theta_2, \dots, \theta_{n-2}$, left unspecified by H_0 (the pendant edges are sometimes fixed by a constraining normalization of tree so that all the leaves are contemporary). Even for simple hypotheses, the **power function** of the test varies with all the parameters, natural and nuisance. This is resolved by using the standard procedure of setting the nuisance parameters, for example, the edge lengths at their maximum likelihood estimates (MLEs).

We consider rooted trees as in Fig. 4.1 because in most phylogenetic studies, biologists are careful to provide outgroups that root the tree with high certainty, this brings down the complexity of the problem by a factor of n , which is well worth while in practical problems.

The first step is often to estimate the parameter τ by $\hat{\tau}$ computed from the data. In the case of parsimony estimation τ represents a branching order, without edge lengths, however, we can always suppose that in this case the edge lengths are the number of mutations between nodes, the general parameter we will be considering will have edge lengths.

In what follows we will consider our parameter space to be partitioned into regions, each region dedicated to one particular branching order $\hat{\tau}$, estimation can thus be pictured as projecting the data set from the data space into a point $\hat{\tau}$ in the parameter space.

The geometrical construction by Billera, Holmes and Vogtmann (denoted hereafter as BHV) [9] makes this picture more precise. The regions become cubes in dimension $(n-2)$ and the boundary regions are lower dimensional. The first thing to decide when making such a topological construction, is what is the definition of a neighbourhood? Our construction is based on a notion of

proximity defined by biologists as nearest neighbour interchange (NNI) moves [52, 78] (also called **Rotation Moves** [75] by combinatorialists), other notions of proximity are also meaningful, in the context of host–parasite comparisons [46] one should use other possible elementary *moves* between neighbouring trees.

This construction enables us to define distances between trees, for both the branching order and the edge enriched trees. With the existence of a distance we are able to define neighbourhoods as balls with a given radius. We will use this distance in much of what follows, but nothing about this distance is unique and many other authors have proposed distances between trees [66].

The boundaries between regions represent an area of uncertainty about the exact branching order, represented by the middle tree in Fig. 4.2. In biological terminology this is called an “unresolved” tree. Biologists call “polytomies” nodes of the tree with more than two branches. These appear as lower dimensional “cube-boundaries” between the regions.

For example, the boundary for trees with three leaves is just a point (Fig. 4.3), while the boundaries between two quadrants in treespace for $n = 4$ are segments (Fig. 4.4).

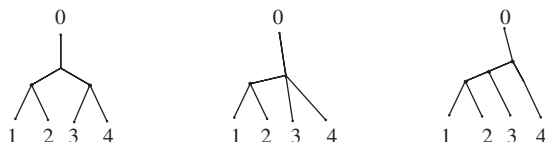


FIG. 4.2. Nearest neighbour interchange (NNI) move, an inner branch becomes zero, then another grows out.

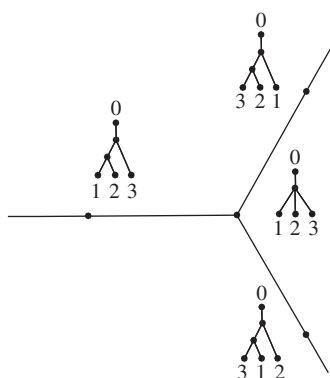


FIG. 4.3. The space of edge enriched trees with three leaves is the union of three half lines meeting at the star tree in the centre, if we limit ourselves to trees with bounded inner edges, the space is the union of three segments of length 1.

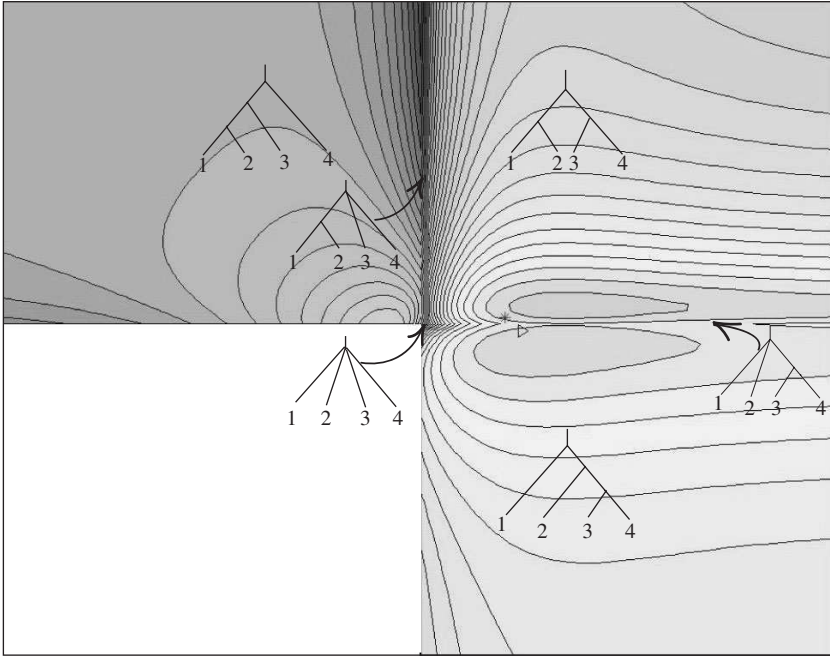


FIG. 4.4. A small part of the likelihood surface mapped onto three neighbouring quadrants of treespace, each quadrant represents one branching order among 15 possible ones for 4 leaves, the true tree that was used to simulate the data is represented as a star close to the horizontal boundary.

4.2.6 Topology of treespace

Many intuitive pictures of treespace have to be revised to incorporate some of its non-standard properties. Many authors describe the landscape of trees as a real line or plane [14], with the likelihood function as an alternative pattern of mountains and valleys, thus if the sea level rises, islands appear [56].

Figure 4.4 is a representation of the likelihood of a tree with four leaves over only 3 of the 15 possible quadrants for data that was generated according to a true tree with one edge very small compared to the other, we see how the phenomenon of “islands” can occur, we also see how hard it would be to make such a representation for trees with many leaves.

This lacks one essential specificity of treespace: it is not embeddable in such a Euclidean representation because it wraps around itself. BHV [9] describe this by defining the *link of the origin* in the following way: all 15 regions corresponding to the 15 possible trees for $n = 4$ share the same origin, we give coordinates to each region according to the edge lengths of their two inner branches, this make each region a square if the tree is constrained to have finite edge lengths. If we take the diagonal line segment $x + y = 1$ in each quadrant, we obtain a graph with an edge for each quadrant and a trivalent vertex for each boundary ray; this graph

is called the *link of the origin*. In the case of 4 leaves, we obtain a well-known graph called the Peterson graph, and in higher dimensions, extensions to what we could call Peterson simplices. One of the redeeming properties of treespace as we have described it is that if a group of trees share several edges we can ignore those dimensions and only look at the subspace composed of the trees without these common edges, thus decreasing the dimension of the relevant comparison space.

The wraparound has important consequences for the MCMC methods based on NNI moves, since a wraparound will ensure a speedup in convergence as compared to what would happen a Euclidean space.

The main property of treespace as proved in BHV [9] is that it is a CAT(0) space, succinctly this can be rephrased in the more intuitive fact that *triangles are thin in treespace*. Mathematical details may be found in BHV [9]: the most important consequences are being a CAT(0) space ensures the existence of convex hulls and distances in treespace [32].

To picture how distances are computed in treespace, Fig. 4.5 shows paths between A and B and between C and D, the latter passes through the star tree and is a cone path that can always be constructed by making all edges zero and then growing the new edges, the distance between two points in tree space is

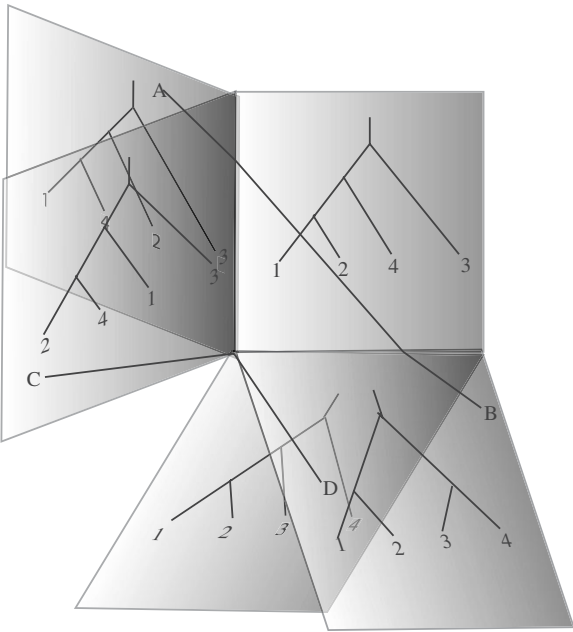


FIG. 4.5. Five of the fifteen possible quadrants corresponding to trees with four leaves and two geodesic paths in treespace, in fact each quadrant contains the star tree and has two other neighbouring quadrants.

computed as the shortest path between the points that stays in treespace, thus the geodesic path between A and B does not pass through the star tree. This computation can be intractable, but in real cases, the problem splits down and the distance can be computed in reasonable time [41].

4.2.7 The data

The data from which the tree is often estimated are usually matrices of aligned characters for a set of n species.

The data can be:

- Binary, often coming from morphological characters

```
Lemur_cat  00000000000001010100000
Tarsius_s  100000100000000010000000
Saimiri_s  100000100000001010000000
Macaca_sy  000000000000000100000000
Macaca_fa  100000100000000010000000
```

- Aligned:

```
6 40
Lemur_cat  AAGCTTCATA GGAGCAACCA TTCTAATAAT CGCACATGGC
Tarsius_s  AAGTTTCATT GGAGCCACCA CTCTTATAAT TGCCCATGGC
Saimiri_s  AAGCTTCACC GGC GCAATGA TCCTAATAAT CGCTCACGGG
Macaca_sy  AAGCTTCTCC GGTGCAACTA TCCTTATAGT TGCCCATGGA
Macaca_fa  AAGCTTCTCC GGC GCAACCA CCCTTATAAT CGCCACGGG
Macaca_mu  AAGCTTTTCT GGC GCAACCA TCCTCATGAT TGCTCACGGA
```

- Gene order (see the Chapters 9 to 13, this volume, for some examples).

An important property of the data is that they come with their own metrics. There is a meaningful notion of proximity for two data sets, whether the data are permutations, Amino Acid or DNA sequences. One of the points we want to emphasize in this chapter is that we often have less data than actually needed given the multiplicity of choices we have to make when making decisions involving trees. Most statistical tests in use suppose that the columns of the data (characters) are independent. In fact we know that this is not true, and in highly conserved regions there are strong dependencies between the characters. There is thus much less information in the data than meets the eye. The data may contain 1000 characters, but be equivalent only to 50 independent ones.

4.2.8 Statistical paradigms

The algorithms followed in the classical *frequentist* context are:

- Estimate the parameter (either in a parametric (ML) way, semiparametric (Distance-based methods), or non-parametric way (Parsimony)).
- Find the sampling distribution of the estimator under the null.

On the other hand Bayesians follow the following procedure

- Specify a Prior Distribution for the parameter.
- Update the Prior using the Data.
- Compute the Posterior Distribution.

Both use the result of the last steps of their procedures to implement the Hypothesis tests. Frequentists use the estimate and the sampling distribution of the tree parameter to do tests, whether parametric or non-parametric. This is the distribution of the estimates $\hat{\tau}$ when the data are drawn at random from their parent population.

In the case of complex parameters such as trees, no analytical results exist about these sampling distributions, so that the Bootstrap [20, 23] is often employed to provide reasonable approximations to such unknown sampling distributions.

Bayesians use the posterior distribution to compute estimates such as the mode of the posterior (MAP) estimator or the expected value of the posterior and to compute Bayesian credibility regions with given level. More important is the fact that usually Bayesians assign a prior probability to the null hypothesis, such as $P(H_0) = 1/2$ and using this prior and the data can compute $P(H_0 \mid \text{Data})$. This computation is impossible in the frequentist context, only computations based on the sampling distribution are allowed.

4.2.9 Distributions on treespace

As we see, in both paradigms the key element is the construction of either the sampling distribution or the posterior distribution, both distributions in treespace. We thus need to understand distributions on treespace. If we had a probability density f over treespace, we could write statements such as equation (3) in Aris-Brosou [4] that integrates the likelihood $\ell(\theta, \mathcal{T} \mid D)$ over a subset of trees $\mathfrak{T}$:

$$h_{0,f} = \int_{\mathfrak{T}} \ell(\theta, \mathcal{T} \mid D) df(\mathcal{T}).$$

This allows the replacement of a composite null hypotheses of equality of a set of trees by an integrated simple hypotheses as suggested by Lehmann's [50] adaptation of the Bayesian procedure. The integral is undefined unless we have such a probability distribution on treespace.

The basic example of a distribution on treespace that we would like to summarize is the sampling distribution, that we will now define in more detail. Suppose the data comes from a distribution $\mathcal{F}$, and that we are given many such data sets, as shown in Fig. 4.6. Estimation of the tree from the data provides a projection onto treespace for each of the data sets, thus we obtain many estimates $\hat{\tau}_k$.

We need to know what this true “theoretical” sampling distribution is in order to build confidence statements about the true parameter.

The true **sampling distribution** is usually inaccessible, as we are not given many sets of data from the distribution $\mathcal{F}$ with which to work. Figure 4.7 shows

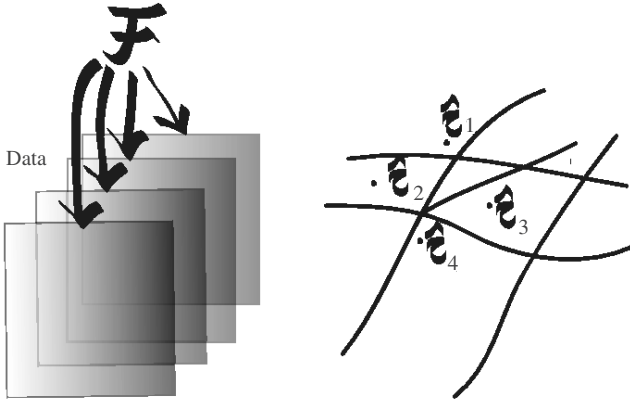


FIG. 4.6. The true sampling distribution lies in a non-standard parameter space.

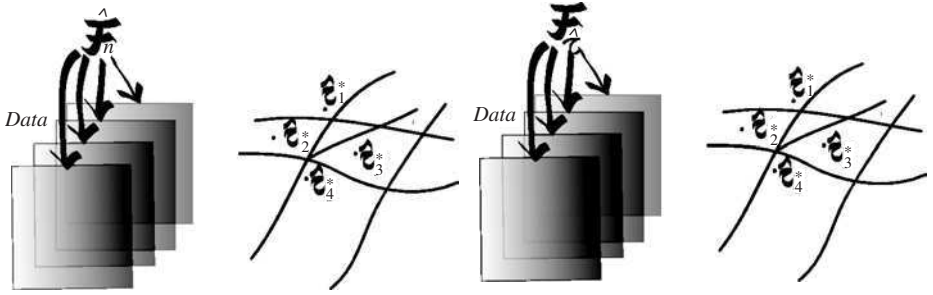


FIG. 4.7. Bootstrap sampling distributions: non-parametric (left), parametric (right).

how the non-parametric bootstrap replaces $\mathcal{F}$ with the **empirical distribution** $\hat{\mathcal{F}}_n$, new data sets are “plausible” perturbations of the original, drawn from the empirical cumulative distribution instead of the unknown $\mathcal{F}$. Data are created by drawing repeatedly from the empirical distribution given from the original data, for each new data set a new tree $\hat{\tau}_k^*$ is estimated, and thus there is a simulated sampling distribution computed by using the multinomial reweighting of the original data [23]. Note that even if we generate a large number of resamples, the bootstrap resampling distribution cannot overcome the fact that it is only an approximation built from one data set. It is actually possible to give the complete bootstrap sampling distribution without using Monte Carlo at all [17], nonetheless the bootstrap remains an approximation as it replaces the unknown distribution $\mathcal{F}$ by the empirical distribution constructed from one sample.

If the data are known to come from a parametric distribution with an unknown parameter such as the edge-weighted tree $\mathcal{T}$, the parametric distribution produces simulated data set by supposing the estimate from the original

data $\hat{T}$ is the true estimate and generating the data from that model as indicated by the right side of Fig. 4.7. This means generating many data sets by simulating sequences from the estimated tree following the Markovian model of evolution.

However, given the large number of possible trees and the small amount of information, both these methods may have problems finding the sampling distribution if it is not simplified. If we consider the simplest possible distribution on trees, we will be using the uniform distribution, however, there are an exponentially growing number of trees. This leads to paradoxes such as the blade of grass type argument [65]: if we consider the probability of obtaining a tree τ_0 we will have conclusions such as $P(\hat{\tau} = \tau_0) = 1/(2n - 3)!!$ this becomes exponentially small very quickly, making for paradoxical statements.¹

Overcoming the discreteness and size of the parameter space. If one wanted to use a sample of size 100 to infer the most likely of 10,000 possible choices, one would need to borrow strength from some underlying structure. Thinking of the choices as boxes that can be ordered in a line with proximity of the boxes being meaningful shows that we can borrow information from “neighbouring” boxes. We will see as we go along that the notion of neighbouring trees is essential to improving our statistical procedures.

We can imagine creating useful features for summarizing the distribution or treespace (either Bayesian posterior or Bootstrapped sampling distributions).

The most common summary in use is the presence or absence of a clade. If we only enumerate those that appeared in the original tree, this would be a vector of length $n - 2$. If we just wanted to give an inventory of all the clades in the data, the number of possible clades is the number of bipartitions where both sets have at least 2 leaves. The complete feature vector in that case would be a vector of length $2^{n-1} - n - 1$. This multidimensional approach can be followed through by doing an analysis of the data as if it were a contingency table and we could keep statements of the kind “clade (1,2) is always present when clade (4,5) is present” thus improving on the basic confidence values currently in use.

Other features might be incorporated into an exponential distribution such as Mallows’ model [57] that was originally implemented for ranked data

$$P(\tau_i) = K e^{-\lambda d(\tau_i, \tau_0)},$$

as described in reference [39]. This distribution uses a central tree τ_0 and a distance d in treespace. Mallows model would work well if we had strong belief in a very symmetrical distribution around a central tree. In reality this does not seem to be the case, so a more intricate mixture model would be required. One could imagine having the mixture of two underlying trees which might have biological meaning. Other distributions of interest are extensions of the Yule process (studied by Aldous [1]) or exponential families incorporating information

¹After choosing a blade of grass in a field, one cannot ask, what were the chances of choosing this blade? With probability one, I was going to choose one [19].

about the estimation method used. The reason for doing this is that Gascuel [29] has shown the influence of the estimation method chosen (parsimony, maximum likelihood, or distance based) on the shape of the estimated tree. We could build different exponential families running through certain important parameters such as “balance”, or tree width as studied by evolutionary biologists who use the concept of tree shape (see [36, 60, 62]).

Some methods for comparing trees measure some property of the data with regards to the tree, such as the minimum number of mutations along the tree to produce the given data (the parsimony score) or the probability of the data given a fixed evolutionary model with parameters $\alpha_1, \alpha_2, \dots, \alpha_k$ and a fixed tree

$$P(D \mid \mathcal{T}_n, \alpha) = L(\mathcal{T}_n).$$

This, considered as a function of $\mathcal{T}_n$ defines the likelihood of $\mathcal{T}_n$. Sometimes this is replaced by the likelihood of a branching pattern τ maximized and the branch lengths $\theta_1, \dots, \theta_{2n-2}$ are chosen to maximize the likelihood.

The lack of a natural ordering in the parameter space encourages the use of simpler statistical parameters. The presence/absence of a given clade, a confidence level, a distance between trees are all acceptable simplifying parameters as we will see. This multiplicity of riches is something that also occurs in other areas of statistics, for instance when choosing between a multiplicity of graphical models. In that domain, researchers use the notion of “features” characterizing shared aspects of subsets of models.

For one particular observed value, say 1.8921 of a real-valued statistic it is meaningless to ask what would the probability $P(Y = 1.8921)$ be equal to, but we can ask the probability of Y belonging to a neighbourhood around the value 1.8921. The definition of features enables the definition of meaningful neighbourhoods of treespace if the features can be defined by a continuous function from treespace to feature space. This has another advantage, as explained in BHV [9] the parameter space is not embedded in either the real line $\mathbb{R}$ nor an euclidean space such as $\mathbb{R}^d$, on the other hand we can choose the features to be real valued.

Returning to testing, one of the problems facing a biologist is that natural comparisons are not nested within each other. Efron [21] carries out a geometrical analysis of the problem of comparing two non-nested simple linear models, and the analysis is already quite intricate. When comparing a small number of models, the number of parameters grows, but the degrees of freedom remain manageable. Yang *et al.* [80] already noticed that comparing tree parameters is akin to model comparison. However, in this case the number of available models (the trees) increases exponentially with the number of species and the data will never be sufficient to choose between them. Classical model comparison methods such as the AIC and BIC cannot be applied in their vanilla versions here. We have exponentially many trees to choose from, and in the absence of a “continuum” and an underlying statistic providing a natural ordering of the models, we will be unable to use even a large data set to compare the multiplicity of possibilities. (Think of trying to choose between 1 million pictures when only a thousand samples from them exist.)

There is, however, a solution. If we think of each model as a box, each with an unknown probability, if the sampling distribution throws K balls into the boxes and K is much smaller than the number of boxes, then we cannot conclude. However, if we have a notion of neighbourhood boxes, we can borrow strength from the neighbouring boxes.

Remember in this image, that if the balls correspond to the trees obtained by a Bootstrap resample, we cannot increase indefinitely the number of balls and hope to fill all possible boxes. The non-parametric Bootstrap cannot provide more information than is available in the sample.

The classical statistical location summary in the case of trees would be the **mean** and the **median**, and thus we could use the Bootstrap to estimate bias as in reference [8]. The notion of **mean** (centre of the distribution as defined using an integral of the underlying probability distribution) supposes that we already have a probability distribution defined on treespace and know how to integrate. These are currently open problems. Associated to this view of a “centre” of a distribution of trees, we can ask the question: *What distribution is the “majority rule consensus” a centre of?* This would enable more meaningful statistical inferences using the consensii that biologists so often favour. The median, another useful location statistic, can be defined by either of the various multivariate extensions of the univariate median to the multivariate median (in particular Tukey’s multivariate median [77]), which we revisit in the multivariate section below.

Usually the best results in hypothesis testing are obtained by using a statistic that is centred and rescaled like the t -statistic, by dividing it by its sampling variance, here this cannot be defined. By analogy we can suppose that it is beneficial to divide by a similar statistic, for instance $\{E_{P_n} d^2(\hat{\tau}, \tau)\}^{-1/2}$ (where d is a distance defined on tree space and E_{P_n} is the expectation with regards to an underlying distribution P_n) is an ersatz-standard deviation.

4.3 Different types of tests involving phylogenies

There are two main types of statistical testing problems involving phylogenies. First, tests involving the tree parameter itself of the form $P(\tau \in \mathcal{M})$ the second type are tests that treat the phylogenetic tree as a nuisance parameter and will be treated in the second paragraph.

4.3.1 Testing τ_1 versus τ_2

The Neyman Pearson theorem ensures that the case of a parametric evolutionary Markovian model the likelihood ratio test as introduced as the Kishino–Hasegawa [34] test will be the most powerful for comparing two prespecified trees. A very clear discussion of the case where one combinatorial tree τ_1 is compared to an alternative τ_2 is given by Goldman *et al.* [30]. In particular the authors explain how important the assumption that the trees were specified prior to seeing the data. The problem of both estimating and testing a tree with the same data is a more complicated problem and needs adjustments for multiple comparisons as

carried out by Shimodaira and Hasegawa [73]. It is definitely the case that the use of the same data to estimate and test a tree is an incorrect procedure.

The use of the non-parametric bootstrap when comparing trees where a satisfactory evolutionary model is known (and may have been used in the estimation of the trees τ_1 and τ_2 to be compared) is not a coherent strategy as the most powerful procedure is to keep the parametric model and use this to generate the resampled data using the parametric bootstrap as implemented by `seqgen` [67] for instance.

4.3.2 *Conditional tests*

Another class of hypothesis tests are those included in what is commonly known as the *Comparative Method* [33, 59]. In this setting, the phylogenetic tree is a nuisance parameter and the interest is in the distribution of variables conditioned on the tree being given. For instance if we wanted to study a morphological trait but subtract the variability that can be explained by the phylogenetic relationship between the species, we may (following Felsenstein [26]), condition on the tree and make a Brownian motion model of the variation of a variable on the tree. More recently, [42] and [55] propose another parametric model, akin to an ordinary linear mixed model. The variability is decomposed into heritable and residual parts, quantifying the residuals conditions out of the phylogenetic information.

Some recent work enables incorporation of incomplete phylogenetic information [43] providing a way of conducting such tests in a parametric setup where the phylogeny is not known. It would also be interesting to have a Bayesian equivalent of this procedure that could enable the incorporation of some information about the tree we want to condition on, without knowing it exactly.

4.3.3 *Modern Bayesian hypothesis testing*

The Bayesian outlook in hypothesis testing is as yet underdeveloped in the phylogenetic literature but the availability of posterior distributions through Monte Carlo Markov chain (MCMC) algorithms makes this type of testing possible in a rigid parametric context [53, 61, 81]. Useful software have been made available [47, 49]. Biologists wishing to use these methods have to take into account the main problem with MCMC (see the review in Huelsenbeck *et al.* [45]):

1. We don't know how long the algorithms have to run to reach stationarity, the only precise theorems [2, 18, 71] have studied very simple symmetric methods, without any Metropolis weighting.
2. Current procedures are based on a restrictive Markovian model of evolution; no study of the robustness of these methods to departure from the Markovian assumptions is available.

One large open question in this area is how to develop non-parametric or semi-parametric priors for Bayesian computations in cases where the Markovian model is not adequate. One possibility is to use both the information on the tree shape

that is provided both by the estimation method and the phylogenetic noise level [35, 37].

4.3.4 *Bootstrap tests*

I have explained in detail elsewhere [39] some of the caveats to the interpretation of bootstrap support estimates as actual confidence values in the sense of hypothesis testing. If we wanted to test only one clade in the tree, we could consider the existence of this clade as a Bernoulli 0/1 variable and try to estimate it through the plug in principle by using the Bootstrap [25], however, if the model used for estimating the tree is the Markovian Model, we should use the parametric bootstrap, generating new data through simulation from the estimated tree [67]. Using the multinomial non-parametric bootstrap would be incoherent. This procedure allows the construction of one confidence value that can be interpreted on its own. However, two common extensions to this are invalid. If we want to have confidence values on all clades at once, we will be entering the realm of multiple testing: we are using the same data to make confidence statements about different aspects of the data, and statistical theory [51] is very clear about the inaccuracies involved in reporting all the numbers at once on the tree.

We cannot reconstruct the complete bootstrap resampling distribution from the numbers on the clades, this is because these numbers taken together do not form a sufficient statistic for the distribution on treespace (this is discussed in detail in reference [40]).

Finally, we cannot compare bootstrap confidence values from one tree to another. This is due to the fact that the number of alternative trees in the neighbourhood of a given tree with a pre-specified size is not always the same. Zharkikh and Li [82] already underlined the importance of taking into account that a given tree may have k alternatives and through simulation experiments, asked the relevant question: *How many neighbours for a given tree?* In fact, through combinatorics developed in BHV [9]'s continuum of trees (see Section 4.1), we know the number of neighbours of each tree in a precise sense. In this geometric construction each tree topology τ_n with n leaves is represented by a cube of dimension $n - 2$ each dimension representing the inner edge lengths which are supposed to be bounded by one. Figure 4.8 shows the neighbourhoods of two such trees with four leaves. Each quadrant will have as its origin the star tree, which is not a true binary tree since all its inner edges have lengths zero. The point on the left represents a tree with both inner edges close to 1, and only has as neighbours, trees with the same branching order. The point on the right has one of its edges much closer to zero, so has two other different branching orders (combinatorial trees) in its neighbourhood.

For a tree with only two inner edges, there is the only one way of having two edges small: to be close to the origin-star tree and thus the tree is in a neighbourhood of 15 others. This same notion of neighbourhood containing 15 different branching orders applies to all trees on as many leaves as necessary but who have

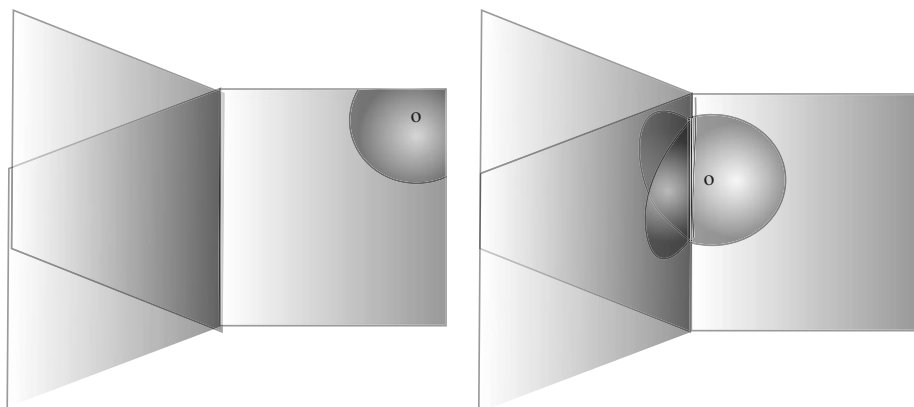


FIG. 4.8. A one tree neighbourhood and a three tree neighbourhood.

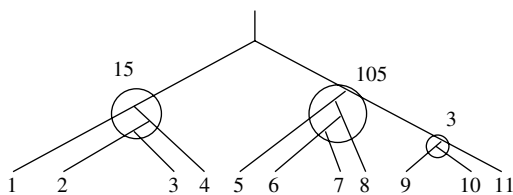


FIG. 4.9. Finding all the trees in a neighbourhood of radius r , each circle shows a set of contiguous edges smaller than r , from left to right we see subtrees with 2, 3, and 1 inner edge respectively.

two contiguous “small edges” and all the other inner edges significantly bigger than 0.

This picture of treespace frees us from having to use simulations to find out how many different trees are in a neighbourhood of a given radius r around a given tree. All we have to do is check the sets of contiguous small edges in the tree (say, smaller than r), for example, if there is only one set of size k , then the neighbourhood will contain $(2k-3)!!$ different branching orders (combinatorial trees).

The circles represented in Fig. 4.9 show how all edges smaller than this radius r define the contiguous edge sets. On the left there are two small contiguous edges, in the middle there are three small contiguous edges and on the right there is only one, underneath each disjoint contiguous set, we have counted the number of trees in the neighbourhood of this contiguous set. Here we have three contiguous components, thus a product of three factors for the overall number of neighbours.

In this case the number of trees within a radius r will be the product of the tree numbers $15 * 105 * 3 = 4725$. In general: If there are m sets of contiguous

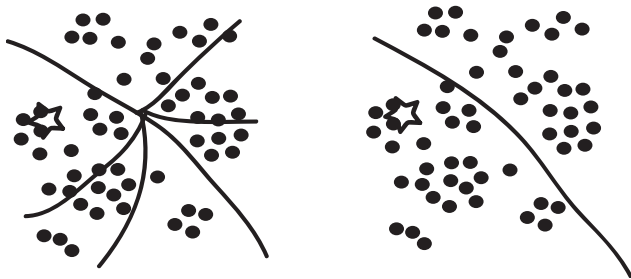


FIG. 4.10. Bootstrap sampling distributions with different neighbourhoods.

edges of sizes $(n_1, n_2, \dots, n_m)$ there will be

$$(2n_1 - 3)!! \times (2n_2 - 3)!! \times (2n_3 - 3)!! \cdots \times (2n_m - 3)!!$$

trees in the neighbourhood. A tree near the star tree at the origin will have an exponential number of neighbours. This explosion of the volume of a neighbourhood at the origin provides for interesting mathematical problems that have to be solved before any considerations about consistency (when the number of leaves increases) can be addressed.

Figure 4.10 aims to illustrate the sense in which just simulating points from a given tree (star) and counting the number of simulated points in the region of interest may not directly inform one on the distance between the true tree (star) and the boundary. The boundary represents the cutoff from one branching order to another, thus separating treespace into regions, each region represents a different branching order. If the distribution were uniform in this region, we would be actually trying to estimate the distance to the boundary by counting the stars in the same region, it is clear that the two different configurations do not give the same answer, whereas the distances are the same. In general we are trying to estimate a weighted distance to the boundary where the weights are provided by the local density.

These differing number of neighbours for different trees show that the bootstrap values cannot be compared from one tree to another. Again, we encounter the problem that “ p -values” depend on the context and cannot be compared across studies. This was implicitly understood by Hendy and Penny in their NN Bootstrap procedure (personal communication).

In any classical statistical setup, p -values suffer from a lack of “qualifying weights”, in the sense that this one number summary, although on a common scale does not come with any information on the actual amount of information that was used to obtain it. Of course this is a common criticism of p -values by Bayesians (for intricate discussions of this important point see [5, 7, 72], for a textbook introduction see [48]). This has to be taken into account here, as the amount of information available in the data is actually insufficient to conclude in a refined way [63]. For once, there are theorems providing bounds to the amount of precision (the size of the tree) that can be inferred from a given data set

(see Chapter 14, this volume). Thus, we should be careful not to provide the equivalent of 15 significant digits for a mean computed with 10 numbers, spread around 100 with a standard deviation of 10 (in this case the standard error would be around 3, so even one significant digit is hubris).

4.4 Non-parametric multivariate hypothesis testing

There is less literature on testing in the non-parametric context; Sitnikova *et al.* [74] who provide interior branch tests and some authors who have permutation test for Bremer [10] support for parsimony trees. In order to be able to design non-parametric tests, we have to leave the realm of the reliance on a molecular clock or even a Markovian model for evolution and explore non-parametric or semiparametric distributions on treespace. To do this we will use the analogies provided from non-parametric multivariate statistics on Euclidean spaces.

4.4.1 Multivariate confidence regions

There is an inherent duality in statistics between hypothesis tests and confidence regions for the relevant parameter

$$P(\mathcal{M}_\alpha \ni \tau) = 1 - \alpha.$$

The complement to $\mathcal{M}_\alpha$ provides the rejection region of the test. It is important to note that in this probabilistic statement, the frequentist interpretation is that the region $\mathcal{M}_\alpha$ is random, built from the random variables observed as data and that the parameter has a fixed unknown value τ . For a fixed region B and a parameter τ , the statement $P(\tau \in B)$ is meaningless for a frequentist. Bayesians have a natural way of building such regions, often called **credibility regions** as they have randomness built into the parameters through the posterior distribution, so finding the region that covers $1 - \alpha$ of the posterior probability is quite amenable once the posterior has been either calculated or simulated. However, all current methods for such calculations are based on the parametric Markovian evolutionary model. If we are unsure of the validity of the Markovian model (or absence of a molecular clock, for an example with an unbeatable title see [79]), we can use symmetry arguments leading to semiparametric or non-parametric approaches.

We have found that there are several important questions to address when studying the properties of tests based on confidence regions [22]. One concerns the curvature of the boundary surrounding a region of treespace; the other the number of different regions in contact with a particular frontier. The latter is answered by the mathematical construction of BHV [9]. However, although the geometric analysis provided in BHV [9] does show that the natural geodesic distances and the edges of convex hulls in treespace are negatively curved, exact bounds on the amount of curvature are not yet available.

In order to provide both classical non-parametric and Bayesian non-parametric confidence regions, we will use Tukey's [77] approach involving the

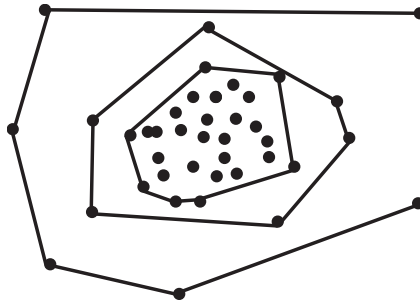


FIG. 4.11. Successive convex hulls built on a scatterplot.

construction of regions based on convex hulls. He suggested peeling convex hulls to construct successive “deeper” confidence regions as illustrated in Fig. 4.11. Liu and Singh [54] have developed this for ordering circular data for instance. Here we can use this as a non-parametric method for estimating the “centre” of a distribution in treespace, as well finding central regions holding say 90% of the points.

Example 1 Confidence regions constructed by bootstrapping.

Instead of summarizing the bootstrap sampling distribution by just presence or absence of clades we can explore whether 90% of bootstrap trees are in a specific region in treespace. We can also ask whether the sampling distribution is centred around the star tree, which would indicate that the data does not have strong treelike characteristics.

Such a procedure would be as follows:

- Estimate the tree from the original data call this, t_0 .
- Generate K bootstrap trees.
- Compute the 90% convex envelope by peeling the successive hulls, until we have a convex envelope containing 90% of the bootstrap trees call this $C_{0.10}$.
- Look at whether $C_{0.10}$ contains the star tree.
- If it does not, the data are in fact treelike. □

Example 2 Are two data sets congruent, suggesting that they come from the same evolutionary process?

This is an important question often asked before combining datasets [11].

This can be seen as a multidimensional two sample test problem. We want to see if the two bootstrap sampling distributions overlap significantly (A and B). Here we use an extension of the Friedman–Rafsky (FR) [28] test. This method is inspired by the Wald–Wolfowitz test, and solves the problem that there is no natural multidimensional “ordering.” First the bootstrap trees from bootstrapping both data sets are organized into a minimal spanning tree following the classical

Minimal Spanning Tree Algorithm (a greedy algorithm is easy to implement).

- Pool the two bootstrap samples of points in treespace together.
- Compute the distances between all the trees, as defined in BHV [9].
- Make a minimal spanning ignoring which data set they came from (labels A and B).
- Colour the points according to the data sets they came from.
- Count the number of “pure” edges, that is the number of edges of the minimal spanning tree whose vertices come from the same sample, call this the test statistic S_0 , if S_0 is very large, we will reject the null hypothesis that the two data sets come from the same process.
(An equivalent statistic is provided by taking out all the edges that have mixed colours and counting how many “separate” trees remain.)
- Compute the permutation distribution of S^* by reassigning the labels to the points at random and recomputing the test statistic, say B times.
- Compute the p -value as the ratio

$$\frac{\#\{S_k^* > S_0\}}{B}.$$

This extends to case of more than two data sets by just looking at the distribution of the pure edges as the test statistic. $\square$

Example 3 Using distances between trees to compute the bootstrap sampling distribution.

By computing the bootstrap distribution we can give an approximation to the distribution of $(d(\hat{T}, \mathcal{T}))$ by $d(\hat{T}^*, \hat{T})$. This is a statement by analogy to many other theorems about the bootstrap, nothing has been proved in this context. However, this analogy is very useful as it also suggests that the better test statistic in this case is: $d(\hat{T}^*, \hat{T})\{\text{var}(d(\hat{T}^*, \hat{T}))\}^{-1/2}$ which should have a near pivotal distribution that provides a good approximation to the unknown distribution of $d(\hat{T}, \mathcal{T})/\{\text{var}(d(\hat{T}, \mathcal{T}))\}^{-1/2}$ equivalent of a “studentized” statistic [23]. As can be seen in Fig. 4.12, this distribution is not necessarily Normal, or even symmetric.

Such a sampling distribution can be used to see if a given tree $\mathcal{T}_0$ could be the tree parameter responsible for this data. If the test statistic

$$\frac{d(\hat{T}, \hat{\mathcal{T}}_0)}{\sqrt{\text{var}(d(\hat{T}^*, \hat{T}))}}$$

is within the 95th percentile confidence interval around $\hat{T}$ we cannot reject that it could be the true $\mathcal{T}$ parameter for this data. $\square$

Example 4 Embedding the data into $\mathbb{R}^d$.

Finally a whole other class of multivariate tests are available through an approximate embedding of treespace into $\mathbb{R}^d$. Assume a finite set of trees: it could be

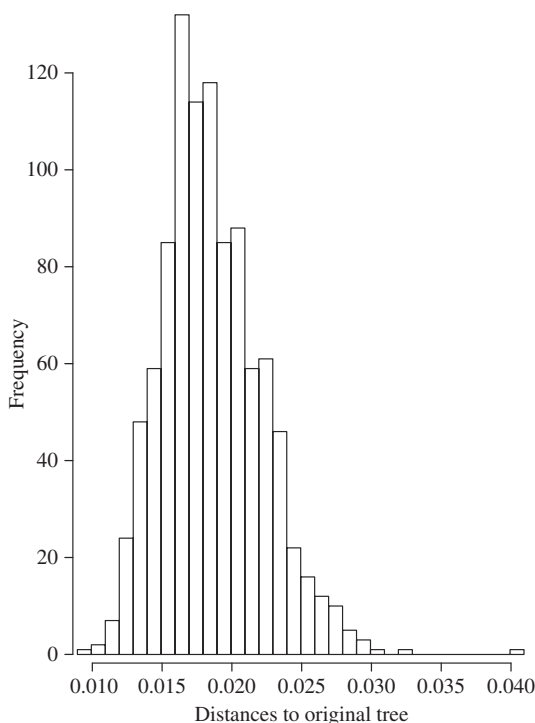


FIG. 4.12. Bootstrap sampling distribution of the distances between the original tree and the bootstrapped trees in the analysis of the *Plasmodium F.* data analysed by Efron *et al.* [22], the distances were computed according to BHV [9].

a set of trees from bootstrap resamples, it could be a pick from a Bayesian posterior distribution, or sets of trees from different types of data on the same species. Consider the matrix of distances between trees and use a multidimensional scaling algorithm (either metric or non-metric) to find the best approximate embedding of the trees in $\mathbb{R}^d$ in the sense of distance reconstruction. Then we can use all the usual multivariate statistical techniques to analyse the relationships between the trees. The likely candidates are

- discriminant analysis that enables finding combinations of the coordinates that reconstruct prior groupings of the trees (trees made from different data sources, molecular, behavioural, phenotypic for instance)
- principal components that provide a few principal directions of variation
- clustering that would point out if the trees can be seen as a mixture of a few tightly clustered groups, thus pointing to a multiplicity in the underlying evolutionary structure, in this case a mixture of trees would be appropriate (see Chapter 7, this volume). $\square$

4.5 Conclusions: there are many open problems

Much work is yet to be done to clarify the meaning of the procedures and tests already in practice, as well as to provide sensible non-parametric extensions to the testing procedures already available.

Here are some interesting open problems:

- Find a test for measuring how close the data are to being treelike, without postulation of a parametric model, some progress on this has been made by comparing the distances on the data to the closest distance fulfilling the four point condition (see Chapter 7, this volume).
- Find a test for finding out whether the data are a mixture of two trees? This can be done with networks as in Chapter 7, this volume, or it can be done by looking at the posterior distribution (see Yang, Chapter 3, this volume) and finding if there is a evidence of bimodality.
- Find satisfactory probability distributions on treespace that enable simple definitions of non-parametric sampling and Bayesian posterior distributions.
- Find the optimal ways of aggregating trees as either expectations for various measures or modes of these distributions.
- Find a notion of differential in treespace to study the influence functions necessary for robustness calculations.
- Quantify how the departure from independence in most biological data influences the validity of using Bootstrap procedures that assume independence.
- Quantify the amount of information in a given data set and find the equivalent number of degrees of freedom needed to fit a tree under constraints.
- Generalize the decomposition into phylogenetic information and non-heritable residuals to a non-parametric setting.

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MIXTURE MODELS IN PHYLOGENETIC INFERENCE

Mark Pagel and Andrew Meade

Conventional models of gene sequence evolution for use in phylogenetic inference presume that sites evolve according to a common underlying model or allow the rates of evolution to vary across sites. In this chapter, we discuss how a general class of approaches known as “mixture models” can be used to accommodate heterogeneity across sites in the patterns of gene sequence evolution. Mixture models fit more than one model of evolution to the data, but do not require prior knowledge of the patterns of evolution across sites. It can be shown that partitioning of gene-sequence data such that different models are applied to different sites is a special case of a more general mixture model, as is the popular gamma rate-heterogeneity model. We apply a mixture model based upon unconstrained general time reversible rate matrices to a 16.4 kb alignment of 22 different genes, to infer the phylogeny of the mammals. The trees we derive broadly agree with a previous study of these data that used a single rate matrix in conjunction with gamma rate heterogeneity. However, the mixture model substantially improves the likelihood, suggests a different placement for some mammalian orders, and repositions the hyrax as the nearest neighbour to elephant as suspected from morphological investigations. The tree is also significantly longer than the tree derived from simpler models. Taken together, these results suggest that mixture models can detect heterogeneity across sites in the patterns and rates of evolution and improve phylogenetic inference.

5.1 Introduction: models of gene-sequence evolution

The conventional likelihood-based approach to inferring phylogenetic trees from aligned gene-sequence or other data is to apply a single substitutional model to all sites. The model is defined by a rate matrix, $\mathbf{Q}$, that specifies the instantaneous rates of change among the possible character states. If the data are gene-sequences, $\mathbf{Q}$ is the familiar 4×4 matrix of possible transitions among the nucleotides. Swofford *et al.* [30] provide a thorough introduction to models of gene-sequence evolution and Bryant, Galtier, and Poursat (Chapter 2, this volume) and Yang (Chapter 3, this volume) discuss them in the context of phylogenetic inference.

The homogeneous-model may adequately represent pseudo-genes, repetitive sequences, or other sequence data whose evolution is governed largely by neutral forces. But there are often good reasons to believe *a priori* that sites may differ in either their rates of evolution or in the pattern of substitutional changes. When the data are nucleotides from a protein coding region, natural selection may constrain variability at some sites more than others (so-called purifying selection), or may positively select some sites, with the result being that sites will, minimally, exhibit different rates of evolution. To accommodate heterogeneity across sites in the rates of evolution, Yang [34] introduced the gamma rate-heterogeneity model. This model presumes that variation in the rates of evolution can be modelled by a gamma probability distribution, and has proved highly successful in improving the characterization of gene-sequence data.

Rate-heterogeneity models are less obviously applicable to cases in which sites do not just vary in their overall rates of evolution, but exhibit distinct patterns of substitution. Ribosomal RNA folds into well-known secondary structures in which stems frequently adopt canonical Watson–Crick base pairing. This gives the expectation that the frequency of transitions at paired stem sites will greatly exceed those of transversions [7, 8, 28]. However, loop regions are not so constrained and no specific prediction is made about their evolution. Because of these varying patterns of evolution across sites, special substitution models have been proposed to characterize ribosomal data [28, 29].

Codon-based substitutional models account for heterogeneities in codon evolution that appear to be independent of the underlying rates of nucleotide substitution [5, 20]. Rates of change of different codons are modelled as arising from changes at the DNA level but also as a function of various chemical properties of the amino acids. Concatenated sequence alignments are another probable source of heterogeneity across sites in the pattern of substitutions. Murphy *et al.* [19] used an alignment of 22 genes comprising 16.4 kb of DNA to infer the mammalian phylogeny. Even an alignment of this size may soon seem routine. The growth of what might be called genomic-phylogenetics, in which large portions of genomes are aligned across species, is creating alignments of unprecedented size. Rokas *et al.* [27], for example, use genomic-phylogenetics to find the phylogeny of yeast, using 106 genes comprising 127,026 sites.

5.2 Mixture models

A common way to accommodate heterogeneity in the pattern of evolution across sites is to partition the data such that a different substitutional model $\mathbf{Q}$ is assigned to different sites; later the information from the different models being combined into a single overall likelihood. This can be helpful when there is a sound prior reason to believe that the partitions follow different evolutionary models, or even necessary if qualitatively different characters, such as gene sequences and morphological traits, are combined in one analysis. Frequently, however, decisions on partitioning the data may be based solely on having fitted different models to different classes of site, and it may often be true

that there is important variability within these classes. Elsewhere, for example, we have shown how partitioning by gene, by codon position, or by the stems and loops in ribosomal data, misses significant evolutionary variation within these categories [24].

A possibly more realistic accounting of the knowledge an investigator brings to a typical data set is to entertain the possibility that more than one model can apply to the same site in the gene or alignment. The likelihood approach is then to sum the individual likelihoods of the various models at each site, weighting the models by the probability that they apply to that site. The probability that a given model applies to a site might be obtained from prior information or the weights can be estimated from the data. Summing over models may be preferred when there is not a clear case for partitioning the data, may allow for unforeseen patterns of evolution to emerge, and has the attractive feature of using all of the data to estimate parameters.

Gelman *et al.* [3] use the term “mixture models” to describe the practice of calculating likelihoods by summing over a range of statistical models for a given data point. Mixture models have received some attention in phylogenetic studies. Koshi and Goldstein [11] employ a mixture model to characterize amino acid sequences, identifying potentially important chemical and structural dimensions of amino acids, and summing at each site over models that measure them. Yang *et al.* [35] used a mixture model to include a distribution of values of the synonymous/non-synonymous substitution ratio at each site, and Huelsenbeck and Nielsen [9] permit site to site variation in rates of transitions and transversions. More recently, Pagel and Meade [24] describe a general mixture model for gene sequence data, and Lartillot and Philippe [13] construct a mixture model that allows for heterogeneity across sites in the equilibrium frequencies of the different amino acids.

5.3 Defining mixture models

In the usual likelihood framework we can define the likelihood of a model of gene sequence evolution as proportional to the probability of the data given the model (see also Bryant, Galtier, and Poursat, Chapter 2 this volume):

$$L(\mathbf{Q}) \propto P(\mathbf{D} \mid \mathbf{Q}),$$

where $\mathbf{Q}$ is the substitution rate matrix that defines the model of evolution, and $\mathbf{D}$ is normally an aligned set of sequence data. The probability of the data in $\mathbf{D}$ is found as the product over sites of the individual probabilities of each site, reflecting our assumption that sites are independent of one another. Considering that we are calculating the likelihood for a specific phylogenetic tree we can write the right-hand side of the above equation as

$$P(\mathbf{D} \mid \mathbf{Q}, T) = \prod_i P(\mathbf{D}_i \mid \mathbf{Q}, T),$$

where the product is over all of the sites in the data matrix and T stands for the specific tree.

A mixture model for gene-sequence or amino acid data modifies this basic framework by including more than one model of evolution $\mathbf{Q}$. The probability of the data is now calculated by summing the likelihood at each site over all of the different $\mathbf{Q}$ matrices. Thus, defining the different matrices as $\mathbf{Q}_1, \mathbf{Q}_2, \dots, \mathbf{Q}_J$ write the probability of the data under the mixture model as

$$P(\mathbf{D} \mid \mathbf{Q}_1, \mathbf{Q}_2, \dots, \mathbf{Q}_J, T) = \prod_i \sum_j w_j P(\mathbf{D}_i \mid \mathbf{Q}_j, T), \quad (5.1)$$

where the summation over j ($1 \leq j \leq J$) now specifies that the likelihood of the data at each site is summed over J separate rate or $\mathbf{Q}$ matrices, the summation being weighted by w 's where $w_1 + w_2 + \dots + w_J = 1.0$. The number of matrices, J , can be determined either by prior knowledge of how many different patterns are expected in the data, or empirically as we illustrate in a later section.

Equation (5.1) is a general statement about how to combine likelihoods from different models of evolution applied to the same data. It says that the observed data at a given site arose with probability w_j from the model implied by the rate parameters in $\mathbf{Q}_j$. One $\mathbf{Q}$ might, for example, contain parameters that conform to the nature of evolution that tends to predominate at coding positions, while another conforms to the patterns seen at silent sites. However, both are allowed to apply with some probability to each site.

5.3.1 Partitioning and mixture models

Equation (5.1) can be used to understand the relationship of partitioning the data to mixture modelling. Partitioning data by applying different models to different sites is equivalent to setting to zero different w 's of a mixture model at different sites. In some cases this partitioning might be justified on empirical grounds that it improves the likelihood of the data. In other cases, such as with secondary structure, or when different kinds of data are combined into a single analysis, the data are partitioned on the basis of an *a priori* expectation.

Partitioning of either sort listed above may need to be carefully justified on a case by case basis. For many sites in both nucleotide or amino acid alignments one model may so dominate that the remaining poor fitting models can safely be ignored (weights set to zero). On the other hand, there may be a significant number of sites for which it is difficult or even impossible to choose the best fitting model [24].

5.3.2 Discrete-gamma model as a mixture model

The popular discrete-gamma model [34] is a mixture model that is constrained to take a specific form. The gamma model supposes that rates of evolution vary across sites with probabilities that follow a gamma distribution. The discretized gamma curve supplies K multipliers ranging from slow (<1) to fast (>1). The discrete-gamma model then sums the likelihood of equation (5.1) over these K categories by, in turn, multiplying the elements of the single $\mathbf{Q}$ matrix by

the separate γ_k ; the different $\mathbf{Q}$'s of equation (5.1) all become multiples of each other in the gamma model:

$$P(\mathbf{D} \mid \mathbf{Q}, \gamma, T) = \prod_i \sum_k w_k P(\mathbf{D}_i \mid \gamma_k \mathbf{Q}, T).$$

The K gamma rates are chosen to divide the continuous gamma distribution into K equally probable parts, such that $w_1 = w_2 = \dots = w_K = 1/K$.

The amount of realism that the gamma model brings to a data set depends upon whether the variability in the data is limited to differences in rates and whether these differences conform to a gamma distribution. The gamma distribution is confined to a class of right-skewed curves, reflecting the assumption that most sites evolve relatively slowly, with a smaller number evolving at higher rates. Other distributions, such as the beta, allow left-skewed, and even U-shaped distributions of rates and are easily incorporated into the above formalism.

The most general mixture model allowing the $\mathbf{Q}$ matrices to adopt any configuration will always perform at least as well as the discrete-gamma (or other distribution) model, and frequently better, although the mixture model will often require more parameters. The performance of the mixture model relative to the gamma arises because the separate $\mathbf{Q}$ matrices of the general model can always be made to conform to those that would arise under the gamma model. In the limiting case when all of the data conform to a single homogeneous process, both the general mixture model and the gamma rate-heterogeneity models simplify to a model based upon a single $\mathbf{Q}$ matrix.

5.3.3 Combining rate and pattern-heterogeneity

A mixture model can be constructed to combine variation across sites in the rates of evolution with variation in the qualitative pattern of evolution. To combine rate and pattern-heterogeneity, rewrite equation (5.1) as

$$P(\mathbf{D} \mid \mathbf{Q}_1, \mathbf{Q}_2, \dots, \mathbf{Q}_J, \gamma, T) = \prod_i \sum_j w_j / K \sum_k P(\mathbf{D}_i \mid \gamma_k \mathbf{Q}_j, T). \quad (5.2)$$

This model fits J separate rate matrices of the pattern-heterogeneity model each of which is scaled by K different rates from the gamma rates model. If both rate and pattern-heterogeneity exist in the data, equation (5.2) allows the rate heterogeneity to be detected by the addition of a single parameter. This reduces the number of parameters in the model, freeing the remaining $\mathbf{Q}$ matrices to detect non-rate related pattern-heterogeneity.

5.4 Digression: Bayesian phylogenetic inference

The mixture models that we discuss in this chapter have been implemented in a Bayesian Markov Chain Monte Carlo (MCMC) method [24] and so we briefly introduce Bayesian inference here. Yang (Chapter 3, this volume) provides a thorough treatment of Bayesian inference methods for phylogenies.

Bayesian methods provide a way to calculate the posterior probability distribution of phylogenetic trees. Given an aligned set of sequence data, $\mathbf{D}$, Bayes rule as applied to phylogenetic inference states that the posterior probability of tree T_i is

$$P(T_i | \mathbf{D}) = \frac{P(\mathbf{D} | T_i)P(T_i)}{\sum_T P(\mathbf{D} | T)P(T)}, \quad (5.3)$$

where $P(T_i | \mathbf{D})$ is the probability of tree T_i given the sequence data $\mathbf{D}$, $P(\mathbf{D} | T_i)$ is the probability or likelihood of the data given tree T_i and $P(T_i)$ is the prior probability of T_i . The denominator sums the probabilities over all possible trees T . Equation (5.3) can be difficult to put into practice. The number of possible different unrooted topologies for n species is $(2n - 5)!/2^{n-3}(n - 3)!$. This means that the summation in the denominator is over a large number of topologies for all but the smallest data sets. In turn, for each of these possible topologies the quantity $P(\mathbf{D} | T_i)$ must be integrated over all possible values of the lengths of the branches of the tree and over the parameters of the model of evolution used to describe the sequence data.

Letting $\mathbf{t}$ be a vector of the branch lengths of the tree and $\mathbf{m}$ a vector of the parameters of the model of sequence evolution, then

$$P(\mathbf{D} | T_i) = \int_t \int_m P(\mathbf{D} | T_i, \mathbf{t}, \mathbf{m})P(\mathbf{t})P(\mathbf{m}) dt d\mathbf{m}, \quad (5.4)$$

where $P(\mathbf{t})$ and $P(\mathbf{m})$ are the prior probabilities of the branch lengths and the parameters of the model.

5.4.1 Bayesian inference of trees via MCMC

The MCMC methods [4] as applied to phylogenetic inference provide a computationally efficient way to estimate the posterior probability distribution of trees. A Markov-chain is constructed, the states of which are different phylogenetic trees ([10, 12, 14, 16, 23, 26, 33] and Chapter 3, this volume). At each step in the chain a new tree is proposed by altering the topology, or by changing branch lengths, or the parameters of the model of sequence evolution. The Metropolis–Hastings algorithm [6, 17] is then used to accept or reject the new tree. A newly proposed tree that improves upon the previous tree in the chain is always accepted (sampled), otherwise it is accepted with probability proportional to the ratio of its likelihood to that of the previous tree in the chain. If such a Markov chain is allowed to run long enough, it reaches a stationary distribution. At stationarity, the Metropolis–Hastings sampling algorithm ensures that the Markov chain “wanders” through the universe of trees, sampling better and worse trees, rather than inexorably moving towards “better” trees as an optimizing approach would do. A properly constructed chain samples trees from the posterior density of trees in proportion to their frequency of occurrence in the actual density. That is, the Markov chain draws a sample of trees that can be used to approximate the posterior distribution. In fact, the stationary distribution simultaneously samples the posterior density of trees, the posterior distributions of the branch

lengths and parameters of the model of sequence evolution. By allowing the chain to run for a very long time—perhaps hundreds of thousands or millions of trees, the continuously varying posterior distribution defined in equations (5.1) and (5.2) can be approximated to whatever degree of precision is desired.

5.5 A mixture model combining rate and pattern-heterogeneity

Pagel and Meade [24] implement the basic mixture model of equations (5.1) and (5.2) including rate and pattern heterogeneity. We use the general time reversible model (GTR) to characterize the transition rates among the four nucleotides [30]. This means that the mixture model has no *a priori* constraints on the patterns it can detect beyond those inherent to a time-reversible process. For phylogenetic inference, this matrix is conventionally specified as the product of a symmetric rate matrix $\mathbf{R}$, and a diagonal matrix called $\mathbf{\Pi}$ (see Chapter 2, this volume, for more). The $\mathbf{R}$ matrix contains the six rate parameters describing symmetrical rates of changes between pairs of nucleotides, and $\mathbf{\Pi}$ contains the four base frequencies (denoted π_i). Their product returns the matrix $\mathbf{Q}$ with up to 12 different transition rates among pairs of nucleotides.

$$\mathbf{Q}_{\text{GTR}} = \mathbf{R}\mathbf{\Pi} = \begin{array}{c} A \\ C \\ G \\ T \end{array} \begin{array}{cccc} A & C & G & T \\ \left[\begin{array}{cccc} -\sum_J q_{AJ}\pi_J & q_{AC}\pi_C & q_{AG}\pi_G & q_{AT}\pi_T \\ q_{AC}\pi_A & -\sum_J q_{CJ}\pi_J & q_{CG}\pi_G & q_{CT}\pi_T \\ q_{AG}\pi_A & q_{CG}\pi_C & -\sum_J q_{GJ}\pi_J & q_{GT}\pi_T \\ q_{AT}\pi_A & q_{CT}\pi_C & q_{GT}\pi_G & -\sum_J q_{TJ}\pi_J \end{array} \right] \end{array}.$$

The $\mathbf{R}$ matrix of the GTR model is conventionally specified by five free rate parameters, with the sixth, the $G \leftrightarrow T$ transition, set to 1.0. Popular models of gene-sequence evolution are simply modifications of $\mathbf{Q}$. For example, the Jukes–Cantor model presumes that all of the transition rates and all the base frequencies are equal.

When using more than one rate matrix in our mixture model (equation (5.1)) we use the conventional five rate-parameter configuration for the first rate matrix, but then allow the successive matrices to have six free rate parameters. We use a common set of base frequency parameters across all rate matrices, estimated from the data, although it is straightforward to estimate these parameters separately for each matrix. In addition to the rate parameters, we estimate a weight term (equation (5.1)) for each rate matrix. Each additional GTR rate matrix in the mixture model therefore requires seven new parameters. Adding gamma rate heterogeneity requires one parameter independently of the number of rate matrices.

5.5.1 Selected simulation results

Mixture models should characterize the substitutional processes better than non-mixture models when the data are heterogeneous in their patterns of evolution. One way this will be manifested is in more accurate estimation of branch lengths. Branch lengths are estimated in units of expected nucleotide substitutions

per site. There will normally be saturation in the data such that a given site has evolved twice or more along a branch. Owing to this, the expectation is that the correct model should return, on average, longer branch lengths than incorrect models. Another feature to investigate is how well a mixture model can retrieve the pattern of substitutions in data known to be derived from more than one evolutionary process.

To test these ideas, Pagel and Meade [24] simulated gene-sequence data under several models of evolution, on a random phylogenetic tree of 50 tips with known branch lengths. Here we report selected results from analyses of data generated according to a model with two GTR rate matrices producing qualitatively distinct patterns of sequence evolution ($2Q$), and a $2Q + \Gamma$ model. Values of the rate parameters in the Q matrices for both models were drawn from a uniform random number generator on the interval $[0, 5]$, and we used a gamma shape parameter of $\alpha = 1.0$ to generate rate heterogeneity. Two-thousand sites were simulated with 1200 being derived from one of the rate matrices, and 800 from the other.

The simulated data were analysed by MCMC methods, drawing a sample of 100 widely spaced trees after convergence of the Markov chain. In Fig. 5.1 we

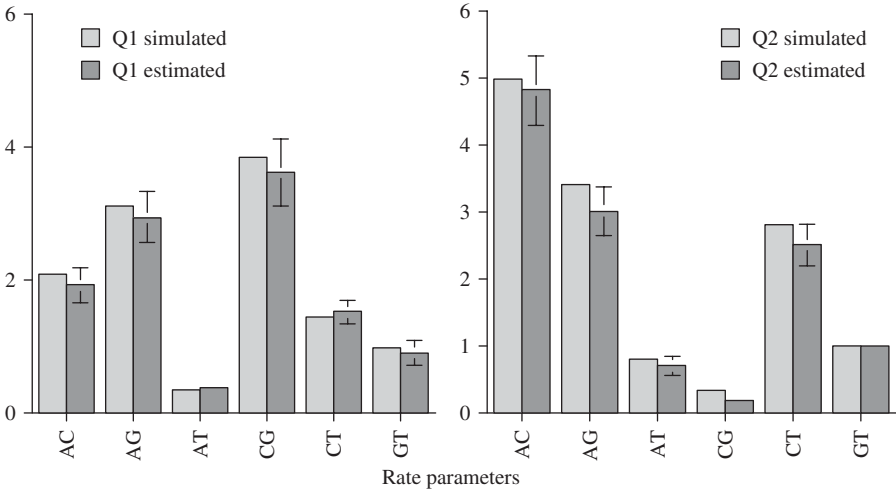


FIG. 5.1. Comparison of the estimated and simulated rate parameters for the $2Q$ model. Left panel shows the results for the matrix designated $Q1$ and the right panel shows the results for $Q2$. Data were simulated on a random tree of 50 tips using two independent rate matrices with random parameter values. Estimated values are the means plus or minus two standard deviations as derived from a MCMC sample of 100 outcomes. Across both matrices the correlation between the simulated and the actual mean of the estimated rates is $r = 0.997$.

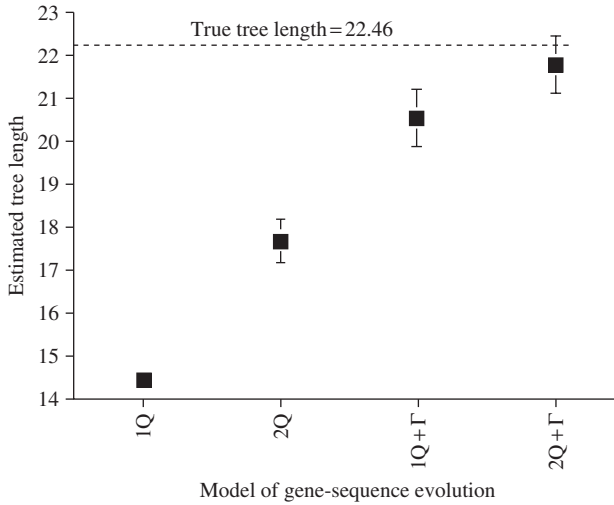


FIG. 5.2. Comparison of the estimated tree lengths obtained from applying different models to gene-sequence data simulated according to a $2Q + \Gamma$ model. Data were simulated on a random tree of 50 tips using two independent rate matrices with random parameter values. Estimated values are the means plus or minus two standard deviations as derived from a MCMC sample of 100 outcomes. Only when the data are analysed with the $2Q + \Gamma$ model do the estimated tree lengths include the real value.

investigate the mixture model's ability to retrieve the known parameters values used to simulate the $2Q$ data. The figure plots the mean values from 100 trees of the rate parameters estimated by the $2Q$ mixture model next to their true values as used in the simulation; the mixture model can retrieve the distinct signature of these two processes, and without prior knowledge. The correlation between the actual and the mean of the estimated values is $r = 0.997$.

We investigated whether by better characterizing the patterns of evolution, the mixture model captures more evolutionary events. This will be manifested in longer tree lengths. Figure 5.2 plots the average tree length derived when several different models of evolution are applied to data simulated from the $2Q + \Gamma$ model. The means are based upon 100 trees sampled from a converged Markov chain. Increasing the complexity of the model increases the average tree length, but only when the $2Q + \Gamma$ model is used to analyse the data do the tree lengths overlap the true length.

5.6 Application of the mixture model to inferring the phylogeny of the mammals

Murphy *et al.* [19] used a data set of 16,397 base pairs comprising 22 genes to infer the phylogeny of the mammals. Their study and two previous molecular

phylogenetic studies by these authors resolved four major mammalian groups that radiated early in the diversification of mammals [15, 18], see also [32]. The four major groups contain about twenty different mammalian orders (such as rodents, primates, bats, carnivores, artiodactyla, and insectivores). Establishing their branching patterns is not only of intrinsic interest, but also necessary to test biogeographical hypotheses and to identify the likely evolutionary processes that gave rise to the diversity of mammalian types.

Owing to the diversity of mammals and the large number of genes in this data set, we might expect considerable heterogeneity in both the rate and pattern of evolution across sites. Murphy *et al.* [19] analysed their data with a GTR + Γ + I model, where the I refers to the use of the invariant sites model. We repeated the analyses of these data using a n GTR + Γ mixture model approach where we allowed the number of independent GTR rate matrices, n , to vary between 1 and 5. We did not fit the invariant sites model, preferring instead to allow the mixture model to find an invariant-like GTR rate matrix should this pattern be a significant one in the data.

5.6.1 Model testing

The conventional likelihood ratio test statistic for comparing models (cf. reference [1]; Chapters 2 and 4, this volume) is not applicable in a Bayesian setting. The asymptotic theory that underpins the likelihood ratio (LR) test presumes that the parameter estimates are at their maximum likelihood values. MCMC methods sample the posterior density of a parameter rather than finding its maximum likelihood estimate, and so a different approach to hypothesis testing is needed.

Bayes factors (cf. reference [3]; Chapters 3 and 4, this volume) are commonly used to compare models in which Bayesian methods are used to estimate the parameters. The Bayes factor for model i compared to model j is the ratio of the marginal likelihood of model i to that of model j . The marginal likelihood is the probability of the data given the model, scaled by the model's prior probability, then integrated over all values of the model parameters. In a phylogenetic setting the marginal likelihood is integrated over trees and values of the rate parameters:

$$P(\mathbf{D} \mid M) = \int_T \int_Q P(\mathbf{D} \mid \mathbf{Q}, T) P(\mathbf{Q}) P(T) d\mathbf{Q} dT.$$

Here we use the term $P(\mathbf{D} \mid M)$ to refer to the marginal probability of the data given some model M , where M includes the parameters of the substitutional process and the phylogenetic trees. Given marginal likelihoods for two different models the log-Bayes factor is defined as:

$$\log \text{BF} = -2 \log \left(\frac{P(\mathbf{D} \mid M_i)}{P(\mathbf{D} \mid M_j)} \right).$$

The interpretation of Bayes factors is subjective. Using the log Bayes factor as defined above, Raftery [25] suggests that a rule of thumb of 2–5 be taken

as “positive” evidence for model i , and greater than 5 as “strong” evidence. Log-Bayes factors of less than 0 provide evidence for model j .

Computing the Bayes factor can be difficult in practice. The converged Markov chain yields the posterior probabilities, not the prior probabilities as specified in the integral. One method to estimate $P(\mathbf{D} \mid M)$ from a converged chain is to calculate the harmonic mean of the posteriors [25]. Although this method converges to $P(\mathbf{D} \mid M)$ as the number of observations in the chain grows large, it can be unstable owing to the occasional result with very small likelihood. As will be seen from the results we report below, differences among the models we report always greatly exceed even the value of 5, and so it seems unlikely that instability in the harmonic mean estimator has influenced our conclusions. Raftery [25] discusses a number of alternative estimators of the Bayes factor and Lartillot and Philippe [13] outline an approach drawing on thermodynamic ideas.

The Bayes factor penalizes more complex models by including prior probability terms for each parameter. The likelihood of the data is multiplied by the set of priors, which, normally being numbers less than 1.0, reduce the marginal likelihood. In all of our MCMC runs, we assigned uniform priors on the interval of 0–100 to parameters of the models of sequence evolution, and all trees were considered equally likely *a priori*.

These priors mean that we can derive an approximation to the Bayes factor to use as a “rule of thumb” in comparing models. We wish to compare models with different numbers of rate matrices. Over many samples, the priors for trees and for gamma rate heterogeneity will approximately cancel as both always appear in the numerator and denominator. Our models then differ only in the numbers of parameters as determined by the $\mathbf{Q}$ matrices, with each additional $\mathbf{Q}$ matrix accounting for six new rate parameters and one weight parameter.

The prior probability of any observation from a uniform 0–100 distribution is 0.01, and thus the prior $P(\mathbf{Q})$ for each additional rate matrix = 0.01^7 . Translating this into a log-Bayes factor, each additional rate matrix “costs” $7 * \log(0.01) = -32.23$. That is, each additional $\mathbf{Q}$ matrix must improve the likelihood by approximately 32 log-units to return a log-Bayes factor of 0.0.

5.7 Results

Table 5.1 reports the average log-likelihoods of fitting the various models to the mammal data, along with the results of the $1\mathbf{Q} + \Gamma + \text{I}$ model that Murphy *et al.* used [19]. In all of our MCMC runs we fitted $n\mathbf{Q} + \Gamma$ mixture models, where n varied between 1 and 5 independent rate matrices. We always used four rate categories in the gamma-rates model. More than five rate matrices did not increase the log-likelihood. We allowed the Markov-chains to reach convergence before sampling 100 trees at widely spaced intervals (10,000 trees) to ensure independence of successive trees. We treated a chain as being at convergence when there was no average improvement in the likelihood for 200,000 iterations. We ran at least five chains for each model, and all runs converged to the same region of tree

TABLE 5.1. Mixture model results for mammals data

Model ^a	Mean log-likelihood	Tree length	Number of parameters ^b	Bayes factor ^c
1 Q + Γ + I ^d	−211110	3.78	7	
1 Q + Γ	−211541 ± 6.53 (−211554)	3.81 ± 0.04	6	n.a.
2 Q + Γ	−210048 ± 6.35 (−210062)	4.06 ± 0.03	13	2922
3 Q + Γ	−209334 ± 7.66 (−209350)	3.79 ± 0.05	20	1354
4 Q + Γ	−209017 ± 9.01 (−209032)	4.01 ± 0.06	27	578
5 Q + Γ	−208915 ± 5.04 (−208921)	4.26 ± 0.01	34	154

^aModels are specified by the number of independent rate matrices (**Q**s) in the mixture model. All models use gamma rate heterogeneity (Γ).

^bSee text combining rate and pattern-heterogeneity for a description of the number of parameters in the mixture model.

^cTest of difference between specified model and the model above it in the table, based upon harmonic means of the likelihoods (in parentheses). We do not compare the 1**Q** + Γ and 1**Q** + Γ + I models because the latter is a maximum likelihood value from Murphy *et al.* [19]. See text Model testing for details of the Bayes Factor test.

^dMurphy *et al.* [19] used a 1**Q** + Γ + I model incorporating invariant sites (I). The log-likelihood and the tree length are taken from the maximum likelihood tree.

space as judged by likelihoods and posterior probabilities of trees. The means and averages for each model are based upon a sample of 100 trees from a single run.

Overall Table 5.1 shows that applying mixture models to the Murphy *et al.* [18, 19] data can return substantial improvements in the log-likelihood. The Bayes factors indicate that these improvements are highly significant, but that the incremental improvement from additional **Q** matrices declines as more are added. Two issues stand-out for analysis. One is whether the original 1**Q** + Γ + I model that Murphy *et al.* [19] used adequately describes the data, and the other is how many rate matrices should be included in the mixture model for these data.

The simple 1**Q** model plus gamma rate heterogeneity returns a log-likelihood of about 430 log-units worse than the Murphy *et al.* maximum likelihood tree derived from the invariant sites model. Fitting a mixture model with two rate matrices plus rate heterogeneity (2**Q**+ Γ model) improves the likelihood by about 1,500 log-units, or about 1,000 log-units improvement over the 1**Q** + Γ + I model, and returns a significantly longer tree.

Before discussing the models with three, four, and five rate matrices, we analyse in Table 5.2 the estimated rate parameters of the two independent rate matrices of the 2**Q** + Γ mixture model. If the 1**Q** + Γ + I model were the correct model for these data we would expect the 2**Q** + Γ mixture model to

TABLE 5.2. Estimated transition rate parameters for $2Q + \Gamma$ model^a applied to the mammalian data

Rate	$A \leftrightarrow C$	$\mathbf{A} \leftrightarrow \mathbf{G}$	$A \leftrightarrow T$	$C \leftrightarrow G$	$\mathbf{C} \leftrightarrow \mathbf{T}$	$G \leftrightarrow T$	Q-weight
Q1	1.57± 0.06	2.27± 0.11	0.91± 0.04	1.57± 0.07	2.21± 0.11	1.0 n.a. ^b	0.44 (0.02)
Q2	0.23± 0.02	2.56± 0.16	0.18± 0.01	0.02± 0.07	3.55± 0.27	0.20± 0.02	0.56 (0.02)

^aValues in the table are the transition rate parameters from the $\mathbf{R}$ matrix of the GTR model. As the estimated base frequencies are all close to 0.25, these values are proportional to the actual rates.

^bThis transition rate is fixed at 1.0. Pairs of nucleotides in bold type are transitions, the remainder are transversions.

yield rate matrices that conform to the invariant sites model. The invariant sites model assumes the existence of an unconstrained rate matrix plus a fixed rate matrix in which all transition rates among different pairs of nucleotides are constrained to be zero.

Table 5.2 shows that neither of the $2Q + \Gamma$ models rate matrices conforms to an invariant sites model. Even though some of the rates in the matrix designated **Q2** are small, all are ten or more standard deviations from zero. Instead of invariance, this second rate matrix suggests a different pattern of evolution to the first matrix, one in which there is a substantial number of sites in which transversions occur but only very slowly, while transitions occur at much higher rates. The **Q2** rate matrix receives a weight of 0.56 indicating that a majority of the sites may be of this slowly evolving class.

5.7.1 How many rate matrices to include in the mixture model?

We do not know in advance how many different rate matrices to estimate, relying instead on the data in combination with Bayes factors to guide that choice. Pagel and Meade [24] show in simulated data that this procedure, in combination with information on the variability of estimated parameters, can correctly identify the number of independent patterns. Figure 5.3 plots the log-likelihoods from Table 5.1 for mixture models with from 1 to 5 rate matrices. The rate of increase in log-likelihood slows noticeably beyond four rate matrices. The Bayes factors (Table 5.1) superficially justify a fifth rate matrix but we suggest that four rate matrices is the better solution for these data. One reason for this is that the Bayes Factor test (like the likelihood ratio statistic) applied to phylogenetic log-likelihoods assumes that all of the sites in the alignment are independent. The true number of independent sites is probably far fewer than the 16.3 thousand in this alignment, and this will inflate the differences in likelihood between models.

We also expect [24] that when sufficient rate matrices have been estimated for a given data set, the parameters of additional matrices will be poorly estimated

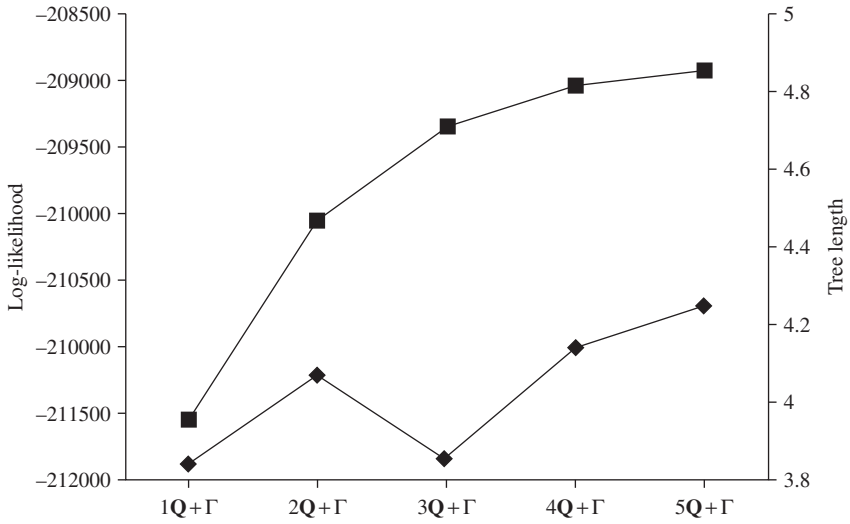


FIG. 5.3. Upper curve: Improvement in the log-likelihood for mixture models with increasing numbers of independent rate matrices ($\mathbf{Q}$). Sharp improvement in the likelihood for small numbers of rate matrices reaches a plateau such that the $5\mathbf{Q} + \Gamma$ model does not substantially improve upon the $4\mathbf{Q} + \Gamma$ model. Lower curve: Total tree lengths associated with each model. The decline in the total tree length between the $2\mathbf{Q} + \Gamma$ and the $3\mathbf{Q} + \Gamma$ model is associated with the dominant tree topology changing from the one on the Fig. 5.4(a) to the one on the Fig. 5.4(b).

and that superfluous matrices will receive small weights (equation (5.2)). One of the rate matrices in the $5\mathbf{Q} + \Gamma$ model receives a weight of 0.02. The standard deviations of the rates for this matrix have an average of 1.70 compared to just 0.15 for the four other rate matrices.

5.7.2 Inferring the tree of mammals

We shall use the $4\mathbf{Q} + \Gamma$ model to infer the tree of mammals, comparing it to the tree that the single rate matrix model produces. We choose this comparison because our single rate matrix model returns the same tree topology as Murphy *et al.* report [19]. Figure 5.4(a) and (b) reports these two trees, both of which are consensus trees derived from 100 trees sampled from the converged Markov chains. The Bayesian posterior probabilities of each node are shown.

The trees are similar in a number of important ways. Both, for example, find the four broad groupings of placental mammals that have emerged from other recent molecular trees of the mammals [15, 32]: the Afrotheria [31], the Xenarthra, the Euarchontoglires, and the Laurasiatheria. The nodes corresponding to these clades are assigned 100% posterior support in both trees.

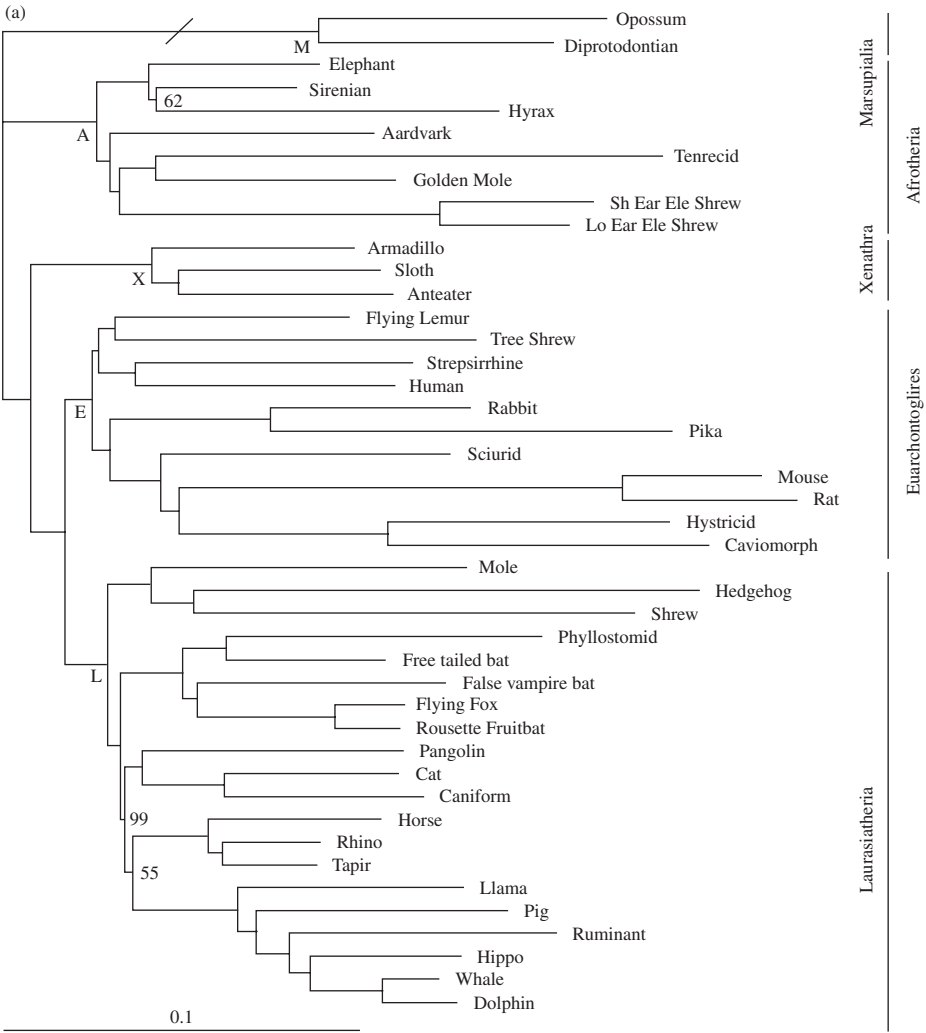


FIG. 5.4. (a) The consensus phylogenetic tree (with maximum likelihood branch lengths) for the $1Q + \Gamma$ model, based upon 100 samples drawn from a converged Markov chain. This topology is virtually identical to the Murphy *et al.* [19] tree. (b) The consensus phylogenetic tree (with maximum likelihood branch lengths) for the $4Q + \Gamma$ model, based upon 100 samples drawn from a converged Markov chain. For both trees, posterior probabilities of internal nodes are labelled, unlabelled nodes have posterior probabilities of 100%. The $4Q + \Gamma$ model alters the position of the hyrax and the sciurid rodent, and indicates that there is more uncertainty about the placement of some mammalian orders than is evident from the simpler model.

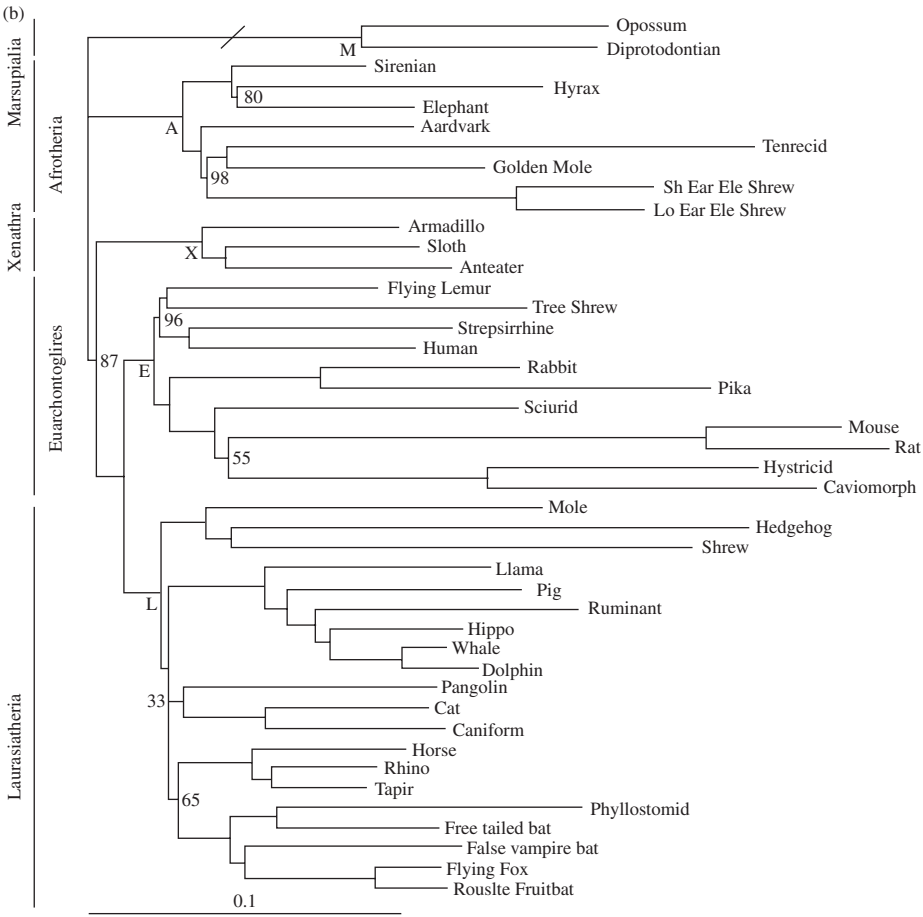


FIG. 5.4. (*continued*)

Both trees also place the root of the placentals between the Afrotheria and the remaining three groups.

The precise branching sequence of the mammalian orders is difficult to identify owing to their rapid diversification. The short time period of this diversification is reflected in the very short branch lengths for many of the deep interior nodes of the tree. It is at these short branches that the two trees of Fig. 5.4 show some topological differences. Within the Laurasiatheria the tree based upon a single rate matrix finds a well supported major division between the bats on the one hand and the canids, whales, ruminants, and perissodactyls (horses and other odd-toed ungulates) on the other. The more complex mixture model has the whale, dolphin, and ruminant group branching off first, with strong posterior support. Placement of the canids, perissodactyls, and bats within the Laurasiathea is less certain, but the model

favours canids branching off separately with the perissodactyls and bats forming a sister group. This latter result agrees with Waddell and Shelleys analysis [32] based upon an independent data set. The two trees in Fig. 5.4 agree on the Euarachontoglires, but the $4\mathbf{Q} + \Gamma$ tree has lower posterior support at several nodes.

The weaker posterior support of the $4\mathbf{Q} + \Gamma$ model within the Laurasiatheira is disappointing but important. Erixon *et al.* [2] find that Bayesian posterior support at nodes of phylogenetic trees is too high when the model of sequence evolution is under-parameterized. The $4\mathbf{Q} + \Gamma$ model's 2,500 or so log-unit improvement over the $1\mathbf{Q} + \Gamma$ model provides quite clear evidence that the latter model is under-parameterized for these data, and may explain its higher posterior probabilities. If this interpretation is correct, then the ordinal branching patterns within some parts of the mammalian tree remain uncertain, and data sets with even greater resolution than the data used here are needed to resolve their branching order. On the other hand, the agreement between the $1\mathbf{Q} + \Gamma$ and $4\mathbf{Q} + \Gamma$ models on the branching orders of the four major mammalian groups gives even greater confidence in those results.

The mixture model suggests a change to the Afrotheria. The $1\mathbf{Q} + \Gamma$ model places the hyrax closer to the aquatic sirenians (sea cows), but the $4\mathbf{Q} + \Gamma$ model shifts the hyrax to be next to the elephants and with reasonably high support. This latter placement is consistent with the widespread suspicion that the small terrestrial hyrax species is the closest living relative to the largest terrestrial animal. It also sends the message that complex models can achieve quite remarkable stability. For the elephant–hyrax–sirenian clade, we recorded the branch length leading to whichever pair of species was placed together in each of the 100 trees derived from the $1\mathbf{Q} + \Gamma$ model and from the $4\mathbf{Q} + \Gamma$ model. As the posteriors show, 62% of these pairs were (sirenian, hyrax) for the simpler model, whereas 80% were (elephant, hyrax) for the more complex model. What is impressive about the change in topology between the two models is that the average length of the branch leading to whichever pair of species is placed together is only 0.003 ± 0.0009 for the $1\mathbf{Q} + \Gamma$ model, and for the $4\mathbf{Q} + \Gamma$ model it is an even shorter 0.0025 ± 0.0009 .

5.7.3 Tree lengths

The mixture models return longer trees (Table 5.1) indicating that they better characterize the substitutional process in the concatenated alignment. The average tree length for the $4\mathbf{Q} + \Gamma$ model does not overlap with the tree lengths from the simpler models, including the $1\mathbf{Q} + \Gamma$ and the $1\mathbf{Q} + \Gamma + \mathbf{I}$ models. Pagel and Meade [24] and Fig. 5.2 above shows that mixture model more accurately estimates branch lengths when the data contain heterogeneity in the patterns of evolution across sites. The results from the mixture models emphasize an important difference between likelihood models and models such as parsimony or minimum distance that prefer trees that imply fewer evolutionary events. One potentially important consequence of producing longer trees is that ancestral timings derived from applying molecular clocks to branch

lengths derived from mixture models may differ from those derived from simpler models.

5.8 Discussion

Mixture models provide a useful way to detect and characterize the evolution of gene or protein sequences that may harbour the signal of more than one evolutionary process. This will often give them advantages over homogeneous process models or models allowing heterogeneity in the rates of evolution.

The mixture models approach differs in philosophy and application from the common practice of partitioning of the data. When the data are all of the same type (e.g. nucleotides) partitioning is equivalent to a mixture model in which it is presumed that the weights for some models are zero at some sites. From the mixture modelling perspective this kind of knowledge will seldom be available, and it is preferable to sum the likelihood of the data at each site over all of the models. The mixture also uses all of the data to estimate each of its parameters, rather than using different partitions of the data to estimate different parameters.

There will undoubtedly be cases where partitioning, either on the basis of empirical or *a priori* information improves the likelihood of the data over that of a mixture model. But it remains a question in need of further study whether the practice of partitioning in general returns better trees or leads to better estimates of the parameters of the models of evolution. For example, Pagel and Meade [24] show that partitioning protein coding data by codon position can miss substantial variability in the pattern of evolution within a particular codon position. A similar situation can arise when ribosomal DNA data are partitioned by their secondary structure into stems and loops.

The mixture model also shows how use of the invariant sites model can miss important patterns of variation in the data. In the invariant sites model one rate matrix is free to vary while the other is fixed with rates of change among nucleotides set to zero. The comparable mixture model also uses two matrices but estimates them from the data. We found that, applied to the mammalian data, neither of the matrices that emerged from the mixture model with two rate matrices conformed to the invariant sites matrix. Rather, one of the matrices yielded very slow rates of transversions, but high rates of transitions, while the other matrix had high rates of change between all pairs of nucleotides.

This mixture model with two matrices plus rate heterogeneity substantially improves the likelihood over the gamma rate variability plus invariant sites model that Murphy *et al.* [19] originally used to analyse these data. What appears to be happening is that some sites evolve slowly, occasionally showing no change at all, whereas others of these slow sites do change in perhaps one or two species. When they do, it is more likely to be a transition, although transversions are also occasionally seen. The invariant sites model can characterize the former class of sites reasonably well, but not those sites that do show changes. By comparison the mixture model rate matrix treats both kinds of site as forming a continuum and therefore provides a better overall fit.

We found that a mixture model based upon four distinct rate matrices, plus gamma rate heterogeneity, provided the best justified fit to the mammalian data, yielding substantial increases in the likelihood over any other simpler model. This model returned a tree that largely agrees with the Murphy *et al.* [18, 19] tree but suggests some changes to the placement of mammalian orders. The mixture model serves to emphasize that the ordinal branching patterns may be less well identified than was previously believed, returning lower posterior support for several nodes. Interestingly, the topology we derive agrees in several respects with Waddell and Shelley's tree [32] derived from independent data.

The mixture model reassigns the hyrax to share an ancestor with the elephant rather than with the sirenian, and improves the support for the placement of hyrax over that observed in the original tree. These results show that mixture models can identify regions of trees in which perhaps too much confidence is placed on the basis of simple models, and they can also sharpen up our confidence in other regions of trees.

We might expect phylogenetically structured data to harbour complex signals of the history of evolution. The mixture model we report here shows that these signals can be detected and characterized, and without imposing patterns on the data. The model can be applied to any kind of aligned data set, including proteins or morphological traits. To the extent that the signals in such data are not lost or overwritten by more recent evolutionary events, investigators can use statistical approaches validly to infer the nature and modes of past evolutionary events and processes [21, 22], complementing experimental and palaeontological methods.

We have implemented the mixture model in a computer program available from www.ams.reading.ac.uk/zoology/pagel.

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HADAMARD CONJUGATION: AN ANALYTIC TOOL FOR PHYLOGENETICS

Michael D. Hendy

A phylogeny (evolutionary tree) on a set of taxa X , is a tree T whose leaves are labelled by the elements of X . When we identify the root R of T ; provide a distribution of nucleotides at R ; and give a stochastic model of the nucleotide substitutions between the vertices of each edge e of T , we can calculate the probability of each possible distribution (which we will refer to as a “pattern”) of nucleotides at the leaves. A set of aligned homologous nucleotide sequences of common length l , is then modelled as l samples drawn sequentially from this distribution, with the pattern of nucleotides at a common site representing one sample. The relative frequencies of each observed pattern provide estimates of these probabilities.

Phylogenetic inference is the process of estimating T (and perhaps some of the parameters of the model) from the observed pattern frequencies, thus inverting the mechanism which generated these patterns. For most models this inversion cannot be analysed directly, even if all the pattern probabilities were known exactly. In this chapter, we consider several simple models of nucleotide substitution where this inversion is possible, where we can derive invertible analytic formulae for each pattern probability. From these the tree T and the some model parameters can be deduced. Our analysis is referred to as Hadamard conjugation (or phylogenetic spectral analysis).

Hadamard conjugation, though limited to a few simple models, provides an analytic tool to give insight into the general phylogenetic inference process. Hadamard conjugation will be described, together with illustrations of how it can be applied to analyse a number of related concepts, such as the inconsistency of maximum parsimony (MP), the determination of maximum likelihood (ML) points, and some other issues of phylogenetic analysis.

6.1 Introduction

Nucleotide sequences are called homologous when they are inferred as being descendants from a common ancestral sequence. Phylogenetics is concerned with the mechanism of estimating their phylogeny (evolutionary tree) which describes their history of descent. A model of nucleotide substitution is a mathematical description of the process by which a particular nucleotide at a site

of a sequence is replaced by a different nucleotide. These models are generally stochastic (probabilistic) with a probability (as a function of time) which is specified for each possible substitution. A matrix containing these substitutions is called a stochastic (or transition) matrix. Here the four states correspond to the nucleotides of DNA or RNA. The model is symmetric when the probability of substituting X by Y is the same as for Y to X , for each pair of states X and Y . Often a model is defined in terms of rates of substitution given in a rate matrix, together with times at each branching point. Although models of nucleotide substitutions on a prescribed tree T giving descendant sequences are easy to specify, the inverse problem of inferring T and the model from the sequences is not generally solvable.

In this chapter, we describe a simple model where this inversion is possible. The inversion employs the relationship known as Hadamard conjugation. We first give an overview of Hadamard conjugation, introduce Hadamard matrices, and the symmetric substitution models of Jukes and Cantor [19], Kimura [20, 21], and Neyman [25], and extended to models where rate variations can be included. Applications are then considered to the tree building methods of maximum parsimony and maximum likelihood. In particular the problem of the inconsistency of maximum parsimony is discussed.

For a set $\{\sigma_0, \sigma_1, \dots, \sigma_n\}$ of aligned homologous nucleotide sequences nucleotide sequences (DNA or RNA) generated on a phylogeny T , we will derive the following relationships (called *Hadamard conjugation*):

$$\mathbf{s} = H^{-1}\text{Exp}(H\mathbf{q}), \quad (6.1)$$

and its inverse

$$\mathbf{q} = H^{-1}\text{Ln}(H\mathbf{s}). \quad (6.2)$$

For the four-state symmetric models of nucleotide substitution introduced by Jukes and Cantor [19], and by Kimura [20, 21], we will find:

- $\mathbf{q} \in \mathbb{R}^{4^n}$ is a vector which encodes T and the model parameters on the edges of T
- $\mathbf{s} \in \mathbb{R}^{4^n}$ is a vector that gives the probabilities of each of the 4^n patterns of nucleotide differences at a site, it can also be called a site likelihood vector
- $H = [h_{ij}]$, with $h_{ij} \in \{-1, 1\}$, is a $4^n \times 4^n$ Hadamard matrix, with inverse $H^{-1} = 4^{-n}H$
- Exp and Ln are functions applied componentwise to vectors, so that for $\mathbf{v} = [v_i]$, we define $\text{Exp}(\mathbf{v}) = [\exp(v_i)]$ and $\text{Ln}(\mathbf{v}) = [\ln(v_i)]$, vectors of the same dimension, with the exponential and natural logarithm functions applied to each component.

6.2 Hadamard conjugation for two sequences

6.2.1 Hadamard matrices—a brief introduction

Jacques Hadamard in 1893 [12] introduced a class of matrices we now call Hadamard matrices.

Definition 1 An $n \times n$ matrix $A = [a_{ij}]$, with $a_{ij} \in \{-1, 1\}$, is Hadamard (order n)

$$\iff A^T A = nI_n.$$

It is easily shown that every Hadamard matrix A of order n has the following properties:

1. Hadamard was able to provide a useful bound for determinants, in particular for every $n \times n$ matrix $B = [b_{ij}]$, where $|b_{ij}| \leq 1$,

$$|\det(B)| \leq |\det(A)| = n^{n/2}.$$

2. The rows and columns of A are orthogonal, so the product $AA^T = nI_n$, where A^T is the transpose (interchanging rows and columns) of A .
3. A Hadamard matrix is easily inverted, with inverse

$$A^{-1} = \frac{1}{n}A^T.$$

4. The order of a Hadamard matrix is either 1, 2, or a multiple of 4.
5. Hadamard matrices of order n can be constructed for:

- $n = 1, 2$
- $n = ab$, when there exist Hadamard matrices of orders a and b (and thus in particular for all powers of 2)
- $n = 4m$, whenever $4m - 1$ is a prime
- and some other special cases.

Hadamard Conjecture. It is conjectured that a Hadamard matrix of order n exists for every multiple of 4. (Currently Hadamard matrices are known for orders $n = 4m$, for $n = 4, 8, 12, 16, \dots, 664$. The smallest case yet to be decided is with $n = 668$.)

Sylvester matrices. A special family of Hadamard matrices known as Sylvester matrices $H_0, H_1, H_2, \dots$, (introduced by J.J. Sylvester [31] in 1867) which can be defined recursively by

$$H_0 = [1], \quad H_1 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

and for $n \geq 1$

$$H_{n+1} = H_1 \otimes H_n = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}.$$

The Kronecker product $A \otimes B$ is the matrix where each entry a_{ij} of A is replaced by $a_{ij}B$, so if A and B were of orders $m \times n$ and $p \times q$, $A \otimes B$ has order $mp \times nq$

and comprises mn blocks, with the (i, j) th block being B multiplied by a_{ij} . Thus:

$$H_2 = H_1 \otimes H_1 = \begin{bmatrix} H_1 & H_1 \\ H_1 & -H_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix},$$

$$H_3 = H_1 \otimes H_2 = \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix},$$

etc. so H_n is a Hadamard matrix of order 2^n . The matrix H of equations (6.1) and (6.2) is $H = H_{2n}$.

Properties of Sylvester matrices. We will find the following properties useful. It is easily seen that for $x, a, b, c \in \mathbb{R}$

$$H_2 \begin{bmatrix} x \\ a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ a \\ b \\ c \end{bmatrix} = \begin{bmatrix} x + a + b + c \\ x - a + b - c \\ x + a - b - c \\ x - a - b + c \end{bmatrix}. \quad (6.3)$$

These components can also be expressed as

$$H_1 \begin{bmatrix} x & a \\ b & c \end{bmatrix} H_1 = \begin{bmatrix} x + a + b + c & x - a + b - c \\ x + a - b - c & x - a - b + c \end{bmatrix}. \quad (6.4)$$

This can be extended recursively to vectors $\mathbf{x}, \mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^{2^n}$ with

$$H_{n+2} \begin{bmatrix} \mathbf{x} \\ \mathbf{a} \\ \mathbf{b} \\ \mathbf{c} \end{bmatrix} = \begin{bmatrix} H_n(\mathbf{x} + \mathbf{a} + \mathbf{b} + \mathbf{c}) \\ H_n(\mathbf{x} - \mathbf{a} + \mathbf{b} - \mathbf{c}) \\ H_n(\mathbf{x} + \mathbf{a} - \mathbf{b} - \mathbf{c}) \\ H_n(\mathbf{x} - \mathbf{a} - \mathbf{b} + \mathbf{c}) \end{bmatrix}, \quad (6.5)$$

which can be re-expressed as

$$H_{n+1} \begin{bmatrix} \mathbf{x} & \mathbf{a} \\ \mathbf{b} & \mathbf{c} \end{bmatrix} H_{n+1} = \begin{bmatrix} H_n(\mathbf{x} + \mathbf{a} + \mathbf{b} + \mathbf{c}) & H_n(\mathbf{x} - \mathbf{a} + \mathbf{b} - \mathbf{c}) \\ H_n(\mathbf{x} + \mathbf{a} - \mathbf{b} - \mathbf{c}) & H_n(\mathbf{x} - \mathbf{a} - \mathbf{b} + \mathbf{c}) \end{bmatrix}. \quad (6.6)$$

6.3 Some symmetric models of nucleotide substitution

6.3.1 Kimura's 3-substitution types model

In 1981 Motoo Kimura [21] introduced his 3-substitution types (K3ST) model of nucleotide substitution. In that model he proposed three independent substitution rates, a rate α for transitions, and rates β and γ for the two types of transversions, as defined in Fig. 6.1(b). We will refer to those three substitution types as tr_α , tr_β , and tr_γ , respectively.

A succession of two or more substitutions between the sequences will not be observed directly, so the number of observed differences underestimates the actual number of substitutions that occurred. Kimura derived a correction to estimate the numbers of each substitution type, and these are called the *expected* numbers of substitutions. Kimura [21] defined parameters P , the probability of observing a transitional difference at a site, and Q and R , the probabilities of observing tr_β and tr_γ transversional differences respectively at a site. Then he derived formulae for the expected numbers of each of these substitutions evolving under a Poisson process. Assuming sequence evolution, as displayed in Fig. 6.1(a), he showed that the expected number of transitions is

$$2\alpha t = -\frac{1}{4} \ln[(1 - 2P - 2Q)(1 - 2P - 2R)/(1 - 2Q - 2R)], \quad (6.7)$$

the expected number of tr_β transversions is

$$2\beta t = -\frac{1}{4} \ln[(1 - 2P - 2Q)(1 - 2Q - 2R)/(1 - 2P - 2R)] \quad (6.8)$$

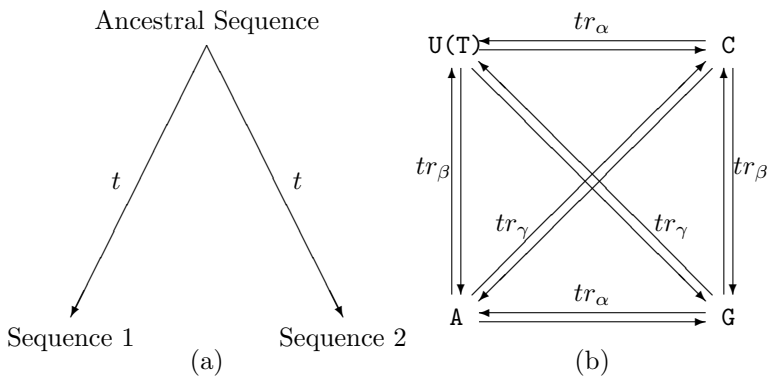


FIG. 6.1. Kimura's 3ST model [21]. (a) The relationship between Sequence 1 and Sequence 2, descendant from an Ancestral Sequence t years before present. (b) The three substitution types tr_α , tr_β , and tr_γ proposed by Kimura. The RNA nucleotides are A (adenine) and G (guanine) are called purines, and U (uracil, replaced by T (thymine) in DNA), and C (cytosine) are called pyrimidines. Substitutions within these two chemical classes (tr_α) are referred to as transitions, and substitutions between the classes (tr_β, tr_γ) are called transversions.

and the expected number of tr_γ transversions is

$$2\gamma t = -\frac{1}{4} \ln[(1 - 2P - 2R)(1 - 2Q - 2R)/(1 - 2P - 2Q)]. \quad (6.9)$$

Hence adding these terms, we find that the expected total number of substitutions is

$$K = -\frac{1}{4} \ln[(1 - 2P - 2Q)(1 - 2P - 2R)(1 - 2Q - 2R)], \quad (6.10)$$

which Kimura refers to as the “evolutionary distance.”

Note, in his derivation, Kimura has assumed that each of the arguments of the logarithm functions in equations (6.7)–(6.10) are positive (for the logarithm to be well-defined). Here we will continue with this assumption.

When we compare the corresponding sites of two homologous DNA or RNA sequences, we can take the proportion of sites with observed differences of each type as estimates for P , Q , and R . Then equations (6.7)–(6.9) give estimates of the expected numbers of substitutions of each type. This estimates the number of substitutions not observed directly as differences, because multiple successive substitutions appear as either one or no substitution between the endpoints. If t were known, then these formulae provide estimates for the rates α , β , and γ . However, can we invert equations (6.7)–(6.9) to express P , Q , and R in terms of αt , βt , and γt ?

Below we will find that equations (6.7)–(6.10) can be formulated as a Hadamard conjugation, and using this formulation, the inversion is easy to derive. For consistency we will adopt a different notation system, using roman letters, p for the probabilities, and q (quantities) for the expected numbers of substitutions, with suffixes indicating type. Thus

$$p_\alpha = P, \quad p_\beta = Q, \quad p_\gamma = R, \quad q_\alpha = 2\alpha t, \quad q_\beta = 2\beta t, \quad q_\gamma = 2\gamma t,$$

so for example p_α is the probability that the nucleotides at the endpoints of the path differ by tr_α , and q_α is the expected number of tr_α substitutions along the path. Further we will write $q_\emptyset = -q_\alpha - q_\beta - q_\gamma$ (so $-q_\emptyset = K$ is the evolutionary distance) and $p_\emptyset = 1 - p_\alpha - p_\beta - p_\gamma$ (so $p_\emptyset$ is the probability that the nucleotides at the endpoints of the path are the same).

With these notational changes, equations (6.7)–(6.10) can be rewritten (expanding the logarithms and rearranging) as

$$\begin{aligned} q_\emptyset &= \frac{1}{4} [\ln(1 - 2p_\alpha - 2p_\gamma) + \ln(1 - 2p_\beta - 2p_\gamma) + \ln(1 - 2p_\alpha - 2p_\beta)], \\ q_\alpha &= \frac{1}{4} [-\ln(1 - 2p_\alpha - 2p_\gamma) + \ln(1 - 2p_\beta - 2p_\gamma) - \ln(1 - 2p_\alpha - 2p_\beta)], \\ q_\beta &= \frac{1}{4} [\ln(1 - 2p_\alpha - 2p_\gamma) - \ln(1 - 2p_\beta - 2p_\gamma) - \ln(1 - 2p_\alpha - 2p_\beta)], \\ q_\gamma &= \frac{1}{4} [\ln(1 - 2p_\alpha - 2p_\gamma) - \ln(1 - 2p_\beta - 2p_\gamma) + \ln(1 - 2p_\alpha - 2p_\beta)], \end{aligned}$$

which can be expressed as the vector equation

$$\begin{bmatrix} q_\emptyset \\ q_\alpha \\ q_\beta \\ q_\gamma \end{bmatrix} = \frac{1}{4} \begin{bmatrix} \ln(1 - 2p_\alpha - 2p_\gamma) + \ln(1 - 2p_\beta - 2p_\gamma) + \ln(1 - 2p_\alpha - 2p_\beta) \\ -\ln(1 - 2p_\alpha - 2p_\gamma) + \ln(1 - 2p_\beta - 2p_\gamma) - \ln(1 - 2p_\alpha - 2p_\beta) \\ \ln(1 - 2p_\alpha - 2p_\gamma) - \ln(1 - 2p_\beta - 2p_\gamma) - \ln(1 - 2p_\alpha - 2p_\beta) \\ \ln(1 - 2p_\alpha - 2p_\gamma) - \ln(1 - 2p_\beta - 2p_\gamma) + \ln(1 - 2p_\alpha - 2p_\beta) \end{bmatrix}. \quad (6.11)$$

Now recalling the Sylvester matrix H_2 (with inverse $H_2^{-1} = \frac{1}{4}H_2$) we see that equation (6.11) can be written in the form of equation (6.1), with $x = 0$, $a = \ln(1 - 2p_\alpha - 2p_\gamma)$, $b = \ln(1 - 2p_\beta - 2p_\gamma)$, and $c = \ln(1 - 2p_\alpha - 2p_\beta)$, giving

$$\mathbf{q} = \begin{bmatrix} q_\emptyset \\ q_\alpha \\ q_\beta \\ q_\gamma \end{bmatrix} = H_2^{-1} \begin{bmatrix} 0 \\ \ln(1 - 2p_\alpha - 2p_\gamma) \\ \ln(1 - 2p_\beta - 2p_\gamma) \\ \ln(1 - 2p_\alpha - 2p_\beta) \end{bmatrix} = H_2^{-1} \text{Ln} \left(\begin{bmatrix} 1 \\ 1 - 2p_\alpha - 2p_\gamma \\ 1 - 2p_\beta - 2p_\gamma \\ 1 - 2p_\alpha - 2p_\beta \end{bmatrix} \right). \quad (6.12)$$

Now, recalling $p_\emptyset = 1 - p_\alpha - p_\beta - p_\gamma$, we see

$$1 - 2p_\alpha - 2p_\gamma = p_\emptyset - p_\alpha + p_\beta - p_\gamma,$$

etc., so

$$\begin{bmatrix} 1 \\ 1 - 2p_\alpha - 2p_\gamma \\ 1 - 2p_\beta - 2p_\gamma \\ 1 - 2p_\alpha - 2p_\beta \end{bmatrix} = \begin{bmatrix} p_\emptyset + p_\alpha + p_\beta + p_\gamma \\ p_\emptyset - p_\alpha + p_\beta - p_\gamma \\ p_\emptyset + p_\alpha - p_\beta - p_\gamma \\ p_\emptyset - p_\alpha - p_\beta + p_\gamma \end{bmatrix} = H_2 \begin{bmatrix} p_\emptyset \\ p_\alpha \\ p_\beta \\ p_\gamma \end{bmatrix}. \quad (6.13)$$

Hence defining

$$\mathbf{p} = \begin{bmatrix} p_\emptyset \\ p_\alpha \\ p_\beta \\ p_\gamma \end{bmatrix},$$

equations (6.12) and (6.13) give us the Hadamard conjugation

$$\mathbf{q} = H_2^{-1} \text{Ln}(H_2 \mathbf{p}), \quad (6.14)$$

which is easily inverted to give

$$\mathbf{p} = H_2^{-1} \text{Exp}(H_2 \mathbf{q}). \quad (6.15)$$

This inversion allows us to give the probabilities in terms of evolutionary distances. Thus

$$H_2 \mathbf{q} = -2 \begin{bmatrix} 0 \\ q_\alpha + q_\gamma \\ q_\beta + q_\gamma \\ q_\alpha + q_\beta \end{bmatrix},$$

so from equation (6.15),

$$p_{\emptyset} = \frac{1}{4}(1 + e^{-2(q_{\alpha}+q_{\gamma})} + e^{-2(q_{\beta}+q_{\gamma})} + e^{-2(q_{\alpha}+q_{\beta})}),$$

$$p_{\alpha} = \frac{1}{4}(1 - e^{-2(q_{\alpha}+q_{\gamma})} + e^{-2(q_{\beta}+q_{\gamma})} - e^{-2(q_{\alpha}+q_{\beta})}),$$

$$p_{\beta} = \frac{1}{4}(1 + e^{-2(q_{\alpha}+q_{\gamma})} - e^{-2(q_{\beta}+q_{\gamma})} - e^{-2(q_{\alpha}+q_{\beta})}),$$

$$p_{\gamma} = \frac{1}{4}(1 - e^{-2(q_{\alpha}+q_{\gamma})} - e^{-2(q_{\beta}+q_{\gamma})} + e^{-2(q_{\alpha}+q_{\beta})}),$$

which when expressed using Kimura's notation is

$$P = \frac{1}{4}(1 - e^{-4(\alpha+\gamma)t} + e^{-4(\beta+\gamma)t} - e^{-4(\alpha+\beta)t}),$$

$$Q = \frac{1}{4}(1 + e^{-4(\alpha+\gamma)t} - e^{-4(\beta+\gamma)t} - e^{-4(\alpha+\beta)t}),$$

$$R = \frac{1}{4}(1 - e^{-4(\alpha+\gamma)t} - e^{-4(\beta+\gamma)t} + e^{-4(\alpha+\beta)t}),$$

the inversion we sought.

We note, following equation (6.4), equations (6.14) and (6.15) can also be expressed in terms of 2×2 matrices, thus

$$\begin{bmatrix} q_{\emptyset} & q_{\alpha} \\ q_{\beta} & q_{\gamma} \end{bmatrix} = H_1^{-1} \text{Ln} \left(H_1 \begin{bmatrix} p_{\emptyset} & p_{\alpha} \\ p_{\beta} & p_{\gamma} \end{bmatrix} H_1 \right) H_1^{-1}, \quad (6.16)$$

and

$$\begin{bmatrix} p_{\emptyset} & p_{\alpha} \\ p_{\beta} & p_{\gamma} \end{bmatrix} = H_1^{-1} \text{Exp} \left(H_1 \begin{bmatrix} q_{\emptyset} & q_{\alpha} \\ q_{\beta} & q_{\gamma} \end{bmatrix} H_1 \right) H_1^{-1}. \quad (6.17)$$

This transformation from observed differences to expected numbers of substitutions is referred to as “distance correction.” These corrections depend on the model under analysis. For the Kimura 3ST model, the observed distance is represented by the probability of difference, $1 - p_{\emptyset}$, and the corrected distance is $K = -q_{\emptyset}$, the expected number of substitutions. Thus

$$d_{\text{obs}} = 1 - p_{\emptyset} = p_{\alpha} + p_{\beta} + p_{\gamma},$$

and Kimura's evolutionary distance $K = -q_{\emptyset}$ is the corrected distance

$$\begin{aligned} d_{\text{corr}} &= -q_{\emptyset} = q_{\alpha} + q_{\beta} + q_{\gamma} \\ &= -\frac{1}{4} \ln[(1 - 2p_{\alpha} - 2p_{\beta})(1 - 2p_{\alpha} - 2p_{\gamma})(1 - 2p_{\beta} - 2p_{\gamma})]. \end{aligned} \quad (6.18)$$

6.3.2 Other symmetric models

By imposing relationships on the parameters we can derive the formulae for some other simpler symmetric models.

When we set $p_\gamma = p_\beta$ we obtain Kimura's two parameter model (K2ST) [20]. Here the probability of a transition is defined to be $P = p_\alpha$, and of a transversion to be $Q = p_\beta + p_\gamma = 2p_\beta$. The corresponding distance correction (from equation (6.18)) for the K2ST model is

$$d_{\text{corr}} = -\frac{1}{2} \ln(1 - 2P - Q) - \frac{1}{4} \ln(1 - 2Q).$$

When we set $p_\gamma = p_\beta = p_\alpha$, we obtain the Jukes–Cantor one parameter model (JC) [19]. For this model the probability of a substitution is $P = p_\alpha + p_\beta + p_\gamma = 3p_\alpha$. The corresponding distance correction (from equation (6.18)) for the Jukes–Cantor model is

$$d_{\text{corr}} = -\frac{3}{4} \ln \left(1 - \frac{4}{3} P \right).$$

There is also a symmetric 2-state model of theoretical interest, called the Neyman (or Cavender/Farris) model [2, 7, 25]. This model postulates just two states (these could be the purines (A and G) and pyrimidines (C and T or U)). We can derive the formulae for Neyman's model by setting $p_\alpha = p_\gamma = 0$ and $P = p_\beta$, with tv_β substitutions occurring at the rate β . The corresponding distance correction (from equation (6.18)) for Neyman's 2-parameter model is therefore

$$d_{\text{corr}} = -\frac{1}{2} \ln(1 - 2P). \quad (6.19)$$

Neyman's model is useful to develop the theory supporting Hadamard conjugation, which can then be extended to the four-state symmetric models. It is also generally useful as it is the simplest continuous time rate model.

6.4 Hadamard conjugation—Neyman model

In this section, we will develop the relationships among three related two-state sequences evolving under the two-state symmetric model of Neyman [25]. This will be a precursor to describing the relationships for four or more sequences, first under the Neyman model, and then for the Kimura 3ST model.

6.4.1 Neyman model on three sequences

Here we consider three sequences σ_A , σ_B , and σ_C , each of 2-state characters, which we will take as purines (R) and pyrimidines (Y). We assume a symmetric model of character substitution across the edges of a phylogenetic tree T , so that on an edge e of T , the probabilities of substitution from states R to Y, and from states Y to R have the same value, p_e (the observed distance). Let q_e be the expected number of substitutions (the corrected distance) across edge e , so from

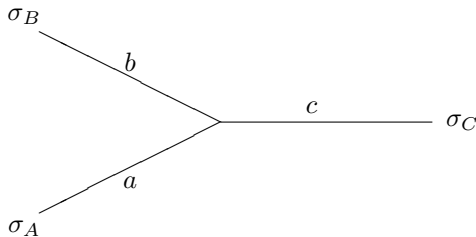


FIG. 6.2. A tree T connecting three sequences, σ_A , σ_B , and σ_C , with edges a , b , and c , as shown. The probability of a substitution between corresponding characters at the endpoints of the edge a , is p_a , etc.

equation (6.19)

$$q_e = -\frac{1}{2} \ln(1 - 2p_e), \quad \text{and} \quad p_e = \frac{1}{2}(1 - e^{-2q_e}), \quad (6.20)$$

and hence the probability that there is no change between the endpoints of e is

$$1 - p_e = \frac{1}{2}(1 + e^{-2q_e}).$$

Given the Neyman model on the tree T of Fig. 6.2, we can derive formulae for the probabilities of different patterns among the characters at a site. We group the characters at a site into one of 4 patterns, by identifying which of σ_A and σ_B contain a character which differs from the character at the “reference” sequence, σ_C , at that site. In particular:

- pattern A identifies a site where the characters at σ_B and σ_C agree, but differ from that at σ_A
- pattern B identifies a site where the characters at σ_A and σ_C agree, but differ from that at σ_B
- pattern C identifies a site where the characters at σ_A and σ_B agree, but differ from that at σ_C
- we identify a pattern $\emptyset$ as a site where all the characters are the same.

Given T , and the probabilities p_e , let s_A , s_B , s_C , and $s_\emptyset$ be the probabilities of generating a site with the corresponding site pattern in the original data. Thus s_A is the probability that the site pattern is either **YRR** or **RYY** (i.e. the character of σ_A differs from the characters of σ_B and σ_C .) Similarly we define s_B to be the probability of the site pattern **RYR** or **YRY**, and s_C to be the probability the site pattern **RRY** or **YYR**.

$s_\emptyset$ is the probability that the character at each leaf is the same. This occurs either when the character at the central vertex is also the same (with probability $(1 - p_a)(1 - p_b)(1 - p_c)$), or when the central vertex has the other character (with

probability $p_a p_b p_c$). Thus

$$\begin{aligned}
 s_\emptyset &= (1 - p_a)(1 - p_b)(1 - p_c) + p_a p_b p_c, \\
 &= \frac{1}{8}[(1 + e^{-2q_a})(1 + e^{-2q_b})(1 + e^{-2q_c})] \\
 &\quad + \frac{1}{8}[(1 - e^{-2q_a})(1 - e^{-2q_b})(1 - e^{-2q_c})], \\
 &= \frac{1}{4}[1 + e^{-2(q_a + q_c)} + e^{-2(q_b + q_c)} + e^{-2(q_a + q_b)}], \\
 &= \frac{1}{4}[1 + e^{-2d_{AC}} + e^{-2d_{BC}} + e^{-2d_{AB}}], \tag{6.21}
 \end{aligned}$$

where $d_{AB} = q_A + q_B$ is the expected number of substitutions between σ_A and σ_B , etc. Similarly, following the derivation of equation (6.21), we find

$$s_A = p_a(1 - p_b)(1 - p_c) + (1 - p_a)p_b p_c = \frac{1}{4}[1 - e^{-2d_{AC}} + e^{-2d_{BC}} - e^{-2d_{AB}}], \tag{6.22}$$

$$s_B = (1 - p_a)p_b(1 - p_c) + p_a(1 - p_b)p_c = \frac{1}{4}[1 + e^{-2d_{AC}} - e^{-2d_{BC}} - e^{-2d_{AB}}], \tag{6.23}$$

and

$$s_C = (1 - p_a)(1 - p_b)p_c + p_a p_b(1 - p_c) = \frac{1}{4}[1 - e^{-2d_{AC}} - e^{-2d_{BC}} + e^{-2d_{AB}}]. \tag{6.24}$$

Equations (6.21)–(6.24) can be expressed succinctly as a Hadamard conjugation.

$$\begin{aligned}
 \mathbf{s} &= \begin{bmatrix} s_\emptyset \\ s_A \\ s_B \\ s_C \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 + e^{-2d_{AC}} + e^{-2d_{BC}} + e^{-2d_{AB}} \\ 1 - e^{-2d_{AC}} + e^{-2d_{BC}} - e^{-2d_{AB}} \\ 1 + e^{-2d_{AC}} - e^{-2d_{BC}} - e^{-2d_{AB}} \\ 1 - e^{-2d_{AC}} - e^{-2d_{BC}} + e^{-2d_{AB}} \end{bmatrix} \\
 &= \frac{1}{4} H_2 \text{Exp} \left(\begin{bmatrix} 0 \\ -2d_{AC} \\ -2d_{BC} \\ -2d_{AB} \end{bmatrix} \right) = H_2^{-1} \text{Exp}(-2\mathbf{d}), \tag{6.25}
 \end{aligned}$$

where $\mathbf{d} = \begin{bmatrix} 0 \\ d_{AC} \\ d_{BC} \\ d_{AB} \end{bmatrix}$. Let $\mathbf{q} = \begin{bmatrix} q_\emptyset \\ q_a \\ q_b \\ q_c \end{bmatrix}$, with $q_\emptyset = -(q_a + q_b + q_c)$, then we see

$$\mathbf{d} = \begin{bmatrix} 0 \\ q_a + q_c \\ q_b + q_c \\ q_a + q_b \end{bmatrix} = -\frac{1}{2} H_2 \mathbf{q}. \tag{6.26}$$

Hence

$$\mathbf{s} = H_2^{-1}\text{Exp}(H_2\mathbf{q}), \tag{6.27}$$

which provided $H_2\mathbf{s} > \mathbf{0}$, inverts to give

$$\mathbf{q} = H_2^{-1}\text{Ln}(H_2\mathbf{s}). \tag{6.28}$$

6.4.2 *Neyman model on four sequences*

In the analysis of Neyman’s model for three sequences we grouped complimentary site patterns (such as RYR and YRY) together. This is equivalent to identifying the pattern of n differences between the $n + 1$ sequences. We can formalize this as follows. Given $n + 1$ aligned homologous two-state sequences we identify a particular sequence σ_0 as the reference sequence, and compare each of the other n sequences, $\sigma_1, \dots, \sigma_n$, to σ_0 , site by site. This comparison produces a set of n sequences $\delta_1, \dots, \delta_n$ of differences, where the j th component of δ_i is $\delta_{ij} = 0$ when the j th characters of σ_0 and σ_i are the same, and $\delta_{ij} = 1$ when they differ. (See the example in Table 6.1.)

The edge-length spectrum for the tree of Fig. 6.3 is

$$\mathbf{q} = \begin{bmatrix} q_\emptyset \\ q_1 \\ q_2 \\ q_{12} \\ q_3 \\ q_{13} \\ q_{23} \\ q_{123} \end{bmatrix} = \begin{bmatrix} -0.7 \\ 0.1 \\ 0.1 \\ 0 \\ 0.2 \\ 0.2 \\ 0 \\ 0.1 \end{bmatrix}.$$

TABLE 6.1. Example of creating the $n = 3$ sequences of differences from $n + 1 = 4$ two-state character sequences. The “pattern” of differences at the site j , is the set of sequences whose character differs from the corresponding character of the reference sequence σ_0

Site no.	1	2	3	4	5	6	7	8	9	10
σ_0	R	Y	Y	R	Y	R	R	Y	Y	Y
σ_1	R	Y	R	R	Y	Y	R	Y	R	R
σ_2	R	R	Y	R	Y	Y	R	Y	Y	Y
σ_3	R	Y	R	Y	Y	Y	Y	Y	R	Y
δ_1	0	0	1	0	0	1	0	0	1	1
δ_2	0	1	0	0	0	1	0	0	0	0
δ_3	0	0	1	1	0	1	1	0	1	0
Pattern	$\emptyset$	{2}	{1, 3}	{3}	$\emptyset$	{1, 2, 3}	{3}	$\emptyset$	{1, 3}	{1}

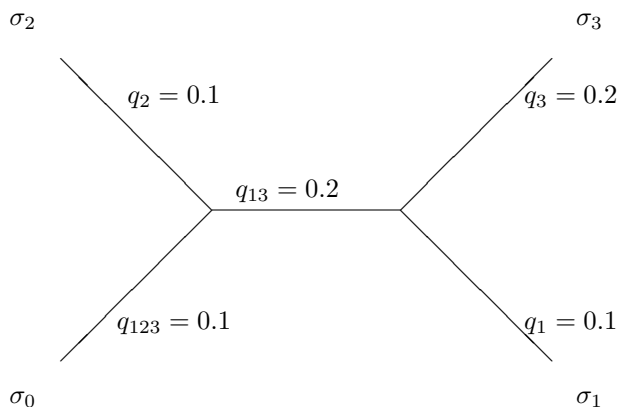


FIG. 6.3. The tree T_{13} on 4 sequences and the induced edge splits with edge weights. The edge length q_{13} refers to the split $\{0, 2\} \mid \{1, 3\}$, which is indexed by the subset *not* containing the reference element 0, and the subscript 13 is used to indicate this subset.

In this vector the splits are indexed by the subsets of $\{1, 2, 3\}$ listed in lexicographic order: $\emptyset$, $\{1\}$, $\{2\}$, $\{1, 2\}$, $\{3\}$, $\{1, 3\}$, $\{2, 3\}$, $\{1, 2, 3\}$. $q_{12} = q_{23} = 0$ as no edge of T induces these splits, and $q_\emptyset = -\sum_{\alpha \neq \emptyset} q_\alpha$ is the negative of the length of the tree.

Equation (6.26) for $n+1 = 3$, showed $H_2 \mathbf{q} = -2\mathbf{d}$, relating the three distances to the three independent edge-lengths ($d_\emptyset = 0$ and $q_\emptyset = -(q_a + q_b + q_c)$ are not free parameters. In the case $n+1 = 4$, there are five independent edge lengths, one for each edge of T , but there are $\binom{4}{2} = 6$ distances.) Equation (6.27) then gives $H_2 \mathbf{s} = e^{-2\mathbf{d}}$.

By comparing corresponding terms of $H_n \mathbf{q}$ and $H_n \mathbf{s}$, first when $n+1 = 4$, and then generally, we will establish the generality of equations (6.27) and (6.28).

Consider the product $H_3 \mathbf{q}$ in the case of the tree T of Fig. 6.3. We find

$$H_3 \mathbf{q} = \begin{bmatrix} 0 \\ -2(q_1 + q_{13} + q_{123}) \\ -2(q_2 + q_{123}) \\ -2(q_1 + q_{13} + q_2) \\ -2(q_3 + q_{13} + q_2) \\ -2(q_1 + q_2) \\ -2(q_2 + q_3 + q_{123}) \\ -2(q_1 + q_2 + q_3 + q_{123}) \end{bmatrix} = -2 \begin{bmatrix} 0 \\ d_{01} \\ d_{02} \\ d_{12} \\ d_{03} \\ d_{13} \\ d_{23} \\ d_{02} + d_{13} \end{bmatrix} \quad (6.29)$$

with each term, except the first, 0, and the last, $-2(q_1 + q_2 + q_3 + q_{123}) = -2(d_{02} + d_{13})$, being a distance between pairs of taxa. We will define $d_\emptyset = 0$,

and for tree T of Fig. 6.3, define $d_{0123} = d_{02} + d_{13}$. Then with

$$\mathbf{d} = \begin{bmatrix} d_0 \\ d_{01} \\ d_{02} \\ d_{12} \\ d_{03} \\ d_{13} \\ d_{23} \\ d_{0123} \end{bmatrix},$$

equation (6.29) becomes

$$H_3 \mathbf{q} = -2\mathbf{d}. \quad (6.30)$$

We will now examine the terms of $\text{Exp}(-2\mathbf{d})$ to show these give the terms of $H_3 \mathbf{s}$. We have equation

$$e^{-2d_0} = e^0 = 1 = s_0 + s_1 + s_2 + s_{12} + s_3 + s_{13} + s_{23} + s_{123},$$

which is the first row of $H_3 \mathbf{s}$. From equations (6.20) we saw that the probability p_{ij} that the characters of σ_i and σ_j differ at a site is $p_{ij} = \frac{1}{2}(1 - e^{-2d_{ij}})$, so

$$e^{-2d_{ij}} = 1 - 2p_{ij}. \quad (6.31)$$

However p_{ij} is the sum of the site pattern probabilities that the states of σ_i and σ_j differ. Thus in particular p_{13} is the sum of the s_α terms for those α which split 1 and 3, that is, those subsets α which contain one, but not both of 1 and 3. Hence

$$p_{13} = s_1 + s_{12} + s_3 + s_{23},$$

which gives

$$e^{-2d_{13}} = 1 - 2p_{13} = s_0 - s_1 + s_2 - s_{12} - s_3 + s_{13} - s_{23} + s_{123}. \quad (6.32)$$

Further p_{02} will be the sum of the s_α terms for the sets α which contain 2 (we reference the split by the subset not containing 0), so

$$p_{02} = s_2 + s_{12} + s_{23} + s_{123},$$

and

$$e^{-2d_{02}} = 1 - 2p_{02} = s_0 + s_1 - s_2 - s_{12} + s_3 + s_{13} - s_{23} - s_{123}.$$

Continuing this analysis for the other terms $e^{-2d_{ij}}$ we find that each agrees with the corresponding term in $H_3 \mathbf{s}$.

Finally we see

$$\begin{aligned} e^{-2d_{0123}} &= e^{-2(d_{02}+d_{13})} = e^{-2d_{02}} e^{-2d_{13}} \\ &= (1 - 2p_{02})(1 - 2p_{13}) = 1 - 2p_{02} - 2p_{13} + 4p_{02}p_{13}. \end{aligned} \quad (6.33)$$

Recall $p_{02} = s_2 + s_{12} + s_{23} + s_{123}$ and $p_{13} = s_1 + s_{12} + s_3 + s_{23}$. We see in Fig. 6.3 that the paths in T from 0 to 2, and from 1 to 3 do not intersect, so the product of the probabilities $p_{02}p_{13}$ gives the probability that the states at 0 and 2 differ,

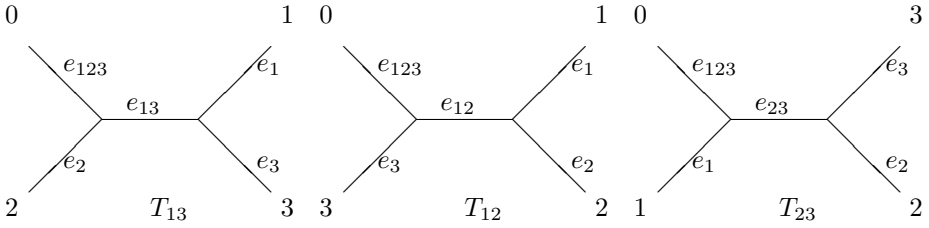


FIG. 6.4. The three unrooted trees on $\{0, 1, 2, 3\}$. The edges are labelled e_α where α is the set of leaf labels separated from 0 by that edge. For convenience we write e_{12} for $e_{\{1,2\}}$, etc., when not ambiguous. These trees are identified by their internal edge label.

and (simultaneously) that the states at 1 and 3 differ. This event is recorded by the s_α terms which simultaneously split both 0 and 2, and 1 and 3, thus $p_{02}p_{13} = s_{12} + s_{23}$. Substituting these in equation (6.33) gives:

$$e^{-2d_{0123}} = s_\emptyset - s_1 - s_2 + s_{12} - s_3 + s_{13} + s_{23} - s_{123}. \quad (6.34)$$

Hence expressing $H_3\mathbf{s}$ in full we find

$$H_3\mathbf{s} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} s_\emptyset \\ s_1 \\ s_2 \\ s_{12} \\ s_3 \\ s_{13} \\ s_{23} \\ s_{123} \end{bmatrix} = \text{Exp} \left(-2 \begin{bmatrix} d_\emptyset \\ d_{01} \\ d_{02} \\ d_{12} \\ d_{03} \\ d_{13} \\ d_{23} \\ d_{0123} \end{bmatrix} \right). \quad (6.35)$$

Hence from equation (6.30) we obtain

$$\mathbf{s} = H_3^{-1} \text{Exp}(H_3\mathbf{q}). \quad (6.36)$$

Now provided $H_3\mathbf{s} > 0$, this can be inverted giving

$$\mathbf{q} = H_3^{-1} \text{Ln}(H_3\mathbf{s}). \quad (6.37)$$

Corresponding derivations for T_{12} and T_{23} (Fig. 6.4) can be achieved by permuting the subscripts $2 \leftrightarrow 3$ and $1 \leftrightarrow 2$. For T_{12} , $q_{13} = q_{23} = 0$, and for T_{23} , $q_{12} = q_{13} = 0$. We must also re-interpret the meaning of d_{0123} , noting that in T_{12} , $d_{0123} = d_{03} + d_{12}$ and in T_{23} , $d_{0123} = d_{01} + d_{23}$. These can be summarized, as in each tree

$$d_{0123} = \min(d_{01} + d_{23}, d_{02} + d_{13}, d_{12} + d_{03}). \quad (6.38)$$

In each case given the edge weight spectrum $\mathbf{q}$, the corresponding sequence spectrum $\mathbf{s}$ can be calculated using equation (6.36).

Example 1 If the edgeweights on T_{13} were $q_1 = 0.1$, $q_2 = 0.1$, $q_3 = 0.2$, $q_{13} = 0.2$, and $q_{123} = 0.1$, then $q_{12} = q_{23} = 0$, and $q_\emptyset = -0.7$. Applying equation (6.36)

$$\mathbf{q} = \begin{bmatrix} -0.7 \\ 0.1 \\ 0.1 \\ 0 \\ 0.2 \\ 0.2 \\ 0 \\ 0.1 \end{bmatrix} \implies \mathbf{s} = \frac{1}{8} H_3 \text{Exp}(H_3 \mathbf{q}) = \begin{bmatrix} 0.528 \\ 0.074 \\ 0.064 \\ 0.019 \\ 0.115 \\ 0.119 \\ 0.019 \\ 0.064 \end{bmatrix}.$$

Hence in particular the probability of a constant site is 0.528, and the probability of a $(\{0, 1\} \mid \{2, 3\})$ split is 0.019 (even though there is no corresponding edge split). Note the values in $\mathbf{s}$ are rounded to three decimal figures. If we use these values as displayed, and apply equation (6.37) we find, displaying to four decimals:

$$\mathbf{s} = \begin{bmatrix} 0.5280 \\ 0.0740 \\ 0.0640 \\ 0.0190 \\ 0.1150 \\ 0.1190 \\ 0.0190 \\ 0.0640 \end{bmatrix} \implies \mathbf{q} = \frac{1}{8} H_3 \text{Ln}(H_3 \mathbf{s}) = \begin{bmatrix} -0.6995 \\ 0.1000 \\ 0.1001 \\ 0.0005 \\ 0.2006 \\ 0.1996 \\ 0.0005 \\ 0.1001 \end{bmatrix}.$$

This illustrates that if the values in the $\mathbf{s}$ vector are not *exactly* the expected sequence probabilities, then the derived $\mathbf{q}$ will not fit any tree exactly. Noting the entries, $q_{12} = q_{23} = 0.0005$ are much smaller than all the other entries, we can make the assumption that these are approximating 0. The splits for the values that are significantly larger, define the edges of T_{13} . $\square$

6.4.3 *Neyman model on $n + 1$ sequences*

We saw that with 4 sequences, $H_3 \mathbf{q} = -2\mathbf{d}$, and $H_3 \mathbf{s} = \text{Exp}(-2\mathbf{d})$, with both these vectors indexed by the even ordered subsets of $X = \{0, 1, 2, 3\}$. In the general case with $n + 1$ sequences, we will define a general “distance” spectrum $\mathbf{d} = -\frac{1}{2} H_n \mathbf{q}$, and show $H_n \mathbf{s} = \text{Exp}(-2\mathbf{d})$ holds generally, as introduced by Hendy and Penny [15].

Let $X = \{0, 1, \dots, n\}$, and let $\mathbb{E}(X)$ be the set of all even ordered subsets of X . Consider the matrix H_n with its rows labelled by the subsets of $X^* = \{1, 2, \dots, n\}$ and the columns labelled by the elements of $\mathbb{E}(X)$. Then one can show (using the recursion $H_n = H_1 \otimes H_{n-1}$) that the element $h_{\alpha\beta}$ of row $\alpha \subseteq X^*$

and column $\beta \in \mathbb{E}(X)$ is

$$h_{\alpha\beta} = (-1)^{|\alpha \cap \beta|}, \quad (6.39)$$

(i.e. $h_{\alpha\beta} = -1 \iff \alpha$ and β have an odd number of common elements).

Pathsets Let T be a tree with leaf set X and edge set $e(T)$. For $i, j \in X$, let $\Pi_{ij}(T)$ be the set of edges connecting leaves i and j in T . For $\beta \in \mathbb{E}(X)$, let

$$\Pi_\beta(T) = \{e_\alpha \in e(T) \mid h_{\alpha\beta} = -1\}.$$

Lemma 1

$$\Pi_{\{i,j\}}(T) = \Pi_{ij}(T)$$

and for $\beta, \gamma \in \mathbb{E}(X)$

$$\Pi_{\beta \triangle \gamma}(T) = \Pi_\beta(T) \triangle \Pi_\gamma(T),$$

where $A \triangle B = (A \cup B) - (A \cap B)$ is the symmetric difference of sets A and B .

Proof $e_\alpha \in E(T)$ separates i from $j \iff$ one, but not both of i and j belong to α , that is, $\iff |\alpha \cap \{i, j\}| = 1$. Thus

$$e_\alpha \in \Pi_{ij}(T) \iff (-1)^{|\alpha \cap \{i,j\}|} = -1 \iff e_\alpha \in \Pi_{\{i,j\}}(T),$$

hence $\Pi_{ij}(T) = \Pi_{\{i,j\}}(T)$.

Further, with $\delta = \beta \triangle \gamma$, and noting $|\alpha \cap \delta| \equiv |\alpha \cap \beta| + |\alpha \cap \gamma| \pmod{2}$, for any $\alpha \subset X$,

$$\begin{aligned} \Pi_\delta(T) &= \{e_\alpha \mid h_{\alpha\delta} = -1\} = \{e_\alpha \mid h_{\alpha\beta}h_{\alpha\gamma} = -1\} \\ &= \{e_\alpha \mid h_{\alpha\beta} = -1\} \triangle \{e_\alpha \mid h_{\alpha\gamma} = -1\} \\ &= \Pi_\beta(T) \triangle \Pi_\gamma(T). \end{aligned}$$

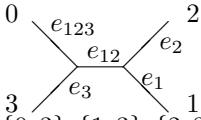
□

Definitions: Summarizing we note

1. T is an X -tree $\iff T$ is a phylogeny with leaf set $X = \{0, 1, 2, \dots, n\}$.
2. $\mathbb{E}(X) = \{\alpha \subseteq X \mid \alpha \text{ is of even order}\}$.
3. For $\alpha \in \mathbb{E}(X)$ the *pathset* $\Pi_\alpha(T)$ is recursively constructed by:
 - $\Pi_\emptyset(T) = \emptyset$
 - $\Pi_{\{i,j\}}(T) = \Pi_{ij}(T)$ is the path in T connecting leaves i and j
 - and for $|\alpha| \geq 4$ and $i, j \in \alpha$, $\Pi_\alpha(T) = \Pi_{ij}(T) \triangle \Pi_{\alpha - \{i,j\}}(T)$.
4. A *weighted X -tree* $(T, \mathbf{q})$ is:
 - a tree T with leaf set X and edge set $e(T)$
 - a vector $\mathbf{q} \in \mathbb{R}^{2^n}$ indexed by the subsets of $X^* = X - \{0\}$ such that:
 - * $q_\beta > 0$ for each edge $e_\beta \in e(T)$
 - * $q_\emptyset = -\sum_{e_\beta \in e(T)} q_\beta$
 - * $q_\alpha = 0$ for all $\alpha \in X^* - e(T) - \{\emptyset\}$.
5. For $e_\beta \in e(T)$, q_β is the *length* of e_β . $\mathbf{q}$ is the *edge-length spectrum*. $\mathbf{q}$ defines T by $e(T) = \{\beta \mid q_\beta > 0\}$. q_β is the *length* of e_β .

6. For $\alpha \in \mathbb{E}(X)$, the *length* of the pathset $(T)_\alpha$ is $d_\alpha = \sum_{e_\beta \in \Pi_\alpha(T)} q_\beta$. (Hence in particular $d_\emptyset = 0$, $d_{\{i,j\}} = d_{ij}$, and for $\alpha, \alpha' \in \mathbb{E}(X)$, with $\alpha \cap \alpha' = \emptyset$, $d_{\alpha \cup \alpha'} = d_\alpha + d_{\alpha'}$.)

Example 2 $X = \{0, 1, 2, 3\}$, $T_{12} =$



$\mathbb{E}(X) = \{\emptyset, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 3\}, \{1, 3\}, \{2, 3\}, \{0, 1, 2, 3\}\}.$

$\Pi_\emptyset(T) = \emptyset, \quad \Pi_{\{0,2\}}(T) = \{e_2, e_{12}, e_{123}\},$

$\Pi_{\{0,1,2,3\}}(T) = \Pi_{\{0,1\}}(T) \triangle \Pi_{\{2,3\}}(T) = \{e_{123}, e_{12}, e_1\} \triangle \{e_2, e_{12}, e_3\}$

$= \{e_1, e_2, e_3, e_{123}\}$

$= (\Pi_{\{0,3\}}(T) \cup \Pi_{\{1,2\}}(T)).$

□

Example 3 Suppose for the tree T of Fig. 6.5, the time scale is in units of 10^6 years, and the sequences are evolving from the root with a substitution rate of $\lambda = 10^{-7}$ substitutions per year. This will induce edge weights of $q_\alpha = \lambda \times t_\alpha$, where t_α is the elapsed total time between the endpoints of edge e_α . From the figure we read $t_1 = 3 \times 10^6$, $t_2 = 4 \times 10^6$, $t_3 = 3 \times 10^6$, $t_4 = 2 \times 10^6$, $t_{13} = 3 \times 10^6$, $t_{123} = 2 \times 10^6$, and $t_{1234} = 2 \times 10^6$. Hence we calculate

$$\mathbf{q} = \begin{bmatrix} -0.95 \\ 0.15 \\ 0.20 \\ 0 \\ 0.15 \\ 0.15 \\ 0 \\ 0.10 \\ 0.10 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0.10 \end{bmatrix} \begin{matrix} \emptyset \\ 1 \\ 2 \\ 12 \\ 3 \\ 13 \\ 23 \\ 123 \\ 4 \\ 14 \\ 24 \\ 124 \\ 34 \\ 134 \\ 234 \\ 1234 \end{matrix}, \quad H_4 \mathbf{q} = \begin{bmatrix} 0 \\ -1.00 \\ -0.80 \\ -1.00 \\ -1.00 \\ -0.60 \\ -1.00 \\ -1.40 \\ -0.40 \\ -1.00 \\ -0.80 \\ -1.40 \\ -1.00 \\ -1.00 \\ -1.40 \\ -1.40 \end{bmatrix} \begin{matrix} \emptyset \\ 01 \\ 02 \\ 12 \\ 03 \\ 13 \\ 23 \\ 0123 \\ 04 \\ 14 \\ 24 \\ 0124 \\ 34 \\ 0134 \\ 0234 \\ 1234 \end{matrix}, \quad \mathbf{d} = \begin{bmatrix} 0 \\ 0.5 \\ 0.4 \\ 0.5 \\ 0.5 \\ 0.3 \\ 0.5 \\ 0.7 \\ 0.2 \\ 0.5 \\ 0.4 \\ 0.7 \\ 0.5 \\ 0.5 \\ 0.7 \\ 0.7 \end{bmatrix} \begin{matrix} \emptyset \\ 01 \\ 02 \\ 12 \\ 03 \\ 13 \\ 23 \\ 0123 \\ 04 \\ 14 \\ 24 \\ 0124 \\ 34 \\ 0134 \\ 0234 \\ 1234 \end{matrix}, \quad \mathbf{s} = \begin{bmatrix} 0.418 \\ 0.072 \\ 0.090 \\ 0.022 \\ 0.072 \\ 0.080 \\ 0.022 \\ 0.060 \\ 0.047 \\ 0.009 \\ 0.017 \\ 0.009 \\ 0.009 \\ 0.017 \\ 0.009 \\ 0.047 \end{bmatrix} \begin{matrix} \emptyset \\ 1 \\ 2 \\ 12 \\ 3 \\ 13 \\ 23 \\ 123 \\ 4 \\ 14 \\ 24 \\ 124 \\ 34 \\ 134 \\ 234 \\ 1234 \end{matrix},$$

where $\mathbf{d} = -\frac{1}{2}H_4\mathbf{q}$ and $\mathbf{s} = H_4^{-1}(-2\mathbf{d})$. (The indices are displayed to the right of each vector.) □

We assume $n + 1$ two-state sequences, $\sigma_0, \dots, \sigma_n$, are indexed by $X = \{0, 1, 2, \dots, n\}$. We set $X^* = \{1, 2, \dots, n\}$, $\mathbb{E}(X) = \{\alpha \subseteq X : |X| \equiv 0 \pmod{2}\}$. Let T be an X -tree with edge set $e(T)$.

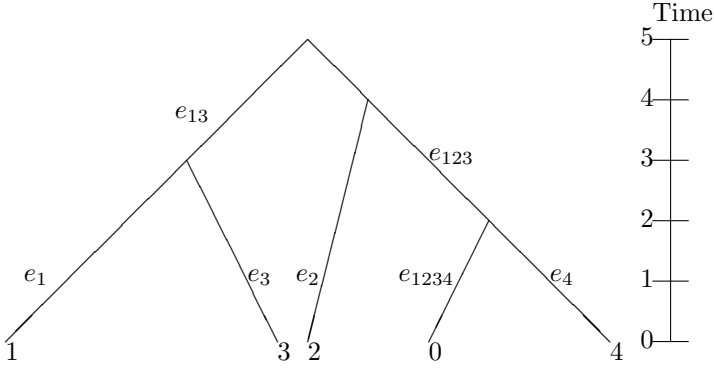


FIG. 6.5. A rooted X -tree T , for $X = \{0, 1, 2, 3, 4\}$. If two-state sequences $\sigma_0, \sigma_1, \sigma_2, \sigma_3, \sigma_4$ at the leaves of T have evolved from a root sequence σ with a rate of 10^{-7} substitutions per site per year, and the time scale shown is in units of 10^6 years, then the edge-length and sequence spectra are given in Example 3.

We assign a probability $p_\alpha < \frac{1}{2}$ to each edge $e_\alpha \in e(T)$, and assume $\sigma_0, \dots, \sigma_n$ have evolved on T under the Neyman model of character substitution, with p_α the probability of that the characters at the endpoints of e_α differ.

We let $q_\alpha = -\frac{1}{2} \ln(1 - 2p_\alpha)$ for each $e_\alpha \in e(T)$, define $q_\emptyset = -\sum_{e_\alpha \in e(T)} q_\alpha$, and set $q_\alpha = 0$ for all remaining $\alpha \subseteq X^*$. The vector $[q_\alpha]_{\alpha \subseteq X^*}$ is called the edge-length spectrum.

For $\alpha \subseteq X^*$, we define s_α to be the probability of the split $\{\alpha, X - \alpha\}$ occurring at a site among the aligned sequences $\sigma_0, \dots, \sigma_n$, and set $\mathbf{s} = [s_\alpha]_{\alpha \subseteq X^*}$ to be the sequence spectrum. For $\alpha, \beta \subseteq X$ let $h_{\alpha\beta} = (-1)^{|\alpha \cap \beta|}$.

The following general properties, given as a series of lemmas, generalize the theory from the specific cases with $n + 1 = 3, 4$ introduced previously. The arguments are developed in the series of papers, [13, 16, 17, 29].

It can be shown that:

Lemma 2

$$H = [h_{\alpha\beta}]_{\alpha, \beta \subseteq X^*} = H_n,$$

is the Sylvester matrix with 2^n rows and columns.

Lemma 3 Given $\beta \in \mathbb{E}(X)$, the path set $\Pi_\beta(T)$ (a set of disjoint paths of T whose endpoints cover β) can be specified by

$$\Pi_\beta(T) = \{e_\alpha \in e(T) \mid h_{\alpha\beta} = -1\}.$$

For $\beta \in \mathbb{E}(X)$, let P_β be the probability that at a site, the number of leaves in β of the leaf set of T coded **R**, is odd. (This also implies that the number of leaves in β coded **Y**, is odd.)

Lemma 4

$$d_\beta = \sum_{e_\alpha \in \Pi_\beta(T)} q_\alpha = -\frac{1}{2} \ln(1 - 2P_\beta). \quad (6.40)$$

For each $\beta \in \mathbb{E}(X)$, d_β is called the length of Π_β . Thus

$$(H_n \mathbf{q})_\beta = \sum_{\alpha \subseteq X^*} h_{\alpha\beta} q_\alpha = \sum_{\alpha \subseteq (X^* - \{\emptyset\})} (h_{\alpha\beta} - 1) q_\alpha = -2 \sum_{e_\alpha \in \Pi_\beta(T)} q_\beta,$$

which implies

Lemma 5

$$(H_n \mathbf{q})_\beta = -2d_\beta. \quad (6.41)$$

Now as $\sum_{\alpha \subseteq X^*} s_\alpha = 1$,

$$(H_n \mathbf{s})_\beta = \sum_{\alpha \subseteq X^*} h_{\alpha\beta} s_\alpha = 1 - \sum_{\alpha \subseteq X^*} (1 - h_{\alpha\beta}) s_\alpha = 1 - 2 \sum_{h_{\alpha\beta} = -1} s_\alpha,$$

so

Lemma 6

$$(H_n \mathbf{s})_\beta = 1 - 2P_\beta.$$

Combining the results of lemmas 4, 5, and 6 we obtain

Lemma 7 $(H_n \mathbf{q})_\beta = \ln((H_n \mathbf{s})_\beta).$

This establishes the general result of Hadamard conjugation for $n + 1$ sequences evolving under the Neyman model.

Theorem 8 $(T, \mathbf{q})$ a weighted X -tree with induced sequence spectrum $\mathbf{s}$, then

$$\begin{aligned} \mathbf{q} &= H_n^{-1} \text{Ln}(\mathbf{H}\mathbf{s}), \\ \mathbf{s} &= H_n^{-1} \text{Exp}(\mathbf{H}\mathbf{q}). \end{aligned}$$

These vector equations can be expressed in terms of the components as:

$$\forall \alpha \subseteq X^*, \quad s_\alpha = 2^{-n} \sum_{\beta \in \mathbb{E}(X)} (-1)^{|\alpha \cap \beta|} \exp \left(\sum_{\gamma \subseteq X^*} (-1)^{|\beta \cap \gamma|} q_\gamma \right), \quad (6.42)$$

$$\forall \gamma \subseteq X^*, \quad q_\gamma = 2^{-n} \sum_{\beta \in \mathbb{E}(X)} (-1)^{|\beta \cap \gamma|} \ln \left(\sum_{\alpha \subseteq X^*} (-1)^{|\alpha \cap \beta|} s_\alpha \right). \quad (6.43)$$

This development has now been extended by Steel and co-workers [29, 33, 34] to a number of more general models of sequence substitution.

6.5 Applications: using the Neyman model

6.5.1 Rate variation

We can calculate the expected sequence spectrum if the sequences have evolved under two or more rate classes. When the sites in each class can be identified,

then they can be analysed independently. If only the sizes of the classes are known, we can still determine the combined edge-length spectrum. For example if x sites have edge-length spectrum $\mathbf{q}^{(1)}$, and y sites have edge-length spectrum $\mathbf{q}^{(2)}$, then the expected sequence spectrum is

$$\mathbf{s} = \frac{x}{x+y}\mathbf{s}^{(1)} + \frac{y}{x+y}\mathbf{s}^{(2)} = H^{-1} \left[\frac{x}{x+y} \text{Exp}(H\mathbf{q}^{(1)}) + \frac{y}{x+y} \text{Exp}(H\mathbf{q}^{(2)}) \right].$$

The Hadamard conjugation can also be extended to some cases where there is a continuous distribution of rates across the sites, such as a Γ distribution, or a mixture of γ and invariant sites. For examples see references [32, 37, 38].

Waddell [35] and Lockhart *et al.* [24] showed that when variation in rates across sites occurs, then a maximum likelihood search using a fixed rates model can be inconsistent.

6.5.2 Invertibility

If $\mathbf{q}^{(1)} = \mathbf{q}^{(1)}(T_1)$ and $\mathbf{q}^{(2)} = \mathbf{q}^{(2)}(T_2)$, assuming no rate heterogeneity, we find

$$\mathbf{s}^{(1)} = H^{-1} \text{Exp}(H\mathbf{q}^{(1)}),$$

$$\mathbf{s}^{(2)} = H^{-1} \text{Exp}(H\mathbf{q}^{(2)}),$$

then

$$\mathbf{s}^{(1)} = \mathbf{s}^{(2)} \iff \mathbf{q}^{(1)} = \mathbf{q}^{(2)} \implies T_1 = T_2,$$

where a tree is defined by its edges with positive edge lengths. Thus any tree can be recovered from its sequence spectrum.

However, Waddell [35] and Baake [1] both constructed examples with rate heterogeneity which showed that it is possible for

$$\frac{x}{x+y}\mathbf{s}^{(1)}(T) + \frac{y}{x+y}\mathbf{s}^{(2)}(T) = \frac{x'}{x'+y'}\mathbf{s}'^{(1)}(T') + \frac{y'}{x'+y'}\mathbf{s}'^{(2)}(T')$$

with $T \neq T'$. In this case two distinct trees can give rise to the same spectrum, and thus given that spectrum, we should not be able to derive the generating tree.

6.5.3 Invariants

For a tree topology $\mathcal{T}$ (i.e. $\mathcal{T}$ is a tree with no values associated with its edges) let $\mathcal{Q}(\mathcal{T})$ be the set of all edge-length spectra on the edges of $\mathcal{T}$, and let

$$\mathcal{S}(\mathcal{T}) = \{\mathbf{s} = H_n^{-1} \text{Exp}(H_n \mathbf{q}) \mid \mathbf{q} \in \mathcal{Q}(\mathcal{T})\}.$$

For the Neyman model $\mathbf{s} \in \mathbb{R}^{2^n}$ is constrained by $\sum s_\alpha = 1$, so appears to have $2^n - 1$ degrees of freedom. However, $\mathbf{s} = H_n^{-1} \text{Exp}(H_n \mathbf{q})$, where $\mathbf{q} = \mathbf{q}(T)$ for some tree T with at most $2n - 1$ edges, each with a single edge length. Thus $\mathbf{s}(T)$ is a function of at most $2n - 1$ parameters for trees $T \in \mathcal{Q}(\mathcal{T})$. Hence there are $2^n - 2n$ constraints on $\mathbf{q}$ corresponding to $q_\beta = 0$, for each $\beta \subseteq X^*$ with $\beta \neq \emptyset$ and $e_\beta \notin e(T)$. Each of these constraints is an “invariant,” a function of the sequence spectrum which is independent of the edge-lengths (but may depend on the choice of $\mathcal{T}$). The study of phylogenetic invariants was introduced by Lake [22], Cavender and Felsenstein [3]. Evans and Speed [6] extended the theory of phylogenetic invariants to the K3ST model.

6.5.4 Closest tree

We cannot expect the observed sequence spectrum $\hat{\mathbf{s}}$ from a finite set of site patterns to estimate the probabilities $\mathbf{s}$ exactly, so we expect $\hat{\mathbf{q}} = H_n^{-1} \text{Ln}(H_n \hat{\mathbf{s}}) \approx \mathbf{q}$ for some $\mathbf{q} \in \mathcal{Q}(\mathcal{T})$ for some topology $\mathcal{T}$, where $\mathcal{Q}(\mathcal{T})$ is the set of all possible edge-length spectra. Lento *et al.* [23] introduced an informative visual display (now referred to as a Lentoplot) which gives a histogram of the largest $\hat{q}_\gamma$ components, ordered by value, together with the sum of the q_δ values for each split δ inconsistent with γ . This is useful in quickly identifying trees strongly supported by the data, and which pairs of splits are in conflict.

For each tree T we can define the “distance” $d(\hat{\mathbf{q}}, T)$ to be

$$\min_{\mathbf{q}(T) \in \mathcal{Q}(\mathcal{T})} |\hat{\mathbf{q}} - \mathbf{q}|.$$

The tree T_c for which $d(\hat{\mathbf{q}}, T_c)$ is minimal is called the “closest tree.” This can be used to select a tree to represent the data. The closest tree method is introduced in reference [14], and generalized by Steel *et al.* [30].

Other methods of fitting $\hat{\mathbf{s}}$ to a model with more desirable statistical properties such as weighted least-squares (WLS) and generalized least-squares (GLS), were introduced by Waddell [35].

6.5.5 Maximum parsimony

Given a set $S = \{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$ of four aligned homologous sequences, and a tree T_α ($T_\alpha \in \{T_{12}, T_{13}, T_{23}\}$, see Fig. 6.4), the “Fitch length” $F(T_\alpha, S)$ is the minimum number of substitutions required for T_α to span S . We find $F(T_\alpha, S)$ is a function of T_α and $\mathbf{s}$, so we can write $F(T_\alpha, \mathbf{s})$ for $F(T_\alpha, S)$. It is easily shown that

$$F(T_{12}, \mathbf{s}) = s_1 + s_2 + s_{12} + s_3 + 2s_{13} + 2s_{23} + s_{123},$$

$$F(T_{13}, \mathbf{s}) = s_1 + s_2 + 2s_{12} + s_3 + s_{13} + 2s_{23} + s_{123},$$

$$F(T_{23}, \mathbf{s}) = s_1 + s_2 + 2s_{12} + s_3 + 2s_{13} + s_{23} + s_{123}.$$

Let

$$K(\mathbf{s}) = s_1 + s_2 + 2s_{12} + s_3 + 2s_{13} + 2s_{23} + s_{123},$$

then

$$F(T_\alpha, \mathbf{s}) = K(\mathbf{s}) - s_\alpha. \quad (6.44)$$

The principle of maximum parsimony [11] selects the tree T_α for which $F(T_\alpha, \mathbf{s})$ is minimal, as the MP tree. In the case of four sequences, the MP tree is T_α where s_α is maximal among $\{s_{12}, s_{13}, s_{23}\}$. When we are given $(T_\alpha, \mathbf{q})$, $\mathbf{s} = H_3^{-1} \text{Exp}(H_3 \mathbf{q})$, the MP tree is selected by comparing s_{12} , s_{13} , and s_{23} .

6.5.6 Parsimony inconsistency, Felsenstein's example

In his classic 1978 paper “Cases in which parsimony or compatibility methods will be positively misleading,” Joseph Felsenstein [8] showed that parsimony is statistically inconsistent, meaning that there are examples of sequence data $\mathbf{s} = H_3^{-1} \text{Exp}(H_3 \mathbf{q})$ generated on a phylogenetic tree T_α for which the MP principle will select a tree $T_\beta \neq T_\alpha$ with increasing probability as sampling error diminishes. In reference [8] he derived a quadratic bounding function for the Fitch length to find specific examples where $(T, \mathbf{q})$ with $s_{23} < s_{12}$, for $\mathbf{s} = H_3^{-1} \text{Exp}(H_3 \mathbf{q}(T_{23}))$ under the Neyman model. As his examples required severe violation of the molecular clock hypothesis, he speculated that for “reasonable data,” inconsistency might not be a problem.

In this example let:

$$x = q_1 = q_2 = -\frac{1}{2} \ln(1 - 2P), \quad X = e^{-2x} = 1 - 2P,$$

$$y = q_3 = q_{23} = q_{123} = -\frac{1}{2} \ln(1 - 2Q), \quad Y = e^{-2y} = 1 - 2Q.$$

Felsenstein considered sequences generated on the weighted tree $T = T_{23}$ similar to that of Fig. 6.6, where the edge weights are $-\frac{1}{2} \ln(1 - 2P)$ for edges e_1 and e_2 , and $-\frac{1}{2} \ln(1 - 2Q)$ for the edges e_3 , e_{23} , and e_{123} . Then

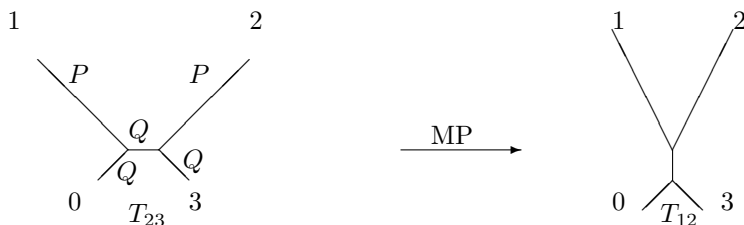


FIG. 6.6. Example of inconsistency of MP. If $P^2 > Q$ then MP will select T_{12} from data generated on T_{23} . This is an example of “long edge attraction,” MP would prefer the tree T_{12} which groups together the two long edges.

we find

$$\mathbf{q} = \begin{bmatrix} -2x - 3y \\ x \\ x \\ 0 \\ y \\ 0 \\ y \\ y \end{bmatrix}, \quad H_3 \mathbf{q} = -2 \begin{bmatrix} 0 \\ x + y \\ x + 2y \\ 2x + y \\ 3y \\ x + 2y \\ x + 2y \\ 2x + 2y \end{bmatrix}, \quad \text{Exp}(H_3 \mathbf{q}) = \begin{bmatrix} 1 \\ XY \\ XY^2 \\ X^2Y \\ Y^3 \\ XY^2 \\ XY^2 \\ X^2Y^2 \end{bmatrix}.$$

so

$$\mathbf{s} = H_3^{-1} \text{Exp}(H_3 \mathbf{q}) = \frac{1}{8} \begin{bmatrix} 1 & 2 & 2 & 1 & 1 & 1 \\ 1 & 0 & 0 & -1 & 1 & -1 \\ 1 & 0 & 0 & -1 & 1 & -1 \\ 1 & -2 & -2 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & -1 & -1 \\ 1 & -2 & 2 & -1 & -1 & 1 \\ 1 & 2 & -2 & -1 & -1 & 1 \\ 1 & 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ XY \\ XY^2 \\ X^2Y \\ Y^3 \\ X^2Y^2 \end{bmatrix}.$$

Hence we find

$$\begin{aligned} s_{12} &= \frac{1}{8}(1 - 2XY - 2XY^2 + X^2Y + Y^3 + X^2Y^2), \\ s_{13} &= \frac{1}{8}(1 - 2XY + 2XY^2 - X^2Y - Y^3 + X^2Y^2), \\ s_{23} &= \frac{1}{8}(1 + 2XY - 2XY^2 - X^2Y - Y^3 + X^2Y^2). \end{aligned}$$

Thus

$$\begin{aligned} 8(s_{23} - s_{13}) &= 4XY - 4XY^2 = 4XY(1 - Y) = 8XYQ > 0, \\ 8(s_{23} - s_{12}) &= 4XY - 2X^2Y - 2Y^3 = 2Y(2X - X^2 - Y^2). \end{aligned}$$

Now noting

$$\begin{aligned} 2X - X^2 - Y^2 &= 2(1 - 2P) - (1 - 2P)^2 - (1 - 2Q)^2 \\ &= 4(Q - P^2 - Q^2) \leq -4Q^2, \quad \text{when } P^2 > Q, \end{aligned}$$

we find

$$F(T_{23}, \mathbf{s}) \geq F(T_{12}, \mathbf{s}) \iff s_{12} \geq s_{23} \iff P^2 \geq Q(1 - Q).$$

Thus, in the example of Fig. 6.6, parsimony is inconsistent as soon as $P^2 > Q(1 - Q)$.

Felsenstein hinted that MP inconsistency might be a consequence of molecular clock violation. Theorem 8 allows us to test this for the Neyman model on four sequences. A binary tree on four leaves can be rooted either on the internal edge or on a pendant edge (as shown in Fig. 6.7). For each of these two trees let

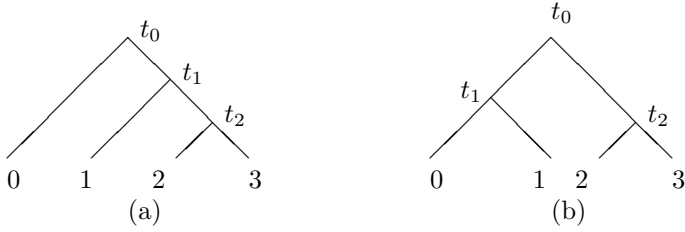


FIG. 6.7. The two possible ways of placing a root on tree T_{23} , with times t_1 , t_2 , and t_3 as shown. If a common substitution rate λ is applied to each edge, then the generated sequence data will satisfy the molecular clock.

λ be the common rate of nucleotide substitution. We denote the corresponding edge-length spectra for each tree as $\mathbf{q}^{(a)}$ and $\mathbf{q}^{(b)}$, and find

$$\mathbf{q}^{(a)} = \lambda \begin{bmatrix} -2t_0 - t_1 - t_2 \\ t_1 \\ t_2 \\ 0 \\ t_2 \\ 0 \\ t_1 - t_2 \\ 2t_0 - t_1 \end{bmatrix}, \quad \mathbf{q}^{(b)} = \lambda \begin{bmatrix} -2t_0 - t_1 - t_2 \\ t_1 \\ t_2 \\ 0 \\ t_2 \\ 0 \\ 2t_0 - t_1 - t_2 \\ t_1 \end{bmatrix}.$$

From these spectra we derive

$$\text{Exp}(H_3 \mathbf{q}^{(a)}) = \begin{bmatrix} 1 \\ X \\ X \\ Y \\ X \\ Y \\ Z \\ XZ \end{bmatrix}, \quad \text{Exp}(H_3 \mathbf{q}^{(b)}) = \begin{bmatrix} 1 \\ Y \\ X \\ X \\ X \\ X \\ Z \\ YZ \end{bmatrix}.$$

where $X = e^{-2\lambda t_0}$, $Y = e^{-2\lambda t_1}$, and $Z = e^{-2\lambda t_2}$. In each case $t_0 > t_1, t_2$, so $X < Y, Z$, and in (a) $t_1 > t_2$ so $Y < Z$. In both cases we find $s_{12} - s_{13} = 0$. In (a) we find $s_{12} - s_{23} = Y - Z < 0$ and in (b) $s_{12} - s_{23} = 2X - Y - Z < 0$. Hence in both cases

$$F(T_{12}, \mathbf{s}) = F(T_{13}, \mathbf{s}) > F(T_{23}, \mathbf{s}),$$

which proves Felsenstein's conjecture for four sequences.

6.5.7 Parsimony inconsistency, molecular clock

Here we demonstrate, that even if the sequences have evolved under a molecular clock hypothesis, it is possible, when $n + 1 = 5$, for MP to be inconsistent,

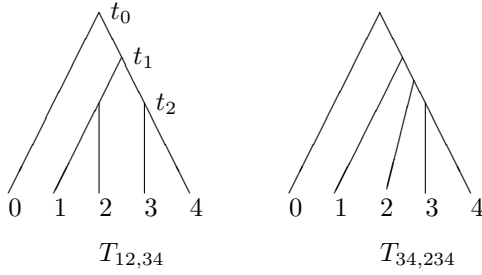


FIG. 6.8. An example where MP can be inconsistent under a molecular clock.

If the substitution rate is $\lambda = 0.05$ on the tree $T_{12,34}$ with the times (before present) set to $t_0 = 20, t_1 = 8, t_2 = 7$, then the Fitch length $F(T_{12,34}, \mathbf{s}) > F(T_{34,234}, \mathbf{s})$, so MP (which selects the tree with minimal Fitch length) will not select the generating tree $T_{12,34}$.

(as first noted by Hendy and Penny [15]). In particular let $T_{12,34}$ and $T_{34,234}$ be the trees of Fig. 6.8. We find comparing Fitch lengths

$$F(T_{12,34}, \mathbf{s}) - F(T_{34,234}, \mathbf{s}) = s_{234} - s_{12}.$$

Now with a rate λ of substitutions per site per unit time, the components of $\mathbf{d}$ are

$$\begin{aligned} d_{\emptyset} &= 0, \\ d_{01} &= d_{02} = d_{03} = d_{04} = 2\lambda t_0, \\ d_{13} &= d_{23} = d_{14} = d_{24} = 2\lambda t_1, \\ d_{12} &= d_{34} = 2\lambda t_2, \\ d_{0123} &= d_{0124} = d_{0134} = d_{0234} = 2\lambda(t_0 + t_2), \\ d_{1234} &= 4\lambda t_2. \end{aligned}$$

Hence setting $X = e^{-4\lambda t_0}$, $Y = e^{-4\lambda t_1}$, and $Z = e^{-4\lambda t_2}$, the components of $\text{Exp}(H_4(-2\mathbf{d}))$ are

$$\begin{aligned} e^{-2d_{\emptyset}} &= 1, \\ e^{-2d_{01}} &= e^{-2d_{02}} = e^{-2d_{03}} = e^{-2d_{04}} = X, \\ e^{-2d_{13}} &= e^{-2d_{23}} = e^{-2d_{14}} = e^{-2d_{24}} = Y, \\ e^{-2d_{12}} &= e^{-2d_{34}} = Z, \\ e^{-2d_{0123}} &= e^{-2d_{0124}} = e^{-2d_{0134}} = e^{-2d_{0234}} = XZ, \\ e^{-2d_{1234}} &= Z^2. \end{aligned}$$

Thus as $\mathbf{s} = \frac{1}{16} H_4 \text{Exp}(H_4(-2\mathbf{d}))$ we find

$$s_{12} = \frac{1}{16}[1 - 4Y + 2Z + Z^2], \quad s_{234} = \frac{1}{16}[1 - 2X + 2XZ - Z^2].$$

In particular, if we set $\lambda = 0.05$, $t_1 = 20$, $t_2 = 11$, and $t_3 = 10$, we calculate (to 4 decimal places) $X = 0.0183$, $Y = 0.1108$, and $Z = 0.1353$. From these we find

$$s_{12} = 0.0529 < s_{234} = 0.0594,$$

which implies for these parameters, $F(T_{12,34}, \mathbf{s}) > F(T_{34,234}, \mathbf{s})$. Thus the generating tree $T_{12,34}$ cannot be the MP tree. This example illustrates that MP is not necessarily consistent under the molecular clock.

To determine the MP tree in this example, we would need to calculate the Fitch lengths of each of the 15 possible binary trees on X . When we do this we discover that there are four trees $T_{12,123}$, $T_{12,124}$, $T_{34,134}$, and $T_{34,234}$, each with equal minimal Fitch length. These are the trees where the long edge from 0 is “attracted” to one of the other (long) pendant edges.

6.5.8 Maximum likelihood under the Neyman model

Felsenstein [9, 10], introduced maximum likelihood as a tool for selecting the “most likely” phylogeny, given some sequence data, and a model for their evolution. We can derive some formulae to describe the likelihood function, given an observed sequence spectrum $\hat{\mathbf{s}}$ and an hypothesized edglength spectrum $\mathbf{s}(T)$. Given an observed sequence spectrum $\hat{\mathbf{s}}$ for a set of taxa $X = \{0, 1, \dots, n\}$, and a weighted X -tree $(T, \mathbf{q})$, the likelihood of $\hat{\mathbf{s}}$ being derived from $T, \mathbf{q}$ is

$$L(\hat{\mathbf{s}} \mid T, \mathbf{q}) = \prod_{\alpha \subseteq X^*} s_{\alpha}^{\hat{s}_{\alpha}},$$

where

$$s_{\alpha} = 2^{-n} \sum_{\beta \in \mathbb{E}(X)} h_{\alpha\beta} e^{-2d_{\beta}}, \quad d_{\beta} = -\frac{1}{2} \sum_{\gamma \subseteq X^*} h_{\beta\gamma} q_{\gamma}.$$

We can derive formulae for the partial derivatives with respect to the set of independent generators $\{q_{\gamma} \mid e_{\gamma} \in e(T)\}$, noting all other $q_{\gamma} = 0$, except for $q_{\emptyset} = -\sum_{e_{\gamma} \in e(T)} q_{\gamma}$. Hence for each $\gamma \mid e_{\gamma} \in e(T)$,

$$\begin{aligned} \frac{\partial d_{\beta}}{\partial q_{\gamma}} &= -\frac{1}{2}(h_{\beta\gamma} - h_{\emptyset\gamma})q_{\gamma} = \frac{1}{2}(1 - h_{\beta\gamma})q_{\gamma}, \\ \frac{\partial s_{\alpha}}{\partial q_{\gamma}} &= 2^{-n} \sum_{\beta \in \mathbb{E}(X)} h_{\alpha\beta}(h_{\beta\gamma} - 1)e^{-2d_{\beta}} \\ &= s_{\alpha \triangle \gamma} - s_{\alpha}. \end{aligned}$$

Hence

$$\frac{\partial L}{\partial q_{\gamma}} = L \sum_{\alpha \subseteq X^*} \frac{\hat{s}_{\alpha}}{s_{\alpha}} \frac{\partial s_{\alpha}}{\partial q_{\gamma}} = L \sum_{\alpha \subseteq X^*} \frac{\hat{s}_{\alpha}}{s_{\alpha}} (s_{\alpha \triangle \gamma} - s_{\alpha}). \quad (6.45)$$

Noting that $\sum_{\alpha \in X^*} \hat{s}_\alpha = 1$, and that all terms in equation (6.45) for which $\hat{s}_\alpha = 0$ will vanish, equation (6.45) can be rewritten as

$$\frac{\partial L}{\partial q_\gamma} = L \sum_{\alpha | \hat{s}_\alpha \neq 0} \frac{\hat{s}_\alpha}{s_\alpha} s_{\alpha \triangle \gamma} - L = L(f_\gamma - 1), \quad (6.46)$$

where $f_\gamma = \sum_{\alpha | \hat{s}_\alpha \neq 0} (\hat{s}_\alpha / s_\alpha) s_{\alpha \triangle \gamma}$.

We find

$$\frac{\partial f_\gamma}{\partial q_\delta} = f_{\gamma \triangle \delta} - \sum_{\alpha | \hat{s}_\alpha \neq 0} \hat{s}_\alpha \frac{s_{\alpha \triangle \gamma} s_{\alpha \triangle \delta}}{s_\alpha^2},$$

so the second derivatives

$$\begin{aligned} \frac{\partial^2 L}{\partial q_\gamma \partial q_\delta} &= \frac{\partial L(f_\gamma - 1)}{\partial q_\delta} \\ &= L(f_\gamma - 1)(f_\delta - 1) + L \left(f_{\gamma \triangle \delta} - \sum_{\alpha | \hat{s}_\alpha \neq 0} \hat{s}_\alpha \frac{s_{\alpha \triangle \gamma} s_{\alpha \triangle \delta}}{s_\alpha^2} \right) \\ &= \frac{1}{L} \frac{\partial L}{\partial q_\gamma} \frac{\partial L}{\partial q_\delta} + \frac{\partial L}{\partial q_{\gamma \triangle \delta}} + L \left(1 - \sum_{\alpha | \hat{s}_\alpha \neq 0} \hat{s}_\alpha \frac{s_{\alpha \triangle \gamma} s_{\alpha \triangle \delta}}{s_\alpha^2} \right). \end{aligned} \quad (6.47)$$

In particular we observe, for any binary tree T , that if T has edges e_γ and e_δ , then $\gamma \triangle \delta$ is also an edge of T . Hence at a turning point of $L(\hat{\mathbf{s}} \mid \mathbf{s}(T, \mathbf{q}))$, $\partial L / \partial q_\alpha = 0$ for each edge e_α of $e(T)$, and the second derivatives are

$$\frac{\partial^2 L}{\partial q_\gamma \partial q_\delta} = L \left(1 - \sum_{\alpha | \hat{s}_\alpha \neq 0} \hat{s}_\alpha \frac{s_{\alpha \triangle \gamma} s_{\alpha \triangle \delta}}{s_\alpha^2} \right).$$

If $(T, \mathbf{q})$ is “balanced” (the edge lengths q_α , for each edge e_α are of similar size), and the data $\hat{\mathbf{s}}$ “fits $(T, \mathbf{q})$ well” ($\hat{\mathbf{s}} \approx \mathbf{s}(T, \mathbf{q})$), then at a turning point

$$\frac{\partial^2 L}{\partial q_\gamma \partial q_\delta} = L \left(1 - \sum_{\alpha | \hat{s}_\alpha \neq 0} \hat{s}_\alpha \frac{s_{\alpha \triangle \gamma} s_{\alpha \triangle \delta}}{s_\alpha^2} \right) \approx L \left(1 - \sum_{\alpha | \hat{s}_\alpha \neq 0} \frac{\hat{s}_{\alpha \triangle \gamma} \hat{s}_{\alpha \triangle \delta}}{\hat{s}_\alpha} \right).$$

If $\hat{s}_\gamma \approx \hat{s}_\delta \approx \hat{s}_{\gamma \triangle \delta}$ and $\hat{s}_\emptyset \gg 0.5$, then $\hat{s}_\emptyset \hat{s}_{\gamma \triangle \delta} (1/\hat{s}_\gamma + 1/\hat{s}_\delta) > 1$, so in that case

$$\frac{\partial^2 L}{\partial q_\gamma \partial q_\delta} < L \left(1 - \sum_{\alpha = \gamma, \delta} \frac{\hat{s}_{\alpha \triangle \gamma} \hat{s}_{\alpha \triangle \delta}}{\hat{s}_\alpha} \right) < 0$$

and the turning point is a local maximum. (To create a turning point which is not a local maximum, these conditions have to be strongly violated.) This gives support for confidence in ML as a tree selection method, provided the data fits the ML tree “closely.”

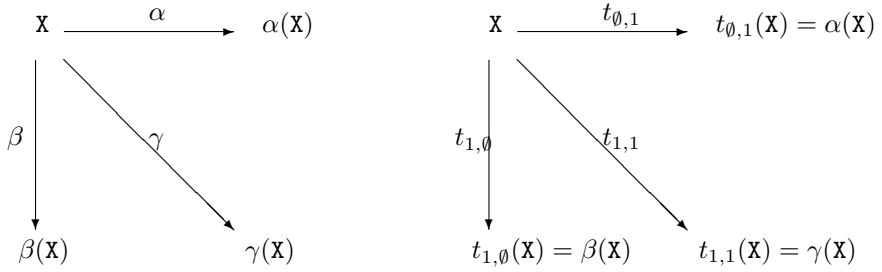


FIG. 6.9. For $X \in \{A, C, G, T(U)\}$, the figure shows the effect of each of the 3 types of substitution.

6.6 Kimura's 3-substitution types model

In this section, we illustrate how the equations for Neyman's model can be extended to give similar equations for Kimura's K3ST model.

6.6.1 One edge

For a single edge (connecting vertices 0, 1) we found the relationship between the three expected numbers of substitutions q_α , q_β , q_γ and the three probabilities of differences at the endpoints of the edge to be (equation (6.11))

$$\mathbf{q} = H_2^{-1} \text{Ln}(H_2 \mathbf{s}), \text{ which inverts to } \mathbf{s} = H_2^{-1} \text{Exp}(H_2 \mathbf{q}),$$

where

$$\mathbf{q} = \begin{bmatrix} q_\emptyset \\ q_\alpha \\ q_\beta \\ q_\gamma \end{bmatrix} \quad \text{and} \quad \mathbf{p} = \begin{bmatrix} p_\emptyset \\ p_\alpha \\ p_\beta \\ p_\gamma \end{bmatrix}$$

with $q_\emptyset = -q_\alpha - q_\beta - q_\gamma$ and $p_\emptyset = 1 - p_\alpha - p_\beta - p_\gamma$. We saw that these can also be expressed as

$$Q = H_1^{-1} \text{Ln}(H_1 P H_1) H_1^{-1}, \text{ which inverts to } P = H_1^{-1} \text{Exp}(H_1 Q H_1) H_1^{-1},$$

where

$$Q = \begin{bmatrix} -q & q_\alpha \\ q_\beta & q_\gamma \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} p & p_\alpha \\ p_\beta & p_\gamma \end{bmatrix}.$$

Before we extend the analysis to more than 2 sequences we will introduce a change in notation, indexing the rows and columns of P and Q by the sets $\emptyset$ and $\{1\}$ (which will usually be written as "1" when used as a subscript). Thus we write

$$Q = \begin{bmatrix} q_{\emptyset\emptyset} & q_{\emptyset 1} \\ q_{1\emptyset} & q_{11} \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} p_{\emptyset\emptyset} & p_{\emptyset 1} \\ p_{1\emptyset} & p_{11} \end{bmatrix},$$

where $q_{\emptyset\emptyset} = q_\emptyset$, $q_{\emptyset 1} = q_\alpha$, $q_{1\emptyset} = q_\beta$ and $q_{11} = q_\gamma$, and $p_{\emptyset\emptyset} = p_\emptyset$, $p_{\emptyset 1} = p_\alpha$, $p_{1\emptyset} = p_\beta$ and $p_{11} = p_\gamma$.

6.6.2 *K3ST for $n + 1$ sequences*

Substitution types It can be shown [29] that for any tree on $n + 1$ leaves:

$$Q = H_n^{-1} \text{Ln}(H_n P H_n) H_n^{-1},$$

$$P = H_n^{-1} \text{Exp}(H_n Q H_n) H_n^{-1},$$

with P and Q suitably defined matrices of 2^n rows and columns indexed by the subsets of $\{1, 2, \dots, n\}$.

For $n + 1 = 3$ sequences we find $4^n = 16$ relative site patterns by listing the differences between the characters of the reference sequence σ_0 and those of σ_i , $i = 1, 2$ at any site.

Example 4 $n + 1 = 3$, $X = \{0, 1, 2\}$, $X^* = \{1, 2\}$. The tree T on these leaves has edges e_1, e_2, e_{12} . The expected numbers of substitutions $q_\alpha, q_\beta, q_\gamma$ can be independently chosen for each edge. The entries in Q are arranged as follows: $Q_{\emptyset,1} = q_\alpha(e_1)$; $Q_{\emptyset,2} = q_\alpha(e_2)$; $Q_{\emptyset,12} = q_\alpha(e_{12})$; $Q_{1,\emptyset} = q_\beta(e_1)$; $Q_{2,\emptyset} = q_\beta(e_2)$; $Q_{12,\emptyset} = q_\beta(e_{12})$; $Q_{1,1} = q_\gamma(e_1)$; $Q_{2,2} = q_\gamma(e_2)$; $Q_{12,12} = q_\gamma(e_{12})$. $Q_{\emptyset,\emptyset}$ is set to -1 times the total number of substitutions (of all types) over all edges of T . All the remaining entries are set to 0, then

$$Q = \begin{bmatrix} -q & q_\alpha(e_1) & q_\alpha(e_2) & q_\alpha(e_{12}) \\ q_\beta(e_1) & q_\gamma(e_1) & 0 & 0 \\ q_\beta(e_2) & 0 & q_\gamma(e_2) & 0 \\ q_\beta(e_{12}) & 0 & 0 & q_\gamma(e_{12}) \end{bmatrix}.$$

Note that the non-zero entries lie on the leading row, column or main diagonal. The rows and columns are indexed by the subsets of $X^* = \{1, 2\}$ in the order $\emptyset, \{1\}, \{2\}, \{1, 2\}$. The $a \subseteq X^* - \{\emptyset\}$ entries of the leading row, main diagonal and leading column are $q_\alpha(e_a)$, $q_\beta(e_a)$, and $q_\gamma(e_a)$ respectively. Setting $q(e_x) = -\sum_{e_x \in E(T)} q_\alpha(e_x)$ then the leading entry of Q is $Q_{\emptyset,\emptyset} = q(e_a) + q(e_b) + q(e_c)$. The remaining entries are all set to 0.

In particular if we set

$$\begin{aligned} q_\alpha(e_1) &= 0.01, & q_\beta(e_1) &= 0.02, & q_\gamma(e_1) &= 0.03, \\ q_\alpha(e_2) &= 0.04, & q_\beta(e_2) &= 0.05, & q_\gamma(e_2) &= 0.06, \\ q_\alpha(e_3) &= 0.07, & q_\beta(e_3) &= 0.08, & q_\gamma(e_3) &= 0.09, \end{aligned}$$

then we have

$$Q = \begin{bmatrix} -0.45 & 0.01 & 0.04 & 0.07 \\ 0.02 & 0.03 & 0 & 0 \\ 0.05 & 0 & 0.06 & 0 \\ 0.08 & 0 & 0 & 0.09 \end{bmatrix}.$$

□

Site Patterns In general we identify 4^n site patterns. The observed frequencies can be recorded in a $2^n \times 2^n$ matrix S , with the following convention. Suppose the nucleotides at a site (in the order of $[\sigma_0, \sigma_1, \sigma_2, \sigma_3, \dots, \sigma_n]^t$) are $[\mathbf{A}, \mathbf{C}, \mathbf{A}, \mathbf{G}, \dots, \mathbf{T}]^t$. At any given site we determine the n “differences” from the reference sequence σ_0 .

TABLE 6.2. Four sample sequences $\sigma_0, \sigma_1, \sigma_2$, and σ_3 , each of length 7, together with the three sequences of differences, and the site patterns (a, b) where $a, b \subseteq \{1, 2, 3\}$

σ_0 :	C	A	T	C	C	A	A
σ_1 :	C	C	T	A	C	C	A
σ_2 :	G	G	T	C	T	A	T
σ_3 :	G	T	T	G	T	C	T
$\sigma_1 - \sigma_0$:	00	11	00	11	00	11	00
$\sigma_2 - \sigma_0$:	10	01	00	00	01	00	10
$\sigma_3 - \sigma_0$:	10	10	00	10	01	11	10
a :	$\{2, 3\}$	$\{1, 3\}$	$\emptyset$	$\{1, 3\}$	$\emptyset$	$\{1, 3\}$	$\{2, 3\}$
b :	$\emptyset$	$\{1, 2\}$	$\emptyset$	$\{1\}$	$\{1, 2\}$	$\{1, 3\}$	$\emptyset$

These are the substitutions required to transform the character at the reference sequence to the corresponding characters of each other sequence. In this case these differences are $[11, 00, 01, \dots, 10]^t$. (The substitution $A \rightarrow C$ is identified by placing $X = A$, and noting $C = 11(A)$, giving the first entry 11, etc.) Then the list of n binary pairs is identified by a pair (a, b) of subsets of X^* , where $a = \{1, \dots, n\}$ is the set of sequences with 1 in the first entry, and $b = \{1, 3, \dots\}$ is the set of sequences with 1 in the second entry. Then the matrix $S = [s_{ab}]_{a, b \subseteq X^*}$ is the matrix with s_{ab} recording the frequency of observing site pattern (a, b) .

Example 5 The sample sequences of Table 6.2 give the 8×8 sequence spectrum matrix

$$S = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Note the entry 2 in row $\{2, 3\}$, column $\emptyset$, counts the repeated pattern $(\{2, 3\}, \emptyset)$ of sites 1 and 7. $\square$

Edge-length and Sequence Spectra (K3ST) $X = \{0, 1, 2, \dots, n\}$, $X^* = \{1, 2, \dots, n\}$, T an X -tree, with each edge $e_a \in e(T)$ having three length parameters $\alpha(a)$, $\beta(a)$, $\gamma(a)$, which are the expected number of α , β , and γ type substitutions across e_a . The “edge-length spectrum” is the matrix

$Q = [q_{ab}]_{a,b \subseteq X^*}$ of 2^n rows and columns indexed by the subsets of X^* with:

$$\begin{aligned} q_{\emptyset a} &= \alpha(a), \quad \forall e_a \in e(T), \\ q_{a\emptyset} &= \beta(a), \quad \forall e_a \in e(T), \\ q_{aa} &= \gamma(a), \quad \forall e_a \in e(T), \\ q_{\emptyset\emptyset} &= - \sum_{e_a \in e(T)} (\alpha(a) + \beta(a) + \gamma(a)), \\ q_{ab} &= 0 \text{ otherwise.} \end{aligned}$$

The “sequence spectrum” is the matrix $S = [s_{ab}]_{a,b \subseteq X^*}$ of 2^n rows and columns indexed by the subsets of X^* , with s_{ab} being the probability of observing the site pattern (a, b) , for $a, b \subseteq X^*$.

As $H_n = [h_{ab}]_{a,b \subseteq X^*}$ and $h_{ab} = (-1)^{|a \cap b|}$ then

$$Q = H_n^{-1} \text{Ln}(H_n P H_n) H_n^{-1}, \quad \text{and} \quad P = H_n^{-1} \exp(H_n Q H_n) H_n^{-1}$$

can be expressed as

$$q_{ab} = 4^{-n} \left[\sum_{c,d \in \mathbb{E}(X)} (-1)^{(|a \cap c| + |b \cap d|)} \ln \left(\sum_{e,f \subseteq X^*} (-1)^{|c \cap e| + |d \cap f|} s_{e,f} \right) \right] \quad (6.48)$$

$\forall a, b \subseteq X^*,$

and

$$s_{ab} = 4^{-n} \left[\sum_{c,d \in \mathbb{E}(X)} (-1)^{(|a \cap c| + |b \cap d|)} \exp \left(\sum_{e,f \subseteq X^*} (-1)^{|c \cap e| + |d \cap f|} q_{e,f} \right) \right], \quad (6.49)$$

$\forall a, b \subseteq X^*.$

If the probabilities s_{ab} are estimated from the observed frequency $\hat{s}_{ab}$ we suppose that $S \approx \hat{S}$ and presume $Q \approx \hat{Q}$, where the entries $\hat{q}_{ab}$ are from entering $\hat{s}_{ab}$ in equation (6.48).

6.7 Other applications and perspectives

In this chapter, we have introduced a few of the potential applications of Hadamard conjugation to understanding phylogenetics. Here we will give a brief description of some other applications, and indicate directions for further research.

Hadamard conjugation provides a mechanism for simulation studies. Samples can be drawn from the expected sequence spectrum $\mathbf{s}$ to provide an observed sequence spectrum $\hat{\mathbf{s}}$. Charleston *et al.* [4] used this approach to examine biases of various tree building methods to variations in sequence length and tree topology. Holland *et al.* [18] undertook an extensive study showing some inaccuracies of tree building methods for data generated under a molecular clock, in particular with the “outgroup” method of locating the root.

In 1994 Steel [28] used Theorem 8 to give a pathological example showing that it is possible for the maximum likelihood function to have more than one

maximum point, which means the standard “hill climbing” algorithm for locating a maximum cannot guarantee to find the optima. In a simulation study Rogers and Swofford and co-workers [27] suggested that this could largely be overcome using multiple random starting points. Chor *et al.* [5] used Hadamard conjugation to obtain examples with infinite sets of multiple optima.

Other applications of Hadamard conjugation to explore the statistical geometry of tree space and the relationships between tree selection processes were developed by Waddell and co-workers [26, 35, 36, 39]. However, many open problems remain, for example is it possible to find the conditions under which the likelihood function has a unique maximum? Is it possible to extend this analysis to more complex models of nucleotide substitution?

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PHYLOGENETIC NETWORKS

Katharina T. Huber and Vincent Moulton

Phylogenetic networks are a generalization of phylogenetic trees that permit the representation of conflicting signal or alternative phylogenetic histories. Networks are clearly useful when the underlying evolutionary history is non-treelike. Recombination, hybridization, lateral gene transfer can all lead to histories that are not adequately modelled by a single tree. Moreover, even in case the underlying history *is* treelike, phenomena such as parallel evolution, model heterogeneity, and sampling error can make it difficult to represent the history by a single tree. In such situations networks can provide a useful tool for representing ambiguity or for simultaneously visualizing a collection of feasible trees. In this chapter, we will review some methods for network reconstruction that are based on the representation of bipartitions or *splits* of the data set in question. As we shall see, these methods are based on a theoretical foundation that naturally generalizes the theory of phylogenetic trees.

7.1 Introduction

Phylogenetic networks are a generalization of phylogenetic trees that permit the representation of conflicting signal or alternative phylogenetic histories. Networks rather than trees are clearly useful when the underlying evolutionary history is non-treelike. Recombination, hybridization, lateral gene transfer can all lead to histories that are not adequately modelled by a single tree. Moreover, even in case the underlying history *is* treelike, phenomena such as parallel evolution, model heterogeneity, and sampling error can make it difficult to represent the history by a single tree. In such situations networks can provide a useful tool for representing ambiguity or for simultaneously visualizing a collection of feasible trees.

Several network methods are available for constructing phylogenetic networks—see [44] for a recent review. In this chapter we will review some methods for network reconstruction that are based on the representation of bipartitions or *splits* of the data set in question (see, for example, Fig. 7.1). As we shall see, these methods have the advantage that they are based on a theoretical foundation that naturally includes the theory of phylogenetic trees, although the networks that they produce require some effort to interpret. Network methods

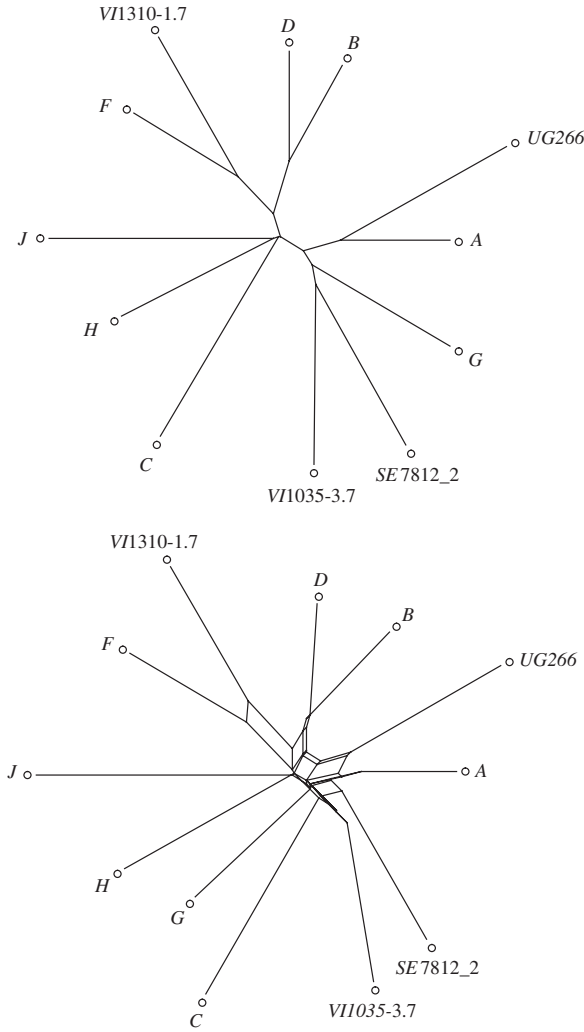


FIG. 7.1. The Neighbor Joining tree [45] and a splits graph [37] for the data set presented in Section 14.6.4 of [46]. This data set consists of HIV virus DNA sequences. The labels UG268, VI1310-1.7, VI1035-3.7, and SE7812.2 all correspond to viruses that are known to be recombinants, whereas the remaining labels correspond to non-recombinant viruses. The topology of the network indicates which recombination might have occurred, which is not easy to deduce by looking at the tree. For example, UG268 is known to be a recombinant of the viruses corresponding to the A and D labels. Both the tree and the splits graph are a graphical representation of a collection of splits. For example, the split that partitions the taxa F, VI1310-1.7 from the rest of the taxa is represented by a branch in the Neighbor Joining tree and by two parallel edges in the splits graph.

not based on splits—such as reticulograms [39] and ancestral recombination graphs [48]—will not be reviewed here.

Although the phylogenetic networks we consider here may be constructed in various ways, in essence they can all be considered as being the end result of two main steps. First, using properties of the data set in question, a collection of splits of the data is derived (usually together with some weight for each split representing its relative support) that hopefully reflects pertinent relationships between the species being studied. Second, using the splits (and associated weights), a phylogenetic network or *splits graph* is constructed that provides a visualization of these splits. For example, the splits graph in Fig. 7.1 is a graphical representation of a collection of splits, where each collection of parallel edges represents a split of the taxa. Note that these two steps are independent: splits may be derived in various ways (using, for example, characters, distances, or other functions of the data), and these may be represented using different networks. However, as we shall see, the choice of which combination of methods is used to derive the network is usually guided by factors such as how the resulting networks are to be interpreted or best visualized.

We now describe the contents of this chapter. In general, network methods can be roughly divided into character and distance based methods (some attempts have been made to develop likelihood based methods, but these have met with limited success due to, for example, difficulties in performing computations and formulating of appropriate models cf. [52]). Accordingly, we divide up this chapter into two parts. In Sections 7.2 and 7.3 we describe median networks and related constructions. These character based constructions are mainly used to analyse intraspecific data. In Section 7.4 we present consensus networks, a generalization of consensus trees that allow the representation of collections of trees using median networks.

In the rest of the chapter we consider some distance-based methods for network construction. In Section 7.5 we discuss how to quantify the treelikeness of a distance matrix using networks constructed on quartets. In Section 7.6 we describe splits graphs, phylogenetic networks that generalize the networks on quartets described in Section 7.5. Finally, in Section 7.7 we present neighbour-net, a method for constructing splits graphs that extends the popular Neighbor Joining (NJ) method for constructing phylogenetic trees.

7.2 Median networks

We begin by considering *median networks*, a class of phylogenetic networks that is regularly used to study intraspecies data. Although these may be directly inferred from collections of splits, we will introduce them using a more intuitive construction. We will then indicate their relationship to splits.

Following the approach in [7], we use an example to illustrate the construction. Starting with a DNA sequence alignment, we remove all constant columns and positions containing more than two character states. This means that we could lose some information depending on the number of such columns

TABLE 7.1. For an alignment in (a), the recoded alignment is given in (b) (see text for details)

Taxa	(a) Alignment	(b) Coding
<i>a</i>	GAGGTTGCCGCCGTA	1111111 (s ₁)
<i>b</i>	AGGGCCGCAGTAGCT	0110110 (s ₂)
<i>c</i>	GAGGTCACCACCATT	1110001 (s ₃)
<i>d</i>	GATGTCGCCGCCGCT	1010110 (s ₄)
<i>e</i>	GAGATCGACACCGCT	1100100 (s ₅)
		6122211

there are in the alignment. Suppose the resulting alignment is as pictured in Table 7.1(a).

This alignment is then recoded into binary (i.e. 0,1) sequences as follows: an arbitrary reference sequence is chosen—in our example *a*—and recoded as the sequence of length fifteen all of whose entries are 1. Next, for each of the remaining sequences *s* we create a binary sequence whose *i*th position ($1 \leq i \leq 15$) will be 1 if the *i*th position of *s* agrees with the *i*th position of the reference sequence, and 0 otherwise. In this way a vertex in the 15-dimensional hypercube is associated with each sequence. Finally, starting from the left of the newly created aligned binary sequences, all repeated columns are removed and the total number of times that each repeat has occurred is recorded, thus reducing the dimension of the hypercube considered without reducing the information contained in the data. The resulting binary sequences appear in the rows of Table 7.1(b); underneath each column the number of times that this column is repeated is recorded.

Now, for any three of the binary sequences *x*, *y*, *z* in Table 7.1(b), their *median* *med*(*x*, *y*, *z*) is computed. This is defined to be the binary sequence of length seven that has value in its *i*th position equal to the majority of values *x*[*i*], *y*[*i*], *z*[*i*], where *s*[*i*] denotes the symbol at the *i*th position of a binary sequence *s*. For example, the sequence 1110110 is the median of sequences **s**₁, **s**₂, and **s**₄ in Table 7.1(b). The sequence *med*(*x*, *y*, *z*) may be regarded as a hypothetical ancestral sequence for the three sequences *x*, *y*, *z*. Based on this interpretation, it is reasonable not to restrict the construction of medians to the original sequences, but to compute medians also using the newly generated hypothetical ancestral sequences. If this process is iteratively applied to all triplets formed from newly generated sequences as well as the original sequences, then it must terminate after a (possibly very large) finite number of steps. The resulting set of binary sequences is called the *median closure* of the original set. It is easy to check that the median closure of the sequences appearing in Table 7.1(b) consists of these sequences together with the following four: **s**₆ = 1110110, **s**₇ = 1110100, **s**₈ = 1110111, and **s**₉ = 1110101.

It is now a simple matter to define the median network associated to the original set of sequences. Its vertex set is the median closure, and two vertices

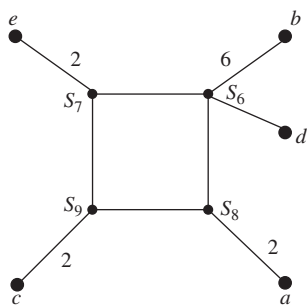


FIG. 7.2. The median network for the data set in Table 7.1(b). For clarity we have indicated which vertices correspond to the sequences $s_5, \dots, s_9$. Note that collections of parallel edges in this network are in one-to-one correspondence with the columns in Table 7.1(b). For example, the two horizontal parallel edges of the square represent column 6, since their removal results in two connected graphs labelled by c, e and a, b, d .

(i.e. sequences) are defined to be adjacent whenever they differ in exactly one position. In Fig. 7.2, we present the median network for the sequences in Table 7.1(b). The labelled vertices (represented by large dots) correspond to (the sequences representing) the taxa; unlabelled vertices (which are represented by smaller dots) correspond to the remaining sequences in the median closure. The weights appearing next to the edges represent the number of times that the column is repeated (columns that are not repeated give rise to no number next to the corresponding edge).

The procedure for generating the median network in Fig. 7.2 can be applied to any set of equally long binary sequences of fixed length. Median networks have been studied for some time within mathematics, where they appear in the setting of *median algebras* (cf., for example, references [4, 8]). They were introduced (in various guises) as a tool for phylogenetic analysis by Guénoche [29], Barthélemy [10] (see also [11]), and Bandelt [1]. Subsequently, they have been extensively employed for the analysis of intraspecific data (cf., for example, [7]). They often run into problems when the level of diversity increases because the networks become too complicated. We will discuss this in the next section. Note that a number of ways have been described for constructing and characterizing median networks (cf., for example, [7, 10, 36]), and that there are programs which allow their automatic construction (e.g. SplitsTree4 [37] and Spectronet [34]).

The median network associated to a given set X of length n binary sequences has several interesting properties, some of the more important of which are:

- (1) The network is necessarily connected, and contains (as subnetworks) all most parsimonious trees for X [7].
- (2) The network is a tree if and only if any two columns of the given binary sequences are *compatible*, that is if, for any two columns i and j , only

- three of the four possible patterns 11, 10, 01, 00 occur for $\mathbf{x}[i]\mathbf{x}[j]$, for $\mathbf{x}$ any sequence in X (see, for example, [10]).
- (3) The network is a hypercube of dimension n if and only if any pair of columns are *incompatible*, that is, not compatible (see, for example, [10, 20]).
 - (4) More specifically, k -cubes contained in the network, where $k \leq n$, are in bijective correspondence with subcollections of pairwise incompatible columns with cardinality k (see, for example, [40]).

As mentioned in the introduction, all phylogenetic networks that we consider in this chapter can be constructed using splits. We now indicate why this is the case for median networks. Consider the median network pictured in Fig. 7.2. As noted in this figure, the columns of the binary sequences in Table 7.1(b) correspond to collections of parallel edges in this network. Now, removing such a collection will result in two connected networks each labelled by the elements in a part of some split of the taxa. For example, if we remove the collection of parallel edges corresponding to column 6, this gives two connected networks labelled by $\{a, b, d\}$ and $\{c, e\}$. Moreover, the split $\{\{a, b, d\}, \{c, e\}\}$ corresponds precisely to the split of the taxa set induced by the pattern of 0's and 1's in column 6.

In general, given a set of taxa X , we can associate a median network directly to any collection of splits of X (e.g. see [20]). This network will represent the collection of splits in that certain collections of “parallel” edges will be in one-to-one correspondence with the splits, in that the removal of any such collection will result in two networks labelled by the parts of the corresponding split. Moreover, Properties (2)–(4) of median networks listed near the end of Section 7.2 can be translated into the language of splits as follows.

Suppose that X is a finite set, and denote a split of X into two parts A and B by $A \mid B$. For short, we call a collection of splits a *split system* (on X). Given a phylogenetic tree with leaves labelled by X , each edge of the tree naturally gives rise to a split; by removing the edge we obtain two trees, each one being labelled by the elements in one part of a split of X (much in the same way as with the network as we just described above). We shall say that a phylogenetic tree *displays* a split if there is an edge in the tree that gives rise to the split, and we shall say that two splits are *compatible* if there is a phylogenetic tree that displays both splits, otherwise we call them *incompatible*.

Then, in this terminology, Property (2) states that the median network corresponding to a split system is a (phylogenetic) tree if and only if every pair of splits in the split system is compatible (in fact, it also follows that in this case, the median network is the necessarily unique phylogenetic tree corresponding to the split system cf. [47]). Moreover, Property (3) states that the median network corresponding to a split system is a hypercube if and only if every pair of splits in the system is incompatible, and Property (4) states that subsets consisting of k pairwise incompatible splits are in bijective correspondence with k -cubes in the network. For this reason, median networks can become quite complex to

visualize in case there is a high degree of incompatibility in the data. In the next section, we present some possible solutions to this problem.

7.3 Visual complexity of median networks

Although we have observed that in some cases the median network can be a tree, we have also seen that—at the opposite extreme—it can also be a hypercube of dimension equal to the length of the binary sequences involved. Hence median networks can be very complex and highly interconnected, in which case the network may not shed much light on the phylogenetic relationships in the data. For this reason, various methods have been proposed for reducing the complexity of median networks, while at the same time attempting to preserve their representation of underlying phylogenetic signals (cf., for example, reference [7] for a method that uses haplotype and column frequency arguments to resolve reticulations). For the purposes of illustration, we briefly present an approach to complexity reduction that was introduced in reference [36].

Suppose that $\mathcal{M} = \mathcal{M}_X$ denotes the median closure of a set X of binary sequences of length n , obtained as described in the previous section. Then it can be easily seen that a binary sequence $\mathbf{s}$ of length n is contained in $\mathcal{M}$ if and only if it satisfies the following property:

(P_2) : For any pair of positions i_1 and i_2 of $\mathbf{s}$, there is a sequence in X that agrees with $\mathbf{s}$ in position i_1 and i_2 .

For a binary sequence $\mathbf{s}$ denote by $\bar{\mathbf{s}}$ the binary sequence with $\bar{\mathbf{s}}[i] \in \{0, 1\} - \mathbf{s}[i]$, for all $i = 1, \dots, n$. Then property (P_2) is naturally generalized to allow the exclusion of binary sequences $\mathbf{s}$ through the consideration of p -tuples of sequence positions, $p \geq 2$.

(P_p) : For any p -tuple of positions $i_1, i_2, \dots, i_p$ of $\mathbf{s}$, there is either a sequence in X that agrees with $\mathbf{s}$ in every position $i_1, i_2, \dots, i_p$, or no sequence in X agrees with $\bar{\mathbf{s}}$ in every position $i_1, i_2, \dots, i_p$.

Clearly, every sequence in X satisfies the first alternative in (P_p) , $p \geq 2$ and so will not be removed. The second alternative in the (P_p) condition tries to retain those sequences that have some support from the sequences in X (and actually arises from a rather abstract description of a complex that can be associated median networks—cf. reference [21] for more details). For example, if X consists of the six binary sequences 000, 011, 110, 101, 100, and 010, then the median closure of these sequences consists of the eight possible binary sequences of length three. However, the medians 001 and 111 do not satisfy (P_3) since $\overline{001} = 110$ and $\overline{111} = 000$ are both sequences in X .

Now put $\mathcal{M}_0 := \mathcal{M}$ and, for $p \geq 1$, recursively define $\mathcal{M}_p$ to consist of all the vertices in $\mathcal{M}_{p-1}$ that, in addition, satisfy property (P_{p+2}) . Then we obtain a filtration of the vertices in the median closure $\mathcal{M}$:

$$\mathcal{M}_0 = \mathcal{M} \supseteq \mathcal{M}_1 \supseteq \mathcal{M}_2 \supseteq \dots \supseteq \mathcal{M}_{n-2}. \quad (7.1)$$

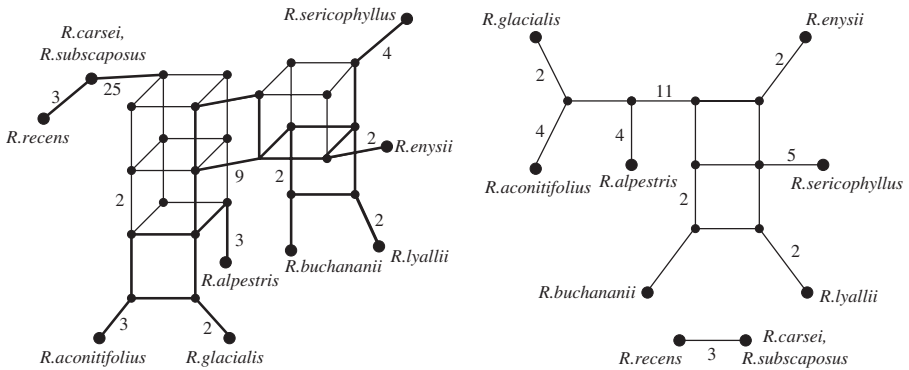


FIG. 7.3. A median network and its corresponding pruned median network G_1 for a data set presented in [36]. This set consisted of DNA sequences obtained from buttercups, where hybridization is believed to have occurred.

The effect of this filtering of $\mathcal{M}$ at step p will be the removal of those sequences from the set $\mathcal{M}$ of potential hypothetical ancestral sequences which exhibit certain i -tuple differences with elements in X for each $2 \leq i \leq p$.

As with median networks, we define the *pruned median network* G_p to be the network with vertex set $\mathcal{M}_p$, and edge set consisting of those pairs of sequences in $\mathcal{M}_p$ that differ in exactly one position. Thus, since $\mathcal{M}_0 = \mathcal{M}$, we see that G_0 is the median network associated to X and, by the set inclusions given in equation (7.1), the pruned median networks provide a hierarchy of subnetworks of the median network. In Fig. 7.3 (left), we present the median network together with the pruned median network G_1 (pictured with bold edges) for a data set presented in reference [35]. Note that the pruned median network is not necessarily connected (though we shall still consider it as a phylogenetic network). Also, unlike median networks, it is possible that certain collections of parallel edges in the pruned median network will no longer represent splits of the data (e.g. the collection of vertical parallel edges in the square consisting of bold edges that connects two 3-cubes in Fig. 7.3 (left)). As explained in reference [35], this can be remedied by recomputing the median network(s) corresponding to the split systems induced by the pruned median network on the subsets of taxa labelling its connected components. We present this network in Fig. 7.3 (right).

Before concluding our discussion on median networks, we note that related network methods include the *netting* method [27] and *statistical parsimony* [53]. The netting and statistical parsimony constructions are both rule based procedures. Netting considers the Hamming distance between aligned binary sequences and constructs a weighted network by first joining two sequences of minimal distance by a weighted edge and then stepwise extending this network by greedily adding in taxa, one at a time, so that at each step in the construction, the Hamming distance between any two already processed taxa equals the graph theoretical distance of these taxa in the network obtained so far. Statistical

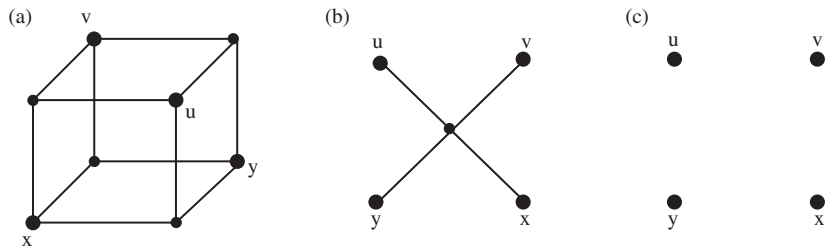


FIG. 7.4. (a) The median, (b) netting, and (c) statistical parsimony networks associated to the binary sequences $\mathbf{u} = 010$, $\mathbf{v} = 001$, $\mathbf{x} = 111$, and $\mathbf{y} = 100$. All edges in the graphs have weight one.

parsimony also proceeds iteratively, using rules that rely on similarities between pairs of haplotypes as well as a probabilistic criterion that reflects the confidence in creating parsimony links between haplotype pairs. The median, netting, and statistical parsimony constructions are strongly related although in general they will not yield the same network (see Fig. 7.4).

Finally, we note that median networks can be generalized so as to construct networks from non-binary sequences (as can the netting and statistical parsimony methods). The resulting *quasi-median networks* [8] have the advantage of retaining information that can be lost in the recoding process described above. However, in general they are much more complex than median networks (see [5] for some mathematical reasons for this explosion in complexity). The *median-joining* method [6] for phylogenetic analysis, that is closely related to netting, employs distance techniques to extract phylogenetic information from quasi-median networks.

7.4 Consensus networks

Quite often phylogenetic methods produce a collection of trees rather than some point estimate of the best tree, since such an estimate with no measure of reliability may not be particularly informative. Examples of methods producing collections of trees include Monte Carlo Markov Chain (MCMC) methods and bootstrapping.

Large collections of trees can be difficult to interpret and draw conclusions from. Thus, when faced with such a collection, it is common practice to construct a *consensus tree*, that is, a tree that attempts to reconcile the information contained within all of the trees. Many ways have been devised for constructing consensus trees (see reference [13] for a recent overview). However, they all suffer from a common limitation: by summarizing all of the given trees by a single output tree, information about conflicting hypotheses is necessarily lost. In this section, we briefly review an approach for visualizing collections of trees utilizing phylogenetic networks that is presented in reference [32], and extends a method that was proposed by Bandelt in reference [2].

As mentioned at the end of Section 7.2, the complexity of visualizing the median network associated to a split system is directly related to the degree of incompatibility in the split system. This is true for phylogenetic networks in general. Hence it is useful to quantify this incompatibility as follows. For k a positive integer, we say that a split system is k -compatible if it contains no subset of $k + 1$ splits that is pairwise incompatible. Clearly, every pair of splits in a k -compatible split system is compatible if and only if $k = 1$, in which case its associated median network is a tree. However, for larger values of k , the associated median network can become progressively more complex. The concept of k -compatibility was introduced and studied in reference [22], and has led to some fascinating mathematical results in extremal set theory (see, for example, [23]). For example, it is well-known that a 1-compatible split system on a set X of cardinality n contains at most $2n - 3$ splits. This result was generalized in reference [24], where it is shown that a 2-compatible split system on X contains at most $4n - 10$ splits (a bound that is, in fact, tight). In general, a k -compatible split system on X contains at most $n(1 + k \log_2(n))$ splits [22]. Hence, for low values of n and k the number of splits in a k -compatible split system on X will not be very large, again making the associated median network (or phylogenetic network) easier to visualize.

We now introduce the concept of a consensus network. Given a collection of phylogenetic trees, two common methods for computing a consensus tree are the *strict consensus* method, which outputs the tree displaying only those splits that are displayed by all of the input trees, and the *majority-rule* consensus method, which outputs the tree displaying only those splits that are displayed in more than half of the input trees. Thus, these two methods can be viewed as members of a one-parameter family of consensus methods which associates a split system S_x to a collection of phylogenetic trees consisting of those splits that are displayed by more than a proportion x of the trees (for strict consensus $x = 1$, and for majority-rule $x = \frac{1}{2}$).

In case x is less than $\frac{1}{2}$, it may no longer be possible to associate a tree to the split system S_x , as S_x may contain some pairs of incompatible splits. However, it is still possible to represent S_x by a phylogenetic network. We call any such network a *consensus network*. Since we have introduced median networks, we will consider the median network associated to S_x . As we pointed out above this network can be quite complex. However, the following attractive property of S_x , that was presented in reference [32], gives a way in which to control this complexity.

Theorem 7.1 *Suppose that we are given N phylogenetic trees and, for $0 < x \leq 1$, that S_x denotes the split system containing those splits that are displayed in $\lceil Nx \rceil$ or more of these trees. Then S_x is $(\lfloor 1/x \rfloor)$ -compatible.*

Thus, for instance, if we only accept splits that appear in more than $\frac{1}{4}$ of the input trees, then $S_{1/4}$ will be 4-compatible, so that, by property (4) of median networks in Section 7.2, the associated median network is guaranteed to contain cubes only of dimension 4 or less.

We conclude this section by presenting an example that illustrates the utility of consensus networks. An MCMC analysis of 37 mamalian mitochondrial sequences was performed under a general time-reversible model with gamma distributed rates across sites to generate a chain of 1,000,000 trees. Of these every hundredth tree was recorded, and the first half of these trees was discarded to provide for a burn in period, leaving 5000 trees in our collection.

Figure 7.5 shows consensus networks corresponding to the split systems S_x for $x = 1$ and $\frac{1}{10}$. As we have explained, collections of parallel edges in the networks are in one-to-one correspondence to splits in S_x . In this figure the length of the edges corresponding to some split in S_x is proportional to the proportion of trees which induce that split (here we use lengths as opposed to weighting the edges as in, for example, Fig. 7.2). In an MCMC analysis the proportion of times a split is induced by a tree in the chain is interpreted as its posterior probability of being induced by the true tree, hence the length of the edges in the network are proportional to their posterior probability. Note that all the pendant edges have posterior probability 1, as they necessarily appear in all of the trees in the collection.

7.5 Treelikeness

In this section, we consider a quartet-based method for evaluating the treelikeness of a distance. As we shall see, this approach has a natural interpretation in terms of phylogenetic networks. It also provides the basis of a more general method for deriving networks from distances that we will present in Section 7.6.

Suppose that X is a set of taxa, and that d is a *distance* on X , that is, an assignment of putative *genetic distances* $d_{xy} \geq 0$ to pairs of elements x, y in X that satisfies $d_{xx} = 0$ and $d_{xy} = d_{yx}$, for all $x, y \in X$. For any four elements x, y, u, v in X , put

$$d_{xy|uv} = d_{xy} + d_{uv}.$$

Then a quartet $q = \{x, y, u, v\}$ in X satisfies the *four-point condition* if the larger two of the three quantities $d_{xy|uv}, d_{xu|yv}, d_{xv|yu}$ are equal. As is well known, d can be represented by a weighted tree with leaves labelled by X (by taking shortest paths between leaves) if and only if every quartet $q = \{x, y, u, v\}$ of X satisfies this condition [17, 55].

In case the distance d is derived from biological data, it will almost never satisfy the four-point condition. Thus, assuming $d_{xy|uv} \leq d_{xu|yv} \leq d_{xv|yu}$ holds, it is natural to consider the ratio

$$\delta = \delta_q = \begin{cases} \frac{d_{xv|yu} - d_{xu|yv}}{d_{xv|yu} - d_{xy|uv}}, & \text{if } d_{xv|yu} - d_{xy|uv} \neq 0, \\ 0, & \text{else,} \end{cases}$$

as a quantification of how far q deviates from being a tree: a value of 0 indicates that q is perfectly treelike, and progressively higher values (up to a maximum value of 1) that it is less and less so.

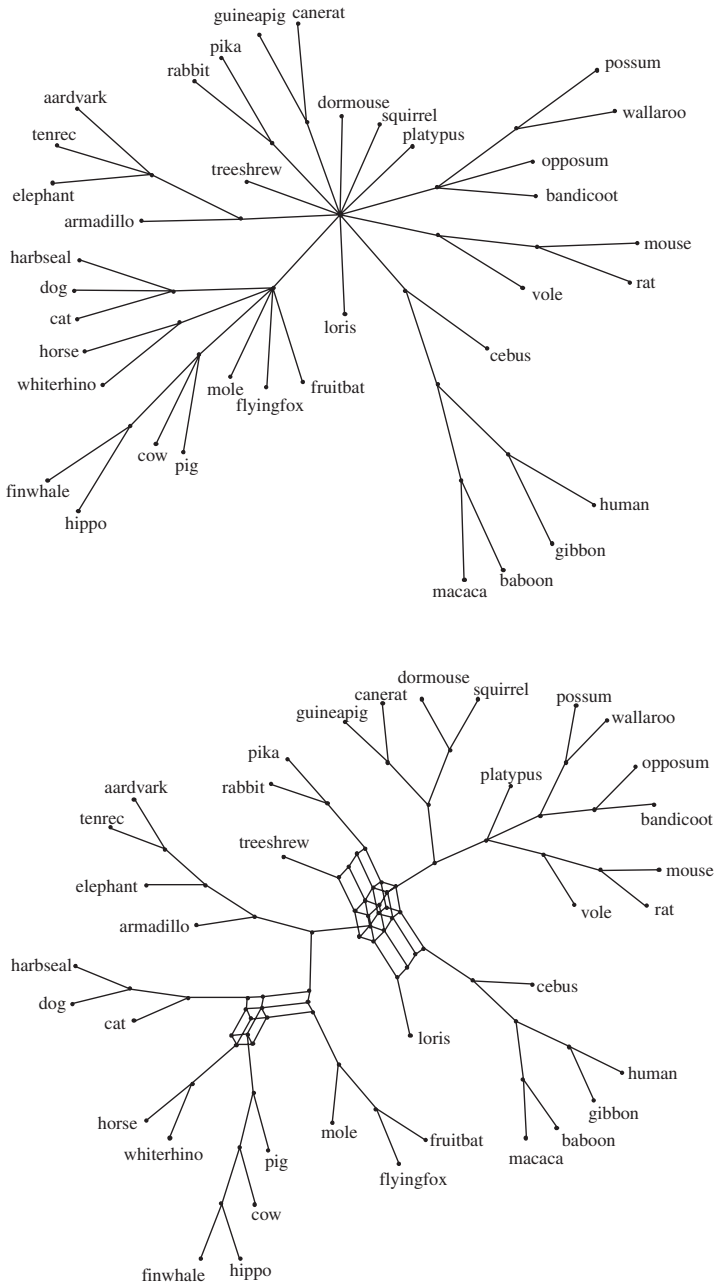


FIG. 7.5. The strict consensus tree, that is, the consensus network with $x = 1$ (top) and the consensus network with $x = \frac{1}{10}$ for the MCMC analysis described in the text.

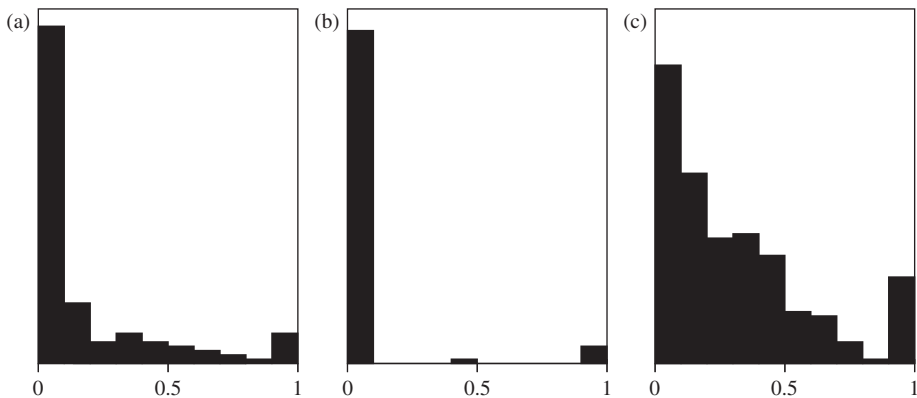


FIG. 7.6. Three δ -plots corresponding to a distance derived from fragment length polymorphism for 42 *Candida albicans* isolates. The x -axis denotes δ -values, whereas the y -axis denotes the number of quartets having δ -values within the indicated range. (a) δ -plot for the complete data set of 42 isolates. (b) δ -plot for a subset of 26 isolates that is suspected to have a treelike evolutionary history. (c) δ -plot for the 16 isolates not in this subset, which are suspected to have a non-treelike evolutionary history. See [32] for more details.

The measure δ for treelikeness was introduced within *statistical geometry* [26]—see [42] for a review. It was also studied in reference [31], where δ -plots were introduced. In such a plot, the δ values for all quartets are displayed in a histogram. The “shape” of the δ -plot (corresponding to the distribution of the δ -values) serves as an indicator of the treelikeness of data set in question (see Fig. 7.6).

In case the distance d is a *metric* on X , that is, it satisfies the triangle inequality $d_{xy} \leq d_{xz} + d_{zy}$ for all $x, y, z \in X$, its restriction to any quartet $q = \{x, y, u, v\}$ of X can be represented by a simple phylogenetic network as pictured in Fig. 7.7. As can be easily seen, $\delta_q = s/l$ in case $l \neq 0$. Hence the degree of the treelikeness of q corresponds to the shape of the rectangle in this network: if δ is small, the rectangle will be long and thin (and so the network will look more treelike), whereas if δ is large the rectangle will be almost a square, and so the network will be less treelike (unless the rectangle is small relative to the length of the pendant edges, in which case the network will approximate a tree with the star topology).

Quartet-mappings [43] (adapted from *likelihood-mappings* [51]) exploit these facts to provide another way to visualize the treelikeness of a distance function. These mappings are constructed as follows. On four taxa there are precisely three fully resolved topologies T_1, T_2, T_3 . Given a set of taxa X and a quartet q of X , a support σ_i is computed for each of the three possible trees T_i on q , $1 \leq i \leq 3$. This support can be either the likelihood of the sequences given the tree, a measure that is used in likelihood-mapping [51], or it can be computed using parsimony

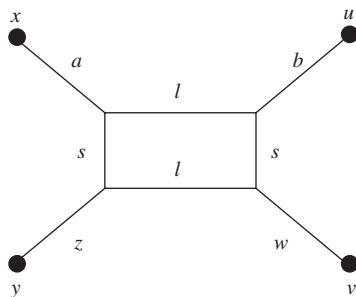


FIG. 7.7. Any distance d restricted to a quartet $q = \{x, y, u, v\}$ with $d_{xv|yu} \geq d_{xu|yv} \geq d_{xy|uv}$ can be represented by the network above, where $a = (d_{xy} + d_{xu} - d_{yu})/2$, $b = (d_{xu} + d_{uv} - d_{xv})/2$, $w = (d_{yv} + d_{uv} - d_{yu})/2$, $z = (d_{xy} + d_{yv} - d_{xv})/2$, $l = (d_{xv} + d_{yu} - d_{xy} - d_{uv})/2$, and $s = (d_{xv} + d_{yu} - d_{xu} - d_{yv})/2$. Note that by construction s and l are non-negative while a , b , w , z are non-negative if and only if d satisfies the triangle inequality.

or distance techniques [43]. A relative support s_i is also computed for each tree T_i , $i = 1, 2, 3$, that is defined by

$$s_i = \frac{\sigma_i}{\sigma_1 + \sigma_2 + \sigma_3}.$$

In particular, $0 \leq s_i \leq 1$ and $s_1 + s_2 + s_3 = 1$.

The main idea behind quartet-mappings is to represent the relative support values s_1, s_2, s_3 as a vector in two-dimensional space (which can be achieved since the three components s_i are dependent). In the quartet-mapping each vector is represented by a point in an equilateral triangle using a barycentric coordinate system (Fig. 7.8). For instance, the three vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, correspond to the tree topologies T_1 , T_2 , and T_3 respectively, giving rise to the three vertices of the triangle, whereas the vector $(1/3, 1/3, 1/3)$, assigning equal weight to all three quartet trees (corresponding to the star tree), gives rise to the central point of the triangle. For an alignment of n sequences, there are $\binom{n}{4}$ possible quartets of sequences, so that a complete quartet-mapping diagram contains $\binom{n}{4}$ points which provide an intuitive picture of how the sequences might have evolved [51].

Besides being of use for analysing treelikeness, quartet-mappings have also been used to analyse the extent of lateral gene transfer [19], and they are also employed in a new tool for visual recombination detection, VisRD [49].

7.6 Deriving phylogenetic networks from distances

In Section 7.5, we saw that we could uniquely associate a phylogenetic network to any metric on four-points. We now see how this can be extended to metrics in general. Suppose that X is a set of taxa and d is a metric on X . We first use d to

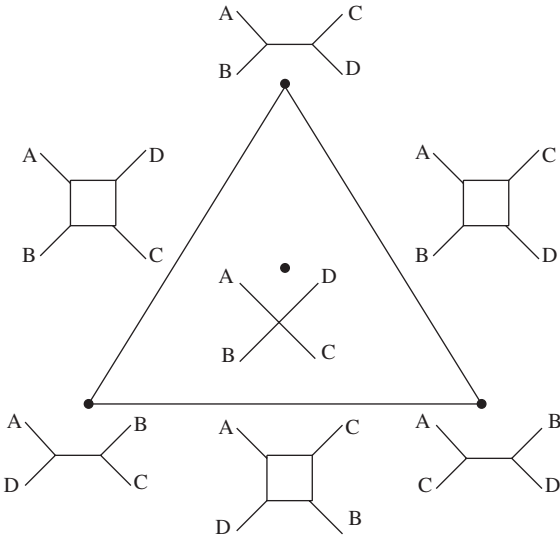


FIG. 7.8. A quartet-mapping. Points in the triangle represent the relative supports for quartet topologies. A point near to one of the vertices of the triangle implies high support for the corresponding tree topology, whereas a point near the edge indicates that a network better supports the data (figure adapted from reference [43, Fig. 1]).

derive a collection of weighted splits of X , and then associate a phylogenetic network to this collection. As mentioned in the introduction, these two steps are independent, and may be performed using different techniques. However, for the purposes of illustration, we begin by presenting the split–decomposition method for deriving collections of weighted splits.

To any quartet of points x, y, u, v in X , associate the quantity

$$\alpha_{xy|uv} = \frac{1}{2}[\max\{d_{xu} + d_{yv}, d_{xv} + d_{yu}, d_{xy} + d_{uv}\} - d_{xy} - d_{uv}].$$

Note that this quantity is precisely the length l of the two horizontal parallel edges of the network presented in Fig. 7.7. Now, for any split $A \mid B$ of X , associate the *isolation index* $\alpha_{A|B}$, which is defined as

$$\alpha_{A|B} = \min_{x,y \in A, u,v \in B} \alpha_{xy|uv}. \tag{7.2}$$

We will be concerned with the collection of splits $\mathcal{S}_d^+$ consisting of all splits $A \mid B$ of X with $\alpha_{A|B} > 0$, that is, the collection of splits having positive isolation index.

The isolation index of a split was introduced by Bandelt and Dress [3] as part of the *split–decomposition* method. This is part of a rich theory concerning finite metric spaces (sometimes called T-theory [25]), and we will not go into the full

details of split-decomposition here. However, we note that the collection $\mathcal{S}_d^+$ has the following useful properties that are proven in [3]:

- If d satisfies the four-point condition, then the isolation index of each split in $\mathcal{S}_d^+$ will be precisely the length of the corresponding edge in the unique tree corresponding to d .
- The collection $\mathcal{S}_d^+$ is *weakly-compatible*, that is, for every three splits S_1, S_2, S_3 in $\mathcal{S}_d^+$, for all $A_i \in S_i$ ($i = 1, 2, 3$), one of the four intersections

$$A_1 \cap A_2 \cap A_3, \quad \overline{A_1} \cap \overline{A_2} \cap A_3, \quad \overline{A_1} \cap A_2 \cap \overline{A_3}, \quad A_1 \cap \overline{A_2} \cap \overline{A_3}$$

is empty. Note that every collection of compatible splits must be weakly compatible.

- The number of splits in $\mathcal{S}_d^+$ is bounded by $\binom{|X|}{2}$, and $\mathcal{S}_d^+$ can be computed efficiently (see, for example, reference [12] for an $O(|X|^5)$ algorithm).

Now, once $\mathcal{S}_d^+$ has been computed, we can represent $\mathcal{S}_d^+$ by a phylogenetic network. We could use a median network, but, as we have seen in Section 7.3, such networks can become quite complex depending on the level of incompatibility between the splits in $\mathcal{S}_d^+$. Moreover, for biological examples (such as the one presented in Fig. 7.1) it has been observed that the collection of splits $\mathcal{S}_d^+$ quite often has a special property that allows for a less complex network representation. In particular, $\mathcal{S}_d^+$ is quite often *circular*, that is, there is an ordering $x_1, x_2, \dots, x_n$ of X such that every split in $\mathcal{S}_d^+$ is of the form $\{x_i, x_{i+1}, \dots, x_j\} \mid (X - \{x_i, \dots, x_j\})$ for some i and j satisfying $1 \leq i \leq j < n$.

Geometrically, circular collections of splits arise when we place the taxa around a circle and consider the splits given by cutting the circle along a line. Dress and Huson (personal communication) have proven that circular collections of splits can always be represented by a planar splits graph (see Fig. 7.9). As with median networks, collections of parallel edges in such phylogenetic networks correspond to splits. Moreover, in case the splits are weighted, the length of the

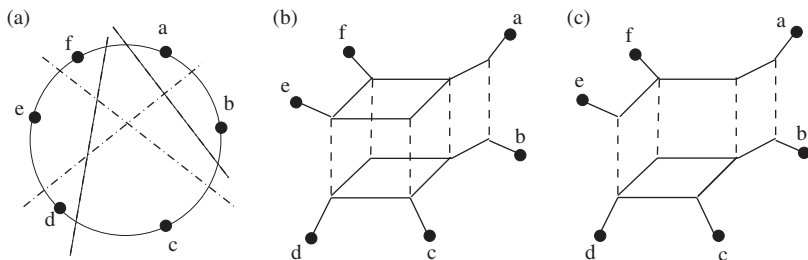


FIG. 7.9. (a) A circular collection of splits of the set $\{a, b, \dots, e\}$ in which splits are represented by dashed lines. (b) The median network representing the collection of splits in (a). (c) A planar splits graph that also represents the splits in (a).

edges are usually drawn with length proportional to the weight of the split to which they correspond. The phylogenetic network in Fig. 7.1 is such a splits graph.

It should be noted that a splits graph gives an approximate representation of the distance d . In particular, the distance between any two taxa in X is approximated by the length of a shortest path between these two taxa in the splits graph (which equals the sum of the isolation indices of the splits in $\mathcal{S}_d^+$ which separate these taxa). For this reason a *fit index* is usually associated to the splits graph that represents the proportion of d that is represented by the graph and is given by

$$\frac{\sum_{x,y \in X} l_{xy}}{\sum_{x,y \in X} d_{xy}},$$

where l_{xy} is the length of the shortest path in the graph between elements x and y of X . It follows from the theory of the split-decomposition that this index always lies between 0 and 1.

The split-decomposition method has a systematic bias which may cause problems with the estimation of edge lengths in splits graphs. In particular, the computation of the isolation index of a split using equation (7.2) involves taking a minimum over quartets that are induced by the split. Thus, one quartet can greatly influence the value of the isolation index, and can lead to under-estimates of edge lengths.

One way to adjust for this problem is to generalize least squares estimation of branch lengths for trees to networks (for more details and examples see [54]). Suppose that the splits in $\mathcal{S}_d^+$ are numbered $1, 2, \dots, m$ and that the taxa in X are numbered $1, 2, \dots, n$. Let $\mathbf{A}$ be the $n(n-1)/2 \times m$ matrix with rows indexed by pairs of taxa, columns indexed by splits, and entry $\mathbf{A}_{(ij)k}$ given by

$$\mathbf{A}_{(ij)k} = \begin{cases} 1, & \text{if } i, j \text{ are separated by split } k, \\ 0, & \text{otherwise.} \end{cases}$$

The matrix $\mathbf{A}$ is the network equivalent of the standard topological matrix for a tree (Chapter 1, this volume). If we represent the distance d by an $n(n-1)/2$ dimensional vector

$$\mathbf{d} = (d_{12}, d_{13}, \dots, d_{(n-1)n})^T,$$

then the corresponding vector of network distances is $\mathbf{A}\mathbf{b}$ where $\mathbf{b}$ is the m -dimensional vector of branch lengths. Since the collection $\mathcal{S}_d^+$ is weakly compatible, it follows that the matrix $\mathbf{A}$ has full rank [3]. Hence ordinary least squares estimates for $\mathbf{b}$ can be computed from the observed distance vector $\mathbf{d}$ using the standard formula

$$\mathbf{b} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{d}. \quad (7.3)$$

In addition, weighted least squares estimates can be computed using

$$\mathbf{b} = (\mathbf{A}^T \mathbf{W} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{W} \mathbf{d}, \quad (7.4)$$

where $\mathbf{W}$ is the $n(n-1)/2 \times n(n-1)/2$ diagonal matrix with $1/\text{var}(d_{ij})$ in entry $\mathbf{W}_{(ij)(ij)}$. These formulae are identical to those used for phylogenetic trees (see Chapter 1, this volume).

Even though least squares estimates for networks can be useful, it can still be problematic in case many splits have isolation index zero. This tends to be the case when dealing with large data sets, since there is a higher chance that at least one quartet leads to rejection of a split according to equation (7.2). In the next section we describe an alternative method for generating splits graphs using an agglomerative approach that makes some progress in solving this problem.

7.7 Neighbour-net

In [16] an agglomerative approach to constructing planar phylogenetic networks is presented. This method, called neighbour-net, is a generalization of the tree-building method Neighbor Joining [45] (see also Chapter 1, this volume). We begin by giving an informal introduction to the neighbour-net algorithm, and then provide more precise details below.

Starting with a set of nodes representing the taxa, NJ works by iteratively selecting pairs of nodes and replacing them by a new composite node. Neighbour-net has one important difference. When pairs of nodes are selected, they are not combined and replaced immediately. Instead, the method waits until a node has been paired up a second time, at which stage three linked nodes are replaced with two linked nodes. In case a node linked to two others remains, a second agglomeration and reduction is performed. This process is illustrated in Fig. 7.10.

With NJ, pairs of nodes are repeatedly amalgamated into a single node until only three nodes remain. If we keep a list of these amalgamations, the NJ

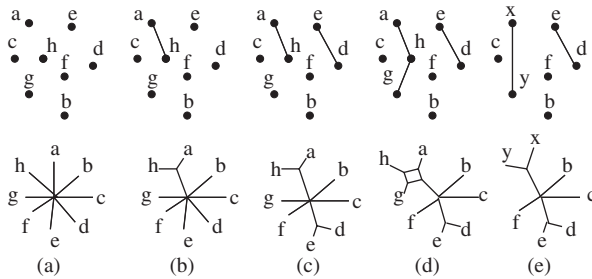


FIG. 7.10. Neighbour-net's agglomeration process. (a) We start with nodes corresponding to a set $a, b, \dots, h$ of taxa. (b) Using a selection criterion similar to NJ, nodes a and h are identified as neighbours. Unlike NJ, a and h are not immediately agglomerated. (c) Nodes e and d are identified as neighbours. (d) Node h is identified as a neighbour of g . Thus, h is now a neighbour of both a and g , which can be represented by a split graph. (e) Since h now has two neighbours, a reduction is performed that replaces a, h, g by x, y .

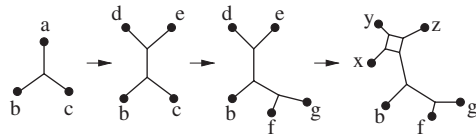


FIG. 7.11. The expansion process for neighbour-net. In the first and second expansions, a is replaced by d, e and c by f, g , respectively. Until this point, the expansion procedure is the same as with NJ. However, in the third expansion, d, e is replaced by x, y, z , leading to a split graph.

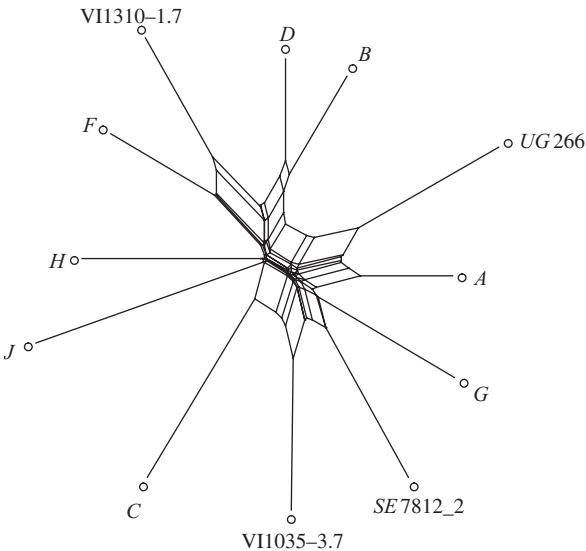


FIG. 7.12. A neighbour-net for the data set whose split-decomposition splits graph is presented in Fig. 7.1.

tree can be constructed by reversing the amalgamation process (Fig. 7.11). In neighbour-net a list of amalgamations is also recorded, though each amalgamation replaces three nodes with two. Reversing the amalgamation process gives the splits that will be represented in the neighbour-net network. In particular, the end-product of the neighbour-net process is a circular collection of splits, which can be represented by a planar splits graph as explained in the previous section.

In Fig. 7.12, we present the neighbour-net for the data set whose split-decomposition network appears in Fig. 7.1 (for more examples with biological interpretations see [16]). As can be seen, the neighbour-net is somewhat more resolved than the splits graph that was obtained using split decomposition.

Data Structures:

- Set Y of active elements, initially X ;
- Distance ρ on Y , initially d ;
- Array of neighbour relations;
- Stack F of five-tuples $[x, y, z, u, v]$ of X encoding agglomerative events;
- Circular ordering $\theta = y_1, y_2, \dots, y_m$ of Y and a non-negative weight β_S^d for each $S \in \mathcal{S}_\theta = \{\{y_p, \dots, y_q\} \mid (Y - \{y_p, \dots, y_q\}): 1 \leq p \leq q < m\}$.

NEIGHBOURNET(d)

1. **while** $|Y| > 3$ **do**
2. *Selection:* use ρ to choose a pair of elements $x, y \in Y$ and make these neighbours.
3. **while** there exists an element $y \in Y$ with two neighbours **do**
4. let x and z denote the neighbours of y ,
5. let u and v be new neighbours,
6. *Reduction:* $Y \leftarrow Y \cup \{u, v\} - \{x, y, z\}$,
7. compute new entries for ρ ,
8. push $[x, y, z, u, v]$ on top of F .
9. **end**
10. **end**
11. let θ be an arbitrary circular ordering of Y .
12. **while** F is non-empty **do**
13. pop $[x, y, z, u, v]$ off the top of F ,
14. replace u, v in θ by x, y, z .
15. **end**
16. *Estimation:* compute a weight β_S^d for each $S \in \mathcal{S}_\theta$.
17. output $\{(S, \beta_S^d): S \in \mathcal{S}_\theta\}$.

FIG. 7.13. The neighbour-net algorithm.

This tends to be the case in general, although it also often happens that many splits are produced that have relatively small weights (probably due to noise).

We now present a more detailed explanation of the neighbour-net algorithm, the formal algorithm for which is given in Fig. 7.13. The algorithm is determined by the formulae used to select nodes for agglomeration in Step 2, to reduce the distance matrix after each agglomeration in Step 6, and estimate the split weights in Step 16.

The selection and reduction criteria are related to those used by NJ. Selection proceeds as follows. Suppose that we have n nodes remaining. At the start of the algorithm, none of the nodes will have neighbours assigned to them. Later on, some pairs of nodes will have been identified as neighbours, but not yet agglomerated. We take these neighbour relations into account when selecting nodes to agglomerate. In particular, neighbouring relations group the n nodes into clusters

$C_1, C_2, \dots, C_m$, $m \leq n$, some of which contain a single node and others which contain a pair of neighbouring nodes. The distance $d(C_i, C_j)$ between two clusters is taken to be the average of the distances between elements in each cluster:

$$d(C_i, C_j) = \frac{1}{|C_i||C_j|} \sum_{x \in C_i} \sum_{y \in C_j} d_{xy}. \quad (7.5)$$

The selection of neighbouring nodes proceeds in two steps. First a pair of clusters that minimize the standard NJ formula is found

$$Q(C_i, C_j) = (m-2)d(C_i, C_j) - \sum_{\substack{k=1 \\ k \neq i}}^m d(C_i, C_k) - \sum_{\substack{k=1 \\ k \neq j}}^m d(C_j, C_k). \quad (7.6)$$

Now, suppose that C_{i^*} and C_{j^*} are two clusters that minimize $Q(C_i, C_j)$. The second step is to choose which nodes $x_i \in C_{i^*}$ and $x_j \in C_{j^*}$ are to be made neighbours. The clusters C_{i^*} and C_{j^*} each contain either one or two nodes. If these clusters were separated out into individual nodes we would end up with $m + |C_{i^*}| + |C_{j^*}| - 2$ clusters in total. Let $\hat{m}$ denote $m + |C_{i^*}| + |C_{j^*}| - 2$. To maintain consistency, this value $\hat{m}$ replaces m in equation (7.6) when we are selecting particular nodes *within* clusters. In particular, we select the node $x_i \in C_{i^*}$ and node $x_j \in C_{j^*}$ that minimizes

$$\hat{Q}(x_i, x_j) = (\hat{m}-2)d(x_i, x_j) - \sum_{\substack{k=1 \\ k \neq i}}^{\hat{m}} d(x_i, C_k) - \sum_{\substack{k=1 \\ k \neq j}}^{\hat{m}} d(x_j, C_k). \quad (7.7)$$

We now explain how reduction is performed. Suppose that node y has two neighbours, x and z . In the neighbour-net agglomeration step, we replace x, y, z with two new nodes u, v . The distances from u and v to another node a are computed using the reduction formulae

$$\begin{aligned} d(u, a) &= \alpha d(x, a) + \beta d(y, a), \\ d(v, a) &= \beta d(y, a) + \gamma d(z, a), \\ d(u, v) &= \alpha d(x, y) + \beta d(x, z) + \gamma d(y, z). \end{aligned}$$

where α, β, γ are non-negative real numbers with $\alpha + \beta + \gamma = 1$. In reference [28] it was observed that a single degree of freedom can be introduced into the reduction formulae for NJ. In the above formulae we have two degrees of freedom, thus allowing the possibility for a variance reduction method in future versions of neighbour-net. Currently $\alpha = \beta = \gamma = \frac{1}{3}$ is used, in direct analogy to NJ.

The final estimation of split weights is performed using least squares (see Section 7.6). This is done using equations (7.3) and (7.4). However, since some negative split weights may result whose omission often leave the remaining splits grossly overestimated, a non-negativity constraint is also employed. Since there is no closed formula for constrained least squares estimates [38], enforcing the

constraint increases computation time considerably, although the result is far cleaner and more accurate.

Before concluding this section, we mention an important property of neighbour-net. As with NJ, if the input to neighbour-net is a treelike distance matrix, neighbour-net will return the splits and branch lengths of the corresponding tree. Moreover, neighbour-net is also consistent for the more general class of *circular* distance matrices (a distance matrix is *circular*—also called *Kalmanson*—if it corresponds to the distance obtained from a circular collection of splits with positive weights by adding the weights of the splits that separate pairs of elements—cf., for example, reference [18]). If the input distance matrix is circular, neighbour-net is guaranteed to return the corresponding circular splits with their split weights. The proof is non-trivial—see [15] for details. This consistency property is one of the main factors that influenced the choice of selection and reduction formulae presented above.

7.8 Discussion

In the chapter, we have seen various methods for constructing phylogenetic networks. As with most phylogeny tools, some care needs to be taken in deciding which network is applicable to the data set in question. As a general guide, splits graphs and neighbour-nets can be used to provide a quick snapshot for most data sets, whereas median and related networks are more suited to low-diversity, intraspecies data. Phylogenetic networks (including splits graphs, neighbour-nets, and consensus networks) can be generated using the SplitsTree4 program [37]. Median networks and various supporting data visualizations can be generated using the program Spectronet [34].

In general, some care needs to be taken in interpreting phylogenetic networks. For example, as we have seen in Fig. 7.9, it is possible to represent a collection of splits by different graphs, and so care must be taken when interpreting internal nodes of such a graph. Even though in median networks we may interpret internal nodes as putative ancestral states, this is not generally the case for all splits graphs [52]. A splits graph represents conflict, and conflicting signals, rather than an explicit history of which reticulations took place. In general, splits graphs should probably be used as a technique for data representation and exploration, much in the same way as a scatter diagram can be used to explore the relationship between two real valued variables. However, in order to go beyond exploration to diagnosis we require a consistent framework for interpretation of splits graphs, particularly if we are to design meaningful significance tests. Recent progress towards this problem has been made by Bryant *et al.* [14], where it is shown that under certain conditions the weights of the splits represented in the network can be interpreted as estimations of splits in certain trees (see Fig. 7.14).

There are still many open problems in connection to phylogenetic networks. For example, even though some progress was made in developing a likelihood setting for splits graphs [50], the results were not completely satisfactory [52]. Also, as we have seen, the concept of consensus networks is a natural generalization of

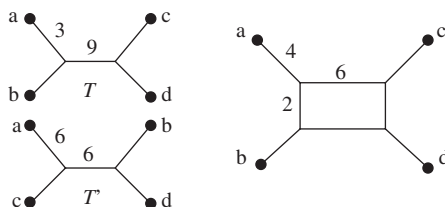


FIG. 7.14. The splits graph can be considered as representing a mixture of the two trees T, T' . The weights assigned to the splits graph are consistent with an alignment where $2/3$ of the sites support T and $1/3$ support T' . For example, the weight 6 of the split $\{a, b\} \mid \{c, d\}$ in the splits graph equals the weight 9 for this split in T multiplied by $\frac{2}{3}$. Also, the split $\{a\} \mid \{b, c, d\}$ appears with weight 3 in T and weight 6 in T' . Hence, the weight of this split in the splits graph is $\frac{2}{3} \times 3 + \frac{1}{3} \times 6 = 4$.

consensus trees, and so it would be of interest to develop the concept of supernetworks as a natural generalization of supertrees. And, of course, it will be important to find good interpretations for such networks—see [33], where some progress has been made in understanding species phylogeny through the construction of consensus networks from collections of gene trees. In this regard, it could be useful to develop tools which allow the user to easily map features of phylogenetic networks back onto the original data. Finally, as mentioned in the introduction, in this chapter we did not review network methods that are not based on splits. However, a rich new theory for phylogenetic networks based on directed acyclic graphs is currently emerging (cf., for example, references [9, 30, 41, 48]), that promises to yield many exciting new mathematical and biological results.

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RECONSTRUCTING THE DUPLICATION HISTORY OF TANDEMLY REPEATED SEQUENCES

Olivier Gascuel, Denis Bertrand, and Olivier Elemento

Tandemly repeated sequences can be found in all genomes that have been sequenced so far. However, their evolution is only beginning to be understood. In this chapter, we present state-of-the-art mathematical concepts and approaches for studying tandemly repeated sequences, from an evolutionary perspective. We describe a tandem duplication model for representing the evolution of these sequences, and shows that it has strong biological support. Then, we provide extensive mathematical and combinatorial characterization of tandem duplication trees and describe several algorithms for inferring tandem duplication trees from aligned and ordered sequences. We finally compare these algorithms using computer simulations and discuss directions for further research.

8.1 Introduction

Repeated sequences constitute an important fraction of most genomes, from the well studied *Escherichia coli* bacterial genome [4] to the human genome [29]. For example, it is estimated that more than 50% of the human genome consists of repeated sequences [29, 44]. As described in Section 8.2, there exist three major types of repeated sequences: transposon-derived repeats, micro- or minisatellites, and large duplicated sequences, the last often containing one or several RNA or protein-coding genes. Micro- or minisatellites arise through a mechanism called slipped-strand mispairing, and are always arranged in tandem: copies of a same basic unit are linearly ordered on the chromosome. Large duplicated sequences are also often found in tandem and, when this is the case, unequal recombination is widely assumed to be responsible for their formation.

In the present chapter, we focus on tandemly arranged duplicated sequences and study their evolution within single genomes. Both the linear order among tandemly repeated sequences, and the knowledge of the biological mechanisms responsible for their generation, suggest a simple model of intra-species evolution by duplication. This model, first described by Fitch in 1977 [15], introduces tandem duplication trees as phylogenies constrained by the unequal recombination mechanism. Although it is a completely different biological mechanism, *slipped-strand mispairing* leads to the same duplication model [33]. The paper

published by Fitch received relatively little attention, probably due to the lack of available sequence data at that time. Rediscovered by Benson and Dong in 1999 [2], tandemly repeated sequences and their suggested duplication model have recently received more focus, providing several new problems and challenges for computer scientists and mathematicians. The main challenge consists of creating algorithms for reconstructing the duplication history of tandemly repeated sequences [9, 10, 11, 26, 49, 59].

As whole-genome sequences accumulate, accurate reconstruction of duplication histories will be useful to elucidate various aspects of genome evolution. They will provide new insights into the mechanisms and determinants of gene and protein domain duplication, often recognized as major generators of novelty at the genome level [34]. Several important gene families, such as immunity-related genes, are arranged in tandem; better understanding their evolution should provide new insights into their duplication dynamics and clues about their functional specialization. Studying the evolution of micro- and minisatellites could resolve unanswered biological questions regarding human migrations or the evolution of bacterial diseases [30]. Also, as we show in this chapter, duplication trees appear to have interesting combinatorial properties [18, 56] and the ability to recognize, count, and enumerate duplication trees provides clues on how to create efficient reconstruction algorithms.

The content of this chapter is organized as follows. In Section 8.2, we describe the different categories of repeated sequences, present the duplication model that was introduced by Fitch, examine its biological validity and discuss its potential limitations. In Section 8.3, we introduce tandem duplication trees as mathematical objects and provide detailed description of their properties. In the same section, we describe exact and approximate approaches for counting tandem duplication trees, as well as recognition and enumeration algorithms. Then, in Section 8.4, we introduce the tandem duplication tree inference problem and describe algorithms that have been proposed for solving this problem. In Section 8.5, we compare these algorithms using simulations, then we provide directions for future research on duplication trees.

8.2 Repeated sequences and duplication model

8.2.1 *Different categories of repeated sequences*

Most *repeated sequences* (approximately 45% of the human genome) are derived from transposable elements [29]. Some DNA transposable elements (e.g. Long INterspersed Elements, or LINEs) are transcribed and the resulting RNAs are translated into functional proteins. In turn, LINE proteins possess the ability to reinsert their own RNA at other places in the genome. Other elements, such as ALU repeats, are transcribed but cannot transpose by themselves; they are therefore thought to rely on other proteins, such as those coded by LINEs, to insert back into their host genome.

Other types of duplicated sequences include simple, short, sequence repeats (from 1 to a few dozens base pairs), organized in tandem. These short

repeats, called *micro- or minisatellites*, do not code for any protein, however they can occur in protein-coding genes. The uncontrolled expansion of some microsatellites has been associated with certain human genetic diseases, such as Huntington's disease [50]. Blocks containing several thousand copies of these short sequences can also be found in the centromeric and telomeric portions of the chromosomes [29]. Micro- and minisatellites are thought to be created during DNA replication, by small-scale biological accidents termed slipped-strand mispairings [33].

On a very different size scale, there also exist segmental duplications, that is, blocks of size ranging from less than 1 kb to several hundred kilobases that have been copied from one region of the genome to another. Often, these blocks contain one or several protein-coding genes, and such duplicated genes are free to evolve independently by accumulating mutations. Repeated rounds of gene duplications followed by mutations create *gene families*, that is, sets of genes with related—but often slightly different, i.e., specialized—functions [34]. Such genes that share a common ancestor as a result of gene duplication are called paralogous genes as opposed to orthologous genes, when common ancestry stems from speciation.

Strikingly, gene families are often not randomly scattered in their host genomes, but organized in clusters. Gene clusters may contain between two to more than a hundred members; these members are said to be arranged in tandem, that is, are adjacent to each other on their chromosome. As detailed below, clusters of tandemly arranged genes are widely viewed as being generated (in eukaryotes) by a mechanism termed unequal recombination [15]. Note that unequal recombination is also responsible for creating repeated protein domains, as found in apolipoprotein A-I [15], and immunoglobulin constant genes [38]. Well studied examples of tandemly arranged genes include HOX genes [58], immunoglobulin and T-cell receptor genes [38], MHC genes [37], and olfactory receptor genes [21]. The evolution of these tandemly repeated genes is not well understood. In the case of genes involved in the immune response, gene duplication, followed by specialization, probably represents an efficient way to generate the diversity that is necessary to respond to a large—and ever changing—spectrum of external aggressions; however, the presence of large number of pseudogenes (i.e. genes which have lost functionality across evolution) within these clusters is not well explained [38]. The variable number of copies of orthologous gene clusters among species, also remains to be explained [38].

8.2.2 *Biological model and assumptions*

Tandemly repeated sequences can be defined as two or more adjacent and often approximate copies (also called segments in the following) of the same DNA fragment. Fitch [15] was the first to propose a duplication model for tandemly repeated sequences, based on *unequal recombination*. Recombination arises during meiosis, just after chromosome replication, when chromosomes line up in tetrad configuration. At that time, homologous non-sister chromatids can

exchange DNA fragments (see [1], p. 1131). It is widely assumed that the presence of repeated segments (LINEs, ALUs, micro- or minisatellites) at distinct places on the chromosomes often misleads pairing mechanisms into unequal pairing between non-sister chromatids. Such unequal pairing followed by recombination creates a tandem duplication on one chromosome and deletes the corresponding DNA fragment from the other chromosome. By increasing the possibilities of mispairing, tandemly repeated sequences increase the likelihood of additional tandem duplications.

In the following, we assume that no segment deletion occurred during the evolution of the studied sequences. This could be seen as a strong assumption regarding the unequal recombination process, but we show in Section 8.2.5 that the tandem duplication model is relatively tolerant to deletion events. Moreover, in the examples we studied (e.g. from the immune system) diversity is an advantage and deletions have low probability to be fixed in the population. This model also assumes that unequal recombination is the only mechanism responsible for generating the repeated sequences. In particular, the model supposes that the repeated sequences did not undergo any gene conversions. *Gene conversion* is a mechanism by which a DNA sequence is replaced by another sequence from a homologous region of the genome. When it occurs, gene conversion does not modify the number of segments in a set of tandemly repeated sequences, but modifies the content of some sequences. However, few examples of gene conversions have been described in the literature; moreover the replaced sequences are usually short. Gene conversion thus appears to be a minor evolutionary event and assuming its absence greatly simplifies the model, while keeping it reasonable from a biological point of view.

8.2.3 *Duplication events, duplication histories, and duplication trees*

The allowed duplication events form the basis of the duplication model proposed by Fitch, which can be described in the following way. According to the unequal recombination mechanism, a duplicated fragment may contain one or several segments. When the duplicated fragment contains a single segment, it is replaced by two adjacent and identical segments. The event is then called *simple duplication event*. Simultaneous duplication of several adjacent segments can also occur as a result of unequal recombination. For example, when the duplicated fragment contains 2 segments, it is replaced by two adjacent and identical copies of itself, resulting in 4 adjacent segments. These duplication events can be generalized to any number of segments, and events involving several segments are called *multiple duplication events*. In all the above cases, each segment is free to evolve independently of the other ones by accumulating mutations.

Assuming we could trace the evolution of a set of tandemly repeated sequences, the duplication history of these sequences could be visualized as a succession of duplication events separated by variable time intervals. An example of such duplication history is given in Fig. 8.1(a), for nine extant segments. It is straightforward to see that a duplication history induces a phylogeny, whose

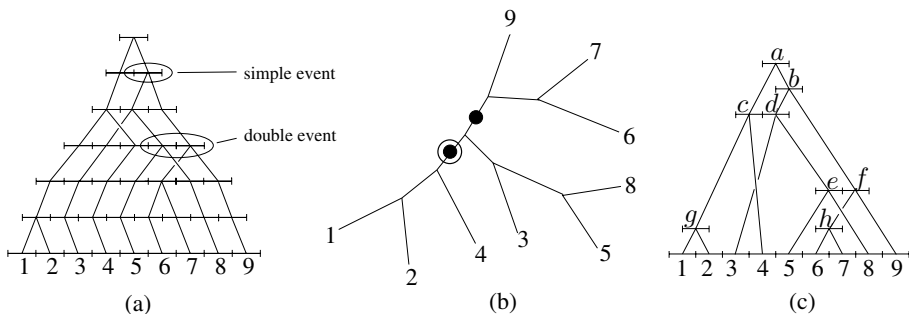


FIG. 8.1. (a) Duplication history; (b) Duplication tree, the two possible root positions are indicated by black dots, and the root in tree (a) is circled; (c) Rooted duplication tree.

leaves are ordered (each leaf is associated to a single segment on the chromosome). The edges of a duplication history are time-valued. The distances in the tree between the root and the leaves are identical: they represent evolutionary time elapsed since the very first duplication event. Moreover, the root of a duplication history is situated somewhere in the phylogeny on the path between the left-most and right-most segments on the chromosome (segments 1 and 9 in Fig. 8.1(a)). However, the presence of multiple duplications events in the duplication history can imply restrictions on potential root positions (Section 8.3.2).

Inferring a duplication history, as described above, is not possible when using only the nucleotide (or protein) sequences of the extant segments. In particular, both the position of the root and the order in which the duplication events occurred cannot be recovered from these sequences. Indeed, the molecular clock hypothesis, which implies that substitution rates are constant among different lineages, is often significantly violated. All that can be obtained from these sequences is an unrooted tree with ordered leaves, which we called “*tandem duplication tree*” (see Fig. 8.1(b)), the term “tandem” being sometimes omitted for brevity. By definition, a duplication tree is compatible with at least one duplication history, and its edges are mutation rate valued. Duplication trees are phylogenies with ordered leaves. However, it is easy to show that not all phylogenies are duplication trees, when assuming any given leaf ordering. Properties of duplication trees, as well as methods for counting and enumerating them will be discussed in Section 8.3.

While a duplication tree is by definition unrooted, potential roots can be positioned somewhere (but not anywhere) in the tree between the left-most and right-most segments on the locus. Traditional phylogenetic tree rooting techniques, such as the midpoint and outgroup methods [48], can be applied to duplication trees in order to infer the probable position of the root. As shown in Fig. 8.1(c), a rooted duplication tree is a rooted phylogeny with ordered leaves, in which duplication events are partially ordered. For example, it is impossible

to determine in Fig. 8.1(c) which one of the two simple duplication events that created segments (1,2) and (6,7) happened first. However, it is possible to assert that the two double duplication events occurred one after another. Also note that although the edges of a rooted duplication tree are mutation rate valued, they are often represented with meaningless lengths to obtain readable drawings, as in Fig. 8.1(c).

8.2.4 *The human T cell receptor Gamma genes*

We applied this duplication model to the variable genes of the *human T cell receptor Gamma* (TRGV) locus [31, 32]. This locus contains 9 tandemly repeated genes, and each segment is approximately 5 kb long. The amount of identity among segments (after alignment) varies from 80 to 95%. We applied a branch-and-bound approach [12, 24] for finding the most parsimonious phylogeny explaining the 9 sequences (the parsimony criterion is well suited to sequences presenting this level of divergence). The branch-and-bound approach we applied is used for general phylogeny problems and is not restricted to duplication trees. However, using a duplication tree recognition algorithm (Sections 8.3.3–8.3.5), we showed that the unique most parsimonious phylogeny obtained for these sequences is also a duplication tree; we also showed that this result remains stable when subjected to bootstrap analysis [11]. The (duplication) tree we obtained, shown in Fig. 8.2(a), possesses interesting properties. Indeed, the number of distinct duplication trees for 9 segments is 5,202, while the number of unrooted phylogenies with 9 leaves is 135,135. It follows that the probability of randomly picking up a duplication tree among all distinct unrooted phylogenies is $5,202/135,135$, or 0.038. This small probability indicates that the identity between the most parsimonious duplication tree and the most parsimonious phylogeny is very unlikely to be due to chance, and provides an important support for the tandem duplication model, at least for the human TRGV genes. Rooting the TRGV duplication tree using both the midpoint and the outgroup method provides additional support. Indeed, the inferred position of the root in the tree, shown in Fig. 8.2(b), corresponds to one of the 4 positions that are allowed according to the duplication model, out of the 15 edges in the tree. Further support is provided by the known polymorphisms of the human TRGV locus. Indeed, simultaneous absence of segments V4 and V5 has been reported in French, Lebanese, Tunisian, Black-African, and Chinese populations [19, 20]. Examination of the TRGV duplication tree shows that V4 and V5 are the result of the most recent double duplication event; simply assuming that this duplication did not occur in some individuals of the above populations predicts (based on a single sequenced locus) this striking human polymorphism.

8.2.5 *Other data sets, applicability of the model*

In reference [10] we described another convincing application of this model to the seven genes of the human IGLC locus [7, 25, 53], which code for the constant region of the human immunoglobulin light chain. A third example is provided by

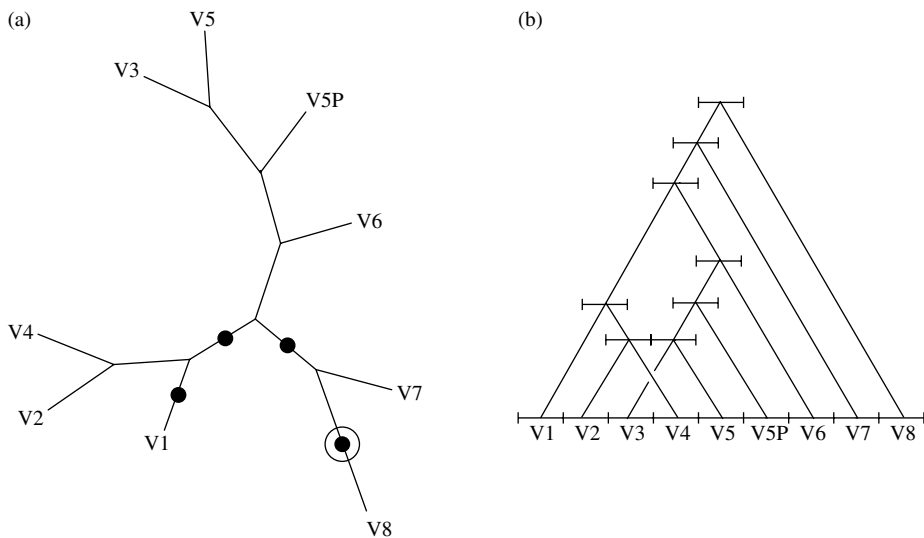


FIG. 8.2. (a) Duplication tree for the nine human TRGV genes. Black dots represent allowed root positions, according to the tandem duplication model; the selected root position is circled. (b) Rooted duplication tree, obtained using both the midpoint and outgroup rooting methods.

the Xa21 disease-resistance genes in rice [46]; while these genes encode proteins that are different from those encoded by the above described human immune genes, they also represent a case where diversity is certainly an advantage. Therefore, gene deletion is also likely to be rare. As for the human TRGV genes, the Xa21 most parsimonious phylogeny is also a duplication tree (see Fig. 8.3). For seven taxa, the probability that an unrooted phylogeny is a duplication tree is approximately 0.222. Although this probability is not as low as that obtained for TRGV, it nonetheless supports the duplication model. Moreover, the position of the root obtained using the midpoint method (no suitable outgroup could be found for these sequences) is also in agreement with the duplication model, according to which the root could be positioned on only 2 edges, out of 11.

Although more data and more systematic analyses would be required to assess the generality of tandem duplication trees, these results provide strong support in favour of our simple duplication model. Note that less supportive examples also exist. For example, we tried to reconstruct the duplication history of the 11 repeats of the UbiA polyubiquitin locus in *Caenorhabditis elegans* [22]. Unfortunately, the five most (and equally) parsimonious duplication trees obtained using exhaustive search were different from the unique most parsimonious phylogeny found using branch-and-bound [11]. This indicates that our model of evolution

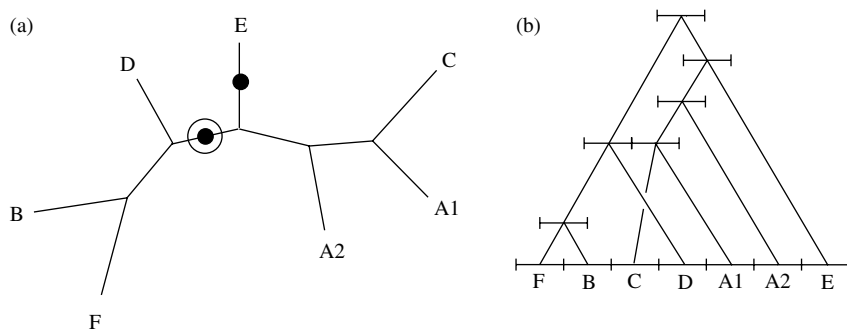


FIG. 8.3. (a) Duplication tree for the seven rice Xa21 genes. Black dots represent allowed root positions, according to the tandem duplication model; the selected root position is circled. (b) Rooted duplication tree, obtained using the midpoint rooting method.

by tandem duplication needs to be refined in some cases, for example, by introducing other mechanisms such as deletions. However, it is also easy to show, using the TRGV duplication tree, that the tandem duplication model is relatively tolerant to deletions. Indeed, removing any of the extant segments in the TRGV duplication tree in Fig. 8.2 results in another duplication tree with 8 segments. However, simultaneous removal of segments V1 and V2 creates a tree which is not a duplication tree. In an evolutionary scenario where all duplications are simple duplication events, deletion of any number of segments always results in another duplication tree. Evolutionary scenarios in which simple duplications are predominant should therefore be resistant to deletions, that is, they should be explained using duplication trees even though some segments were deleted in the course of evolution.

For these reasons, the duplication model we defined, although simple, should have a large applicability range, particularly when diversity of the studied sequences is an evolutionary advantage.

8.3 Mathematical model and properties

We described in the previous section the biological process that gives rise to tandem duplication trees, and we provided evidence supporting this model for tandemly repeated genes. In this section, we give a formal definition of tandem duplication trees and review their main mathematical properties. We also provide formulae for the number of duplication histories and duplication trees. As seen in Section 8.2, the proportion of duplication trees among the set of all phylogenies gives a simple and powerful way to estimate the evidential support of the duplication model. Moreover, counting these combinatorial objects allows for a better understanding of their properties and gives insight into the computational difficulties of their inference from data (Section 8.4).

8.3.1 Notation

As explained above, the duplication process is analogous to speciation, and a rooted (unrooted) duplication tree is mathematically speaking a rooted (unrooted) phylogeny. Let $s_1, s_2, \dots, s_n$ denote the extant duplicated segments, and T be the duplication tree that links these segments. T is a fully resolved phylogeny of the n segments, that is, T is a tree with n leaves which are bijectively labelled by the segments. The internal (non-root) vertices of T have degree 3. When T is rooted it has one more internal vertex with degree 2 that defines the root. The tree root is denoted as ρ and represents the common ancestor of all extant segments. T then captures the ancestral relationship of the duplicated segments.

T is associated to a leaf ordering, denoted as $O = (s_1, s_2, \dots, s_n)$, which expresses the order in which the segments appear on the extant locus being studied. Segments are ordered from left to right, and for any segment pair u, v from O , we use notation $u < v$ to express that u is before v in O , and $(u, v) \subseteq O$ when u and v are adjacent and $u < v$. This notation is also used when u, v is a leaf pair of T , as leaves are bijectively labelled with the segments, and $(u_j, u_{j+1}, \dots, u_k) \subseteq O$ means $(u_i, u_{i+1}) \subseteq O$ for every $i, j \leq i < k$.

T is also associated to a partition of internal nodes into duplication “events” (or “blocks” following [49]), which groups the duplications that have jointly occurred in the course of evolution. We distinguish “simple” duplication events that contain a unique internal node (e.g. b and g in Fig. 8.1(c)) and “multiple” duplication events which group a series of adjacent and simultaneous duplications (e.g. (c, d) in Fig. 8.1(c)). When the tree is rooted, every internal node u is unambiguously associated to one parent and two child nodes; moreover, one child of u is “left” and the other one is “right,” which is denoted as $l(u)$ and $r(u)$, respectively, and is further discussed. When the tree is unrooted, some ambiguities are possible, but duplications from multiple events are still oriented as we know that these duplications occurred after the initial root duplication (see also below for more).

8.3.2 Root position

Contrary to phylogenies which can be rooted on any edge, the root position is strongly constrained in duplication trees. In any possible history, the direct ancestor of s_1 was in left-most position in the ancestral locus, and, recursively, all ancestors of s_1 were in left-most position until tree root ρ . In the same way, all ancestors of segment s_n were in last position until ρ (Fig. 8.1(a)). This implies that the intersection of the paths from s_1 to ρ and from s_n to ρ only contains ρ . Then in any duplication tree the root must be situated on the path from the left-most to the right-most segment. Consider now any multiple duplication event. Such an event represents segments that were simultaneously present during evolution, which implies that the tree root is an ancestor of these segments. The first occurring multiple duplication events then marks the limits of the possible root locations on the path connecting the left-most and the right-most segment

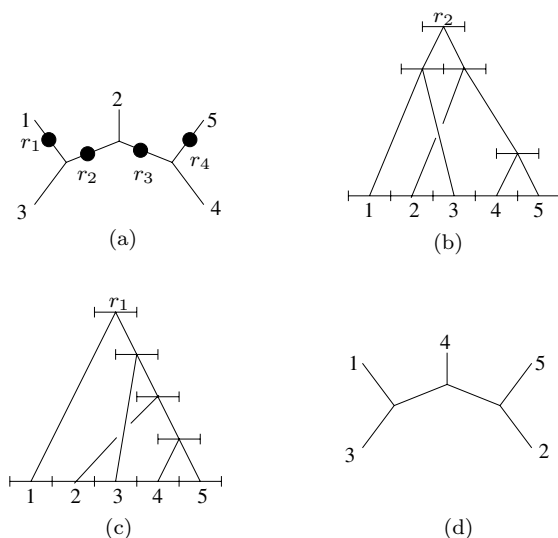


FIG. 8.4. Not all potential root positions lead to valid rooted duplication trees; tree (a) can be rooted at position r_1 , r_2 , r_3 , and r_4 on the path in the tree from 1 to 5; r_2 (b) is valid, while all other positions are not, for example, r_1 (c). Not all phylogenies with ordered leaves are duplication trees, for example, none of the possible root positions of (d) leads to a valid rooted duplication tree.

(Fig. 8.1). For example, when the initial duplication is followed by a double duplication event, the root is “trapped” and only one root position is valid (Fig. 8.4(b)). On the other hand, when the path connecting the left-most and the right-most segment only contains simple duplication events, the root can be placed everywhere along this path. Although the number of potential root placements on an unrooted duplication tree can vary, as shown below, the average number of possible root locations over all duplication trees of $n > 2$ segments, is exactly 2 [18, 56].

8.3.3 Recursive definition of rooted and unrooted duplication trees

A duplication tree is a phylogeny with ordered leaves, which is induced by at least one duplication history. This suggests a recursive definition, which progressively reconstructs a possible history, given a phylogeny T and a leaf ordering O . We define a *cherry* (l, u, r) as a pair of leaves (l and r) separated by a single node u in T (see Fig. 8.5), and we call $C(T)$ the set of cherries of T . This recursive definition reverses evolution: it searches for a “visible duplication event” (i.e. a duplication event in which none of the duplicated segments was subsequently duplicated), “agglomerates” this event and checks whether the “reduced” tree is

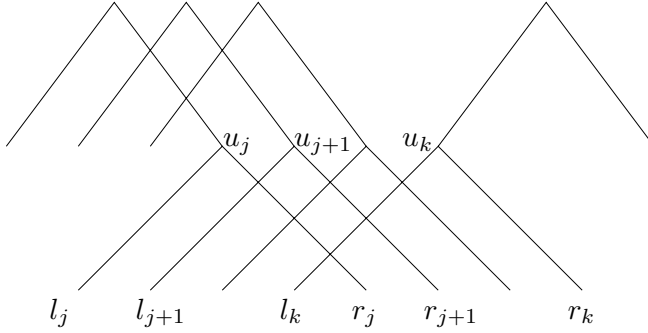


FIG. 8.5. Partial representation of a rooted duplication tree. The set of cherries $(l_j, u_j, r_j), (l_{j+1}, u_{j+1}, r_{j+1}), \dots, (l_k, u_k, r_k)$ forms a “visible duplication event” that can be agglomerated into $u_j, u_{j+1}, \dots, u_k$ to form a “reduced duplication tree”.

a duplication tree. In case of rooted trees, we have:

(T, O) defines a duplication tree with root ρ if and only if:

1. (T, O) only contains ρ ; or
2. there is in $C(T)$ a series of cherries $(l_j, u_j, r_j), (l_{j+1}, u_{j+1}, r_{j+1}), \dots, (l_k, u_k, r_k)$ with $k \geq j$ and $(l_j, l_{j+1}, \dots, l_k, r_j, r_{j+1}, \dots, r_k) \subseteq O$ (see Fig. 8.5), such that (T', O') defines a duplication tree with root ρ , where T' is obtained from T by removing $l_j, l_{j+1}, \dots, l_k, r_j, r_{j+1}, \dots, r_k$, and O' is obtained by replacing $(l_j, l_{j+1}, \dots, l_k, r_j, r_{j+1}, \dots, r_k)$ by $(u_j, u_{j+1}, \dots, u_k)$ in O .

The definition for unrooted trees is quite similar:

(T, O) defines an unrooted duplication tree if and only if:

1. (T, O) contains 1 segment; or
2. same as for rooted trees with (T', O') now defining an unrooted duplication tree.

For example, it can be checked using these definitions that tree (d) in Fig. 8.4 is not an unrooted duplication tree, that tree (a) in Fig. 8.4 only admits one possible root position, and that tree (b) in Fig. 8.1 is an unrooted duplication tree.

8.3.4 From phylogenies with ordered leaves to duplication trees

Those definitions provide simple recursive algorithms to check whether any given phylogeny with ordered leaves is a duplication tree. In case of success, these algorithms can also be used to reconstruct the duplication events: at each step the series of internal nodes above denoted as $(u_j, u_{j+1}, \dots, u_k)$ is a duplication event. The order in which the duplication events are reconstructed is unimportant as every internal node belongs to one and only one event in a duplication tree.

When the tree is rooted, l_i is the left child of u_i and r_i its right child, for every $i, j \leq i \leq k$. When the tree is unrooted, this property only holds in case of multiple duplication events, but it is still possible to define the orientation of a simple event when it belongs to a root-to-leaf path for all possible root positions. In the rooted case, the algorithm also reconstructs a duplication history that is compatible with the given phylogeny and leaf ordering; the duplication events of the history are in the reverse order in which they are reconstructed by the algorithm, and the successive values of O correspond to the successive states of the ancestral locus. Changing the order in which the events are reconstructed changes the duplication history, and all compatible duplication histories can be obtained this way. Finally, the algorithm for the rooted case can be used to draw duplication trees in a bottom-up way, as shown in Figs 8.1(c), 8.2(b), 8.3(b) and 8.4(b).

As we shall see (Section 8.4) recognizing phylogenies with ordered leaves that are duplication trees is an important issue for duplication tree inference. The above algorithms iteratively find a visible duplication event, which requires a computing time in $O(n)$, and then reduce T and O thus decreasing n by at least one unit. The total time complexity is then in $O(n^2)$, for the rooted as for the unrooted case. In reference [18] we propose an improved implementation in $O(n)$. The principle consists of searching for the left-most visible duplication, scanning the segments from left to right, never moving to useless points and storing the location of cherries. A “partially” visible event is a series of cherries $(l_j, u_j, r_j), (l_{j+1}, u_{j+1}, r_{j+1}), \dots, (l_p, u_p, r_p)$ with $(l_j, l_{j+1}, \dots, l_p) \subset O$, $(r_j, r_{j+1}, \dots, r_p) \subset O$, and $l_p < r_j$ but $(l_p, r_j) \not\subset O$. The algorithm remembers the endpoints of already encountered partially visible events, so that after finding a visible event, the algorithm can continue the investigation of a partially visible event without returning to its starting segment. In this way, the algorithm always moves from left to right, unless a visible event is agglomerated, in which case it jumps to its left-most segment. Thus, the number of steps is $O(n)$, and so is the time complexity of the whole algorithm.

8.3.5 Top-down approach and left-right properties of rooted duplication trees

The above algorithms are bottom-up as they proceed from leaves to root of the tree. Top-down approaches, as proposed by Tang *et al.* [49] and Zhang *et al.* [59], start from the root of the tree, and progressively identify the duplication events until the leaves are reached. These algorithms exploit basic properties of the l (left) and r (right) operators, which must be satisfied when T and O define a rooted duplication tree. Let (T, O) be a rooted duplication tree and u be a node of T . We define the left-most descendant of u by: $L(u) = u$ if u is a leaf, else $L(u) = L(l(u))$. In the same way, we define the right-most descendant of u : $R(u) = u$ if u is a leaf, else $R(u) = R(r(u))$. We then have the following properties:

1. $L(u)$ is the leaf descending from u with smallest label in O , and $R(u)$ is the leaf descending of u with largest label.
2. Unless u is a leaf: $L(u) = L(l(u)) < L(r(u))$ and $R(l(u)) < R(r(u)) = R(u)$.

3. When $e = (u_j, u_{j+1}, \dots, u_k)$, $k \geq 1$, is a duplication event, then $L(u_j) < L(u_{j+1}) < \dots < L(u_k) < R(u_j) < R(u_{j+1}) < \dots < R(u_k)$.

The algorithm proposed by Tang *et al.* [49] proceeds as follows. It uses (1) to compute L and R for every node, and then (2) to identify left and right children of every node in T . Note that (2) does not always hold for non-duplication trees, in which case the algorithm returns NO. Both computations are achieved in $O(n)$ using simple tree traversals. After this preprocessing step, the algorithm reconstructs the duplication events starting from the tree root ρ . It uses at each step an ordering of the nodes, denoted as G , which corresponds to a possible ancestral locus, just as O and O' in algorithms of Section 8.3.3. In a rooted duplication tree any ancestral locus G must satisfy:

4. Let G be equal to $(u_1, u_2, \dots, u_p)$ ($1 \leq p \leq n$), then $L(u_1) < L(u_2) \dots < L(u_p)$ and $R(u_1) < R(u_2) \dots < R(u_p)$.

The algorithm starts with $G=(\rho)$, searches for an event $e = (u_j, u_{j+1}, \dots, u_k) \subseteq G$ satisfying (3) and such that G' satisfies (4), where G' is obtained from G by replacing e by $(l(u_j), l(u_{j+1}), \dots, l(u_k), r(u_j), r(u_{j+1}), \dots, r(u_k))$. When such an event is found the algorithm continues with G' in place of G , otherwise NO is returned. The algorithm successfully terminates when G becomes equal to the extant locus O . This algorithm is then closely related to our algorithm of Section 8.3.3 and has the same properties (event identification, reconstruction of a possible history, tree drawing), but it proceeds in a top-down instead of a bottom-up way. This algorithm can be implemented in $O(n^2)$. A faster $O(n)$ algorithm is proposed in reference [59]. This algorithm is top-down, but it detects several multiple duplication events at each step and progressively reduces and modifies the tree until only simple duplications remain. However, this algorithm, just as the previous one [49], only applies to rooted trees; this represents a limitation since inferred trees are usually unrooted. Applying the algorithm to the $O(n)$ rooted trees obtained by rooting the tree on each edge between the left-most and right-most segment overcomes this limitation, but increases time complexity to $O(n^2)$.

8.3.6 Counting duplication histories

Let $DH(n)$ denote the number of duplication histories with n segments. A locus containing n segments can be obtained from any of $(n - 1)$ simple duplication events from a locus containing $(n - 1)$ segments or from any of $(n - 3)$ double events from a locus containing $(n - 2)$ segments, etc. Therefore, $DH(n)$ is given by the following recursive formula [11]:

$$DH(n) = \sum_{k=1}^{\lfloor n/2 \rfloor} (n - 2k + 1)DH(n - k) \quad \text{when } n > 1, \quad \text{and } DH(1) = 1.$$

8.3.7 Counting simple event duplication trees

Let $\text{RDT}(n)$ and $\text{DT}(n)$ denote the number of rooted and unrooted duplication trees, respectively. Moreover, let $1\text{-RDT}(n)$ and $1\text{-DT}(n)$ denote the number of rooted and unrooted duplication trees, respectively, which only contain simple duplication events. In the rooted case, such trees are identical to standard binary search trees, as commonly used in computer science [6]. Any such tree with n leaves is composed of two (binary search) subtrees with k and $n - k$ leaves ($1 \leq k \leq n-1$). Using the *Catalan recursion* [52], we then have $1\text{-RDT}(1) = 1$ and [10]:

$$\begin{aligned} 1\text{-RDT}(n) &= \sum_{k=1}^{n-1} 1\text{-RDT}(k) \times 1\text{-RDT}(n-k), \text{ when } n > 1, \\ &= \frac{(2n)!}{n!(n+1)!}, \\ &\approx \frac{4^n}{\sqrt{\pi n^{3/2}}}. \end{aligned}$$

As already discussed (Section 8.3.2), a duplication tree that only contains simple events can be rooted anywhere along the path between the two most distant segments. We then root these trees on the parent edge of the last segment (s_n) and count the number of such rooted trees. In these trees the left subtree is a rooted tree with $n - 1$ segments that only contains simple events. Then we have the following simple equality [10]:

$$1\text{-DT}(n+1) = 1\text{-RDT}(n).$$

8.3.8 Counting (unrestricted) duplication trees

Following preliminary analysis by Fitch [15] and computer estimation by Elemento *et al.* [11], the general case for $\text{DT}(n)$ and $\text{RDT}(n)$ was solved by Gascuel *et al.* [18]. The main results are summarized here, but additional results can be found in references [18, 56]. We first provide a recursive formula for $\text{RDT}(n)$ and then show that $\text{RDT}(n) = 2\text{DT}(n)$ when $n > 2$. We use for that purpose a different non-biological way of generating/agglomerating duplication events and trees. Let T and O define a rooted duplication tree and consider the left-most visible event. Using notation defined in Section 8.3.3, O is then of the form $s_1, s_2, \dots, l_1, l_2, \dots, l_k, r_1, r_2, \dots, r_k, s_p, s_{p+1}, \dots, s_n$, where the given event is from l_1 to r_k , and where there is no visible event before l_1 . Let m be number of segments situated to the right of this duplication event, that is, $m = n - p + 1$ if $r_k \neq s_n$, otherwise $m = 0$; we denote as $\text{RDT}(n, m)$ the set of all such trees having n leaves and m segments to the right of the left-most visible event. The agglomerating scheme involves removing l_k and r_k in T , with u_k being now a leaf, while in O , l_k is removed and r_k is replaced by u_k . We then obtain a rooted duplication tree with $n - 1$ leaves. This scheme is clearly equivalent to that of Section 8.3.3 as after k steps the whole duplication event is agglomerated.

The generating scheme is as follows. Let (T, O) define an element of $\text{RDT}(n, m)$ and use above notation. There are two main possibilities:

1. we duplicate any segment between s_1 and r_k ; or
2. when $m > 0$, we extend the left-most visible event by inserting a new segment l_{k+1} between l_k and r_1 and by creating a new cherry (l_{k+1}, u_{k+1}, s_p) where u_{k+1} is a new node that is inserted on the parent edge of s_p .

In case (1) the new tree belongs to $\text{RDT}(n+1, j)$, $m \leq j \leq n-1$, and in case (2) to $\text{RDT}(n+1, m-1)$. Figure 8.6 provides an illustration of this. It is easily seen that the agglomerating scheme reverts the generating scheme: if (T', O') is obtained from (T, O) by agglomeration, then (T, O) is one of the trees that can be obtained from (T', O') using the generating scheme. This implies that every rooted duplication tree can be generated from the 2-leaf tree and that the generating path is unique.

Let $p(n, q, m)$, $2 \leq q \leq n$, and $0 \leq m \leq q-2$, be the number of rooted trees with n segments that can be generated from a single tree in $\text{RDT}(q, m)$. From above remarks we have:

$$p(n, q, m) = \sum_{j=\max(0, m-1)}^{q-1} p(n, q+1, j).$$

Using this equation, the recurrence for rooted duplication trees can then be written as:

$$\text{RDT}(n) = p(n, 2, 0) = p(n, 3, 0) + p(n, 3, 1),$$

and when $q \geq 3$ and $0 \leq m \leq q-2$:

$$\begin{aligned} p(n, n, m) &= 1, \\ p(n, q, 0) &= p(n, q, 1), \\ p(n, q, q) &= p(n, q+1, q-1), \\ p(n, q, m) &= p(n, q+1, m-1) + p(n, q, m+1). \end{aligned}$$

Based on the size of $\text{RDT}(n, m)$ sets we simplified the above equations into a double recurrence [18], which was further improved by Yang and Zhang [56] to obtain the following simple recurrence:

$$\begin{aligned} \text{RDT}(n) &= \sum_{k=1}^{\lfloor (n+1)/3 \rfloor} (-1)^{k+1} \binom{n+1-2k}{k} \text{RDT}(n-k), \quad n > 2, \\ \text{RDT}(1) &= \text{RDT}(2) = 1. \end{aligned}$$

Consider now the case of unrooted trees. Just as for $1\text{-DT}(n)$, we place the root on the right-most possible root location and count the number of such rooted trees. As explained above (Section 8.3.2), the right-most possible root location is either just above the last segment s_n , or just above the first multiple duplication event that is above s_n . A relevant feature of the generating scheme

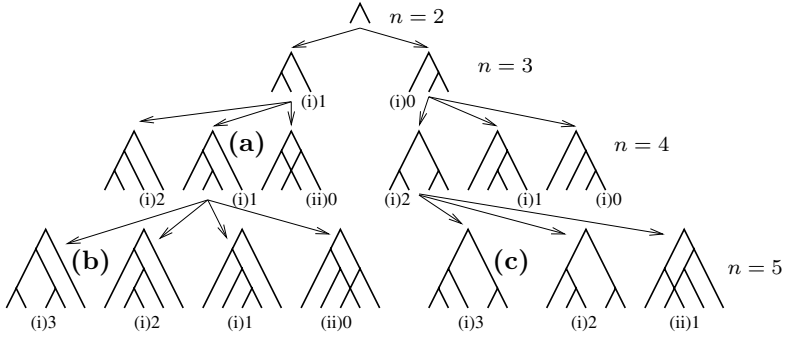


FIG. 8.6. Generating/agglomerating scheme. The extant segments are ordered from left to right. For every tree the type of generating move (i.e. (i) or (ii)) is indicated as well as the value of m (i.e. the number of segments on the right of the left-most visible duplication). For example, tree (b) is obtained from tree (a) by duplicating the left-most segment, that is, a type (i) move, and $m = 3$; tree (b) then belongs to $\text{RDT}(5, 3)$, just as tree (c).

is that all trees that are generated from the left child of the 2-leaf tree satisfy this requirement (Fig. 8.6). On the other hand, no descendant of the 2-leaf tree right child is rooted on the right-most position, as its root is always above the simple duplication that occurred just after the initial duplication. We then have $\text{DT}(n) = p(n, 3, 1)$, and using the above recurrence:

$$\text{DT}(n) = p(n, 3, 1) = \frac{1}{2}(p(n, 3, 0) + p(n, 3, 1)) = \frac{1}{2}\text{RDT}(n).$$

The same result was derived using a non-counting proof in reference [56]. Moreover, we used generating functions to obtain the following asymptotic expression (see [18] for more details):

$$\text{DT}(n) \approx d \left(\frac{27}{4} \right)^n n^{-3/2}, \quad \text{where } d \approx 0.00168809016.$$

This has to be compared with the number of phylogenies [43], that is, $\approx (1/2\sqrt{2})(2/e)^n n^{n-2}$, which grows much faster. For example, when $n = 9$ the proportion of duplication trees among phylogenies is about 3.85×10^{-2} , while with $n = 15$ it is only about 2×10^{-5} .

Moreover, this non-biological generating scheme can be used in a number of computational tasks, for example, to enumerate rooted or unrooted duplication trees, or for random tree generation. To generate random duplication trees with n segments, we first compute by dynamic programming all $p(n, q, m)$ values for $2 \leq q \leq n$ and $0 \leq m \leq q - 2$. To obtain a uniform distribution on rooted trees, we start from the 2-leaf tree and use the generating scheme by drawing at each stage from among the possible moves with a probability distribution that is proportional to the number of trees with n segments that can be generated from

these moves, as given by the $p(n, q, m)$ values. To uniformly randomly generate unrooted trees, we proceed in the same way but starting from the left child of the initial 2-leaf tree and, finally, removing the root of the tree that has been generated.

8.4 Inferring duplication trees from sequence data

8.4.1 Preamble

Data consist of an alignment of n segments with length q , and of the order O of the segments along the locus. Most studies consider DNA sequences, that is, segments are written using alphabet $\Sigma = \{A, T, G, C\}$, but most methods could deal with protein sequences, particularly distance-based methods. Gaps can be removed from the alignment, as often done in phylogenetic analysis, or kept and treated as a fifth character denoted as “-”. Note that the alignment has been created before tree construction and that the problem is not to build simultaneously the alignment and the tree, a much more complicated task [54]. Only Jaitly *et al.* [26] discuss simultaneous construction of alignments and trees as a possible extension of their approximation algorithm. In case of distance-based methods, aligned sequences are used to estimate the matrix of pairwise evolutionary distances between the segments, using any standard distance estimator, for example, Kimura two-parameter [27] or more sophisticated ones [48]. The computing time required for estimating all pairwise distances is in $O(n^2q)$, and the obtained distance matrix is used as input to the distance-based reconstruction algorithms.

Most studies address the inference of trees that only contain simple duplication events. Indeed, this task is simpler than dealing with the general case, as any such tree with leaves labelled by $O = (s_1, s_2, \dots, s_n)$ is composed of two subtrees with leaves labelled by $(s_1, \dots, s_m)$ and $(s_{m+1}, \dots, s_n)$, respectively, these two subtrees being themselves simple event duplication trees. As we shall see, this opens the way to dynamic programming and exact or approximation algorithms. Parsimony and distance-based approaches have been proposed, but, to the best of our knowledge, no probabilistic method (Chapter 2, this volume) has been published so far, even when this would be a natural and likely accurate way to infer duplication trees. In the following, we first address the computational hardness of duplication tree inference, then show that the inference of simple event duplication trees is easy with distances, and, finally, describe two parsimony and distance-based heuristic to infer unrestricted duplication trees. A review of various algorithmic and combinatorial aspects of tandemly repeated sequences is also provided by Rivals [36].

Before ending this preamble, we have to mention that standard phylogenetic reconstruction algorithms can often be used to infer duplication trees. For example, the two trees of Section 8.2 were built using DNAPENNY [24] from the PHYLIP package [12]. Indeed, when the data strictly conform to the duplication model, any phylogeny program should output a duplication tree, which can then be recognized, completed with its duplication events and drawn as

explained in Sections 8.3.3 and 8.3.4. In turn, finding a duplication tree when using any phylogeny inference method provides strong support for the duplication model. However, phylogeny algorithms are based on heuristics and often recover multiple equally optimal trees; in some situations, the duplication model may also be over-simplified and only a rough approximation of evolutionary processes. We then expect that the output phylogeny frequently does not strictly conform to the duplication tree constraints. A natural approach [59] is to perturb this phylogeny by small topological rearrangements until it becomes a duplication tree, but this approach becomes hazardous when the number of segments is large and when the initial phylogeny is far from any duplication tree (Section 8.5).

8.4.2 Computational hardness of duplication tree inference

In reference [26], the authors show that finding the optimal simple event duplication tree according to the parsimony criterion is NP-hard, just as is the phylogeny problem with parsimony [16]. This result does not prove that the same holds for unrestricted duplication trees, as a larger solution space sometimes makes the problems easier, but it is commonly believed that unrestricted duplication trees are more difficult to infer than restricted ones. However, NP-hardness only holds when both n (the number of segments) and q (the length of each segment, after multiple alignment) are unbounded. When q is fixed, Benson and Dong [2] describe a simple dynamic algorithm (close to that of Section 8.4.3) that find the most parsimonious (restricted) tree in times $O(|\Sigma|^q n^3)$, which makes it applicable for q of about 5 when Σ equals $\{A, T, G, C, -\}$, that is, applicable to micro-satellites. When n is fixed, we simply enumerate all possible trees, for example, using the generating scheme of Section 8.3.8. This brute force approach is often applicable, even in the unrestricted setting, as duplication trees are much less numerous than phylogenies. It was used by Elemento *et al.* [11] to deal with the human TRGV locus (see Section 8.2), the nine genes (i.e. 5,202 trees) being processed in a few minutes on a standard computer. Such an approach is then often suitable for tandemly repeated genes which contain around a dozen units or less.

We show below that finding the optimal simple event duplication tree can be done in polynomial time when using distances and the minimum-evolution (ME) criterion. However, the hardness of inferring unrestricted trees from distances remains an open question. Moreover, it is not known whether our result applies to other distance-based criteria in the restricted case. Note that for phylogenies the same questions are still partly open.

Because of the hardness of the task, a natural approach is to search for *approximation algorithms*. Benson and Dong [2] and Tang *et al.* [49] describe two different 2-approximation algorithms for the inference of simple event duplication trees using parsimony. Such an algorithm outputs a tree whose parsimony value is always less than twice that of the most parsimonious tree. This finding is an extension of a well-known result in phylogenetics, where the 2-approximation is built from a minimum spanning tree [43]. However, in the case of simple event

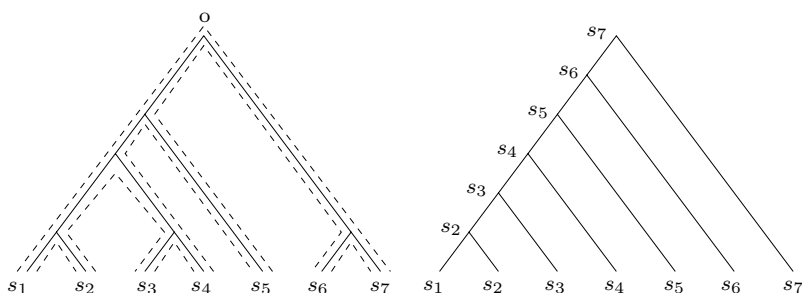


FIG. 8.7. (a) An optimal tree with parsimony P^* . (b) A caterpillar tree; when the internal nodes are loaded with segments $s_2, s_3, \dots, s_7$, the parsimony of this tree is less than $2P^*$.

duplication trees, this result is not of any practical help (even if important from a theoretical standpoint), as we shall see from Benson and Dong's construction [2]. Consider an optimal tree with parsimony P^* , and perform a depth-first traversal of this tree; every edge is run twice (once in each direction) and then the cost of this traversal is $2P^*$ (see Fig. 8.7(a) for an illustration of this). Writing down this traversal we get a tour of the form $\dots s_1 \dots s_2 \dots s_3 \dots s_n \dots s_1 \dots$, where internal nodes are not indicated. Because of the triangle inequality, the cost of this tour (i.e. $2P^*$) is higher than the cost of the spanning tree $s_1 - s_2 - s_3 - \dots - s_n$. But the cost of this spanning tree is itself higher than the parsimony of a caterpillar tree (a caterpillar tree is a tree in which each internal node is adjacent to at least one leaf node). Indeed, the cost of the caterpillar tree is equal to that of the spanning tree, when the internal nodes are loaded with $s_2, s_3, \dots, s_n$, as shown in Fig. 8.7(b). An optimal loading of the internal nodes, as computed by Fitch-Hartigan algorithm [14, 23], then gives a parsimony lower than the cost of the spanning tree, that is, lower than $2P^*$. In other words, always outputting a caterpillar tree, whatever the sequence data, gives a 2-approximation algorithm, which is clearly unsatisfactory.

To improve this approximation ratio of 2, Jaitly *et al.* [26] and Tang *et al.* [49] described polynomial time approximation schemes (PTAS) for the problem of inferring simple event duplication trees using parsimony. A PTAS is an algorithm which, for every $\epsilon > 0$, returns a solution whose cost is at most $(1 + \epsilon)$ times the cost of the optimal solution, and which runs in time bounded by a polynomial (depending on ϵ) in the input size. The two proposed PTAS are very similar and combine dynamic programming on growing intervals (just as in Section 8.4.3) with previous results on the problem of tree alignment of multiple sequences with a given phylogeny [54, 55]. Even though having a PTAS is positive from a theoretical standpoint, it again does not seem to be helpful in practice. For example, PTAS by Jaitly *et al.* [26] requires a computing time in $O(n^{11})$ to guaranty a ratio of 1.5. Those authors suggest that this impressive

time complexity is due to a rough analysis and they display favourable performance of their PTAS in comparison with Benson and Dong [2] heuristic algorithm (Section 8.4.4). However, we have not been able to reproduce their observations when using simulated data such as those described in Section 8.5. The same was observed by Wang and Gusfield [54] for the tree alignment problem, their PTAS being clearly outperformed by the simple heuristic of Sankoff *et al.* [41].

8.4.3 Distance-based inference of simple event duplication trees

We address in this section the simple event duplication tree problem, when using as input the matrix of pairwise evolutionary distances between the segments. Our construction is based on the minimum-evolution principle, which involves selecting the tree whose estimated length is minimal among all possible trees. Tree length estimation is based on ordinary least-squares (OLS) fitting, and it is known that under this setting the *minimum-evolution principle* is consistent (see [8, 39] and Chapter 1, this volume), that is, if the distance matrix exactly corresponds to a given tree with positive edge lengths, then this tree is the shortest tree. Using this principle makes the simple event duplication tree problem easy, as we describe an algorithm that selects the shortest tree among all possible simple event duplication trees and runs in polynomial time. We first introduce notation, then provide the recurrence formula for tree length estimation on which our algorithm is based. Implementation details are given in reference [10].

The distance matrix is denoted as $\Delta = (\Delta_{s_i s_j})$, where $\Delta_{s_i s_j}$ is the estimated evolutionary distance between the segments s_i and s_j . The average distance between two non-intersecting subtrees I and J is $\Delta_{IJ} = (1/|I||J|) \sum \Delta_{s_i s_j}$, where s_i and s_j are leaves (segments) in I and J , respectively. Δ being given (but omitted for the sake of simplicity), we denote $\hat{l}(u, v)$ the OLS length estimate of edge (u, v) , and $\hat{l}(T)$ the length estimate of tree T , that is, the sum of length estimates of every edge of T . By extension we denote $\hat{l}(X)$ the length estimate of any subtree X of T . Finally, letting X be a rooted subtree, $\bar{X}$ represents the average of path length estimates between the root of X and its leaves.

The OLS edge-length estimation can be obtained from local computations, which explains the simplicity of the problem at hand, when combined with the fact that the leaf set of any simple event duplication subtree is an interval of O . Using notation of Fig. 8.8, we have [51]:

$$\hat{l}(a, u) = \frac{1}{2}(\Delta_{AB} + \Delta_{AC} - \Delta_{BC}) - \bar{A}.$$

As can be seen from this formula, the $\hat{l}(a, u)$ estimate does not depend on the topology of B and C , but only on the average distances between A , B , and C , and on the estimated lengths of the edges in A . In the same way, the edge length estimates within A do not depend on the topology associated to the segments that are outside A . We can then compute $\hat{l}(A)$ and $\bar{A}$ without knowing the rest of the tree, and the same holds for B and C by symmetry. Moreover, it is easily

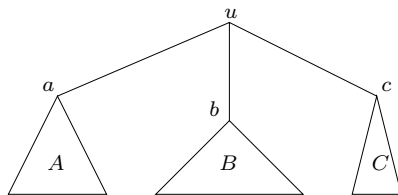


FIG. 8.8. Any unrooted simple event duplication tree is composed of three subtrees that we denote A , B , and C ; the corresponding leaf sets (also denoted as A , B , and C , for the sake of simplicity) are adjacent intervals of O ; a , b , and c denote the roots of subtrees A , B , and C , respectively.

seen from the above equation that the total tree length estimate is given by:

$$\hat{l}(T) = \frac{1}{2}(\Delta_{AB} + \Delta_{AC} + \Delta_{BC}) + (\hat{l}(A) - \bar{A}) + (\hat{l}(B) - \bar{B}) + (\hat{l}(C) - \bar{C}). \quad (8.1)$$

Assuming now that A is composed of two subtrees A_1 and A_2 , we obtain in the same way:

$$\begin{aligned} \hat{l}(A) - \bar{A} &= (\hat{l}(A_1) - \bar{A}_1) + (\hat{l}(A_2) - \bar{A}_2) + \frac{1}{2}\Delta_{A_1A_2} \\ &+ \frac{1}{2}\left(\frac{|A_2| - |A_1|}{|A|}\right)\Delta_{A_1(B \cup C)} + \frac{1}{2}\left(\frac{|A_1| - |A_2|}{|A|}\right)\Delta_{A_2(B \cup C)}. \end{aligned} \quad (8.2)$$

Equation (8.1) consists of four independent terms: $(\hat{l}(A) - \bar{A})$, $(\hat{l}(B) - \bar{B})$, $(\hat{l}(C) - \bar{C})$, and the remaining term. To minimize the total tree length $\hat{l}(T)$, we adopt a divisive strategy which consists of partitioning O into three subsets A , B , and C , then of independently computing the topology which minimizes $\hat{l}(X) - \bar{X}$ for each of these subsets, and finally of applying equation (8.1). The optimal tree is given by the optimal partition. Identically, to obtain the optimal topology for X ($X = A, B$, or C), we need to evaluate every partitioning of X into two subsets X_1 and X_2 , then to independently compute the topology for X_1 and X_2 which minimizes $\hat{l}(X) - \bar{X}$ and finally to select the partitioning of X which minimizes equation (8.2).

These computations are achieved by *dynamic programming*. We compute the optimal value of $\hat{l}(X) - \bar{X}$ and the corresponding partitioning for every growing interval $X = (s_i, \dots, s_j)$ of O . If $j = i$, then $\hat{l}(X) - \bar{X} = 0$. If $j = i + 1$, then there is only one possible partitioning and $\hat{l}(X) - \bar{X}$ is directly obtained from equation (8.2). When $j > i + 1$, we evaluate every partitioning $(s_i, \dots, s_m)$, $(s_{m+1}, \dots, s_j)$, $i \leq m < j$; each subinterval has already been processed and we apply equation (8.2) to compute $\hat{l}(X) - \bar{X}$ for every partitioning and find the best one. We stop when having the optimal value and partitioning for every

interval of length $n - 2$. We then apply equation (8.1) and step back through the optimal interval partitionings to construct the shortest tree. This algorithm can be implemented in $O(n^3)$ time using preprocessing and simple data structures [10].

8.4.4 *A simple parsimony heuristic to infer unrestricted duplication trees*

Parsimony-based inference of duplication trees is computationally difficult (Section 8.4.2). Benson and Dong [2] describe a simple heuristic applying to various settings, which they detail for the special case of simple event duplication trees. We describe here this heuristic for the more general case where multiple duplication events are allowed. This heuristic uses an agglomerative approach, which is very common in distance-based phylogeny reconstruction (e.g. Neighbor Joining [40]) and was also employed in the first parsimony inference algorithms. The principle consists of searching for a series of cherries forming a visible duplication (Section 8.3.3), computing the ancestral segment of every selected cherry, replacing both leaves of every cherry by its ancestral segment, and iterating the process until 1 segment remains. The algorithmic scheme is then very close to that described in Section 8.3.3, the difference being that visible duplication events are now selected from the segments.

Let l and r be 2 segments of O , and let $l[p]$ and $r[p]$ be the value of the p th site of l and r , respectively. The parsimony distance between l and r is then simply equal to the number of sites p where $l[p] \neq r[p]$, and the value of the ancestral sequence u is given by: if $l[p] = r[p]$, then $u[p] = l[p]$, else $u[p] = \{l[p], r[p]\}$. In the latter case $u[p]$ can be equal to $l[p]$ or to $r[p]$ but in both cases the parsimony cost is 1. Let now l and r be any two given segments, taken from O or computed during the course of the algorithm; $l[p]$ and $r[p]$ are then sets of possible values included in alphabet Σ , the original segments of O having only one possible value per site (unless the alignment itself contains ambiguities). The ancestral sequence and the parsimony distance is then given by Fitch and Hartigan [14, 23]: if $l[p] \cap r[p] \neq \emptyset$ then $u[p] = l[p] \cap r[p]$ and the parsimony cost is 0 for site p , else $u[p] = l[p] \cup r[p]$ and the parsimony cost is 1 for site p ; the parsimony distance between l and r is equal to the sum of the parsimony costs for all the sites. This more general definition clearly includes our initial setting where l and r were taken from O , and is sufficient to describe Benson and Dong's heuristic in a simple way.

Given O and the aligned segments, we search for a visible duplication event, that is, a series of segment pairs (l_i, r_i) , $j \leq i \leq k$, such that $(l_j, \dots, l_i, \dots, l_k, r_j, \dots, r_i, \dots, r_k)$ is included in O (see also Section 8.3.3). Among all possibilities, we select the series such that the average parsimony distance between each (l_i, r_i) pair is minimum. We then compute the ancestral sequences u_i as indicated above, create the cherries (l_i, u_i, r_i) in the tree being constructed, and replace $(l_j, \dots, l_i, \dots, l_k, r_j, \dots, r_i, \dots, r_k)$ by $(u_j, \dots, u_i, \dots, u_k)$ in O and in the alignment. This process is repeated until one segment remains, and the parsimony of the resulting tree is equal to the sum of the parsimony distances for all of

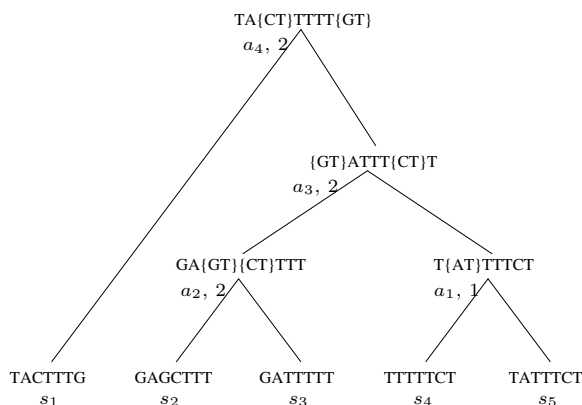


FIG. 8.9. Sample execution of Benson and Dong algorithm. For each internal node, we indicate the reconstructed ancestral sequence, the order between successive agglomerations ($a_1, a_2, \dots, a_n$) and their parsimony cost. The shown example is based on Jaitly *et al.* [26] and corresponds to the tree that was recovered using their PTAS. It is interesting to note that, at the second step, there are two possible agglomerations with cost 2 (the other one would agglomerate s_3 with a_1). The agglomeration that was selected in this example (s_2 with s_3) yields a tree with final cost 7, while agglomerating s_3 with a_1 yields a tree with cost 8 (cf. Jaitly *et al.* [26]). As suggested by Benson and Dong [2], this example clearly shows that exploring alternatives with identical cost can sometimes lead to more parsimonious trees.

the cherries that have been created during the course of the algorithm (Fig. 8.9). Note that the final segment cannot be interpreted as the tree root, as other roots are possible (except special cases, see Section 8.3.2) with the same parsimony value. Also, as ties can occur in visible event selection step (e.g. Fig. 8.9), Benson and Dong use some backtracking to search more extensively the solution space and then find more parsimonious duplication trees.

At each stage the number of segment comparisons is in $O(n^2)$ and computing the cost of every possible visible duplication events is in $O(n^3)$; the whole algorithm then requires $O(n^3q + n^4)$ as the number of steps is in $O(n)$. However, this algorithm can be accelerated to $O(n^2q + n^4)$ by only comparing segment pairs where one of both segments is a new ancestral segment computed during the previous step.

8.4.5 Simple distance-based heuristic to infer unrestricted duplication trees

In this section, we show that the very same greedy heuristic as described above can be adapted to distances. The input is now the matrix Δ of pairwise evolutionary distances between the segments, and the segment ordering O . The first algorithm of this kind, called WINDOW, was proposed in reference [49]. To select

at each step the series of segment pairs (l_i, r_i) forming a visible duplication event (see notation in Sections 8.3.3 and 8.4.4), this algorithm simply uses the entries in Δ . The selection criterion (to be minimized) is the average of the distances in Δ between all (l_i, r_i) pairs of the series. Once the best series has been selected, the cherries (l_i, u_i, r_i) are created in the tree being constructed, the l_i s and r_i s are replaced by the u_i s in Δ and O , and the new distances in Δ are computed using:

$$\Delta_{u_i u_{i'}} = \frac{1}{4} (\Delta_{l_i l_{i'}} + \Delta_{l_i r_{i'}} + \Delta_{l_{i'} r_i} + \Delta_{r_i r_{i'}})$$

and

$$\Delta_{u_i v} = \frac{1}{2} (\Delta_{l_i v} + \Delta_{r_i v}),$$

when v does not belong to the u_i s.

The algorithm continues until one segment remains. The time complexity is in $O(n^4)$, as for Benson and Dong's algorithm except that segment comparisons are no longer performed within the algorithm but before when computing the distance matrix (which also requires $O(n^2q)$).

The WINDOW algorithm is closely related to UPGMA and WPGMA ([45] and Chapter 1, this volume) as it simply uses the distance between the segments to select the pairs to be agglomerated. It is well known in phylogenetics that this approach can be inconsistent when the molecular clock hypothesis is not satisfied [48]. Even when duplication histories deal with relatively recent time, this hypothesis does not always hold, specially when dealing with gene families containing pseudo-genes that are not functional and often evolve much faster than other genes [11]. Therefore, we used a more suitable pair selection criterion to build an improved algorithm which we called DTSCORE [9].

This criterion is the same as that employed by ADDTREE [42] and relies on the four-point condition [5, 57] and Chapter 1, this volume. If we consider a quartet of different segments $\{s_i, s_j, s_k, s_l\}$ and assume that Δ perfectly fits a tree T with positive edge lengths, then the smallest sum among $(\Delta_{s_i s_j} + \Delta_{s_k s_l})$, $(\Delta_{s_i s_k} + \Delta_{s_j s_l})$, and $(\Delta_{s_i s_l} + \Delta_{s_j s_k})$ defines the two external pairs in the restriction of T to those 4 segments. For example, if $(\Delta_{s_i s_j} + \Delta_{s_k s_l})$ is the smallest sum, then (s_i, s_j) and (s_k, s_l) are the external pairs. Moreover, a pair (s_i, s_j) is a cherry of T when it is external for every other pair (s_k, s_l) . The score S is then defined as follows:

$$S(s_i, s_j) = \sum_{\{s_k, s_l\} \cap \{s_i, s_j\} = \emptyset} \left(H \left[(\Delta_{s_i s_k} + \Delta_{s_j s_l}) - (\Delta_{s_i s_j} + \Delta_{s_k s_l}) \right] \right. \\ \left. H \left[(\Delta_{s_i s_l} + \Delta_{s_j s_k}) - (\Delta_{s_i s_j} + \Delta_{s_k s_l}) \right] \right),$$

where H is the heaviside function: $H(x) = 1$ if $x > 0$, else $H(x) = 0$.

When Δ perfectly fits T , $S(s_i, s_j)$ is maximal (i.e. equal to $(n-2)(n-3)/2$) if and only if (s_i, s_j) is a cherry of T , and all cherries of T have a maximal score.

This property is needed for duplication tree inference as we are searching for visible duplication events which might contain several cherries. In contrast, Neighbor Joining criterion [17, 40, 47] does not possess this property (only the pair with best value is guaranteed to be a cherry) and is not suited for duplication trees.

The pair scores are used to compute the fitness of every possible duplication event. The average score for all pairs in a given event can be used, just as in Benson and Dong's and WINDOW algorithms. However, better results are obtained when using the minimum of those scores, and further improvements are obtained by combining both solutions in a lexicographic way, first considering the minimum of the scores and then the average in case of tie.

The whole DTSCORE algorithm can be summarized as follows. At each step, scores are computed for all pairs of segments; these scores are used to evaluate the fitness of every possible duplication event and the best event is selected; this event is agglomerated as in WINDOW algorithm; the process is repeated until only one segment remains. All scores can be computed in $O(n^4)$ by updating them from step to step, instead of recalculating them from scratch, and the time complexity of the whole algorithm is then in $O(n^4)$ just as with WINDOW. More implementations tricks are detailed in reference [9].

8.5 Simulation comparison and prospects

The duplication tree reconstruction methods presented in Section 8.4 are very different, and comparing them is a difficult task. We used computer simulations, as in reference [9] and many phylogenetic reconstruction studies. We uniformly randomly generated unrestricted (i.e. possibly containing multiple duplication events) duplication trees with 12 and 24 leaves (Section 8.3.8) and assigned lengths to the edges of these trees using the coalescent model [28]. We then obtained molecular clock trees (MC), which might be unrealistic in numerous cases, for example, when the sequences being studied contain pseudo-genes which evolve much faster than fully functional genes. Therefore, we generated non-molecular clock trees (NO-MC) from the previous ones, by independently multiplying every edge length in these trees by an exponentially distributed random variable (see [9] for more details). The trees so obtained (MC and NO-MC) have a maximum leaf-to-leaf divergence in the range [0.1, 0.7], and in NO-MC trees the ratio between the longest and shortest root-to-leaf lineages is of about 3.0 on average. Both values are in accordance with real data, for example, gene families (Sections 8.2.4, 8.2.5). SEQGEN [35] was used to produce a 1,000 bp-long nucleotide multiple alignment from each of the generated trees, using F84 model of substitution [13], and a distance matrix was computed by DNADIST [12] from this alignment using the same substitution model. One thousand trees (and then 1,000 sequence sets and 1,000 distance matrices) were generated per tree size and per condition. These data sets were used to compare the ability of the various methods to recover the original trees, from the sequences or from the distance matrices depending on the method being tested. Two criteria were

TABLE 8.1. Simulation comparison of 5 inference methods

	12 MC		24 MC		12 NO-MC		24 NO-MC	
	%tr	%ev	%tr	%ev	%tr	%ev	%tr	%ev
NJ	53.3	92.9	11.9	87.5	44.6	90.8	9.0	86.0
GS	54.5	85.7	12.7	67.4	46.8	82.8	9.6	65.3
GMT	34.9	87.6	4.2	78.4	25.7	82.9	2.2	71.9
WINDOW	52.8	92.5	12.8	87.1	26.4	85.5	3.3	75.8
DTSCORE	63.4	94.4	24.6	90.4	55.2	92.5	18.8	89.0

MC, molecular clock trees; NO-MC, no-molecular clock trees; %tr, percentage of correctly reconstructed duplication trees; %ev, percentage of recovered duplication events.

measured: %tr, percentage of trees (out of 1,000) being correctly reconstructed; %ev, percentage of duplication events in the true tree being recovered by the inferred trees. To measure the latter, it is necessary to root the inferred tree, and we simply used the (allowed) root position corresponding to best criterion value.

Using this simulation protocol, we compared: NJ [40], GREEDY-SEARCH (GS) [59] when starting from the NJ tree (Section 8.4.1), GREEDY-MANY-TRHIST (GMT) [2] as described in (Section 8.4.4) (i.e. without backtracking), WINDOWS [49] and DTSCORE [9] (Section 8.4.5). Results are displayed in Table 8.1. They clearly indicate that DTSCORE performs better than all the other tested methods. NJ performs relatively well, but it often outputs trees that are not duplication trees, which is unsatisfactory. GS only slightly improves over NJ regarding the proportion of correctly reconstructed trees, but considerably degrades the number of recovered duplication events, which is likely explained by the blind search it performs to transform NJ trees into duplication trees. GMT results are also relatively poor, possibly due to the fact that we did not implement any backtracking as recommended in reference [2]. As expected from its assumptions, WINDOW performs better in the MC case, where it can be seen as the second best method, than in the NO-MC one.

Even though DTSCORE simulation results are satisfactory, we do not believe it is the last word as it derives from ADDTREE [42], which is now outperformed by a number of more recent phylogeny inference methods. Better algorithms to reconstruct duplication trees are certainly possible, both in terms of time complexity and accuracy. Topological rearrangements specially designed for tandem duplication trees represent an interesting direction for further research (see [3] for a first attempt), as rearrangements have proven very efficient in classical phylogeny reconstruction [48]. It is unclear whether the dynamic programming approach we described in Section 8.4.3 for the simple duplication tree problem can be extended to handle multiple duplication events or not. Extending this approach or proving NP-hardness results represents another possible direction for

further research. Successful classical phylogeny approaches based on minimum-evolution principle (Chapter 1, this volume), maximum likelihood (Chapter 2, this volume), and Bayesian inference (Chapter 3, this volume) could also be adapted to the duplication tree problem. On the biology side, it would be relevant to study a large number of loci containing repeated segments, so as to provide even more support for the current duplication model. The availability of several fully completed and annotated genomes makes this kind of study possible. Such studies might also lead to refinements to the duplication model, to take into account additional evolutionary events as deletions or conversions. Appropriate reconstruction algorithms taking these events into account already represent an important direction for further research.

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CONSERVED SEGMENT STATISTICS AND REARRANGEMENT INFERENCES IN COMPARATIVE GENOMICS

David Sankoff

The statistical treatment of chromosomal rearrangement has evolved along with the biological methods for producing pertinent data. We trace the development of conserved segment statistics, from the mouse linkage/human chromosome assignment data analysed by Nadeau and Taylor in 1984, through the comparative gene-order information on organelles (late 1980s) and prokaryotes (mid-1990s), to higher eukaryote genome sequences, whose rearrangements have been studied without prior gene identification. Each new type of data suggested new questions and led to new analyses. We focus on the problems introduced when small sequence fragments are treated as noise in the inference of rearrangement history.

9.1 Introduction

The history of modelling and quantitative analysis for comparative genomics has been largely determined by the kinds of experimental data available at various periods (see the timeline in Table 9.1). For over 80 years, recombination-based linkage maps have been used for studying genome rearrangements. Through studies of giant salivary gland chromosomes in *Drosophila*, band structure became a valuable tool 70 years ago, allowing the visualization of inverted segments and the localization of their breakpoints with the microscope, and enabling the first rearrangement-based phylogeny.

Cytogenetics blossomed in the intervening years but modern banding techniques for human and other eukaryotic chromosomes are little more than 30 years old. These soon led to phylogenies for primates and a number of other groups. The last 30 years also saw the development of radiation hybrid methodology as well as a number of sequence-level molecular biological techniques, first for gene assignment to chromosomes, then for constructing chromosomal maps of genes and other features at increasing levels of resolution. Complete genome sequencing resulted in the first complete virus map in 1975 and the first complete organelle map in 1981, and increasing number of these became available for comparative work in the mid-1980s. It is less than 10 years since whole genome sequences of

TABLE 9.1. Availability of comparative genomic data

1921	Recombination maps	Sturtevant [49]
1933	Chromosome bands (<i>Drosophila</i>)	Painter [30]
1970	Chromosome bands (<i>human</i>)	Caspersson <i>et al.</i> [9]
1975	Radiation hybrid	Goss and Harris [16]
1975	Virus genome sequences	Sanger <i>et al.</i> [35]
1981	Organelle genome sequences	Anderson <i>et al.</i> [1]
1995	Prokaryotic genome sequence	Fleischmann <i>et al.</i> [13]
1996	Eukaryotic genome sequences	Goffeau <i>et al.</i> [15]
2001	Human genome sequence	[21, 54]

prokaryotes could be compared, and it is within the last 5 years that comparative genomics of eukaryotes could be based on whole genome sequences.

There are at least two mathematically oriented literatures in comparative genomics that go beyond traditional quantitative reports of intensive variables such as base composition or codon usage or summary variables such as genome size or gene content. One is the statistical analysis of genetic maps to quantitatively characterize the chromosomal segments conserved in both of two genomes being compared, as well as the breakpoints between these segments, dating to the fundamental paper of Nadeau and Taylor [28]. The other is the algorithmic inference of rearrangement processes, highlighted by the remarkable work of Hannenhalli and Pevzner [17–19], based on the comparison of complete gene orders. In this chapter, we review statistical analyses based on recombination distance and on gene order, as well as, very briefly, algorithms based on complete gene order, before focusing on the convergence of statistics and algorithmics in the comparison of whole genome sequences.

9.2 Genetic (recombinational) distance

At the time Nadeau and Taylor developed their approach to conserved segment statistics, distance along chromosomes was quantified in terms of linkage disequilibrium in recombination experiments, measured in centimorgans. They observed that some genes known to be located on the same human chromosome had homologous genes clustered on the same mouse chromosome linkage map, generally in the same order or in exactly the inverse order. Their insight was that the position of the mouse genes in these clusters could be used to determine the average size μ of conserved segments and hence the total number n of conserved segments, where the known total genome length is $|G| = n\mu$. Since the different clusters generally did not overlap, they made the assumption that each cluster represented a sample of the genes in a single conserved segment. The data to be considered was then of the form represented in Fig. 9.1.

Then the simplest form of the inference, though not exactly in Nadeau and Taylor's terms, is as follows: Let $x_1 < \dots < x_h$ be order statistics based on h independent samples from a uniform distribution on $[a, b]$. There are a number

$$a \quad x_1 \quad x_2 \quad \cdots \quad x_h \quad b$$

FIG. 9.1. Genes of known position $x_1 < \cdots < x_h$ in a conserved segment with unknown endpoints (breakpoints) a and b .

of ways of estimating $b - a$: the maximum likelihood estimate is $x_h - x_1$ but for small h this is obviously very biased towards underestimation. An unbiased estimate of $b - a$, but one which is only defined for $h \geq 2$, is

$$\widehat{(b - a)} = \frac{h + 1}{h - 1}(x_h - x_1). \quad (9.1)$$

Nadeau and Taylor could not calculate μ by simply averaging the estimates for the different segments, for two reasons. First, they could not observe those segments containing no mapped genes, and for data quality reasons, they did not consider segments containing only one gene. (In any case, the length estimators do not give meaningful estimates for segments containing one gene.) Second, the expected number of genes observed in a segment is proportional to the length of that segment, which itself approximately follows an exponential distribution, assuming a uniform distribution of breakpoints. Thus, the set of observed segment estimates must be fit to an exponential length distribution, conditioned on the probability that a segment of a specific length contains at least two mapped genes. The parameter of this distribution is an estimate $\hat{\mu}$ of μ , the average segment length. An estimate of the number of segments is then $\hat{n} = |G|/\hat{\mu}$.

9.3 Gene counts

We can use the uniformly distributed breakpoints component of the Nadeau–Taylor procedure to estimate n without first estimating the size in centimorgans of the observed segments, simply by counting the number of genes in each observed segment.

We model the genome as a single long unit broken at $n - 1$ random breakpoints into n segments, within each of which gene order has been conserved with reference to some other genome. Little is lost in not distinguishing between breakpoints and concatenation boundaries separating two successive chromosomes [38]. If the total number of genes is m , the marginal probability that a segment contains r genes, $0 < r < m$, has been shown [43] to be:

$$P(r) = \left(1 + \frac{m}{n + 1}\right)^{-1} \frac{\binom{m}{r}}{\binom{n + m}{r}}. \quad (9.2)$$

We cannot directly compare the theoretical distribution $P(r)$ with n_r , the number of segments observed to contain r genes, since we cannot observe n_0 , the number of segments containing no identified genes, and hence n is unknown.

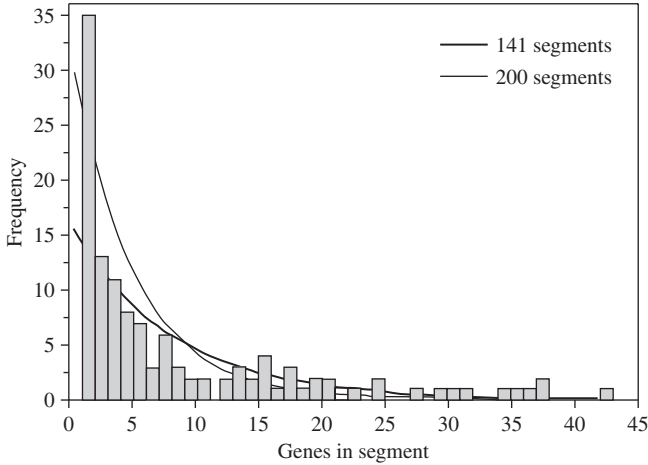


FIG. 9.2. Comparison of relative frequencies $n_r, r > 0$ of segments containing r genes, with predictions of the Nadeau–Taylor model, for MLE $\hat{n} = 141$ and Kolmogorov–Smirnov-based estimator $\hat{n} = 200$ [41].

We can, however, compare the frequencies n_r with the predicted frequencies $\hat{n}P(r), r > 0$, for various estimators $\hat{n}$, as illustrated in Fig. 9.2, our first analysis based on the $m = 1423$ human–mouse orthologies documented in 1996.

The largest discrepancy is the comparison between n_1 and $\hat{n}P(1)$, due at least in part to error in the identification of orthologous genes or other experimental error in chromosome assignment, but also possibly to a genuine shortfall in the model when predicting the number of short segments. We will return to this question in Section 9.8 on genome sequences.

9.4 The inference problem

It might seem undeniable that the number of segments n_r observed to contain r genes, for $r = 1, 2, \dots, m$ would be useful data for inference about the Nadeau–Taylor model, in particular about n , the unknown number of segments. It is remarkable, then, that to estimate n from m and n_r , only the number of non-empty segments $a = \sum_{r>0} n_r$ is important, since for practical purposes it behaves like a sufficient statistic for the estimation of n [41], although sufficiency is not strictly satisfied [20].

To estimate n , we study $P(a, m, n)$ the probability of observing a non-empty segments if there are m genes and n segments. Combinatorial arguments give:

$$P(a, m, n) = \frac{\binom{m-1}{a-1} \binom{n}{a}}{\binom{n+m-1}{m}}, \quad (9.3)$$

which is a constrained hypergeometric distribution with mean and variance:

$$\mu_a = \frac{mn}{m+n-1}, \quad \sigma_a^2 = \frac{n(n-1)m(m-1)}{(n+m-2)(n+m-1)^2}. \quad (9.4)$$

Note that this model reduces to a classical occupancy problem of statistical mechanics ([12], p. 62).

The maximum likelihood estimate $\hat{n}$, given m and a , is the value of n which maximizes P . For given m and n , the expectation and the variance of $\hat{n}$ can be calculated making use of the probability distribution in equation (9.3), except in the special case of few data ($m \leq n$) and every gene in a separate segment ($a = m$), where the estimates are undefined.

Substituting a for μ_a in equation (9.4) gives Parent's estimator [31]

$$\hat{n} = \frac{a(m-1)}{m-a}, \quad (9.5)$$

which, when rounded to the nearest integer, coincides with the maximum likelihood estimator over the range of a , m , and n likely to be experimentally interesting, as long as some segments contain at least two genes.

Alternatives, extensions, and generalizations of the Nadeau–Taylor and the gene count approaches have been investigated by a number of researchers. Schoen has shown that high marker (e.g. gene) density and high translocation/inversion ratios greatly improve the accuracy of estimation [46, 47]. Waddington *et al.* have developed the theory in the direction of allowing different densities of breakpoints for each chromosome [55] and have compared various approaches for their performance in avian genomes, with their distinctly bimodal distribution of chromosome sizes [56]. The evolution of chromosome sizes have been studied analytically, through simulation and empirically [4, 10, 38]. Marchand [27] initiated the statistical study of inhomogeneities in breakpoint densities and gene densities on the chromosome. Housworth and Postlethwait [20] showed how the number of observed conserved syntenies, that is, pairs of chromosomes—one in each genome—that share at least one ortholog, has some better statistical properties than the number of observed segments.

9.5 What can we infer from conserved segments?

The comparative study of whole-genome maps makes no formal reference to the processes that create the breakpoints while progressively fragmenting the conserved segments, except for an implicit assumption that the number of breakpoints and segments increases roughly in parallel with the number of rearrangement events affecting either of the two genomes being compared. In observing the order of segments along the chromosomes in one genome while noting to which chromosomes they correspond in the other genome, however, we can extract additional information about the relative proportion of intra-chromosomal and inter-chromosomal events that gave rise to this pattern. Considering only autosomes, that is, setting aside the sex chromosomes, which

are essentially excluded from inter-chromosomal exchanges, let the total number of segments on a human chromosome i be

$$n^{(i)} = t + u + 1, \quad (9.6)$$

where t is the number due to inter-chromosomal transfers, and u the number due to local rearrangements. Under a random exchange model we can try to predict how often two or more segments from the same mouse chromosome will co-occur on the same human chromosome through inter-chromosomal events. By then compiling co-occurrence frequencies from the empirical comparison of the two genomes, we can estimate the relative proportion of intra-chromosomal and inter-chromosomal events.

We label the ancestral chromosomes $1, \dots, c$, ignoring for the moment that there may have been changes in the number of chromosomes due to fusions and/or fissions in the human or mouse lineages or both. We model each chromosome as a linear segment with identified left-hand and right-hand endpoints. A reciprocal translocation between two chromosomes h and k consists of breaking each one, at some interior point, into two segments, and rejoining the four resulting segments such that two new chromosomes are produced, each containing a left-hand part of one of the original chromosomes and the right-hand part of the other. We label each new chromosome according to which left-hand it contains, but for each of its constituent segments, we retain the information of which ancestral chromosome it derived from.

At the outset, assume the first translocation on the human lineage involves ancestral chromosome i . We assume that its partner can be any of the $c - 1$ other ancestral autosomes with equal probability $1/c - 1$, so that the probability that the new chromosome labelled i contains no fragment of ancestral chromosome h , where $h \neq i$, is exactly $1 - (1/c - 1)$. For small t , after chromosome i has undergone t translocations, the probability that it contains no fragment of the ancestral chromosome h is approximately $(1 - (1/c - 1))^t$, with some correspondingly small corrections, for example, to take into account the event that h previously translocated with one or more of the t chromosomes that then translocated with i , and that a secondary transfer to i of material originally from h thereby occurred.

Then the probability that the new (i.e. human) chromosome i now contains at least one fragment from h is approximately $1 - (1 - (1/c - 1))^t$ and the expected number of ancestral chromosomes with at least one fragment showing up on human chromosome i is

$$E(c^i) \approx 1 + (c - 1) \left[1 - \left(1 - \frac{1}{c - 1} \right)^t \right], \quad (9.7)$$

where the leading 1 counts the fragment containing the left-hand endpoint of the ancestral chromosome i itself. More refined models are described in [42].

We assume that our random translocation process is stochastically reversible. This assumption should not introduce much error as long as chromosome sizes

do not deviate too much from their stationary distribution. Then we can treat the mouse genome as ancestral and the human derived (or vice versa), instead of considering them as diverging independently from a common ancestor. Now $E(c^i)$ represents the expected number of mouse chromosomes with at least one fragment showing up on human chromosome i .

As t increases for all the chromosomes, so that each human chromosome contains segments from several mouse chromosomes, equation (9.7) could wrongly predict c^i , since a translocation with chromosome j might transfer fragments of several ancestral chromosomes, possibly not including j and possibly of the same origin contained in chromosome i . Nevertheless, substituting c^i for $E(c^i)$ in equation (9.7) gives us

$$\hat{t} = \frac{\log(c-1) - \log(c-c^i)}{\log(c-1) - \log(c-2)}, \quad (9.8)$$

a good first estimate of t , where $c = 19$, the number of mouse autosomes. To illustrate, for the 22 human autosomes, a 100 kb resolution construction [53] indicates 350 autosomal segments, while the sum of the c^i is 109. Applying equation (9.8) to each chromosome and summing the 22 values of $\hat{t}$ gives a total of 130 segments. In other words, for $130 - 109 = 21$ segments, two (or more) segments from the same mouse chromosome are found on the same human chromosome *because of independent translocational events*. By equation (9.6), this leaves unaccounted for

$$\begin{aligned} \sum u &= \sum n^{(i)} - \sum t - 22 \\ &= 350 - 130 - 22 \\ &= 198 \end{aligned}$$

segments, which must be attributed to local rearrangements such as inversion.

Table 9.2 shows the results of these calculations for this and a number of other maps of various levels of resolution, based on genomic sequence or gene maps. Of interest in the genome sequence-based results is the relative stability of the estimates of the number of reciprocal translocations or other inter-chromosomal events versus the great increase in local rearrangements over the analyses based on gene maps. This reflects the discovery of high numbers of smaller-scale local arrangements recognizable from genomic sequence [8, 25] compared to gene maps. As resolution increases, a greater proportion of these local rearrangements have no effect on gene order and more of the conserved segments identified will contain no genes. At the same time, many of the conserved segments identified in the recent gene maps contain a number of genes in a relatively small stretch of sequence, too short to even show up as a conserved segment in the sequence-based analyses (cf. [5, 52]). Thus the congruence apparent between large conserved segments in the genome sequence and the gene map data breaks down as we zoom down to smaller segments, with small segments of conserved sequence containing no genes and the small segments containing genes passing beneath the radar of sequence-based analyses.

TABLE 9.2. Inference of inter- and intra-chromosomal rearrangements based on number of conserved segments and number of segment-sharing autosome pairs in the two genomes

Resolution of comparative map	Autosomal segments $\sum n^{(i)}$	Segment-sharing chromosome pairs $\sum c^i$	Inter- chromosomal $\sum t$	Intra- chromosomal $\sum u$
100 Kb [53]	350	109	130	198
300 Kb [8]	370	107	128	220
1 Mb [14]	270	100	117	131
200 genes [48]	192	99	114	59
12000 genes [7], NCBI	213	113/120	137/149	64/41

Sources: [53] based on UCSC Genome Browser, [8, 14] on anchor-sequence constructions, [48] on outdated human mouse homology data cited in [7, 40] on MGI 2004 Oxford grid cells containing at least three/two genes for the c^i , and on NCBI Human Mouse Homology Map for the $n^{(i)}$.

Many of the single-gene orthologies on comparative maps are undoubtedly due to paralogy and other errors in assignment, but a significant proportion will certainly prove to be valid, opening questions about the nature of the processes creating them. The repertoire of inversions, reciprocal translocations, and Robertsonian translocations, popular with modellers may have to be expanded to include such processes as transpositions or jump translocations, within and between chromosomes and non-tandem duplication processes, with or without loss of functionality.

9.6 Rearrangement algorithms

We can derive much more detailed inferences about the processes responsible for a particular comparative map if we are willing to work within the framework of a sufficiently restrictive model, though we must then be vigilant that our results are really consequences of the data rather than simply artifacts of the model restrictions. The types of chromosomal rearrangement most often modelled are inversions, reciprocal translocations, and fissions and fusions, including Robertsonian translocations. The basic aim is to efficiently transform a given genome, represented as a set of idealized disjoint chromosomes made up of an ordered subset of genes, into another given genome made up of the same genes but differently partitioned among chromosomes, in a minimum number of steps d . The algorithm outputs d and a sequence of d rearrangements that carry out the desired transformation. The literature on this problem area (highlighted by the Hannenhalli–Pevzner discoveries [17–19] and reviewed in reference [37]) is extensive and has seen much recent progress (cf. [2, 29, 50, 51, 58] and Chapter 10, this volume), and we will not go into details here. Some points will be important

in the ensuing sections:

1. Each reciprocal translocation or inversion increases or decreases the number of segments by at most two; that is, it adds or removes at most two breakpoints between adjacent segments. (Other rearrangements, such as transpositions or “jump translocations” can change the number of segments by three, but these are thought to be rare.)
2. In the Hannenhalli–Pevzner algorithms and their improvements, virtually all moves decrease the number of segments, that is, decrease the number of breakpoints, by two or one.
3. In general, there are a large number of optimal solutions.
4. The algorithms consider all operations to have the same cost, independent of whether they are inversions or translocations, and independent of how many genes are in their scope. This is essential to the algorithms. If we wish to modify the problems or to change the objective function, the mathematical basis of the algorithm is lost.

We will return to other aspects of these algorithms in Section 9.8.

9.7 Loss of signal

To what extent does the sequence of rearrangements reconstructed by rearrangement algorithms actually reflect the true evolutionary history? It is well-known that past a threshold of θn , where n is the number of genes and θ is in the range of $\frac{1}{3}$ to $\frac{2}{3}$, the inferred value of d tends to underestimate the number of events that actually occurred [22–24].

Whether any signal is conserved as to the actual individual events themselves, and which ones, is even more problematic.

Lefebvre *et al.* [26] carried out the following test: for a genome of size $n = 1,000$, they generated u inversions of size l at random (for $l = 5, 10, 15, 20, 50, 100, 200$), and then reconstructed the optimal inversion history, for a range of values of u . Typically, for small enough values of u , the algorithm reconstructed the true inversion history, although inversions that do not overlap may be reconstructed in any order. Above a certain value of u , however, depending on l , the reconstructed inversions manifest a range of sizes, as illustrated in Fig. 9.3, reflecting the ability of the algorithm to find alternative solutions, and eventually solutions where $d < u$, with the concomitant decay of the evolutionary signal.

For each l , they then calculated

$$\bar{s}_l = \max(u \mid \text{reconstruction has at most 95\% error})$$

$$\underline{s}_l = \min(u \mid \text{reconstruction has at least 5\% error})$$

where any inversion having length different from l is considered to be an error. Figure 9.4 plots $\bar{s}$ and $\underline{s}$ as a function of l and shows how quickly the detailed evolutionary signal decays for large inversions. Only for very small inversions is a clear signal preserved long after longer ones have been completely obscured.

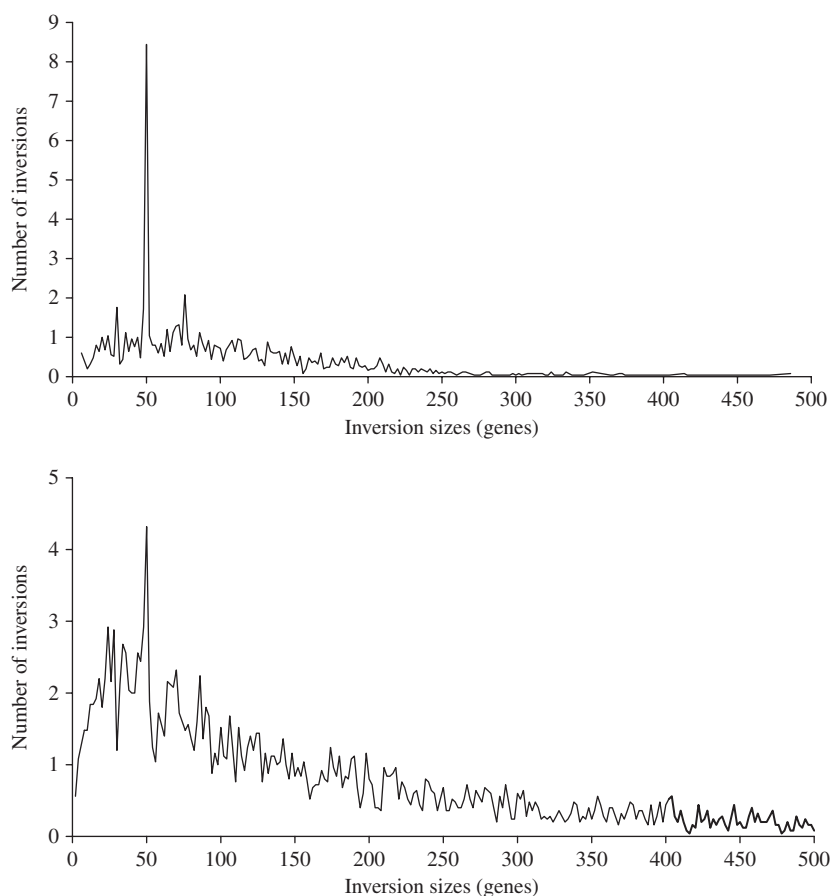


FIG. 9.3. Frequency of inversion sizes inferred by the algorithm for random genomes obtained by performing u inversions of size $l = 50$. Top: $u = 80$. Bottom: $u = 200$.

9.8 From gene order to genomic sequence

Gene order rearrangement algorithms can handle many thousands of genes in reasonable computing time. Faced with large nuclear genome sequences, particularly from the higher eukaryotes, however, uncertainties in global alignments, lack of complete consensus inventories of genes, and the difficulties of distinguishing among paralogs widely distributed across the genome, constitute apparently insurmountable impediments to the direct application of the algorithms.

9.8.1 The Pevzner–Tesler approach

In comparing drafts of the human and mouse genomes, Pevzner and colleagues [8,32–34] adopt an ingenious stratagem to leap-frog the global alignment,

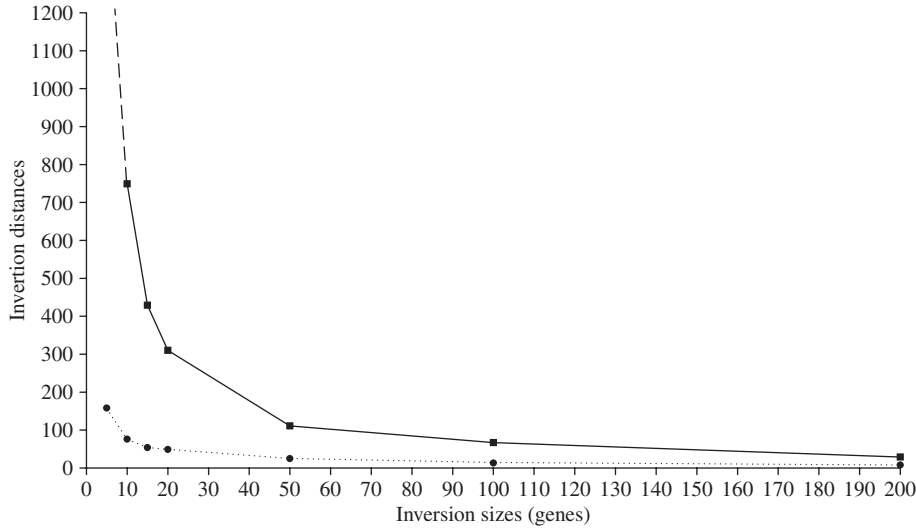


FIG. 9.4. Solid line: Values of $\bar{s}$. Dotted line: Values of $\bar{g}$.

gene finding, and ortholog identification steps. In their first study, on the human–mouse comparison, they analysed almost 600,000 relatively short (average length 340 bp) anchors of highly aligned sequence fragments as a starting point for building blocks of conserved synteny, and then amalgamated neighbouring sub-blocks using a variety of criteria to avoid disruptions due to “microrearrangements” less than 1 Mb. This procedure inferred a set of 281 blocks larger than 1 Mb, which is basically what is reported in references [14] and [8]. (The latter also improve the resolution down to 300 kb—see Table 9.2.) They then used the order of these blocks on the 23 chromosomes as input to a gene order rearrangement algorithms in order to reconstruct optimal sequences of d inversions and translocations to account for the divergent arrangements of the two genomes.

9.8.2 The re-use statistic r

One of the key results reported by Pevzner and Tesler pertains to the “re-use” of the breakpoints between the b' syntenic blocks on the c chromosomes used as input to their rearrangement algorithms. Basic to the combinatorial optimization approach to inferring genome rearrangements are the bounds $b/2 \leq d \leq b$, where $b = b' - c$. (This type of bound was first found in 1982 [57].) We define breakpoint re-use as $r = 2d/b$. Then

$$1 \leq r \leq 2.$$

The lower value $r = 1$ is characteristic of an evolutionary trajectory where each inversion or translocation breaks the genome at two sites specific to that particular rearrangement; no other inversion or translocation breaks the genome at

either of these sites. High values of r , near $r = 2$, are characteristic of evolutionary histories where each rearrangement after the first one breaks the genome at one new site and at one previously broken site. In their comparison of the human and mouse genomes, Pevzner's group find that r is somewhere between 1.6 and 1.9, depending on the resolution of their syntenic block construction and on whether they discard telomeric blocks or not, and argue that this is evidence that evolutionary breakpoints are concentrated in fragile regions covering a relatively small proportion of the genome.

Now it is easily shown that for random permutations of size n , the expected value of b is very close to n [39], and it is an observed property [23] of such permutations that the number of inversions needed to sort them (d) is also very close to n , and thus breakpoint re-use is close to 2. Without getting into the substantive claim about fragile regions persisting across the entire mammalian class, for which the evidence is controversial [11, 25, 36, 44], we may ask what breakpoint re-use in empirical genome comparison really measures: a bonafide tendency for repeated use of breakpoints or simply the degree of randomness of one genome with respect to the other at the level of syntenic blocks.

9.8.3 *Simulating rearrangement inference with a block-size threshold*

To see whether a high inferred rate of breakpoint re-use necessarily reflects a real phenomenon or is an artifact of methodology, Sankoff and Trinh [45] generated model genomes with NO breakpoint re-use ($r = 1$), then mimicked the Pevzner–Tesler imposition of a block-size threshold by discarding random parts of the genome before applying the Hannenhalli–Pevzner algorithm to the remainder of the genome to infer d and hence r .

Each genome consisted of a permutation of length $n = 1,000$ or $n = 100$ terms generated by applying d “two-breakpoint” inversions to the identity permutation $(12, \dots, n)$. A two-breakpoint inversion is one that disrupts two hitherto intact adjacencies in the starting (i.e. identity) permutation. At each step, the two breakpoints were chosen at random among the remaining original adjacencies. This represents the extreme hypothesis of no breakpoint re-use at all during evolution, which is not unreasonable given the 3×10^9 distinct dinucleotide sites available in a mammalian genome.

Of course, the terms are just abstract elements in the permutation and have no associated size, and indeed the Hannenhalli–Pevzner procedures do not involve any concept of block size. Thus, one way to imitate the effect of imposing a block-size threshold involves simply deleting a fixed proportion of the terms at random, the same terms from both the starting and derived genomes, relabelling the remaining terms according to their order in the starting (identity) genome, and applying the Hannenhalli–Pevzner algorithm.

It can be shown that before any deletions, the Hannenhalli–Pevzner algorithm will recover exactly d inversions. At each step it will find a configuration of form $\dots gh| - (i - 1) \dots - (h + 1) | ij \dots$ and will “undo” the inversion between h and i , removing two breakpoints. There being $b = 2d$ breakpoints, breakpoint re-use is 1.0.

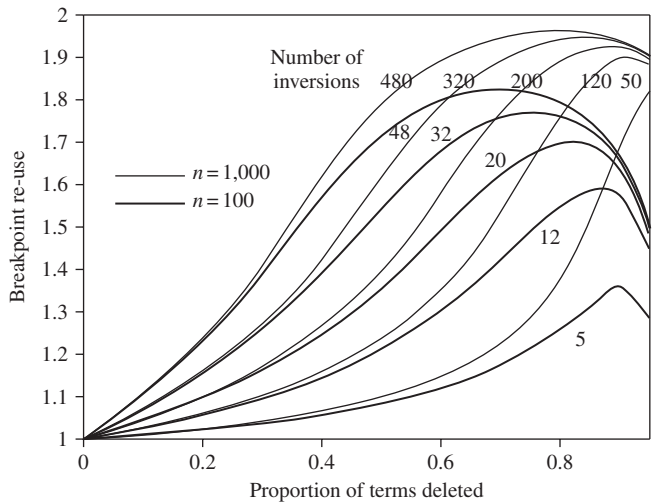


FIG. 9.5. Effect of deleting random terms on breakpoint re-use, as a function of proportion of terms deleted, for various levels of rearrangement of the genome.

What happens as terms are deleted? Suppose $j \neq i + 1$ in the above example, and i is deleted. Then the two-breakpoint inversion from $-(i - 1)$ to $-(h + 1)$ is no longer available to undo. An inversion that erases the breakpoint between h and $-(i - 1)$ will not eliminate a second breakpoint. So while the distance d drops by 1, the number of breakpoints b also only drops by 1, and r increases.

The probability that one, two, or more two-breakpoint inversions are “spoiled” in this way depends on the number of terms deleted.

Figure 9.5 shows how r increases with the proportion of terms deleted, for different values of d , for $n = 100$ and $n = 1,000$.

Note

- r increases more rapidly for more highly rearranged genomes
- the initial rate of increase of r depends only on d/n
- the increase in r levels off well below $r = 2$ and then descends sharply. The maximum level attained increases with n .

The first of these is readily explained. In more rearranged permutations, the deletion of term i is more likely to cause the configuration change described above, that is, $\cdots gh|-(i - 1) \cdots -(h + 1)|ij \cdots$, simply because it is more likely that $j \neq i + 1$.

The third observation is also easily understood. For large n , the re-use rate r approaches 2 for random permutations. As n decreases, however, expected re-use drops as indicated in Table 9.3. As more and more terms are dropped from a permutation, it loses its “structure,” that is, the pairs of breakpoints involved

TABLE 9.3. Expected re-use for random permutations as a function of n . Estimated from samples of size 500

n	r
5	1.53
25	1.83
50	1.90
100	1.94
250	1.97

in the original inversions are wholly or partially deleted, and the remaining permutation becomes essentially random. We may consider that after a curve in Fig. 9.5 attains its maximum, it is entering into the “noisy” region where the historical signal becomes thoroughly hidden.

9.8.4 *A model for breakpoint re-use*

This section explains the second observation above about the pertinence of d/n for the initial shape of the curves. Suppose a genome G has b breakpoints with respect to $12 \cdots n$ and the inversion distance is $d = d_2 + d_1$, where d_1 and d_2 represent the number of one-breakpoint inversions and two-breakpoint inversions required to sort G optimally. Then $2d_2 + d_1 = b$.

Suppose now that we delete one gene i at random and relabel genes $j = i + 1, \dots, n$ as $j = i, \dots, n - 1$, respectively. The number of breakpoints changes, and quantities b , d_1 , d_2 , and d can change only if the original gene i was flanked by two breakpoints. The probability of this event is $b(b-1)/n(n-1)$. The various configurations in which the two breakpoints may be involved, their probabilities and the effects of deleting i on d_1 and d_2 (b always decreases by 1, except in case 21 where it decreases by 2) are summarized in Table 9.4 and discussed in some detail in reference [45].

These considerations are, of course only valid insofar as the inversions associated with the endpoints are directly available in G (in fact, some are set up by other inversions later, during the sorting of G), but they give us an idea of the dynamics of the situation and motivate the deterministic model:

$$\begin{aligned}
 d_2(t+1) &= d_2(t) + \frac{b(t)(b(t)-1)}{(n-t)(n-t-1)}(-p_{21}(t) - p_{22}(t) - p_3(t)) \\
 &= d_2(t) - \frac{2d_2(2d_2-1) + 4d_1d_2}{(n-t)(n-t-1)},
 \end{aligned}$$

$$\begin{aligned} d_1(t+1) &= d_1(t) + \frac{b(t)(b(t)-1)}{(n-t)(n-t-1)}(p_{22}(t) + p_3(t) - \max[0, p_1(t)]) \\ &= d_1(t) + \frac{(d_2-2)/(d_2-1)2d_2(2d_2-1) + 4d_1d_2 - \max[0, d_1(d_1-1)]}{(n-t)(n-t-1)}, \end{aligned}$$

$$b(t+1) = 2d_2(t+1) + d_1(t+1),$$

where t ranges from 0 to n and with initial conditions $b(0) = 2d_2(0) = 2d(0)$ and $d_1(0) = 0$. (NB All the d terms on the RHS of the recurrence should be understood as indexed by t .)

Figure 9.6 shows how the recurrence models closely the average evolution of r as the number of terms randomly deleted increases, particularly at the outset, before there are large numbers of one-breakpoint inversions in the Hannenhalli–Pevzner reconstruction. As d_1 increases, the model renders less well the changing

TABLE 9.4. Probabilities and usual effects of discarding gene i in various configurations, given it is flanked by two breakpoints

Case	Configuration	Probability	Effect on	
			d_1	d_2
11	$-(i+1) i j$	$\frac{d_1}{b(b-1)}$	-1	0
12	$g -(i-1)\cdots h i j\cdots-(i+1) k$	$\frac{d_1(d_1-2)}{4b(b-1)}$	-1	0
	$g -(i-1)\cdots h i -(k-1)\cdots j k$	$\frac{d_1(d_1-2)}{2b(b-1)}$	-1	0
	$g h\cdots-(g+1) i -(k-1)\cdots j k$	$\frac{d_1(d_1-2)}{4b(b-1)}$	-1	0
21	$-(i+1) i -(i-1)$	$\frac{1}{d_2-1} \frac{2d_2(2d_2-1)}{b(b-1)}$	0	-1
22	$g -(i-1)\cdots-(g+1) i $	$\frac{d_2-2}{d_2-1} \frac{2d_2(2d_2-1)}{b(b-1)}$	+1	-1
	$-(k-1)\cdots-(i+1) k$			
3	$g -(i-1)\cdots-(g+1) i -(k-1)\cdots j k$	$\frac{d_1d_2}{b(b-1)}$	+1	-1
	$g -(i-1)\cdots-(g+1) i j\cdots-(i+1) k$	$\frac{d_1d_2}{b(b-1)}$	+1	-1

Note: Probabilities include those of inverted or nested versions (not listed) of configurations shown. Special cases of configurations with order $O(1/n)$ probabilities not distinguished, for example, $g|-(i-1)\cdots h|i|h+1\cdots-(i+1)|k$.

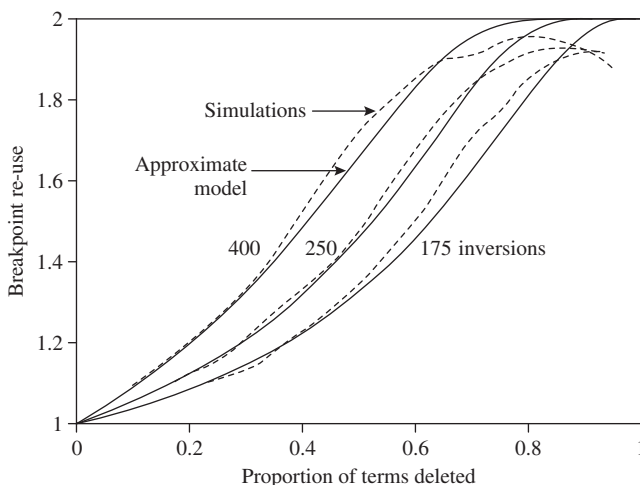


FIG. 9.6. Plot of r predicted by the recurrence compared to true value estimated by simulation.

structure of optimal reconstructions. Finally, the loss of historical signal in the noisy zone for the reconstructions is not built into the model, which thus attains $r = 2$ as the last terms of the permutation are deleted rather than the values in Table 9.3.

Let $\theta = t/n$ represent the proportion of terms deleted. Formally, since $r = 2d/b$, and d is constant in a neighbourhood of $t = 0$, while $db/dt \approx -(b/n)^2$, we can write that $dr/d\theta|_{\theta=0} = 2d/n$. This explains the coincidence between the curves for $n = 100$ and $n = 1,000$ in Fig. 9.5.

9.8.5 A measure of noise?

After investigating the effect of threshold size on r , albeit indirectly by varying the rate of random deletion of blocks, Sankoff and Trinh [45] carried out simulations that showed how amalgamations exacerbate the re-use artifact caused by deleting small blocks.

Though Pevzner and Tesler used r to infer relative susceptibility of genomic regions to rearrangement, the simulations described in this section show that it serves rather to measure the loss of signal of evolutionary history, due to the imposition of thresholds for retaining syntenic blocks and for repairing micro-rearrangements. Indeed, breakpoint re-use of the same magnitude as found by Pevzner's group may very well be artifacts of the use of thresholds in a context where NO re-use actually occurred. Indeed, while this may not have been their goal, Pevzner and Tesler have invented a statistic that is a measure of the noise affecting a genomic rearrangement process at the sequence level. Given some information about the parameters of rearrangement, the number of blocks and the size of the thresholds, the re-use rate tells us whether we can have

confidence in evolutionary signal reconstructed, whether it must be considered largely random, or whether we are in the “twilight” zone between the two.

9.9 Between the blocks

The syntenic blocks are reconstructed by algorithms that bridge the gaps between neighbouring well-aligned regions on both genomes, such as the Pevzner–Tesler method described in Section 9.8.1 above or the approach used in the UCSC Genome Browser [25].

Generally the two syntenic blocks on either side of a breakpoint on, say, a human chromosome do not abut directly, but are rather separated by a short region where there is little similarity with the mouse genome. The obverse of analysing the order of the reconstructed syntenic blocks as in Section 9.8 is the investigation of these regions, the largely unaligned stretches of genomic DNA left over once the blocks are identified.

Pevzner and Tesler interpret the lack of sustained human–mouse similarity in the regions containing breakpoints as suggestive of the “fragility” of these regions, their susceptibility to frequent rearrangement, in line with their claimed inference of breakpoint re-use. Previous documentation of evolutionary subtelomeric translocational hotspots and pericentromeric duplication and/or transpositional hotspots [11] can be adduced to support the strong hypothesis that potential breakpoints are largely restricted to a limited number (e.g. <500) of very small regions in the genome, and that this regional susceptibility is conserved over considerable evolutionary time scales. Further lines of evidence for this viewpoint include the high rates of recurrence of certain breakpoints in the clinical study of tumor cell karyotypes, and the existence of certain physically fragile regions in human chromosomes under laboratory conditions.

But how can we reconcile the apparently contradictory notions of evolutionarily conserved fragility of breakpoint regions and the lack of human–mouse similarity in these regions? If conserved fragility is based on some substantial primary sequence signal, why is this not picked up by the alignment protocol and how is it conserved if the region is being churned by rearrangements? There are, of course, many possible answers: the signals may be too short, they may be removed by repeat-masking prior to the reconstruction of the syntenic blocks, they may involve conserved secondary but not primary structures, they may involve GC-poorness or other gross sequence characteristics, or they may even be determined by unknown epigenetic considerations. There is no evidence, however, for any of these nor, as we argued in Section 9.8, for the contention that the breakpoint regions contain multiple breakpoints.

This notion of “fragile regions” or *a priori* proclivity for breakage as interpretation of the evidence is rejected in reference [53], where a combination of the following three factors is suggested to explain the limited amounts of similarity in the neighbourhood of breakpoints.

1. The algorithms [25,32] that reconstruct the syntenic blocks bridge gaps as long as appropriate similarity exists at both ends of the gap. A rearrangement

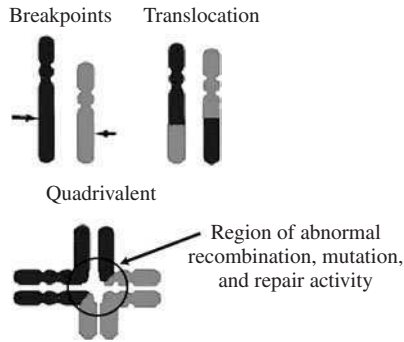


FIG. 9.7. Effect of meiotic non-alignment of regions surrounding breakpoints in heterokaryotypes.

event with one breakpoint within a gap destroys the match between the homologies at each end. This effect would show up only *after* the breakage event.

2. For a rearrangement to become established in a population, the process of meiosis has to tolerate the coexistence of different rearrangement haplotypes through many generations of heterokaryotypy. The mechanism of this tolerance may be seen in quadrivalent meiotic figures (in the case of reciprocal translocations), as depicted in Fig. 9.7, and in looped figures (in the case of inversions). Though there does not appear to be any direct molecular cytogenetic evidence, it is hypothesized that there is an increase of aberrant processes, such as recombination errors, deletion, duplication, or retroposition in the necessarily unapposed chromosomal regions in the immediate vicinity of breakpoints in such figures, during the heterokaryotypy period before the rearrangement becomes fixed. Note that this process is operative only *after* the rearrangement event, and is consistent with breakpoints occurring randomly over virtually the entire genome and not confined to a small number of regions.

3. To the extent that breakage occurs disproportionately in intergenic regions, these tend to undergo more rapid sequence evolution than regions containing exons and introns. This is not the same as the fragile regions hypothesis: the number of intergenic regions is almost two order of magnitudes greater than the supposed number of fragile regions, and the intergenic regions cover most of the genome! Rather, accelerated intergenic sequence evolution would compound, *after* breakage, the effects of the preceding two paragraphs. Note that in general the breakpoint regions contain many genes [25] and, depending on the criteria used to delimit the regions, parts of genes.

9.9.1 *Fragments*

The largely unaligned region between two syntenic blocks on a human chromosome usually contains a number of smaller regions (or fragments) that are aligned with regions on various mouse chromosomes. As depicted in Fig. 9.8,

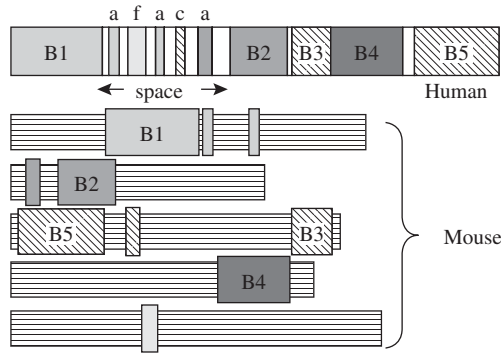


FIG. 9.8. Hypothetical human chromosome with breakpoint region (space) containing three types of small fragment. Shading of syntenic blocks B1–B5 and fragments keyed to aligned portions of mouse chromosomes. a = archipelago, c = compatriot, f = foreigner.

these fragments fall into three categories:

1. If a fragment is aligned with a region on the same mouse chromosome as one of the two adjacent syntenic blocks on the left or right of the space, it is said to be in the *archipelago*.
2. Fragments aligned with regions on other mouse autosomes sharing syntenic blocks with the same human chromosome are called *compatriots*. (Recall that the X chromosome generally does not participate in inter-chromosomal exchanges.)
3. Fragments aligned with regions on mouse chromosomes, including X, sharing no syntenic blocks with the same human chromosome, are *foreigners*.

Trinh *et al.* [53] undertook a statistical assessment of the three types in the hopes of revealing the formative processes of the breakpoint regions.

Based on the construction in the UCSC Genome Browser comparison of the mouse and human genomes, and using a 100 Kb threshold for the minimum size of a syntenic block, they extracted 320 inter-block spaces on the human genome for analysis, excluding pericentromeric spaces subject to repetitive segmental duplication and/or transposition [3]. Their median length was 120 Kb, about the same as the shortest blocks. For about half the spaces, the two adjacent syntenic blocks were from different mouse chromosomes. The spaces contained 12,930 smaller aligned fragments as identified by the browser, and these were labelled as archipelago ($N = 4,139$), compatriot ($N = 2,706$), or foreigner ($N = 6,085$).

The archipelago fragments are considerably longer than the compatriot and foreigner fragments as can be seen from the distributions of fragment length in Fig. 9.9. The median length of the archipelago fragments is twice as large as

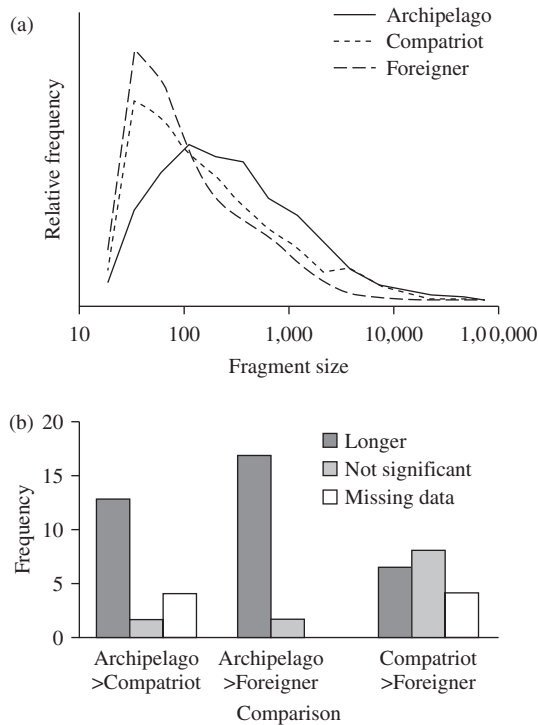


FIG. 9.9. (a) Length distribution for fragment categories. (b) Number of chromosomes for which the null hypotheses of identical size fragments is rejected or accepted.

either of the other two in most chromosomes. The disparity prevails throughout the genome as can be seen in the plot of the number of chromosomes for which a one-tailed Kolmogorov–Smirnov test rejects the null hypothesis that the different types of fragment have the same distribution of lengths.

Figure 9.9 also shows that the compatriot fragments are systematically longer than the foreigner ones, though the difference is less marked than that between either of these categories and the archipelago. Fourteen of the 18 chromosomes for which there are sufficient data have longer mean fragment size for compatriots than foreigners, and eight of these are significantly so at the 5% level.

Trinh *et al.* [53] also showed that:

1. Archipelago fragments tended to be much more frequent in an inter-block space than compatriot fragments, in proportion to the number of different mouse chromosomes in which the two types could originate. In turn, compatriots tended to be much more frequent than foreigners, again relative to the number of different mouse chromosomes in which the two types could originate.

2. The proportion of the inter-block space covered by archipelago fragments is much greater than that of the compatriots, which in turn is greater than that of the foreigners.
3. The archipelago fragments in spaces defined by blocks from two different mouse chromosomes, though somewhat interspersed, tended to segregate towards the corresponding block.
4. The archipelago fragments tend to correspond to regions in the mouse chromosome close to the homolog of the adjacent block. The compatriot fragments tend to correspond to regions in the mouse chromosome close to the homolog of one of the blocks on the same human chromosome.

These observations about the different kinds of fragments suggest that they derive from at least three separate types of process. All or most of the foreigners but a smaller proportion of the compatriots and a much smaller proportion of the archipelago, probably come from some common processes such as retroposition of mRNA, or small jumping translocation or transposition events originating randomly across the genome and correlating roughly with chromosome size. Compatriots represent either a greater propensity for retroposition to the same chromosome originating, due to geometrical considerations (mRNA is more concentrated around the chromosome from which it is transcribed) or, in some lesser proportion, from some intra-chromosomal shuffling process, such as inversion or transposition. Finally, the larger archipelago blocks seem to be hived off the large syntenic blocks on either side, and are the results, in some proportion, of two types of process. One is the residual similarity exceeding whatever thresholds are required by the alignment algorithms. These islands of similarity “peeking through” the noise may be either a natural consequence of the variable degree of similarity across all regions of the genome, or indicate the sporadic way the algorithms fail near breakpoints, or both. Second, these fragments may be chunks of the two surrounding syntenic blocks that have been thrown from near the ends of these blocks into the space by the same processes of local rearrangement that affect the interior of the blocks. That the archipelago fragments corresponding to two syntenic blocks are partially interspersed is evidence that such rearrangement continues to occur post-rearrangement, and that they are not solely the residues of decaying measures of similarity.

One process that is not invoked in explaining these statistics is the repeated use of the same breakpoints by several large-scale genomic rearrangements. The archipelago fragments only attest to local rearrangements, and the numerous small compatriot and foreigner fragments, including many fragments of X chromosome origin, do not seem like the residue of repeated large-scale rearrangements.

9.10 Conclusions

Genome rearrangement analysis has not scaled up directly to genomic sequences, not because of any computational difficulty, but because this new information is not as neat as the gene order data of organelles. Whatever the loss

of evolutionary signal from divergent organellar or prokaryotic genomes, this problem is compounded in nuclear genomes by the difficulties of gene finding and ortholog identification at the gene level, and the lack of congruence of genomic sequence rearrangement and gene order rearrangement. Whereas the former involves movement of material that may not involve any genes, the latter may sometimes operate on gene-containing fragments too short to be picked up by syntenic block construction algorithms.

Genome sequence data has thus proved to be more of a problem for comparative genomics than a solution to old problems.

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THE INVERSION DISTANCE PROBLEM

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Among the many genome rearrangement operations, signed inversions stand out for many biological and computational reasons. Inversions, also known as reversals, are widely identified as one of the common rearrangement operations on chromosomes, they are basic to the understanding of more complex operations such as translocations, and they offer many computational challenges. From the first formulation of the inversion distance problem, ca. 1992, to its first polynomial solution in 1995, to the several simplifications of the solution in recent years, there is not yet a simple, complete, and elementary treatment of the subject. This is the goal of this chapter.

10.1 Introduction and biological background

In the last 10 years, beginning with Sankoff [20], many papers have been devoted to the subject of computing the *inversion distance* between two permutations. An *inversion* of an interval from p_i to p_j transforms a permutation P into P' :

$$\begin{aligned} P &= (p_1 \quad \cdots \quad p_i \quad p_{i+1} \quad \cdots \quad p_j \quad \cdots \quad p_n), \\ P' &= (p_1 \quad \cdots \quad p_j \quad \cdots \quad p_{i+1} \quad p_i \quad \cdots \quad p_n). \end{aligned}$$

The inversion distance between two permutations is the minimum number of inversions that transform one into the other.

From a problem of unknown complexity, it eventually graduated to an NP-hard problem [9], but an interesting variant was proven to be polynomial [12]. In the *signed* version of the problem, each element of the permutation has a plus or minus sign, and an inversion of an interval from p_i to p_j transforms P to P' :

$$\begin{aligned} P &= (p_1 \quad \cdots \quad p_i \quad p_{i+1} \quad \cdots \quad p_j \quad \cdots \quad p_n), \\ P' &= (p_1 \quad \cdots \quad -p_j \quad \cdots \quad -p_{i+1} \quad -p_i \quad \cdots \quad p_n). \end{aligned}$$

Permutations, and their inversions, are useful tools in the comparative study of genomes. The genome of a species can be thought of as a set of ordered

sequences of genes—the *chromosomes*—each gene having an orientation given by its location on the DNA double strand. Different species often share similar genes that were inherited from common ancestors. However, these genes have been shuffled by evolutionary events that modified the content of chromosomes, the order of genes within a particular chromosome, and/or the orientation of a gene. Assigning the same index to similar genes appearing along a chromosome in two different species, and using negative signs to model changes in orientation, yields two signed permutations. The inversion distance between these permutations can thus be used to compare species.

Computing the inversion distance of signed permutations is a delicate task since some inversions unexpectedly affect deep structures in permutations. In 1995, Hannenhalli and Pevzner proposed the first polynomial algorithm to solve it [12], developing along the way a theory of how and why some permutations were particularly resistant to sorting by inversions. It is of no surprise that the label *fortress* was assigned to specially acute cases.

Hannenhalli and Pevzner relied on several intermediate constructions that have been subsequently simplified [7, 13], but grasping all the details remained a challenge. Before Bergeron [3], all the criteria given for choosing a *safe* inversion involved the construction of an associated permutation on $2n$ points, and the analysis of cycles and/or connected component of the graph associated with this permutation.

Moreover, most papers tended to mix two different problems, as pointed out in references [1, 13]: the computation of the *number* of necessary inversions, and the reconstruction of one possible sequence of inversions that realizes this number. The first problem was finally proved to be of linear time complexity [1], but this approach still used many of the Hannenhalli–Pevzner constructions. However, the existence of a linear-time solution was a strong incentive to try to present the computation in an elementary way, which led to the recognition of the central role played by *subpermutations* in the theory [4, 6, 11].

In this chapter, we present an elementary treatment of the sorting by inversions problem. We give a complete proof of the Hannenhalli–Pevzner *duality theorem* in terms of the elements of a given signed permutation, efficient, and simple algorithms to compute the inversion distance, and simple procedures for the construction of optimal inversion sequences.

In the next section, we introduce the basic definitions and describe the sorting by inversions problem. In Section 10.3 we introduce several concepts, such as cycles and components, which are central to the solution of this problem. The relations between components are used to construct a tree associated to a signed permutation. This tree is the basis of a simple proof of the Hannenhalli–Pevzner *duality theorem* presented in Section 10.4. Finally, in Section 10.5 we present algorithms to identify the components, to count the number of cycles, and to construct the tree associated to a signed permutation.

The last section contains a glossary of the terminology used in this chapter.

10.2 Definitions and examples

A *signed permutation* is a permutation on the set of integers $\{0, 1, 2, \dots, n\}$ in which each element has a sign, positive or negative. For convenience,¹ we will assume that all permutations begin with 0 and end with n . For example:

$$P_1 = (0 \quad -2 \quad -1 \quad 4 \quad 3 \quad 5 \quad -8 \quad 6 \quad 7 \quad 9).$$

Since integers represent genes and signs represent the orientation of a gene on a particular chromosome, we will refer to the underlying gene as an *unsigned element* of the permutation.

A *point* $p \cdot q$ is defined by a pair of consecutive elements in the permutation. For example, $0 \cdot -2$ and $-2 \cdot -1$ are the first two points of P_1 . When a point is of the form $i \cdot i + 1$, or $-(i + 1) \cdot -i$, it is called an *adjacency*, otherwise it is called a *breakpoint*. For example, P_1 has two adjacencies, $-2 \cdot -1$ and $6 \cdot 7$. All other points of P_1 are breakpoints.

We will make an extensive use of intervals of consecutive elements in a permutation. An interval is easily defined by giving its *endpoints*. The *elements* of the interval are the elements between the two endpoints. When the two endpoints are equal, the interval contains no elements. A non-empty interval can also be specified by giving its first and last element, such as $(i \dots j)$, called the *bounding elements* of the interval.

An *inversion* of an interval of a signed permutation is the operation that consists of inverting the order of the elements of the interval, while changing their signs. For example, the inversion of the interval of P_1 whose endpoints are $-2 \cdot -1$ and $5 \cdot -8$ yields the permutation P'_1 :

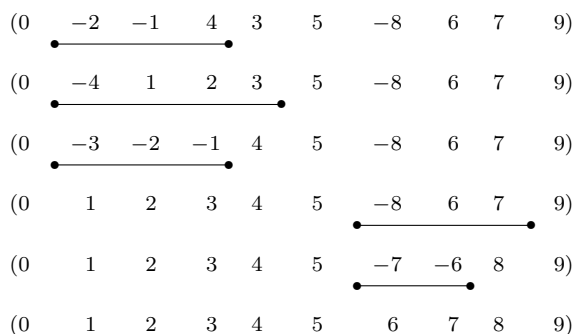
$$\begin{array}{cccccccccc} P_1 = & (0 & -2 & -1 & 4 & 3 & 5 & -8 & 6 & 7 & 9), \\ & & & \bullet & \text{-----} & \bullet & & & & & \\ P'_1 = & (0 & -2 & -5 & -3 & -4 & 1 & -8 & 6 & 7 & 9). \end{array}$$

The inversion of an interval modifies the points of a signed permutation in various ways. Points $p \cdot q$ that are inside the interval are transformed to $-q \cdot -p$, the endpoints of the interval exchange their flanking elements, and points that are outside the interval are unaffected.

The *inversion distance* $d(P)$ of a permutation P is the minimum number of inversions needed to transform P into the identity permutation. Finding one sequence of inversions that realizes this distance is called the *sorting by inversions problem*. For example, $d(P_1) = 5$, and Fig. 10.1 shows a sequence of inversions that realizes this distance.

A sequence of inversions, applied to a permutation P , is called an *optimal sorting sequence* if it transforms P into the identity permutation, and if its length

¹This assumption simplifies the theory and is coherent with biological applications in which whole chromosomes do not have a global orientation: only local changes of orientation are relevant.

FIG. 10.1. Sorting $P_1 = (0 -2 -1 4 3 5 -8 6 7 9)$ by inversions.

$$\begin{array}{ll}
 P = (0 -2 -1 & 4 \quad 3 \quad 5 \quad \overline{-8 \quad 6 \quad 7} \quad 9) & Q^{-1} \circ P = (0 \quad 1 \quad -3 \quad \overline{-7 \quad -6 \quad 2} \quad \overline{-8 \quad 4 \quad 5} \quad 9) \\
 (0 -2 -1 & \overline{4 \quad 3 \quad 5} \quad \overline{-7 \quad -6 \quad 8} \quad 9) & (0 \quad 1 \quad -3 \quad \overline{-7 \quad -6 \quad 2} \quad \overline{-5 \quad -4 \quad 8} \quad 9) \\
 (0 -2 -1 & \overline{-5 \quad -3 \quad -4} \quad \overline{-7 \quad -6} \quad 8 \quad 9) & (0 \quad 1 \quad -3 \quad \overline{-2 \quad 6 \quad 7} \quad \overline{-5 \quad -4} \quad 8 \quad 9) \\
 (0 -2 & \overline{-1 \quad -5} \quad \overline{6 \quad 7 \quad 4 \quad 3} \quad 8 \quad 9) & (0 \quad 1 \quad \overline{-3 \quad -2} \quad \overline{4 \quad 5 \quad -7 \quad -6} \quad 8 \quad 9) \\
 (0 -2 & \overline{5 \quad 1 \quad 6 \quad 7 \quad 4 \quad 3} \quad 8 \quad 9) & (0 \quad 1 \quad \overline{2 \quad 3 \quad 4 \quad 5 \quad -7 \quad -6} \quad 8 \quad 9) \\
 Q = (0 -2 & 5 \quad 1 \quad 6 \quad 7 \quad \overline{-3 \quad -4} \quad 8 \quad 9) & Q^{-1} \circ Q = (0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad \overline{6 \quad 7 \quad 8 \quad 9})
 \end{array}$$

FIG. 10.2. Transforming permutation $P_1 = (0 -2 -1 4 3 5 -8 6 7 9)$ into permutation $Q = (0 -2 5 1 6 7 -3 -4 8 9)$ is simulated by transforming permutation $Q^{-1} \circ P_1$ into $Q^{-1} \circ Q$, where $Q^{-1} = (0 3 -1 -6 -7 2 4 5 8 9)$.

is $d(P)$. An inversion that belongs to an optimal sorting sequence is called a *sorting inversion*.

In general, the *inversion distance* between two arbitrary permutations P and Q is the minimum number of inversions that transform one into the other. One can always reduce this problem to a problem of inversion distance to the identity permutation by composing² the permutations P and Q with the inverse permutation of one of them, say Q^{-1} . Any sequence of inversions that transforms $Q^{-1} \circ P$ into $Q^{-1} \circ Q$ can be applied to the original problem. An example is given in Fig. 10.2.

Historical notes. Surprisingly, inversions of segments of chromosomes have been identified in close species by Sturtevant [23] early in the last century. It then took decades of biological experiments to accumulate sufficient data to compare gene order of a vast array of species. For simple chromosomes, such as mitochondria, the sequence of genes is now known for several hundred species. See Chapter 9, this volume, for more details.

²Here, composition is understood as the standard composition of functions. Dealing with signed permutations requires the additional axiom that $P(-a) = -P(a)$.

In 1982, Watterson *et al.* [26] first formulated the problem of finding the minimum number of inversions required to bring one configuration of genes into another. It took more than 10 years until Kececioğlu and Sankoff [14] developed the first approximation algorithm for the problem of sorting an unsigned permutation by inversions. They also conjectured that this problem is NP-hard. Indeed, this was shown in 1997 by Caprara [9]. Bafna and Pevzner [2] initiated the study of signed permutations in order to model the orientation of genes. In 1995, Hannenhalli and Pevzner [12] gave the first polynomial-time algorithm for the problem of sorting a signed permutation by inversions using the concepts developed by Bafna and Pevzner. A clear distinction between the problem of computing the inversion distance and finding an optimal sorting sequence was worked out by Kaplan *et al.* [13] and Bader *et al.* [1]. Currently, the most efficient algorithms to solve the inversion distance problem are linear, while the most efficient algorithms to find optimal sorting sequences are not [19, 24].

Since many optimal sorting sequences exist, recently Siepel [22] studied the problem of finding all optimal sequences and gave a polynomial-time algorithm to find all sorting inversions of a permutation.

10.3 Anatomy of a signed permutation

In the following, we define several concepts central to the analysis of signed permutations, and study the effect of inversions on these structures. First, we consider the elementary intervals and cycles in Sections 10.3.1 and 10.3.2, and then we treat the components of a permutation in Sections 10.3.3 and 10.3.4.

10.3.1 Elementary intervals and cycles

Let P be a signed permutation on the set $\{0, 1, 2, \dots, n\}$ that begins with 0 and ends with n . Any element i of P , $0 < i < n$, has a *right* and a *left* point.

Definition 10.1 *For each pair of unsigned elements $(k, k + 1)$, $0 \leq k < n$, define the elementary interval I_k associated to the pair to be the interval whose endpoints are:*

1. *The right point of k , if k is positive, otherwise its left point.*
2. *The left point of $k + 1$, if $k + 1$ is positive, otherwise its right point.*

Elements k and $k + 1$ are called the extremities of the elementary interval.

An elementary interval can contain zero, one, or both of its extremities. For example, in Fig. 10.3, interval I_0 contains one of its extremities, interval I_3 contains both, and interval I_5 contains none. Empty elementary intervals, such as I_1 and I_6 , correspond to adjacencies in the permutation.

When the extremities of an elementary interval have different signs, the interval is said to be *oriented*, otherwise it is *unoriented*. Oriented intervals are exactly those intervals that contain one of their extremities.

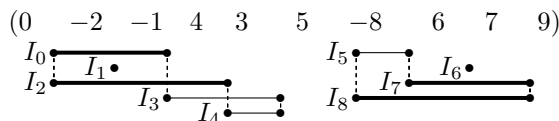


FIG. 10.3. Elementary intervals and cycles of a permutation. Oriented intervals are represented by thick lines, and unoriented intervals by thin lines. Vertical dashed lines join intervals that meet at breakpoints, tracing the cycles.

Oriented intervals play a crucial role in the problem of sorting by inversions since they can be used to create adjacencies. Namely, we have:

Proposition 10.2 *Inverting an oriented interval I_k creates, in the resulting permutation, either the adjacency $k \cdot k + 1$ or the adjacency $-(k + 1) \cdot -k$.*

Proof Suppose that k is positive, then $k + 1$ must be negative for the interval I_k to be oriented. If $k + 1$ succeeds k , then the interval will contain $k + 1$ but not k , and inverting it will create the adjacency $k \cdot k + 1$. If $k + 1$ precedes k , then the interval will contain k but not $k + 1$, and inverting it will create the adjacency $-(k + 1) \cdot -k$. The case when k is negative is treated similarly. $\square$

For example, inverting the oriented elementary interval I_8 in permutation P_1 of Fig. 10.3 creates the adjacency $8 \cdot 9$.

When a point is the endpoint of two elementary intervals, these are said to *meet* at that point.

Proposition 10.3 *Exactly two elementary intervals meet at each breakpoint of a permutation.*

Proof From Definition 10.1, the right and left point of each element of the permutation is used once as an endpoint of an elementary interval, thus each breakpoint is used twice. $\square$

Therefore, by Proposition 10.3, starting from an arbitrary breakpoint, one can follow elementary intervals on a unique path that eventually comes back to the original breakpoint. More formally:

Definition 10.4 *A cycle is a sequence $b_1, b_2, \dots, b_k$ of points such that two successive points are the endpoints of an elementary interval, including b_k and b_1 . Adjacencies define trivial cycles consisting of a single point.*

For example, as shown in Fig. 10.3, permutation P_1 has four cycles, two of them are trivial, and the other two contain, respectively, 4 and 3 breakpoints.

Cycles are conveniently defined with breakpoints, but one can always focus on the elementary intervals that are defined by the breakpoints of a cycle. The following property, on the number of oriented intervals of a cycle, will be useful to prove results on the number of cycles of a permutation.

Lemma 10.5 *A cycle always contains an even number of oriented intervals.*

Proof Let J_i be the interval that connects b_i to the next breakpoint in a cycle $b_1, b_2, \dots, b_k$. Define e_i to be the number of extremities of J_i contained in it, either 0, 1, or 2, and consider the sum: $E = \sum_{i=1}^k e_i$. We will show that E is an even number, implying that the number of oriented intervals is even.

The idea is to construct the sum E by considering the contribution of each breakpoint of the cycle. Follow the breakpoints in the order $b_1, b_2, \dots, b_k$. A given breakpoint can either join two disjoint intervals, or two stacked intervals. In this last case, the breakpoint is a *turning* point of the cycle. Each turning point $p \cdot q$ contributes 1 to the number E , since either p or q is inside both intervals, and the other is outside both intervals. Each breakpoint $p \cdot q$ that joins two disjoint intervals contributes 0 or 2 to the number E , since p is inside its interval if and only if q is. However, the number of turning points of a cycle must be even, therefore E is even. $\square$

A last fundamental relation between elementary intervals is the *overlap* relation.

Definition 10.6 *Two elementary intervals I and J overlap if each contains exactly one of the extremities of the other.*

The overlap relation is often easily detectable, like the overlap of the intervals I_2 and I_1 in Fig. 10.4. Intervals that meet at a breakpoint can overlap or not. For example, intervals I_0 and I_2 overlap since I_0 contains element -3 , and I_2 contains element 1; on the other hand intervals I_0 and I_3 do not overlap, despite the fact that they meet at breakpoint $0 \cdot 4$.

A common way to represent the overlap relation between elementary intervals is the *overlap graph* $\mathcal{O}$ with black and white vertices standing, respectively, for oriented and unoriented elementary intervals. Two vertices are connected

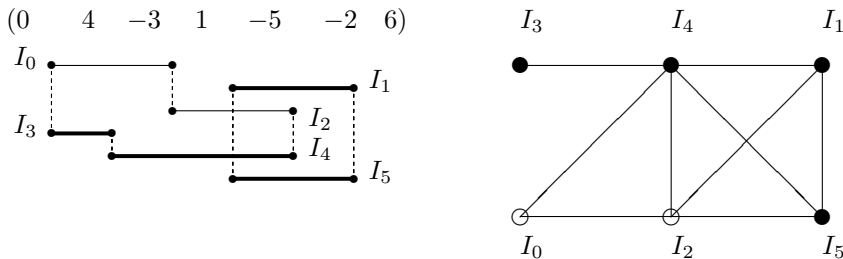


FIG. 10.4. A permutation and its overlap graph $\mathcal{O}$. Only two elementary intervals are unoriented, I_0 and I_2 , corresponding to white vertices of the graph $\mathcal{O}$. Intervals I_0 and I_2 overlap since I_0 contains element -3 , and I_2 contains element 1; on the other hand intervals I_0 and I_3 do not overlap, despite the fact that they meet at breakpoint $0 \cdot 4$.

in $\mathcal{O}$ if and only if the corresponding intervals overlap. The right hand side of Fig. 10.4 gives an example of such a graph.

10.3.2 Effects of an inversion on elementary intervals and cycles

One of the cornerstones of the sorting by inversions problem is to study the effects of an inversion on elementary intervals and cycles. The first result, due to reference [15], is the effect of an inversion on the number of cycles. It is based on the fact that, for all points except the endpoints of an inversion, the elementary intervals that meet at those points will still meet at that point after the inversion.

Proposition 10.7 *An inversion can only modify the number of cycles by $+1$, 0 , or -1 .*

Proof An inversion exchanges the elements of two points of a permutation. If these two points belong to the same cycle, then either the cycle is split in two, or is conserved but with different breakpoints. If the two points belong to different cycles, then these cycles are merged. Figure 10.5 gives an illustration of the three cases. $\square$

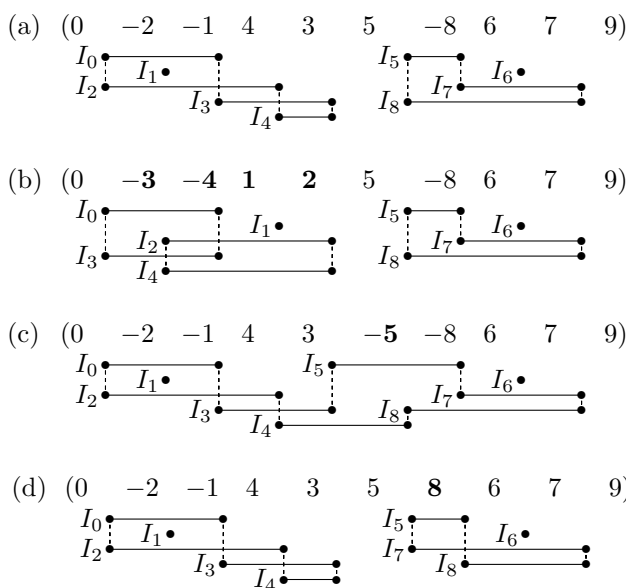


FIG. 10.5. Effects of inversions on cycles. The original permutation, again P_1 , is shown in (a). In (b), the inversion of interval $(-2, -1, 4, 3)$ splits the cycle of length 4 of the original permutation. In (c), the inversion of element 5 merges the two long cycles of the original permutation. Finally, in (d), the inversion of element 8 leaves the number of cycles unchanged.

By Propositions 10.2 and 10.7, inverting an oriented interval always splits a cycle, since an adjacency is a trivial cycle. The identity permutation on the set $\{0, 1, 2, \dots, n\}$ is the only one with n cycles, all adjacencies. Since at most one cycle can be added by an inversion, Proposition 10.7 implies a first lower bound to the inversion distance of a permutation:

Lemma 10.8 *Let c be the number of cycles of a signed permutation P on the set $\{0, 1, 2, \dots, n\}$. Then $d(P) \geq n - c$.*

The next important observation is an easy consequence of the overlap relation. If I and J overlap, then inverting the interval I will change the orientation of J , since only one extremity of J will change sign.

When two intervals J and K overlap an interval I , the effect of inverting I complements the overlap relation between J and K : if J and K overlapped before the inversion, they do not overlap after it; if J and K did not overlap before the inversion, they overlap after it.

Formally, we have:

Proposition 10.9 *Let G_I be the subgraph of the overlap graph formed by vertex I and its adjacent vertices. Consider the inversion of elementary interval I .*

1. *If I is unoriented, the effect on the overlap graph is to change the colour of all vertices in $G_I - \{I\}$, and complement the edges of $G_I - \{I\}$.*
2. *If I is oriented, the effect on the overlap graph is to change the colour of all vertices in G_I , and complement the edges of G_I .*

Proof 1. If the elementary interval I is unoriented, either both or none of the extremities of I are contained in the interval I , thus inverting the interval I does not change the orientation of the vertex I . Let vertex J be adjacent to I , then I contains exactly one of the extremities of J , and inverting the interval I changes the sign of one extremity of J . Thus, J changes orientation. If vertices J and K are adjacent to I , then one extremity of J and one of K are contained in I . If J and K are overlapping, then inverting the elementary interval I will invert the order of the extremities of J and K that are contained in I . The elementary intervals J and K will either be disjoint, or one will be contained in the other. Thus, they are not overlapping in the resulting permutation. A similar argument shows that if J and K are not overlapping, then they will overlap after the inversion.

2. Inverting the oriented elementary interval I creates the isolated vertex I , since it creates an adjacency by Proposition 10.2. Thus each edge incident to I is erased. The complementation of the edges and the orientation of $G_I - \{I\}$ is similar to the unoriented case. $\square$

10.3.3 Components

Elementary intervals and cycles are organized in higher structures called *components*. These were first identified in reference [11] as *subpermutations* since

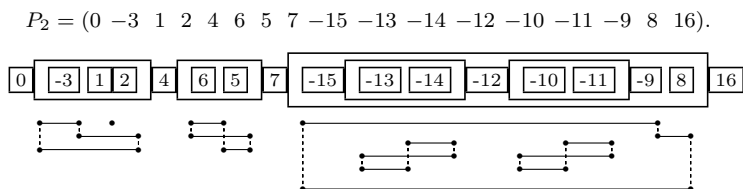


FIG. 10.6. A permutation and the boxed representation of its components. Endpoints of elementary intervals, and thus cycles, belong to exactly one component.

they are intervals that contain a permutation of a set of consecutive integers, and later studied in more detail in reference [4] as *framed common intervals*.

Definition 10.10 Let P be a signed permutation on the set $\{0, 1, 2, \dots, n\}$. A component of P is an interval from i to $(i + j)$ or from $-(i + j)$ to $-i$, for some $j > 0$, whose set of unsigned elements is $\{i, \dots, i + j\}$, and that is not the union of two such intervals. Components with positive, respectively negative, bounding elements are referred to as direct, respectively reversed, components.

For example, consider the permutation P_2 of Fig. 10.6. It has six components: four of them are direct, $(0 \dots 4)$, $(4 \dots 7)$, $(7 \dots 16)$, and $(1 \dots 2)$; and two of them are reversed, $(-15 \dots -12)$ and $(-12 \dots -9)$. Note that a component, such as the adjacency $1 \cdot 2$, can contain only two elements.

Components of a permutation can be represented by a boxed diagram, such as in Fig. 10.6, in which bounding elements of each component have been boxed, and elements between them are enclosed in a rectangle. Elements which are not bounding elements of any component are also boxed.

Components organize hierarchically the points, elementary intervals, and cycles of a permutation.

Definition 10.11 A point $p \cdot q$ belongs to the smallest component that contains both p and q .

Note that this does not prevent the elements p and q to belong, separately, to other components, such as point $7 \cdot -15$ in the permutation of Fig. 10.6.

Proposition 10.12 The endpoints of an elementary interval belong to the same component, thus all the points of a cycle belong to the same component.

Proof Consider an elementary interval I_k and any component C of the form

$$(i \dots i + j) \quad \text{or} \quad (-(i + j) \dots -i),$$

such that $i \leq k < i + j$. We will show that both endpoints of I_k are contained in C . This is obvious if k is different from i and $k + 1$ is different from $i + j$, since both k and $k + 1$ will be in the interior of the component. If $k = i$, then k and i have the same sign, and the first endpoint of I_k belongs to the component.

If $k+1 = i+j$, then $k+1$ and $i+j$ have the same sign, and the second endpoint of I_k belongs to the component.

Thus endpoints of I_k are either both contained, or not, in any given component, and the result follows. $\square$

A component can have more than one cycle. For example, the permutation of Fig. 10.4 has one component $(0 \dots 6)$ consisting of two cycles. Finally, components can be classified according to the nature of the points they contain:

Definition 10.13 *The sign of a point $p \cdot q$ is positive if both p and q are positive, it is negative if both p and q are negative. A component is unoriented if it has one or more breakpoints, and all of them have the same sign, otherwise the component is oriented.*

For example, the unoriented components of the permutation of Fig. 10.6 are $(4 \dots 7)$, $(-15 \dots -12)$, and $(-12 \dots -9)$. All the elementary intervals whose endpoints belong to the same unoriented component are unoriented intervals. Therefore, it is impossible to create an adjacency in an unoriented component with only one inversion. On the other hand, an oriented component contains at least one oriented interval, thus at least two, by Lemma 10.5 and Proposition 10.12.

In order to optimally solve the sorting problem, it is necessary to understand the relationship between the components of a permutation. The following definitions and propositions establish these relationships.

Proposition 10.14 ([6]) *Two different components of a permutation are either disjoint, nested with different endpoints, or overlapping on one element.*

Proof First note that two components that share an endpoint must be both direct or both reversed.

Consider two direct components C and C' of the form

$$C = (i \dots i+j) \quad \text{and} \quad C' = (i' \dots i'+j').$$

Suppose the components C and C' are nested with $i = i'$ and $j' < j$. Since C' is a component, it contains all unsigned elements between its bounding elements i' and $i'+j'$, and hence the interval $(i'+j' \dots i+j)$ contains all unsigned elements between $i'+j'$ and $i+j$. This contradicts the fact that the component C is not the union of two shorter components. The case where the components C and C' are reversed can be treated similarly.

Suppose that the components $C = (i \dots i+j)$ and $C' = (i' \dots i'+j')$ are direct and overlap with more than one element. We can assume that

$$i < i' < i+j < i'+j'.$$

Since all unsigned elements between i' and $i'+j'$ are greater than i' , the interval $(i \dots i')$ must contain all unsigned elements between i and i' . Thus, C is the

union of two shorter components, which leads to a contradiction. Again, the reverse case follows by a similar argument. $\square$

When two components overlap on one element, we say that they are *linked*. Successive linked components form a *chain*. A chain that cannot be extended to the left or right is called *maximal*. Note that a maximal chain may consist of a single component. If one component of a chain is nested in a component A , then all other components of the chain are also nested in A .

The nesting and linking relations between components turn out to play a major role in the sorting by inversions problem. Another way of representing these relations is by using the following tree:

Definition 10.15 *Given a permutation P on the set $\{0, 1, \dots, n\}$ and its components, define the tree T_P by the following construction:*

1. *Each component is represented by a round node.*
2. *Each maximal chain is represented by a square node whose (ordered) children are the round nodes that represent the components of this chain.*
3. *A square node is the child of the smallest component that contains this chain.*

For example, Fig. 10.7 represents the tree T_{P_2} associated to permutation P_2 of Fig. 10.6.

It is easy to see that, if the permutation begins with 0 and ends with n , the resulting graph is a single tree with a square node as root. The tree is similar to the PQ-tree used in different context such as the consecutive ones test [8]. The following properties of paths in T_P are elementary consequences of the definition of T_P .

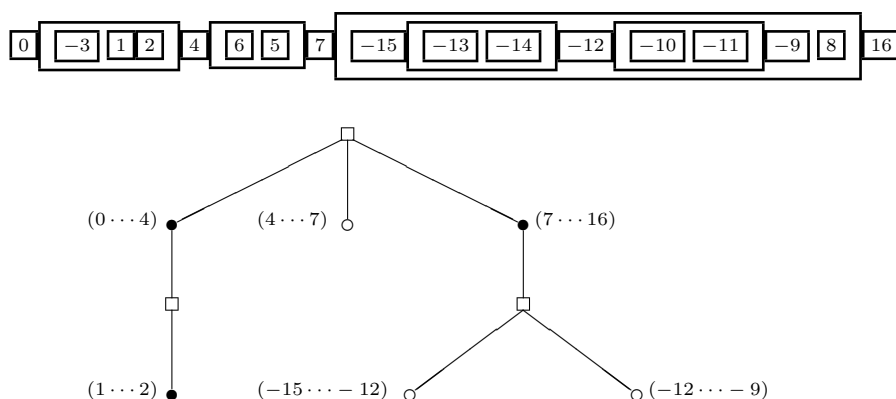


FIG. 10.7. The tree T_{P_2} associated to permutation P_2 of Fig. 10.6. White round nodes correspond to unoriented components, and black round nodes correspond to oriented components.

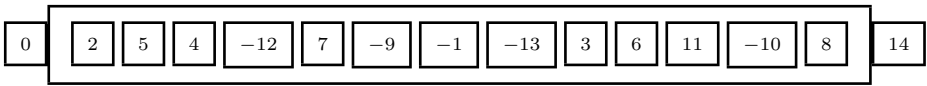
Proposition 10.16 *Let C be a component on the (unique) path joining components A and B in T_P , then C contains either A or B , or both.*

1. *If C contains both A and B , it is unique.*
2. *No component of the path contains both A and B if and only if A and B are included in two components that are in the same chain.*

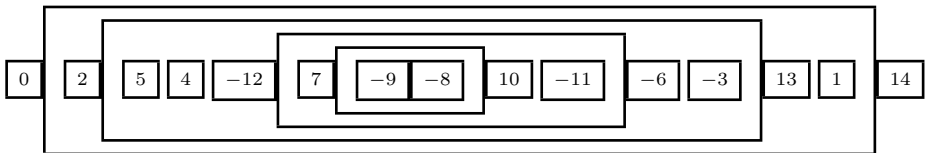
Proof Consider the smallest component D that contains components A and B . If it is on the path that joins A and B , then any other component that contains A and B is an ancestor of D , therefore not on the path. If D is not on the path that joins A and B , then the least common ancestor of components A and B is a square node q that is a child of the round node representing D , thus A and B are included in two components that are in the chain represented by q . $\square$

10.3.4 *Effects of an inversion on components*

We saw, in Proposition 10.7, that an inversion can modify the number of cycles of a permutation by at most 1. On the other hand, an inversion can create or destroy any number of components. For example, inverting the interval $(-1, \dots, 8)$ in the following permutation



creates the adjacency $-9 \cdot -8$ and yields a permutation with four new components:



As we will see in the next section, creating oriented components, or adjacencies, is generally considered a good move towards optimally sorting a permutation. However, the creation of unoriented components should be avoided. Luckily, few inversions have that effect.

The next three propositions describe the effects of inversions whose endpoints are in unoriented components. These are classical results from the Hannenhalli–Pevzner theory.

Proposition 10.17 *If a component C is unoriented, no inversion with its two endpoints in C can split one of its cycles, nor create a new component.*

Proof First note that Lemma 10.5 implies that the number of positive—or negative—extremities of intervals of a cycle must be even, since each oriented interval has a positive and a negative extremity.

If a component C is unoriented, then all the breakpoints of its cycles have the same sign. An inversion with its two endpoints in one of the cycles of C will introduce exactly two new breakpoints which are neither positive nor negative. If a cycle of C is split, those two breakpoints must belong to different cycles c_1 and c_2 . In each of these cycles, the remaining breakpoints are either positive or negative. Thus, the number of positive extremities of the intervals of c_1 and of c_2 would be odd numbers.

Suppose an inversion creates a new component D , then one bounding element of D has to be inside the inverted interval, and the other one outside the inverted interval, otherwise the component D would have existed before the inversion. Therefore, the bounding elements of the component D have different sign, which contradicts the definition of a component. $\square$

Proposition 10.18 *If a component C is unoriented, the inversion of an elementary interval whose endpoints belong to C orients C , and leaves the number of cycles of the permutation unchanged.*

Proof Inverting an elementary interval changes the sign of the elements of the inverted interval. Therefore, component C will be oriented. Since the endpoints of an elementary interval belong to the same cycle, the inversion cannot merge cycles. By Proposition 10.17, the inversion of I cannot split a cycle. Therefore, the number of cycles remains unchanged. $\square$

Orienting a component as in Proposition 10.18 is called *cutting* the component. Such an inversion is seldom a sorting inversion since it is possible, with a single inversion, to get rid of more than one unoriented component. The following proposition describes how to *merge* several components, and the relation of this operation to paths in T_P .

Proposition 10.19 *An inversion that has its two endpoints in different components A and B destroys, or orients, all components on the path from A to B in T_P , without creating new unoriented components.*

Proof Note first that an inversion with endpoints in different components A and B must merge two cycles, one from each component, into a new cycle c . If A and B are unoriented, cycle c contains at least one oriented interval.

Suppose that a new component D is created by such an inversion, then the bounding elements of D must be both outside the inverted interval. Indeed, if both bounding elements of D are inside the inverted interval, D existed in the original permutation. If one bounding element of D is outside the interval, then component D must contain at least one endpoint of the inverted interval in order to be affected by the inversion. Since the two endpoints of the inverted interval belong to the same cycle c , the second endpoint of the interval must also be in component D , thus the second bounding element of D is also outside the interval.

Thus, the only component eventually created by an inversion with endpoints in different components is the union of two or more linked components. Since

linked components have bounding elements with the same sign, the sign of the former links will be different from the sign of the bounding elements of the new component, thus it will be oriented.

By Proposition 10.16, if there is a component C on the path from A to B and that contains both, then A and B are not included in linked components, thus no new component can be created by the inversion. Since C is the smallest component that contains the new cycle c , C will be oriented.

Finally, suppose that a component C is on the path from A to B and contains either A or B , but not both. Then the inversion changes the sign of one of the bounding elements of C , and C will be destroyed. $\square$

Proposition 10.19 thus states that one can get rid of many unoriented components with only one inversion. This idea is exploited in the next section to compute the inversion distance of a permutation.

Historical notes. In 1984, Nadeau and Taylor [18] introduced the notion of breakpoints of a permutation. One decade later, Kececioğlu and Sankoff [14] brought in the *breakpoint graph* in their analysis of the sorting by inversions problem. Later, Bafna and Pevzner [2] extended the breakpoint graph to signed permutations.

The most common version of the breakpoint graph³ is based on an unsigned permutation of $2n$ elements defined as follows: replace any positive element x of a signed permutation by $2x - 1, 2x$ and any negative element $-x$ by $2x, 2x - 1$. The breakpoint graph is an edge-coloured graph whose set of vertices are the elements $(p_0, \dots, p_{2n-1})$ of this unsigned permutation.

For each $0 \leq i < n$, vertices p_{2i} and p_{2i+1} are joined by a black edge, and elements $2i$ and $2i + 1$ of the permutation are joined by a grey edge. Thus, each vertex of the breakpoint graph has exactly two incident edges. This allows the unique decomposition of the breakpoint graph into *cycles*. The *support* of a grey edge is the interval of elements between and including the endpoints. Two grey edges *overlap* if their supports intersect without proper containment. The *overlap graph* is the graph whose vertices are the grey edges of the breakpoint graph and whose edges join overlapping grey edges.

In the traditional analysis of the sorting by inversions problem, the cycles of the breakpoint graph, and the connected components of the overlap graph, play an important role. The elementary intervals, cycles and overlap graph of this section are equivalent to the traditional concepts, but directly defined on the elements of the permutation. The components of Definition 10.10 correspond to the connected components of the overlap graph.

It is also worth mentioning that Setubal and Meidanis [21] obtained many combinatorial results on the effects of inversions on a permutation, generalizing results such as Proposition 10.17.

³For a more detailed presentation of the breakpoint graph, see Chapter 11, this volume.

10.4 The Hannenhalli–Pevzner duality theorem

In this section, we develop a formula for computing the inversion distance of a signed permutation. There are two basically different problems: the contribution of oriented components to the total distance is treated in Section 10.4.1, and the general formula is given in Section 10.4.2.

10.4.1 *Sorting oriented components*

We will show that sorting oriented components can be done by choosing oriented inversions that do not create new unoriented components. For example, the inversion of the oriented interval I_3 in the following permutation creates a new unoriented component $(0\ 2\ 1\ 3)$. In the resulting positive permutation, no inversion can create an adjacency, or split a cycle.

$$\begin{array}{ccccccc} (0 & 2 & & -3 & -1 & 4), \\ & & \bullet & \xrightarrow{\hspace{1.5cm}} & \bullet & & \\ (0 & 2 & & 1 & 3 & 4). \end{array}$$

However, one can invert the oriented interval I_0 , and the resulting component(s) remain oriented, thus allowing the sorting process to continue.

$$\begin{array}{ccccccc} (0 & 2 & & -3 & -1 & 4), \\ & \bullet & \xrightarrow{\hspace{1.5cm}} & \bullet & & & \\ (0 & 1 & 3 & -2 & 4). \end{array}$$

Choosing oriented inversions that do not create new unoriented components, called *safe* inversions, can be done by trial and error: choose an oriented inversion, perform it, then test for the presence of new unoriented components. However, it is possible to do much better. Several different criteria exist in the literature, and we give here the simplest one, which also provides a proof of existence of safe inversions in any oriented component.

Definition 10.20 *The score of an inversion is the number of oriented elementary intervals in the resulting permutation.*

Theorem 10.21 ([3]) *The inversion of an oriented elementary interval of maximal score does not create new unoriented components.*

Proof Consider a permutation P and its overlap graph. Suppose that vertex I has maximal score, and that the inversion induced by I creates a new unoriented component C containing more than one vertex. At least one of the vertices in C must have been adjacent to I , since the only edges affected by the inversion are those connecting vertices adjacent to I .

Let J be a vertex formerly adjacent to I and contained in C , thus J is oriented in P .

By Proposition 10.9, the scores of I and J can be written as:

$$\begin{aligned} \text{score}(I) &= T + U - O - 1, \\ \text{score}(J) &= T + U' - O' - 1, \end{aligned}$$

where T is the total number of oriented vertices in the overlap graph, U and O are the numbers of unoriented, respectively oriented, vertices adjacent to I , and U' and O' are the numbers of unoriented, respectively oriented, vertices adjacent to J .

All unoriented vertices formerly adjacent to I must have been adjacent to J . Indeed, an unoriented vertex adjacent to I and not to J will become oriented, and connected to J , contrary to the assumption that C is unoriented. Thus, $U' \geq U$.

All oriented vertices formerly adjacent to J must have been adjacent to I . If this was not the case, an oriented vertex adjacent to J but not to I would remain oriented, again contradicting the fact that C is unoriented. Thus, $O' \leq O$.

Now, if both $O' = O$ and $U' = U$, vertices I and J have the same set of vertices, and complementing the subgraph of I and its adjacent vertices will isolate both I and J . Therefore, we must have $score(J) > score(I)$, which is a contradiction. $\square$

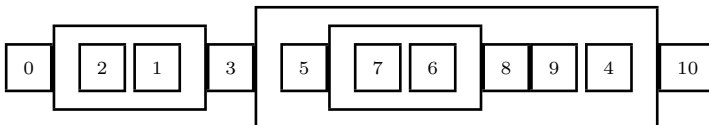
Corollary 10.22 *If a permutation P on the set $\{0, \dots, n\}$ has only oriented components and c cycles, then $d(P) = n - c$.*

Proof By Lemma 10.8, we have $d(P) \geq n - c$ since any inversion adds at most one cycle, and the identity permutation has n cycles. Any oriented inversion adds one cycle, thus Theorem 10.21 guarantees that there will be always enough oriented inversions to sort the permutation. $\square$

Corollary 10.22 implies that it is possible to compute the inversion distance of some permutations without actually sorting them: counting cycles is the important step, and is easily done, as we will show in Section 10.5. It is in this respect that the problem of computing the inversion distance differs from the problem of finding an optimal sorting sequence. There is no need to identify safe inversions in order to compute the distance.

10.4.2 Computing the inversion distance

In the preceding section, we have determined the number of inversions needed to sort a permutation which contains only oriented components. If a permutation has unoriented components, we first have to orient or destroy them. It is desirable to use as few inversions as possible for this task. Consider, for example, the following permutation which has three unoriented components. It is possible to get rid of all three of them by inverting the interval $(1 \dots 7)$ that merges the two components $(0 \dots 3)$ and $(5 \dots 8)$.



In the following, we will use the tree T_P defined in Section 10.3.3 in order to compute the minimum number of inversions required to orient unoriented

components of a given permutation. The basic idea is to cover the unoriented components of T_P with paths that indicate which pairs of components should be merged together.

Definition 10.23 *A cover $\mathcal{C}$ of T_P is a collection of paths joining all the unoriented components of P , and such that each terminal node of a path belongs to a unique path.*

By Propositions 10.18 and 10.19, each cover of T_P describes a sequence of inversions that orients all the components of P . A path that contains two or more unoriented components, called a *long* path, corresponds to merging the two components at its terminal nodes. In Fig. 10.7, for example, a path joining components $(4 \dots 7)$ and $(-12 \dots -9)$ would destroy these components, along with component $(7 \dots 16)$. A path that contains only one component, a *short* path, corresponds to cutting the component.

The *cost* of a cover is defined to be the sum of the costs of its paths, given that:

- (1) the cost of a short path is 1;
- (2) the cost of a long path is 2.

An *optimal* cover is a cover of minimal cost. Define t as the cost of any optimal cover of T_P .

The following theorem shows that the cost of an optimal cover is precisely the number of extra inversions needed to optimally sort a signed permutation containing unoriented components.

Theorem 10.24 ([5]) *If a permutation P on the set $\{0, \dots, n\}$ has c cycles, and the associated tree T_P has minimal cost t , then we have*

$$d(P) = n - c + t.$$

Proof We first show $d(P) \leq n - c + t$. Let $\mathcal{C}$ be an optimal cover of T_P . Apply to P the sequence of m merges and q cuts induced by the cover $\mathcal{C}$. Note that $t = 2m + q$. By Proposition 10.12, the resulting permutation P' has $c - m$ cycles, since merging two components always merges two cycles, and cutting components does not change the number of cycles. Thus, by Corollary 10.22, $d(P') = n - c + m$. Since $m + q$ inversions were applied to P , we have:

$$d(P) \leq d(P') + (m + q) = n - c + 2m + q = n - c + t.$$

In order to show that $d(P) \geq n - c + t$, consider any sequence of length d that optimally sorts the permutation. By Proposition 10.7, d can be written as

$$d = s + m + q,$$

where s is the number of inversions that split cycles, m is the number of inversions that merge cycles, and q is the number of inversions that do not change the number of cycles. Since the m inversions remove m cycles, and the s inversions add

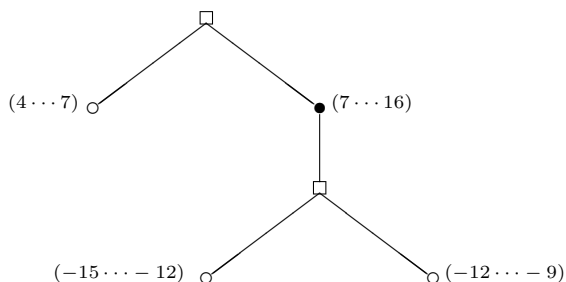


FIG. 10.8. The tree T' associated to the tree T_{P_2} of Fig. 10.7.

s cycles, we must have:

$$c - m + s = n, \quad \text{implying } d = n - c + 2m + q.$$

The sequence of d inversions induces a cover of T_P . Indeed, any inversion that merges a group of components traces a path in T_P , of which we keep the shortest segment that includes all unoriented components of the group. Of these paths, suppose that m_1 are long paths, and m_2 are short paths. Clearly we have $m_1 + m_2 \leq m$. The $q' \leq q$ remaining unoriented components are all cut. Thus

$$2m_1 + m_2 + q' \leq 2m_1 + 2m_2 + q' \leq 2m + q.$$

Since we have $t \leq 2m_1 + m_2 + q'$, we get $d \geq n - c + t$. □

The last task is to give an explicit formula for t . Let T' be the smallest unrooted subtree of T_P that contains all unoriented components of P . Formally, T' is obtained by recursively removing from T_P all dangling oriented components and square nodes. All leaves of T' will thus be unoriented components, while internal round nodes may still represent oriented components. For example, the tree T' of Fig. 10.8 is obtained from the tree T_{P_2} of Fig. 10.7. It contains three unoriented components and one oriented one.

Define a *branch* of a tree as the set of nodes from a leaf up to, but excluding, the next node of degree ≥ 3 . A *short* branch of T' contains one unoriented component, and a *long* branch contains two or more unoriented components. For example, the tree of Fig. 10.8 has three branches, and all of them are short. We have:

Theorem 10.25 *Let T' be the unrooted subtree of T_P that contains all the unoriented components as defined above.*

1. *If T' has $2k$ leaves, then $t = 2k$.*
2. *If T' has $2k + 1$ leaves, one of them on a short branch, then $t = 2k + 1$.*
3. *If T' has $2k + 1$ leaves, none of them on a short branch, then $t = 2k + 2$.*

Proof Let $\mathcal{C}$ be an optimal cover of T' , with m long paths and q shorts ones. By joining any pair of short paths into a long one, $\mathcal{C}$ can be transformed into an optimal cover with $q = 0$ or 1.

Any optimal cover has only one path on a given branch, since if there were two, one could merge the two paths and lower the cost. Thus if a tree has only long branches, there always exists an optimal cover with $q = 0$.

Since a long path covers at most two leaves, we have $t = 2m + q \geq l$, where l is the number of leaves of T' . Thus cases (1) and (2) are lower bounds. But if $q = 0$, then t must be even, and case (3) is also a lower bound.

To complete the proof, it is thus sufficient to exhibit a cover achieving these lower bounds. Suppose that $l = 2k$. If $k = 1$, the result is obvious. For $k > 1$, suppose T' has at least two nodes of degree ≥ 3 . Consider any path in T' that contains two of these nodes, and that connects two leaves A and B . The branches connecting A and B to the tree T' are incident to different nodes of T' . Thus cutting these two branches yields a tree with $2k - 2$ leaves. If the tree T' has only one node of degree ≥ 3 , the degree of this node must be at least 4, since the tree has at least four leaves. In this case, cutting any two branches yields a tree with $2k - 2$ leaves.

If $l = 2k + 1$ and one of the leaves is on a short branch, select this branch as a short path, and apply the above argument to the rest of the tree. If there is no short branch, select a long branch as a first (long) path. $\square$

For example, the permutation

$$P_2 = (0 \ -3 \ 1 \ 2 \ 4 \ 6 \ 5 \ 7 \ -15 \ -13 \ -14 \ -12 \ -10 \ -11 \ -9 \ 8 \ 16)$$

has 6 cycles, as shown in Fig. 10.6. Its associated tree T' , see Fig. 10.8, can be covered by one long path and one short path, since it has three leaves, all of them on short branches. Thus:

$$d(P_2) = n - c + t = 16 - 6 + 3 = 13.$$

Historical notes. There exist different criteria to choose a safe inversion. Hannenhalli and Pevzner [12] proved the existence of a safe inversion in any oriented component. Their algorithm suggests an exhaustive search for a safe inversion by trial and error, and runs in $\mathcal{O}(n^3)$ time. Berman and Hannenhalli [7] halved the number of candidates for every successive trial and bounded the number of trials by $\mathcal{O}(\log(n))$ yielding an algorithm to find a safe inversion in $\mathcal{O}(n\alpha(n))$ time, where $\alpha(n)$ is the inverse Ackermann function. Kaplan *et al.* [13] introduced the concept of a *happy clique* and developed an algorithm that finds a safe inversion in $\mathcal{O}(n)$ time. Bergeron [3] worked with an adjacency matrix to represent the overlap graph, with an additional score vector. The search for a safe inversion is simply the vertex with maximal score, and the update of the overlap graph is done with bit-vector operations.

The inversion distance formula given in Theorem 10.24 was first developed by Hannenhalli and Pevzner [12] in 1995. They introduced the notions of *hurdles*

and *fortresses* in order to express the inversion distance in terms of breakpoints, cycles, and hurdles.

In the literature the notion of *hurdle* is handled in various ways: Hannenhalli and Pevzner [12] define *minimal hurdles* as unoriented components which are minimal with respect to the order induced by span inclusion. In addition, the greatest element is a hurdle, called *greatest hurdle*, if it does not separate any two minimal hurdles. Kaplan *et al.* [13] do not distinguish between minimal and greatest hurdles since they order the elements of unoriented components on a circle. They define a *hurdle* as an unoriented connected component whose elements occur consecutively on the circle. Regardless of the precise definition of a hurdle, hurdles can be classified as follows: A *simple hurdle* is defined as a hurdle whose elimination decreases the number of hurdles, otherwise the hurdle is called a *super-hurdle*. A *fortress* is a permutation that has an odd number of hurdles, all of which are super-hurdles.

Let P be a permutation on the set $\{0, \dots, n\}$, Hannenhalli and Pevzner proved the following:

$$d(P) = \begin{cases} n - c + h + 1, & \text{if } P \text{ is a fortress,} \\ n - c + h, & \text{otherwise.} \end{cases}$$

where c is the number of cycles and h is the number of hurdles of permutation P .

10.5 Algorithms

In this section, we present algorithms to compute the inversion distance of a permutation P based on Theorems 10.24 and 10.25. The overall procedure consists of three parts. First, the number of cycles c is computed by a left-to-right scan of P , then the components of P are computed by an algorithm originally presented in reference [4], and finally the tree T_P is created by a simple pass over the components of P , followed by a trimming procedure yielding T' .

The number of cycles is computed in linear time by Algorithm 1. The idea is to mark each point of P as follows. The points of P are processed in left-to-right order, and each time an unmarked point is detected, all points on its cycle are marked, and the number of cycles is incremented by one. Adjacencies are treated as a limiting case. In order to do this efficiently, we need to know the endpoints of each elementary interval, and the pair of intervals that meet at each point. Figure 10.9 gives an example, along with tables containing the necessary information.

The second part of the overall procedure is the computation of the components, shown in Algorithm 2. The input of this algorithm is a signed permutation P , separated into an array of unsigned elements $\pi = (\pi_0, \pi_1, \dots, \pi_n)$ and an array of signs $\sigma = (\sigma_0, \sigma_1, \dots, \sigma_n)$.

Direct and reversed components are identified independently. Here we trace the algorithm only for direct components. In order to find these components, an array M is used, defined as follows: $M[i]$ is the nearest unsigned element of π that precedes π_i , and is greater than π_i , and n if no such element exists.

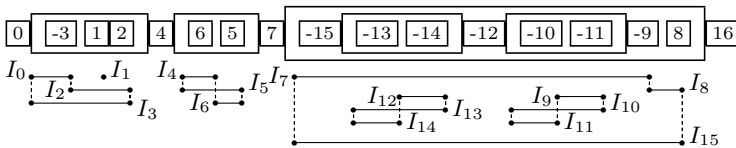
Algorithm 1 (Compute the number of cycles)

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1: a point  $\pi_{p-1} \cdot \pi_p$  is represented by the index  $p$  of its right element
2:  $\text{marked}[1, \dots, n]$  is an array of  $n$  boolean values, initially set to FALSE

3:  $c \leftarrow 0$  (* counter for the number of cycles *)
4: for  $p \leftarrow 1, \dots, n$  do
5:   if not  $\text{marked}[p]$  then
6:      $i \leftarrow$  one of the two intervals meeting at point  $p$ 
7:     while not  $\text{marked}[p]$  do
8:        $\text{marked}[p] \leftarrow \text{TRUE}$ 
9:        $i \leftarrow$  the interval meeting  $i$  at point  $p$ 
10:       $p \leftarrow$  the other endpoint of  $i$ 
11:    end while
12:     $c \leftarrow c + 1$ 
13:  end if
14: end for

```



Elementary interval	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}	I_{11}	I_{12}	I_{13}	I_{14}	I_{15}
First endpoint	1	3	4	1	5	7	6	8	16	14	12	13	11	9	10	8
Second endpoint	2	3	2	4	6	5	7	15	15	13	14	12	10	11	9	16

Point	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
First interval	I_0	I_0	I_1	I_2	I_4	I_4	I_5	I_7	I_{13}	I_{12}	I_{12}	I_{10}	I_9	I_9	I_7	I_8
Second interval	I_3	I_2	I_1	I_3	I_5	I_6	I_6	I_{15}	I_{14}	I_{14}	I_{13}	I_{11}	I_{11}	I_{10}	I_8	I_{15}

FIG. 10.9. Detecting cycles in permutation P_2 using Algorithm 1. Starting at the first point of P_2 , we identify the cycle consisting of the elementary intervals I_0 , I_2 , and I_3 . The next iteration is skipped because the second point was marked during the traversal of the first cycle. Eventually, all six cycles are recovered.

For example, the array M of permutation P_2 is:

$P_2 = (0 \ -3 \ 1 \ 2 \ 4 \ 6 \ 5 \ 7 \ -15 \ -13 \ -14 \ -12 \ -10 \ -11 \ -9 \ 8 \ 16)$,
 $M = (16 \ 16 \ 3 \ 3 \ 16 \ 16 \ 6 \ 16 \ 16 \ 15 \ 15 \ 14 \ 12 \ 12 \ 11 \ 9 \ 16)$.
 M is computed using a stack M_1 as shown in lines 5–10 of Algorithm 2.

To find the direct components (lines 11–14 of Algorithm 2), a second stack S_1 stores potential left boundary elements s , which are then tested by the following criterion: $(\pi_s \dots \pi_i)$ is a direct component if and only if:

Algorithm 2 (Find the components of signed permutation $P = (\pi, \sigma)$)

```

1:  $M_1$  and  $M_2$  are stacks of integers; initially  $M_1$  contains  $n$  and  $M_2$  contains 0
2:  $S_1$  and  $S_2$  are stacks of integers; initially  $S_1$  contains 0 and  $S_2$  contains 0
3:  $M[0] \leftarrow n$ ,  $m[0] \leftarrow 0$ 
4: for  $i \leftarrow 1, \dots, n$  do
    (* Compute the  $M[i]$  *)
5:   if  $\pi[i-1] > \pi[i]$  then
6:     push  $\pi[i-1]$  on  $M_1$ 
7:   else
8:     pop from  $M_1$  all entries that are smaller than  $\pi[i]$ 
9:   end if
10:   $M[i] \leftarrow$  the top element of  $M_1$ 
    (* Find direct components *)
11:  pop the top element  $s$  from  $S_1$  as long as  $\pi[s] > \pi[i]$  or  $M[s] < \pi[i]$ 
12:  if  $\sigma[i] = +$  and  $M[i] = M[s]$  and  $i - s = \pi[i] - \pi[s]$  then
13:    report the component  $(\pi_s \dots \pi_i)$ 
14:  end if
    (* Compute the  $m[i]$  *)
15:  if  $\pi[i-1] < \pi[i]$  then
16:    push  $\pi[i-1]$  on  $M_2$ 
17:  else
18:    pop from  $M_2$  all entries that are larger than  $\pi[i]$ 
19:  end if
20:   $m[i] \leftarrow$  the top element of  $M_2$ 
    (* Find reversed components *)
21:  pop the top element  $s$  from  $S_2$  as long as  $(\pi[s] < \pi[i]$  or  $m[s] > \pi[i])$  and  $s > 0$ 
22:  if  $\sigma[i] = -$  and  $m[i] = m[s]$  and  $i - s = \pi[i] - \pi[s]$  then
23:    report the component  $(-\pi_s \dots -\pi_i)$ 
24:  end if
    (* Update stacks *)
25:  if  $\sigma[i] = +$  then
26:    push  $i$  on  $S_1$ 
27:  else
28:    push  $i$  on  $S_2$ 
29:  end if
30: end for

```

- (1) both σ_s and σ_i are positive,
- (2) all elements between π_s and π_i in π are greater than π_s and smaller than π_i , the latter being equivalent to the simple test $M[i] = M[s]$, and
- (3) no element “between” π_s and π_i is missing, that is, $i - s = \pi_i - \pi_s$.

For example, the component $(4 \dots 7)$ will be found in iteration $i = 7$ because:

- (1) both 4 and 7 are positive,

- (2) all elements between 4 and 7 are greater than 4 (since element 4 is still stacked on S_1 when $i = 7$) and smaller than 7 (since $M[4] = 16 = M[7]$), and
 (3) $i - s = 7 - 4 = \pi_i - \pi_s$.

Similarly, for the detection of reversed components, we use a stack M_2 to compute $m[i]$, the nearest unsigned element of π that precedes π_i and is smaller than π_i , and a stack S_2 that stores potential left boundary elements of reversed components.

The classification of components as oriented or unoriented can be done by a slight modification of Algorithm 2, without affecting the running time. We need an extra array o to store the signs of the points of the permutation P (for ease of notation shifted down by one position). For $0 \leq i < n$, the entries of the array o are initially defined as follows:

$$o[i] = \begin{cases} +, & \text{if } \sigma_i = + \text{ and } \sigma_{i+1} = +, \\ -, & \text{if } \sigma_i = - \text{ and } \sigma_{i+1} = -, \\ 0, & \text{otherwise.} \end{cases}$$

For example, the initial array o of permutation P_2 is:

$$o = (0 \quad 0 \quad + \quad + \quad + \quad + \quad + \quad 0 \quad - \quad - \quad - \quad - \quad - \quad - \quad 0 \quad +).$$

Now we define a function $f: \{-, 0, +\}^2 \rightarrow \{-, 0, +\}$ as:

$$f(x_1, x_2) = \begin{cases} x_1, & \text{if } x_1 = x_2, \\ 0, & \text{otherwise.} \end{cases}$$

Then, in the modified algorithm, whenever an index s is removed from the stack such that index r becomes the top of the stack, $o[r]$ will be replaced by $f(o[r], o[s])$. We also replace the entry of the left bounding element of an identified direct component by $+$, and the entry of the left bounding element of an identified reversed component by $-$. This way, when a direct component $(\pi_s \dots \pi_i)$ is reported in line 13 of Algorithm 2, the signs of all its points are folded by repeated application of function f to the leftmost point s of the component. Its orientation can easily then be derived: $(\pi_s \dots \pi_i)$ is unoriented if and only if

- (1) $s + 1 \neq i$ (the component contains one or more breakpoints); and
 (2) $o[s]$ equals $+$ or $-$ (all its points have the same sign).

The correctness of this algorithm follows from the fact that all the indices of elements of an unoriented component are stacked on the same stack, and that all its points have the same sign. If a component C contains other components, these will be identified before C , and are treated as single positive or negative elements. Since the bounding elements of oriented components have the same sign, each oriented component has at least two points for which $o(i) = 0$, and at least one index on each stack for which $o(i) = 0$.

In order to understand the third part of the overall procedure, note that Algorithm 2 reports the components in left-to-right order with respect to their

Algorithm 3 (Construct T_P from the components $C_1, \dots, C_k$ of P)

```

1: create a square node  $q$ , the root of  $T_P$ , and a round node  $p$  as the child of  $q$ 
2: for  $i \leftarrow 1, \dots, n-1$  do
3:   if there is a component  $C$  starting at position  $i$  then
4:     if there is no component ending at position  $i$  then
5:       create a new square node  $q$  as a child of  $p$ 
6:     end if
7:     create a new round node  $p$  (representing  $C$ ) as a child of  $q$ 
8:   else if there is a component ending at position  $i$  then
9:      $p \leftarrow$  parent of  $q$ 
10:     $q \leftarrow$  parent of  $p$ 
11:   end if
12: end for

```

right bounding element. For each index i , $0 \leq i \leq n$, at most one component can start at position i , and at most one component can end at position i . Hence, it is possible to create a data structure that tells, in constant time, if there is a component beginning or ending at position i and, if so, reports such components. Given this data structure, it is a simple procedure to construct the tree T_P in one left-to-right scan along the permutation. Initially one square root node and one round node representing the component with left bounding element 0 are created. Then, for each additional component, a new round node p is created as the child of a new or an existing square node q , depending if p is the first component in a chain or not. For details, see Algorithm 3.

To generate tree T' from tree T_P , a bottom-up traversal of T_P recursively removes all dangling round leaves, that represent oriented components, and square nodes, including the root if it has degree 1. Given the tree T' , it is easy to compute the inversion distance: perform a depth-first traversal of T' and count the number of leaves and the number of long and short branches, including the root if it has degree 1. Then use the formula from Theorem 10.25 to obtain t , and the formula from Theorem 10.24 to obtain d .

Altogether we have:

Theorem 10.26 *Using Algorithms 1, 2, and 3, the inversion distance $d(P)$ of a permutation P on the set $\{0, \dots, n\}$ can be computed in linear time $O(n)$.*

Historical notes. Traditionally, the inversion distance is computed by using the formula of Hannenhalli and Pevzner. As the hurdles and fortresses are detectable from connected component analysis, the most delicate part is to compute the connected components. The existing algorithms solve this problem in different ways. The initial algorithm of Hannenhalli and Pevzner [12], restricted to the computation of the inversion distance, runs in quadratic time by constructing the overlap graph. In 1996, Berman and Hannenhalli [7] developed a faster algorithm for computing the connected components, yielding an algorithm

to compute $d(P)$ in $\mathcal{O}(n \cdot \alpha(n))$ time. They used a Union/Find structure to maintain the connected components of the overlap graph, without constructing the graph itself. In 2001, Bader *et al.* [1] gave the first linear time algorithm for computing the inversion distance. By scanning the permutation twice, their algorithm constructs another graph, called the *overlap forest*, which has exactly one tree per connected component of the overlap graph.

10.6 Conclusion

This chapter gave an elementary presentation of the results of the classical Hannenhalli–Pevzner theory on the inversion distance problem. Most of the results are obtained by working directly on the elements of the permutation, instead of relying on intermediate constructions. This effort yielded a simpler equation for the distance, an increased understanding of the effects of inversions on a permutation, and the development of very elementary algorithms.

Looking at the problem from this point of view led to some interesting variants of genome comparison tools. The concept of *conserved intervals* [4, 6], for example, can be used to measure the similarity of a set of permutations. It is a direct offspring of the crucial role played by components in the inversion problem.

This work is also a first step in the simplification of the problem of comparing multi-chromosomal genomes. Rearrangement operations between these genomes include, among others, inversions, translocations, fusions, and fissions of chromosomes. The algorithmic treatment of this problem relies on the properties of the sorting by inversions problem, and currently involves half a dozen parameters [12]. The initial solution contained gaps that took years to be closed [19, 25]. A linear algorithm for the translocation distance problem [11] is given in reference [16].

Another crucial extension is the ability to handle insertions, deletions, and duplications of genes. This extension is much harder, but much more important for biological applications. Indeed, one of the main driving forces of genome evolution are segment duplications. Recent work on this problem can be found in [10, 17], and is surveyed in the Chapters 11 and 12, this volume.

Glossary

<i>adjacency</i>	pair of consecutive integers, Section 10.2
<i>bounding elements</i>	first and last elements of an interval, Section 10.2
<i>branch</i>	set of nodes from a leaf to the next node of degree ≥ 3 , Section 10.4
<i>breakpoint</i>	a point that is not an adjacency, Section 10.2
<i>chain</i>	a sequence of components overlapping on one element, Section 10.3.3

<i>component</i>	Definition 10.10
<i>cover</i>	Definition 10.23
<i>cycle</i>	Definition 10.4
<i>direct component</i>	Definition 10.10
<i>elementary interval</i>	Definition 10.1
<i>endpoints</i>	first and last points of an interval, Section 10.2
<i>extremities</i>	Definition 10.1
<i>long branch</i>	branch containing more than one unoriented component, Section 10.4
<i>oriented component</i>	Definition 10.13
<i>oriented interval</i>	elementary interval whose inversion creates an adjacency, Section 10.3.1
<i>overlapping intervals</i>	Definition 10.6
<i>point</i>	pair of consecutive elements, Section 10.2
<i>reversed component</i>	Definition 10.10
<i>safe inversion</i>	oriented inversion that does not create new unoriented components, Section 10.4.1
<i>score</i>	Definition 10.20
<i>sign of a point</i>	Definition 10.13
<i>short branch</i>	branch containing one unoriented component, Section 10.4
<i>sorting inversion</i>	an inversion that belongs to an optimal sorting sequence, Section 10.2
<i>sorting sequence</i>	sequence of inversions that transform a permutation into the identity permutation, Section 10.2
<i>tree T_P</i>	Definition 10.15
<i>unoriented component</i>	Definition 10.13
<i>unoriented interval</i>	elementary interval whose inversion does not create an adjacency, Section 10.3.1

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GENOME REARRANGEMENTS WITH GENE FAMILIES

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The genome rearrangement approach to comparative genomics infers divergence history in terms of global genomic mutations. The major focus in the last decades has been to infer the most economical scenario of elementary operations transforming one linear order of genes into another. Implicit in most of these studies is that each gene has exactly one copy in each genome. This hypothesis is clearly unsuitable for divergent species containing several copies of highly paralogous gene, for example, multigene families. In this chapter, we review the different algorithmic methods that have been considered to account for multigene families in the genome rearrangement context, and in the phylogenetic context.

Another fundamental question raised by duplicated genes is: given a genome with multigene families, how can we reconstruct an ancestral genome containing unique gene copies? This question has been widely studied by our group. We review the algorithmic methods considered in the case of genome-wide doubling events, and duplications at a regional level.

11.1 Introduction

With the accumulating number of sequenced genomes, it becomes possible to analyse and compare genomes based on their overall content in genes and other genetic elements. This genomic approach is an alternative to the traditional one based on the comparison of gene sequences. In particular, whole genome alignment methods have recently been developed and applied to the comparison of the human and mouse genomes [12, 20]. Other methods have been used to detect regions of conserved synteny and orthologous genes between two genomes [60, 72]. These analysis allow to formally represent a chromosome as a linear order of its building blocks or genes. The problem of comparing two genomes is then abstracted as one of comparing two permutations defined on a set of objects. This approach infers divergence history, not in terms of local mutations, but in terms of more global genomic mutations, involving the displacement, insertion, and duplication of chromosomal segments of various sizes.

The genome rearrangement approach has been widely studied in the last decade. The major focus has been to infer the most economical scenario of elementary operations transforming one linear order of genes into another. In this

context, inversion (also called “reversal”) has been the most studied rearrangement event [7, 9, 15, 16, 38, 40, 41], followed by transpositions [5, 39, 48, 73] and translocations [6, 37, 57, 71]. All these studies are based on the assumption that the compared genomes have the same genes, each one appearing exactly once in each genome. In reference [25], we considered the case of genomes with different gene contents, and generalized the Hannenhalli and Pevzner theory to include insertions and deletions of gene blocks. However, the assumption of unique gene copies remains a necessary condition to the development of efficient methods for rearrangement distance computation. While this hypothesis may be appropriate for small genomes, for example, viruses and organelles, it is clearly unsuitable for divergent species containing several copies of highly paralogous and orthologous genes, scattered across the genome. In this case, it is important to introduce the possibility of having different copies of the same gene, for example, *multigene families*. We discuss this issue in Section 11.4.

The first method that has been considered to account for multigene families in the genome rearrangement context is the *exemplar approach* [62]. The basic idea is to remove all but one member of each gene family in each of the two genomes being compared, so as to minimize a rearrangement distance. More recently, Marron *et al.* [46] presented a straightforward approach by enumerating all the possible assignments of orthologs between two genomes. In contrast with genome rearrangement, gene families have been widely considered in the phylogenetic context, where the goal is to reconstruct the correct evolutionary topology for a set of taxa given a set of gene trees. For this purpose, the “reconciliation” approach, consisting in projecting a gene tree T_g onto a “true” species tree $\mathcal{T}$, has been used to infer gene duplication and gene loss events [17, 33, 45, 58], or horizontal gene transfer [34]. We describe these methods in Section 11.5.

Another fundamental question raised by duplicate genes is: what is the ancestral copy of each gene family? More generally, given a genome with many multigene families, how can we reconstruct an ancestral genome containing unique gene copies? This question is strongly related to the evolutionary model giving rise to duplicate genes. In the last paragraph, we have mentioned the “gene duplication and loss” model, and the “horizontal transfer” model. Other models have been proposed to account for the origin of gene duplications. They fall into two categories: genome-wide doubling events, and duplications at a regional level.

Evidence of whole genome duplication has shown up across the eukaryote spectrum and is particularly prevalent in plants [3, 32, 49, 54, 69]. Originally, a duplicated genome contains two identical copies of each chromosome, but through genomic rearrangements, this simple doubled structure is disrupted, and all that can be observed is a succession of conserved segments, each segment appearing exactly twice in the genome. In a series of papers [28–30], we have developed algorithms for reconstructing the ancestral doubled genome minimizing the number of rearrangement events required to derive the observed order of genes along the present-day chromosomes. We considered different

genome structures (synteny blocks, ordered and signed genes, circular genomes, multichromosomal genomes), and different rearrangement events (reversals, translocations, both reversals and translocations). In Section 11.6, we present the general methodology common to most of these models.

Tandem duplications are the most easily recognized segment duplications. The mathematical problem of reconstructing the history of such duplications has been extensively studied [24,78]. Chapter 8, reviews different approaches and mathematical concepts for studying tandem duplications from an evolutionary perspective. Another important regional event by which gene duplications can occur has been referred to as *duplication transposition* [55]. In this model, entire regions are duplicated from one location of the genome to another. Studies from human genomic sequence indicate that many of these segments have been duplicatively transposed in very recent evolutionary time [23]. Many of these duplications play a role in both human disease and human evolution [47]. In reference [26], we considered the problem of reconstructing an ancestral genome of a modern genome, arising through duplication transpositions and reversals. We used our approach to reconstruct gene orders at the ancestral nodes of a species tree $\mathcal{T}$, given the gene trees of each gene family. We present this approach in Section 11.7.

We begin this presentation by formalizing the notion of a genome in Section 11.2. We then briefly introduce the rearrangement distance problem and the Hannenhalli and Pevzner approach in Section 11.3.

11.2 The formal representation of the genome

In contrast to prokaryotes that tend to have single, often circular chromosomes, the genes in plants, animals, yeasts, and other eukaryotes are partitioned among several chromosomes. The number of chromosomes is generally between 10 and 100, though it can be as low as 2 or 3, or much higher than 100. In particular, fern species exhibit some of the largest chromosome numbers, which is a result of polyploidy. For example, Adder's tongue fern (*Ophiglossum*) has a base number of 120 chromosomes, the diploid species has 240 chromosomes, and a related species has 1,200 chromosomes.

The genome rearrangement approach to comparative genomics focuses on the general structure of a chromosome, rather than on the internal nucleic structure of each gene. This approach assumes that the problems of determining the identity of each gene, and its homologs (paralogs and orthologs) among a set of genomes, have been solved, so that a gene is simply labelled by a symbol indicating the class of homologs to which it belongs. We have to point here that this gene annotation step is far from being trivial. In many cases, the similarity scores given by the local alignment tools are too ambiguous to conclude to a homology. Distinguishing between paralogs (evolution by duplication, possible loss of function), and orthologs (evolution by speciation, potentially the same function) is even harder. In this chapter, as in most papers accounting for multigene families in a genome rearrangement context, *paralogs* will refer to homologs detected in

Chro.1:	{a ₁ , a ₃ , b ₁ , c ₂ }	a ₁ b ₁ a ₃ c ₂	+a ₁ -b ₁ -a ₃ +c ₂
Chro.2:	{a ₂ , b ₂ , c ₁ }	b ₂ a ₂ b ₂ c ₁ a ₂	-b ₂ +a ₂ -b ₂ +c ₁ +a ₂
Chro.3:	{a ₄ , b ₃ , c ₃ , c ₄ , c ₅ }	c ₃ b ₃ a ₄ c ₄ c ₅	+c ₃ +b ₃ -a ₄ +c ₄ -c ₅
	Synteny sets	Ordered, unsigned genes	Ordered, signed genes

FIG. 11.1. The different levels of chromosomal structures considered in the genome rearrangement literature.

the same genome, and *orthologs* will refer to homologs detected among different genomes.

Three levels of chromosomal structures have been studied in the literature (Fig. 11.1). The syntenic structure just indicates the content of genes among the set of chromosomes of a genome. Two genes located on the same chromosome are said to be *syntenic*. The genome rearrangement approach based on syntenic structures infers divergence history in term of interchromosomal movements such as reciprocal translocation, fusion, and fission (see Section 11.3). Intrachromosomal movements can be detected only if the order of genes in chromosomes is known. In that case, a chromosome is represented as a linear sequence of genes. In the most realistic version of the rearrangement problem, a sign (+ or -) is associated with each gene, representing its transcriptional orientation. This orientation indicates on which of the two complementary DNA strands the gene is located. The distance problems in which this level of structure is known and taken into account are called “signed,” in contrast to the situation where no directional information is used, the “unsigned” case.

Note that the mathematical developments in the genome rearrangement field do not depend on the fact that the objects in a linear order (or a synteny set) describing a chromosome are genes. They could as well be blocks of genes contiguous in the two (or N) species being compared, conserved chromosomal segments in comparative genetic maps or the results of any decomposition of the chromosome into disjoint ordered fragments, each identifiable in the two (or in all N) genomes.

11.3 Genome rearrangement

Gene orders can be compared according to a variety of criteria. The *breakpoint distance* between two genomes G and H measures the number of disruptions between conserved segments in G and H , that is the number of pairs of genes a, b that are adjacent in one genome (contains the segment “ $a\ b$ ”) but not in the other (contains neither “ ab ”, nor “ $-b-a$ ”). This metric, introduced by Watterson *et al.* [75], is easily computed in time linear in the length of the genomes. It has been successfully used for inferring phylogenetic trees [10, 51]. Other metrics, *rearrangement distances*, are based on specific models of evolution. They measure the minimal number of genomic mutations (genome rearrangements) necessary to transform one linear order of genes into another. The

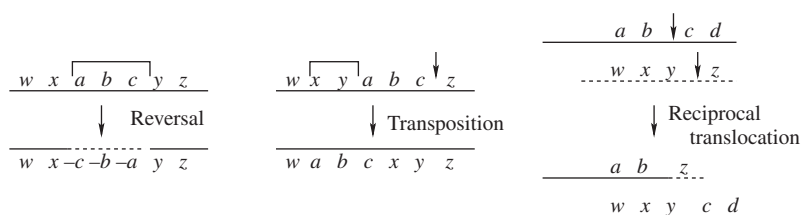


FIG. 11.2. Intrachromosomal (reversal, transposition) and interchromosomal (reciprocal translocation) rearrangement events.

rearrangement operation that has been considered most often is the *inversion* (reversal) of a chromosomal segment. The inversion distance has also been used to infer phylogenetic trees [50]. As the inversion distance underestimates the true evolutionary distance, a corrected distance (EDE distance) has been devised to better estimate the actual number of inversions. Such a correction has also been considered for the breakpoint distance [74] (see also Chapters 12 and 13, this volume).

Other rearrangement operations that have been considered in the genome rearrangement literature are *transposition* of a segment from one site to another in one chromosome, and *translocation* (exchange) of terminal segments between two chromosomes. The *fusion* of two chromosomes or *fission* of one chromosome (one chromosome cut in two disjoint parts) are two special cases of translocation. A *reciprocal translocation* is just a translocation that is neither a fusion, nor a fission (Fig. 11.2).

More recently, Bergeron and Stoye [8] have introduced a new measure of similarity between a set of genomes, based on the number of *conserved segments* in the genomes (see also Chapter 10, this volume).

From a combinatorial point of view, the differences between the synteny, the signed version, and the unsigned version of the rearrangement problem are fundamental. In the unordered version of the problem, computing the synteny distance (minimal number of translocations required to transform one genome into another) has been shown to be NP-complete [19]. This is also the case for the reversal distance in the ordered, unsigned version of the problem [14]. However, the problem becomes tractable for ordered and signed genomes. The exact polynomial algorithm of Hannenhalli and Pevzner (hereafter HP) for sorting signed permutations by reversals [37, 38] was a breakthrough for the formal analysis of evolutionary genome rearrangement. Moreover, they were able to extend their approach to include the analysis of translocations [37]. Different optimizations and simplifications of the original method have been proposed in the literature [4, 7, 9, 40] (see also Chapter 10, this volume). We further extended the HP approach to include insertions and deletions of gene blocks, allowing to compare genomes with different gene contents [25]. We sketch the HP approach in the next paragraph.

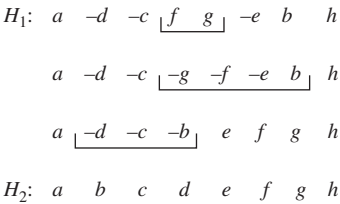


FIG. 11.3. Transforming H_1 to H_2 by three reversals. Here $\mathcal{B} = \{a,b,c,d,e,f,g,h\}$.

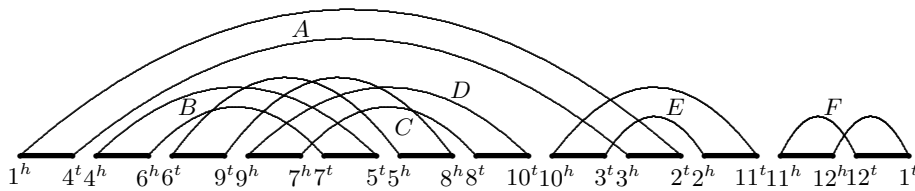


FIG. 11.4. Graph $\mathcal{G}_{12}$ corresponding to circular genomes (i.e. first gene is adjacent to last gene) $H_1 = +1 \ +4 \ -6 \ +9 \ -7 \ +5 \ -8 \ +10 \ +3 \ +2 \ +11 \ -12$ (black edges) and $H_2 = +1 \ +2 \ +3 \ + \cdots +12$ (grey edges). A , B , C , D , E , and F are the 6 cycles of $\mathcal{G}_{12}$. $\{A, E\}$, $\{B, C, D\}$, and $\{F\}$ are the three components of $\mathcal{G}_{12}$.

The Hannenhalli and Pevzner theory. Let H_1 and H_2 be two genomes defined on the same gene set $\mathcal{B}$, where each gene appears exactly once in each genome. The problem is to find the minimum number of rearrangement operations necessary to transform H_1 into H_2 (Fig. 11.3).

The HP algorithms for sorting by reversals, translocations, or both reversals, and translocations, all depend on a bicoloured graph $\mathcal{G}_{12}$ constructed from H_1 and H_2 , in the following way: if gene x in H_1 or H_2 has positive sign, replace it by the pair $x^t x^h$ in the considered permutation, and if it is negative, by $x^h x^t$. Then the vertices of $\mathcal{G}_{12}$ are just the x^t and the x^h for all x in $\mathcal{B}$. Any two vertices which are adjacent in some chromosome in H_1 , other than x^t and x^h deriving from the same x , are connected by a black edge (thick lines in Fig. 11.4), and any two vertices adjacent in H_2 , by a grey edge (thin lines in Fig. 11.4). In the case of a single chromosome, the black edges may be displayed linearly according to the order of the genes in the chromosome (Fig. 11.4). For a genome containing N chromosomes, N such linear orders are required.

Each vertex of $\mathcal{G}_{12}$ is incident to exactly one black and one grey edge, thus there is a unique decomposition into c_{12} disjoint cycles of alternating edge colours. This is precisely the reason for dedoubling each vertex x into x^t and x^h . Note that $c_{21} = c_{12} = c$ is maximized when $H_1 = H_2$, in which case each cycle has one black edge and one grey edge.

A rearrangement operation ρ , either a reversal or a translocation, is determined by the two points where it “cuts” genome H_1 , which correspond to two black edges. Rearrangement operations may change the number of cycles of the graph, and minimizing the number of operations can be seen in terms of increasing the number of cycles as fast as possible. Let $\Delta(c)$ be the difference between the number of cycles before and after applying the rearrangement operation ρ . Reference [41] showed that $\Delta(c)$ may take on values 1, 0, or -1 , in which cases they called ρ *proper*, *improper*, or *bad*, respectively. Roughly speaking, an operation acting on two black edges in two different cycles will be bad, while one acting on two black edges within the same cycle may be proper or improper, depending on the type of cycle and the type of edges considered.

Key to the HP approach are the graph components. A *component* of $\mathcal{G}_{12}$ is a maximal set of crossing cycles (cycles containing grey edges that “cross,” for example cycles B and C in Fig. 11.4), excluding the case of a cycle of length 2. A component is termed *good* if it can be transformed to a set of cycles of length 2 by a series of proper operations, and *bad* otherwise. Bad components are called *minimal subpermutations* in the translocations-only model, *hurdles* in the reversals-only model, and *knots* in the combined model.

The HP formulae for all three models may be summarized as follows:

$$\text{HP: RO}(H_1, H_2) = b(\mathcal{G}_{12}) - c(\mathcal{G}_{12}) + m(\mathcal{G}_{12}) + f(\mathcal{G}_{12}),$$

where $\text{RO}(G, H)$ is the minimum number of rearrangement operations (reversals and/or translocations), $b(\mathcal{G}_{12})$ is the number of black edges, $c(\mathcal{G}_{12})$ the number of cycles and $m(\mathcal{G}_{12})$ the number of bad components of $\mathcal{G}_{12}$, and $f(\mathcal{G}_{12})$ is a correction of 0, 1, or 2 depending on the set of bad components (see Chapter 10, this volume, for more details).

Generally speaking, bad components are rare, so the number of cycles of $\mathcal{G}_{12}$ is the dominant parameter in the HP formula, if $b(\mathcal{G}_{12})$ is considered as a constant. In other words, the more cycles there are, the fewer reversals we need to transform H_1 into H_2 .

Biological applications. In a series of recent papers, Pevzner and co-authors have applied the breakpoint graph and genome rearrangement algorithms for inversions and translocations to several mammalian genomes including human, mouse, cat, cattle, and rat [11, 52, 60, 61]. The human and mouse genomes comparison revealed evidence for more rearrangements than thought previously, involving a large number of micro-rearrangements. The rearrangement scenarios obtained for these genomes gave them arguments in favour of a new model of chromosome evolution that they called the *fragile breakpoint model*. In contrast with the previously adopted Nadeau–Taylor *random breakage model*, this new model postulates that the breakpoints mainly occur within relatively short fragile regions (hot spots of rearrangements). However, this new model remains controversial [67].

We also applied the breakpoint graph to test the mechanism of reversals in bacterial genomes. More precisely, we used a specially designed implementation

of the HP theory to test the hypothesis that, in bacteria, most reversals act on segments surrounding one of the two endpoints of the replication axis [2]. We also found a large excess of short inversions, especially those involving a single gene, in comparison with a random inversion model [42].

11.4 Multigene families

Implicit in the rearrangement literature, and in most tree reconstruction methods based on gene orders, is that both genomes being compared contain an identical set of genes and the one-to-one orthologies between all pairs of corresponding genes in the two genomes have previously been established. This hypothesis is clearly unsuitable, since almost all genomes which have been studied contain genes that are present in two or more copies. These copies may be identical, or found to have a high similarity with a BLAST-like search. They may be adjacent on a single chromosome, or dispersed throughout the genome. As an example, Li *et al.* [43] find that duplicated genes account for about 15% of the protein genes in the human genome. Another analysis of eucaryotic genome sequences accounts for 10–16% duplicated genes in the yeast genome, and about 20% in the worm genomes [44, 76].

Several models have been proposed to account for the origin of gene duplications: tandem repeat through slippage during recombination (Chapter 8, this volume), gene conversion, horizontal transfer, hybridization, and whole genome duplication [22, 63]. These models fall into two categories: genome-wide doubling events, and duplications at a regional level.

Whole genome duplication is perhaps the most spectacular mechanism giving rise to multigene families. Normally a lethal accident of meiosis, if genome doublings can be resolved in the organism and eventually fixed as a normalized diploid state in a population, it constitutes a duplication of the entire genetic material. Although the creative role of polyploidy in the evolution of a species is controversial [56], it may have a considerable effect on evolution, as whole new physiological pathways may emerge, involving novel functions for many of the duplicated genes.

Genome doubling is widespread in plants. In particular, many familiar crop species, including oats, wheat [49], maize, and rice [1, 32], have shown traces of genome duplication. Following the complete sequencing of all *Saccharomyces cerevisiae* chromosomes, the prevalence of gene duplication has led to the hypothesis that this yeast genome is also the product of an ancient doubling [68, 77]. Traces of genome duplication have also shown up across the eukaryote spectrum. More than 200 million years ago, the vertebrate genome may have undergone two duplications [3, 54], though at least one of these remains controversial [31].

In contrast, local duplications involve the duplication of small portions of chromosomes, either in tandem, or transposed to new locations within the genome. Duplicated segments may be as short as single genes, though not all the repeated segments contain genes or parts of genes.



FIG. 11.5. A duplication-transposition event.

Duplication transposition [55] is one of the most important regional event by which gene duplications can occur. In this model, entire regions are duplicated from one location of the genome to another (Fig. 11.5). Studies from human genomic sequence indicate that many of these segments have been duplicatively transposed in very recent evolutionary time [23]. Many of these duplications play a role in both human disease and human evolution [47]. O’Keefe and Eichler [55] have identified two patterns of segment duplication in the human genome: intrachromosomal duplication, and interchromosomal duplication. In this last case, material located on some chromosome is copied to the pericentromeric or subtelomeric regions of another chromosome.

In both cases of duplication (local and global), the presence of multigene families greatly complicates the analysis of chromosomal rearrangements. It is no longer clear how to obtain the basic datum for rearrangement analysis: the word “*caba*” is not a permutation of the word “*abc*”.

11.5 Algorithms and models

In contrast with the abundance of mathematical, algorithmic, and combinatorial methods that have been developed to compare genomes with identical gene contents, few approaches have been considered to account for multigene families. This is probably due to the significant combinatorial difficulty that is added in this case.

In this section, we introduce some approaches that have been developed to account for gene duplicates in the genome rearrangement and phylogenetic context. We then focus, in the two following sections, on the reconstruction of the evolutionary history of a single genome containing multigene families.

11.5.1 Exemplar distance

Sankoff [62] has formulated a generalized version of the genome rearrangement problem where each gene may be present in many copies. The idea is to delete, from each gene family, all copies except one in each of the compared genomes G and H . This preserved copy, called *the exemplar*, represents the common ancestor of all copies in G and H . The criteria for deleting gene copies is to form two permutations having the minimal distance. Sankoff considers two distance measures: the breakpoint distance and the reversal distance.

The underlying evolutionary model is that the most recent common ancestor F of genomes G and H has single gene copies (Fig. 11.6). After divergence, the gene a in F can be duplicated many times in the two lineages leading to G and H , and appear anywhere in the genomes. Each genome is then subject to

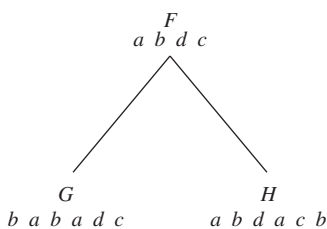


FIG. 11.6. The evolutionary model considered in the exemplar analysis. Using the breakpoint distance as a criterion, the chosen exemplar are the underlined ones.

rearrangement events. The key idea is that, after rearrangements, the *true exemplar*, that is the direct descendent of a in G and H , will have been displaced less frequently than the other gene copies. The true exemplar strings can thus be identified as those that have been less rearranged with respect to each other than any other pair of reduced genomes.

Even though finding the exemplar has been shown NP-hard [13], Sankoff [62] developed a branch-and-bound algorithm that has been shown practical enough for simulated data. The strategy is to begin with empty strings, and to insert successively one pair of homologous genes from each gene family, one after the other. At each step, the chosen pair of exemplars is the one which least increases the distance when inserted into the partial exemplar string already constructed. The gene families are processed in increasing order of their sizes: singletons first, then families of size three, four, and so on.

Sankoff considers a branch and bound strategy. At each step (for each next gene family), all pairs in the family are tested to see how much they increase the distance when the two members are inserted into the partial exemplar strings. The chosen exemplar pair is the one which least increases the distance. A backtracking step from the family currently being considered occurs whenever all its remaining unused pairs have too large test values, that is test values that would increase the distance beyond the current best value.

Discussion and biological applications. A natural application of the exemplar approach is to identify orthologies between two genomes containing families of paralogous genes. Unfortunately, as far as we know, the algorithm has only been tested on simulated data. In references [46, 70], a straightforward approach by enumerating all the possible assignments of orthologs between two genomes has been considered. However, this approach is applicable only to genomes with a very small number of duplicated genes, as the number of possible assignments grows exponentially with the number of paralogs. Very recently, Chen *et al.* [18] introduced a new approach to ortholog assignments that considers both sequence similarity and genome rearrangement. The method has been tested on the X chromosomes of human, mouse, and rat. They reported a relatively coherent assignment of orthologs compared to GenBank annotations.

To conclude, from a practical, as well as a combinatorial point of view, finding efficient algorithmic methods to assign true orthologs remains an open problem.

11.5.2 *Phylogenetic analysis*

Information on gene families have been extensively used in the context of inferring a phylogenetic tree for a set of N taxa, given a set of gene trees.

In a phylogenetic context, a gene family is the set of all occurrences (orthologs and paralogs) of a gene a in all the N species. Using standard phylogenetic procedures, one can end up with a gene tree for each of these families. For various reasons, two or more gene trees may not always agree. The question then arises on how to reconstruct the correct species tree. Suppose now the species tree is also known. Then the problem is to explain how the gene trees could have arisen with respect to the species tree.

In this context, the *gene duplication/loss model* has largely been considered in the literature [17,33,45,58]. It explains the potential non-congruences between trees by the duplication and loss of genes in some lineages. The *reconciliation method* is based on a particular projection of a gene tree into the species tree, which allows to situate duplications in the gene tree and locate them with respect to the speciation events in the species tree.

Hallett and Lagergren [34] used the reconciliation method in the context of another evolutionary model involving horizontal gene transfer. They have investigated a problem where a number of (conflicting) gene trees are to be mapped to a species tree in such a way as to minimize the number of transfer events implied.

In reference [66], we investigated the problem of inferring ancestral genomes of a phylogenetic tree when the data genomes contain multiple gene copies. More precisely, given:

- a (correct) phylogenetic tree $\mathcal{T}$ on N species
- N permutations corresponding to the gene orders in the N genomes
- a (correct) gene tree for each gene family
- a distance d between two gene orders containing only unique genes,

the problem is to find, in each ancestral genome (internal node) of $\mathcal{T}$,

- its set of genes, as well as
- their relationships with respect to genes in the immediate ancestor
- the order of these genes in the genome
- among each set of sibling genes (offspring of the same copy in the immediate ancestor), one gene, designated as the exemplar,

such that the sum of the branch lengths of the tree $\mathcal{T}$ is minimal. The length of the branch connecting a genome G to its immediate ancestor A is $d(G', A)$, where G' is the genome built from G by deleting all but the exemplar from each family.

For this purpose, we integrated three approaches to genomic evolution: 1. the theory of gene tree/species tree reconciliation, 2. genome rearrangement theory

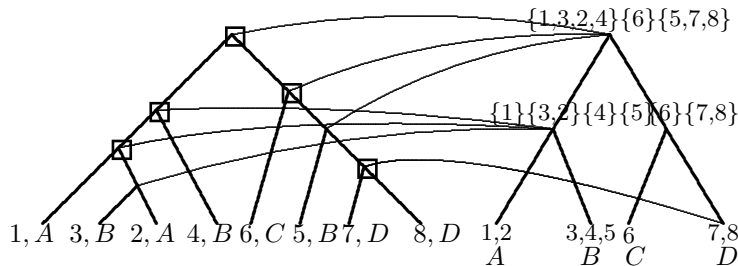


FIG. 11.7. Projection P from a gene tree $\mathcal{T}_a$ (left) to the species tree $\mathcal{T}$ (right). Numbers correspond to the different copies of gene a ; letters refer to the species (i.e. the genome). Each node e of $\mathcal{T}_a$ is projected to the node of $\mathcal{T}$ corresponding to the more recent common ancestor of all genomes containing at least one copy of a which is a descendant of e in $\mathcal{T}_a$. A duplication (drawn as a square) is deduced at each node of $\mathcal{T}_a$ that has the same projection than one of its offsprings. P induces a number of groupings at each internal node of $\mathcal{T}$, as indicated by the sets enclosed in braces. Each set refers to a single gene whose descendants are just the copies listed in the grouping.

and particularly its extension to include multigene families, and 3. breakpoint-based phylogeny and genome reconstruction.

The first step is to assign the right number of copies of each gene at each internal node of $\mathcal{T}$. The reconciliation approach is used for this purpose (Fig. 11.7).

The next step is to attribute the right gene order at each internal node of $\mathcal{T}$. Starting with an initial assignment of gene orders, recomputation of the internal nodes is carried out one by one, each time using the most recently computed versions of the neighbouring internal nodes. Iteration continues until no improvement can be made at any node. At each step, gene order at each internal node X is obtained by using the median and the exemplar approach, as described below.

Given three genomes A , B , and C with unique gene copies, the *median problem* is to find a genome X that minimizes $d(A, X) + d(B, X) + d(C, X)$ for a distance d . An efficient heuristic exists for the breakpoint distance, even for pairs of genomes containing genes that are not common to both genomes [64,65]. Applying this heuristic on $\mathcal{T}$ requires to compute pairwise distances between a genome G with single gene copies (can be X is comparing with A or B , and C if comparing with X), and a genome H with multiple gene copies (A or B or X). This computation can be done by choosing the right “exemplar” (see Section 11.5.1) in the corresponding genome with multiple gene copies. More precisely, we apply the exemplar method to genome pairs (A, X) , (B, X) , and (X, C) , and choose an exemplar gene from each grouping. The alternating application of exemplar and median analysis is shown in Fig. 11.8.

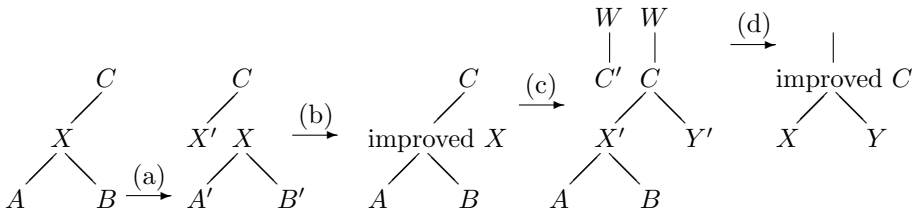


FIG. 11.8. Alternating application of exemplar and median analysis. (a) and (c): Exemplar extraction; (b) and (d): calculation of the median.

Biological applications. The reconciliation approach has been used for the reconstruction of the vertebrate phylogeny. Page and Cotton [59] analysed 118 vertebrate gene families and obtained a species tree minimizing the number of duplications that is in agreement with other data. They also localized 1,380 gene duplications in the 118 gene family data set, showing that gene duplication is an important feature of vertebrate evolution.

In contrast, the duplication, rearrangement, and reconciliation approach remains, for the moment, mostly theoretical, awaiting of appropriate data before being applied.

11.6 Genome duplication

Right after a whole genome duplication event, a doubled genome contains two identical copies of each chromosome. However, during evolution, this simple doubled structure is disrupted through intrachromosomal movements and reciprocal translocations. Even after a considerable time, however, we can hope to detect a number of scattered chromosome segments, each of which has one apparent double, so that the two segments contain a certain number of paralogous genes in a parallel order.

The main methodological question addressed in this field is: how can we reconstruct some or most of the original gene orders at the time of genome duplication, based on traces conserved in the ordering of those duplicate genes still identifiable? Some of the contributions to this methodology consider synteny blocks [28], and signed ordered genomes [27,29,30,68]. In this section, we describe the general method used in the latter case, for three different models of evolution: reversals only for circular genomes [29], translocations only, and both reversals and translocations for multichromosomal genomes [27,30].

11.6.1 Formalizing the problem

Given a modern rearranged duplicated genome G , the problem is to calculate the minimum number of rearrangement operations required to transform G into an unknown *perfect duplicated genome* H (or simply *duplicated genome*), that has to be found. In the case of a multichromosomal genome, H is of a set of pairs of identical chromosomes (Fig. 11.9). In the case of a circular genome, H is of

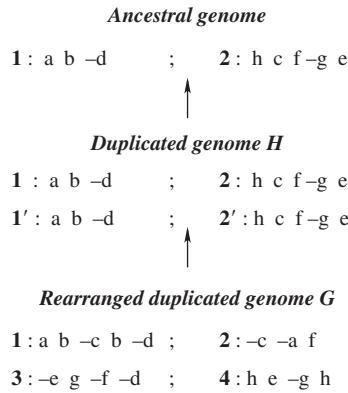


FIG. 11.9. After duplication, a genome of two chromosomes contains two pairs of identical chromosomes. After genomic rearrangements, we observe pairs of genes scattered across the genome.

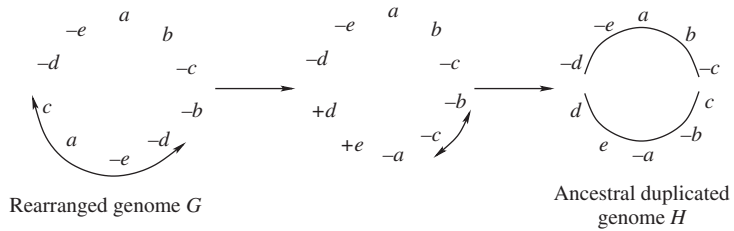


FIG. 11.10. Obtaining a circular duplicated genome H from a modern rearranged duplicated genome G after two reversals.

the form CC or $C-C$, where C is a string containing exactly one occurrence of each gene (Fig. 11.10).

11.6.2 Methodology

To make use of the Hannenhalli and Pevzner (hereafter HP) graph structure, we introduce, arbitrarily, a distinction within each pair of identical genes, labelling one occurrence x_1 and the other x_2 . In the case of linear chromosomes, to ensure the constraint of fixed endpoints required by the HP theory, we add a new initial “gene” O_{i1} and a new final “gene” O_{i2} to each chromosome C_i . This also ensures that all translocations, including those which reduce (by *fusion*), or augment (by *fission*) the number of chromosomes in the genome, can be treated as reciprocal translocations.

The general approach is to estimate the ancestral duplicated genome H by one whose comparison with G minimizes the HP formula (Section 11.3). Since the ancestral genome H is unknown, we can start only with the *partial graph* of black

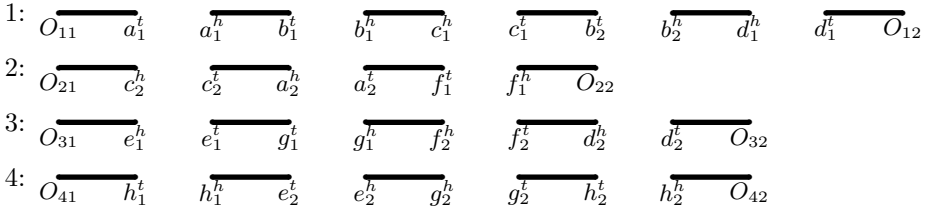


FIG. 11.11. The partial graph corresponding to genome G of Fig. 11.9.

edges that is, adjacencies in G (Fig. 11.11), and we must complete this graph with an optimal set of grey edges. Though the three evolutionary models have different behaviour related to the particular kind of genome (multichromosomal or circular), and operation (translocations and/or reversals) considered, the key concepts are the same for the three models.

Valid edges. The first step is to complete the graph with *valid* grey edges. Denote by $x, \bar{x}$ the two occurrences of the same gene (i.e. x_1 and x_2). We must add to the partial graph a set Γ of grey edges, such that every vertex is incident to exactly one black and one grey edge, and such that the resulting genome is a perfect duplicated one. For a set of grey edges to be *valid* (give rise to a duplicated genome H), the following conditions should be satisfied:

- in the case of a multichromosomal genome G , Γ should contain no edge of the form $(x, \bar{x})$; in the case of a circular genome, at most one edge of the form $(x, \bar{x})$ can be present
- if the edge (x, y) is in Γ , then $(\bar{x}, \bar{y})$ is also in Γ
- in the case of a multichromosomal genome G , the resulting genome H should not contain any circular chromosome; in the case of a circular genome G , the resulting genome H should also be a single circular chromosome.

The graph obtained by adding a valid set of grey edges is called a *complete graph*. To end up with a duplicated genome H giving rise to the minimal number of rearrangement operations, the complete graph should minimize the HP formula (Section 11.3). The key idea is to decompose the partial graph into a set of subgraphs that can be completed independently.

Decomposition into subgraphs. We group the black edges into subsets of minimal size, such that the two copies of each vertex (x_1^t and x_2^t , or x_1^h and x_2^h) are in the same subset (Fig. 11.12).

However, some of these groupings (or *natural graphs*) cannot be completed independently. For example, in Fig. 11.12, there is no way to construct a set of valid grey edges linking the vertices of $\mathcal{S}_2$. This would necessarily give rise to an invalid edge of the form $(x, \bar{x})$. The natural graphs that are problematic are those containing an odd number of edges. We thus amalgamate pairs of such graphs into *supernatural graphs*. In the example of Fig. 11.12, graphs $\mathcal{S}_2$ and

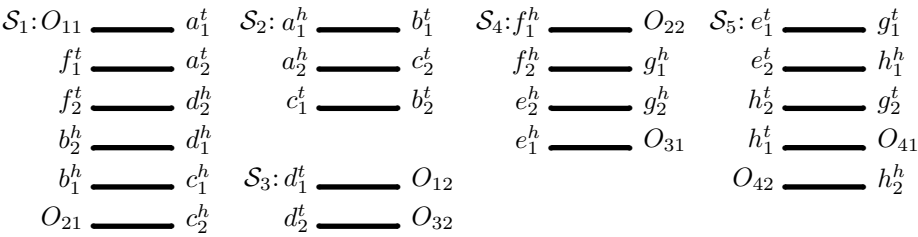


FIG. 11.12. The (unique) natural graph decomposition of the partial graph in Fig. 11.11.

$\mathcal{S}_5$ are amalgamated into $\mathcal{S}_{25}$. The set $\{\mathcal{S}_1, \mathcal{S}_{25}, \mathcal{S}_3, \mathcal{S}_4\}$ is a decomposition into supernatural graphs.

In the ensuing discussion, we start with any decomposition of the partial graph into a set $\mathcal{SN}$ of supernatural graphs. As the dominant parameter in the HP formula is the number of cycles, we begin by considering a set of valid grey edges maximizing the number of cycles of a complete graph.

Upper bound on the number of cycles. Let $\mathcal{S}$ be a supernatural graph containing $\mathcal{S}_b$ black edges. We can show that:

1. If $\mathcal{S}$ is obtained by amalgamating two natural graphs, then a complete graph of $\mathcal{S}$ contains at most $\mathcal{S}_b/2$ cycles.
2. If $\mathcal{S}$ is a natural graph of even size, then a complete graph of $\mathcal{S}$ contains at most $\mathcal{S}_b/2 + 1$ cycles.

It follows that the maximum number of cycles of a complete graph is

$$\frac{b}{2} + \gamma(G),$$

where b is the number of black edges of the partial graph, and $\gamma(G)$ is the number of natural graphs of even size.

Maximizing the number of cycles. A *fragment* of the genome G is a linear substring of G . For example, $F_1 = g - f - d$ is a fragment (of chromosome 3) of genome G given in Fig. 11.9. In the case of a multichromosomal genome, we have to be careful, during the construction of grey edges, not to end up with a circular fragment (of genome H). Suppose we have reached a certain step in the construction, and $\mathcal{F}$ is the fragment set obtained at this stage. As the construction proceeds, whenever a grey edge (x, y) is created, the fragment containing x and the one containing y are joined together. Figure 11.13 describes the situations that create a circular fragment.

During the construction of a complete graph, we also have to be careful not to end up with a *bad graph*, that is a graph in which any set of grey edges linking its remaining vertices is guaranteed to create at least one circular fragment.

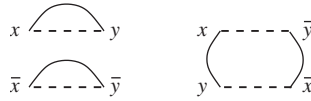


FIG. 11.13. Left and right figures represent the two situations where constructing the grey edge (x, y) creates a circular fragment. Dotted lines represent fragments already obtained for genome H .

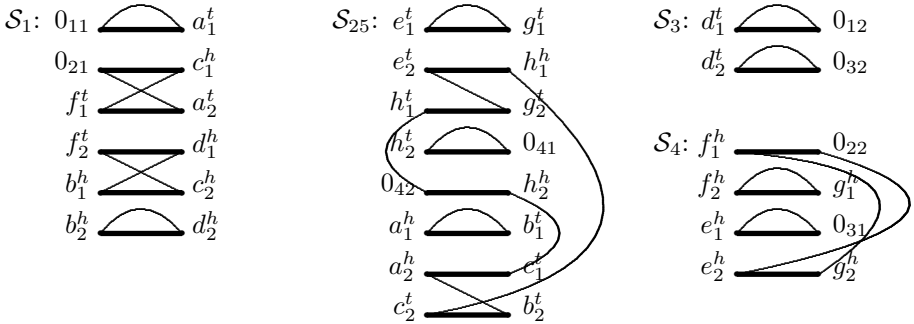


FIG. 11.14. A complete graph corresponding to the natural graphs of Fig. 11.12 constructed by algorithm *dedouble*. The resulting genome H is directly deduced from the grey edges. Namely, it contains the 4 chromosomes:
 (1) $a_1 b_1 - d_1$; (2) $a_2 b_2 - d_2$; (3) $h_1 c_2 f_2 - g_1 e_1$; (4) $h_2 c_1 f_1 - g_2 e_2$.

An algorithm *dedouble*, linear in the number of genes, has been described [26] that constructs, at each step, a valid pair of grey edges. Moreover the number of cycles of the resulting graph is maximal over all complete graphs (see the previous paragraph). An example of such a complete graph is shown in Fig. 11.14.

Bad components. It remains to minimize the number of bad components of a complete graph. Even if the concept of bad components is different for each of the three evolutionary models considered here (translocations only, reversals only, or both reversals and translocations), it is always related to the notion of “subpermutation” introduced by Hannenhalli [36].

Given two genomes H_1 and H_2 defined on the same gene set, where each gene appears exactly once in each genome, a *subpermutation* (SP) of H_1 is a subsequence $S = u_1 u_2, \dots, u_{p-1} u_p$ of H_1 such that $T = u_1 P(u_2, \dots, u_{p-1}) u_p$ is a subsequence of H_2 , where P is not identity permutation. A *minimal subpermutation* (minSP) is an SP not containing any other SP (Fig. 11.15).

For the problem of rearrangement by translocations [36], all minSP’s are bad components of a HP graph. For the problem of rearrangement by reversals, or by reversals and translocations, some SP’s can still be solved by proper operations, while others require *bad* operations to be solved. The *hurdles* in the case of

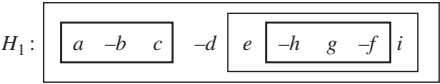


FIG. 11.15. The subpermutations of H_1 for H_2 being the identity permutation $a\ b\ c\ d\ e\ f\ g\ h\ i$. Bold rectangles indicate the minSPs.

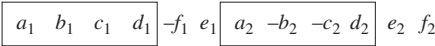


FIG. 11.16. A local SP.

reversals [38], and the *knots* in the case of reversals and translocations [37] are the bad (intrachromosomal) minSP's.

Returning to genome duplication, we want to determine the minimal number of such (bad) minSP's in a complete graph. The notion of a *local SP* is similar to the notion of an SP, but restricted to one genome. The precise definition requires to take the “dummy endpoints” (the $O_{i,1}$ and $O_{i,2}$ for each chromosome i) into account, and to distinguish between the mutichromosomal and circular case. Here is a simplified definition.

Definition 11.1 *Let $S = x_1x_2\cdots x_{n-1}x_n$ be a subsequence of G . S is a local SP of G if there exists another subsequence of G of the form $\overline{S} = \overline{x_1}P(\overline{x_2},\dots,\overline{x_{n-1}})\overline{x_n}$, where P is a permutation other than the identity. A local SP is minimal if it does not contain any subsequence corresponding to another local SP (Fig. 11.16).*

Even if the genome G does not contain any local SP, algorithm *dedouble* can give rise to a complete graph containing SPs. However, in that case, there is an easy correction to the algorithm that allows to obtain a complete graph with a maximal number of cycles and no SPs, that is a complete graph minimizing the HP formula. The minimal number $\text{RO}(G)$ of rearrangement operations (inversions, translocations, inversions and translocations) required to transform G into a duplicated genome is then deduced from the HP formula (Section 11.3) and the result of Paragraph 11.6.2:

$$\text{RO}(G) = \frac{b}{2} - \gamma(G),$$

where b is the number of black edges of the partial graph, and $\gamma(G)$ its number of natural graphs of even size. In the general case (G containing local SPs):

$$\text{RO}(G) = \frac{b}{2} - \gamma(G) + m(G) + \phi(G),$$

where $m(G)$ is the number of local SP's of G , and $\phi(G)$ is a correction factor that depends on the model considered (multichromosomal or circular case). Note that all these parameters depend solely on G .

11.6.3 *Analysing the yeast genome*

Following the complete sequencing of all *Saccharomyces cerevisiae* chromosomes, the prevalence of gene duplication has led to the hypothesis that this yeast genome is the product of an ancient doubling. Wolfe and Shields [77] proposed that the yeast genome is a degenerate tetraploid resulting from a genome duplication 10^8 years ago. They identified 55 duplicated regions, representing 50% of the genome.

As the permutations representing the sixteen chromosomes of the yeast genome do not contain any local subpermutation, the method for sorting by reversals + translocations does not involve any reversal. With this method, a perfect duplicated genome is obtained with a minimal number of 45 translocations.

11.6.4 *An application on a circular genome*

The mitochondrial genome of the liverwort plant *Marchantia polymorpha* is rather unusual in that many of its genes are manifested in two or three copies [53]. It is very unlikely that these arose from genome doubling, since this would not account for the numerous triplicates, nor is it consistent with comparative data on mitochondrial genomes. Nevertheless, it provides a convenient small example to test our method. A somewhat artificial map was extracted from the GenBank entry, deleting all singleton genes and one gene from each triplet (the two genes furthest apart were saved from each triplet). This led to a “rearranged duplicated genome” with 25 pairs of genes. A single supernatural graph emerged from the analysis. This produced a minimum of 25 inversions, which is what one would expect from a random distribution of the duplicate genes on the genome. Any trace of genome duplication, were this even biologically plausible, has been obscured.

11.7 Duplication of chromosomal segments

Duplication at a regional level consists in the doubling of chromosomal segments or genes, either in tandem or transposed to other regions in the genome (Fig. 11.5). In reference [26], we investigated the problem of reconstructing an ancestral genome of a modern circular one by considering an evolutionary model based on regional duplications and reversals. For a genome G with gene families of different sizes, the implicit hypothesis is that G has an ancestor containing exactly one copy of each gene, and that G has evolved from this ancestor through a series of duplication transpositions, and substring reversals. The question is: how can we reconstruct an ancestral genome giving rise to the minimal number of duplication, transpositions, and reversals? We formalize the problem in Section 11.7.1, and sketch the method in the following sections. The idea is to reduce the problem to a series of sub-problems involving genomes with at most two copies of each gene. We present this simplified version in Section 11.7.2, and

the general case in Section 11.7.3. Finally, in Section 11.7.4, we show how to use this method in the context of recovering gene orders at the ancestral nodes of a phylogenetic tree.

11.7.1 Formalizing the problem

A genome is said to be *ambiguous* if it contains at least one gene in more than one copy, and *non-ambiguous* otherwise. A *duplication* is defined as an operation that transforms a genome $G = ABCD$ into $G' = ABCBD$ or $G' = ABC - BD$, where A, B, C, D are four substrings of G , and $-B$ is the reverse of B . If C is the empty string, then it is a tandem duplication.

The problem is to find the minimal number $\mathbf{RD}(G)$ of reversals and duplications that transforms an unknown non-ambiguous genome H into G , and exhibit a possible sequence of such mutations. The key idea is to reduce the problem to a series of sub-problems involving simplified data. A *semi-ambiguous genome* G is an ambiguous genome such that each gene appears at most twice in G . A gene that has only one copy in G is called a *singleton*, otherwise it is called a *duplicated gene*.

A *repeat* is a maximal substring of G that is present twice in the genome. We denote by $D(G)$ the number of repeats of G . For example, the following genome:

$$\underbrace{+a - b + c}_{S_1} + x \underbrace{+d - e}_{S_2} \underbrace{+e - d}_{\overline{S_2}} \underbrace{+a - b + c}_{\overline{S_1}} + y$$

contains two repeats.

We consider the following evolutionary model for semi-ambiguous genomes: a semi-ambiguous genome G has an ancestor H containing exactly one copy of each gene, and G has evolved from H through a series of duplications, giving rise to an *intermediate ancestral genome* I , which is a genome containing exactly the same genes as those in G in the same number of copies, followed by a series of reversals (Fig. 11.17). The problem is to reconstruct an intermediate ancestral genome I such that $D(I) + R(G, I)$ is minimal over all possible ancestral genomes, where $D(I)$ is the number of repeats of I , and $R(G, I)$ is the reversal distance between G and I . Indeed, it is straightforward to recover, from I , a genome H giving rise to $D(I)$ duplications. Thus, the only ancestral genome which is of interest is I . In the rest of the discussion, an *ancestral genome* of G will refer to a genome containing exactly the same genes than G , in the same number of copies.

The constraint to have all duplications first and then all reversals can be seen as a restriction. However, the semi-ambiguous genome problem is just a sub-problem of the general ambiguous genome one. The general model of evolution for ambiguous genomes is then a mix of reversals and duplications: a series of duplications, followed by a series of reversals, followed by a series of duplications and so on.

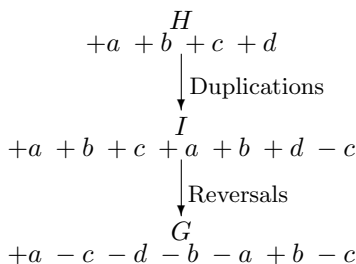


FIG. 11.17. G has evolved from a genome H through two duplications, giving rise to an ancestral genome I , followed by a series of reversals. I has two repeats: $\{+a + b, +c\}$; G has 3 repeats: $\{+a, +b, +c\}$.

11.7.2 Recovering an ancestor of a semi-ambiguous genome

We use a method that mimics in many ways the technique we have developed previously to find an ancestral duplicated genome (see Section 11.6). It is based on the HP graph for sorting signed permutations by reversals. The problem is to complete the *partial graph* representing G by an appropriate set of grey edges representing an ancestral genome I of G , so that the final complete graph minimizes $\text{RO}(G, I) = D(I) + R(G, I)$, where $R(G, I)$ is the reversal distance between G and I calculated by the HP formula (Section 11.3). As the dominant parameter in the HP formula is the number of cycles, we begin by constructing a valid set of grey edges representing an ancestral genome I such $D(I) - c(G, I)$ is minimal, where $c(G, I)$ the number of cycles of the graph.

The main difference with the genome duplication problem is that some genes can appear in single copies (gene d in genome G of Fig. 11.17), and should be considered differently. In reference [26], we have developed a linear algorithm that resembles in many aspects the one described in the previous section. The partial graph is subdivided into a set of natural subgraphs, that are completed independently. However, we can end up with more than one circular sequence. A correction is then described that transforms this set into a single circular genome, representing a possible ancestral genome I .

11.7.3 Recovering an ancestor of an ambiguous genome

What to do in the general case of a genome containing genes in more than two copies? One possibility is to try all possible pairings of duplicated genes, and choose the one that gives rise to the minimal number of reversals/duplications. Such a method is, of course, highly exponential, and does not take into account any meaningful biological information. This could be avoided if one has a preliminary information about the evolutionary relationship between all genes of a gene family, summarized by a gene tree (Fig. 11.18).

For our purpose, we need to know both the tree topology, and the approximate time of divergence events. We can then subdivide the set of internal nodes into subsets corresponding to the same historical time t .

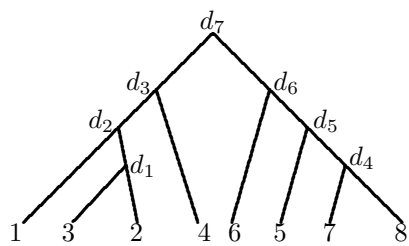


FIG. 11.18. A gene tree for a gene family of size 8. Leaves represent gene copies, and internal nodes represent gene duplication events.

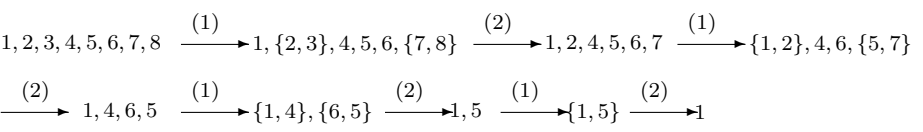


FIG. 11.19. Possible steps in processing the gene family represented by the tree of Fig. 11.18: (1) Gene pairing; (2) Algorithm *Complete-Graph*.

Let G be an ambiguous genome, and suppose we have b gene trees summarizing the results of independent phylogenetic analysis within each of the b multigene families of G . The general algorithm used to reconstruct a non-ambiguous genome from G follows a number of steps, each step subdivided into two procedures (Fig. 11.19).

1. Gene pairing: Consider the most recent divergence event in each tree, and pair the corresponding leafs.
2. Algorithm *Complete-Graph*: Apply the algorithm described in the preceding section to the semi-ambiguous genome G' obtained from (1). The resulting non-ambiguous genome H contains exactly one copy of each of the genes paired in step (1).

11.7.4 Recovering the ancestral nodes of a species tree

Given a species tree $\mathcal{T}$, the N genomes of these species that may contain multi-gene families, and the gene trees summarizing the results of independent phylogenetic analysis within each multigene family, how to reconstruct gene orders at the ancestral nodes of $\mathcal{T}$? As discussed in Section 11.5.2, a method to solve this problem has been developed in reference [66]. It integrates three approaches to genomic evolution: reconciliation, exemplar analysis, and breakpoint-based phylogeny. As shown in Fig. 11.7, the reconciliation approach gives rise to gene groupings at each internal node of the species tree, each grouping referring to a single gene whose descendants are just the copies listed in the grouping. The exemplar approach is then used to choose an exemplar gene from each set.

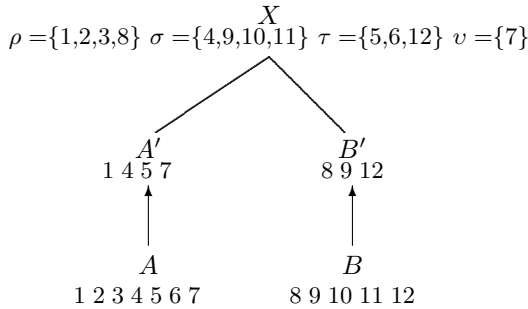


FIG. 11.20. Subtree consisting of genomes A, B , and their common immediate ancestor X . Each grouping represents a gene copy whose descendants in A and B are just the copies listed between braces. A' and B' represent the non-ambiguous ancestors of A and B .

In this context, our duplication/reversal model can be used to replace the exemplar approach. Indeed, each grouping can be seen as a gene family on its own. Then, instead of choosing an exemplar from this group, the method described above can be used to recover the ancestral genomes containing single gene copies. Figure 11.20 is an example of groupings obtained for three adjacent nodes (A, B, X) of a species tree. The genome A contains 7 copies of a gene a . As copies 1, 2, 3 are grouped in ρ , and 5, 6 are grouped in τ , the sets $\{1, 2, 3\}, \{4\}, \{5, 6\}, \{7\}$ are considered separately, and the method described in Section 11.7.3 is used to recover an ancestral genome A' of A with single gene copies. These copies can replace the “exemplars”.

The advantage of this method is that, in addition to finding an ancestral genome, it produces a possible sequence of rearrangements, which is not the case of the exemplar approach. Another advantage is that it is designed to reconstruct the evolutionary history of a single genome. As the exemplar approach is designed to compare two genomes, it should be applied to a “good” node (a leaf, or an internal node already optimized) and a “bad” node (an initial assignment, or a node that is not optimized). In contrast, our new approach is applied to only good nodes.

11.8 Conclusion

When genes are present in multiple copies in the compared genomes, analysing the complexity of genomic distances and devising exact and heuristic algorithms for them remains a challenge for computer scientists. In particular, no clear measure of genomic distance has been defined in that case. The exemplar approach of [62] consists in reducing the problem to a classical genome rearrangement one by deleting all but one member of each gene family: the copy that best reflects the original position of the ancestral gene in the common ancestor of the two genomes being compared. Another possibility for choosing the true orthologs in

two genomes would be to keep the copies that are found in the same “relative order” or in the same “clusters” in two or more genomes. Probabilistic models for determining the significance of gene clusters have been developed by Durand and Sankoff [21]. Their study take into account incomplete clusters, as well as multigene families.

We reviewed a series of methods developed by our group to infer the ancestral genome of a modern one that has evolved through local or global duplication. This work represents the use of computational biology techniques first developed for comparative genomics, as tools for the internal reconstruction of the evolutionary history of a single genome. An important future development in this field would be to consider a more complete model accounting for the specificity of the different sites in a genome, in particular the centromeric and telomeric regions that are subject to rapid genomic changes [22]. Duplications among subtelomeric regions appear to be widespread among eukaryotes, and many ambiguities in the mapping of orthologous yeast genes, which occur specifically near the telomere.

Gene families have also been considered in the phylogenetic context with specific evolutionary models involving duplication/loss or hybridization. However, phylogenetic analysis based on gene order is a difficult field, and the methods that have been developed to account for gene families are still theoretical, based on simplified models, and hardly applicable to real data.

To conclude, from a practical, as well as combinatorial point of view, finding efficient methods to assign true orthologs and account for gene duplicates in the genome rearrangement and phylogenetic context remains a current research field.

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RECONSTRUCTING PHYLOGENIES FROM GENE-CONTENT AND GENE-ORDER DATA

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Gene-order data have been used successfully to reconstruct organellar phylogenies; they offer low error rates, the potential to reach farther back in time than through DNA sequences (because genome-level events are rarer than DNA point mutations), and immunity from the so-called gene-tree versus species-tree problem (caused by the fact that the evolutionary history of specific genes is not isomorphic to that of the organism as a whole). They have also provided deep mathematical and algorithmic results dealing with permutations and shortest sequences of operations on these permutations. Recent developments include generalizations to handle insertions, duplications, and deletions, scaling to large numbers of organisms, and, to a lesser extent, to larger genomes; and the first Bayesian approach to the reconstruction problem. We survey the state-of-the-art in using such data for phylogenetic reconstruction, focusing on recent work by our group that has enabled us to handle arbitrary insertions, duplications, and deletions of genes, as well as inversions of gene subsequences. We conclude with a list of research questions (mathematical, algorithmic, and biological) that will need to be addressed in order to realize the full potential of this type of data.

12.1 Introduction: phylogenies and phylogenetic data

12.1.1 *Phylogenies*

A phylogeny is a reconstruction of the evolutionary history of a collection of organisms. It usually takes the form of a tree, where modern organisms are placed at the leaves and edges denote evolutionary relationships. In that setting, “species” correspond to edge-disjoint paths. Figure 12.1 shows three phylogenetic trees, in different display formats.

Phylogenies have been and still are inferred from all kinds of data: from geographic and ecological, through behavioural, morphological, and metabolic, to the current data of choice, namely molecular data [74]. Molecular data have the significant advantage of being exact and reproducible, at least within experimental error, not to mention fairly easy to obtain. Each nucleotide in a DNA or RNA sequence (or each codon) is, by itself, a well defined *character*, whereas morphological data (a flower, a dinosaur bone, etc.), for instance, must first

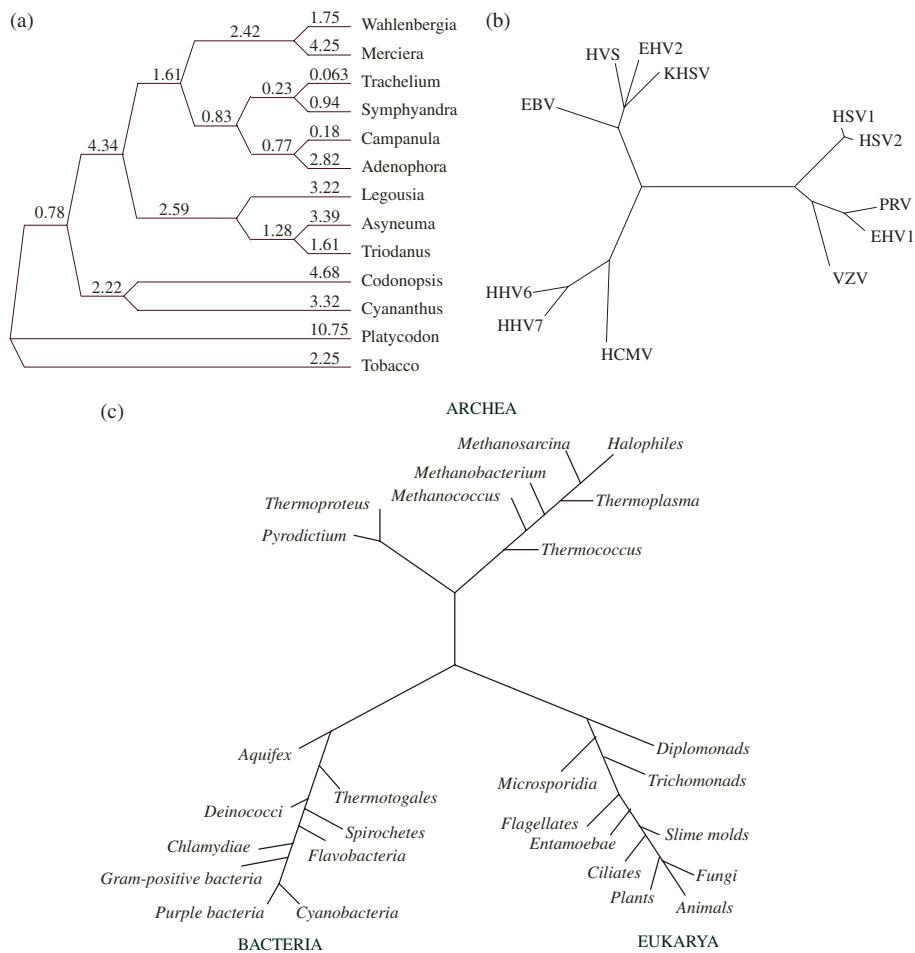


FIG. 12.1. Various phylogenetic trees, in different formats: (a) 12 plants from the Campanulaceae family [14]; (b) Herpes viruses affecting humans [43]; (c) one possible high-level view of the Tree of life.

be encoded into characters, with all the attending problems of interpretation, discretization, etc.

The predominant molecular data have been and continue to be sequence data: DNA or RNA nucleotide or codon sequences for a few genes. A promising new kind of data is gene-order data, where the sequence of genes on each chromosome is specified.

Sequence Data. In sequence data, characters are individual positions in the string and so can assume one of a few states: 4 states for nucleotides or 20 states

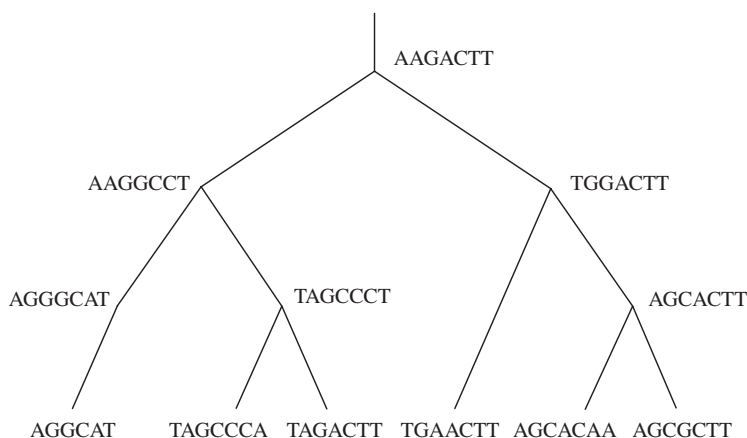


FIG. 12.2. Evolving sequences down a given tree topology.

for amino-acids. Such data evolve through *point mutations*, that is, changes in the state of a character, plus *insertions* (including *duplications*), and *deletions*. Figure 12.2 shows a simple evolutionary history, from the ancestral sequence at the root to modern sequences at the leaves, with evolutionary events occurring on each edge. Note that this history is incomplete, as it does not detail the events that have taken place along each edge of the tree. Thus, while one might reasonably conclude that, in order to reach the leftmost leaf, labelled AGGCAT, from its parent, labelled AGGGCAT, one should infer the deletion of one nucleotide (one of the three G's in the parent), a more complex scenario may in fact have unfolded. If one were to compare the leftmost leaf with the rightmost one, labelled AGCGCTT, one could account for the difference with two changes: starting with AGGCAT, insert a C between the two G's to obtain AGCGCAT, then mutate the penultimate A into a T. Yet the tree itself indicates that the change occurred in a far more complex manner: the path between these two leaves in the tree goes through the series of sequences

$$\text{AGGCAT} \leftrightarrow \text{AGGGCAT} \leftrightarrow \text{AAGGCCT} \leftrightarrow \text{AAGACTT} \leftrightarrow \text{TGGACTT} \leftrightarrow \text{AGCACTT} \leftrightarrow \text{AGCGCTT}$$

and each arrow in this series indicates at least one evolutionary event.

Preparing sequence data for phylogenetic analysis involves the following steps: (1) finding *homologous* genes (i.e. genes that have evolved from a common ancestral gene—and most likely fulfil the same function in each organism) across all organisms; (2) retrieving and then aligning the sequences for these genes (typical genes yield sequences of several hundred base pairs) across the entire set of organisms, in order to identify gaps (corresponding to insertions or deletions) and matches or mutations; and finally, (3) deciding whether to use all available data

at once for a *combined analysis* or to use each gene separately and then *reconcile* the resulting trees.

Sequence data are by far the most common form of molecular data used in phylogenetic analyses. The main reason is simply availability: large amounts of data are easily available from databases such as GenBank, along with search tools (such as BLAST) and annotations; moreover, the volume of such data grows at an exponential pace—indeed, it is outpacing the growth in computer speed (Moore’s law). A second reason is the widespread availability of analysis tools for such data: packages such as PAUP* [73], MacClade [37], Mesquite [40], Phylip [18], MEGA [32], MrBayes [28], and TNT [21], all available either freely or for a modest fee, are in widespread use and have provided biologists with satisfactory results on many datasets. Finally, the success of these packages is due in good part to the fact that sequence evolution has long been studied, both in terms of the biochemistry of nucleotides and of the biological mechanisms of change, so that accepted models of sequence evolution provide a reasonable framework within which to define computational optimization problems.

Sequence data do suffer from a number of problems. A fairly minor problem is simple experimental errors: in the process of sequencing, some base pairs are misidentified (miscalled), currently with a probability of the order of 10^{-2} . A more serious limitation is the relatively fast pace of mutation in many regions of the genome; combined with the fact that each position can assume one of only a few values, this fast pace results in *silent changes*—changes that are subsequently reversed in the course of evolution, leaving no trace in modern organisms. (Using amino-acid sequences, with 20 possible states per character, only modestly alleviates this problem.) In consequence, sequence data must be selected to fit the problem at hand: very stable regions to reconstruct very old events, highly variable regions to reconstruct very recent history, etc. This specialized nature may cause difficulties when attempting to reconstruct a phylogeny that includes both recent and ancient events, since such an attempt would require mixing variable and conserved regions in the analysis, triggering the next and most important problem. The evolution of any given gene (or region of the sequence) need not be identical to that of the organism—this is the *gene tree* versus *species tree* problem [39, 57]. Thus a combined analysis, based on the use of all available genes, risks running into internal contradictions and the loss of resolution, whereas one based on individual genes will typically yield different trees for the different genes, trees that must then be reconciled through a process known as *lineage sorting*. Sequence data also suffer from computational problems: most prominently, the problem of multiple sequence alignment is currently only poorly solved—indeed, most systematists will align sequence data by hand, or at least edit by hand the alignments proposed by the software. Less importantly, at least in a relative sense, current phylogenetic reconstruction methods used with sequence data do not scale well, whether in terms of accuracy or running time.

Gene-content and gene-order data. The data here are lists of genes in the order in which they are placed along one or more chromosomes. Nucleotide data are

not part of this picture: instead, each gene along a chromosome is identified by some name, a name shared with its homologs on other chromosomes (or, for that matter, on the same chromosome, in case of gene duplications). The entire gene order forms a *single* character, but one that can assume a huge number of states—a chromosome with n genes presents a character with $2^n \cdot n!$ states (the first term is for the strandedness of each gene and the second for the possible permutations in the ordering). A typical single circular chromosome for the chloroplast organelle of a *Guillardia* species (taken from the NCBI database) is shown in Fig. 12.3. A gene order evolves through *inversions*, sometimes also called reversals (well documented in chloroplast organelles [31, 58]), and

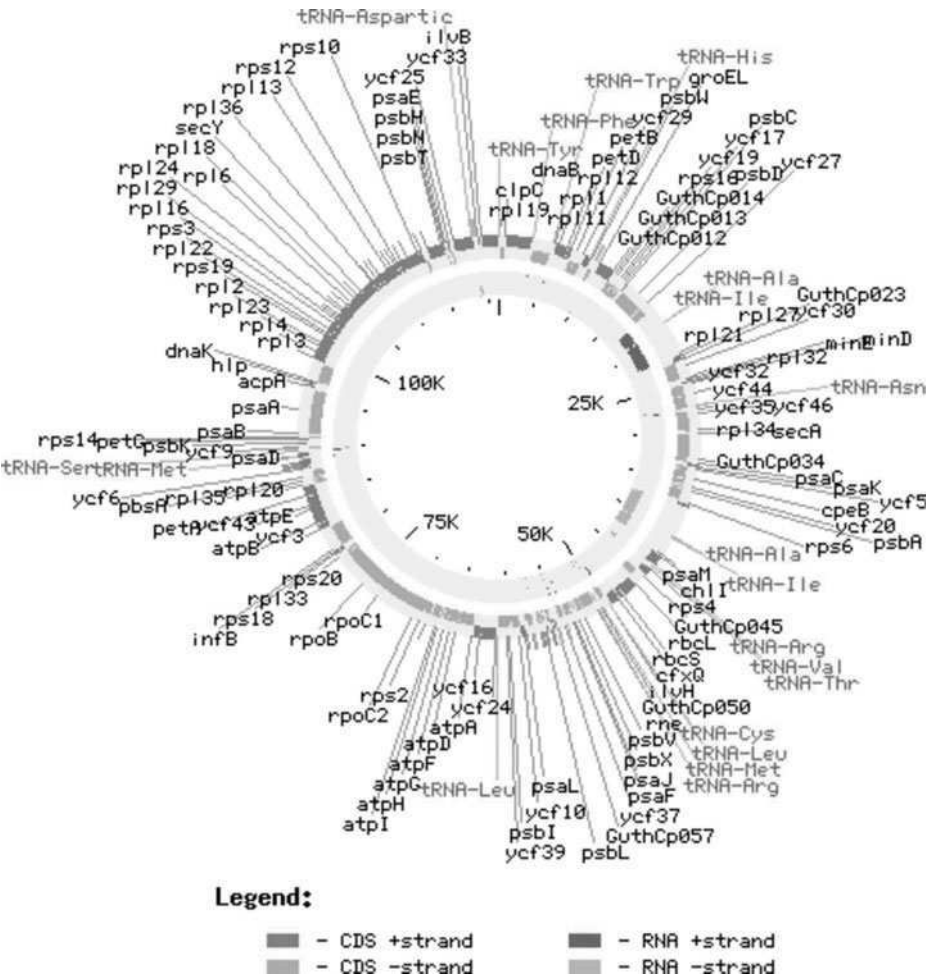


FIG. 12.3. The chloroplast chromosome of *Guillardia* (from NCBI).

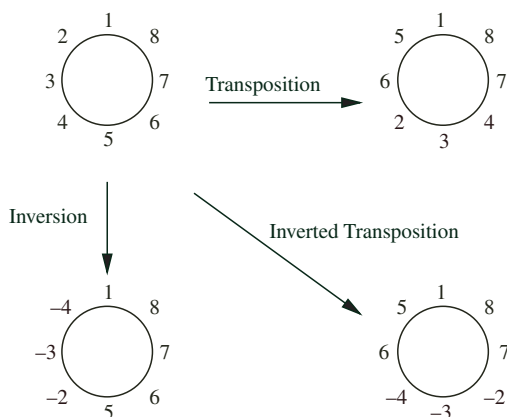


FIG. 12.4. The three rearrangement operations operating on a single circular chromosome, all operating on the gene subsequence (2, 3, 4).

perhaps also *transpositions* and *inverted transpositions* (strongly suspected in mitochondria [7, 8]); these three operations are illustrated in Fig. 12.4. (Other, more complex rearrangements may well be possible, particularly in the context of DNA repair of radiation damage.) These operations do not affect the gene content of the chromosome.

In the case of multiple chromosomes, other operations come into play. One such operation is *translocation*, which moves a piece of one chromosome into another—in effect, it is a transposition between chromosomes. Other operations that are applicable to multiple chromosome evolution include *fusion*, which merges two chromosomes into one, and *fission*, which divides a single chromosome into two. In multichromosomal organisms, colocation of genes on the same chromosome, or *synteny*, is an important evolutionary attribute and has been used in phylogenetic reconstruction [54, 67, 68]. Finally, two additional evolutionary events affect both the gene content and, indirectly, the gene order: *insertions* (including *duplications*) and *deletions* of single genes or sequences of genes.

In order to conduct a phylogenetic analysis based on gene-order data, we must identify homologous genes (including duplications) within and across the chromosomes. As the system under study is much more complex than sequence data, we may also have to refine the model to fit specific collections of organisms; for instance, bacteria often have conserved *clusters* of genes, or *operons*—genes that stay together throughout evolution, but not in any specific order—while most chloroplast organelles exhibit a characteristic partition of their chromosome into four regions, two of which are mirror images of each other (the “inverted repeat” structure). Figure 12.5 shows a typical evolutionary scenario based on inversions alone; compare with Fig. 12.2.

The use of gene-order and gene-content data in phylogenetic reconstruction is relatively recent and the subject of much current research. Such data present

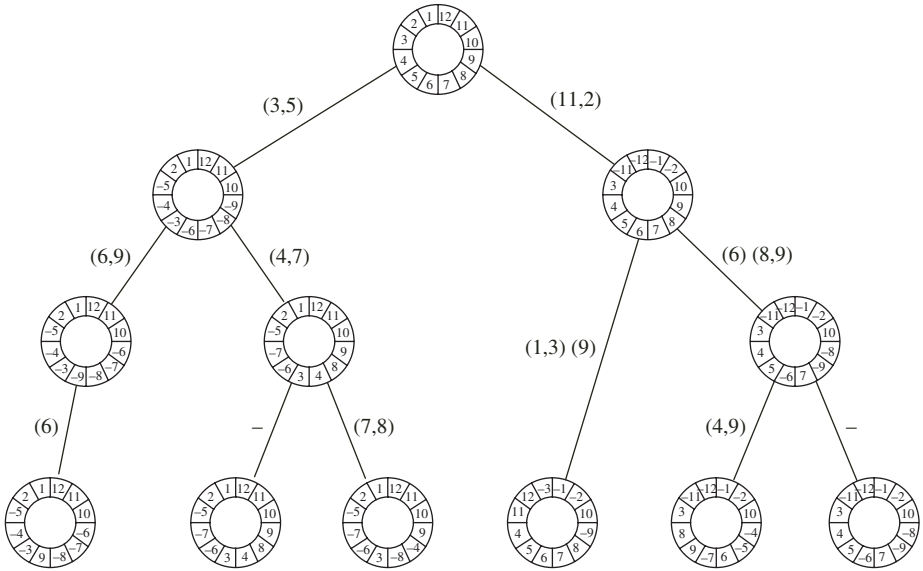


FIG. 12.5. Evolving gene orders down a given tree topology; each edge is labelled by the inversions that took place along it.

several advantages: (1) because the entire genome is studied at once, there is no gene tree versus species tree problem; (2) there is no need for alignment; and (3) gene rearrangements and duplications are much rarer events than nucleotide mutations (they are “rare genomic events” in the sense of Rokas and Holland [61]) and thus enable us to trace evolution farther back than sequence data.

On the other hand, there remain significant challenges. Foremost among them is the lack of data: mapping a full genome, while easier than sequencing the full genome, remains much more demanding than sequencing a few genes. Table 12.1 gives a rough idea of the state of affairs around 2003. The bacteria are not well sampled: for obvious reasons, most of the bacteria sequenced to date are human pathogens. The eukaryotes are the model species chosen in genome projects: human, mouse, fruit fly, worm, mustard plant, yeast, etc.; although their number

TABLE 12.1. Existing whole-genome data ca. 2003 (approximate values)

Type	Attributes	Numbers
Animal mitochondria	1 chromosome, 40 genes	500
Plant chloroplast	1 chromosome, 140 genes	100
Bacteria	1–2 chromosomes, 500–5,000 genes	150
Eukaryotes	3–30 chromosomes, 2,000–30,000 genes	10

TABLE 12.2. Main attributes of sequence and gene-order data

	Sequence	Gene-order
Evolution	Fast	Slow
Data type	A few genes	Whole genome
Data quantity	Abundant	Sparse
# Char. states	Tiny	Huge
Models	Good	Primitive
Computation	Easy	Hard

is quickly growing (with several more mammalian genomes nearing completion), coverage at this level of detail will probably never exceed a small fraction of the total number of described organisms.

This lack of data in turn gives rise to another problem: there is no good model of evolution for the gene-order data—for instance, we still do not have firm evidence for transpositions, much less any notion of relative prevalence of the various rearrangement, duplication, and loss events. This lack of a good model combines with a third problem, the extreme (at least in comparison with sequence data) mathematical complexity of gene orders, to create major computational challenges.

Sequence versus gene-order data. Table 12.2 summarizes the characteristics of sequence data and gene-order data. At present, there is every reason to expect that whole-genome data will remain limited to a small subset of the organisms for which we will have some sequence data: sequencing one gene is fast and inexpensive, whereas sequencing a complete eukaryotic genome is a major enterprise. Yet gene-order data remain worth studying: not only will the advantages discussed earlier enable us to provide valuable cross-checking for sequence-derived phylogenies (or even provide a framework around which to build a sequence-derived phylogeny), but the rapid pace of change in genomic technology may yet enable us to sequence entire genomes rapidly and at low cost.

12.1.2 *Phylogenetic reconstruction*

Methods for phylogenetic reconstruction from sequence data can be roughly classified as (1) *distance-based* methods, such as Neighbor Joining (NJ); (2) *parsimony-based* methods, such as implemented in PAUP*, Phylip, MEGA, TNT, etc.; and (3) *likelihood-based* methods, including Bayesian methods, such as implemented in PAUP*, Phylip, fastDNAmI [56], MrBayes, GAML [35], etc. In addition, *metamethods* can be used to scale up any of these three *base* methods: metamethods decompose the data in various ways and rely on one or more base methods to reconstruct trees for the subsets they produce. Metamethods include *quartet-based* methods (see [70]) and *disk-covering* methods [29, 30, 55, 62, 76]—about which we will have more to say. We will use the same categories when

discussing methods for reconstruction from gene-order data, so we give a brief characterization of each category.

Phylogenetic distances. As our discussion of the phylogeny presented in Fig. 12.2 indicates, the distance between two taxa (as represented by sequence or gene-order data) can be defined in several ways. First, we have the *true evolutionary distance*, that is, the actual number of evolutionary events (mutations, deletions, etc.) that separate one datum (gene or genome) from the other. This is the distance measure we would really want to have, but of course it cannot be inferred—as our earlier discussion made clear, we cannot infer such a distance even when we know the correct phylogeny and have correctly inferred ancestral data (at internal nodes of the tree). What we can define precisely and compute (in most cases) is the *edit distance*, the minimum number of permitted evolutionary events that can transform one datum into the other. Since the edit distance will invariably underestimate the true evolutionary distance, we can attempt to *correct* the edit distance according to an assumed model of evolution in order to produce the *expected true evolutionary distance*, or at least an approximation thereof—see Chapters 6 and 13, this volume for a discussion of distance correction.

Distance-based methods. Distance-based methods use edit distances or expected true evolutionary distances and typically proceed by grouping (as siblings) taxa (or groups of taxa) whose normalized pairwise distance is smallest. They usually run in *low polynomial time*, a significant advantage over all other methods. Most such methods only reconstruct the tree topology—they do not estimate the character states at internal nodes within the tree. The prototype in this category is the *Neighbor Joining (NJ)* method [63], later refined to produce BIONJ [20] and *Weighbor* [10]. When each entry in the distance matrix equals the true evolutionary distance (i.e. the distance along the unique path between these two taxa in the true tree), NJ is guaranteed to produce the true tree; moreover, NJ is statistically consistent—that is, it produces the true tree with probability 1 as the sequence length goes to infinity [3], under those models for which statistically consistent distance estimators exist. Chapter 1, this volume discusses distance-based methods.

Parsimony-based methods. These methods aim to minimize the total *number of character changes* (which can be weighted to reflect statistical evidence). Characters are assumed to evolve independently—so each character makes an independent contribution to the total. In order to evaluate that contribution, parsimony methods all reconstruct ancestral sequences at internal nodes. In contrast to NJ and likelihood methods, parsimony methods are not always statistically consistent. However, it can be argued that trees reconstructed under parsimony are not substantially less accurate than trees reconstructed using statistically consistent methods, given the restriction on the amount of data and the lack of fit between models and real data. Finding the most parsimonious tree is known to be NP-hard, but scoring a single fixed tree is easily accomplished in linear time; at present, provably optimal solutions are limited to datasets of

20–30 taxa, while good approximate solutions can be obtained for datasets of several hundred taxa; the latest results from our group [62] indicate that we can achieve the same quality of reconstruction on tens of thousands of taxa within reasonable time.

Likelihood-based methods. Likelihood-based methods assume some specific model of evolution and attempt to find the tree, and its associated model parameters, which together maximize the probability of the observed data. Thus a likelihood method must both *estimate model parameters* on a given fixed tree and also search through tree space to find the best tree. Chapter 2, this volume, discusses likelihood methods.

Likelihood-based methods are usually (but, perhaps surprisingly, not always) statistically consistent, although, of course, that consistency is meaningless if the chosen model does not match the biological reality. Likelihood methods are the slowest of the three categories and also prone to numerical problems, because the likelihood of typical trees is extremely small—with just 20 taxa, the average likelihood in the order of 10^{-21} , going down to 10^{-75} with 50 taxa. Identifying the tree of maximum likelihood (ML) is presumably NP-hard, although no proof has yet been devised; indeed, even computing the likelihood a fixed tree under a fixed model cannot currently be done in polynomial time [71]. Thus optimal solutions are limited to trees with fewer than 10 taxa, while good approximations are possible for perhaps 100 taxa.

Bayesian methods deserve a special mention among likelihood-based approaches; they compute the posterior probability that the observed data would have been produced by various trees (in contrast to a true maximum likelihood method, which computes the probability that a fixed tree would produce various kinds of data at its leaves). Their implementation with Markov Chain Monte-Carlo (MCMC) algorithms often run significantly faster than pure ML methods; moreover, the moves through state space can be designed to enhance convergence rates and speed up the execution. Chapter 3, this volume, discusses Bayesian approaches.

12.2 Computing with gene-order data

As indicated earlier, gene-order data present significant mathematical challenges not encountered when dealing with sequence data. Many evolutionary events may affect the gene order and gene content of a genome; and each of these events creates its own challenges, not least of which is the computation of a pair-wise genomic distance. Armed with algorithms for computing distances, we can proceed to phylogenetic reconstruction, starting with scoring a single tree in terms of its total evolutionary distance.

12.2.1 Genomic distances

We begin with distances between genomes with equal gene content: in this case, the only operations allowed are rearrangements.

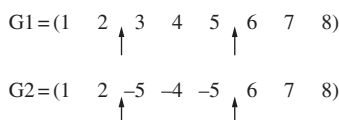


FIG. 12.6. Breakpoints.

Breakpoint distance. A *breakpoint* is an adjacency present in one genome, but not in the other. Figure 12.6 shows two breakpoints between two genomes—note that the gene subsequence 3 4 5 is identical to $-5 -4 -3$, since the latter is just the former read on the complementary strand. The *breakpoint distance* is then the number of breakpoints present; this measure is easily computed in linear time, but it does not directly reflect rearrangement events—only their final outcome. In particular, it typically underestimates the true evolutionary distance even more than an edit distance does.

Inversion distance. Given two signed gene orders of equal content, the *inversion distance* is simply the edit distance when inversion is the only operation allowed. Even though we have to consider only one type of rearrangement, this distance is very difficult to compute. For *unsigned* permutations, in fact, the problem is NP-hard. For signed permutations, it can be computed in linear time [4], using the deep theoretical results of Hannenhalli and Pevzner [23].

The algorithm is based on the *breakpoint graph*. Refer to Fig. 12.7 for an illustration. We assume without loss of generality that one permutation is the identity. We represent gene i by two vertices, $2i - 1$ and $2i$, connected by an edge; think of that edge as oriented from $2i - 1$ to $2i$ when gene i appears with positive sign, but oriented in the reverse direction when gene i appears with negative sign. Now we connect these edges with two further sets of edges, one for each genome—one represents the identity (i.e. it simply connects vertex j to vertex $j + 1$, for all j) and is shown with dashed arcs in Fig. 12.7, and the other represents the other genome and is shown with solid edges in the figure. The crucial concept is that of alternating cycles in this graph, that is, cycles of even length in which every odd edge is a dashed edge and every even one is a solid edge. Overlapping cycles in certain configurations create structures known as *hurdles* and a very unique configuration of such hurdles is known as a *fortress*.

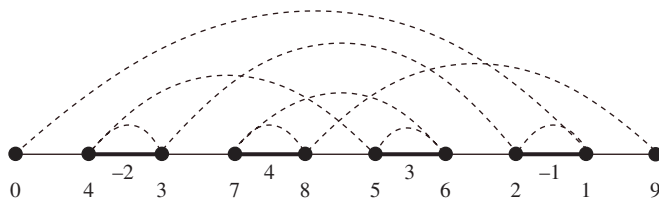


FIG. 12.7. The breakpoint graph for the signed permutations of Fig. 12.6.

Hannenhalli and Pevzner proved that the inversion distance between two signed permutations of n genes is given by

$$n - \# \text{cycles} + \# \text{hurdles} + (\text{fortress})$$

In Chapter 10, this volume, Bergeron *et al.* offer an alternate formulation of this result, within a framework based on certain nested intervals.

Generalized gene-order distance. The restriction that no gene be duplicated and that all genomes contain exactly the same set of genes is clearly unrealistic, even in the case of organellar genomes. However, accounting for additional evolutionary events such as duplications, insertions, and deletions is proving very difficult. One extension has been present since the beginning: in the second of their two seminal papers [24], Hannenhalli and Pevzner showed that their framework (cycles, hurdles, etc.) could account for both insertions and multichromosomal events, namely translocations, fusions, and fissions. Bourque and Pevzner [9] designed a heuristic approach to phylogenetic reconstruction for multichromosomal organisms under inversions, translocations, and fissions and fusions, based upon the work of Tesler [78]; they used the GRAPPA (Genome Rearrangement Analysis under Parsimony and other Phylogenetic Algorithms) implementation [53] of the linear-time algorithm [4] for inversion and confirmed the findings of Moret *et al.* [48] that inversion-based reconstruction of ancestral genomes outperforms breakpoint-based reconstruction of same.

More recently, El-Mabrouk [17] showed how to compute a minimum edit sequence in polynomial time when both inversions and deletions are allowed; Liu *et al.* [36] then showed that the distance itself can be computed in linear time. Because edit sequences are symmetric, these results also apply to combinations of inversions and *non-duplicating* insertions. In the same paper, El-Mabrouk showed that her method could provide a bounded approximation to the edit distance in the presence of both deletions and (non-duplicating) insertions. Sankoff [64] had earlier proposed a heuristic approach to the problem of duplications, suggesting that a single copy—the *exemplar*—be kept, namely that copy whose use minimized the number of other operations. Unfortunately, finding the exemplar, even for a single gene, is an NP-hard problem [11]. Marron *et al.* [41] gave the first bounded approximation algorithm for computing an edit sequence (or distance) in the presence of inversions, duplications, insertions, and deletions; a similar approach was used by Tang *et al.* [77] in the context of phylogenetic reconstruction. Most recently, Swenson *et al.* [72] gave an extension of the algorithm of Marron *et al.*, one that closely approximates the true evolutionary distance between two arbitrary genomes under any combinations of inversions, insertions, duplications, and deletions; they also showed that this distance measure is sufficiently accurate to enable accurate phylogenetic reconstruction by simply using Neighbor Joining on the distance matrix.

Work on transposition distances has been limited to equal-content genomes with no duplications and, even then, only to approximations, all with guaranteed ratio 1.5. The first approximation is due to Bafna and Pevzner [5], using much the

same framework defined for the study of inversions; the approach was recently simplified, then extended to include inverted transpositions by Hartman [25, 26]. Work on transposition distance is clearly lagging behind work on inversion distance and remains to be integrated with it and extended to genomes with unequal content.

In a different vein, Bergeron and Stoye [6] defined a distance estimate based on the number and lengths of conserved gene clusters; this distance is well suited to prokaryotic genomes (where gene clusters and operons are common), but it still requires that duplicate genes be removed.

Estimating true pairwise evolutionary distances. We give a brief overview of the results of Swenson *et al.* [72]. In earlier work [41], the same group had shown that any shortest edit sequence could always be rewritten to that all insertions and duplications take place first, followed by all inversions, followed by all deletions. In order to estimate pairwise evolutionary distances between arbitrary genomes, it remains to handle duplications; this is done gene by gene by computing a mapping from the genome with the smaller number of copies of that gene to that with the larger number of copies, using simple heuristics. Deletions and inversions are computed quite accurately, using extensions to the work of El-Mabrouk [17], while insertions (which now include any “excess” duplicates not matched in the first phase) are computed by retracing the sequence of inversions and deletions. The result is a systematic overestimate of the edit distance, but a very accurate estimate of the true evolutionary distance. Figure 12.8 presents some results from simulations in which evolutionary events were selected through a mix of 70% inversions, 16% deletions, 7% insertions, and 7% duplications with inversions having a mean length of 20 and a standard deviation of 10, and deletions, insertions, and duplications having a mean length of 10 with a standard deviation of 5. The top two examples come from datasets of 16 taxa with 800 genes, with expected pairwise distances of 20 through 160 events (left) and 40 through 320 events (right); the bottom example comes from a dataset of 57 taxa with 1,200 genes and expected pairwise distances from 20 to 280 events. The distance computation, which has a randomized component (to break ties in the assignment of duplicate genes), was run 10 times with different seeds. The figure indicates clearly that the distance estimate is highly accurate up to saturation, which occurs only at very large distances (around 250 events for a genome of 800 genes).

12.2.2 *Evolutionary models and distance corrections*

In order to use gene-order and gene-content data, we need a reasonable model of evolution for the gene order of a chromosome—and here we lack sufficient data for the construction of strong models. To date, biologists have strong evidence for the occurrence of inversions in chloroplasts—and have at least two possible models for the creation of inversions (one through DNA breakage and misrepair, the other through loops traversed in the wrong order during replication). Since DNA breakage is relatively common and particularly pronounced as a result

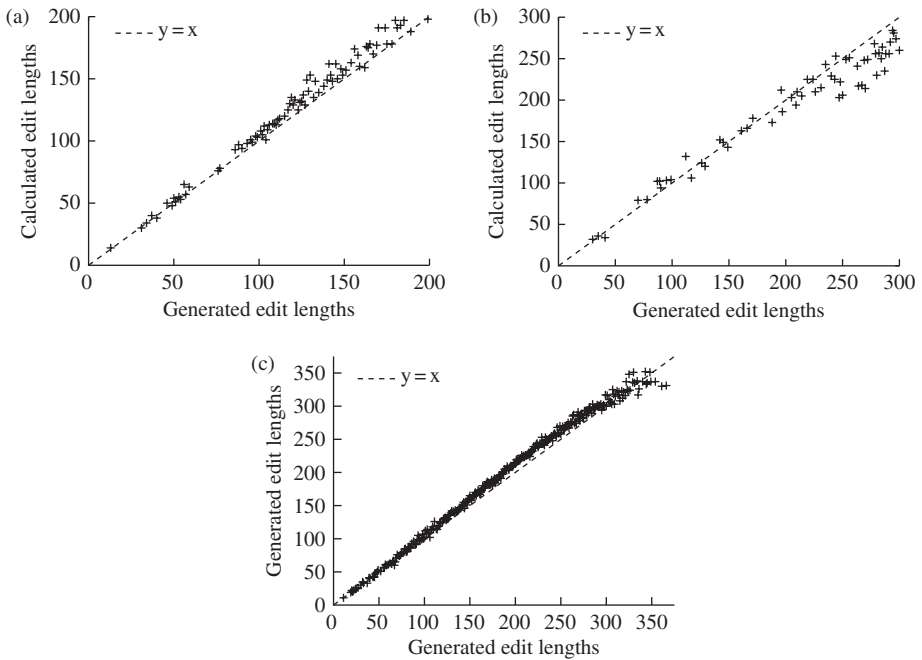


FIG. 12.8. Generated pairwise edit length versus reconstructed length for three simulated datasets; an exact estimate follows the indicated line $y = x$. (a) 16 taxa, 800 genes, 160 max. exp. dist. (b) 16 taxa, 800 genes, 320 max. exp. dist. (c) 57 taxa, 1,200 genes, 240 max. exp. dist.

of radiation damage, other rearrangements due to misrepair appear at least possible. Sankoff [65] has given statistical evidence for a distinction between short and long inversions: short inversions tend to preserve clusters (and thus could be common in prokaryotes), whereas long inversions tend to preserve runs of genes (and thus could be more common in eukaryotes); in a subsequent study of prokaryotic data [34], an ad hoc computational investigation gave additional evidence that short inversions play a significant role in prokaryotic organisms. However, even if we limit ourselves to (short and long) inversions, the respective probabilities of these two events remain unknown.

While we do not yet have a strong model of genome evolution through rearrangements, we do know that edit distances must underestimate true evolutionary distances, especially as the distances grow large. As is discussed in detail in Chapter 13, this volume, it is possible to devise effective schemes to convert the edit distance into an estimate, however rough, of the true evolutionary distance. Figure 12.9 illustrates the most successful of these attempts: working from a scenario of uniformly distributed inversions, Moret *et al.* [49] collected data on the inversion distance versus the number of inversions actually used

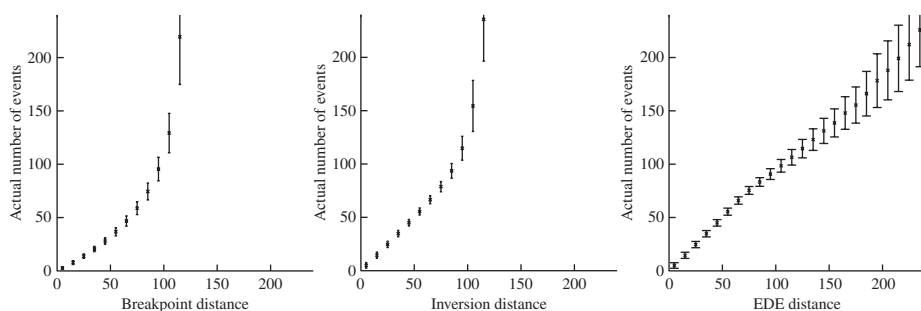


FIG. 12.9. Edit distances versus true evolutionary distances and the EDE correction.

in generating the permutations (the middle plot), then produced a formula to correct the underestimate, with the result, the EDE distance, shown in the third plot. (The first plot shows that the breakpoint distance is even more subject to underestimation than the inversion distance.) The use of EDE distances in lieu of inversion distances leads to more accurate phylogenetic reconstructions with both distance methods and parsimony methods [49, 50, 79, 80].

12.2.3 Reconstructing ancestral genomes

Reconstructing ancestral genomes is an integral part of both parsimony- and Bayesian-based reconstruction methods and may also have independent interest. In a parsimony context, we want to reconstruct a signed gene order at each internal node in the tree so as to minimize the sum of genomic distances over all edges of the tree. Unfortunately, this optimization problem is NP-hard even for just three leaves and for the simplest of settings—equal gene content, no duplication, and breakpoint distance [59] or inversion distance [12]. Computing such a gene order for three leaves is the *median problem* for signed genomes: given three genomes, produce a new genome that will minimize the sum of the distances from it to the other three. In the case of breakpoint distances, Sankoff and Blanchette [66] showed how to convert this problem to the *Travelling Salesperson Problem*; Figure 12.10 illustrates the process. Each gene gives rise to a pair of cities connected by an edge that must be included in any solution; the distance between any two cities not forming such pairs is simply the number of genomes in which the corresponding pair of genes is not consecutive (and thus varies from 0 to 3, a limited range that was put to good use in the fast GRAPPA implementation [53]).

No equivalently simple formulation in terms of a standard optimization problem is known for more general genomic distances. Yet even the simple inversion distance gives rise to significantly better results than the breakpoint distance, in terms of computational demands and topological accuracy [48, 49, 51, 76] as well as of the accuracy of reconstructed ancestral genomes [9, 48]. For inversion distances, exact algorithms have been proposed [13, 69] that work well for

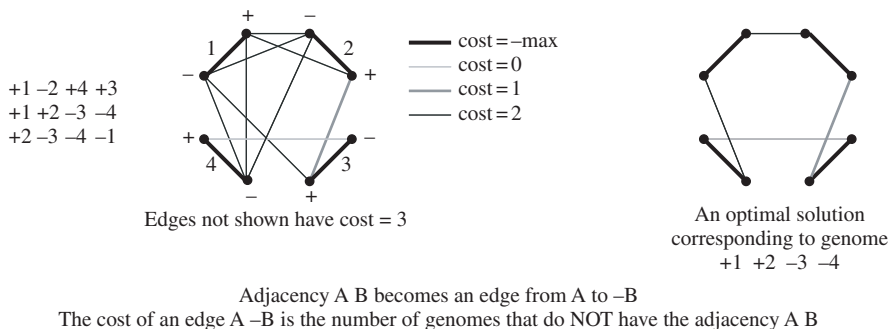


FIG. 12.10. Reducing the breakpoint median to a TSP instance.

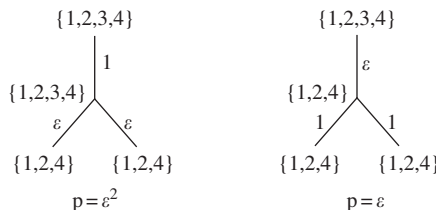


FIG. 12.11. Determining the gene content of the median.

small distances (of fewer than 15 inversions). Tang and Moret [75] showed that the median problem under inversions, deletions, and insertions or duplications could be solved exactly for small numbers of deletions and duplications, using a few simple assumptions; they recently extended that work for somewhat larger changes in gene content [77]. Their approach first determines the gene content of the median, then computes an ordering through those genes via an optimization procedure. The basic assumptions are that (1) no change is reversed and (2) changes are independent and of low probability. These two assumptions, common in phylogenetic work (e.g. see [38, 42]), imply that simultaneous identical changes on two edges are vanishingly unlikely compared to the reverse change on the third edge—since the simultaneous changes have a probability on the order of ε^2 , for a small ε , compared to a probability of ε for a change on a single edge, as illustrated in Fig. 12.11. The results obtained by Tang and Moret on a small, but difficult dataset of just seven chloroplast genomes from red and green algae and land plants are shown in Fig. 12.12. Part (a) shows the reference phylogeny obtained through combined likelihood and maximum parsimony (MP) analyses of the codon sequences of several cpDNA genes; it should be noted that the placement of *Mesostigma* is unclear from the data. Part (b) shows the phylogeny obtained by Tang and Moret, which is consistent with the reference phylogeny. Part (c) shows the phylogeny obtained by using the simple Neighbor Joining method on the distance matrix computed from

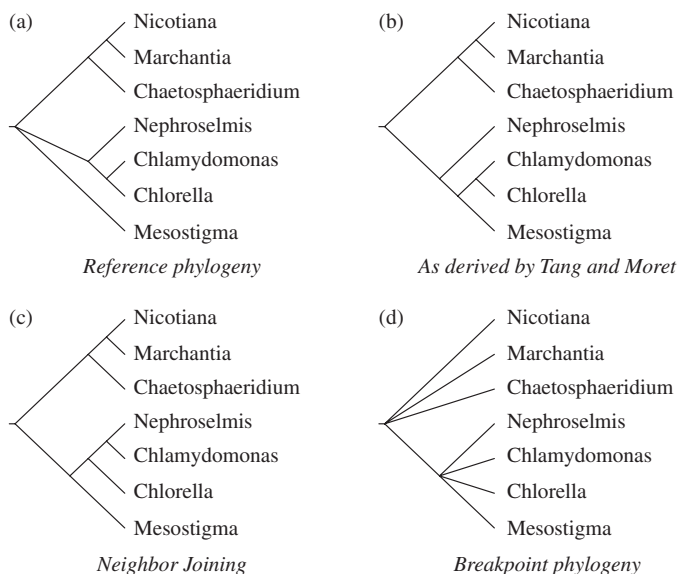


FIG. 12.12. Phylogenies on the seven taxon cpDNA dataset [77].

the seven genomes with equalized gene content: the method produced a false positive. Finally, part (d) shows the tree built by using breakpoint distances on equalized gene contents: note that the tree is nearly a star, with just one resolved edge.

In the presence of very large differences in gene content and of many duplicates, the problem is much harder. For one thing, given three genomes with these characteristics, the number of possible optimum medians is very large—indicating that a biologically sound reconstruction will require external constraints to select from these many choices. Knowing the direction of time flow (as is the case after the tree has been rooted) simplifies the problem somewhat—at least it makes the question of gene content much simpler to resolve [16], but it is fair to say that, at present, we simply lack the tools to reconstruct ancestral data for complex nuclear genomes.

In a completely different vein, El-Mabrouk (see Chapter 11, this volume) has shown how to reconstruct ancestral genomes in the presence of a single duplication event, one, however, that duplicated the entire genome just once.

12.3 Reconstruction from gene-order data

Phylogenetic reconstruction methods from gene-order data fall within the same general categories as methods for sequence data, to wit: (1) distance-based methods, (2) parsimony-based methods, and (3) likelihood-based methods, all with the possibility of using a metamethod on top of the base method.

In Chapter 13, this volume, Wang and Warnow give a detailed discussion of distance-based methods. Likelihood methods are represented to date by a single effort, from Larget *et al.* [33], in which a Bayesian approach showed evidence of success on a couple of fairly easy datasets; the same approach, however, failed to converge on a harder dataset analysed by Tang *et al.* [77]. We thus focus here on approaches based on parsimony, which have seen more development. These approaches fall into two subcategories: encoding methods, which reduce the gene-order problems to sequence problems, and direct methods, which run optimization algorithms directly on the gene-order data.

12.3.1 *Encoding gene-order data into sequences*

As we shall see in Section 12.3.2, direct optimization approaches have running times that are exponential in both the number of genomes and the number of genes, so that analyses of even small datasets (containing only 10 or 20 genomes) may remain computationally intractable. Therefore an approach that, while remaining exponential in the number of genomes, takes time polynomial in the number of genes, may be of significant interest. Since sequence-based methods have such characteristics, a simple idea is to reduce the gene-order data to sequence data through some type of encoding. Our group developed two such methods.

The first method, *Maximum Parsimony on Binary Encodings (MPBE)* [14, 15], produces one character for each gene adjacency present in the data—that is, if genes i and j occur as the adjacent pair ij (or $-j-i$) in one of the genomes, then we set up a binary character to indicate the presence or absence of this adjacency (coded 1 for presence and 0 for absence). The position of a character within the sequence is arbitrary, as long as it is the same for all genomes. By definition, there are at most $2n^2$ characters, so that the sequences are of lengths polynomial in the number of genes. Thus, analyses using maximum parsimony will run in time polynomial in the number of genes, but may require time exponential in the number of genomes. However, while a parsimony analysis relies on independence among characters, the characters produced by MPBE are emphatically dependent; moreover, translating the evolutionary model of gene orders into a matching model of sequence evolution for the encodings is quite difficult. This method suffers from several problems: (1) the ancestral sequences produced by the reconstruction method may not be valid encodings; (2) none of the ancestral sequences can describe adjacencies not already present in the input data, thus limiting the possible rearrangements; and (3) genomes must have equal gene content with no duplication.

The second method is the MPME method [79], where the second “M” stands for *Multistate*. In this method, we have exactly one character for each signed gene (thus $2n$ characters in all) and the state of a character is the signed gene that follows it in the gene ordering (in the direction indicated by the sign), so that each character can assume one of $2n$ possible states. Again, the position of each character within the sequence is arbitrary as long as it is consistent across all genomes, although it is most convenient to think of the i th character

(with $i \leq n$) as associated with gene i , with the $n + i$ th character associated with gene $-i$. For instance, the circular gene order $(1, -4, -3, -2)$ gives rise to the encoding $(-4, 3, 4, -1, 2, 1, -2, -3)$. Our results indicate that the MPME method dominates the MPBE method (among other things, the MPME method is able to create ancestral encodings that represent adjacencies not present in the input data). However, it still suffers from some of the same problems, as it also requires equal gene content with no duplication and it too can create invalid encodings. In addition it introduces a new problem of its own: the large number of character states quickly exceeds the computational limits of popular MP software. In any case, both MPBE and MPME methods are easily surpassed by direct optimization approaches.

12.3.2 *Direct optimization*

Sankoff and Blanchette [66] proposed to reconstruct the *breakpoint phylogeny*, that is, the tree and ancestral gene orders that together minimize the total number of breakpoints along all edges of the tree. Since this problem includes the breakpoint median as a special case, it is NP-hard even for a fixed tree. Thus they proposed a heuristic, based on iterative improvement, for scoring a fixed tree and simply decided to examine all possible trees; the resulting procedure, **BPAnalysis**, is summarized in Fig. 12.13. Sankoff and Blanchette used this method to analyse a small mitochondrial dataset. This method is expensive at every level: first, its innermost loop repeatedly solves the breakpoint median problem, an NP-hard problem; second, the labelling procedure runs until no improvement is possible, thus using a potentially large number of iterations; and finally, the labelling procedure is used on every possible tree topology, of which there is an exponential number. The number of unrooted, unordered trees on n labelled leaves is $(2n - 5)!!$, where the double factorial denotes the fact that only every other factor is used—that is, we have $(2n - 5)!! = (2n - 5) \cdot (2n - 7) \cdot (2n - 9) \cdot \dots \cdot 5 \cdot 3$. For just 13 genomes, we obtain 13.5 billion trees; for 20 genomes, there are so many trees that merely counting to that value would take thousands of years on the fastest supercomputer.

Realizing this problem (we estimated that running **BPAnalysis** on an easy set of 13 chloroplast genomes would take several centuries), we reimplemented the

```

For each possible tree do
  Initially label all internal nodes with gene orders
  Repeat
    For each internal node  $v$ , with neighbours labelled  $A$ ,  $B$ , and  $C$ , do
      Solve the median problem on  $A$ ,  $B$ , and  $C$  to yield label  $M$ 
      If relabelling  $v$  with  $M$  improves the score of  $T$ , then do it
  until no internal node can be relabelled

```

FIG. 12.13. **BPAnalysis**.

TABLE 12.3. Speedups for various algorithm engineering techniques

Technique used	Speedup obtained
Improving tree lower bound	500×
Reducing memory usage	10×
Better median solver	10×
Hand-tuning code	5×
“Layering” approach	5×
Improving median lower bound	2×

strategy of Blanchette and Sankoff, but made extensive use of *algorithmic engineering* techniques [46] to speed it up—most notably in the use of lower bounds to avoid scoring most of the trees—and added the use of inversion distances in order to produce *inversion phylogenies*. The various techniques we used are listed in Table 12.3. In the case of the 13-taxon dataset, for instance, our bounding and ordering strategies eliminate all but 10,000 of the 13.5 billion trees. The tree lower bound is based on the triangle inequality that must be obeyed by any metric: in any ordering of the leaves of the tree, half of the sum of the pairwise distances between consecutive leaves must be a lower bound on the total length of the tree edges in the optimal tree. We take advantage of the unordered nature of the trees to compute the largest possible lower bound through swaps of the two children whenever such a swap leads to a larger value. The layering approach precomputes lower bounds for *all* trees and stores the trees in buckets according to increasing values of the lower bound; it then goes through the trees bucket by bucket, starting with those with the smallest lower bound taking advantage of (1) the high correlation between lower bound and final score and (2) the low cost of bounding compared to the high cost of scoring. Reducing memory usage is accomplished by predeclaring all necessary space and re-using much of it on the fly; and hand-tuning code includes hand-unrolling loops, precomputing common expressions, choosing branch order, and, in general, carefully optimizing any inner loop that profiles too high.

The resulting code, GRAPPA [53], with our best bounding and ordering schemes, can analyse the same 13-taxon dataset in 20 minutes on a laptop [49]—a speedup by a factor of about 2 million. Moreover, this speedup can easily be increased by the use of a large cluster computer, since GRAPPA is fully parallel and gets a nearly perfect speedup; in particular, running the code on a 512-processor machine yielded a 1-billion-fold speedup.

However, a speedup by any constant factor, even a factor as large as a billion, can only add a constant to the size of datasets that can be analysed with this method: every added taxon multiplies the total number of trees, and thus the running time, by twice the number of taxa. For instance, whereas GRAPPA can solve a 13-taxon dataset in 20 min, it would need over 2 million years to solve a 20-taxon dataset! In effect, the direct optimization method is, for now, limited

to datasets of about 15 taxa; to put it differently: in order to scale direct optimization to larger datasets, we need to decompose those larger datasets into chunks of at most 14 taxa each.

12.3.3 *Direct optimization with a metamethod: DCM-GRAPPA*

Tang and Moret [76] succeeded in scaling up GRAPPA from its limit of around 15 taxa to over 1,000 taxa with no loss of accuracy and at a minimal cost in running time (on the order of 1–2 days). They did so by adapting a metamethod, the Disk-covering method (DCM), to the problem at hand, producing DCM-GRAPPA.

Disk-covering methods are a family of divide-and-conquer methods devised by Warnow and her colleagues. All DCMs are based on the idea of decomposing the set of taxa into overlapping “tight” subsets, using a base reconstruction method on the subsets to obtain trees, then combining the trees thus obtained to produce a tree for the entire dataset. There are three DCM variants to date, differing in their method of decomposition and their measure of tightness for subsets. The first DCM published, DCM-1 [29], is based on a distance matrix. It creates a graph in which each vertex is a taxon and two taxa are connected by an edge if their pairwise distance falls below some predetermined threshold; this graph is then triangulated and its maximum cliques computed (the former is done heuristically, the second exactly, both in polynomial time) to yield the desired subsets. Thus this method produces overlapping subsets in which no pair of taxa is farther apart than the threshold. The second DCM method, DCM-2 [30], also creates a threshold graph, but then computes a graph separator for it and produces subsets, each of which is the union of the separator and one of the isolated subgraphs. Finally, the third DCM method, DCM-3 [62], uses a *guide tree* to determine the decomposition and is best used in an iterative setting, with the tree produced at each iteration serving as guide tree for the next iteration. When used with sequence data, all three DCM variants use tree refinement methods to reduce the number of polytomies in the trees returned for each subset and for the entire dataset. When used for maximum parsimony analysis on sequences with the TNT package as its base method, the recursive and iterative version of DCM3 can easily analyse biological datasets of over 10,000 taxa, producing trees with parsimony scores within 0.01% of optimal in less than a day of computation [62].

Tang and Moret [76] used DCM-1 to produce DCM-GRAPPA. Because gene-order data produces very few polytomies, they did not need any tree refinement phase. However, because the size of the subsets cannot be constrained beforehand, they needed to use the DCM recursively in order to keep decomposing subsets until no subset held more than 14 taxa; a recursive decomposition is a natural enough idea, but poses difficult questions, such as the relationship between the size threshold used at one level of the recursion and that used at the level below. On simulated data (there are no biological gene-order datasets of such sizes), they found that DCM-GRAPPA scaled gracefully to well over 1,000 taxa (in 2 days of computation) and retained the high accuracy of the base method, GRAPPA—with fewer than 3% of the edges in error.

12.3.4 *Handling unequal gene content in reconstruction*

The method used by Tang and Moret [75] for computing the median of three known genomes in the presence of unequal gene content is not directly applicable to phylogenetic reconstruction in the style of **GRAPPA**, because the latter cannot rely on known gene orders for the three neighbours—certainly not initially, when internal nodes must be assigned gene orders in some rough manner, and not during the process, when every internal gene order is subject to replacement by a new median. To overcome this problem, Tang *et al.* [77] begin by computing the gene *content* of each internal node and then only proceed to assign and iterate over gene orders. Gene contents are assigned starting from the leaves (with known gene contents), using the principle illustrated in Fig. 12.11: if two sibling leaves both contain gene X , then so does their parent, while, if neither leaf contains X , then neither does their parent. When one leaf contains gene X and the other does not, gene X is noted as *ambiguous* for the parent; such ambiguities are resolved through propagation of constraints and iterative improvement, much in the style of the basic optimization heuristic of **GRAPPA**. This approach to the handling of unequal gene orders and duplications can be incorporated within **DCM-GRAPPA**, yielding a method for the analysis of large datasets with arbitrary gene content.

12.4 Experimentation in phylogeny

Before we conclude our survey, we should say a few words about experimentation with phylogenetic reconstruction algorithms. While computer scientists have long evaluated algorithms in terms of their asymptotic running time and performance guarantees, it is only in the last 10 years that more formal approaches to the experimental assessment of algorithms have emerged, under the collective name of *experimental algorithmics*. Experimental algorithmics (see [19, 45, 47] and the *Journal of Experimental Algorithmics* at www.jea.acm.org) is an emerging discipline that deals with how to test algorithms empirically to obtain reliable characterizations of their performance as well as deepen our understanding of their properties in order to refine them. Because it is based on experimental data, experimental algorithmics can seek inspiration from the physical sciences, but it must adapt to the specific goal—not to understand one phenomenon, but to generalize findings to an infinite range of possible instances.

In phylogenetic reconstruction, an assessment must take into account the accuracy of the reconstruction (in terms of the chosen optimization criterion but also, and more importantly, in terms of the biological significance of the results) as well as the scaling up of resource consumption (time and space). In turn, conducting such an assessment requires the use of a carefully designed set of benchmark datasets [52].

12.4.1 *How to test?*

First, how do we choose test sets? *Biological datasets* test performance where it matters, but they can be used only for ranking, are too few to permit quantitative

evaluations, and are often hard to obtain. Moreover, the analysis of any large biological dataset will be hard to evaluate: one cannot just walk up to one's colleague in systematics with a 10,000-taxon tree in hand and ask her whether the tree is biologically plausible! Thus biological datasets are good for anecdotal reports and for "reality checks." In the latter capacity, of course, they are indispensable: no simulation can be accurate enough to replace real data. *Simulated datasets* enable absolute evaluations of solution quality (because the model, and thus the "true" answer, is known) and can be generated in arbitrarily large numbers to ensure statistical significance. Thus a combination of large-scale simulations and reasonable numbers of biological datasets is the only way to obtain valid characterizations of algorithms for phylogenetic reconstruction. The simulations must be based on the best possible models of the application at hand—in our case, we need accurate models of speciation and extinction, of gene duplication, gain, and loss, and of genome rearrangements.

12.4.2 *Phylogenetic considerations*

A typical simulation study runs as follows:

- (1) generate a rooted binary tree (according to a chosen model of speciation and extinction) with the appropriate number of leaves—this is known as the *model tree*;
- (2) assign a "length" (i.e. number of evolutionary events) to each edge of the tree according to a chosen model of divergence;
- (3) place a genome of suitable size and composition at the root;
- (4) evolve the genomes down the tree, that is, transform the parent genome along each edge to its children according to the number of evolutionary events on that edge and to the chosen model of genome evolution;
- (5) collect the genomes thus generated at the leaves and use them as input to the reconstruction algorithm under test; and
- (6) compare the topology (and, if desired, the internal genomes) of the reconstructed tree with that of the model tree.

This sequence of operations is run many times for the same parameter values (number of taxa, size of genomes, parameters of the model of genome evolution, distribution of edge lengths, etc.) to ensure statistical significance. Naturally, a range of parameters is also explored. Thus the computational requirements are significant—keeping in mind that even a single reconstruction can prove quite expensive in terms of running time.

In the many years of experimental work we have conducted, we have found a number of useful guidelines, summarized below:

- Tree shape plays a surprisingly large role. Thus we need a reasonable model of speciation (and extinction), one that certainly goes beyond the simplistic models of uniform distributions or birth–death processes. Of course, the shape of the true trees is unknown and, in any case, depends on the selection of genomes (tight clades will show very different shapes from that of the

entire Tree of Life, for instance), so that good simulations will need to use a selection of parameters.

- The evolutionary models for divergence and genome evolution are important. In particular, most reconstruction methods exhibit poor accuracy when the *diameter* of the dataset (the ratio of the largest to the smallest pairwise distance in the dataset) is large. Methods aimed at minimizing inversion distances may not perform as well on datasets where the predominant events are transpositions. Large numbers of duplications or very large gene losses also confuse most reconstruction methods. Thus the challenge is to devise an evolutionary model with few parameters that is easily manipulated analytically and computationally and produces realistic data.
- Testing a large range of parameters and using many runs for each setting to estimate variance are essential parts of any testing strategy. In the huge parameter space induced by even the simplest of models, it is all too easy to fall within an uncharacteristic region and draw entirely wrong conclusions about the behaviour of the algorithm. Of course, the size of the parameter space makes it difficult to sample well.

That tree shape plays such a role was an unexpected finding. Most studies to date have used either a uniform model (popular in computer science) or a birth–death model (so-called Yule trees, popular in biology). Several authors [1, 2, 22, 27, 44] noted that published phylogenies exhibit a shape distribution that deviates from either model: in terms of balance (relative size or height of the two children of a node), published trees tend to be more balanced than uniformly distributed trees, but less balanced than birth–death trees. We subsequently found that simple strategies such as Neighbor Joining do very well on datasets generated from birth–death trees and, with all other parameters held unchanged, quite poorly on datasets generated from uniformly distributed trees. Aldous [1, 2] proposed a model with a single balance parameter, the β -*splitting* model, that, according to the value of the parameter β , can generate perfectly balanced trees, birth–death trees, uniformly distributed trees, down to “caterpillar” (or “ladder”) trees (in which each internal node has a leaf as one of its children) and recommended a particular parameter setting to match the balance factors of published phylogenies. Unfortunately, that model lacks a biological foundation—it is a purely combinatorial model; moreover, the single parameter cannot localize tree structure—it acts on the entire tree at once. Heard [27] had earlier published a model with a strong biological foundation, in which the speciation rate is inherited and also subject to variation; again, depending on the setting of the speciation parameters (inheritance and variability), most distributions of tree balance can be produced. Heard’s model, because it is founded on the birth–death process, has the added advantage of producing edge lengths (in terms of elapsed times), from which the number of evolutionary events can be inferred in terms of various evolutionary models. We have used both Aldous’ and Heard’s models in our simulations, with the most convincing results coming from Heard’s model.

Many problems of biological verisimilitude appear at every stage, but perhaps most importantly in the process of generating genome rearrangements. Most studies to date, including ours, have used a simple process in which inversions (and, if included, transpositions and inverted transpositions) are generated uniformly at random. However, most chromosomes have internal structure that might prevent the occurrence of certain events (for instance, inversion might not be possible across a centromere) or favour the occurrence of others (for instance, there might be “hotspots” in the chromosome that are frequently involved as the endpoint of inversions or transpositions—for recent evidence of such, see [60]). The length of inversions and transpositions is an important question that has recently been considered in models of genomic evolution [65], in phylogenetics [34], and in comparative genomics—the latter of particular importance in the evolution of cancerous cells, where many short rearrangements are common.

Finally, a thorny issue in all optimization problems is the issue of robustness. NP-hard optimization problems, such as MP and (presumably) ML, often exhibit very brittle characteristics; little is known about the space of trees in the neighbourhood of the true tree in phylogenetic reconstruction or about the effect on this space of the choice of parameters in the models.

12.5 Conclusion and open problems

Gene-content and gene-order data are being produced at increasing rates for many simple organisms, from organelles to bacteria, and in a few model eukaryotes. In phylogenetic work, such data have been found to carry a very strong and robust phylogenetic signal—reconstructions using such data, both in simulations and with biological datasets, provide information consistent with the best analyses run on sequence data, robust in the face of small changes, and less sensitive to mixes of small and large evolutionary distances than any sequence-based analysis. Moreover, these techniques scale well to large datasets (at least to 1,000 taxa, but most likely many more). That these data do so well in spite of the primitive tools available to date (simplistic models, limited optimization frameworks, enormous computational demands) bodes well and justifies a call for more research, particularly on the following topics.

- Tree models. Heard’s model [27] is promising and perhaps even sufficient, but the effect of its various parameters on the accuracy and complexity of phylogenetic reconstruction needs to be better understood.
- Evolutionary models for genomes. As mentioned above, there are many questions and very few answers to date. For the time being, one can run simulations under many different models and verify that certain solutions work better than others; as new data emerge, however, one can expect improvements in the models.
- Extensions of the theory pioneered by Hannenhalli and Pevzner, beyond the work of El-Mabrouk, Marron *et al.*, and Hartman, to handle transpositions alone, transpositions and inversions, length-dependent rearrangements, position-dependent rearrangements, and duplications.

- Good combinatorial formulations of the median problem for inversions and for more general cases and, by extension, of the problem of assigning ancestral gene orders to a fixed tree in order to minimize the total number of evolutionary events (as weighted by the model of evolution). In particular, handling of large multichromosomal genomes, by integrating advances such as MGR and DCM-GRAPPA, would enable the use of gene-order data in the reconstruction of eukaryotic phylogenies.
- Tighter bounds on tree scores under the optimization model, so as to scale up the optimization to the largest possible datasets.
- Integration of the above within a DCM-like framework, in order to scale the computations to (nearly) arbitrarily large datasets. Our group recently made significant progress on this issue using integer linear programming.

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DISTANCE-BASED GENOME REARRANGEMENT PHYLOGENY

Li-San Wang and Tandy Warnow

Evolution operates on whole genomes through mutations, such as inversions, transpositions, and inverted transpositions, which rearrange genes within genomes. In this chapter, we survey distance-based techniques for estimating evolutionary history under these events. We present results on the distribution of genomic distances under the Generalized Nadeau–Taylor model, a Markovian model that allows an arbitrary mixture of the three types of mutations, and the derivation of three statistically-based evolutionary distance estimators based on these results. We then demonstrate by simulation that the use of these new distance estimators with methods such as Neighbor Joining and Weighbor can result in improved reconstructions of evolutionary history.

13.1 Introduction

The genomes of some organisms have a single chromosome or contain single chromosome organelles (such as mitochondria [5, 25] or chloroplasts [10, 24, 25, 27]) whose evolution is largely independent of the evolution of the nuclear genome for these organisms. Evolutionary events can alter these orderings through rearrangements such as inversions and transpositions, collectively called genome rearrangements. These events fall under the general category of “rare genomic changes,” and are thought to have great potential for clarifying deep evolutionary histories [28]. In the last decade or so, a few researchers have used such data in their phylogenetic analyses [3, 5–7, 10, 24, 27, 31].

Of the various techniques for estimating phylogenies from gene order data, only distance-based methods are polynomial time. The first study that used distance-based methods to reconstruct phylogenies from gene orders was done by Blanchette *et al.* [5]. Their study gave a phylogenetic analysis using the Neighbor Joining (NJ) [29] method applied to a matrix of “breakpoint distances” defined on a set of mitochondrial genomes for six metazoan groups. However, as this chapter will show, breakpoint distances do not provide particularly accurate estimations of evolutionary distances, and better estimations of trees can be obtained using other distance estimators.

The rest of the chapter is organized as follows. Section 13.2 provides the background on genome rearrangement evolution and describes the Generalized Nadeau–Taylor model. In Section 13.3 we discuss distance-based phylogeny reconstruction. We describe three new distance estimators for genome rearrangement evolution in Sections 13.4 and 13.5. We report on simulation studies evaluating the accuracy of these estimators, and of phylogenies estimated using these estimators on random tree topologies in Section 13.6. Finally, in Section 13.7 we discuss recent extensions to the Generalized Nadeau–Taylor model and discuss some relevant open problems in phylogeny reconstruction that arise.

13.2 Whole genomes and events that change gene orders

In this chapter, we will study phylogeny reconstruction on whole genomes under the assumption that all genomes have exactly one copy of each gene; thus, all genomes have exactly the same gene content.

13.2.1 *Inversions and transpositions*

The events we consider do not change the number of copies of a gene, but only scramble the order of the genes within the genomes. Thus we will not consider events such as duplications, insertions, or deletions, but will restrict ourselves to inversions (also called “reversals”) and transpositions.

Inversions operate by picking up a segment within the genome and reinserting the segment in the reverse direction; thus, the order and strandedness of the genes involved change. A transposition has the effect of moving a segment from between two genes to another location (between two other genes), without changing the order or strandedness of the genes within the segment. If the transposition is combined with an inversion, then the order and strandedness change as well—this is called an inverted transposition. Examples of these events are shown in Fig. 13.1.

(a)	1	2	3	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	9	10
(b)	1	2	3	<u>-8</u>	<u>-7</u>	<u>-6</u>	<u>-5</u>	<u>-4</u>	9	10
(c)	1	2	3	9	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	10
(d)	1	2	3	9	<u>-8</u>	<u>-7</u>	<u>-6</u>	<u>-5</u>	<u>-4</u>	10

FIG. 13.1. Examples of genome rearrangements. Genome (a) is the starting point for all the events we demonstrate. Genome (b) is obtained by applying an inversion to Genome (a). Genome (c) is obtained by applying a transposition to Genome (a). Genome (d) is obtained by applying an inverted transposition to Genome (a). In each of these events we have affected the same target segment of genes (genes 4 through 8, underlined in Genome (a)), and indicated its location (also by underlining) in the resultant genome.

13.2.2 Representations of genomes

In order to analyse gene order evolution mathematically, we represent each genome (whether linear or circular) as a signed permutation of $(1, 2, \dots, n)$, where n is the number of genes and where the sign indicates the strand on which the gene occurs. Thus, a circular genome can be represented as a signed circular permutation, and a linear genome can be represented as a signed linear permutation. In the case of circular genomes, we use linear representations by beginning at any of its genes, in either orientation. We consider two such representations of a circular genome equivalent. As an example, the circular genome given by the linear ordering $(1, 2, 3, 4, 5)$ is equivalently represented by the linear orderings $(2, 3, 4, 5, 1)$ and $(-2, -1, -5, -4, -3)$. As an example of how an inversion acts, if we apply an inversion on the segment 2, 3 to $(1, 2, 3, 4, 5)$, we obtain $(1, -3, -2, 4, 5)$. For an example of a transposition, if we then apply a transposition moving the segment $-2, 4$ to between 1 and -3 , we obtain $(1, -2, 4, -3, 5)$.

For the rest of the chapter we focus on circular genomes unless stated otherwise (our simulations show that all results can be directly applied to linear genomes without any significant difference in accuracy).

13.2.3 Edit distances between genomes: inversion and breakpoint distances

The kinds of distances we are most interested in estimating are evolutionary distances—the number of events that took place in the evolutionary history between two genomes. However, the two common ways of defining distances between genomes are breakpoint distances and inversion distances, neither of which provides a good estimate of evolutionary distances. We obtain our evolutionary distance estimators (described later in the chapter) by “correcting” these two distances.

Inversion distance. The inversion distance between genomes G and G' is the minimum number of inversions needed to transform G into G' (or vice-versa, as it is symmetric); we denote this distance by $d_{\text{INV}}(G, G')$. The first polynomial time algorithm for computing this distance was obtained by Hannenhalli and Pevzner [15], and later improved by Kaplan *et al.* [16] and Bader *et al.* [2] (the latter obtained an optimal linear-time algorithm). See Chapter 10, this volume, for a review of these algorithms.

Breakpoint distance. Another popular distance measure between genomes is the breakpoint distance [4]. A breakpoint occurs between genes g and g' in genome G' with respect to genome G if g is not followed immediately by g' in G . As an example, consider circular genomes $G = (1, 2, -3, 4, 5)$ and $G' = (1, 2, 3, -5, -4)$. There is a breakpoint between 2 and 3 in G' , since 2 is not followed by 3 in G , but there is no breakpoint between -5 and -4 in G' (since G can be equivalently written as $(-1, -5, -4, 3, -2)$). The breakpoint distance between two genomes is the number of breakpoints in one genome with respect to the other, which is

clearly symmetric; we denote this distance by $d_{\text{BP}}(G, G')$. In the example above the breakpoint distance is 3.

13.2.4 The Nadeau–Taylor model and its generalization

The Nadeau–Taylor model [22] assumes that only inversions occur (i.e. no transpositions or inverted transpositions occur), and all inversions have the same probability of occurring. This assumption that inversions are equiprobable was inspired by the observation made in reference [22] that the length of conserved segments between the human and mouse genomes (relative to each other) seems to be uniformly randomly distributed.

In reference [40] we proposed a generalized version of the Nadeau–Taylor model which allows for transpositions and inverted transpositions to occur. In the Generalized Nadeau–Taylor (GNT) model, all inversions have equal probability, as do different transpositions and inverted transpositions. Each model tree thus has parameters w_I , w_T , and w_{IT} , where w_I is the probability that a rearrangement event is an inversion, w_T is the probability that a rearrangement event is a transposition, and w_{IT} is the probability that a rearrangement event is an inverted transposition. Because we assume that all events are of these three types, $w_I + w_T + w_{IT} = 1$. Given a model tree, we will let $X(e)$ be the random variable for the number of evolutionary events that takes place on the edge e . We assume that $X(e)$ is a Poisson random variable with mean λ_e ; hence, λ_e can be considered the length of the edge e . We also assume that events on one edge are independent of the events on other edges. Thus, the GNT model requires $O(m)$ parameters, where m is the number of genomes (i.e. leaves): the length λ_e of each edge e , and the triplet w_I, w_T, w_{IT} . We let $\text{GNT}(w_I, w_T, w_{IT})$ denote the set of model trees with the triplet (w_I, w_T, w_{IT}) . Thus, the Nadeau–Taylor model is simply the $\text{GNT}(1, 0, 0)$ model.

13.3 Distance-based phylogeny reconstruction

There are many methods for reconstructing phylogenies, such as *maximum parsimony* (MP) and *maximum likelihood* (ML), which are computationally intensive. In this chapter, we focus on phylogeny reconstruction techniques that are polynomial time. For gene order phylogeny reconstruction, the fast methods are primarily *distance-based* methods. We briefly review the basic concepts here, and direct the interested reader to Chapter 1, this volume, on distance-based methods for a more in-depth discussion.

13.3.1 Additive and near-additive matrices

Suppose we have a phylogenetic tree T on m leaves, and we assign a positive length $l(e)$ to each edge e in the tree. Consider the $m \times m$ matrix (D_{ij}) defined by $D_{ij} = \sum_{e \in P_{ij}} l(e)$, where P_{ij} is the path in T between leaves i and j . This matrix is said to be “additive.” Interestingly, given the matrix (D_{ij}) , it is possible to construct T and the edge lengths in polynomial time, up to the location of the root [41, 42], provided that we assume that T has no nodes of degree two.

The connection between this discussion and the inference of evolutionary histories is obtained by setting $l(e)$ to be the actual number of changes on the edge e . Then, $D_{ij} = \sum_{e \in P_{ij}} l(e)$ is the actual number of events (in our case, inversions, transpositions, and inverted transpositions) that took place in the evolutionary history relating genomes i and j .

Since estimations of evolutionary distances have some error, the matrices (d_{ij}) given as input to distance-based methods generally are not additive. Therefore, we may wish to understand the conditions under which a distance-based method will still correctly reconstruct the tree, even though the edge lengths may be incorrect. Research in the last few years has established that various methods, including Neighbor Joining [1], will still reconstruct the true tree as long as $L_\infty(D, d) = \max_{ij} |D_{ij} - d_{ij}|$ is small enough, where (d_{ij}) is the input matrix and (D_{ij}) is the matrix for the true tree (see [1, 17] and Chapter 1, this volume).

Consequently, methods such as Neighbor Joining which have some error tolerance will yield correct estimates of the true tree, as long as each D_{ij} can be estimated with sufficient accuracy.

13.3.2 *The two steps of a distance-based method*

Using these observations, it is clear why distance-based methods have these two steps:

- **Step 1:** Estimate “evolutionary distances” (expected or actual number of changes) between every pair of taxa, producing matrix (d_{ij}) .
- **Step 2:** Use a method (such as Neighbor Joining) to infer an edge-weighted tree from (d_{ij}) .

The second step is fairly standard at this point, with Neighbor Joining [29] the most popular of the distance-based methods. However, the first step is very important as well. Extensive simulation studies under DNA models of site substitution have shown that phylogenies obtained using distance-based methods (such as Neighbor Joining) applied to statistically based distance estimation techniques are closer to the true tree than when used with uncorrected distances. If, however, the evolutionary model obeys the molecular clock, so that the expected number of changes is proportional to time, then statistically based estimations of distance are unnecessary—correct trees can be reconstructed by applying simple reconstruction methods such as UPGMA [33] applied to Hamming distances. However, since the molecular clock assumption is not generally applicable, better distance estimation techniques are necessary for phylogeny reconstruction purposes.

The use of breakpoint distances and inversion distances in whole genome phylogeny reconstruction is problematic because these typically underestimate the actual number of events; therefore, they are not statistically consistent distance-estimators under the GNT model. This theoretical observation, coupled with empirical results, motivates us to produce statistically based distance estimators for the GNT model.

13.3.3 Method of moments estimators

The distance estimators we describe in this chapter are all method of moments estimators. Let X be a real-valued random variable whose distribution is parametrized by p ; as a result $E[X]$ is a function f of p . The estimator $\hat{p} = f^{-1}(x)$, where x is the observed value for the mean of X , is a method of moments estimator of the parameter p . In our case, since there is only one observation for X , the mean of X is simply the observed value for X . Method of moments estimators are common in many statistical applications, and generally have good accuracy; see any standard statistics textbook (such as Section 7.1 in reference [9]) for details.

In the context of gene order phylogeny, we have developed two functions which estimate the expected breakpoint distance produced by k random events under the GNT(w_I, w_T, w_{IT}) model, for each way of setting w_I, w_T , and w_{IT} . One of these two functions is provably correct, and the other is approximate (with provable error bounds), but both have almost identical performance in simulation. We also have a function which estimates the expected inversion distance produced by k random inversions (i.e. random events in the GNT(1, 0, 0) model).

Each of these functions is invertible, and thus can be used to estimate the number of events in the evolutionary history between two genomes in a simple way. For example, given the function $f(k)$ for the expected breakpoint distance produced by k random events in the GNT(w_I, w_T, w_{IT}) model on n genes (see Section 13.5), we can define a distance estimation technique, which we call IEBP, for “Inverting the Expected Breakpoint Distance” as follows:

- **Step 1:** Given genomes G and G' , compute their breakpoint distance d .
- **Step 2:** Using the assumed values for w_I, w_T , and w_{IT} , compute $f^{-1}(d)$.
This is the estimate of the evolutionary distance between G and G' .

We demonstrate this technique in Fig. 13.2.

We have also developed a distance estimation technique called Empirically Derived Estimator (EDE), for the “Empirically Derived Estimator,” which estimates the evolutionary distance between two genomes by inverting the expected inversion distance. (See Section 13.4 for the derivation of EDE.)

In the next sections, we describe these three distance-estimators: **Exact-IEBP**, which is based upon an exact formula for the expected breakpoint distance, **Approx-IEBP**, which is based upon an approximate formula (with guaranteed error bounds) for the expected breakpoint distance, and **EDE**, which is based upon a heuristic for the expected inversion distance. All three estimators improve upon both breakpoint and inversion distances as evolutionary distance estimators, and produce better phylogenetic trees, especially when the datasets come from model trees with high evolutionary diameters (so that the datasets are close to saturation). Of the three, **Exact-IEBP** and **Approx-IEBP** have the best accuracy with respect to distance estimation, but surprisingly phylogeny reconstruction based upon **EDE** is somewhat more accurate than phylogeny reconstruction based upon the other estimators.

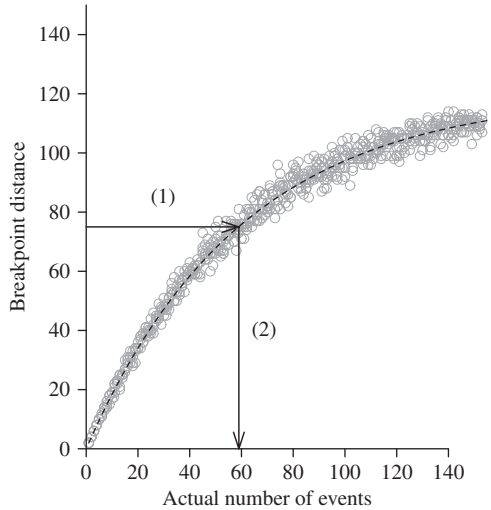


FIG. 13.2. Illustration of the IEBP technique, a method of moments estimator.

The backdrop is the scatter plot of simulations with 120 genes, inversion-only evolution. The dashed line is the expected breakpoint distance (the function f in the paragraph describing IEBP), as a function of the number of inversions. In the first step we compute the breakpoint distance d (the y -axis coordinate); in the second step we find $f^{-1}(d)$ as the estimate of the actual number of inversions.

In the next sections we provide the derivations for these three evolutionary distance estimators. We begin with EDE because it is the simplest to explain, and the mathematics is the least complicated.

13.4 Empirically Derived Estimator

Our first method of moments estimator is EDE, which is based upon inverting the expected inversion distance produced by random inversions. Because our technique in deriving EDE is empirical (i.e. we do not have theory to establish any performance guarantees for EDE's distance estimation), we call it the "Empirically Derived Estimator." However, despite the lack of provable theory, of our three evolutionary distance estimators, EDE produces the best results whether we use Neighbor Joining or Weighbor [8] (a variant of Neighbor Joining that uses the variance of the evolutionary distance estimators as well). EDE is quite robust, and performs well even when the model does not permit inversions. The results in this section are taken from [20,39].

13.4.1 The method of moments estimator: EDE

The EDE estimator is based upon inverting the expectation of the inversion distance produced by a sequence of random inversions under the GNT(1,0,0)

model. Thus, to create EDE we have to find a function which will estimate the expected inversion distance produced by a sequence of random inversions. Theoretical approaches (i.e. actually trying to analytically solve the expected inversion distance produced by k random inversions) proved to be quite difficult, and so we studied this under simulation. Our initial studies showed little difference in the behaviour under 120 genes (typical for chloroplasts) and 37 genes (typical of mitochondria), and in particular suggested that it should be possible to express the normalized expected inversion distance as a function of the normalized number of random inversions. Therefore, we attempted to define a simple function $Q(k/n)$ that approximates $E[d_{\text{INV}}(G_0, G_k)/n]$ well, for k the number of random inversions, n the number of genes, G_0 the initial genome, and G_k the result of applying k random inversions to G_0 . This function Q should have the following properties:

- (1) $0 \leq Q(x) \leq x$, since the inversion distance is always less than or equal to the actual number of inversions;
- (2) $\lim_{x \rightarrow \infty} Q(x) \simeq 1$, as simulation shows the normalized expected inversion distance is close to 1 when a large number of random inversions is applied;
- (3) $Q'(0) = 1$, since a single random inversion always produces a genome that is inversion distance 1 away;
- (4) $Q^{-1}(y)$ is defined for all $y \in [0, 1]$, so that we may invert the function.

We use $nQ(x)$ to estimate $E[d_{\text{INV}}(G_{nx}, G_0)]$, the expected inversion distance after nx inversions are applied. The non-linear formula

$$Q(x) = \frac{ax^2 + bx}{x^2 + cx + b}$$

satisfies constraints (2)–(4).

The quantity $\lim_{x \rightarrow \infty} Q(x) = a$ in constraint (2) has the following interpretation. When a large number of random inversions are being applied to a genome G , the resultant genome should look random with respect to G . This quantity is very close to one as n , the number of genes in G , increases, but for finite n , a does not equal 1. Nonetheless, by simply setting $a = 1$ the formula produces very accurate results in practice.

The estimation of b and c amounts to a least-squares non-linear regression. We found that setting $b = 0.5956$ and $c = 0.4577$ produced a good fit to the empirical data. However, with this setting for a , b , and c , the formula does not satisfy the first constraint. Hence, we modify the formula to ensure that constraint (1) holds, and obtain:

$$Q^*(x) = \min\{x, Q(x)\} = \min\left\{x, \frac{ax^2 + bx}{x^2 + cx + b}\right\}.$$

Please refer to Fig. 13.3 for our simulation study evaluating the performance of this formula in fitting the expectation.

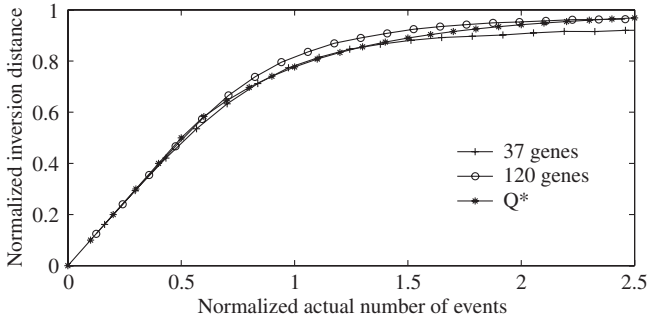


FIG. 13.3. Comparison of the regression formula Q^* for the expected inversion distance in EDE with simulated data. Both the x - and y -axis coordinates are normalized—both are divided by the number of genes.

EDE's algorithm. We can define a method of moments estimator EDE, using the function Q^* , as follows:

- **Step 1:** Given genomes G and G' , compute the inversion distance d .
- **Step 2:** Return $k = n(Q^*)^{-1}(d/n)$, where n is the number of genes.

As the number of actual events must be an integer, another way to obtain an estimate of the evolutionary distance is to choose either $\lfloor k \rfloor$ and $\lceil k \rceil$. However, in practice there is almost no difference in the accuracy of the tree inferred whether we use the inverted function or the closest integer criterion to compute the EDE distance matrix.

We summarize the EDE distance estimator as follows.

Definition 13.1 Let G and G' be two genomes with genes $\{1, 2, \dots, n\}$. Define

$$Q^*(x) = \min\{x, Q(x)\} = \min\left\{x, \frac{x^2 + 0.5956x}{x^2 + 0.4577x + 0.5956}\right\}.$$

The EDE distance between G and G' is

$$\text{EDE}(G, G') = n(Q^*)^{-1}\left(\frac{d}{n}\right),$$

where $d = d_{\text{INV}}(G, G')$ is the inversion distance between G and G' .

EDE therefore is a method of moments estimator of the actual number of inversions that took place in transforming G into G' under the GNT(1,0,0) model (i.e. inversion-only evolution).

Let m be the number of genomes and let n be the number of genes. Computing the inversion distance between each pair of genomes takes only $O(n)$ time, for a total of $O(nm^2)$ time. Once the inversion distance matrix is computed, as the formula Q^* used in EDE is directly invertible, computing the entire EDE distance matrix takes an additional $O(m^2)$ time.

Note that EDE, our first method of moments estimator, was derived on the basis of a simulation study involving 120 genes under an inversion-only evolutionary model. Therefore, the distance estimated by EDE is independent of the model condition: we will get the same estimated distance no matter what we know about the model conditions. Despite this rigidity in EDE's structure and origin, we can apply EDE to any pair of genomes and use it to estimate evolutionary distances. Interestingly, we will see that EDE is quite robust to model violations, and can be used with methods such as Neighbor Joining to produce highly accurate estimations of phylogenies. See Section 13.6 for experimental results evaluating the accuracy of EDE and of distance-based tree reconstruction methods using EDE in simulation.

13.4.2 The variance of the inversion and EDE distances

In order to use EDE with methods such as Weighbor, we need also to have an estimate for the variance of the EDE distance. We therefore developed an estimator (presented in [39]) for the standard deviation of the normalized inversion distance produced by nx random inversions, where n is the number of genes. The approach we used to obtain this estimate is similar to the approach we used to derive EDE.

The variance of the inversion distance. The first step is to obtain the variance of the inversion distance. After several experiments with simulated data, we decided to use the following regression formula:

$$\sigma_n(x) = n^q \frac{ux^2 + vx}{x^2 + wx + t}.$$

The constant term in the numerator is zero because we know $\sigma_n(0) = 0$. As we did in our derivation of the EDE technique, we make the assumption that the actual number of inversions is no more than $3n$.

Note that

$$\begin{aligned} \ln \left(\frac{1}{3n} \sum_{i=0}^{3n} \sigma_n \left(\frac{i}{n} \right) \right) &= q \ln n + \ln \left(\frac{1}{3n} \sum_{i=0}^{3n} \frac{u(i/n)^2 + v(i/n)}{(i/n)^2 + w(i/n) + t} \right) \\ &\simeq q \ln n + \ln \left(\frac{1}{3} \int_0^3 \frac{ux^2 + vx}{x^2 + wx + t} dx \right) \end{aligned}$$

is linear in $\ln n$. Thus we can obtain q as the slope in the linear regression using $\ln n$ as the independent variable and $\ln((1/3n) \sum_{i=0}^{3n} \sigma_n(i/n))$ as the dependent variable. Our simulation results, shown in Fig. 13.4(a), suggest that $\ln((1/3n) \sum_{i=0}^{3n} \sigma_n(i/n))$ indeed is (almost) linear in $\ln n$.

After obtaining $q = -0.6998$, we applied non-linear regression to obtain u , v , w , and t , using the simulated data for 40, 80, 120, and 160 genes, and obtained the values $q = -0.6998$, $u = 0.1684$, $v = 0.1573$, $w = -1.3893$, and $t = 0.8224$. The resultant functions are shown as the solid curves in Fig. 13.4(b).

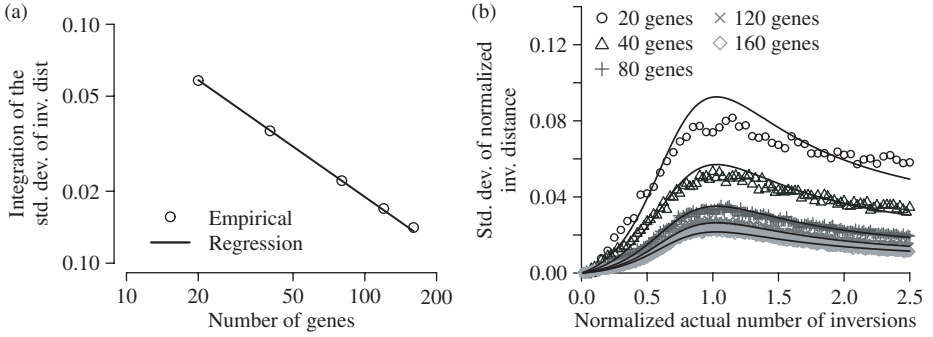


FIG. 13.4. (a) Regression of coefficient q (see Section 13.4); for every point corresponding to n genes, the y coordinate is the average of all data points in the simulation. (b) Simulation (points) and regression (solid lines) of the standard deviation of the inversion distance.

Estimating the variance of EDE. The variance of EDE can now be obtained using a common statistical technique called the *delta method* [23] as follows. Assume Y is a random variable with variance $\text{Var}[Y]$, and let $X = f(Y)$. Then $\text{Var}[X]$ can be approximated by $(dX/dY)^2 \text{Var}[Y]$.

To apply the delta method to EDE, we set Y to be the normalized inversion distance between genomes G and G' (i.e. the inversion distance divided by the number of genes), and set $X = Q^{-1}(Y)$ (we do not use Q^* since it is not differentiable in its entire range).

Let G and G' be two genomes with genes $\{1, 2, \dots, n\}$. Let $x = \text{EDE}(G, G')/n$. Since $(d/dY)Q^{-1}(Y) = (Q'(Q^{-1}(Y)))^{-1}$, the variance of the EDE distance can be approximated as

$$\text{Var}[\text{EDE}(G, G')] \simeq n^2 \left(\frac{1}{Q'(x)} \right)^2 \left(n^{-0.6998} \frac{0.1684x^2 + 0.1573x}{x^2 - 1.3893x + 0.8224} \right).$$

Here $Q(x)$ is the function defined in Section 13.4, upon which Q^* , the expected inversion distance, is based.

13.5 IEBP: “Inverting the expected breakpoint distance”

Exact-IEBP and Approx-IEBP are two method of moments estimators based upon functions for estimating the expected breakpoint distance produced by k random events under the $\text{GNT}(w_I, w_T, w_{IT})$ model, where w_I , w_T , and w_{IT} are given. Thus, “IEBP” stands for “inverting the expected breakpoint distance.” Exact-IEBP is based upon an exact calculation of the expected breakpoint distance, and Approx-IEBP is based upon an approximate estimation of the expected breakpoint distance which we can prove has very low error. In order to use IEBP (Exact- or Approx-) with Weighbor, we also developed a technique for estimating the variance of the IEBP distance; this is presented in Section 13.5.3.

13.5.1 *The method of moments estimator, Exact-IEBP*

We begin with the derivation of the expected breakpoint distance produced by a sequence of random events under the $\text{GNT}(w_I, w_T, w_{IT})$ model. By linearity of expectation and symmetry of the model, it suffices to find the distribution of the presence/absence of a single breakpoint (a zero-one variable).

We consider how a circular genome evolves under the Generalized Nadeau–Taylor model (the analysis for linear genomes can be obtained easily using the same techniques). Let the number of genes in the genome be n . We start with genome $G_0 = (1, 2, \dots, n)$, and we let G_k denote the genome obtained after k random rearrangement events are applied under the Generalized Nadeau–Taylor model.

We begin by defining a character L on circular genomes which will have states in $\{\pm 1, \pm 2, \dots, \pm(n-1)\}$. The state of this character on a genome G' is defined as follows:

1. In G' , do genes 1 and 2 have the same sign, or different signs? If it is the same sign, then $L(G')$ is positive, and otherwise $L(G')$ is negative.
2. We then count the number of genes between 1 and either 2 or -2 in G' (depending upon which one appears in G' 's representation when we use gene 1 in its positive strand), and add 1 to that value; this is $|L(G')|$.

We present some examples of how L is defined on different genomes with 6 genes. If $G' = (1, 2, 4, 5, -3, 6)$ then $L(G') = 1$, while if $G' = (1, -2, 3, 4, 5, 6)$ then $L(G') = -1$. A somewhat harder example is $G' = (1, 5, 3, -2, 4, 6)$, for which $L(G') = -3$ (gene 2 is the third gene to follow gene 1, and it is located on the other strand).

The following lemma shows the number of rearrangement events transforming G into genome G' only depends on $L(G)$, $L(G')$, and the number n of genes. Thus, the distribution of a breakpoint is a $(2n-2)$ -state Markov chain, and we use the character L defined above to assign states to genomes. We sketch the proof for the transposition-only situation.

To facilitate the proof, we formally characterize transpositions on circular genomes. A transposition on G has three indices, a , b , c , with $1 \leq a < b \leq n$ and $2 \leq c \leq n$, $c \notin [a, b]$, and operates on G by picking up the interval $g_a, g_{a+1}, \dots, g_{b-1}$ and inserting it immediately after g_{c-1} . Thus the genome $G = (g_1, g_2, \dots, g_n)$ (with the additional assumption of $c > b$) is replaced by

$$(g_1, \dots, g_{a-1}, g_b, g_{b+1}, \dots, g_{c-1}, g_a, g_{a+1}, \dots, g_{b-1}, g_c, \dots, g_n).$$

Lemma 13.2 ([38]) *Let n be the number of genes. Let $\iota_n(u, v)$, $\tau_n(u, v)$, and $\nu_n(u, v)$ be the minimum number of inversions, transpositions, and inverted transpositions, respectively, that bring a genome in state u to state v . Assume*

the genome is circular. Then

$$\begin{aligned}\iota_n(u, v) &= \begin{cases} \min\{|u|, |v|, n - |u|, n - |v|\} & \text{if } uv < 0, \\ 0 & \text{if } u \neq v, uv > 0, \\ \binom{|u|}{2} + \binom{n-|u|}{2} & \text{if } u = v; \end{cases} \\ \tau_n(u, v) &= \begin{cases} 0 & \text{if } uv < 0, \\ (\min\{|u|, |v|\})(n - \max\{|u|, |v|\}) & \text{if } u \neq v, uv > 0, \\ \binom{|u|}{3} + \binom{n-|u|}{3} & \text{if } u = v; \end{cases} \\ \nu_n(u, v) &= \begin{cases} (n-2)\iota_n(u, v) & \text{if } uv < 0, \\ \tau_n(u, v) & \text{if } u \neq v, uv > 0, \\ 3\tau_n(u, v) & \text{if } u = v. \end{cases}\end{aligned}$$

Proof The formula for ι is first shown in reference [32]. Here we sketch the proof for τ .

Assume that the current genome is in state u . Let v be the new state of the genome after the transposition with indices (a, b, c) , $1 \leq a < b < c \leq n$. Since transpositions do not change the sign, $\tau_n(u, v) = \tau_n(-u, -v)$ and $\tau_n(u, v) = 0$ if $uv < 0$. Therefore we only need to analyse the case where $u, v > 0$.

We first analyse the case when $u = v$. Suppose that either $a \leq u < b$ or $b \leq u < c$. In the first case, we immediately have $v = u + (c - b)$, therefore $v - u = c - b > 0$. In the second case, we have $v = u + (a - b)$, therefore $v - u = a - b < 0$. Both cases contradict the assumption that $u = v$, and the only remaining possibilities that makes $u = v$ are when $1 \leq u = v < a$ or $c \leq u = v \leq n - 1$. This leads to the third line in the $\tau_n(u, v)$ formula.

Next, the total number of solutions (a, b, c) for the following two problems is $\tau_n(u, v)$ when $u \neq v$ and $u, v > 0$:

- (1) $u < v$: $b = c - (v - u)$, $1 \leq a \leq u < b < c \leq n, u < v \leq c$;
- (2) $u > v$: $b = a + (u - v)$, $1 \leq a < b \leq u < c \leq n, a \leq v < u$.

In the first case $\tau_n(u, v) = u(n - v)$, and in the second case $\tau_n(u, v) = v(n - u)$. The second line in the $\tau_n(u, v)$ formula follows by combining the two results. $\square$

We now derive the distribution of the Markov chain. To simplify the formulas, we index all vectors and matrices by the states $\{\pm 1, \pm 2, \dots, \pm(n - 1)\}$. Let G_k be the result of applying k random rearrangements to genome G_0 under GNT(w_I, w_T, w_{IT}). We first obtain the transition matrix.

Lemma 13.3 *Let M_I , M_T , and M_{IT} be the transition matrices of the Markov chain when only inversions, transpositions, or inverted transpositions occur, respectively. We let w_I be the probability of an inversion, w_T be the probability of a transposition, and w_{IT} be the probability of an inverted transposition*

(with $w_I + w_T + w_{IT} = 1$). Then

$$(a) \quad M_I[u, v] = \frac{\iota_n(u, v)}{\binom{n}{2}}, \quad M_T[u, v] = \frac{\tau_n(u, v)}{\binom{n}{3}}, \quad M_{IT}[u, v] = \frac{\nu_n(u, v)}{3\binom{n}{3}}.$$

(b) The transition matrix M of the breakpoint Markov chain is

$$M = w_I M_I + w_T M_T + w_{IT} M_{IT}.$$

Proof Results in (a) follow from Lemma 13.2 together with the observation that there are $\binom{n}{2}$ distinct inversions, $\binom{n}{3}$ distinct transpositions, and $3\binom{n}{3}$ distinct inverted transpositions. $\square$

Theorem 13.4 Let M be the transition matrix of the breakpoint Markov chain as described above. Then

$$E[d_{BP}(G_0, G_k)] = n(1 - M^k[1, 1]).$$

Proof Let L be the character defined for the Markov chain (i.e. $L(G')$ is the state of genome G') and let x_k be the distribution vector of $L(G_k)$. Because $L(G_0) = 1$, we can set x_0 as follows:

$$\begin{aligned} x_0[1] &= 1, \\ x_0[u] &= 0, \quad u \in \{-1, \pm 2, \dots, \pm(n-1)\}. \end{aligned}$$

Since $x_k = M^k x_0$,

$$\begin{aligned} \Pr(L(G_k) = 1) &= (M^k x_0)[1, 1] = M^k[1, 1] \\ \Rightarrow E[d_{BP}(G_0, G_k)] &= n \Pr(L(G_k) \neq 1) = n(1 - M^k[1, 1]). \end{aligned} \quad \square$$

We summarize the **Exact-IEBP** distance as follows.

Definition 13.5 Assume the evolutionary model is $GNT(w_I, w_T, w_{IT})$. Let G and G' be two genomes with genes $\{1, 2, \dots, n\}$. Let

$$Y(k) = n(1 - M^k[1, 1]),$$

where M is defined in Lemma 13.3. The **Exact-IEBP** distance is the non-negative integer k that minimizes $|Y(k) - d|$:

$$Exact\text{-}IEBP(G, G') = \underset{\text{integer } k \geq 0}{\operatorname{argmin}} |Y(k) - d|,$$

where $d = d_{BP}(G, G')$ is the breakpoint distance between G and G' .

Thus, **Exact-IEBP** is a method of moments estimator of the actual number of evolutionary events under the GNT model, which uses assumed values of w_I , w_T , and w_{IT} .

Note the following. First, computing the expected breakpoint distance produced by k random events is done recursively, and the calculation takes $O(n^2 k)$ time. Second, because breakpoints are not independent, extending the approach in order to study higher order statistics such as the variance is difficult. To see

why breakpoints are not independent, consider the following argument. If breakpoints were independent, then the probability of having breakpoint distance 1 would be positive, as it is a product of n positive values. Since no two genomes can differ by one breakpoint, this is impossible.

Let m be the number of genomes, and n be the number of genes in each genome. Computing the breakpoint distance matrix takes $O(m^2n)$ time total. To compute the **Exact-IEBP** distance matrix the first step of the algorithm is to compute $Y(k)$, the expected breakpoint distance produced by k random events, for each k between 1 and $3n$. This amounts to $3n$ (transition) matrix-(state probability) vector multiplications, and uses $O(n^3)$ time. To invert $Y(k)$ (as a method of moments estimator requires) we use binary search in $O(\log n)$ time (we assume the number of rearrangement events never exceed $3n$). Because there are at most n different breakpoint distance values, computing the **Exact-IEBP** distance matrix when the breakpoint distance is known takes $O(n^3 + m^2 + \min\{m^2, n\} \log n)$ time.

13.5.2 The method of moments estimator, **Approx-IEBP**

In this section, we present an approximate version of **Exact-IEBP**, which we call **Approx-IEBP** (see [40] for the details). Rather than exactly computing the expected number of breakpoints produced by a sequence of random events in the GNT model, we compute an approximation of that value. Because we allow an approximation, we can obtain the estimation faster; thus, the main advantage over **Exact-IEBP** is the running time. Fortunately, we are able to provide very good error bounds on the estimation. Our simulation results, shown later in this chapter, also show that **Approx-IEBP** is almost as accurate as **Exact-IEBP**, and that trees inferred based upon either version of **IEBP** are almost indistinguishable. The technique we used to obtain **Approx-IEBP** is based upon an analysis using 2-state Markov chains. We describe that approach here.

Without loss of generality, consider the 2-state stochastic process indicating the presence of a breakpoint between genes 1 and 2. We let 0 denote the absence of a breakpoint between 1 and 2 (i.e. that gene 2 immediately follows gene 1), and we let 1 indicate the presence of the breakpoint (i.e. that gene 1 is not immediately followed by gene 2). The 2-state stochastic process is shown in Fig. 13.5. While the transitional probability s of jumping from state 0 to 1 in one step is a constant, the transitional probability u of jumping from state 1 to 0 in one step depends on both the sign of gene 2 and the number of genes between the two genes. Thus, no Markov chain with only these two states (presence or absence of a breakpoint) can completely specify the stochastic process. However, we can always find tight bounds on u .

Lemma 13.6 *Let G_0 be a signed circular genome with n genes. Let the model of evolution be $GNT(w_I, w_T, w_{IT})$. The transitional probability s of jumping from state 0 to state 1 after a rearrangement event occurs is given by*

$$s = \frac{2 + w_T + w_{IT}}{n}$$

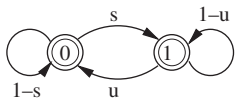


FIG. 13.5. The two-state stochastic process of the breakpoint between genes 1 and 2 under the Generalized Nadeau–Taylor model.

and the transitional probability u of jumping from state 1 to state 0 after a rearrangement event occurs is between 0 and u_H , where

$$u_H = \frac{2(n-2) + 4w_T(n-2) + 2w_{IT}n}{n(n-1)(n-2)}.$$

Based on these bounds we can devise two 2-state Markov chains with different values of u (s is always fixed) so that the probability of having a breakpoint can be bounded. A good approximation of the expected breakpoint distance can then be obtained by taking the product of n with the average of the two probabilities of having a breakpoint.

Theorem 13.7 (From [40]) Assume the genome is signed and circular, and the evolutionary model is $GNT(w_I, w_T, w_{IT})$. Let B_k be the random variable for the presence of a breakpoint between genes 1 and 2 after k rearrangement events. Let

$$L(k) = s \left(\frac{1 - (1 - s - u_H)^k}{s + u_H} \right),$$

and

$$H(k) = s \left(\frac{1 - (1 - s)^k}{s} \right) = 1 - (1 - s)^k.$$

Then for any integer $k \geq 0$, $L(k) \leq \Pr(B_k = 1) \leq H(k)$. The function

$$F(k) = \frac{n}{2} (L(k) + H(k))$$

provides an approximation of the expected breakpoint distance between G_0 and G_k with small absolute and relative error:

$$|F(k) - E[d_{BP}(G_0, G_k)]| = O(1),$$

and

$$\phi^{-1} \leq \frac{F(k)}{E[d_{BP}(G_0, G_k)]} \leq \phi,$$

where $\phi = 1 + O(1/n)$.

We summarize the **Approx-IEBP** distance as follows.

Definition 13.8 Assume the evolutionary model is $GNT(w_I, w_T, w_{IT})$. Let G and G' be two genomes with genes $\{1, 2, \dots, n\}$. Let $d = d_{BP}(G, G')$ be the breakpoint distance between G and G' . Let F be the function defined in Theorem 13.7.

The *Approx-IEBP* distance is the non-negative integer k minimizing $|F(k) - d|$:

$$\text{Approx-IEBP}(G, G') = \underset{\text{integer } k \geq 0}{\operatorname{argmin}} |F(k) - d|.$$

Thus, *Approx-IEBP* is a method of moments estimator which estimates the actual number of rearrangement events between two genomes in the GNT model. Like *Exact-IEBP*, it requires values for w_I , w_T , and w_{IT} .

Let m be the number of genomes, and n be the number of genes in each genome. Computing the breakpoint distance matrix takes $O(m^2n)$ time total. To compute the *Approx-IEBP* distance matrix, we invert $F(k)$, the estimate of the expected breakpoint distance in *Approx-IEBP*, for each pairwise breakpoint distance between two genomes. Computing $F(k)$ takes constant time for each k . To invert $F(k)$ for each pairwise breakpoint distance (as a method of moments estimator requires) we use binary search, which takes $O(\log n)$ time (we assume the number of rearrangement events never exceed $3n$). Because there are at most n different breakpoint distance values, computing the *Approx-IEBP* distance matrix when the breakpoint distance is known takes $O(m^2 + \min\{n, m^2\} \log n)$ time.

13.5.3 The variance of the breakpoint and IEBP distances

In this section, we show how to calculate the variance of the breakpoint distance, so that we can use *IEBP* with methods such as *Weighbor*.

The variance of the breakpoint distance. To estimate the variance of the breakpoint distance, we have to examine at least two breakpoints at the same time. To use a straightforward approach like *Exact-IEBP* we have to analyse a Markov chain with $O(n^3)$ states, where n is the number of genes in each genome. However, if we are willing to relax the model a bit, we can get a good approximation of the variance, and in fact of all the moments of the breakpoint distance under the GNT model, through the use of a “box model.” We present this box model here (see [39] for the full details).

Assume all genomes are circular, and that the genome before random rearrangements that occur is $(1, \dots, n)$. Note that if the number of genes is sufficiently large, once the breakpoint between genes i and $i + 1$ is created, it is unlikely that a later rearrangement event will bring the two genes back together.

We let $G' = G_k$ denote the genome obtained by k rearrangement events. As k increases, G' changes, and so new breakpoints appear in G with respect to G' . We will let each box represent the presence of a breakpoint in G relative to G' . Thus, for $i = 1, 2, \dots, n - 1$, box i will be empty if there is no breakpoint in G between genes i and $i + 1$, and non-empty otherwise. We let box n indicate the presence or absence of a breakpoint between n and 1.

The box model for the inversion-only scenario. To illustrate the box model, we begin with the GNT(1, 0, 0) model in which only inversions occur. We start with n empty boxes, and repeat the following procedure k times. In each iteration we choose two distinct boxes (since an inversion creates two breakpoints). For

each box chosen, if the box is empty, we put a ball in it, and otherwise we do not change anything. We let b_k denote the number of non-empty boxes obtained after k iterations. Under our assumption that breakpoints do not disappear, this is an estimate of the number of breakpoints produced by k random inversions.

Let

$$S(x_1, x_2, \dots, x_n) = \frac{x_1 x_2 + x_1 x_3 + \dots + x_{n-1} x_n}{\binom{n}{2}}.$$

Consider $S^k(x_1, x_2, \dots, x_n)$, the expansion of S to the k th power. Each term in the expansion corresponds to a particular combination of choosing two boxes k times so that the total number of times box i is chosen is the power of x_i , for each $1 \leq i \leq n$. The coefficient of that term is the total probability of these ways. For example, the coefficient of $x_1^3 x_2 x_3^2$ in S^k (when $k = 3$) is the probability of choosing box 1 three times, box 2 once, and box 3 twice. Let $u_{i,k}$ be the sum of the coefficients of all terms taking the form $x_1^{a_1} x_2^{a_2} \dots x_i^{a_i}$ ($a_j > 0$, $1 \leq j \leq i$), in the expansion of S^k . Then $\binom{n}{i} u_{i,k}$ is the probability i boxes are non-empty after k iterations. This is due to the symmetry in S , in the sense that S is not changed by permuting $\{x_1, x_2, \dots, x_n\}$. Let A_j be the value of S when we make the following substitutions: $x_1 = x_2 = \dots = x_j = 1$ and $x_{j+1} = x_{j+2} = \dots = x_n = 0$. For integers j , $0 \leq j \leq n$, we have

$$\sum_{i=0}^j \binom{j}{i} u_{i,k} = S^k(\underbrace{1, 1, 1, \dots, 1}_{j \text{ 1's}}, 0, \dots, 0) = A_j^k.$$

Let

$$Z_a = \sum_{i=0}^n i(i-1) \dots (i-a+1) \binom{n}{i} u_{i,k} = \sum_{i=a}^n n(n-1) \dots (n-a+1) \binom{n-a}{i-a} u_{i,k}$$

for all a , $1 \leq a \leq n$. However, each Z_a can be represented as a linear combination of A_i , for $0 \leq i \leq n$. To obtain the variance of b_k we only need Z_1 and Z_2 .

Lemma 13.9

- (a) $Z_1 = n u_{1,k} = n(A_n^k - A_{n-1}^k)$.
- (b) $Z_2 = n(n-1) u_{2,k} = n(n-1)(1 - 2A_{n-1}^k + A_{n-2}^k)$.

We then have the following theorem.

Theorem 13.10 ([39]) *Let b_k be the number of nonempty boxes in the box model after k iterations. The expectation and variance of b_k are*

$$\begin{aligned} E[b_k] &= n(1 - A_{n-1}^k), \\ \text{Var}[b_k] &= nA_{n-1}^k - n^2 A_{n-1}^{2k} + n(n-1)A_{n-2}^k, \end{aligned}$$

where

$$A_{n-1} = 1 - \frac{2}{n},$$

and

$$A_{n-2} = \frac{(n-3)(n-2)}{n(n-1)}.$$

Proof The first identity follows immediately from the fact that $E[b_k] = Z_1$ and that $A_n = 1$. To prove (b), note

$$\begin{aligned} E[b_k(b_k - 1)] &= Z_2 = n(n-1)(1 - 2A_{n-1}^k + A_{n-2}^k) \\ \Rightarrow E[b_k^2] &= E[b_k(b_k - 1)] + E[b_k] = Z_2 + Z_1 \\ &= n(n-1)(1 - 2A_{n-1}^k + A_{n-2}^k) + n(1 - A_{n-1}^k) \\ &= n^2 - n(2n-1)A_{n-1}^k + n(n-1)A_{n-2}^k \\ \Rightarrow \text{Var}[b_k] &= E[b_k^2] - (E[b_k])^2 \\ &= n^2 - n(2n-1)A_{n-1}^k + n(n-1)A_{n-2}^k - n^2(1 - A_{n-1}^k)^2 \\ &= nA_{n-1}^k - n^2A_{n-1}^{2k} + n(n-1)A_{n-2}^k. \quad \square \end{aligned}$$

A natural idea is to use A_{n-1} as an estimate of the expected breakpoint distance in computing IEBP. The estimate is quite accurate when n is large, though unlike Approx-IEBP the formula does not have provable error bounds.

The box model for the general case. Though we assumed only inversions occur in the derivation of Theorem 13.10, it is only reflected in our definition of S . The derivation of Theorem 13.10 only requires S is symmetric, that is, that S is not changed when we permute $x_1, \dots, x_n$. Therefore, it is easy to extend the result to the general case, that is, to $\text{GNT}(w_I, w_T, w_{IT})$: at each iteration, with probability w_I we choose two boxes, and with probability $w_T + w_{IT}$ we choose three boxes (since each transposition and inverted transposition creates at most three breakpoints). Therefore, we can prove the following generalization.

Corollary 13.11 ([39]) *Let b_k be the number of non-empty boxes in the box model after k iterations. Assume in each iteration, with probability w_I two boxes are picked at random, and with probability $w_T + w_{IT} = 1 - w_I$ three boxes are picked at random. The expectation and variance of b_k are*

$$\begin{aligned} E[b_k] &= n(1 - A_{n-1}^k), \\ \text{Var}[b_k] &= nA_{n-1}^k - n^2A_{n-1}^{2k} + n(n-1)A_{n-2}^k, \end{aligned}$$

where

$$A_{n-1} = 1 - \frac{3 - w_I}{n},$$

and

$$A_{n-2} = \frac{(n-3)(n-4+2w_I)}{n(n-1)}.$$

Proof We set S as follows:

$$S = \frac{w_I}{\binom{n}{2}} \left(\sum_{1 \leq i_1 < i_2 \leq n} x_{i_1} x_{i_2} \right) + \frac{w_T + w_{IT}}{\binom{n}{3}} \left(\sum_{1 \leq i_1 < i_2 < i_3 \leq n} x_{i_1} x_{i_2} x_{i_3} \right).$$

The values A_{n-1} , A_{n-2} in Theorem 13.10 are changed according to S .

$$A_{n-1} = \frac{w_I \binom{n-1}{2}}{\binom{n}{2}} + \frac{(w_T + w_{IT}) \binom{n-1}{3}}{\binom{n}{3}} = 1 - \frac{3 - w_I}{n},$$

$$A_{n-2} = \frac{w_I \binom{n-2}{2}}{\binom{n}{2}} + \frac{(w_T + w_{IT}) \binom{n-2}{3}}{\binom{n}{3}} = \frac{(n-3)(n-4+2w_I)}{n(n-1)}.$$

□

The variance of the IEBP distance. We begin by observing that **Exact-IEBP** and **Approx-IEBP** have almost identical performance, and so we will refer to them collectively as **IEBP**.

Let G and G' be two genomes with genes $\{1, 2, \dots, n\}$. Let $D_b = \text{IEBP}(G, G')$ and $J(k) = E[b_k]$ be the expected number of nonempty boxes in the box model. The variance of the IEBP distance can be approximated using the delta method (see Section 13.4.2) together with the expectation and variance of the box model:

$$\text{Var}[\text{IEBP}(G, G')] \simeq \left(\frac{1}{J'(D_b)} \right)^2 \text{Var}[J(D_b)].$$

13.6 Simulation studies

In this section, we report on the accuracy of the various techniques for defining distances between genomes (both the original inversion and breakpoint distances, and also **EDE**, **Exact-IEBP**, and **Approx-IEBP**). All these studies are based upon simulation under the GNT model, for various settings of the model parameters. All model trees are drawn from the uniform distribution.

We also report on the accuracy of trees reconstructed using either Neighbor Joining or Weighbor under these various distances. We test these distance estimators under optimal conditions—where the true model parameters are known—as well as under conditions where the true model parameters are incorrectly specified. We explore performance on datasets containing 40 or 160 genomes (i.e. moderate and large size), and examine performance for both 37 and 120 genes (typical values for mitochondria and chloroplast genomes, respectively).

13.6.1 Accuracy of the evolutionary distance estimators

In this section, we report on our simulation studies evaluating the performance of the evolutionary distance estimators, by comparison to breakpoint and inversion distances.

In our simulations we see that distances estimated by **Exact-IEBP** and **Approx-IEBP** have almost identical error (there is a slight advantage of **Exact-IEBP** over **Approx-IEBP**, but it is fairly negligible); therefore, we refer to them collectively as **IEBP**.

The results of our simulations show how using either breakpoint and inversion distances is problematic: compared to **IEBP** and **EDE**, breakpoint and inversion distances are highly biased when the number of rearrangement events is large. The inversion distance is a good evolutionary distance estimator when the underlying evolutionary model is inversion-only and the rates of evolution are low (see Fig. 13.6), but is in general not as accurate as either **EDE** or **IEBP** under an inversion-only model.

We also explored the robustness of our estimators by simulating evolution under models other than inversion-only, or by giving incorrect parameter values to **IEBP**. In these cases we see that all five estimators (**BP**, **INV**, **EDE**, **Exact-IEBP**, and **Approx-IEBP**) become less accurate; thus, none of these estimators, including our new ones, is robust to model violations (data not shown).

On the other hand, inaccuracy in distances may not lead to inaccuracy in the trees that are constructed using those distances, *provided* that the estimated distances are just scalar multiples of the evolutionary distances. This is because any such matrix is still an additive matrix for the same underlying tree, but with different edge lengths. Therefore, the estimated distances can be evaluated according to whether they scale linearly with the number of events. Our simulations (data not shown) reveal that all the distance estimators initially scale linearly, implying that all are able to reconstruct good trees when the evolutionary rate is low enough (as indicated by the evolutionary diameter in the dataset). Interestingly, each of the three evolutionary distance estimators seem to scale linearly for a long initial range (**IEBP** more so than **EDE**), even when their assumptions about the model are violated. The worst with respect to linear scaling is clearly **BP**, as seen in Fig. 13.6. These observations may suggest that trees reconstructed from breakpoint distances will have the worst accuracy, especially close to saturation, than trees reconstructed from other methods, and that trees reconstructed from **IEBP** or **EDE** should have the greatest accuracy.

13.6.2 Accuracy of NJ and Weighbor using **IEBP** and **EDE**

As we saw in the previous section, the best estimator of evolutionary distances is **IEBP** (whether **Approx-** or **Exact-**), but **EDE** is also quite accurate, and each is more accurate (except under unusual circumstances) than **INV** and **BP**. The question we investigate in this section is whether the improvement in accuracy of the distance estimators corresponds to an improvement in the accuracy of the resultant phylogenies, as predicted.

We see that the accuracy of trees computed by Neighbor Joining using either **Exact-IEBP** or **Approx-IEBP** is essentially unchanged, and we similarly see unchanged behaviour for Weighbor. Therefore, we will collectively call both

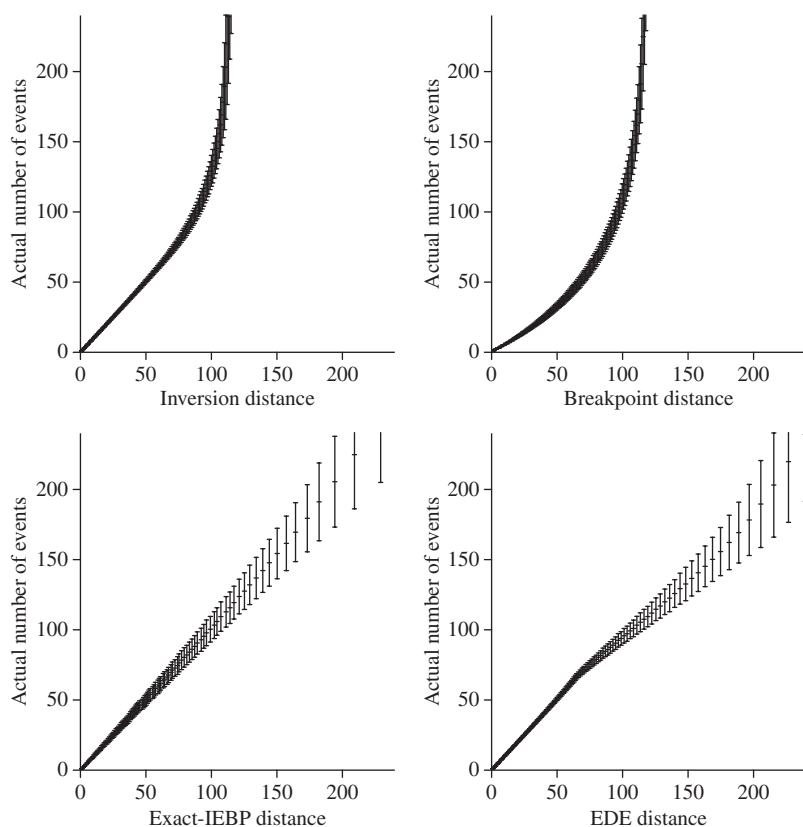


FIG. 13.6. The distribution of genomic distances under the Nadeau–Taylor model (i.e. $\text{GNT}(1,0,0)$, or inversion-only evolution). The number of genes is 120, the x -axis is the measured distance, and the y -axis is the actual number of rearrangement events (inversions in this case). For each vertical line, the middle point is the mean, and the top and bottom tips of the line represent one standard deviation away from the mean. In computing **Exact-IEBP** we use correct values of w_I , w_T , and w_{IT} .

distances **IEBP**. In particular, the results shown in Fig. 13.7 for **Exact-IEBP** apply to **Approx-IEBP** as well.

Model tree generation. In our simulations we produce model trees under the GNT model with 40 or 160 leaves. These model trees have topologies drawn from the uniform distribution on trees leaf-labelled by $1, 2, \dots, m$, where $m = 40$ or 160.

For each model tree we must define branch lengths λ_e , where λ_e is the expected number of changes on the edge. We define these branch lengths in

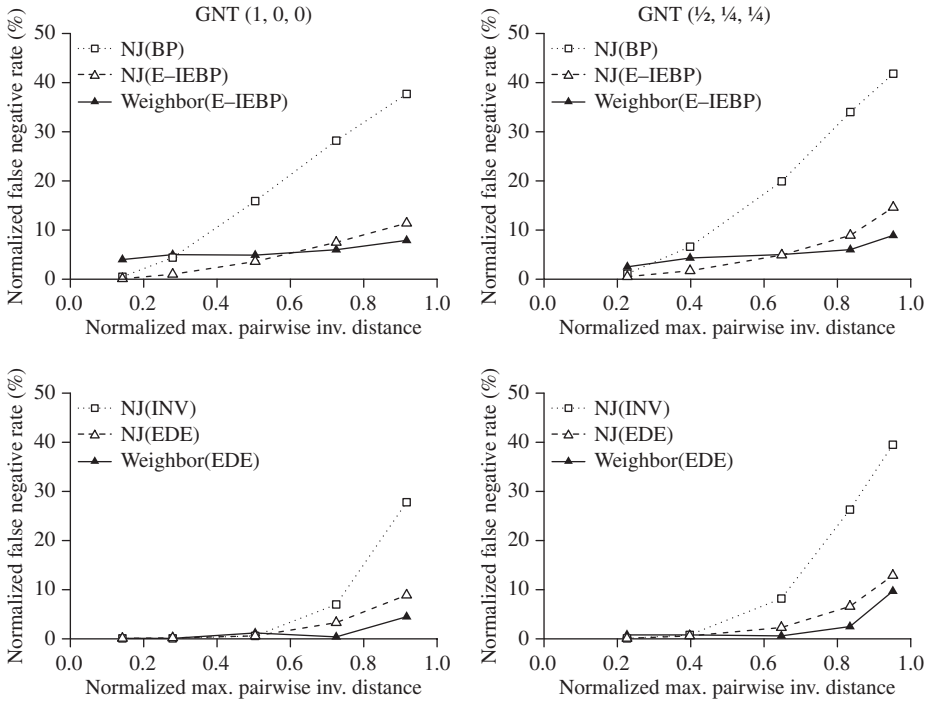


FIG. 13.7. Simulation study of false negative rates of distance-based tree reconstruction methods on 160 circular genomes with 120 genes: (Top) Breakpoint distance based methods, (Bottom) Inversion distance based methods. The x -axis is the normalized diameter (maximum inversion distance between all pairs of genomes) of the dataset, and the y -axis is the false negative rate. The model of evolution is (left) the Nadeau-Taylor model (i.e. $\text{GNT}(1, 0, 0)$), or (right) the GNT model with half inversions, one-fourth transpositions and one-fourth inverted transpositions (i.e. $\text{GNT}(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$). In computing **Exact-IEBP** we use correct values of w_I , w_T , and w_{IT} .

two steps: we assign an initial length, and then we scale all edge lengths to obtain a fixed target maximum path length D for the tree. This maximum path length is defined by $\Delta = \max_{i,j} D_{ij}$, where $D_{ij} = \sum_{e \in P_{ij}} \lambda_e$ and P_{ij} is the path in T between leaves i and j . This value Δ is called the “evolutionary diameter” of T . Our initial assignment of lengths is obtained by choosing random positive integers between 1 and 18 for each edge independently. Then, for each target value of Δ , we scale the edge lengths to obtain the desired evolutionary diameter. The target diameters are drawn from $0.1n$, $0.2n$, $0.4n$, $0.8n$, $1.6n$, and $3.2n$, where n is the number of genes; these settings result in datasets which have maximum normalized inversion lengths ranging from approximately 0.1 up to almost 1, the maximum possible.

Performance criteria. We study the performance of trees reconstructed using these five distances (BP, INV, EDE, Approx-IEBP, and Exact-IEBP). We used Neighbor Joining [29], the most frequently used distance-based method, and Weighbor [8], for comparative purposes. We evolved genomes down different GNT model trees, using different values for w_I , w_T , and w_{IT} , thus producing synthetic data (genomes) at the leaves of the trees. During each run, we noted which edges of the model tree have had no events on them (these are the “zero-event” edges); these edges are not included in the comparison to the reconstructed trees. We then computed distances between the genomes, using the five different distance estimators. (Since IEBP requires values for w_I , w_T , and w_{IT} , in order to test robustness we included incorrect as well as correct values for these parameters.) Each distance matrix was then given to Neighbor Joining and Weighbor, thus producing trees for each matrix. These trees were then compared to the *true tree* (the model tree minus the zero-event edges) for topological accuracy.

This accuracy was measured as follows. Each edge e in a tree T defines a bipartition $\pi_e = A \mid B$ on the leaves of T in the obvious way (deleting e separates S into two sets A and B); we let $C(T) = \{\pi_e : e \in E(T)\}$. However, we do not include zero-event edges in the character encoding. Similarly we can define the set $C(T')$, where T' is the inferred tree. The set of *false positives* is $C(T') - C(T)$, and the set of *false negatives* is $C(T) - C(T')$. The false negative and false positive *rates* are obtained by dividing the number of false negatives and false positives, respectively, by $n - 3$ (the number of internal edges in a binary tree on n leaves). The false negative rate is informative of the true tree edges that are found in the inferred tree (i.e. the true positive rate). A low false negative rate does not indicate that the inferred tree obtained is highly resolved and close to the true tree, but only that it does not miss many edges in the true tree. Therefore, when the true tree has very low resolution, a low false positive rate is not indicative of a highly resolved accurate inferred tree. The false negative rate will be most significant when the true tree is close to fully resolved, that is, when the datasets are close to saturation. Our experiments examine performance under all rates of evolution, but the performance under higher rates of evolution allows us to observe whether tree reconstruction can be done accurately when every edge is expected to have changes on it.

Results. In Figs 13.7 and 13.8 we present a sample of the simulation study, showing the accuracy of Neighbor Joining and Weighbor trees constructed using the different distance estimators.

Our model trees have 160 leaves, and we evolve genomes with 120 genes down the model trees. The model conditions include both an inversion-only scenario (GNT(1,0,0)) and a scenario with half inversions and half transpositions/-inverted transpositions (GNT(.5,.25,.25)).

We gave IEBP correct parameter values for w_I , w_T , w_{IT} in this experiment. The model trees have rates of evolution that range from low to almost saturated, as indicated by the x -axis which measures the normalized maximum

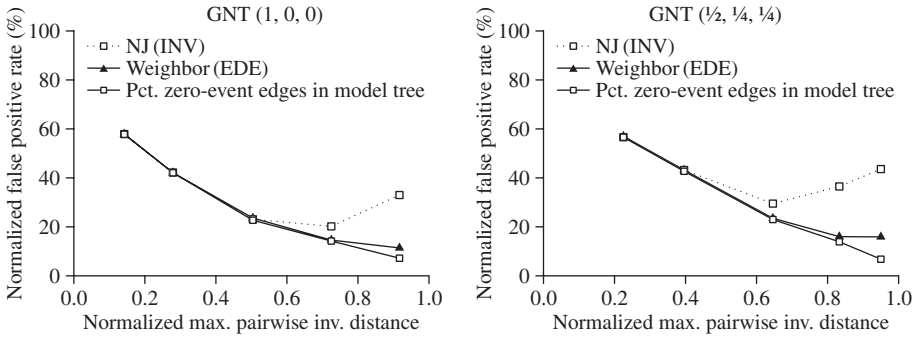


FIG. 13.8. Simulation study of the false positive rates of NJ(INV) and Weighbor(EDE) on 160 circular genomes with 120 genes. We do not include the false positive rates of NJ(EDE) because the curve is very close to that of Weighbor(EDE). The x -axis is the normalized diameter (maximum inversion distance between all pairs of genomes) of the dataset, and the y -axis is the false positive rate. The model of evolution is (left) the Nadeau–Taylor model (i.e. GNT(1, 0, 0)), or (right) the GNT model with half inversions, one-fourth transpositions, and one-fourth inverted transpositions (i.e. GNT($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{4}$)). Refer to Section 13.6.1 for how these figures are generated.

inversion distance in the dataset. For each experimental setting, we bin the datasets according to their diameters (maximum pairwise inversion distance between any two genomes). The x - and y -axis coordinates of each point in the figure are the average diameter and average false negative rates of the corresponding bin, respectively.

False positive rates. Trees returned by Neighbor Joining or Weighbor are always binary. However, since true trees may not be binary (due to the presence of zero-event edges), some false positive edges may be artifacts. In fact, in our experiments, except when quite close to saturation, the true tree will in general be quite unresolved. As a result, any reconstruction method that always returns binary trees will necessarily have a high false positive rate, since the false positive rate must be at least as high as the percentage of edges missing in the true tree. However, in our experiments we see that the false positive rates we obtain generally are not much higher than the percentage of missing edges, indicating quite good performance (see Fig. 13.7).

False negative rates. We see clearly from Fig. 13.8 that for extremely low evolutionary diameters, all methods can reconstruct a good estimate of the true tree, but as the diameter increases, the false negative rates increase for all methods. We also see that overall NJ(BP) has the worst performance, and that Weighbor(IEBP) is generally inferior to the other methods (even when it is given the correct parameter values, for a reason we do not understand). On the other

hand, Weighbor(EDE) is extremely accurate, even when the model condition is not inversion-only. Second best is NJ(EDE), which is also quite accurate even when the model condition is not inversion-only. Thus, although we saw that EDE is not robust to model violations with respect to estimating distances correctly, its apparent linear scaling with the actual distance makes it a good technique for phylogeny reconstruction.

Some of the other trends are also worth noting:

1. As the number of genes increases, the inferred trees become more accurate, at all evolutionary diameters (data not shown). Thus, inferring phylogenies from chloroplast genomes (which contain on average 120 genes) is more reliable than inferring phylogenies from mitochondrial genomes (which contain on average 37 genes).
2. As the number of taxa increases, the inferred trees become less accurate, at all evolutionary diameters (data not shown).
3. Neighbor Joining trees are more accurate when based upon corrected distances (IEBP or EDE) than uncorrected distances (breakpoint or inversion distance). The distinction is the greatest when the dataset has a high evolutionary diameter (i.e. when the dataset contains some pair of genomes that look almost random with respect to each other).
4. NJ(IEBP) and Weighbor(IEBP) perform comparably with incorrect values for the parameters as with correct values; however, Weighbor(IEBP) is not particularly accurate, and neither is as good as NJ(EDE) or Weighbor(EDE).
5. In general, Weighbor(EDE) seems to provide better estimates of evolutionary history than all other methods we examined, especially when the number of genomes and genes are large, and the evolutionary rate is high, but NJ(EDE) is a close second. Both give highly accurate estimations of phylogenies even when the model is not inversion-only.

These observations are specifically for the uniform tree topology case, but most of them hold for other models, including birth–death trees generated by the r8s program [30]. In particular, Weighbor(EDE) is still the most accurate of these methods.

We conclude this section with the following observation. Perhaps the most significant indicator of the difficulty of a dataset is its evolutionary diameter: if the diameter is low, all methods will get a good estimate of the tree, even if the distance estimation is based upon incorrect assumptions, but for the largest diameters (approaching saturation), only Weighbor and NJ on EDE distances are reliably accurate.

13.7 Summary

We have shown that statistically based estimations of evolutionary distances can be quite robust to some model violations, and can help make phylogeny reconstructions much more accurate—especially when the dataset is close to

saturation. However, one of the interesting observations to come out of our experiments is that the accuracy of a phylogeny reconstruction is *usually, but not always* improved by having a better estimate of the evolutionary distance. For example, NJ(EDE) gives better estimates of trees than NJ(IEBP), although IEBP gives more accurate estimates of distances than EDE. Clearly, the interplay between phylogeny reconstruction methods themselves, and the distance estimates, cannot be simply summarized and explained.

Several problems for the GNT model are still open. First, the distribution of the inversion distance is still unknown, as are its expectation and variance. Results along these lines will help us understand why Neighbor Joining based upon the inversion distance gives better results in phylogeny reconstruction than Neighbor Joining based upon the breakpoint distance. Also several studies suggest minimum evolution methods also produce highly accurate trees (see Chapter 1, this volume) for DNA sequence evolution. It will be interesting to see whether minimum evolution methods produce accurate trees for gene-order data.

A maximum likelihood approach for genome rearrangement phylogeny estimation is another approach that will be interesting to explore. MCMC methods are also interesting, but have not been able to scale to reasonable dataset sizes [18]. Maximum likelihood distance estimation is another interesting area to investigate, and it is unknown if the method of moments estimators used for correcting breakpoint and inversion distances are maximum likelihood distance estimators. Another challenging problem is to estimate w_I , w_T , and w_{IT} from the data.

The models we have studied have all presumed that evolutionary events occur with probabilities that only depend upon the type of event. Therefore, a main research question is to explore the estimation of evolutionary distances under newer models of genome evolution. Such models might assume that the probability of the rearrangement events may depend upon the lengths of the affected segments (see [26] for one such model), or may make other assumptions that incorporate hotspots or break the chromosome into distinct regions and require events to stay within these regions [37]. Also of interest are models which allow for deletions, duplications, and other events which change the gene content and not just the gene order. Calculations of distances in these models are much more complicated; initial results along these lines have been obtained by El-Mabrouk, Moret, and others (see [11–14, 19, 34] and Chapters 11 and 12, this volume). Similarly, models which handle multiple chromosomes, and which allow for translocations, need to be considered, and there is much less that has been established for this multichromosomal case than for the unequal gene content case [35, 36].

Finally, as we have noted, the reconstructions of trees we obtain can have a high false positive rate, due to the high incidence of zero-event edges in the model tree (and hence low resolution in the true tree). Determining which edges in the reconstructed tree are valid, and which are not, is a general problem facing phylogenetic analysis. In DNA systematics, bootstrapping and other techniques can be used to assess the confidence in a given edge, and so potentially identify the false positive edges. However, in gene order phylogeny it is not possible to

perform bootstrapping, since there is only one character. Consequently, other techniques would need to be used to identify false positives.

One potential approach would be to use GRAPPA (see [21], and also Chapter 12, this volume) to try to identify the false positives, as follows. First we could assign genomes (i.e. signed circular permutations of $(1, 2, \dots, n)$) to internal nodes in order to minimize the total number of events on the tree, and then we could contract all edges that are assigned the same genomes at the endpoints. Such a technique might be able to identify edges on the tree that have no events on them, but is most likely to succeed when the reconstructed tree is a refinement of the true tree. In our experiments, since the false negative rate is either 0 or close to 0, this would be the case. Future research will investigate this as a potential second phase in the phylogenetic analysis.

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HOW MUCH CAN EVOLVED CHARACTERS TELL US ABOUT THE TREE THAT GENERATED THEM?

Elchanan Mossel and Mike Steel

In this chapter, we review some recent results that shed light on a fundamental question in molecular systematics: how much phylogenetic “signal” can we expect from characters that have evolved under some Markov process? There are many sides to this question and we begin by describing some explicit bounds on the probability of correctly reconstructing an ancestral state from the states observed at the tips. We show how this bound sets upper limits on the probability of tree reconstruction from aligned sequences, and we provide some new extensions that allow site-to-site rate variation or a covarion mechanism. We then explore the relationship between the number of sites required for accurate tree reconstruction and other model parameters—such as the number of species, and substitution probabilities, and we describe a phase transition that occurs when substitution probabilities exceed a critical value. In the remainder of this chapter we turn to models of character evolution where the state space is assumed to be either infinite or very large. These models have some relevance to certain types of genomic data (such as gene order) and here we again investigate how many characters are required for accurate tree reconstruction.

14.1 Introduction

As biologists delve deeper into the evolutionary history of life they often find that sequence data provides conflicting or unclear phylogenetic information. For DNA sequences that have a high site substitution rate the problem of *site saturation* is well known, whereby certain sequences are essentially random with respect to each other due to the number of substitutions that have occurred during their evolution from a common ancestral sequence. For other sorts of data—such as gene-order data, where genomes have undergone much reshuffling—a similar eventual randomization and loss of information also occurs.

The phenomenon of randomization, and the rate at which it occurs, have been well studied in the probability literature—see for example Diaconis [10]. In this setting it is often useful to regard the stochastic process as a random walk on a group. For example, card shuffling, or (unsigned) gene-order rearrangement may be viewed as a random walk on the symmetric group on n elements (i.e. the group consisting of all $n!$ permutations on n elements, under composition) while

site substitution in DNA sequences of length k may be regarded as a random walk on the group $(\mathbb{Z}_2 \times \mathbb{Z}_2)^k$ (since the three types of DNA substitutions—transitions and the two types of transversions—together with an identity forms a group under the operation of composition that is isomorphic to group with elements $(0, 0), (0, 1), (1, 0), (1, 1)$ under componentwise addition; a link that was first noted and exploited by Evans and Speed [14]). An alternative setting to a “random walk on a group” is to consider a random walk on a finite regular connected graph, and most of the examples we have just mentioned can also be viewed from this perspective. Either setting—a random walk on a group, or a random walk on a graph—is just a special type of ergodic Markov chain, for which the usual questions arise, such as what is the limiting distribution, and how fast does the chain approach this limit? Often there is an abrupt transition from non-random to random in a sense that can be formalized and proved. For example, with binary sequences of length k (where k is large) under a model of independent site substitution, this transition occurs when each site has undergone approximately $\frac{1}{4} \log_e(n)$ substitutions—beyond this point the derived sequence quickly becomes essentially random with respect to the first (for a precise rendition of this statement see [10], theorem 3, p. 28). A similar type of transition for gene-order rearrangement under random inversions was recently derived by Durrett [11].

While these questions have been well understood for Markov chains, they have been less thoroughly investigated for the more general setting of Markov processes on trees.

The situation here is interesting for the following reason—as the tree gets larger each leaf tends to become further from the root (and so conveys less information about the ancestral root state) yet the number of leaves also gets larger. It is, *a priori*, not clear whether the gain in information provided by more leaves compensates for the losses experienced by each leaf. This question is also familiar in biology—does the sampling of more species provide a strategy for coping with site saturation? As we will see, these questions are relevant not just for reconstructing ancestral character states, but also for inferring phylogenetic trees.

Evolution processes may be often viewed as Markov processes on trees. These processes are in turn a special family of Markov random fields on trees, the study of which is an important branch of *statistical physics*—see [20] for general background and [13, 23, 30, 34, 38] for results regarding Markov processes on trees. The theory of Markov random fields (and processes) on trees is used to investigate problems such as ancestral reconstruction of states, which is familiar in both biology and physics. In contrast, the problem of reconstructing the tree topology, which is well-studied in biology, seems not to have been addressed in the statistical physics literature.

In this chapter, we survey some of the recent advances in the information-theoretic treatment of Markov processes on trees. We begin by dealing with Markov processes on a fixed (small) state space—for example, nucleotide sequence data. Here we describe information-theoretic limits that place bounds on the extent to which ancestral states and deep divergences can be resolved from

sequence data. We also consider the question of how much sequence information is required to accurately reconstruct a tree, a question where there remains an interesting unresolved issue. We then turn to the analysis of characters on state spaces that are large or infinite, and which exhibit a somewhat different (and more tractable) behaviour. Along the way we will indicate how such character data may be relevant to the analysis of genomic data such as gene order.

14.2 Preliminaries

In this section, we describe some background and notation concerning phylogenetic trees and Markov processes on trees—readers familiar with these topics may wish to skim over this material.

14.2.1 *Phylogenetic trees*

Throughout this chapter X is a finite set and we will let $n = |X|$. A *phylogenetic X -tree* (or more, briefly, a *phylogenetic tree*) is a tree $T = (V, E)$ having leaf set X , and for which the interior vertices are unlabelled and of degree at least 3. If in addition each interior vertex has degree exactly 3 we say that T is *trivalent*. In evolutionary biology, the set X typically represents the extant species (or sequences) while the remaining vertices of the tree represent speciation events (or unknown ancestral sequences). Trivalent trees (also sometimes called “fully-resolved”) are regarded as the most informative as they contain no “polytomies” (vertices of degree >3 that generally represent uncertainty as to the actual order of speciation).

Two phylogenetic X -trees T and T' are regarded as equivalent if the identity map on X , regarded as a bijection from the set of leaves of T to the leaves of T' extends to a graph isomorphism between the two trees. Thus, for example, there are precisely three trivalent (and one non-trivalent) phylogenetic X -trees for any set X of size 4. Less formally, two phylogenetic X -trees are equivalent if they describe the same graphical relationships between the species in X , even though the trees might be drawn differently in the plane.

We are also interested in *rooted* phylogenetic X -trees. Briefly, a rooted phylogenetic tree is obtained from a phylogenetic tree by either distinguishing some interior vertex as a root, or by subdividing an interior edge and calling the new degree-two vertex a root. We denote the root of a rooted phylogenetic tree T by ρ , and direct all edges away from the root. For a rooted phylogenetic tree T we will use throughout this chapter the word *topology* to denote the associated unrooted phylogenetic tree (obtained from T by suppressing the root, and if it is of degree 2 identifying its two incident edges). A rooted phylogenetic tree is said to be *binary* if each non-leaf vertex has precisely two outgoing arcs. Thus a phylogenetic tree is binary precisely if its topology is trivalent. For more background on the mathematics of phylogenetic trees the reader is referred to reference [46].

14.2.2 *Markov processes on trees*

Let $\mathcal{C}$ be the set of character states (such as $\mathcal{C} = \{0, 1\}$, $\mathcal{C} = \{A, C, G, T\}$, or $\mathcal{C} = \{20 \text{ amino acids}\}$). In keeping with biological convention we will often refer

to a site aligned across a set of species X as a *character* on X ; mathematically it is simply a function from X to $\mathcal{C}$. To model the evolution of characters on a rooted phylogenetic tree T by a Markov process we associate to each directed edge e of T a matrix $M(e)$ of transition probabilities, and to the root vertex of T we associate a distribution π of states (see [12] or [48] for a more formal description of the model).

Many of the standard models in biology satisfy $M(e) = \exp(t(e)Q)$, where $Q = (q_{i,j})_{i \in \mathcal{C}, j \in \mathcal{C}}$ is the transition rate matrix and $t(e)$ represents the “length” of the edge e over which the Markov process operates. Furthermore, π is generally taken to be the equilibrium distribution that satisfies $\pi Q = 0$, so as to induce a stationary Markov process.

The simplest 2-state model is the symmetric *Cavender–Farris–Neyman (CFN) model*

$$Q = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

For this model the probability $p(e)$ of a substitution on any edge e of the tree is given by

$$p(e) = \frac{1}{2}(1 - \exp(-2t(e))). \quad (14.1)$$

With 4 states a slightly more general class of models is the Tajima and Nei’s “equal input” model

$$Q = \begin{pmatrix} -(a+b+c) & a & b & c \\ d & -(b+c+d) & b & c \\ d & a & -(a+c+d) & c \\ d & a & b & -(a+b+d) \end{pmatrix}.$$

In case $a = b = c = d (= r, \text{ say})$ this is known as the *Jukes–Cantor model*:

$$Q = \begin{pmatrix} -3r & r & r & r \\ r & -3r & r & r \\ r & r & -3r & r \\ r & r & r & -3r \end{pmatrix}. \quad (14.2)$$

Both of these models lead to reversible Markov processes. See [17], and Chapters 2 and 6 of this volume, for various other families of substitution matrices Q appearing in biology.

A further embellishment of most contemporary models of nucleotide substitution is the inclusion of *site specific rates* (Chapter 5, this volume). That is, one has a distribution $\mathcal{D}$ on some real-valued parameter (the “rate” of evolution of a site) and each site i in the sequence evolves at a rate λ_i that is chosen independently from this distribution. We refer to the distribution that assigns rate 1 to each site with probability 1 as the *degenerate distribution*.

The substitution process is therefore defined by a transition rate matrix Q , a distribution $\mathcal{D}$ of site specific rates, a rooted phylogenetic tree $T = (V, E, \rho)$,

a collection of edge lengths $t: E \rightarrow \mathcal{R}_+$ and a probability distribution π on the states at the root vertex of T .

A *configuration* $\sigma: V \rightarrow \mathcal{C}$ is a labelling of the vertices of T by $\mathcal{C}$. We will write σ_v for the value of σ at the vertex $v \in V$. The distribution of σ_ρ is given by π . If u is v 's parent, then the conditional distribution of σ_v given σ_u at site i is given by the matrix $M(e) = \exp(\lambda_i t(e) Q)$, where $e = (u, v)$. We will denote the collection of leaves of the tree T by ∂T and the value of a configuration σ at the leaves by σ_∂ (which is a character on X —that is, a function from X into the set $\mathcal{C}$).

14.3 Information-theoretic bounds: ancestral states and deep divergences

In this section, we describe explicit and easily computable upper bounds on the information that extant sequences provide concerning (1) ancestral sequences and (2) the branching pattern deep inside a tree. These bounds are in a sense the simplest bounds that can be put on the reconstruction of ancestral states.

For a leaf v , let $\text{path}(v)$ be the set of edges on the path connecting v to the root ρ , and let

$$t(v) = \sum_{e \in \text{path}(v)} t(e).$$

The *molecular clock assumption* is that $t(v)$ takes the same value for each v ; we do not make this assumption anywhere in this chapter, even though we will refer to sums of $t(e)$ values as (elapsed) “time.”

Let π be the prior distribution of the root character, and let

$$\Delta = \sup_f \mathbb{P}[f(\sigma_\partial) = \sigma_\rho] \quad (14.3)$$

be the optimal probability of reconstructing the value of σ_ρ given σ_∂ , where the sup is taken over all functions. Assuming that the parameters of the model (i.e. T , the $t(e)$ values and the root state distribution π) are known, it follows from a classic result (e.g. see theorem 17.2 of reference [22]) that an optimal choice of f is the maximum posterior probability (MAP) estimator—that is, given σ_∂ one select the root state(s) j to maximize

$$\mathbb{P}[\sigma_\partial \mid \sigma_\rho = j] \cdot \pi[\sigma_\rho = j]$$

—a task that can be carried out by an efficient (polynomial-time in n) dynamic programming approach.

It also follows from standard information-theoretic theory (theorem 17.3 of reference [22]) that the following lower bound on Δ applies:

$$\Delta \geq 2^{-H(\sigma_\rho \mid \sigma_\partial)}, \quad (14.4)$$

where $H(\sigma_\rho \mid \sigma_\partial)$ is the *conditional entropy* of σ_ρ given σ_∂ is defined by

$$H(\sigma_\rho \mid \sigma_\partial) = - \sum_{i, \sigma} \mathbb{P}[\sigma_\rho = i, \sigma_\partial = \sigma] \log_2(\mathbb{P}[\sigma_\rho = i \mid \sigma_\partial = \sigma]).$$

Note that in general one cannot expect to recover the root state with probability close to 1. Consider, for example, the Jukes–Cantor model. Even given the state of the children of the root there is a non-negligible probability that mutation events occurred along the two edges adjacent to the root and conditioned on this event the state of the root is independent from the rest of the character.

As we will see in Section 14.4, there are various asymptotic results in statistical physics dealing with the limiting behaviour of $H(\sigma_\rho \mid \sigma_\partial)$ and Δ , but the bounds on Δ in most of these results are not explicit. A notable exception is [13] where a bound on Δ for the CFN model is given in terms of “electrical-resistance” of an electrical network defined on the tree.

However, our main interest here is in providing explicit *upper* bounds on Δ , which we now describe. As the rate of substitution increases and/or the temporal separation of the root of the tree from the leaves increases, we would expect it to become increasingly difficult to recover the root state—a phenomenon well known to biologists as “site saturation.” However, it will be important (particularly for later results) to quantify this rate of decay of information. The following result, which is a slight extension of a result from [36], follows by easy adaptations of coupling arguments appearing earlier in statistical physics, see, for example [34]. We let $M_{\mathcal{D}}(x) = \mathbb{E}_{\mathcal{D}}[e^{\lambda x}]$ the moment generating function of the site specific rate distribution $\mathcal{D}$. Note that, for the degenerate site specific rate distribution we have $M_{\mathcal{D}}(x) = e^x$.

Theorem 14.1 *Consider a Markov model on a tree T , with transition rate matrix Q , edge lengths $t(e)$ (for each edge e of T), and site specific rate distribution $\mathcal{D}$. Let*

$$q_j = \min_{i \neq j} q_{i,j}, \quad q = \sum_j q_j. \quad (14.5)$$

Then the optimal reconstruction probability Δ for the root state satisfies

$$\Delta \leq \max_i \pi[\sigma_\rho = i] + \sum_{v \in \partial T} M_{\mathcal{D}}(-qt(v)). \quad (14.6)$$

Note that the first term in equation (14.6) is precisely the estimate one would make if one had no knowledge of the character states at the leaves of T . Thus Theorem 14.1 says that the improvement over this “trivial” method decays as the expected exponential of $-qt(v)$. Notice also that Theorem 14.1 assumes that T and the values $t(e)$ are all known exactly—if they are not, then the bound on Δ described applies *a fortiori*.

The proof of Theorem 14.1 utilizes the method of coupling where one relates one stochastic process to another that is easier to analyse (e.g. see [1] for background on coupling for Markov chains). The style of argument employed here has been applied to the study of percolation (see [34], and [2, 43] for background). We outline this argument now. First, we establish the result for the special case of constant site specific rate, where each site is assigned rate λ with probability

1. The substitution rate from state i to state j is given by $q_{i,j}$. Recalling equation (14.5), we may define the process equivalently as follows. Given the current state i ,

(J1) jump to state j with rate λq_j ;

(J2) jump to state j with rate $\lambda(q_{i,j} - q_j)$.

The coupling argument relates this process (involving both (J1) and (J2)) to the simpler process involving just (J1). The crucial point here is that (J1) is performed independently of the state i . For edge $e = (u, v)$, let $D(e)$ be the event that a transition of type (J1) occurs along the edge e . Note that the events $D(e)$ are independent for different edges and that $\mathbb{P}[D(e)^c] = \exp(-q\lambda t(e))$, where $D(e)^c$ denotes the (complimentary) event that $D(e)$ does not occur. Moreover, conditioned on $D(e)$, σ_v is independent of σ_ρ . For a leaf v , let $D(v)$ be the event that a transition of type (J1) occurs along an edge $e \in \text{path}(v)$. Then

$$\mathbb{P}[D(v)^c] = \prod_{e \in \text{path}(v)} \mathbb{P}[D(e)^c] = \prod_{e \in \text{path}(v)} e^{-q\lambda t(e)} = e^{-q\lambda t(v)}.$$

Finally, let D be the event that $D(v)$ holds for all leaves $v \in \partial T$. Then

$$\mathbb{P}[D^c] \leq \sum_{v \in \partial T} \mathbb{P}[D(v)^c] = \sum_{v \in \partial T} e^{-q\lambda t(v)}. \quad (14.7)$$

Note that conditioned on D , σ_∂ and σ_ρ are independent.

To prove the bound on reconstruction of equation (14.6), note that if we are not given σ_∂ (or any other information on σ_ρ), then the best reconstruction function f satisfies $f \equiv j$, where j maximizes $\pi[\sigma_\rho = i]$ over all i , and this function has success probability $\max_i \pi[\sigma_\rho = i]$. Now let f be any reconstruction procedure and note that, conditional on the event D , σ_ρ is independent of σ_∂ and therefore

$$\begin{aligned} \mathbb{P}[f(\sigma_\partial) = \sigma_\rho] &\leq \mathbb{P}[D^c] + \mathbb{P}[D]\mathbb{P}[f(\sigma_\partial) = \sigma_\rho \mid D] \\ &\leq \mathbb{P}[D^c] + \mathbb{P}[D] \max_i \pi[\sigma_\rho = i] \leq \mathbb{P}[D^c] + \max_i \pi[\sigma_\rho = i], \end{aligned}$$

and so

$$\mathbb{P}[f(\sigma_\partial) = \sigma_\rho] \leq \max_i \pi[\sigma_\rho = i] + \sum_{v \in \partial T} e^{-q\lambda t(v)}. \quad (14.8)$$

Now, consider the case of a general site specific rate distribution $\mathcal{D}$. Clearly, Δ is the expected value (with respect to $\mathcal{D}$) of the conditional probability $\mathbb{P}[f(\sigma_\partial) = \sigma_\rho \mid \lambda]$ which we may identify with the LHS of equation (14.8). Consequently,

$$\Delta \leq \mathbb{E}_{\mathcal{D}}[\max_i \pi[\sigma_\rho = i]] + \mathbb{E}_{\mathcal{D}}\left[\sum_{v \in \partial T} e^{-q\lambda t(v)}\right] = \max_i \pi[\sigma_\rho = i] + \sum_{v \in \partial T} M_{\mathcal{D}}(-q\lambda t(v))$$

as required. $\square$

Example To illustrate Theorem 14.1 let us consider the simplest model on four states, namely the Jukes–Cantor model defined by equation (14.2) with a degenerate site specific rate distribution and a molecular clock. For this model the equilibrium distribution for states is uniform, so it is natural to take $\pi[\sigma_\rho = i] = \frac{1}{4}$ for all four choices of i . Now suppose we wish to infer the ancestral state at a vertex in a tree that was present t years ago, using the states observed now among the n extant descendant species. Theorem 14.1 provides the following bound on Δ :

$$\Delta \leq \frac{1}{4} + ne^{-qt}$$

and we may identify the product $\frac{3}{4}qt$ with the expected number of substitutions that occur on any path from the root to a leaf. For example, if the substitution rate is constant at (say) 1 substitution per million years, and we have a tree with $n = 100$ leaves whose root is at least 10 million years in the past then $\Delta \leq \frac{1}{4} + 0.0002$ so a character tells us virtually nothing to help us estimate the state that occurred at the root.

Notice that some restriction must be placed on the entries of Q for a bound such as that given by equation (14.6) to be useful. For example, consider a process with three states, with $\pi[\sigma_\rho = i] = \frac{1}{3}$ for each value of i , and with

$$Q = \begin{pmatrix} -2r & r & r \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

for which $q = 0$. Then it can be checked that $\Delta \geq \frac{2}{3}$, however, we also have $\max_i \pi[\sigma_\rho = i] = \frac{1}{3}$ so that for this example, Δ is always bounded away from $\max_i \pi[\sigma_\rho = i]$.

However, Theorem 14.1 can be extended to provide some (exponential-decay) bounds similar to equation (14.6) for certain choices of Q for which $q = 0$. A case in point is the class of “covarion-type” models (see [19, 42, 53]) in which each state can either be in an “on” mode or an “off” mode. A state that is “on” is free to change to other “on” states, or to turn “off” (at various rates), while a state that is “off” is only free to turn “on” (at some rate). For two base states and therefore a total of 4 states, namely $0_{\text{on}}, 1_{\text{on}}, 0_{\text{off}}, 1_{\text{off}}$ the corresponding rate matrix Q can be written as:

$$Q = \begin{pmatrix} -(r_1 + u) & r_1 & u & 0 \\ r_2 & -(r_2 + u) & 0 & u \\ v & 0 & -v & 0 \\ 0 & v & 0 & -v \end{pmatrix} \quad (14.9)$$

and for this matrix it is immediately clear that $q = 0$.

In order to obtain bounds for such models, it is better to apply the coupling argument directly to the matrices $M(e)$. Note that, for simplicity, we will also assume all the “on” sites undergo substitution at the same rate ($\lambda = 1$). Given any real matrix A let $m_j(A) = \min_i A_{i,j}$ and $m(A) = \sum_j m_j(A)$. Write

$m_j(e)$ for $m_j(M(e))$ and $m(e)$ for $m(M(e))$. On the edge e , the transition process can be described equivalently as follows: Given the current state i ,

(J1) jump to state j with probability m_j ;

(J2) jump to state j with probability $m_{i,j} - m_j$.

Note that, as before, (J1) is performed independently of the state i . Repeating the above argument we thus obtain the following bound on the reconstruction probability

$$\Delta \leq \max_i \pi[\sigma_\rho = i] + \sum_{v \in \partial T} \prod_{e \in \text{path}(v)} (1 - m(e)). \quad (14.10)$$

For a given tree and substitution matrices we may apply bound (14.10) directly. However, unlike Theorem 14.1, here it is not enough to know for all leaves the total time elapsing from the root. Instead, all the edge lengths are needed.

More can be said if the process described by Q is ergodic (maybe with 0 entries) so that, for $\epsilon > 0$, $\exp(\epsilon Q)$ has all its entries positive. Let us assume that the length of all branches is at least ϵ and let $\alpha = \sqrt{1 - m(\exp(\epsilon Q))}$ and note that $\alpha < 1$.

Note that if A and B are two stochastic matrices, then

$$1 - m(AB) \leq (1 - m(A))(1 - m(B)).$$

Thus, if $t > \epsilon$, then

$$1 - m(\exp(tQ)) \leq 1 - m\left(\exp\left(\epsilon \left\lfloor \frac{t}{\epsilon} \right\rfloor Q\right)\right) \leq (\alpha^2)^{\lfloor t/\epsilon \rfloor} \leq (\alpha^2)^{t/2\epsilon} = \alpha^{t/\epsilon}.$$

Substituting this into bound (14.10) we obtain that

$$\Delta \leq \max_i \pi[\sigma_\rho = i] + \sum_{v \in \partial T} \alpha^{t(v)/\epsilon}. \quad (14.11)$$

Note the similarity between this expression and the one in Theorem 14.1. In particular, in order to apply this bound it suffices to know for each leaf the total time elapsed from the root.

Example Consider the case where the rates in matrix (14.9) are given by $r_1 = r_2 = u = v = 1$ per 1 million years. Note that since Q is symmetric the stationary distribution is given by the uniform distribution.

Assume furthermore that length of all branches is at least $\epsilon = 0.25$. Using numerical analysis software (e.g. Mathematica) we find that

$$\exp(0.25 \times Q) = \begin{pmatrix} 0.646645 & 0.156621 & 0.175773 & 0.020962 \\ 0.156621 & 0.646645 & 0.020962 & 0.175773 \\ 0.175773 & 0.020962 & 0.801456 & 0.001809 \\ 0.020962 & 0.175773 & 0.001809 & 0.801456 \end{pmatrix}.$$

Therefore $m_1 = m_2 = 0.0209616$ and $m_3 = m_4 = 0.00180925$. Thus, $m(\exp(0.25 \times Q)) = 2 \times 0.0209616 + 2 \times 0.00180925 = 0.0419232$ and $\alpha = \sqrt{1 - m} = 0.978814$.

Suppose we now have a tree with $n = 100$ leaves and we want to infer the ancestral state of a state that was present t million years ago. We thus obtain from equation (14.11) that

$$\Delta \leq \max_i \pi[\sigma_\rho = i] + n\alpha^{t/\epsilon} = \frac{1}{4} + 100\alpha^{4t}.$$

In particular if $t = 100$ million years, the probability of reconstructing the ancestral state correctly is at most $0.25 + 0.000190498$. So again, the character reveals essentially no information about the ancestral state.

14.3.1 Reconstructing deep divergences

Theorem 14.1 allows one to place bounds on the extent to which sequences can resolve a divergence event deep inside a phylogeny. Consider, for example, four monophyletic groups of taxa for which we have aligned sequences of length k . We may wish to determine which of the three possible phylogenetic trees connect these four groups, as illustrated on the left of Fig. 14.1.

Clearly, it will only help us in this task if we know the tree topologies of each of the four monophyletic groups together with their $t(e)$ values. Each sequence site provides a portion of information concerning the “deep” tree structure (i.e. which of the three possible phylogenetic trees connect the four subtrees) and it is possible to explicitly bound the information that the entire sequences provide concerning this divergence. In this way one can set explicit lower bounds on the number of sites would be needed in order to resolve a deep divergence. One such bound was described, for the CFN model, in reference [47]. Here we describe a more general approach from [36] that applies to a wider range of models and settings.

Let $T(s)$ denote the topology of tree T up to time s from the root, and let $T^c(s)$ denote the forest consisting of the subtrees from time s to the present (including the associated edge lengths). In other words, $T(s)$ describes all divergences up to time s , while $T^c(s)$ describes all divergences (and their relative separations) from time s , as illustrated on the centre and right of Fig. 14.1.

Consider the problem of reconstructing $T(s)$ (given $T^c(s)$) from a sequence of characters that are generated by a common Markov process on T , where the

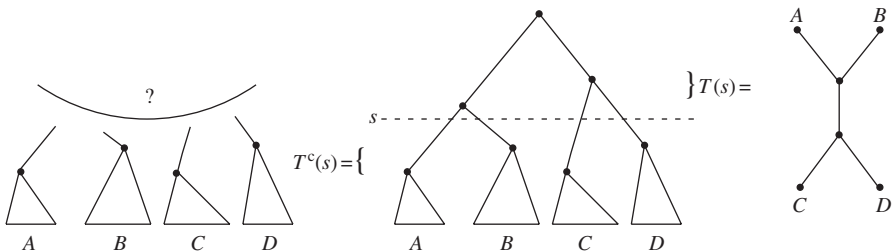


FIG. 14.1. Left: An example of a deep divergence involving four subtrees. Centre and Right: The forest $T^c(s)$ and the tree $T(s)$.

prior distribution on $T(s)$ is given by a measure μ . The prior μ is on $T(s)$ with its edge lengths. However, for a tree topology T , we will write $\mu[T(s) = T]$ for the prior probability that the topology of $T(s)$ is given by T .

Note that in the following result (Theorem 14.2) we do not need to assume independence between sites that evolve according to this process on T .

Let us denote $\sigma^1, \dots, \sigma^k$ a sequence of k identically generated configurations. We will also denote the values of the configuration σ^i at the leaves by σ_∂^i . Similarly, we denote by σ_ρ^i the value of the configuration σ^i at the root ρ . Suppose furthermore, that the characters evolve as in Theorem 14.1 with substitution matrix Q ; and we have a site specific rate distribution $\mathcal{D}$. Let $\Delta^T(s)$ be the probability of reconstructing, given $T^c(s)$ (with its associated $t(e)$ values) the tree topology up to time s ,

$$\Delta^T(s) = \sup_f \mathbb{P}[f((\sigma_\partial^j)_{j=1}^k) = T(s) \mid T^c(s)]. \quad (14.12)$$

The sup is taken over all functions, and as before, the optimal choice of f is the maximum posterior probability (MAP) estimator, which given $(\sigma_\partial^j)_{j=1}^k$ selects a tree T' to maximize

$$\int 1_{\{T(s)=T'\}} \mathbb{P}[(\sigma_\partial^j)_{j=1}^k \mid T(s), T^c(s)] d\mu(T(s)),$$

(where $1_{\{T(s)=T'\}}$ is 1 or 0 depending on whether the topology of $T(s)$ is T' or not). Clearly the probability of reconstructing T from $(\sigma_\partial^j)_{j=1}^k$ is less or equal to $\Delta^T(s)$; this latter quantity, which is the probability of correctly determining the “deep” part of the tree, can be bounded as follows.

Theorem 14.2 *Suppose that k sites evolve under a Markov process with a site specific rate distribution $\mathcal{D}$. Then, for any $s > 0$ we have:*

$$\Delta^T(s) \leq \max_T \mu[T(s) = T] + k \sum_{v \in \partial T} M_{\mathcal{D}}(-q(t(v) - s)), \quad (14.13)$$

where q is given by equation (14.5).

Outline of the proof. The argument follows similar lines to the proof of Theorem 14.1. For character i we say that event D_i occurs if, for all $v \in \partial T$ there exists a time $t \geq s$ at which a transition of type **(J1)** occurs at least once on the path connecting v to the root of the component of $T^c(s)$ that contains v . By the proof of Theorem 14.1 it follows that

$$\mathbb{P}[D_i^c \mid \lambda_i] \leq \sum_{v \in \partial T} e^{-\lambda_i q(t(v) - s)},$$

where λ_i is the rate (chosen from $\mathcal{D}$) that site i evolves at. Consequently,

$$\mathbb{P}[D_i^c] \leq \sum_{v \in \partial T} M_{\mathcal{D}}(-q(t(v) - s)),$$

and so, by the Bonferroni inequality,

$$\mathbb{P}[(\cap_{i=1}^k D_i)^c] \leq k \sum_{v \in \partial T} M_{\mathcal{D}}(-q(t(v) - s)).$$

Now, conditional on $\cap_{i=1}^k D_i$, the two random variables $(\sigma_s^i)_{i=1}^k$ and $(\sigma_{\partial}^i)_{i=1}^k$ are independent, and therefore, $T(s)$ and $(\sigma_{\partial}^i)_{i=1}^k$ are independent. As in Theorem 14.1 we conclude that

$$\begin{aligned} \Delta^T(s) &\leq \mathbb{P}[(\cap_{i=1}^k D_i)^c] + \mathbb{P}[(\cap_{i=1}^k D_i)] \max_T \mu[T(s) = T] \\ &\leq k \sum_{v \in \partial T} M_{\mathcal{D}}(-q(t(v) - s)) + \max_T \mu[T(s) = T], \end{aligned}$$

as required.

Example To illustrate Theorem 14.2 let us consider again the Jukes–Cantor model defined by equation (14.2), with a degenerate site specific rate distribution and molecular clock. Suppose we have four monophyletic groups of taxa—each with 100 extant species, and with a well-specified tree with edge lengths—and we wish to determine which of the three possible trees (choices for $T(s)$) describes how the trees are joined ancestrally (as in Fig. 14.1). In the absence of any prior information it is natural to take $\mu[T(s) = T'] = \frac{1}{3}$ for each of the three possible trivalent trees T' . Suppose it is believed that all four lineages existed as far back as (at least) 1 billion years ago, and taking, for example, a site substitution rate $(3r)$ of one substitution per 50 million years, we have for any leaf v that $qt(v) = 4rt(v) = \frac{4}{3} \cdot (3r)t(v) = \frac{4}{3} \cdot 20$. Theorem 14.2 then gives $\Delta^T(s) \leq \frac{1}{3} + 100ke^{-26.7}$ which implies that at least 700 million sites (!) would be required in order to have any hope of estimating the ancestral divergence with probability more than about 0.5. This is perhaps not too surprising given that the expected number of substitutions per site along the path from the root to any leaf is 20. $\square$

Remarks

- (1) As noted above, Theorem 14.2 applies even when the sequence sites are not independent. It is possible to extend this theorem further to allow the sites to evolve according to different Markov processes.
- (2) In order to get a feeling for the asymptotic behaviour of equation (14.13), fix s and assume that the tree has $n = e^{\beta t}$ leaves, all at time t . Here we take the asymptotics where $t \rightarrow \infty$ (and therefore $n \rightarrow \infty$), while s, q , and β are all constants. Also we assume a degenerate site specific rate distribution. Then

$$\sum_{v \in \partial T} e^{-q(t(v) - s)} = \exp(sq) \exp(-t(q - \beta)).$$

Therefore if $q > \beta$, then by equation (14.13) if we want to reconstruct the topology up to time s with high probability, that is, $\Delta^T(s) \geq \max_T \mu[T(s) = T] + \delta$, where $\delta > 0$ then we need that

$$k \geq \delta \exp(-sq) \exp(t(q - \beta)) = \delta \exp(-sq) n^{q/\beta - 1}.$$

So the number of characters required grows polynomially with n .

14.3.2 Connection with information theory

Similar bounds to the ones we have described so far can also be stated and derived using classical information theory. First we briefly recall the concept of mutual information. For random variables X and Y the *mutual information* between X and Y is defined by

$$I(X; Y)(= I(Y; X)) := \sum_{x,y} \mathbb{P}[X = x, Y = y] \log_2 \left(\frac{\mathbb{P}[X = x, Y = y]}{\mathbb{P}[X = x]\mathbb{P}[Y = y]} \right).$$

Formally, $I(X; Y)$ is the Kullback–Leibler separation of the joint distribution of X, Y and the product distribution of X and Y . Consequently, $I(X; Y) \geq 0$ with equality if and only if X and Y are independent. Informally $I(X; Y)$ measures the amount of information that Y carries about X (or conversely that X carries about Y). When $I(X; Y)$ is small then the best method for inferring Y from X does little better than the best method that simply ignores X —a precise formalization of this claim is Fano’s inequality (see [8] for more details).

The quantity I has some generic properties that make it useful for analysing the information loss of Markov processes. For example, suppose that X, Y , and Z be random variables such that X and Z are independent given Y . Then $I(X; Z) \leq \min\{I(X; Y), I(Y; Z)\}$ (the “data processing lemma”) and $I((X, Z); Y) \leq I(X; Y) + I(Z; Y)$ (the “subadditivity property”). By exploiting these properties one can derive information-theoretic analogues of Theorems 14.1 and 14.2 which we will now briefly describe. For convenience we will deal just with the degenerate site distribution in both cases. In the setting of Theorem 14.1 it can be shown that

$$I(\sigma_{\partial}; \sigma_{\rho}) \leq \log_2 |\mathcal{C}| \sum_{v \in \partial T} e^{-qt(v)}.$$

Similarly, in the setting of Theorem 14.2 it can be shown that

$$I(T(s); (\sigma_{\partial}^j)_{j=1}^k \mid T^c(s)) \leq k \sum_{v \in \partial T} e^{-q(t(v)-s)}.$$

For further details, and applications of these results, see [36].

The results described in this section may give the impression that phylogenetic information decays in a smooth fashion according to an interplay of time, substitution rate, and numbers of leaves in the tree. However, as we explain in the next section there are underlying transitions in this behaviour.

14.4 Phase transitions in ancestral state and tree reconstruction

There is an interesting change (“phase transition”) in the behaviour of Markov models of character evolution on trees as the probability of substitution on edges of the trees passes a certain critical value. This has been well studied in statistical physics and in information theory, in the context of broadcasting on trees.

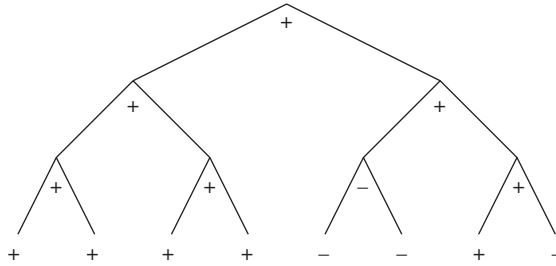


FIG. 14.2. A character of the CFN process on a binary phylogenetic tree on $8 = 2^3$ leaves at distance 3 from the root.

But it is also relevant to biology—particularly in attempting to recover information (ancestral states, branching order) deep within a tree, from observing the character states at the leaves.

The transition is most easily explained, and has been most studied for the case of the 2-state symmetric process (the CFN model described above).

To illustrate this transition between what is called the “ordered” and “unordered” phases of a Markov process on a tree, suppose we have a rooted binary phylogenetic tree T that has $n = 2^m$ leaves that are at distance m from the root vertex, as indicated in Fig. 14.2.

Under the CFN model (and with a degenerate site specific rate distribution) let

$$\theta(e) := \det(M(e)) = \det(\exp(t(e)Q)),$$

where, for any square matrix M , $\det(M)$ denotes the determinant of M (the product of the eigenvalues of M). A classic identity in linear algebra, Jacobi’s identity, states $\det \exp(M) = \exp(\text{tr}(M))$ where $\text{tr}(M)$ is the trace of M (the sum of the diagonal entries of M , which also equals the sum of the eigenvalues of M). Thus,

$$\theta(e) = \exp(t(e)\text{tr}(Q)) = e^{-2t(e)}.$$

By equation (14.1) we have $\theta(e) = 1 - 2p(e)$. Now suppose that each edge of T has the same $t(e)$ value, say t , and thereby the same $\theta(e)$ value, namely $\theta = \exp(-2t)$.

Let us further suppose that the distribution π of states at the root is uniform (i.e. a fair coin toss) and that we wish to use the states $\sigma_\partial = (\sigma_\partial^i)$ at the leaves of T to estimate the state σ_ρ at the root. This gives rise to an interesting contest as m (the height of the tree) increases—first, each leaf is becoming increasingly far from the root, and so the information that it carries about the ancestral root state decays to 0 with increasing m . On the other hand, the number of leaves grows (exponentially) with m , and so although each leaf carries less information, it might be hoped that together they compensate for their individual losses. Which factor wins out depends critically on the value of θ . Evans *et al.* [13] established that, for $2\theta^2 < 1$ the mutual information $I(\sigma_\partial, \sigma_\rho)$ converges to 0, as m tends to infinity (this result was first proven independently by Bleher *et al.* [3]

in a different formulation). Thus, eventually (as the root becomes increasingly “deep” in the tree) it becomes impossible to estimate the root state with any better success than a blind guess, when θ lies in this region. On the other hand, when $2\theta^2 > 1$ then $I(\sigma_\partial; \sigma_\rho)$ is bounded away from 0, so that information about the root “survives” to the leaves, no matter how large the tree grows. In this case maximum likelihood estimation (MLE) or majority rule estimation (i.e. select the root state that corresponds to the majority state at the leaves) suffices to recover some information right up to (but not including [40]) the critical value $2\theta^2 = 1$.

Notice that this critical value translates to a common $t(e)$ value of $t = \frac{1}{4} \log(2)$ and thereby to a common $p(e)$ value of $p = \frac{1}{2}(1 - (1/\sqrt{2}))$.

Curiously, the maximum parsimony (MP) approach for ancestral state reconstruction (i.e. select the root state that requires the fewest transitions to account for the leaf states) recovers information under the CFN model for values of p only up to $\frac{1}{8}$ [5].

The situation for r -states models and for non-symmetric 2-state processes is more subtle. There is not any general criteria for deciding when the mutual information $I(\sigma_\partial; \sigma_\rho)$ is converging to 0 and when is it bounded away from 0. In fact, such criteria do not even exist for symmetric processes on more than 2 states or for general processes on 2 states. In the general setting, there are various conditions which imply that the mutual information either converges to 0 or is bounded away from 0. However, these conditions are not sharp. We describe an example of both types of conditions now.

Suppose that $M(e) = M$ for all e . Since M is a stochastic matrix, 1 is an eigenvalue of M . Let $\{1 = \lambda_1, \dots, \lambda_r\}$ denote the set of eigenvalues of M and let $\theta = \max\{|\lambda_2|, \dots, |\lambda_r|\}$ (note that for the CFN model, this is consistent with the previous definition of θ). A “spectral criterion” (see [26, 38]) implies that for any M if $2\theta^2 > 1$ then $I(\sigma_\partial; \sigma_\rho)$ is bounded away from zero for all trees. This result is not tight in general (see [25, 34, 38]).

In order to illustrate the spectral criterion consider the Jukes–Cantor model defined by equation (14.2). Note that the eigenvalues of Q are 0 (with multiplicity 1) and $-4r$ (with multiplicity 3). Thus if $M = \exp(tQ)$, then the eigenvalues of M are 1 and e^{-4rt} . Therefore, if the stochastic matrix $M = \exp(tQ)$ satisfies

$$A := 2e^{-8rt} > 1, \quad (14.14)$$

then by the spectral criterion $I(\sigma_\partial; \sigma_\rho)$ is bounded away from zero for all trees. This should be compared to Theorem 14.2 which implies that if

$$B := 2e^{-4rt} < 1 \quad (14.15)$$

and the tree is of depth at least d , then the probability of reconstructing the ancestral states is bounded by $\frac{1}{4} + (2e^{-4rt})^d$. The expression $\frac{1}{4} + (2e^{-4rt})^d$ converges to $\frac{1}{4}$ when $d \rightarrow \infty$. Thus, condition (14.14) implies that then $I(\sigma_\partial; \sigma_\rho)$ is bounded away from zero for all trees, while condition (14.15) implies that $I(\sigma_\partial; \sigma_\rho)$ converges to 0 as $d \rightarrow \infty$.

In the other direction, various conditions are derived in [29, 30, 34, 38] that imply that $I(\sigma_\rho; \sigma_\partial)$ converges to 0 for various processes. The simplest of these conditions is given in reference [34]—this condition is closely related to the one given in Theorem 14.2. The results in [29, 30, 38] give sharper bounds for symmetric processes on more than 2 states and for general 2-state processes.

Let us illustrate how Proposition 4.2 of [38] translates to the “Jukes–Cantor” setting. This proposition specialized for the Jukes–Cantor model asserts that if

$$C := \frac{2e^{-8rt}}{\frac{1}{2} + (e^{-4rt}/2)} < 1 \quad (14.16)$$

then $I(\sigma_\partial; \sigma_\rho)$ converges to 0 as $d \rightarrow \infty$. Simple algebra shows that $A \leq C \leq B$. Thus (14.16) gives a weaker condition than (14.15) (and therefore a stronger result) implying that $I(\sigma_\partial; \sigma_\rho)$ converges to 0.

14.4.1 The logarithmic conjecture

Suppose we generate k characters independently and according to the CFN model (with degenerate site specific rate distribution), and ask how large k should be in order that, with probability at least $1 - \epsilon$ we can correctly recover from these characters the topology of the underlying phylogenetic tree. Let $k_{\min}(\epsilon)$ be the smallest value of k that achieves this last property. Clearly $k_{\min}(\epsilon)$ depends on features of the generating tree, in particular the number n of leaves, and the assignment of $t(e)$ values to the edges of this tree (it also depends on ϵ , however, we will regard this as a fixed small number). Any dramatic “shortening” of an interior edge, or “lengthening” of an exterior edge (i.e. making the $t(e)$ value small or large, respectively) will cause $k_{\min}(\epsilon)$ to diverge and so we will assume that each binary phylogenetic tree has all its $t(e)$ values in some fixed interval $[l_n, u_n]$ which may depend on n . The questions of interest are then to determine the dependence of $k_{\min}(\epsilon)$ on n and the values (l_n, u_n) . Essentially this question provides another formalization of the question “how much phylogenetic information is contained in characters that evolve according to a simple Markov model.” The authors of [12] showed that,

$$k_{\min}(\epsilon) \leq c'_\epsilon \cdot \frac{\log(n)}{l_n^2} \cdot \exp(u_n \delta_n(T)), \quad (14.17)$$

where c'_ϵ is a constant (dependent only on ϵ) and $\delta_n(T)$ is a function (only) of the phylogenetic tree T and that grows slowly with n . Specifically, $\delta_n(T)$ is at most a constant times $\log(n)$, but is typically (i.e. on average) $O(\log(\log(n)))$. It is a measure of how many edges of the tree separate the “deepest” vertex from its nearest leaf.

Thus if we were to regard l_n and u_n as constants (independent of n) then $k_{\min}(\epsilon)$ is at worst polynomial in n , and more typically a power of $\log(n)$ (improving an alternative bound described in reference [15]). We have not mentioned the tree reconstruction method used to establish bound (14.17); it is a polynomial time (in $n = |X|$) algorithm, and chosen more for tractability of analysis than

for any supposed superior performance; a comparable analysis for maximum likelihood seems more difficult [52].

An obvious question arises: is the bound on $k_{\min}(\epsilon)$ described by bound (14.17) and the consequent relationship between $k_{\min}(\epsilon)$ and n (for l_n, u_n fixed) optimal? Certainly $k_{\min}(\epsilon)$ must grow at least as fast as (a constant times) $\log(n)$, by elementary counting arguments. This applies under any model of sequence evolution on a bounded state space and any tree reconstruction method [51]. The essence of this argument is the following: there are $(2n-4)!/(n-2)!2^{n-2}$ trivalent phylogenetic X -trees and r^{nk} collections consisting of n aligned sequences of length k on an r -letter alphabet, and so if $k = o(\log(n))$ then for sufficiently large n there exist more trivalent phylogenetic X -trees than r -letter sequences of length k .

Also an inverse square dependence of $k_{\min}(\epsilon)$ on l_n is necessary, even when $n = 4$, as shown in reference [52]. However, there is reason to believe that bound (14.17) is not optimal, provided that u_n is less than the critical transition value (viz. $\frac{1}{4} \log_e(2)$) between the ordered and unordered states, discussed above. This has led to the following conjecture, which promises a remarkable strengthening of bound (14.17) under a further restriction.

Conjecture 14.3 *Consider the CFN model for binary characters, and suppose that $u_n \leq u < \frac{1}{4} \log_e(2)$. Then*

$$k_{\min}(\epsilon) \leq c_{\epsilon, u} \cdot \frac{\log(n)}{l_n^2},$$

where $c_{\epsilon, u}$ is a constant that depends only on ϵ and u .

Conjecture 14.3 is clearly true for trees for which $\delta_n(T)$ is bounded—these are trees for which no vertex is very far from a leaf (e.g. the class of “caterpillar trees” which are the trivalent phylogenetic trees for which every interior vertex is adjacent to a leaf). However, for trees that have “deep” vertices such as the complete balanced binary phylogenetic tree that has all its $n = 2^m$ leaves at distance m from a fixed central edge, bound (14.17) is polynomial in n . Yet precisely in this “worst case” setting the bound promised by Conjecture 14.3 holds—this was recently established in reference [33], using an entirely different approach from [12]. The paper [33] also showed that the restriction on u_n is necessary for Conjecture 14.3 to hold, for when u_n is allowed to take larger values, polynomial dependence of $k_{\min}(\epsilon)$ on n can result.

Conjecture 14.3 has been extended to a much more general conjecture in reference [33] concerning the transition from logarithmic to polynomial dependence of $k_{\min}(\epsilon)$ on n for a range of Markov models at the corresponding transition from the ordered to unordered phase of the process.

14.4.2 Reconstructing forests

Given that it may be difficult to reconstruct deep parts of a tree (e.g. in the region where Conjecture 14.3 does not apply) a more modest task may be to reconstruct

most of the tree that is not “deep.” A natural question then is whether this can be achieved using a small (logarithmic in n) number of sites.

Note that for the rooted binary tree on 2^m leaves, where all the leaves are at distance m from the root, only an $O(2^{-s})$ fraction of the vertices is at distance s or more from the set of leaves. This is true in general for binary trees—only a $O(2^{-s})$ of the vertices are at distance s or more from the set of leaves. In other words, for all binary trees the “deep” part consists of exponentially small fraction of the tree. Therefore reconstructing the part which is not “deep” still contains a lot of information on (recent) divergences.

It turns out that the answer to the above question is positive. In reference [37] it is shown that a logarithmic number of characters suffices to reconstruct a forest containing most edges of the tree. Moreover, Mossel [37] gives a formula relating the “depth” of the forest that can be recovered from a given number of characters.

14.5 Processes on an unbounded state space: the random cluster model

For the remainder of this chapter we will investigate the phylogenetic information that is provided by models which have a large state space. In this section, we deal with a slightly idealized “random cluster” model, in which the underlying state space might be regarded as being infinite—it has the property that any substitution always gives rise to a new state. We will see that this simple model is quite tractable and leads to a result (Theorem 14.4) that is much cleaner than anything that has been established yet for even the CFN model. We will apply this result in the final section of this chapter to investigate a class of models on large but finite state spaces.

For the type of Markov model on a small state space that we have dealt with so far the subsets of the vertices of a phylogenetic tree T that are assigned particular states do not generally form connected subtrees of T (in biological terminology this is because of “homoplasy”—the evolution of the same state more than once in the tree, either due to reversals or convergent evolution).

However, increasingly there is interest in genomic characters such as gene order where the underlying state space may be very large [18, 31, 32, 44]. For example, the order of k genes in a signed circular genome can take any of $2^k(k-1)!$ values. In these models whenever there is a change of state—for example a re-shuffling of genes by a random inversion (of a consecutive subsequence of genes)—it is likely that the resulting state (gene arrangement) is a unique evolutionary event, arising for the first time in the evolution of the genes under study. Indeed Markov models for genome rearrangement such as the (generalized) Nadeau–Taylor model [31, 41] confer a high probability that any given character generated is homoplasy-free on the underlying tree, provided the number of genes is sufficiently large relative to $n = |X|$ [45]. Here the phrase “homoplasy-free” refers to the condition that a character has parsimony score equal to the number

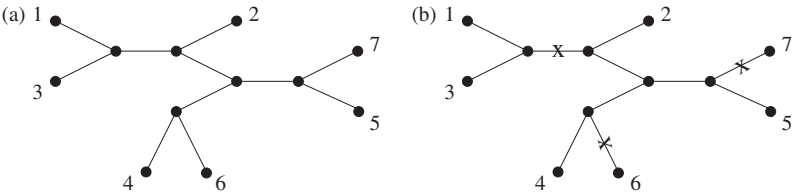


FIG. 14.3. (a) A trivalent phylogenetic X -tree T for $X = \{1, 2, \dots, 7\}$; (b) For the random cluster model, cutting the edges of T that are marked by a cross induces the character χ on X given by $\chi = \{\{1, 3\}, \{2, 4, 5\}, \{6\}, \{7\}\}$.

of states it takes at the leaves minus 1; this condition has a natural interpretation in biology, since it is equivalent to requiring that the character could have evolved on the tree without reversals or convergent evolution (for details of that connection see [45, 46]). In this setting a “random cluster” model which we will describe here is the appropriate (limiting case) model, and may be viewed as the phylogenetic analogue of what is known in population genetics as the “infinite alleles model” of Kimura and Crow [27].

Thus for this section we consider the size of the state space to be infinite (or at least very large, and perhaps variable with n). Some of the arguments described above are no longer valid in this setting. For example, the simple argument in Section 14.4.1 that showed that $k_{\min}(\epsilon)$ must grow at least as fast as the function $\log(n)$ does not apply when the size of state space is infinite, or finite but variable with $n = |X|$. Indeed it has recently been shown that for any trivalent phylogenetic X -tree T there is an associated set of just *four* characters for which T is the only phylogenetic X -tree on which each character in that collection has a homoplasy-free evolution (see [24, 45]). Thus it is reasonable to ask whether $O(1)$ characters might suffice to reconstruct T under a simple random model. We will see that the answer to this question is “no,” but clearly we need a different type of argument.

Consider the following random process on a phylogenetic tree T . For each edge e let us independently either cut this edge—with probability $p(e)$ —or leave it intact. The resulting disconnected graph (forest) G partitions the vertex set $V(T)$ of T into non-empty sets according to the equivalence relation that $u \sim v$ if u and v are in the same component of G . This model thus generates random partitions of $V(T)$, and thereby of X by connectivity, and we will denote these partitions of $V(T)$ and X using the symbols $\bar{\chi}$ and χ , respectively. Figure 14.3(b) illustrates this process.

For an element $x \in X$ we will let $\chi(x)$ denote the equivalence class containing x . We call the resulting probability distribution on partitions of X the *random cluster model* with parameters (T, p) where p is the map $e \mapsto p(e)$.

In keeping with the biological setting we will call an arbitrary partition χ of X a *character* (on X). Let $\mathbb{P}[\chi \mid T, p]$ denote the probability of generating a character χ under the random cluster model with parameters (T, p) . We say

a subset C of the set $E(T)$ of edges of T is a *cutset* for χ on T if the partition χ of X equals that induced by the components of $(V(T), E(T) - C)$. Then

$$\mathbb{P}[\chi \mid T, p] = \sum_C \prod_{e \in C} p(e) \prod_{e \in E(T) - C} (1 - p(e)), \quad (14.18)$$

where the summation is over all cutsets C for χ on T . Note that the number of terms in the summation described by equation (14.18) can be exponential with $n = |X|$. However, by modifying the well-known dynamic programming approach for computing the probability of a character on a tree according to a finite state Markov process (see e.g. [16]) one can compute $\mathbb{P}[\chi \mid T, p]$ in polynomial time in $n = |X|$.

Note that the probability distribution described by equation (14.18) models the evolution of characters under the assumptions that any substitution is always to a new state, and with independence between substitution events on different edges of the tree. We will relate this model to more explicit models of character evolution (on a finite but large state space) in the next section.

Suppose we generate a sequence $\Pi = (\chi_1, \dots, \chi_k)$ of k such independent characters on X where the generating pair (T, p) is unknown. We wish to reconstruct T with probability at least $1 - \epsilon$ from Π .

The following theorem, from reference [39], describes how the required value of k is related to the size of T and properties of p , and illustrates the logarithm-polynomial phase transition that occurs depending on whether or not the $p(e)$ values are all less than $\frac{1}{2}$.

Theorem 14.4 *Let $0 < l \leq u < 1$ and $0 < \epsilon < 1$ be fixed constants. Consider the random cluster model on any collection of the parameters (T, p) where T is a trivalent phylogenetic tree, and $l \leq p(e) \leq u$ for all edges e of T . Let k be the number of characters generated i.i.d. under this model, and $k_{\min}(\epsilon)$ be the minimal k such that the tree can be correctly reconstructed from the characters with probability at least $1 - \epsilon$. Then, if n denotes the number of leaves of T :*

(i) $k_{\min}(\epsilon)$ grows logarithmically with n if $u < \frac{1}{2}$. In particular, if

$$k \geq \frac{2(1-u)^4}{l(1-2u)^4} \log \left(\frac{n}{\sqrt{\epsilon}} \right),$$

then the tree can be reconstructed correctly with probability $1 - \epsilon$. Furthermore, there is a polynomial-time (in n) algorithm for reconstructing T from the generated characters.

(ii) $k_{\min}(\epsilon)$ can grow polynomially with n if $l > \frac{1}{2}$. In particular, for all h , if

$$k \leq \frac{\epsilon(1-l)^h}{6} \left(\frac{n}{3} \right)^{-\log_2(2-2l)}, \quad (14.19)$$

then there exists a distribution on trivalent phylogenetic X -trees, such that if T is drawn according to the distribution, $p(e) = l$, for all edges of the trees, and characters are generated by (T, p) , then the probability

of correctly reconstructing T given the k characters is bounded above by $\epsilon + 3^{-3 \times 2^h}$.

Theorem 14.4 shows that the situation with the random cluster model differs from a bounded-state spaces model such as the CFN model, in two respects. First, the critical value is $\frac{1}{2}$ instead of $\frac{1}{2}(1 - 1/\sqrt{2})$. This corresponds to the fact that in statistical physics models on the binary tree, the critical value for the extremality of the free measure or the Ising model is $\frac{1}{2}(1 - 1/\sqrt{2})$, see [3, 13, 35], while the critical value for uniqueness of Gibbs measure, or the critical value for percolation is $\frac{1}{2}$, see [21, 43]. In reference [36] it is shown that for any Markov model, if the substitution rate is high then k depends polynomially on n .

The second respect in which the random cluster model differs from the symmetric two state model, is that for the random cluster model, the dependence of k on l has exponent -1 rather than -2 , see [12, 33, 52].

We now provide a brief outline of the proof of part (i) of Theorem 14.4, since it combines a combinatorial result with a probabilistic percolation argument; full details can be found in reference [39]. Recall that a *quartet tree* is a trivalent phylogenetic X -tree for $|X| = 4$. We can represent any quartet tree by the notation $xy \mid wz$ where x, y are leaves that are adjacent to one interior vertex, while w, z are leaves that are adjacent to the other interior vertex. For any trivalent phylogenetic X -tree, T let $\mathcal{Q}(T)$ denote the set of quartet trees induced by T by selecting subsets of X of size 4. It is a fundamental result from reference [7] that T is uniquely determined by $\mathcal{Q}(T)$. Suppose that T is a trivalent phylogenetic X -tree. We say that T *displays* a quartet tree $xy \mid wz$ (respectively, a set $\mathcal{Q}$ of quartet trees) if $xy \mid wz \in \mathcal{Q}(T)$ (respectively, if $\mathcal{Q} \subseteq \mathcal{Q}(T)$). For example, the tree T in Fig. 14.3(a) displays the quartet tree $12 \mid 47$. For any three distinct vertices a, b, c of T let $\text{med}(a, b, c)$ denote the *median vertex* of the triple a, b, c ; that is, the unique vertex of T that is shared by the paths connecting a and b , a and c , and b and c .

A collection $\mathcal{Q}$ of quartet trees is a *generous cover* of T if $\mathcal{Q} \subseteq \mathcal{Q}(T)$ and if, for all pairs of interior vertices u, v there exists a quartet tree $xx' \mid yy' \in \mathcal{Q}$ for which $u = \text{med}(x, x', v)$ and $v = \text{med}(u, y, y')$. Given a sequence $\mathcal{C} = (\chi_1, \chi_2, \dots, \chi_k)$ of characters on X , let

$$\mathcal{Q}(\mathcal{C}) = \{xx' \mid yy' : \exists i \in \{1, \dots, k\} : \chi_i(x) = \chi_i(x') \neq \chi_i(y) = \chi_i(y')\}.$$

One of the main steps in the proof of Theorem 14.4 is to establish the following purely combinatorial result:

Proposition 14.5 *If $\mathcal{Q}$ is a generous cover of a trivalent phylogenetic X -tree T then T is the only phylogenetic X -tree that displays $\mathcal{Q}$.*

Using a percolation-style argument, one can then show that provided k is at least as large as that specified in Theorem 14.4(i), an i.i.d sequence $\mathcal{C}$ of k characters will, with probability at least $1 - \epsilon$ have the property that $\mathcal{Q}(\mathcal{C})$ is a generous cover of T . In fact, Theorem 14.4(i) remains valid (using only a slight modification of the proof) even if the i.i.d. condition is weakened to requiring only

that the k characters are generated independently, with possibly different choices of p , provided that for each character the associated p satisfies the constraint $0 < l \leq p(e) \leq u < \frac{1}{2}$ for each edge e .

A further extension of Proposition 14.5, along with a simulation-based study of the random cluster model has been described recently by Dezulian and Steel [9].

14.6 Large but finite state spaces

Finally, we turn to the question of how many characters one needs to reconstruct a large tree if the characters evolve under a Markov model on a large but finite state space. The results of this section are based on reference [49] where further details can be found.

As mentioned earlier, many processes involving simple reversible models of change can be modelled by a random walk on a regular graph. To explain this connection, suppose there are certain “elementary moves” that can transform each state into some “neighbouring” states. In this way we can construct a graph from the state space, by placing an edge between state α and state β precisely if it is possible to go from either state to the other in one elementary move. The graph so obtained is said to be *regular*, or more specifically *d-regular* if each state is adjacent to the same number d of neighbouring states.

For example, aligned sequences of length N under the r -state Poisson model can be regarded as a random walk on the set of all sequences of length N over R ; here an elementary move involves changing the state at any one position to some other state (chosen uniformly at random from the remaining $r - 1$ states). Thus the associated graph has r^N vertices and it is $(r - 1)N$ -regular.

As another example, consider a simple model of (unsigned) genome rearrangement where the state space consists of all permutations of length N (corresponding to the order of genes $1, \dots, N$) and an elementary move consists of an inversion of the order of the elements of the permutation between positions i and j , where this pair is chosen uniformly at random from all such pairs between $\{1, \dots, N\}$. In this case the state space has size $N!$ and the graph is d -regular for $d = \binom{N}{2}$.

Both of the graphs we have just described have more structure than mere d -regularity. To describe this we recall the concept of a Cayley graph. Suppose we have a (non-abelian or abelian) group $\mathcal{G}$ together with a subset S of elements, with the properties that $1_{\mathcal{G}} \notin S$ and $s \in S \Rightarrow s^{-1} \in S$. Then the *Cayley graph* associated with the pair $(\mathcal{G}, S)$ has vertex set $\mathcal{G}$ and an edge connecting g and g' whenever there exists some element $s \in S$ for which $g = g' \cdot s$. To recover the above graph on aligned sequences of length N over an r -letter alphabet, we may take $\mathcal{G}$ as the (abelian) group $(\mathbb{Z}_r)^N$ (the product of N copies of the cyclic group of order r) and the set S of all N -tuples that are the identity element of $\mathbb{Z}_r$ except on any one coordinate. To recover the graph described above for unsigned genome rearrangements we may take $\mathcal{G}$ to be the (non-abelian) group consisting of all $N!$ permutations on N letters and S to be the elements corresponding to inversions.

The demonstration that such graphs are Cayley graphs has an important consequence—it implies that they also have the following property. A graph G is said to be *vertex-transitive* if, for any two vertices u and v there is an automorphism of G that maps u to v . Informally, a graph is vertex-transitive if it “looks the same, regardless of which vertex one is standing at.” Clearly a (finite) vertex-transitive graph must be d -regular for some d , and it is an easy and standard exercise to show that every Cayley graph is vertex-transitive (however not every vertex-transitive graph is a Cayley graph, and not every regular graph is vertex-transitive).

Suppose that R is a group, and for some subset S (closed under inverses and not containing the identity element of R) we have a rate matrix Q (as discussed earlier) for which $Q_{\alpha\beta} = q$ if and only if there exists some element $s \in S$ for which $\beta = \alpha \cdot s$, otherwise for any distinct pair α, β we have $Q_{\alpha\beta} = 0$. Such a process we will call a *group walk process (on the generating set S)*. Group walk processes have a useful property on trees: for each arc $e = (u, v)$ of $T = (V, E)$ consider the event $\Delta(e)$ that the state that occurs at v is different from the state that occurs at u (i.e. there has been a net transition across the edge). Then using the fact that the Cayley graph for (R, S) is vertex transitive, it is easily shown that the events $(\Delta(e), e \in E)$ are independent.

For these models we then ask how likely it is that a character evolves without homoplasy on a tree. This question has been investigated for the 2-state Poisson model (and pairs of taxa) by Chang and Kim [4]. Here we consider more general processes on a larger state space, and for many taxa—consequently we obtain bounds rather than the exact expressions that are possible in the simpler setting of [4]. The proof of the following lemma is straightforward (for details, see [49]).

Lemma 14.6 *Let $(X_t; t \geq 0)$ be a group walk process on generating set of size d . Then, for any two distinct states α, β , and any values $s, t \geq 0$,*

$$\mathbb{P}[X_{t+s} = \beta \mid X_t = \alpha] \leq \frac{1}{d}.$$

The following result shows that for a group-walk process if the size of the generating set is much larger than $2n^2$ (where n is the number of species) then any character generated on a tree with n species will almost certainly be homoplasy-free on that tree.

Proposition 14.7 *Suppose characters evolve on a phylogenetic tree T according to a group walk process on a generating set of size d . Let $p(T)$ denote the probability that the resulting randomly generated character χ is homoplasy-free on T . Then*

$$p(T) \geq 1 - \frac{(2n-3)(n-1)}{d},$$

where $n = |X|$.

Proof Consider a general Markov process on T with state space R . Suppose that for each arc (u, v) of T and each pair α, β of distinct states in R , the conditional probability that state β occurs at v given that α occurs at u is at most p . Then, from reference [45] (Proposition 7.1) we have $p(T) \geq 1 - (2n - 3)(n - 1)p$. By Lemma 14.6 we may take $p = 1/d$. The result now follows. $\square$

We are now ready to state a result for certain Markov processes on large (but finite!) state spaces, which brings together several ideas presented above. Informally, Theorem 14.8 states that for a group walk process, a growth of around $n^2 \log(n)$ in the size of the generating set is sufficient (with all else held constant) for producing a sequence of homoplasy-free characters that define T .

Let $p_{\min} = \min\{\mathbb{P}[\Delta(e)] : e \in E\}$, $p_{\max} = \max\{\mathbb{P}[\Delta(e)] : e \in E\}$, and for any $\epsilon > 0$ let

$$c_\epsilon = \frac{1 + \log(1/\sqrt{\epsilon})}{\beta\epsilon} \quad (14.20)$$

where $\beta = p_{\min}(1 - 2p_{\max})^4 / (1 - p_{\max})^4$.

Theorem 14.8 *Suppose characters evolve i.i.d. on a binary phylogenetic tree T according to a group walk process on a generating set of size d , where*

$$d \geq c_\epsilon \cdot n^2 \log(n)$$

with c_ϵ given by equation (14.20) and with $p_{\max} < \frac{1}{2}$. Then with probability at least $1 - 2\epsilon$ we can correctly reconstruct the topology of T by generating an appropriate (and logarithmic in n) number of characters.

Proof Let us generate $k = \lceil 2\beta^{-1} \log(n/\sqrt{\epsilon}) \rceil$ characters under a group walk process on a rooted phylogenetic tree. Consider the event H that all of these characters are homoplasy-free on T . By Proposition 14.7 we have

$$\mathbb{P}[H] \geq \left(1 - \frac{(2n - 3)(n - 1)}{d}\right)^k \geq \left(1 - \frac{2}{c_\epsilon \log(n)}\right)^{2\beta^{-1} \log(n/\sqrt{\epsilon})}.$$

Now, applying the inequality $(1 - x)^y \geq 1 - xy$ for $x, y > 0$, and straightforward algebra gives

$$\mathbb{P}[H] \geq 1 - \frac{\epsilon(1 + \log(1/\sqrt{\epsilon}))^{-1}}{\log(n)} \left[\log(n) + \log\left(\frac{1}{\sqrt{\epsilon}}\right) \right] \geq 1 - \epsilon$$

using $\log(n) \geq 1$.

To relate the group walk process to the random cluster model we use a simple coupling argument. First consider any linear extension of the partial order induced by T on its vertices—that is, impose a timescale on the tree. To each vertex v assign the pair (α, i) where α is the state of the group walk process at v , and $i \in \{0, 1, 2, \dots\}$ indicates how often this state has arisen from another state earlier in the tree. Note that this model always generates new states (it is

homoplasy-free). Furthermore, let $A(e)$ denote the event that the two vertices in e have different states under the coupled process. Notice that $A(e)$ occurs precisely when the two vertices in e have different states under the original process. Thus, the collection $\{A(e) : e \in E\}$ is a collection of independent events. Consequently this coupled process is precisely the random cluster model, and we may identify β with the expression $l(1 - 2u)^4 / (1 - u)^4$ in Theorem 14.4(i). Furthermore the probability that T will be correctly reconstructed from k characters produced by the coupled process is at least $1 - \epsilon$ by Theorem 14.4 (recalling that $p_{\max} < \frac{1}{2}$).

On the other hand the original k characters induce the same partitions as the derived characters whenever event H holds, and $\mathbb{P}[H] \geq 1 - \epsilon$. Consequently, by the Bonferroni inequality, the joint probability that event H holds and that the k characters produced by the coupled process recovers T is at least $1 - 2\epsilon$. Thus the probability that the original k characters recover T is at least this joint probability, and so at least $1 - 2\epsilon$, as claimed. $\square$

We end this section with a remark of caution. Theorem 14.8 suggests a possible (but flawed) way to avoid the well-known statistical inconsistency of phylogenetic reconstruction methods such as maximum parsimony or maximum compatibility under simple models of site substitution. This approach is based on grouping sites into s -tuples. For example, if the underlying process is the symmetric 2-state model, then if we take s -tuples of sites we obtain a group walk process on a generating set of size $2^s - 1$. So by taking s large enough Theorem 14.8 suggests that methods like compatibility (or maximum parsimony) should be able to correctly reconstruct a tree with high probability. The flaw in this argument is that as s increases, so too does $p_{\max}$ and once $p_{\max}$ exceeds $\frac{1}{2}$, Theorem 14.8 is no longer valid. In fact one can show that for the 2-state symmetric model and a tree with four leaves, the branch length requirements for the statistical consistency of maximum parsimony on site data is identical to that on s -tuple site data (theorem 3A of [50]).

14.7 Concluding comments

Stochastic models of character evolution have come to play a central role in phylogenetic genomics. In this chapter, we have investigated various aspects of the broad question: how much information do characters tell us about their evolutionary history? There are two ways to investigate this issue—one approach that is popular in biology and bioinformatics is to use simulation (e.g. generating sequences on a tree and then comparing the tree that is reconstructed from these sequences with the original tree) and several authors have taken this approach (e.g. [6]). Here we have adopted an analytical approach, that gives provable and general bounds on quantities of interest. This approach has both advantages and disadvantages over a simulation-based one. Its main disadvantage is that the bounds we obtain may sometimes be far from being “exact,” and in that case simulations provide a useful way to detect and quantify this (as was undertaken for the random cluster model in reference [9]). However, analytic

approaches have some unique advantages: the results are generic and not specific to particular parameter settings in the simulation runs; indeed some of the bounds we presented apply to *any* tree reconstruction method, something that would be difficult or impossible to investigate using simulations. Furthermore, analytic results can show precisely the functional dependency between quantities of interest (e.g. the dependence of k on $\log(n)$ in the random cluster model). Analytic approaches have one more virtue—the arguments used provide insight into how information is preserved and lost, and what methods might be most effective in recovering it.

Turning to possible future work, we expect that the conjecture(s) outlined in Section 14.4.1 will likely be resolved in the near future. Another challenging mathematical task would be to provide explicit lower bounds on the tree reconstruction probability for maximum likelihood—the only known bounds (from a more general result in reference [52] that is not specific to phylogeny) involve exponential dependence of k on n and can likely be improved considerably in the phylogenetic setting. For applications, it would also be useful to be able to estimate reconstruction probabilities (or the sequence length required to achieve a given reconstruction probability) from data, perhaps by exploiting bootstrap statistics. Although there has been some empirical approaches to this problem (e.g. [28]) little has been rigorously established.

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